Google Ranking Rules

Every ranking signal and patent-derived rule in this repository, in one document

848 defensible ranking signals (58 core, 790 supporting) · 1,047 contextual signals · 5,456 patent rules · 27 key patents

Sources: 2024 Google Content Warehouse API leak (14,022 attributes), 5,456 Google search patents (77,663 extracted claims)

How to read this document

Signals (Parts I to III) come from the leak. Each entry gives the stored field as **`Module.field`**, the rule in plain language (the classified meaning), Google's own internal description where one exists, and the pipeline stages the module feeds. Within each category, high-confidence classifications are listed first.

Patent rules (Parts IV and V) are the SEO action derived from each patent. A patent proves Google described a method, not that it ships, so every rule carries its filing and grant dates. Full claims for any patent are available with **`python3 scripts/explain.py <patent number>`**.

Tier counts across all leak attributes

Tier	Attributes
core	58
supporting	790
contextual	1,047
not_a_signal	7,580
unclassified	4,547

Not-a-signal and unclassified attributes are infrastructure or undocumented fields and are not rules; search them with scripts/search_signals.py.

The evidence rules

1. Never assert a ranking behavior without citing a field, a patent, or both.

2. An undocumented attribute proves existence, not function. Entries below marked "no Google description" are existence-only evidence.

3. No weights leaked. Any impact percentage anyone quotes is invented.

4. A patent proves Google described a method, not that the method ships. Check the filing date.

5. Say when you do not know. What you ruled out is part of the deliverable.

Limitations

The leak is a 2024 snapshot of an internal API surface; fields may be experimental or deprecated. Tiers and plain meanings are a model classification (91.8% agreement with an independent second pass on signal versus not-signal, 79.6% on exact tier; see data/classification_report.md). Patent rules are model-written paraphrases; pull the original from Google Patents before quoting.

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Part I. Core ranking signals (58)

The fields the classification judged most directly tied to ranking. Start here.

Spam & demotions (29)

`CompressedQualitySignals.exactMatchDomainDemotion`
Rule: A stored demotion for domains that exactly match the query words.
Google's description: Page quality signals converted from fields in proto QualityBoost in quality/q2/proto/quality-boost.proto. To save indexing space, we convert the float values in [0, 1] to integers in range [0, 1023] (use 10 bits). exact_match_domain_demotion: converted from QualityBoost.emd.boost.
type integer() · confidence high · stages: 3. Content Quality & Site-Level Scoring; 8. Spam, Penalty & Trust Systems; 11. Re-Ranking & Twiddler System

`CompressedQualitySignals.navDemotion`
Rule: A stored demotion for pages flagged by the navigational demotion system.
Google's description: nav_demotion: converted from QualityBoost.nav_demoted.boost.
type integer() · confidence high · stages: 3. Content Quality & Site-Level Scoring; 8. Spam, Penalty & Trust Systems; 11. Re-Ranking & Twiddler System

`CompressedQualitySignals.scamness`
Rule: A model score for how scam-like the page looks, used in quality scoring.
Google's description: Scam model score. Used as one of the web page quality qstar signals. Value range from 0 to 1023.
type integer() · confidence high · stages: 3. Content Quality & Site-Level Scoring; 8. Spam, Penalty & Trust Systems; 11. Re-Ranking & Twiddler System

`CompressedQualitySignals.serpDemotion`
Rule: A demotion value applied to the page during Q* scoring.
Google's description: serp demotion: applied in Qstar.
type integer() · confidence high · stages: 3. Content Quality & Site-Level Scoring; 8. Spam, Penalty & Trust Systems; 11. Re-Ranking & Twiddler System

`IndexingDocjoinerAnchorSpamInfo.spamPenalty`
Rule: The overall penalty applied for anchor spam.
Google's description: Combined penalty for anchor demotion.
type number() · confidence high · stages: 4. Link Analysis & Authority Computation; 8. Spam, Penalty & Trust Systems

`IndexingDocjoinerAnchorStatistics.badbacklinksPenalized`
Rule: Whether the page was penalized for bad backlinks, which suppresses its anchor score.
Google's description: Whether this doc is penalized by BadBackLinks, in which case we should not use improvanchor score in mustang ascorer.
type boolean() · confidence high · stages: 4. Link Analysis & Authority Computation

`IndexingDocjoinerAnchorStatistics.penguinPenalty`
Rule: The page-level Penguin link spam penalty, 0 good to 1 bad.
Google's description: Page-level penguin penalty (0 = good, 1 = bad).
type number() · confidence high · stages: 4. Link Analysis & Authority Computation

`PerDocData.DocLevelSpamScore`
Rule: A 0-127 spam score for the individual document.
Google's description: The document spam score is represented as a 7-bits, going from 0 to 127.
type integer() · confidence high · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.KeywordStuffingScore`
Rule: A 0-127 score for how much the page repeats keywords unnaturally.
Google's description: The keyword stuffing score is represented in 7 bits, going from 0 to 127.
type integer() · confidence high · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.SpamWordScore`
Rule: A 0-127 score for how many spammy words appear on the page.
Google's description: The spamword score is represented in 7-bits, going from 0 to 127.
type integer() · confidence high · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.spambrainData`
Rule: SpamBrain scores computed at the host level for this site.
Google's description: Host-v1 sitechunk level scores coming from spambrain.
type GoogleApi.ContentWarehouse.V1.Model.SpamBrainData.t · confidence high · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.spambrainDomainSitechunkData`
Rule: SpamBrain scores computed at the domain level for this site.
Google's description: Domain sitechunk level scores coming from spambrain.
type GoogleApi.ContentWarehouse.V1.Model.SpamBrainData.t · confidence high · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.spambrainTotalDocSpamScore`
Rule: SpamBrain's overall spam score for the document, from 0 to 1.
Google's description: The document total spam score identified by spambrain, going from 0 to 1.
type number() · confidence high · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.uacSpamScore`
Rule: A 0-127 spam score where 64 or more marks the page as UAC spam.
Google's description: The uac spam score is represented in 7 bits, going from 0 to 127. Threshold is 64. Score >= 64 is considered as uac spam.
type integer() · confidence high · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PornFlagData.pornWithHighConfidence`
Rule: A flag marking an image as confidently pornographic, which can demote it on non-porn queries.
Google's description: This field is set to true when we are pretty confident that the image is porn (with higher precision than the img_porn_moderate restrict). In particular, it means that the image might be demoted for non-porn queries when SafeSearch is Off.
type boolean() · confidence high

`QualityNsrNsrData.clutterScore`
Rule: A site-level penalty for loading many distracting or annoying extra resources.
Google's description: Delta site-level signal in Q* penalizing sites with a large number of distracting/annoying resources loaded by the site (see go/clutter-v0).
type number() · confidence high · stages: 3. Content Quality & Site-Level Scoring; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityNsrNsrData.spambrainLavcScore`
Rule: A site-level spam score produced by the SpamBrain system.
Google's description: The SpamBrain LAVC score, as of July 2022. See more information at go/cloverfield-lavc-deck.
type number() · confidence high · stages: 3. Content Quality & Site-Level Scoring; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`SmartphonePerDocData.violatesMobileInterstitialPolicy`
Rule: Whether the page breaks the mobile interstitial rules and should be demoted.
Google's description: Indicates if the page is violating mobile interstitial policy and should be demoted. See go/interstitials-ranking-dd for details.
type boolean() · confidence high · stages: 12. Personalization, Local & Context

`IndexingDocjoinerAnchorPhraseSpamInfo.phraseAnchorSpamPenalty`
Rule: The overall penalty applied for anchor spam.
Google's description: Combined penalty for anchor demotion.
type number() · confidence medium · stages: 4. Link Analysis & Authority Computation

`PerDocData.GibberishScore`
Rule: A 0-127 score for how much of the page reads as nonsense text.
Google's description: The gibberish score is represented in 7 bits, going from 0 to 127.
type integer() · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.IsAnchorBayesSpam`
Rule: Flag saying the anchor-text Bayes classifier judged this document spam.
Google's description: Is this document considered spam by the anchor bayes classifier?
type boolean() · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.ScaledLinkAgeSpamScore`
Rule: A 0-127 score based on the age of links pointing at the page.
Google's description: End DEPRECATED ---------- Link age score is represented as a 7-bit integer, going from 0 to 127.
type integer() · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.ScaledSpamScoreYoram`
Rule: A 0-127 spam score for the document under the Yoram scheme.
Google's description: Spamscores are represented as a 7-bit integer, going from 0 to 127.
type integer() · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.spamCookbookAction`
Rule: Actions triggered when the page matches a spam Cookbook recipe.
Google's description: Actions based on Cookbook recipes that match the page.
type GoogleApi.ContentWarehouse.V1.Model.SpamCookbookAction.t · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.spamrank`
Rule: A 0-65535 measure of how likely the page is to link to known spammers.
Google's description: The spamrank measures the likelihood that this document links to known spammers. Its value is between 0 and 65535.
type integer() · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.spamtokensContentScore`
Rule: A content score used to decide whether a page is user-generated-content spam.
Google's description: For SpamTokens content scores. Used in SiteBoostTwiddler to determine whether a page is UGC Spam. See go/spamtokens-dd for details.
type number() · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.urlPoisoningData`
Rule: Data about URL poisoning used to hold back spam documents.
Google's description: Contains url poisoning data for suppressing spam documents.
type GoogleApi.ContentWarehouse.V1.Model.UrlPoisoningData.t · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`SpamBrainData.versionedData`
Rule: The spam classifier scores for a result, one per model version.
Google's description: Versioned scores of SB classifiers
type list(GoogleApi.ContentWarehouse.V1.Model.SpamBrainScore.t) · confidence medium · stages: 8. Spam, Penalty & Trust Systems

`SpamBrainScore.sbScore`
Rule: SpamBrain's spam score for this document under a given model version.
Google's description: The value corresponding to this version.
type number() · confidence medium · stages: 8. Spam, Penalty & Trust Systems

Content quality (13)

`CompressedQualitySignals.babyPandaDemotion`
Rule: A stored quality demotion value in the Panda family.
Google's description: baby_panda_demotion: converted from QualityBoost.rendered.boost.
type integer() · confidence high · stages: 3. Content Quality & Site-Level Scoring; 8. Spam, Penalty & Trust Systems; 11. Re-Ranking & Twiddler System

`CompressedQualitySignals.babyPandaV2Demotion`
Rule: A newer quality demotion applied on top of the Panda demotion.
Google's description: New BabyPanda demotion, applied on top of Panda. This is meant to replace |baby_panda_demotion|.
type integer() · confidence high · stages: 3. Content Quality & Site-Level Scoring; 8. Spam, Penalty & Trust Systems; 11. Re-Ranking & Twiddler System

`CompressedQualitySignals.lowQuality`
Rule: A low quality score for the site, applied during Q* scoring.
Google's description: S2V low quality score: converted from quality_nsr.NsrData, applied in Qstar. See quality_nsr::util::ConvertNsrDataToLowQuality.
type integer() · confidence high · stages: 3. Content Quality & Site-Level Scoring; 8. Spam, Penalty & Trust Systems; 11. Re-Ranking & Twiddler System

`CompressedQualitySignals.pandaDemotion`
Rule: The encoded Panda site quality demotion for the page.
Google's description: This is the encoding of Panda fields in the proto SiteQualityFeatures in quality/q2/proto/site_quality_features.proto. The encoding/decoding is performed using functions from quality_coati::coati_util.
type integer() · confidence high · stages: 3. Content Quality & Site-Level Scoring; 8. Spam, Penalty & Trust Systems; 11. Re-Ranking & Twiddler System

`CompressedQualitySignals.unauthoritativeScore`
Rule: A score for how unauthoritative the page looks, used in quality scoring.
Google's description: Unauthoritative score. Used as one of the web page quality qstar signals.
type integer() · confidence high · stages: 3. Content Quality & Site-Level Scoring; 8. Spam, Penalty & Trust Systems; 11. Re-Ranking & Twiddler System

`PerDocData.OriginalContentScore`
Rule: A score for how original the page's content is, stored for pages with little content.
Google's description: The original content score is represented as a 7-bits, going from 0 to 127. Only pages with little content have this field. The actual original content score ranges from 0 to 512. It is encoded with quality_q2::OriginalContentUtil::EncodeOriginalContentScore(). To decode the value, use quality_q2::OriginalContentUtil::DecodeOriginalContentScore().
type integer() · confidence high · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityNsrNsrData.nsrOverrideBid`
Rule: A manual value that replaces the site quality score in ranking when set.
Google's description: This signal is used to unconditionally override NSR as a bid in Q*. Should only be used in case of emergency (see go/nsr-override-bid). To have any effect, the value should be present and greater than 0.001.
type number() · confidence high · stages: 3. Content Quality & Site-Level Scoring; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityNsrNsrData.smallPersonalSite`
Rule: A promotion score for small personal sites and blogs.
Google's description: Score of small personal site promotion go/promoting-personal-blogs-v1
type number() · confidence high · stages: 3. Content Quality & Site-Level Scoring; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityNsrPQData.contentEffort`
Rule: A model's estimate of how much effort went into creating the page's content.
Google's description: LLM-based effort estimation for article pages (see landspeeder/4311817).
type list(GoogleApi.ContentWarehouse.V1.Model.QualityNsrVersionedFloatSignal.t) · confidence high · stages: 3. Content Quality & Site-Level Scoring

`WWWDocInfo.qualityWithoutAdjustment`
Rule: The document's quality score before adjustments.
Google's description: Quality score (also known as QScore, see go/qscore-faq).
type number() · confidence high

`CompressedQualitySignals.productReviewPDemoteSite`
Rule: A confidence value used to demote or promote a site for product reviews.
Google's description: Product review demotion/promotion confidences. (Times 1000 and floored)
type integer() · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 8. Spam, Penalty & Trust Systems; 11. Re-Ranking & Twiddler System

`CompressedQualitySignals.ugcDiscussionEffortScore`
Rule: A score for the effort behind user-generated discussion content on the page.
Google's description: UGC page quality signals. (Times 1000 and floored)
type integer() · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 8. Spam, Penalty & Trust Systems; 11. Re-Ranking & Twiddler System

`QualityProductProductSiteDataLocaleData.boostFactor`
Rule: A multiplier used to boost a shopping site.
Google's description: Site boosting multiplier.
type number() · confidence medium

Links & authority (6)

`CompressedQualitySignals.anchorMismatchDemotion`
Rule: A stored demotion value for pages whose anchor text does not match the page.
Google's description: anchor_mismatch_demotion: converted from QualityBoost.mismatched.boost.
type integer() · confidence high · stages: 3. Content Quality & Site-Level Scoring; 8. Spam, Penalty & Trust Systems; 11. Re-Ranking & Twiddler System

`CompressedQualitySignals.authorityPromotion`
Rule: A stored promotion value that raises pages judged authoritative.
Google's description: authority promotion: converted from QualityBoost.authority.boost
type integer() · confidence high · stages: 3. Content Quality & Site-Level Scoring; 8. Spam, Penalty & Trust Systems; 11. Re-Ranking & Twiddler System

`CompressedQualitySignals.siteAuthority`
Rule: A site-level authority score applied during Q* scoring.
Google's description: site_authority: converted from quality_nsr.SiteAuthority, applied in Qstar.
type integer() · confidence high · stages: 3. Content Quality & Site-Level Scoring; 8. Spam, Penalty & Trust Systems; 11. Re-Ranking & Twiddler System

`GDocumentBase.PagerankNS`
Rule: The production PageRank score for the page, computed by the nearest seeds method.
Google's description: Pagerank-NearestSeeds is a pagerank score for the doc, calculated using NearestSeeds method. This is the production PageRank value teams should use.
type integer() · confidence high · stages: 2. Document Processing & Indexing

`CompositeDocForwardingDup.rawPagerank`
Rule: The raw pagerank of the duplicate address.
Google's description: Raw pagerank of the url.
type integer() · confidence medium

`PerDocData.homepagePagerankNs`
Rule: The pagerank of the site's homepage, copied onto this document.
Google's description: The page-rank of the homepage of the site. Copied from the cdoc.doc().pagerank_ns() of the homepage.
type integer() · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

Images & video (5)

`ImageContentQueryBoostQueryBoost.boost`
Rule: A multiplier that raises or lowers an image's score for a particular query.
Google's description: Score multiplier (fully normalized 1 is nop).
type number() · confidence medium

`WWWResultInfoSubImageDocInfo.dimeScore`
Rule: A model score for how engaging an image is.
Google's description: EQ* v3: Deep Image Engagingness(DImE) go/deep-image-engagingness-slides
type number() · confidence medium

`WWWResultInfoSubImageDocInfo.eqStar`
Rule: A score for the emotional quality of an image, such as how inspiring it looks.
Google's description: EQ* is a unified signal to capture the emotional quality (e.g. inspiration, lifestyle, context, etc.) of an image. For more information, please refer to go/image-inspiration-ranking-framework.
type number() · confidence medium

`WWWResultInfoSubImageDocInfo.estRelevance`
Rule: An estimate of how useful the image is for the query, from off-topic to very useful.
Google's description: Estimated Image Relevance ranging between 0.0 (Off-Topic) to 1.0 (Very Useful).
type number() · confidence medium

`WWWResultInfoSubImageDocInfo.tqStar`
Rule: A score for the technical quality of an image, such as sharpness and exposure.
Google's description: TQ* is a signal to capture the technical quality (e.g. exposure, sharpness, composition, etc.) of an image. For more information, please refer to go/tqstar.
type number() · confidence medium

Clicks & engagement (3)

`CountryCountryAttachment.metroNavboost`
Rule: Per-metro NavBoost multipliers to apply to the page.
Google's description: Metro navboost: list of (NavBoost feature V2, navboost float) pairs.
type list(GoogleApi.ContentWarehouse.V1.Model.CountryMetroNBFeature.t) · confidence medium · stages: 12. Personalization, Local & Context

`CountryMetroNBFeature.navboost`
Rule: The multiplier to apply to a result for a given locale and query.
Google's description: This is the multiplier to apply to the result for this locale & query. NOTE: This is for serving purposes only and should not be populated in the index.
type number() · confidence medium

`ImageQualityNavboostImageQualityClickSignals.gveClicks`
Rule: Counts of clicks judged to be good ones for this image.
Google's description: Good clicks that are defined using GVE (go/gve-history).
type number() · confidence medium

Freshness (1)

`WeboftrustLiveResultDocBoostData.boostLevel`

Rule: How much a page's score is boosted while its live result is timely, from zero to one.

Google's description: Identifies the degree to which the existence of this LiveResult should boost a query's score (when the query is performed within the hot_times range). This field is always in the range [0,1]. A missing field, a value of 0, or a value outside the legal range indicates that no boosting is performed. A value of 1 indicates that the maximum level of boosting will be applied. This field will be updated from time to time based on CTR and other signals.

type number() · confidence high · stages: 8. Spam, Penalty & Trust Systems

Personalization & local (1)

`IndexingDupsLocalizedLocalizedCluster.boostSourceBlocker`

Rule: Whether the page may be a boost target but must never be boosted itself.

Google's description: Defined as a source-blocker, a result which can be a boost target but should itself not be boosted (e.g. roboted documents). For more details on source and target blocking, please read through the code for quality/twiddler/impls/PROTECTED/local_result_twiddler_v2.cc

type boolean() · confidence medium

Part II. Supporting ranking signals (790)

Fields that contribute to ranking alongside the core signals. Together with Part I these are the defensible ranking signals.

Content quality (150)

`DocProperties.title`
Rule: The page title taken from the title tag.
Google's description: Extracted from the title tag of the content. This is typically extracted by TitleMetaCollector defined at google3/segindexer/title-meta-collector.h. Please see its documentation for the format and other caveats.
type String.t · confidence high

`IndexingMobileVoltCoreWebVitals.cls`
Rule: The page's Cumulative Layout Shift score, a measure of visual stability while loading.
Google's description: Cumulative Layout Shift.
type String.t · confidence high

`IndexingMobileVoltCoreWebVitals.fid`
Rule: The page's First Input Delay, how long it takes to react to a first user interaction.
Google's description: First Input Delay.
type String.t · confidence high

`IndexingMobileVoltCoreWebVitals.lcp`
Rule: The page's Largest Contentful Paint time, how long the main content takes to appear.
Google's description: Largest Contentful Paint.
type String.t · confidence high

`QualitySalientTermsSalientTerm.salience`
Rule: How important a term is as a descriptor of the document, from zero to one.
Google's description: salience is the importance of the term as a descriptor in [0, 1] (the higher the more important). This field takes precedence over weight field below. NOTE: Please do not access this field directly. Use quality_salient_terms::utils::GetSalience() from salient_terms_utils.h instead.
type number() · confidence high

`QualitySalientTermsSalientTermSet.salientTerm`
Rule: The list of terms that best describe the document, most important first.
Google's description: salient_term is the list of terms that are good descriptors, sorted in decreasing order of weight.
type list(GoogleApi.ContentWarehouse.V1.Model.QualitySalientTermsSalientTerm.t) · confidence high

`QualityTimebasedPageType.isForumPage`
Rule: Whether the page was classified as a forum page.
Google's description: Set to true if this page is classified as a forum page.
type boolean() · confidence high · stages: 7. Freshness & Temporal Signals

`QualityTimebasedPageType.isQnaPage`
Rule: Whether the page was classified as a question-and-answer page.
Google's description: Set to true if this page is classified as a question answers page.
type boolean() · confidence high · stages: 7. Freshness & Temporal Signals

`BlogPerDocData.userQualityScore`
Rule: Quality scores for the author and for the short post itself.
Google's description: User and doc quality scores for updates (aka microposts).
type integer() · confidence medium

`BookCitationPerDocData.discretizedCitationScore`
Rule: A bucketed score for how strongly the page cites its main book.
Google's description: the discretized citation score for the main book. we map the raw score 1.0-20.0 to 0 - 127
type integer() · confidence medium

`BookCitationPerDocData.secondDiscretizedCitationScore`
Rule: A bucketed score for the page's second book citation.
Google's description: the discretized score for the second citation
type integer() · confidence medium

`CompositeDoc.labelData`
Rule: Labels attached to the document, each with a confidence value.
Google's description: This field associates a document to particular labels and assigns confidence values to them.
type GoogleApi.ContentWarehouse.V1.Model.QualityLabelsGoogleLabelData.t · confidence medium · stages: 2. Document Processing & Indexing

`CompressedQualitySignals.pqData`
Rule: Encoded page-level page quality signals.
Google's description: Encoded page-level PQ signals.
type integer() · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 8. Spam, Penalty & Trust Systems; 11. Re-Ranking & Twiddler System

`CompressedQualitySignals.pqDataProto`
Rule: Page-level quality signals kept outside the encoded page quality field.
Google's description: Stripped page-level signals, not present in the encoded field 'pq_data'.
type GoogleApi.ContentWarehouse.V1.Model.QualityNsrPQData.t · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 8. Spam, Penalty & Trust Systems; 11. Re-Ranking & Twiddler System

`CompressedQualitySignals.productReviewPReviewPage`
Rule: The estimated chance that the page is a product review page.
Google's description: Fields product_review_p_review_page and product_review_p_uhq_page are for promoting/demoting HQ/LQ review pages in NGS. See go/pr-boosts for details. The possibility of a page being a review page.
type integer() · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 8. Spam, Penalty & Trust Systems; 11. Re-Ranking & Twiddler System

`CompressedQualitySignals.productReviewPUhqPage`
Rule: The estimated chance that the page is a high quality review page.
Google's description: The possibility of a page being a high quality review page.
type integer() · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 8. Spam, Penalty & Trust Systems; 11. Re-Ranking & Twiddler System

`DocProperties.avgTermWeight`
Rule: The average weighted font size of words in the page body.
Google's description: The average weighted font size of a term in the doc body
type integer() · confidence medium

`DocProperties.badTitle`
Rule: A flag that the page's title is missing or meaningless.
Google's description: Missing or meaningless title
type boolean() · confidence medium

`DocProperties.numTokens`
Rule: An approximate count of words, tags and punctuation in the page.
Google's description: The number of tokens, tags and punctuations in the tokenized contents. This is an approximation of the number of tokens, tags and punctuations we end up with in mustang, but is inexact since we drop some tokens in mustang and also truncate docs at a max cap.
type integer() · confidence medium

`FatcatCompactTaxonomicClassificationCategory.discreteWeight`
Rule: How strongly this topic category applies, on a 0 to 127 scale.
Google's description: go/petacat-faq#how-should-i-interpret-classification-weights Discrete to reduce size. Range is [0,127], corresponding to [0.0,1.0].
type integer() · confidence medium

`FatcatCompactTaxonomicClassificationCategory.id`
Rule: The numeric id of the topic category assigned, such as Sports.
Google's description: The category's ID, e.g. 20 for /Sports in the go/verticals4 taxonomy.
type integer() · confidence medium

`IndexingMobileInterstitialsProtoDesktopInterstitialsDetails.isSmearedSignal`
Rule: Flag saying the interstitial signal was estimated from other URLs rather than measured on this page.
Google's description: Indicates whether the signal value is "smeared", e.g. extrapolated from other URLs.
type boolean() · confidence medium

`IndexingMobileInterstitialsProtoInterstitialBasicInfo.absoluteBox`
Rule: The on-page position and size of a detected interstitial overlay, in page pixels.
Google's description: Stores the geometry of detected interstitial in absolute page pixels.
type GoogleApi.ContentWarehouse.V1.Model.HtmlrenderWebkitHeadlessProtoBox.t · confidence medium

`NSRVersionedItem.value`
Rule: A site quality (NSR) value stored for one particular version of that measure.
Google's description: The NSR value corresponding to this version.
type number() · confidence medium

`PerDocData.TagPageScore`
Rule: A 0-100 score for how much the page looks like a thin tag listing page.
Google's description: Tag-site-ness of a page, repesented in 7-bits range from 0 to 100. Smaller value means worse tag page.
type integer() · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.bodyWordsToTokensRatioBegin`
Rule: The ratio of real words to tokens in the opening part of the page body.
Google's description: The body words over tokens ratios for the beginning part and whole doc. NB: To save space, field body_words_to_tokens_ratio_total is not set if it has the same value as body_words_to_tokens_ratio_begin (e.g., short docs).
type number() · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.commercialScore`
Rule: A score for how commercial the page is; above zero means it sells something.
Google's description: A measure of commerciality of the document Score > 0 indicates document is commercial (i.e. sells something) Computed by repository/pageclassifiers/parsehandler-commercial.cc
type number() · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.desktopInterstitials`
Rule: A signal about interstitials shown on desktop, used by the VOLT ranking change.
Google's description: Contains desktop interstitials signal for VOLT ranking change.
type GoogleApi.ContentWarehouse.V1.Model.IndexingMobileInterstitialsProtoDesktopInterstitials.t · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.fireflySiteSignal`
Rule: Site-level signal data used by the Firefly ranking change.
Google's description: Contains Site signal information for Firefly ranking change. See
type GoogleApi.ContentWarehouse.V1.Model.QualityCopiaFireflySiteSignal.t · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.nsrDataProto`
Rule: A container of site-level quality signals not stored in the separate NSR fields.
Google's description: Stripped site-level signals, not present in the explicit nsr_* fields, nor compressed_quality_signals.
type GoogleApi.ContentWarehouse.V1.Model.QualityNsrNsrData.t · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.originalTitleHardTokenCount`
Rule: How many hard tokens the original title contained.
Google's description: The number of hard tokens in the title.
type integer() · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.titleHardTokenCountWithoutStopwords`
Rule: How many hard tokens the title has once stopwords are excluded.
Google's description: Number of hard tokens originally in title without counting the stopwords.
type integer() · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.voltData`
Rule: Page experience signals used by the VOLT ranking change.
Google's description: Contains page UX signals for VOLT ranking change. See
type GoogleApi.ContentWarehouse.V1.Model.IndexingMobileVoltVoltPerDocData.t · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.ymylHealthScore`
Rule: A classifier score for how much the page is health content in a sensitive area.
Google's description: Stores scores of ymyl health classifier as defined at go/ymyl-classifier-dd. To use this field, you MUST join g/pq-classifiers-announce and add your use case at
type integer() · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.ymylNewsScore`
Rule: A classifier score for how much the page is news content in a sensitive area.
Google's description: Stores scores of ymyl news classifier as defined at go/ymyl-classifier-dd. To use this field, you MUST join g/pq-classifiers-announce and add your use case at
type integer() · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityFringeFringeQueryPriorPerDocData.encodedChardXlqHoaxPrediction`
Rule: A packed model estimate that the page is a hoax.
Google's description: An encoding of the Chard XLQ-hoax prediction in [0,1].
type integer() · confidence medium

`QualityFringeFringeQueryPriorPerDocData.encodedChardXlqTranslatedPrediction`
Rule: A packed low-quality estimate for translated content on the page.
Google's description: An encoding of the Chard XLQ prediction on translated content in [0,1].
type integer() · confidence medium

`QualityFringeFringeQueryPriorPerDocData.encodedChardXlqYmylPrediction`
Rule: A packed low-quality estimate for pages about money or health topics.
Google's description: An encoding of the Chard XLQ-YMYL prediction in [0,1].
type integer() · confidence medium

`QualityFringeFringeQueryPriorPerDocData.encodedDaftScore`
Rule: A packed score for how much the document is about a fringe topic.
Google's description: An encoding of the Document About Fringe Topic (daft) score in [0, 1].
type integer() · confidence medium

`QualityFringeFringeQueryPriorPerDocData.encodedDocumentFringeVulnerability`
Rule: A packed estimate of how likely this page is to surface fringe content.
Google's description: An estimate of the vulnerability of this doc to show fringe content, based on the context around the document. Can be interpreted as a 'safe' QScore threshold to use (see go/doc-fringe-vulnerability for more info). Encoded for compactness and to restrict visibility. Please contact fringe-ranking@ to get access to quality_fringe::DocumentFringeVulnerabilityEncoding to decode this field.
type integer() · confidence medium

`QualityFringeFringeQueryPriorPerDocData.encodedPredictedXlqScoreAndConfidence`
Rule: A packed predicted lowest-quality rating for the page plus its confidence.
Google's description: A combined encoding of the pXLQ score in [0,1] and the confidence with which that score should be interpreted in [0,1].
type integer() · confidence medium

`QualityFringeFringeQueryPriorPerDocData.encodedProximityScore`
Rule: How similar the page is to known fringe-prone example pages.
Google's description: A score in [0, 1] representing the similarity of this doc to known fringe-vulnerable 'seeds'. See go/fringe-proximity for more information. Encoded for compactness and to restrict visibility.
type integer() · confidence medium

`QualityFringeFringeQueryPriorPerDocData.encodedPseudoraterPxlqScore`
Rule: An automated rater's low-quality severity score for the page.
Google's description: An encoding of the XLQ pseudorater severity score translated into pXLQ score space in [0,1]. The encoding includes the pseudorater version and confidence.
type integer() · confidence medium

`QualityNsrKetoKetoVersionedData.siteFrac`
Rule: What share of the site's pages went into computing the site score.
Google's description: What fraction of the site went into the computation of the site_score. Should be a number in (0, 1].
type number() · confidence medium

`QualityNsrKetoKetoVersionedData.siteScore`
Rule: A site-wide aggregated keto quality score.
Google's description: Site-level aggregated keto score.
type number() · confidence medium

`QualityNsrNSRVersionedData.value`
Rule: The site quality (NSR) value for one version.
Google's description: The corresponding NSR value.
type number() · confidence medium

`QualityNsrNsrData.articleScore`
Rule: How article-like the site's content is, per a classifier.
Google's description: Score from article classification of the site.
type number() · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityNsrNsrData.chardEncoded`
Rule: A site-wide predicted content quality score, stored as an integer.
Google's description: Site-level chard score: site quality predictor based on content.
type integer() · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityNsrNsrData.chardScoreEncoded`
Rule: Versioned site-wide predicted content quality scores.
Google's description: Site-level Chard (encoded as an int).
type list(GoogleApi.ContentWarehouse.V1.Model.QualityNsrVersionedIntSignal.t) · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityNsrNsrData.chardScoreVariance`
Rule: How much predicted content quality varies across the site's pages.
Google's description: Site-level Chard Variance for all pages of a site.
type list(GoogleApi.ContentWarehouse.V1.Model.QualityNsrVersionedFloatSignal.t) · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityNsrNsrData.isCovidLocalAuthority`
Rule: Whether the site counts as a local authority for COVID information.
Google's description: Bit to determine whether the site has the local authority covid signal, as computed by go/covid-local-authority
type boolean() · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityNsrNsrData.isElectionAuthority`
Rule: Whether the site counts as an authority on elections.
Google's description: Bit to determine whether the site has the election authority signal, as computed by go/election-authority
type boolean() · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityNsrNsrData.priorAdjustedNsr`
Rule: How far the site's quality sits above or below average for its group.
Google's description: NSR - prior. Estimate of whether the site is above/below average NSR in its slice.
type list(GoogleApi.ContentWarehouse.V1.Model.QualityNsrVersionedFloatSignal.t) · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityNsrNsrData.racterScores`
Rule: Site-level scores from a classifier for auto-generated content.
Google's description: Site-level AGC classification score (see also go/project-racter-overview).
type list(GoogleApi.ContentWarehouse.V1.Model.QualityNsrVersionedFloatSignal.t) · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityNsrNsrData.siteQualityStddev`
Rule: How widely page quality ratings vary across the site.
Google's description: Estimate of site's PQ rating stddev--spread of the page-level PQ ratings of a site. Note this is different from nsr_variance which predicts error of NSR itself from the aggregated site-level rating.
type number() · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityNsrNsrData.titlematchScore`
Rule: How well the site's page titles match the queries users type.
Google's description: Titlematch score of the site, a signal that tells how well titles are matching user queries.
type number() · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityNsrNsrData.tofu`
Rule: A site-wide content-based quality prediction.
Google's description: Site-level tofu score: site quality predictor based on content.
type number() · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityNsrNsrData.vlq`
Rule: A score from a model that spots low-quality video sites.
Google's description: Score of the Video LQ model.
type number() · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityNsrNsrData.vlqNsr`
Rule: A site quality value from a model aimed at low-quality video sites.
Google's description: NSR from a headroom model targeting low-quality video sites.
type number() · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityNsrPQData.chard`
Rule: A page-level predicted content quality score.
Google's description: URL-level chard prediction (encoded as an int).
type integer() · confidence medium · stages: 3. Content Quality & Site-Level Scoring

`QualityNsrPQData.chardScoreEncoded`
Rule: Versioned page-level predicted content quality scores.
Google's description: URL-level Chard (encoded as an int).
type list(GoogleApi.ContentWarehouse.V1.Model.QualityNsrVersionedIntSignal.t) · confidence medium · stages: 3. Content Quality & Site-Level Scoring

`QualityNsrPQData.deltaSubchunkAdjustment`
Rule: A page-level adjustment to site quality based on the page's topic group.
Google's description: Total deltaNSR adjustment based on subchunks. This is a page-level adjustment (subchunks are retrieved based on the page classification).
type number() · confidence medium · stages: 3. Content Quality & Site-Level Scoring

`QualityNsrPQData.predictedDefaultNsr`
Rule: A predicted default site quality score computed for the page.
Google's description: Predicted default NSR score computed in Goldmine via the NSR default predictor (go/default-nsr-predictor).
type list(GoogleApi.ContentWarehouse.V1.Model.QualityNsrVersionedFloatSignal.t) · confidence medium · stages: 3. Content Quality & Site-Level Scoring

`QualityNsrPQData.rhubarb`
Rule: A page quality score based on how the page differs from its site.
Google's description: Site-URL delta signals based quality score computed in Goldmine via the Rhubarb model (go/rhubarb-dd).
type list(GoogleApi.ContentWarehouse.V1.Model.QualityNsrVersionedFloatSignal.t) · confidence medium · stages: 3. Content Quality & Site-Level Scoring

`QualityNsrPQData.tofu`
Rule: A page-level content-based quality prediction.
Google's description: URL-level tofu prediction.
type number() · confidence medium · stages: 3. Content Quality & Site-Level Scoring

`QualityNsrPQData.unversionedRhubarb`
Rule: The page-level quality predictor's difference score.
Google's description: The delta score of the URL-level quality predictor.
type number() · confidence medium · stages: 3. Content Quality & Site-Level Scoring

`QualityNsrPQData.vlq`
Rule: The page's score from the low-quality video model.
Google's description: URL-level score of the VLQ model.
type number() · confidence medium · stages: 3. Content Quality & Site-Level Scoring

`QualityNsrPQDataSubchunkData.deltaNsr`
Rule: How much this topic group shifts the site quality score.
Google's description: Subchunk delta in nsr.
type number() · confidence medium

`QualityNsrPQDataSubchunkData.pageWeight`
Rule: How strongly this page belongs to the topic group.
Google's description: Weight with which this document belong to this subchunk (greater than 0).
type number() · confidence medium

`QualityProductProductSiteDataLocaleData.gobiSite`
Rule: Marks a site as one from a shopping domain eligible for boosting on category queries.
Google's description: Whether this is a gobi site, ie, a site from a gobi domain that should be boosted for a category query with this gobi domain. For example, amazon.com is a gobi store domain for category query [hdtv] but some sites (like askville.amazon.com) from amazon.com should not be boosted.
type boolean() · confidence medium

`QualityRankembedMustangMustangRankEmbedInfo.additionalFixedPointEncodings`
Rule: Extra encoded copies of the document's ranking embedding.
Google's description: Each of the repeated elements in this field has the same scheme as 'fixed_point_encoding'; See go/rankembed-astro-rule-set-dd
type list(String.t) · confidence medium

`QualityRankembedMustangMustangRankEmbedInfo.compressedDocumentEmbedding`
Rule: The document's ranking embedding stored in compressed form.
Google's description: Each uint64 encodes 8 8-bit values for the quantized document embedding
type GoogleApi.ContentWarehouse.V1.Model.QualityRankembedMustangMustangRankEmbedInfoCompressedEmbedding.t · confidence medium

`QualityRankembedMustangMustangRankEmbedInfo.fixedPointEncoding`
Rule: The document's ranking embedding packed into one encoded string.
Google's description: This field replaces the above 3 "per-encoding-type-fields", where the encoding type (and the embedding type) are part of the encoding, and is stored in the first byte. The remaining bytes are the same as the previous 3 fields, but shifted by 1 byte. - byte[0]: encoding type & embedding type - byte[1....]: similar to the above depending on the encoding type.
type String.t · confidence medium

`QualitySalientTermsDocData.confidence`
Rule: How much data was available when the document's important terms were computed.
Google's description: confidence is a measurement of how much data we had to compute the SalientTermSet. Range: [0.0, 1.0]
type number() · confidence medium

`QualitySalientTermsDocData.headVolumeRatio`
Rule: The share of all term occurrences taken by the document's top terms.
Google's description: head_volume_ratio is the ratio of the sum of term frequency of the top K terms over the volume of all terms. Range: [0.0, 1.0]. K is defined by Accumulator2Params::head_size.
type number() · confidence medium

`QualitySalientTermsDocData.virtualVolume`
Rule: A measure of how much term data was available for the document.
Google's description: virtual_volume is a measurement of how much data we had to compute the SalientTermSet. Range: [0.0, +infinity)].
type number() · confidence medium

`QualitySalientTermsSalientTerm.label`
Rule: The term itself, either normalized or as originally written.
Google's description: label can be two things depending on where this message is. When right under a SalientTermSet, it is the normalized term returned by quality_salient_terms::utils::NormalizeTerm() from salient_terms_utils.h. When under another SalientTerm message, it is the original term as found in a signal (see original_term field).
type String.t · confidence medium

`QualitySalientTermsSalientTerm.originalTerm`
Rule: The different spellings under which this term was found on the page.
Google's description: original_term are the different ways we found this normalized term in the signals. They are in increasing idf order (the most common version first). An empty string means that this original term is the same as the label field in the parent SalientTerm message. NOTE: Please do not access this field directly. Use quality_salient_terms::utils::OriginalTermsIterator from salient_terms_utils.h instead.
type list(GoogleApi.ContentWarehouse.V1.Model.QualitySalientTermsSalientTerm.t) · confidence medium

`QualitySalientTermsSignalData.confidence`
Rule: How much this source of term data is trusted, from zero to one.
Google's description: The measurement of how much we trust this signal. Range: [0.0, 1.0] This field is available is both debug and non-debug mode.
type number() · confidence medium

`QualityShoppingShoppingAttachment.datasetModelArticleScore`
Rule: A classifier score for how much the page looks like an article.
Google's description: Score from the blockbert article classifier model. go/article-understanding-project
type integer() · confidence medium

`QualityShoppingShoppingAttachment.datasetModelForumListScore`
Rule: A classifier score for how much the page looks like a forum or Q and A page.
Google's description: From forum and qna confidence score * 100,
type integer() · confidence medium

`QualityShoppingShoppingAttachment.datasetModelMultiProductScore`
Rule: A classifier score for how much the page looks like a multi-product page.
Google's description: From indexing.ml.PageType.confidence * 100 (DatasetModelAnnotation in cdoc) go/sdu-shopping-page-intro
type integer() · confidence medium

`QualityShoppingShoppingAttachment.expiredShoppingPageScore`
Rule: A score for how likely the shopping page is out of date or expired.
Google's description: From indexing.badpages.CollapserInfo.expired_shopping_page_score * 100
type integer() · confidence medium

`QualityShoppingShoppingAttachment.shoppingSiteScore`
Rule: A classifier score for how much the site looks like a shopping site.
Google's description: From ShoppingSiteClassifier.score * 100
type integer() · confidence medium

`QualityShoppingShoppingAttachment.shoppingSiteScoreShopfab`
Rule: A second shopping-site classifier score for the site.
Google's description: From ShoppingSiteClassifierShopfab.score * 100
type integer() · confidence medium

`QualityShoppingShoppingAttachmentProduct.relevanceEmbedding`
Rule: A relevance embedding carried over from the product annotation.
Google's description: Relevance embedding from ShoppingAnnotation.Product
type list(GoogleApi.ContentWarehouse.V1.Model.QualityRankembedMustangMustangRankEmbedInfo.t) · confidence medium

`QualityTravelGoodSitesData.totalScore`
Rule: A quality score for the site that decides which site type it is.
Google's description: Site quality score, which determines the site type.
type number() · confidence medium

`ReneEmbedding.floatValues`
Rule: The numbers that make up the embedding vector.
Google's description: Values of the embedding vector.
type list(number()) · confidence medium

`RepositoryWebrefDocumentMetadata.salientTerms`
Rule: The important terms of the page, copied in when a flag is on.
Google's description: The salient terms for this document. Only set if --webref_doc_metadata_copy_salient_terms is true. Same motivation as the title field above.
type GoogleApi.ContentWarehouse.V1.Model.QualitySalientTermsSalientTermSet.t · confidence medium · stages: 6. Entity & Knowledge Graph Integration

`ResearchScienceSearchSourceUrlDocjoinInfo.petacatInfo`
Rule: Topic classifications assigned to the web page by the Petacat classifier.
Google's description: Petacat classifications for the web document. Normally the results from calling Petacat come in a PetacatResponse, which is very flexible and extensible. This proto takes most of the flexibility away - only rephil clusters, taxonomic classifications, and binary classifications, with discretized weights.
type GoogleApi.ContentWarehouse.V1.Model.FatcatCompactDocClassification.t · confidence medium

`ResearchScienceSearchSourceUrlDocjoinInfo.scholarInfo`
Rule: Scholar-derived per-page data passed into web search.
Google's description: Science per-doc data for inclusion in websearch.
type GoogleApi.ContentWarehouse.V1.Model.ScienceIndexSignal.t · confidence medium

`ResearchScienceSearchSourceUrlDocjoinInfo.title`
Rule: The title of the document.
Google's description: The title of the document.
type String.t · confidence medium

`SmartphonePerDocData.isSmartphoneOptimized`
Rule: Whether the page displays properly on smartphones.
Google's description: Indicates if the page is rendered in a friendly manner on smartphones. We use this field as tri-state: "unset" means the rendering result classification is not available, and "set as false" means that the page is rendered in unfriendly manner on smartphones. See also go/modena-ranking.
type boolean() · confidence medium · stages: 12. Personalization, Local & Context

`TeragoogleDocumentInfo.averageTermWeight`
Rule: Average weight of the terms in the document, used to scale term weighting.
Google's description: The average weight of terms in the document. If not available, there will be no term weight averaging: font sizes will be taken literally from the document HTML.
type integer() · confidence medium

`VideoContentSearchVideoInfo.nsr`
Rule: The stored site quality score for the page hosting the video.
Google's description: NSR for the video page document.
type number() · confidence medium

`VideoContentSearchVideoInfo.salientTermSet`
Rule: The set of terms judged most important on the page.
Google's description: Salient term set from the document. This message contains a lot of data and dependencies, so sub-messages are disabled in model evaluation in scorer.
type GoogleApi.ContentWarehouse.V1.Model.QualitySalientTermsSalientTermSet.t · confidence medium

`BlogPerDocData.numMentions`
Rule: How many at-mentions of other users appear in the post.
Google's description: For replies/reshares. num_mentions = number of times the pattern @foo appears in the document.
type integer() · confidence low

`BookCitationPerDocData.previewable`
Rule: Whether the page shows a preview or excerpt of the book.
Google's description: Is there a preview or excerpt of the book on this document?
type boolean() · confidence low

`CompositeDoc.badSslCertificate`
Rule: Present when the page or its redirects have a bad security certificate.
Google's description: This field is present iff the page has a bad SSL certificate itself or in its redirect chain.
type GoogleApi.ContentWarehouse.V1.Model.IndexingBadSSLCertificate.t · confidence low · stages: 2. Document Processing & Indexing

`CompositeDoc.richcontentData`
Rule: A record that parts of the page's content were inserted, removed, or replaced.
Google's description: If present, indicates that some content was inserted, deleted, or replaced in the document's content (in CompositeDoc::doc::Content::Representation), and stores information about what was inserted, deleted, or replaced.
type GoogleApi.ContentWarehouse.V1.Model.IndexingConverterRichContentData.t · confidence low · stages: 2. Document Processing & Indexing

`CompressedQualitySignals.experimentalNsrTeamData`
Rule: Experimental site quality data kept by the site quality team.
Google's description: This field is
type GoogleApi.ContentWarehouse.V1.Model.QualityNsrExperimentalNsrTeamData.t · confidence low · stages: 3. Content Quality & Site-Level Scoring; 8. Spam, Penalty & Trust Systems; 11. Re-Ranking & Twiddler System

`CompressedQualitySignals.experimentalNsrTeamWsjData`
Rule: A second experimental site quality data set kept by the site quality team.
Google's description: This field is
type list(GoogleApi.ContentWarehouse.V1.Model.QualityNsrExperimentalNsrTeamWSJData.t) · confidence low · stages: 3. Content Quality & Site-Level Scoring; 8. Spam, Penalty & Trust Systems; 11. Re-Ranking & Twiddler System

`CompressedQualitySignals.experimentalQstarDeltaSignal`
Rule: An experimental value measuring a change in a Q* scoring signal.
Google's description: This field is
type number() · confidence low · stages: 3. Content Quality & Site-Level Scoring; 8. Spam, Penalty & Trust Systems; 11. Re-Ranking & Twiddler System

`CompressedQualitySignals.experimentalQstarSignal`
Rule: An experimental page-level value tested in Q* scoring.
Google's description: This field is
type number() · confidence low · stages: 3. Content Quality & Site-Level Scoring; 8. Spam, Penalty & Trust Systems; 11. Re-Ranking & Twiddler System

`CompressedQualitySignals.experimentalQstarSiteSignal`
Rule: An experimental site-level value tested in Q* scoring.
Google's description: This field is
type number() · confidence low · stages: 3. Content Quality & Site-Level Scoring; 8. Spam, Penalty & Trust Systems; 11. Re-Ranking & Twiddler System

`ContentAttributions.freshdocsOutgoing`
Rule: Content the page credits to other sources, found on the fresh document path.
Google's description: Selected outgoing attributions extracted on FreshDocs.
type list(GoogleApi.ContentWarehouse.V1.Model.ContentAttributionsOutgoingAttribution.t) · confidence low

`ContentAttributions.offlineOutgoing`
Rule: Content the page credits to other sources, found by offline batch jobs.
Google's description: Selected outgoing attributions extracted via offline MR jobs.
type list(GoogleApi.ContentWarehouse.V1.Model.ContentAttributionsOutgoingAttribution.t) · confidence low

`ContentAttributions.onlineOutgoing`
Rule: Content the page credits to other sources, found during online indexing.
Google's description: Selected outgoing attributions extracted online on Alexandria.
type list(GoogleApi.ContentWarehouse.V1.Model.ContentAttributionsOutgoingAttribution.t) · confidence low

`FatcatCompactBinaryClassification.binaryClassifier`
Rule: Which yes or no classifier produced this label for the page or site.
Google's description: Either binary_classifier will be set, using the enum above, or binary_classifier_name will be set, if it is not one of the classifiers in the enum - never both.
type String.t · confidence low

`FatcatCompactBinaryClassification.discreteFraction`
Rule: For a site label, what share of the site the label applies to.
Google's description: A CompactDocClassification will not usually have a weight. For a CompactSiteClassification, this value will be 0...127 corresponding to 0.0...1.0, indicating fraction of the site that this label applies to
type integer() · confidence low

`FatcatCompactRephilClustersCluster.discreteWeight`
Rule: How strongly a semantic topic cluster applies to the page, on a 0 to 127 scale.
Google's description: 0...127 corresponds to 0.0 - 1.0
type integer() · confidence low

`NlpSaftDocument.title`
Rule: The title of the document.
Google's description: Optional document title.
type String.t · confidence low · stages: 9. Query Understanding & Intent Classification

`PairwiseQVersionedItem.confidenceValue`
Rule: How confident the stored PairwiseQ score is for this version.
Google's description: The PairwiseQ confidence value corresponding to this version.
type number() · confidence low

`PairwiseQVersionedItem.value`
Rule: A stored PairwiseQ score value for a given version.
Google's description: The PairwiseQ value corresponding to this version.
type number() · confidence low

`PerDocData.BookCitationData`
Rule: Book citation information attached to a web page, about ten bytes per page.
Google's description: the book citation data for each web page, the average size is about 10 bytes
type GoogleApi.ContentWarehouse.V1.Model.BookCitationPerDocData.t · confidence low · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.pageregions`
Rule: An encoded map of where the different regions of the page start and end.
Google's description: String that encodes the position ranges for different regions of the document. See "indexer/pageregion.h" for an explanation, and how to decode the string
type String.t · confidence low · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.productSitesInfo`
Rule: Stored information about product sites.
Google's description: This field stores information about product sites.
type GoogleApi.ContentWarehouse.V1.Model.QualityProductProductSiteData.t · confidence low · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.travelGoodSitesInfo`
Rule: Stored information about travel sites judged good.
Google's description: This field stores information about good travel sites.
type GoogleApi.ContentWarehouse.V1.Model.QualityTravelGoodSitesData.t · confidence low · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityNsrNsrData.healthScore`
Rule: A site-level score within the categorical health grouping.
Google's description: Categorical signals.
type number() · confidence low · stages: 3. Content Quality & Site-Level Scoring; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityNsrNsrData.ketoVersionedData`
Rule: The site's keto quality scores across versions.
Google's description: Keto data.
type list(GoogleApi.ContentWarehouse.V1.Model.QualityNsrKetoKetoVersionedData.t) · confidence low · stages: 3. Content Quality & Site-Level Scoring; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityNsrNsrData.site2vecEmbedding`
Rule: A vector representation of the site.
Google's description: Site2vec embeddings.
type list(GoogleApi.ContentWarehouse.V1.Model.QualityNsrNsrDataEmbedding.t) · confidence low · stages: 3. Content Quality & Site-Level Scoring; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityNsrNsrData.site2vecEmbeddingEncoded`
Rule: A compressed vector representation of the site for use in serving.
Google's description: Encoded site2vec embedding (to be used in superroot) since the full embeddings take too much space.
type list(GoogleApi.ContentWarehouse.V1.Model.QualityNsrNsrDataEncodedEmbedding.t) · confidence low · stages: 3. Content Quality & Site-Level Scoring; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityNsrNsrData.siteAutopilotScore`
Rule: The site-wide aggregate of per-page autopilot scores.
Google's description: Aggregated value of url autopilot scores for this sitechunk.
type number() · confidence low · stages: 3. Content Quality & Site-Level Scoring; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityNsrNsrData.versionedData`
Rule: Site quality values kept per version for testing new releases.
Google's description: Versioned map of NSR values for experimenting with the next release.
type list(GoogleApi.ContentWarehouse.V1.Model.QualityNsrNSRVersionedData.t) · confidence low · stages: 3. Content Quality & Site-Level Scoring; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityNsrPQData.keto`
Rule: The page's keto quality score.
Google's description: Keto score.
type list(GoogleApi.ContentWarehouse.V1.Model.QualityNsrVersionedFloatSignal.t) · confidence low · stages: 3. Content Quality & Site-Level Scoring

`QualityNsrPQDataSubchunkData.confidence`
Rule: How confident the system is in this topic group's numbers.
Google's description: Confidence associated with the chunk.
type number() · confidence low

`QualityNsrPQDataSubchunkData.type`
Rule: Which topic group this chunk represents, such as health.
Google's description: Type of this chunk. Eg, ymyl_health, d2v, etc.
type String.t · confidence low

`QualityRankembedMustangMustangRankEmbedInfo.scaledFixedPoint4Encoding`
Rule: A four-bit-per-value encoded copy of the document's ranking embedding.
Google's description: byte[0]: version - bytes[1...4]: scalar - bytes[5,...]: the values, one byte per 2 values
type String.t · confidence low

`QualityRankembedMustangMustangRankEmbedInfo.scaledFixedPoint8Encoding`
Rule: An eight-bit-per-value encoded copy of the document's ranking embedding.
Google's description: byte[0]: version - bytes[1...4]: scalar - bytes[5,...]: the values, one byte per value
type String.t · confidence low

`QualityRankembedMustangMustangRankEmbedInfo.scaledShiftedFixedPoint4Encoding`
Rule: A shifted and scaled compact encoding of the document's ranking embedding.
Google's description: byte[0]: version - bytes[1...4]: scalar - bytes[5...8]: shift - bytes[9,...]: the values, one byte per 2 values
type String.t · confidence low

`QualityRankembedMustangMustangRankEmbedInfoCompressedEmbedding.packedValue`
Rule: The embedding numbers packed together to save storage space.
Google's description: using fixed64 instead of uint64 saves ~14% is storage
type list(String.t) · confidence low

`QualitySalientTermsDocData.language`
Rule: The main language of the document's important-terms set.
Google's description: language is the main language of this SalientTermSet.
type String.t · confidence low

`QualitySalientTermsDocData.signalData`
Rule: Per-source details, such as body text, anchors and clicks, behind the term set.
Google's description: signal_data contains signal-specific (e.g., body, anchors, clicks) data for this SalientTermSet.
type list(GoogleApi.ContentWarehouse.V1.Model.QualitySalientTermsSignalData.t) · confidence low

`QualitySalientTermsSalientTerm.signalTerm`
Rule: Per-source detail for this term, such as body text, anchors or clicks.
Google's description: signal_term contains extra signal-specific (e.g., body, anchors, clicks) data for this term.
type list(GoogleApi.ContentWarehouse.V1.Model.QualitySalientTermsSignalTermData.t) · confidence low

`QualitySalientTermsSalientTermSet.docData`
Rule: Document-level details attached to the set of important terms.
Google's description: doc_data contain additional salient-term-set-level data.
type GoogleApi.ContentWarehouse.V1.Model.QualitySalientTermsDocData.t · confidence low

`QualitySalientTermsSignalData.source`
Rule: Which source the term data came from, such as body text, anchors or clicks.
Google's description: source is the type of the signal of this SignalData.
type String.t · confidence low

`QualitySalientTermsSignalTermData.source`
Rule: Which source this term data came from.
Google's description: source is the type of the signal of this SignalTermData.
type String.t · confidence low

`ReneEmbedding.clusterList`
Rule: Grouping information about the parts of this embedding.
Google's description: Token information of the embedding vector if available.
type GoogleApi.ContentWarehouse.V1.Model.ReneEmbeddingClusterList.t · confidence low

`ReneEmbedding.compressedValue`
Rule: The embedding stored in a compressed form.
Google's description: Compressed value of the embedding. For example, different versions of starburst signals use different compressor to generate the value.
type String.t · confidence low

`ReneEmbedding.denseFeatureFloatValues`
Rule: Extra feature numbers some models output alongside the embedding.
Google's description: Optional feature map output from the model. Note: this is not applicable to some models.
type list(number()) · confidence low

`ReneEmbedding.floatValuesBeforeProjection`
Rule: The embedding numbers before they were shrunk to fewer dimensions.
Google's description: Optional embedding vector prior to projection. Note: this is not applicable to some models.
type list(number()) · confidence low

`ReneEmbeddingCluster.squaredDistance`
Rule: How far this embedding sits from the center of its cluster.
Google's description: Squared distance of the current embedding to the cluster center.
type number() · confidence low

`ReneEmbeddingClusterList.clusters`
Rule: The list of clusters this embedding falls into.
Google's description: Clusters.
type list(GoogleApi.ContentWarehouse.V1.Model.ReneEmbeddingCluster.t) · confidence low

`RepositoryWebrefDocumentMetadata.title`
Rule: The page title, copied in when a flag is on.
Google's description: The title of the document. Only set if --webref_doc_metadata_set_title is true. The idea is that we can use this to more easily learn things like: title contains "restaurants" -> more likely to be a list page.
type String.t · confidence low · stages: 6. Entity & Knowledge Graph Integration

`ResearchScienceSearchReconciledMetadata.topSalientTermLabel`
Rule: The most prominent terms describing what the page body is about.
Google's description: Top salient term labels that describe the dataset document body.
type list(String.t) · confidence low

`ScienceIndexSignal.VisiblePrefixTerms`
Rule: An estimate of how much of the document most readers actually get to see.
Google's description: Length of document prefix that most users are likely to see. Only filled in when we index subscription fulltext but most users see abstracts. This is a conservative guesstimate - e.g., ACM shows fulltext to university/company subscribers (including Google employees) based on user's IP address, but we don't know subscriber IPs, so ACM's PDF pages would have ~500 in this field (estimated length of abstract).
type integer() · confidence low

`ShingleInfoPerDocData.numShingles`
Rule: How many text fingerprints the document contains.
Google's description: Total number of shingles in the document.
type integer() · confidence low

`ShingleInfoPerDocData.source`
Rule: The list of other sources the document's text fingerprints came from.
Google's description: A list of all sources.
type list(GoogleApi.ContentWarehouse.V1.Model.ShingleSource.t) · confidence low

`ShingleSource.numShingles`
Rule: How many of the document's text fingerprints came from this source.
Google's description: Number of shingles originating from this source.
type integer() · confidence low

`SmartphonePerDocData.maximumFlashRatio`
Rule: How much of the rendered area the largest Flash object takes up.
Google's description: The ratio of the area of the largest Flash to the render area.
type float() · confidence low · stages: 12. Personalization, Local & Context

Entities & authors (137)

`PerDocData.webrefEntities`
Rule: The WebRef entities that the document is about.
Google's description: WebRef entities associated to the document. See go/webref for details.
type GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefWebrefMustangAttachment.t · confidence high · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

RepositoryWebrefDetailedEntityScores.connectedness
Rule: How strongly an entity is related to the other entities on the page, which partly shapes topicality.
Google's description: Represents how much the entity is connected/related to the other entities in the document. This signal partially influences the topicality score, but it is not totally aligned with it: an entity can be very related to the rest of the document, but not central for understanding it. Likewise, an entity can be central to understand a document, but not very related to the rest of the document. The value is in [0, 1].
type number() · confidence high

RepositoryWebrefDetailedEntityScores.isAuthor
Rule: Whether this entity is the author of the page.
Google's description: True if the entity is the author of the document. This was mainly developed and tuned for news articles (e.g. /m/02x27qn on "www.vogue.com/article/flint-town-netflix") but is also popluated for other content (e.g. scientific articles). Important: the semantics of this field may change in the future or it might be removed and replaced with a different API. If you want to use this field, please reach out to ke-authors@ first.
type boolean() · confidence high

RepositoryWebrefDetailedEntityScores.isPublisher
Rule: Whether this entity is the publisher of the page.
Google's description: True if the entity is the publisher of the page (e.g. CNN on "
type boolean() · confidence high

RepositoryWebrefDetailedEntityScores.normalizedTopicality
Rule: The share of the page's overall topic that this entity accounts for.
Google's description: Representation of the topicality score that is normalized in [0, 1] and which sum over all entities in the document is 1. It represents the "proportion" of the document that talks about the entity. This score is less human interpretable as the bucketized topicality score (EntityAnnotations.topicality_score), but is more suited for some usages like aggregations.
type number() · confidence high

RepositoryWebrefDetailedEntityScores.relevanceScore
Rule: A machine-learned score for how relevant the entity is to the page.
Google's description: Relevance score generated by a Machine Learning entity classifier. This signal is similar to topicality, but machine learning based and supported by EntitySignals, not Webref. See
type number() · confidence high

RepositoryWebrefEntityAnnotations.confidenceScore
Rule: Overall confidence that the entity really is mentioned in the page or query.
Google's description: The overall confidence that the entity is annotated somewhere in the document or query. For WebRef it is computed as a function of the mention confidences weighted by the importance of each mention, where for documents a mention is of greater importance if it occurs in the title, h1 or anchors. For QRef it is just the maximum of the confidence over all mentions. NOTE: You probably want to use the mention-level segment_mentions.mention.confidence_score field instead of this one.
type number() · confidence high

RepositoryWebrefEntityAnnotations.segmentMentions
Rule: All mentions of the entity, grouped by where they occur such as title or anchors.
Google's description: All mentions of a given concept grouped by segments. For Webref, there are many different kinds of segment, such as content, title and anchors; while for QRef, there is only one segment called CONTENT. For QRef this field contains the primary output of the annotator, and for WebRef it together with topicality_score does.
type list(GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefSegmentMentions.t) · confidence high

RepositoryWebrefEntityAnnotations.topicalityScore
Rule: How closely the entity matches the overall topic of the page.
Google's description: The WebRef topicality score of the entity for this document. This score indicates how related is the entity to the overall topic of the document. See
type number() · confidence high

RepositoryWebrefMention.confidenceScore
Rule: How sure the annotator is that this exact stretch of text refers to that entity.
Google's description: A probabilistic score describing how certain the annotator is that this exact range in the document or query refers to the entity. Probability that a mention is correct corresponds to confidence score roughly as follows: 0.3 -> 75% 0.5 -> 87% 0.7 -> 89% 0.9 -> 94% 1.0 -> 98% However if you consider all mentions with a score above 0.3, then most of these will have scores close to 1, so the overall precision of these mentions is around 95%.
type number() · confidence high · stages: 6. Entity & Knowledge Graph Integration

RepositoryWebrefMention.priorProbability
Rule: How likely the mention's wording refers to that entity before context is considered.
Google's description: The prior probability of the entity for this mention.
type number() · confidence high · stages: 6. Entity & Knowledge Graph Integration

`RepositoryWebrefWebrefEntities.entity`
Rule: The entities found on the page with their confidence and topicality scores.
Google's description: The annotated entities, with associated confidence scores and metadata. This is the primary output of WebRef/QRef. In case of Webref output, entities are sorted by decreasing topicality score.
type list(GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefWebrefEntity.t) · confidence high · stages: 6. Entity & Knowledge Graph Integration

`RepositoryWebrefWebrefMustangAttachment.confidenceE2`
Rule: Confidence scores for each entity found on the page.
Google's description: The confidence scores of all entities in the encoded_mid array. For space reasons this is also represented as a fixed-point integer with two decimal precision. Convert it to confidence_score using the following formula: float confidence_score = confidence_e2 / 100.0f Should have the same number of elements as the encoded_mid field.
type list(integer()) · confidence high

`RepositoryWebrefWebrefMustangAttachment.encodedMid`
Rule: The entities found on the page, listed in order of how central they are.
Google's description: The int64-encoded MIDs of the entities in the document sorted by topicality score. To convert it to the string form use metaweb::MidToString(encoded_mid) defined in metaweb/util/mid/mid.h Should have the same number of elements as the topicality_e2 field.
type list(String.t) · confidence high

`RepositoryWebrefWebrefMustangAttachment.isAuthorIndex`
Rule: Which of the page's entities are its authors.
Google's description: The indices of all the reference entities in encoded_mid that are authors of the page. Not packed as in most cases when populated, it contains 1 element.
type list(integer()) · confidence high

`RepositoryWebrefWebrefMustangAttachment.isPublisherIndex`
Rule: Which of the page's entities are its publishers.
Google's description: The indices of all the reference entities in encoded_mid that are publishers of the page (e.g. /m/cnn on "
type list(integer()) · confidence high

`RepositoryWebrefWebrefMustangAttachment.topicalityE2`
Rule: How central each entity is to the page, scored 0 to 100.
Google's description: The topicality scores of all entities in the encoded_mid array. For space reasons this is represented as a fixed-point integer with two decimal points precision. Convert it to topicality_score using the following formula: float topicality_score = topicality_e2 / 100.f Should have the same number of elements as the encoded_mid field.
type list(integer()) · confidence high

`CompressedQualitySignals.topicEmbeddingsVersionedData`
Rule: Versioned topic embedding data for the page that scorers can read directly.
Google's description: Versioned TopicEmbeddings data to be populated later into superroot / used directly in scorers.
type list(GoogleApi.ContentWarehouse.V1.Model.QualityAuthorityTopicEmbeddingsVersionedItem.t) · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 8. Spam, Penalty & Trust Systems; 11. Re-Ranking & Twiddler System

`ImageDataMultibangEntitiesMultibangEntity.score`
Rule: How strongly the image is tied to that real-world thing.
Google's description: Multibang score.
type number() · confidence medium

`ImageQualitySensitiveMediaOrPeopleEntities.peopleEntitiesId`
Rule: Identifiers of the five people the document is most about.
Google's description: hrid of the 5 people entities with higher topicality in the document
type list(String.t) · confidence medium

`ImageUnderstandingIndexingLabel.entityId`
Rule: Knowledge Graph entity identifier for a label found in the image.
Google's description: KG entity id.
type String.t · confidence medium

`NlpSaftDocument.entity`
Rule: The list of entities that were recognised in the document text.
Google's description: Entities in the document.
type list(GoogleApi.ContentWarehouse.V1.Model.NlpSaftEntity.t) · confidence medium · stages: 9. Query Understanding & Intent Classification

`NlpSaftDocument.focusEntity`
Rule: The single entity that the page is mainly about, such as a Wikipedia subject.
Google's description: Focus entity. For lexicon articles, like Wikipedia pages, a document is often about a certain entity. This is the local entity id of the focus entity for the document.
type integer() · confidence medium · stages: 9. Query Understanding & Intent Classification

`NlpSaftEntity.mention`
Rule: The places in the document where this entity is mentioned.
Google's description: Mentions of the entity in the document.
type list(GoogleApi.ContentWarehouse.V1.Model.NlpSaftMention.t) · confidence medium

`NlpSaftEntity.salience`
Rule: How central this entity is to what the document is about.
Google's description: Score indicating the saliency (centrality) of this entity to the document.
type float() · confidence medium

`NlpSaftEntity.type`
Rule: The categories this entity belongs to, such as person, place or TV personality.
Google's description: Entity types of the entity. These can include SAFT types (/saft/location, /saft/art, /saft/other/living_thing, etc), collections types (/collection/tv_personalities, /collection/statistical_regions, etc), and more. This refers to the type of the entity itself: in "She is on TV", "She" refers to a specific actor, with type "/collection/tv_personalities". Cf. Mention.Type, which is the type of the referring mention.
type list(GoogleApi.ContentWarehouse.V1.Model.NlpSaftEntityType.t) · confidence medium

`PerDocData.queriesForWhichOfficial`
Rule: The query, country and language combinations for which this page counts as the official page.
Google's description: The set of (query, country, language) triples for which this document is considered to be the official page. For example, www.britneyspears.com would be official for ("britney spears", "us", 0) and others (0 is English).
type GoogleApi.ContentWarehouse.V1.Model.OfficialPagesQuerySet.t · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityFringeFringeQueryPriorPerDocData.sensitiveEntitiesMids`
Rule: Identifiers of entities on the page that refer to people or groups of people.
Google's description: MIDs of entities that represent a person or a group of people (aka sensitive entities). See b/290268614 for reference why this is needed in replacement to sensitive_entities_indices.
type list(String.t) · confidence medium

`RepositoryWebrefCategoryAnnotationHitCatSource.confidence`
Rule: How confident the system is that a category describes the page.
Google's description: Confidence of the category in the range [0.0, 1.0). If a page has the category "NBA" with high confidence we also expect it to have the "Basketball" with high confidence. Categories with a confidence lower than 0.05 are omitted. This can be interpreted as a confidence of 0, i.e. a strong signal that the category is not relevant for the page. For some categories this score is calibrated per-category to estimate the true precision. E.g., 70% of documents retrieved within the co
type number() · confidence medium

`RepositoryWebrefCategoryAnnotationHitCatSource.cumulativeConfidence`
Rule: A calibrated confidence for the category aimed at a target precision.
Google's description: Calibrated cumulative confidence guaranteeing maxmial recall for a precision target. E.g., At least 90% of documents retrieved with cumulative_confidence >= 0.9 will be relevant. This score is always calibrated per-category to estimate the true cumulative precision and is not set for uncalibrated categories.
type number() · confidence medium

`RepositoryWebrefCategoryInfo.allTypes`
Rule: All entity types with their sources and confidences, used by entity scoring.
Google's description: Contains all types relevant for this entity, along with their provenances and confidences. This field basically replicates information above in a unified way, so that Refcon scroing can make better use of it. If present, Refcon scoring will use all_types, and ignore other fields like freebase_type. Note: there is some basic conflict resolution applied when all_types are computed (implemented in type-extractor.cc, IsLikelyConflictingFreebase).
type list(GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefFreebaseType.t) · confidence medium · stages: 6. Entity & Knowledge Graph Integration

`RepositoryWebrefDetailedEntityScores.docScore`
Rule: How well a page scores for an entity, used to compare pages for that entity.
Google's description: How well the document scores for the entity. The score is unnormalized, and serves as a relative ranking signal between different documents for an entity.
type number() · confidence medium

`RepositoryWebrefDetailedEntityScores.isReferencePage`
Rule: Whether the page's full address matches the entity, making it a reference page.
Google's description: Set to true iff the entity matches the full URL of the document, meaning that it is a reference page or related page of the entity.
type boolean() · confidence medium

RepositoryWebrefDocumentMetadata.isDisambiguationPage
Rule: Whether the page is a known disambiguation page.
Google's description: Set to true if the document is a known disambiguation page, e.g.
type boolean() · confidence medium · stages: 6. Entity & Knowledge Graph Integration

RepositoryWebrefEntityAnnotations.isImplicit
Rule: Whether the entity is only implied, with no plain mention in the page text.
Google's description: An entity is marked as implicit if there is no explicit mention of the entity in the content of the page. For instance, all mentions of the entity are in query, url and/or anchors; or the entity has only implicit content mentions.
type boolean() · confidence medium

RepositoryWebrefEntityAnnotations.topicalityRank
Rule: The entity's place in the list when entities are sorted by topicality.
Google's description: Rank of the entity when sorted by topicality score.
type integer() · confidence medium

RepositoryWebrefEntityJoin.linkInfo
Rule: All scored relationships this entity has to other entities.
Google's description: Contains all links (with scores) that Webref knows for this entity. Links are relationships between entities. The data in this field is very important for the quality of the model.
type list(GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefGlobalLinkInfo.t) · confidence medium · stages: 6. Entity & Knowledge Graph Integration

RepositoryWebrefEntityJoin.nameInfo
Rule: All scored names known for this entity.
Google's description: Contains all names (with scores) that Webref knows for this entity. The data in this field is very important for the quality of the model.
type list(GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefGlobalNameInfo.t) · confidence medium · stages: 6. Entity & Knowledge Graph Integration

RepositoryWebrefEntityLinkSource.score
Rule: How much more often two entities appear together than chance would suggest.
Google's description: Score in [0, \infty) that represents how relatively likely it is to see that entity cooccurring with the main entity (in the entity join). A value of 1.0 means that the two entities are basically independent. The higher the more likely (relatively to the individual entity probabilities) they are to cooccur.
type number() · confidence medium

RepositoryWebrefEntityNameScore.confidence
Rule: How sure the system is that this name really refers to the entity.
Google's description: Confidence that this name is a trusted name of the entity. A reasonable threshold for name trust is 0.6. A name can be trusted and still have very low score_ratio, esp. if it is ambiguous (e.g. 'mercury') and/or not the dominant interpretation (e.g. "siberian husky" -> /m/06krnsr (a book)).
type number() · confidence medium · stages: 6. Entity & Knowledge Graph Integration

RepositoryWebrefEntityNameScore.score
Rule: The overall score for this entity as a meaning of the name.
Google's description: The absolute score of that entity. score = artificial_score + volume_based_score
type number() · confidence medium · stages: 6. Entity & Knowledge Graph Integration

RepositoryWebrefEntityNameScore.scoreRatio
Rule: This entity's share of the total score across all entities sharing the name.
Google's description: Ratio between this entity score and the total score over all entities. This is including the "open world" information if it was estimated.
type number() · confidence medium · stages: 6. Entity & Knowledge Graph Integration

RepositoryWebrefEntityNameScore.volumeBasedScore
Rule: The part of a name score that comes from counted sources such as clicks and anchors.
Google's description: Absolute score that comes from quantitative sources such as navboost clicks, anchors, etc. artificial_score = score - volume_based_score
type number() · confidence medium · stages: 6. Entity & Knowledge Graph Integration

RepositoryWebrefEntityScores.commonNgramProb
Rule: How likely the entity's name is just a common word or phrase.
Google's description: Probability that the entity is a common ngram (e.g. from dictionary).
type number() · confidence medium

RepositoryWebrefEntityScores.entityIdf
Rule: How rare an entity is across documents.
Google's description: The final (model omega) idf of an entity. Equals log2(1 / probability of the entity to appear in a document). This probability is currently estimated from its names (i.e. it is a sum of the name frequency weighted by P(entity | name)).
type number() · confidence medium

RepositoryWebrefExtendedEntityNameScore.scoreRatio
Rule: The entity's share of the name score for a given region or site.
Google's description: Score ratio for the entity, same as the EntityNameScore score ratio.
type number() · confidence medium

RepositoryWebrefFatcatCategory.score
Rule: How heavily this category weighs among the categories assigned.
Google's description: The relative weight of the category within a distribution.
type number() · confidence medium

RepositoryWebrefLinkInfo.aggregatedScore
Rule: The combined score for an entity to entity link across all its sources.
Google's description: The score aggregated from all sources.
type number() · confidence medium

RepositoryWebrefMention.detailedMentionScores
Rule: A breakdown of additional scores for an entity mention.
Google's description: Additional detailed scores about this mention.
type GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefDetailedMentionScores.t · confidence medium · stages: 6. Entity & Knowledge Graph Integration

RepositoryWebrefMention.trustedNameConfidence
Rule: How confident the system is that this wording is a trusted name for the entity.
Google's description: Confidence that this name is a trusted name of the entity. This is set only in case the confidence is higher than an internal threshold (see ConceptProbability).
type number() · confidence medium · stages: 6. Entity & Knowledge Graph Integration

RepositoryWebrefNameInfo.aggregatedScores
Rule: The combined score for a name across all its sources.
Google's description: The score aggregated from all sources.
type GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefAggregatedEntityNameScores.t · confidence medium

RepositoryWebrefNameInfo.scores
Rule: Scores that depend on the name alone, independent of any entity.
Google's description: The name-specific scores. These scores only depend on the name and are independent of the entity.
type GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefNameScores.t · confidence medium

RepositoryWebrefNameScores.completeWorldVolume
Rule: An estimate of how much total usage a name has, including unknown entities.
Google's description: Approximates volume of this name including entities we don't have in our set. This is useful to estimate the size of the "open world". For example, this field can be equal to the total number of clicks for the query. Note that for now, we ignore the number of clicks but just use the idf if available.
type number() · confidence medium

RepositoryWebrefNameScores.idfScore
Rule: How rare this name is across the corpus.
Google's description: The idf of this name.
type number() · confidence medium

RepositoryWebrefNameScores.totalScore
Rule: The overall amount of data backing this name from a source.
Google's description: The total score of that name/source. It describes how much data we have for that name/source. It can for example be the sum of all the entity scores for this name.
type number() · confidence medium

RepositoryWebrefNgramContext.weight
Rule: How much weight the context carries on the document, based on repeat occurrences.
Google's description: The weight of the context on the document; depends on how many times we saw the string in the document.
type number() · confidence medium

RepositoryWebrefNgramMention.score
Rule: The average score of the mention across all places the n-gram appeared.
Google's description: The average score the mention was given across all occurrences of the n-gram.
type number() · confidence medium

RepositoryWebrefPreprocessingNameEntityScores.priorScore
Rule: How much evidence there is that this spelling refers to that entity.
Google's description: An unnormalized measure of how much evidence we have that this name variant refers to the key entity. Should be comparable to all scores from the same source for: - other entities having the same name variant - the open world score computed for this name variant
type float() · confidence medium

RepositoryWebrefRefconRefconNameInfo.namePrior
Rule: The baseline likelihood that this name is used for the entity.
Google's description: Name prior to use. We read this from the prior for the "unknown" language.
type number() · confidence medium

RepositoryWebrefReferencePageScores.bookScore
Rule: A score for whether the page is a good book reference page.
Google's description: Stores score for later offline voting to choose reference pages. If zero, it's not a good book reference page.
type number() · confidence medium

RepositoryWebrefReferencePageScores.referencePageScore
Rule: A score used to pick which page is the reference page for an entity.
Google's description: Reference page score used to select the reference page owner.
type number() · confidence medium

RepositoryWebrefReferencePageScores.selected
Rule: Whether this entity was chosen as the owner of the reference page.
Google's description: True if the entity is selected as the reference page owner.
type boolean() · confidence medium

RepositoryWebrefReferencePageScores.singleTopicness
Rule: Score from 0 to 1 saying how much the page is about just this one entity.
Google's description: A score in [0, 1] which indicates the single topicness of the entity.
type number() · confidence medium

RepositoryWebrefReferencePageScores.singleTopicnessV2
Rule: Newer version of the single-topicness score meant to replace the old one.
Google's description: This should have the same semantic as single_topicness, and should replace it in the long term.
type number() · confidence medium

RepositoryWebrefSegmentMentions.mention
Rule: List of every spot in this text segment where the entity was mentioned.
Google's description: A list of all the places the entity in question was annotated within this segment. The (indexing.annotations.goldmine) option is for Goldmine AnnotationsFinder to include Mention only when segment_type="CONTENT"
type list(GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefMention.t) · confidence medium

RepositoryWebrefSimplifiedCompositeDoc.documentMentionSpans
Rule: The spans of document text that were tagged with entities.
Google's description: The composite doc spans which were annotated with entities by Webref.
type GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefRefconDocumentMentionSpans.t · confidence medium

RepositoryWebrefWebrefEntities.category
Rule: Topic categories assigned to the page or query.
Google's description: Categories of the document or query. This replaces the category_score found under EntityAnnotations.
type list(GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefCategoryAnnotation.t) · confidence medium · stages: 6. Entity & Knowledge Graph Integration

RepositoryWebrefWebrefEntity.annotations
Rule: All the markings of this entity found on the page.
Google's description: All annotations of this entity on the given document.
type GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefEntityAnnotations.t · confidence medium

RepositoryWebrefWebrefMustangAttachment.categoryConfidenceE2
Rule: How confident the system is in a category assigned to the page, from 0 to 100.
Google's description: The confidence of the category. In the range [0, 100].
type list(integer()) · confidence medium

RepositoryWebrefWebrefMustangAttachment.categoryEncodedMid
Rule: Category entity IDs attached to the page.
Google's description: See go/category-annotations-api about the story behind various types of category annotations that are provided using the catmid token and category_encoded_mid fields below. Some of these annotation types are experimental, so please contact related-entities@ if you consider using this data. For production uses, please: 1. Add your use-case to go/hits-clients. 2. Subscribe to hits-users@ to receive general updates and info about deprecations. To convert it to the string form use metaweb::MidToStri
type list(String.t) · confidence medium

RepositoryWebrefWebrefMustangAttachment.referencePageIndex
Rule: Marks the page as the main reference page for one or more entities.
Google's description: Only populated when the document is a reference page for an entity. Contains the indices of all reference entities in the encoded_mid and topicality_e2 arrays. Not packed as in most cases when populated, it contains 1 element.
type list(integer()) · confidence medium

ResearchScienceSearchSourceUrlDocjoinInfo.sporeGraphMid
Rule: Entities recognized on the page that exist in the science knowledge graph.
Google's description: A set of entities from WebRef annotations that are in SPORE_GRAPH.
type list(String.t) · confidence medium

`ResearchScienceSearchSourceUrlDocjoinInfo.webrefEntity`
Rule: Entities recognized on the page, copied from the main document record.
Google's description: A set of entities copied from WebRefEntities on cDoc.
type list(GoogleApi.ContentWarehouse.V1.Model.ResearchScienceSearchSourceUrlDocjoinInfoWebrefEntityInfo.t) · confidence medium

`ResearchScienceSearchSourceUrlDocjoinInfoWebrefEntityInfo.mid`
Rule: The knowledge graph identifier of an entity found on the page.
Google's description: The KG identifier of the WebrefEntity.
type String.t · confidence medium

`VideoContentSearchVideoInfo.webrefEntities`
Rule: People, places and things identified in the document by the entity service.
Google's description: Represents a collection of entities returned by the WebRef service. This message contains a lot of data and dependencies, so sub-messages are disabled in model evaluation in scorer.
type GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefWebrefEntities.t · confidence medium

`ImageDataMultibangEntitiesMultibangEntity.entityId`
Rule: The identifier of a real-world thing the image is linked to.
Google's description: Entity ID.
type String.t · confidence low

`ImageQualitySensitiveMediaOrPeopleEntities.mediaEntitiesId`
Rule: Identifiers of media things such as films or bands linked to the document.
Google's description: Similar to above, but for media entities (movie name, fictional character, musical band)
type list(String.t) · confidence low

`NlpSaftDocument.annotatedPhrase`
Rule: Phrases marked up in the text that are not clean references to a known entity.
Google's description: Annotated phrases in the document that are not semantically well-defined mentions of entities.
type list(GoogleApi.ContentWarehouse.V1.Model.NlpSaftAnnotatedPhrase.t) · confidence low · stages: 9. Query Understanding & Intent Classification

`NlpSaftDocument.author`
Rule: The name or names of the people credited as authors of the document.
Google's description: Document author(s).
type list(String.t) · confidence low · stages: 9. Query Understanding & Intent Classification

`NlpSaftDocumentTopic.name`
Rule: The name of a topic assigned to the document.
Google's description: Topic name or identifier.
type String.t · confidence low

`NlpSaftDocumentTopic.score`
Rule: How strongly the document matches an assigned topic.
Google's description: Topic score.
type number() · confidence low

`NlpSaftEntity.name`
Rule: The main name used for this entity.
Google's description: Representative entity name.
type String.t · confidence low

`NlpSaftEntity.profile`
Rule: The knowledge profile attached to this entity.
Google's description: Profile for entity.
type GoogleApi.ContentWarehouse.V1.Model.NlpSaftEntityProfile.t · confidence low

`NlpSaftEntityType.name`
Rule: The name of the category an entity belongs to, such as person or place.
Google's description: The type name, like "/saft/person". See README.entity-types for the inventory of SAFT type tags.
type String.t · confidence low

`NlpSaftEntityType.score`
Rule: How confident the system is that an entity belongs to this category.
Google's description: A score for this type.
type number() · confidence low

`NlpSaftMention.resolution`
Rule: What a mention itself refers to, which can differ from the entity it belongs to.
Google's description: Mention-level resolution. This is used for encoding the meaning of the mention rather than the entity. For example, definite references and appositions are resolved to the mid for the concept rather than the entity.
type GoogleApi.ContentWarehouse.V1.Model.NlpSaftMentionResolution.t · confidence low

`NlxDataSchemaMention.entity`
Rule: Link from a name mentioned in text to the entity it refers to.
Google's description: The entity that this mention refers to.
type GoogleApi.ContentWarehouse.V1.Model.MultiscalePointerIndex.t · confidence low

`OrionDocEntitiesProto.encodedEntity`
Rule: Encoded list of entities found in a document.
Google's description: This is encoded using EntityCandidate::Encode
type list(integer()) · confidence low

`PerDocData.knexAnnotation`
Rule: Stored k'nex annotations attached to fresh documents.
Google's description: For indexing k'nex annotations for FreshDocs.
type GoogleApi.ContentWarehouse.V1.Model.SocialPersonalizationKnexAnnotation.t · confidence low · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.v2KnexAnnotation`
Rule: Stored second-generation k'nex annotations for the document.
Google's description: For indexing v2 k'nex, see/go/knex-v2-doc-annotation for details.
type GoogleApi.ContentWarehouse.V1.Model.QualitySherlockKnexAnnotation.t · confidence low · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualitySherlockKnexAnnotationItem.calibratedScore`
Rule: A calibrated zero-to-one score for an entity annotation on the document.
Google's description: in [0, 1].
type number() · confidence low

`QualitySherlockKnexAnnotationItem.equivalentMid`
Rule: The knowledge-graph identifier of an equivalent entity.
Google's description: in /m/ or /g/.
type String.t · confidence low

`QualitySherlockKnexAnnotationItem.score`
Rule: A zero-to-one score for an entity annotation on the document.
Google's description: in [0, 1].
type number() · confidence low

`QualityViewsExtractionClusterInfo.clusterSiblingMid`
Rule: Identifiers of other entities belonging to the same group.
Google's description: The mids of cluster members that are part of the same cluster. Note that cluster members may end up having their own interpretation (EntityInfo which includes a ClusterInfo) or not (eg because they do not explain the full query, and so aqua does not output an interpretation for them). The latter case (a cluster member is output only as part of this field), is equivalent to its score being 0.
type list(String.t) · confidence low

`RepositoryWebrefAnnotatedCategoryInfo.mid`
Rule: The knowledge graph identifier of the entity that stands for this category.
Google's description: The mid of the entity representing the category.
type String.t · confidence low

`RepositoryWebrefCategoryAnnotation.hitcat`
Rule: The sources that claim this page belongs to a category.
Google's description: Sources asserting the category. In the future we may have one calibrated confidence score.
type GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefCategoryAnnotationHitCatSource.t · confidence low

`RepositoryWebrefCategoryInfo.salientCategory`
Rule: Topic categories that often appear together with this entity across documents.
Google's description: verticals4 categories that cooccur with this entity, aggregated over D2E. See where we read from CompactDocClassification in
type list(GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefFatcatCategory.t) · confidence low · stages: 6. Entity & Knowledge Graph Integration

`RepositoryWebrefCategoryInfo.wpCategory`
Rule: Category classification taken from Wikipedia categories.
Google's description: WPCat CategotyResult classification.
type list(GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefFreebaseType.t) · confidence low · stages: 6. Entity & Knowledge Graph Integration

`RepositoryWebrefDetailedEntityScores.profileUrl`
Rule: A link to the author's profile page on the site.
Google's description: Contains the link to the author's profile url on the platform.
type String.t · confidence low

RepositoryWebrefEntityAnnotations.detailedEntityScores
Rule: Extra detail on how the entity relates to the page, such as being its publisher.
Google's description: Additional information about how the entity relates to the page, for example whether it is a business entity which published the page.
type GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefDetailedEntityScores.t · confidence low

RepositoryWebrefEntityJoin.contextNameInfo
Rule: Scored names related to the entity that are not used to find mentions.
Google's description: The context names (with scores) of this entity. The difference to regular names (aka name_info) is that context names are not used for finding mentions in a document as they consist of names somehow related to the entity (e.g. name "fisherman s wharf" for the entity "Gary Danko"). Used for reconciling freebase and oyster.
type list(GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefGlobalNameInfo.t) · confidence low · stages: 6. Entity & Knowledge Graph Integration

RepositoryWebrefEntityJoin.extraData
Rule: Extra metadata about the entity gathered from its sources.
Google's description: Additional metadata about the entity, that can be derived from the "raw data" (composite doc, domain specific data...), or come from other sources. Despite its name, this field often contains quite important information.
type GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefExtraMetadata.t · confidence low · stages: 6. Entity & Knowledge Graph Integration

RepositoryWebrefEntityJoin.refconNameInfo
Rule: Names and name details used by the entity reconciliation system.
Google's description: Contains names and names metadata used by Refcon.
type list(GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefRefconRefconNameInfo.t) · confidence low · stages: 6. Entity & Knowledge Graph Integration

RepositoryWebrefEntityNameScore.extendedScoreRatio
Rule: Name scores for particular regions where they differ from the language-wide score.
Google's description: Stores region specific score ratios for the entity when it is significantly different from the language version above.
type list(GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefExtendedEntityNameScore.t) · confidence low · stages: 6. Entity & Knowledge Graph Integration

RepositoryWebrefEntityNameScore.nameMetadata
Rule: Combined details about a name gathered from the various name sources.
Google's description: Metadata about this name aggregated from name signals.
type GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefPreprocessingNameEntityMetadata.t · confidence low · stages: 6. Entity & Knowledge Graph Integration

RepositoryWebrefEntityNameScore.rangeMetadata
Rule: Details about the stretch of text that this entity was matched to.
Google's description: Metadata of segment range, which is annotated by this entity.
type list(GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefRangeMetadata.t) · confidence low · stages: 6. Entity & Knowledge Graph Integration

RepositoryWebrefEntityNameSource.entityScore
Rule: All entity name scores contributed by one source.
Google's description: All the entity-name scores from that source. Keyed by the EntityNameScore.entity_id field.
type list(GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefEntityNameScore.t) · confidence low

RepositoryWebrefExplainedRangeInfoExplainedRange.mention
Rule: The mention record showing where the text occurs in the document.
Google's description: SegmentMention describing the occurrence of the token in the document.
type GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefSegmentMention.t · confidence low

RepositoryWebrefExtraMetadata.categoryInfo
Rule: The category types that apply to an entity.
Google's description: Information about category types of the entity.
type GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefCategoryInfo.t · confidence low

RepositoryWebrefExtraMetadata.entityScores
Rule: Extra numeric scores attached to the entity.
Google's description: Additional scores for the entity.
type GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefEntityScores.t · confidence low

RepositoryWebrefExtraMetadata.equivalentEntityId
Rule: Other ids meaning the same real-world thing, so query and page mentions can match.
Google's description: An entity in KG that represents the same (or equivalent) entity in the real world. In particular, this is used for mid-forwarding: when de-duping entities in KG, the old ids represent the exact same entity as the one they were merged with. So when we see one id in the query and the other in a document, they are treated as the same entity.
type list(GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefWebrefEntityId.t) · confidence low

`RepositoryWebrefExtraMetadata.latentEntities`
Rule: Other entities strongly implied by this one, such as a player's club.
Google's description: A list of entities that are latent given this entity. For example, "Lionel Messi" can have the latent entity "FC Barcelona". The latent entity links are materialized in an offline pipeline using r/w/scripts/latent_entities/latent-entities.pq. For more information, see go/latent-entities.
type GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefLatentEntities.t · confidence low

`RepositoryWebrefFatcatCategory.id`
Rule: The identifier of a topic category from the vertical classification system.
Google's description: The category ID from verticals4. See go/verticals4 and where we read them in
type integer() · confidence low

`RepositoryWebrefFreebaseType.score`
Rule: An optional score for how well a type fits the entity.
Google's description: Optional score. Not present in KG directly but e.g. in WPCat.
type number() · confidence low

`RepositoryWebrefFreebaseType.typeName`
Rule: The readable name of an entity type, such as a person or a business industry.
Google's description: e.g.: "/business/industry", "/book/book_subject", "/people/person"... HRID of the type. May be missing for optimisation reasons in Refcon, but is present for Qref/Webref. At least one of type_name or type_fprint will be present at all times
type String.t · confidence low

`RepositoryWebrefGlobalLinkInfo.isBoostedPrimaryWeightLink`
Rule: Flags an entity link that in some cases gets extra weight in primary scoring.
Google's description: Whether this is a BoostedPrimaryWeight link. For these links in some cases a higher weight is used for primary scoring.
type boolean() · confidence low · stages: 6. Entity & Knowledge Graph Integration

`RepositoryWebrefLatentEntity.broaderImportance`
Rule: How strongly a broader entity relates to this one, one score per source.
Google's description: The relatedness score of the two entities corresponding to each source above.
type list(number()) · confidence low

`RepositoryWebrefLinkInfo.source`
Rule: The link's score broken down by each source that supplied it.
Google's description: The per-source scores.
type list(GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefEntityLinkSource.t) · confidence low

`RepositoryWebrefMdvcMetadataPerVertical.expandedOutputIds`
Rule: Extra concept ids that get annotated whenever this entity is annotated.
Google's description: Concept ids to annotate whenever this entity is annotated.
type list(String.t) · confidence low

`RepositoryWebrefMdvcMetadataPerVertical.resolutionPriority`
Rule: Priority deciding which entity resolutions get annotated when several are possible.
Google's description: Resolution priority for this entity. If there are many possible resolutions to a MDVC understanding, only the ones with the highest priority are annotated.
type integer() · confidence low

`RepositoryWebrefNameInfo.ngramData`
Rule: Word frequency data for the name, such as how rare it is.
Google's description: N-gram data (e.g. n-gram IDF).
type GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefUniversalNgramData.t · confidence low

`RepositoryWebrefNameInfo.source`
Rule: The name's score broken down by each source.
Google's description: The per-source scores.
type list(GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefEntityNameSource.t) · confidence low

`RepositoryWebrefNameScores.contextFringeScore`
Rule: How obscure a full query context name is, on a zero to one scale.
Google's description: The fringe score in [0, 1] for this context name. This is only populated for full-query context names.
type number() · confidence low

`RepositoryWebrefNameScores.contextWeight`
Rule: How much evidence exists for this context name.
Google's description: The amount of evidence available for this context name.
type number() · confidence low

`RepositoryWebrefNameScores.openWorldVolumeModifier`
Rule: An adjustment added to the estimated open world usage of a name.
Google's description: 'Raw' open world is computed based on ngram count data. For some names however, we modify the open world based on entities associated with them. This field contains the additive open world modifier. open_world_volume = raw open world + open_world_volume_modifier
type number() · confidence low

`RepositoryWebrefNgramContext.mention`
Rule: The entities annotated in this piece of context text.
Google's description: The entities that were annotated on the context
type list(GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefNgramMention.t) · confidence low

`RepositoryWebrefPreprocessingNameEntityScores.volumeBasedScore`
Rule: The part of a name's prior score that comes from countable sources.
Google's description: Prior score come from source that is quantifiable. artificial_score = prior_score - volume_based_score.
type float() · confidence low

`RepositoryWebrefPreprocessingNameVariantSignals.scores`
Rule: The shared score block for a name variant from one source.
Google's description: Common prior_score/trust proto This field is shared by all sources providing this kind of data, the information has to be considered in context with the source.
type GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefPreprocessingNameEntityScores.t · confidence low

`RepositoryWebrefPreprocessingOriginalNamesOriginalName.score`
Rule: How good this original spelling is, based on source authority and frequency.
Google's description: Score estimating how good this original name is: - some sources are considered more authoritative than others (e.g. KG) - a name found in more sources is better.
type float() · confidence low

`RepositoryWebrefSupportTransferRule.mentionsOnly`
Rule: Setting that lets pages backing one entity also count as backing a related entity.
Google's description: STBR doesn't just create annotations for STBR targets. It also can modify search result support (useful in PostRef) in order for web pages supporting the STBR source to also support the STBR target. Otherwise in PostRef annotations for STBR targets are going to be demoted, resulting in the STBR rule potentially not affecting anything. Regarding treatment of this result support, STBR has 3 possible modes that for historical reasons are represented as 2 booleans - mentions_only and support_share.
type boolean() · confidence low

`RepositoryWebrefTripleAnnotation.confidenceScore`
Rule: How confident the system is in a fact extracted from the page.
Google's description: Triple annotation confidence_score (value between 0 and 1). Higher values correspond to higher confidence.
type number() · confidence low

`RepositoryWebrefWebrefAnnotationStats.docWeight`
Rule: How much weight this document gets when its annotation data is pooled with others.
Google's description: The relative weight of the document, used when aggregating information from multiple documents.
type number() · confidence low

`RepositoryWebrefWebrefAnnotationStats.ngramContext`
Rule: Scores for phrases found around an entity mention, weighted by document weight.
Google's description: Extracted n-grams context scores (in cdoc language, weighted by doc_weight) output if webref_populate_annotation_ngrams is enabled.
type list(GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefNgramContext.t) · confidence low

`RepositoryWebrefWebrefEntity.annotatedRelationship`
Rule: Links recorded between entities annotated on the same page.
Google's description: Information about links (e.g. implications) between the annotated entities.
type list(GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefWebrefEntityRelationship.t) · confidence low

`RepositoryWebrefWebrefEntity.collections`
Rule: The knowledge graph type groups this entity belongs to.
Google's description: Information about the collections of this entity.
type GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefWebrefEntityCollections.t · confidence low

`RepositoryWebrefWebrefEntityRelationship.linkWeight`
Rule: Strength of the link between two entities found on the page.
Google's description: The weight of the link.
type number() · confidence low

`SocialPersonalizationKnexAnnotationItem.mid`
Rule: The identifier of the topic a piece of content or a user profile is tagged with.
Google's description: This is an MID in the /t/ namespace. We will migrate them to /g/ in Q3'15.
type String.t · confidence low

Spam & demotions (97)

`IndexingDocjoinerAnchorPhraseSpamInfo.phraseAnchorSpamCount`
Rule: How many spam phrases appeared in anchors across unique domains.
Google's description: How many spam phrases found in the anchors among unique domains.
type number() · confidence high · stages: 4. Link Analysis & Authority Computation

`IndexingDocjoinerAnchorPhraseSpamInfo.phraseAnchorSpamFraq`
Rule: Share of the page's anchors that were spam phrases.
Google's description: Spam phrases fraction of all anchors of the document.
type number() · confidence high · stages: 4. Link Analysis & Authority Computation

`IndexingDocjoinerAnchorSpamInfo.phraseCount`
Rule: How many spam phrases appeared in anchors across unique domains.
Google's description: Following fields record signals used in anchor spam classification. How many spam phrases found in the anchors among unique domains.
type number() · confidence high · stages: 4. Link Analysis & Authority Computation; 8. Spam, Penalty & Trust Systems

`IndexingDocjoinerAnchorSpamInfo.phraseFraq`
Rule: Share of the page's anchors that were spam phrases.
Google's description: Spam phrases fraction of all anchors of the document.
type number() · confidence high · stages: 4. Link Analysis & Authority Computation; 8. Spam, Penalty & Trust Systems

`IndexingDocjoinerAnchorSpamInfo.spamProbability`
Rule: The model's estimated chance that the page's anchors are spam.
Google's description: Predicted probability of spam.
type number() · confidence high · stages: 4. Link Analysis & Authority Computation; 8. Spam, Penalty & Trust Systems

`IndexingDocjoinerAnchorStatistics.spamLog10Odds`
Rule: Log-odds score that the page's anchor set looks spammy.
Google's description: The log base 10 odds that this set of anchors exhibits spammy behavior. Computed in the AnchorLocalizer.
type number() · confidence high · stages: 4. Link Analysis & Authority Computation

`IndexingDocjoinerAnchorTrustedInfo.matchedScore`
Rule: Score for how closely a trusted site's anchor text resembles the spammy anchors.
Google's description: Difference in KL-divergence from spam and non-spam anchors. Value >0 indicate that anchor text from this trusted source is similar to anchors classified as spam which means that spammy anchors are legitimate.
type number() · confidence high

`WWWDocInfo.pornStats`
Rule: Adult-content statistics used in porn demotion and filtering.
Google's description: 'porn_stats' is used in porn demotion and filtering. See classifier/porn/public/porn-attachments.h.
type integer() · confidence high

`AnchorsAnchorSource.spamrank`
Rule: Spam rank of the linking page on a 16-bit scale.
Google's description: uint16 scale
type integer() · confidence medium · stages: 4. Link Analysis & Authority Computation

`AnchorsAnchorSource.spamscore2`
Rule: A spam score for the linking page on a 0 to 127 scale.
Google's description: 0-127 scale
type integer() · confidence medium · stages: 4. Link Analysis & Authority Computation

`BlogPerDocData.clientSpamminess`
Rule: How spammy the posting tool used to create a short post looks.
Google's description: This score captures how spammy the client is that the micropost was created with. The higher the score the worse.
type integer() · confidence medium

`BlogPerDocData.spamScore`
Rule: A spam score for the blog post.
Google's description: Blog scoring signals.
type integer() · confidence medium

`ClassifierPornReferrerCounts.porn`
Rule: How many referring pages are classified as porn or adult.
Google's description: Number of referrers which are classified as porn and as adult.
type integer() · confidence medium

`ClassifierPornReferrerCounts.total`
Rule: The total number of referring pages counted.
Google's description: Total number of referrers.
type integer() · confidence medium

`ClassifierPornSiteData.numberOfPages`
Rule: How many pages of the site were used to compute these adult content scores.
Google's description: The number of pages that were used to compute the scores and ratios.
type String.t · confidence medium

`ClassifierPornSiteData.sitePornRatio`
Rule: The share of the site's pages that are porn or soft porn.
Google's description: The ratio of porn/softporn of the site this page belongs to.
type number() · confidence medium

`ClassifierPornSiteData.violenceScore`
Rule: An aggregate of how violent the pages on the site score.
Google's description: The cubic mean aggregation of violence page scores in the site.
type number() · confidence medium

`CompositeDoc.porninfo`
Rule: Adult content data about the page used for safe search classification.
Google's description: Porn related data used for image and web search porn classification as well as for diagnostics purposes.
type GoogleApi.ContentWarehouse.V1.Model.ClassifierPornDocumentData.t · confidence medium · stages: 2. Document Processing & Indexing

`ImageData.brainPornScores`
Rule: Scores from image pixels for adult, violent, medical and spoof content.
Google's description: A [0..1] SafeSearch scores based on image pixels, using Google Brain: porn, csai, violence, medical, and spoof. For porn only, if available prefer final_porn_score as it should be more precise than brain_porn_scores.porn_score.
type GoogleApi.ContentWarehouse.V1.Model.ImageSafesearchContentBrainPornAnnotation.t · confidence medium

`ImageData.clickMagnetScore`
Rule: How likely the image is to attract clicks from unrelated or bad queries.
Google's description: A score in (0, 1] to indicate how likely this image is considered as a click magnet based on clicks received from bad queries.
type number() · confidence medium

`ImageData.finalPornScore`
Rule: A combined adult-content score built from image features, referrers and query clicks.
Google's description: A [0..1] porn score based on some image-level features (like content score, referrer statistics, navboost queries, etc.). See class RiflePornScorer for more details.
type number() · confidence medium

`ImageData.isUnwantedContent`
Rule: A flag marking the image as unwanted for the search index.
Google's description: Field to indicate the image is unwanted for search index. The data is propagated from amarna to alexandria to be annotated in the cdoc. Refer to image/repository/proto/unwanted_content.proto for more info.
type boolean() · confidence medium

`ImageData.pornFlagData`
Rule: Adult-content flags used when web and image indexes are combined.
Google's description: Used by the segindexer for combined www+image indices.
type GoogleApi.ContentWarehouse.V1.Model.PornFlagData.t · confidence medium

`ImageRegionsImageRegions.finalPornScore`
Rule: Final adult-content probability score assigned to the image.
Google's description: The final_porn_score for the image.
type number() · confidence medium

`ImageRegionsImageRegions.finalViolenceScore`
Rule: Final violence probability score assigned to the image.
Google's description: The final_violence_score for the image.
type number() · confidence medium

`ImageRegionsImageRegions.isIuInappropriate`
Rule: Whether the image is flagged as inappropriate for Images Universal.
Google's description: True if the image is iu-inappropriate.
type boolean() · confidence medium

`ImageRegionsImageRegions.racyScore`
Rule: Score for how sexually suggestive the image is.
Google's description: The racy_score of the image.
type number() · confidence medium

`ImageRepositoryAnimatedImagePerdocData.aggregatedPornScores`
Rule: Combined adult-content scores computed across the frames of an animated image.
Google's description: Aggregated porn scores for animated images. Aggregated using max sampling rate / max duration. Note the plan is to fold these scores into existing summarized scores, for cases where these scores are available. See tracking bug b/63580795.
type GoogleApi.ContentWarehouse.V1.Model.ImageSafesearchContentBrainPornAnnotation.t · confidence medium

ImageRepositoryContentBasedVideoMetadata.safesearchVideoContentSignals
Rule: SafeSearch classifier scores for the video's content across sensitive categories.
Google's description: Contains SafeSearch video classification outputs which are vertical_name/float pairs.
type GoogleApi.ContentWarehouse.V1.Model.SafesearchVideoContentSignals.t · confidence medium

ImageRepositoryContentBasedVideoMetadata.videoPornScore
Rule: Confidence score that the video contains pornographic content.
Google's description: The video porn confidence score extracted from Whisper featureSet: "video_labels:whisper_v3", with CR2 label: "/cr2/1".
type number() · confidence medium

ImageRepositoryContentBasedVideoMetadata.videoPornScoreV4
Rule: Newer model confidence score that the video contains pornographic content.
Google's description: The video porn confidence score extracted from WhisperV4 featureSet: "VideoFeatures.whisper_v4_labels", with CR2 label: "/tns/porn".
type number() · confidence medium

ImageSafesearchContentBrainPornAnnotation.iuInappropriateScore
Rule: Probability the image is inappropriate for general image results.
Google's description: This field contains the probability that an image is inappropriate for Images Universal, according to this policy: go/iupolicy.
type number() · confidence medium

ImageSafesearchContentBrainPornAnnotation.racyScore
Rule: Score for how sexually suggestive the image is.
Google's description: This score is related to an image being sexually suggestive.
type number() · confidence medium

ImageSafesearchContentBrainPornAnnotation.semanticSexualizationScore
Rule: Graded score for how sexualized the image content is.
Google's description: This score is related to racy/sexual images where scores have semantic meaning from 0 to 1.
type number() · confidence medium

ImageSafesearchContentOCRAnnotation.pornScore
Rule: How adult or racy the text found inside the image is.
Google's description: Image OCR racyness/pornyness, computed by porn query classifier.
type number() · confidence medium

ImageSafesearchContentOCRAnnotation.prominentOffensiveScore
Rule: Offensiveness of the image's text weighted by how prominent that text is.
Google's description: Same as offensive_score, but weighted by prominence.
type number() · confidence medium

ImageSafesearchContentOCRAnnotation.prominentVulgarScore
Rule: Vulgarity of the image's text weighted by how prominent that text is.
Google's description: Same as vulgar_score, but weighted by prominence.
type number() · confidence medium

ImageSafesearchContentOCRAnnotation.qbstOffensiveScore
Rule: Offensiveness score for salient terms found in the image's text.
Google's description: The score produced by offensive salient terms model.
type number() · confidence medium

ImageSearchImageLicenseInfo.safesearchFlags
Rule: Bit flags recording which SafeSearch filters apply to the image.
Google's description: A bitwise-OR of SafeSearch filtering flags. If present, the flags will be a bitwise-AND between this value and all the classifier_porn::query::Vertical enums. If the value is -1, it indicates there is some error with SafeSearch classifier. The default value 0 means no filtering flags are set.
type integer() · confidence medium

IndexingDocjoinerAnchorPhraseSpamInfo.phraseAnchorSpamDays
Rule: The number of days over which most of those spam phrases appeared.
Google's description: Over how many days 80% of these phrases were discovered.
type number() · confidence medium · stages: 4. Link Analysis & Authority Computation

IndexingDocjoinerAnchorPhraseSpamInfo.phraseAnchorSpamDemoted
Rule: How many anchors were demoted.
Google's description: Total number of demoted anchors.
type integer() · confidence medium · stages: 4. Link Analysis & Authority Computation

IndexingDocjoinerAnchorPhraseSpamInfo.phraseAnchorSpamEnd
Rule: When the burst of spammy anchor text ended.
Google's description: Time when anchor spam spike ended with padding.
type integer() · confidence medium · stages: 4. Link Analysis & Authority Computation

`IndexingDocjoinerAnchorPhraseSpamInfo.phraseAnchorSpamProcessed`
Rule: How many anchors were looked at in total.
Google's description: Total number of observed anchors.
type integer() · confidence medium · stages: 4. Link Analysis & Authority Computation

`IndexingDocjoinerAnchorPhraseSpamInfo.phraseAnchorSpamRate`
Rule: Average number of spam anchors found per day.
Google's description: Average daily rate of spam anchor discovery.
type number() · confidence medium · stages: 4. Link Analysis & Authority Computation

`IndexingDocjoinerAnchorPhraseSpamInfo.phraseAnchorSpamStart`
Rule: When the burst of spammy anchor text started.
Google's description: Time when anchor spam spike started with padding.
type integer() · confidence medium · stages: 4. Link Analysis & Authority Computation

`IndexingDocjoinerAnchorSpamInfo.anchorFraq`
Rule: Share of the anchor history that falls in the demoted period.
Google's description: Ratio of spam demoted period to all anchor period.
type number() · confidence medium · stages: 4. Link Analysis & Authority Computation; 8. Spam, Penalty & Trust Systems

`IndexingDocjoinerAnchorSpamInfo.demoted`
Rule: How many anchors were demoted.
Google's description: Following field record details of anchor demotion in action. How many anchors were demoted.
type integer() · confidence medium · stages: 4. Link Analysis & Authority Computation; 8. Spam, Penalty & Trust Systems

`IndexingDocjoinerAnchorSpamInfo.demotedAll`
Rule: Whether every anchor in the period was demoted or only the spammy ones.
Google's description: Demoted all anchors in the period or only anchors classified as spam.
type boolean() · confidence medium · stages: 4. Link Analysis & Authority Computation; 8. Spam, Penalty & Trust Systems

`IndexingDocjoinerAnchorSpamInfo.demotedEnd`
Rule: When the anchor demotion period ends.
Google's description: End date of the demotion period.
type integer() · confidence medium · stages: 4. Link Analysis & Authority Computation; 8. Spam, Penalty & Trust Systems

`IndexingDocjoinerAnchorSpamInfo.demotedStart`
Rule: When the anchor demotion period starts.
Google's description: Start date of the demotion period.
type integer() · confidence medium · stages: 4. Link Analysis & Authority Computation; 8. Spam, Penalty & Trust Systems

`IndexingDocjoinerAnchorSpamInfo.phraseDays`
Rule: The number of days over which most of those phrases appeared.
Google's description: Over how many days 80% of these phrases were discovered.
type number() · confidence medium · stages: 4. Link Analysis & Authority Computation; 8. Spam, Penalty & Trust Systems

`IndexingDocjoinerAnchorSpamInfo.phraseRate`
Rule: Average number of spam anchors found per day.
Google's description: Average daily rate of spam anchor discovery.
type number() · confidence medium · stages: 4. Link Analysis & Authority Computation; 8. Spam, Penalty & Trust Systems

`IndexingDocjoinerAnchorSpamInfo.processed`
Rule: Total number of anchors processed for this page.
Google's description: Total number of processed anchors.
type integer() · confidence medium · stages: 4. Link Analysis & Authority Computation; 8. Spam, Penalty & Trust Systems

`IndexingDocjoinerAnchorSpamInfo.trustedDemoted`
Rule: How many trusted anchors went into the spam probability.
Google's description: Number of trusted anchors used in computation of spam probability.
type integer() · confidence medium · stages: 4. Link Analysis & Authority Computation; 8. Spam, Penalty & Trust Systems

`IndexingDocjoinerAnchorSpamInfo.trustedMatching`
Rule: How many trusted anchors used text matching the spam terms.
Google's description: Number of trusted anchors with anchor text matching spam terms.
type integer() · confidence medium · stages: 4. Link Analysis & Authority Computation; 8. Spam, Penalty & Trust Systems

`IndexingDocjoinerAnchorStatistics.anchorSpamInfo`
Rule: Bundle of anchor spam signals and penalties for the page.
Google's description: This structure contains signals and penalties of AnchorSpamPenalizer. It replaces phrase_anchor_spam_info above, that is deprecated.
type GoogleApi.ContentWarehouse.V1.Model.IndexingDocjoinerAnchorSpamInfo.t · confidence medium · stages: 4. Link Analysis & Authority Computation

`IndexingDocjoinerAnchorStatistics.penguinEarlyAnchorProtected`
Rule: Whether early good anchors shield the page from a link penalty.
Google's description: Doc is protected by goodness of early anchors.
type boolean() · confidence medium · stages: 4. Link Analysis & Authority Computation

`IndexingDocjoinerAnchorStatistics.penguinTooManySources`
Rule: Whether the page was left unscored because it had too many linking sources.
Google's description: Doc not scored because it has too many anchor sources. END: Penguin related fields.
type boolean() · confidence medium · stages: 4. Link Analysis & Authority Computation

`IndexingDocjoinerAnchorStatistics.phraseAnchorSpamInfo`
Rule: Bundle of signals about a spike in spammy anchor phrases.
Google's description: Following signals identify spike of spammy anchor phrases. Anchors created during the spike are tagged with LINK_SPAM_PHRASE_SPIKE.
type GoogleApi.ContentWarehouse.V1.Model.IndexingDocjoinerAnchorPhraseSpamInfo.t · confidence medium · stages: 4. Link Analysis & Authority Computation

`IndexingDocjoinerAnchorTrustedInfo.phrasesScore`
Rule: How many of the site's anchors were classified as spam by their text.
Google's description: Count of anchors classified as spam using anchor text.
type number() · confidence medium

`IndexingMobileInterstitialsProtoDesktopInterstitials.urlTree`
Rule: Tree of URL patterns on the host that show intrusive interstitials.
Google's description: URL tree of interstitial patterns belong to the host, to be used as site-level signal in Index Signals. A pattern may contain a payload InterstitialPatternPayload, which will indicate the violated interstitial types of this pattern.
type GoogleApi.ContentWarehouse.V1.Model.IndexingUrlPatternUrlTreeUrlTree.t · confidence medium

`IndexingMobileInterstitialsProtoDesktopInterstitials.violatesDesktopInterstitialPolicy`
Rule: Whether the page breaks the desktop interstitial rules.
Google's description: Overall policy violation status. If this is true, at least one of the InterstitialSignal below indicates a violation.
type boolean() · confidence medium

`PerDocData.hostAge`
Rule: The earliest date any page on this host was first seen, used to sandbox fresh spam.
Google's description: The earliest firstseen date of all pages in this host/domain. These data are used in twiddler to sandbox fresh spam in serving time. It is 16 bit and the time is day number after 2005-12-31, and all the previous time are set to 0. If this url's host_age == domain_age, then omit domain_age Please use //spam/content/siteage-util.h to convert the day between epoch second. Regarding usage of Sentinel values: We would like to check if a value exists in scoring bundle while using in Ranklab AST. For t
type integer() · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.spamMuppetSignals`
Rule: Signals about the site being hacked, joined in at query time.
Google's description: Contains hacked site signals which will be used in query time joins. As of Oct'19, the field is stored in a separate corpus. It'll only be populated for in-flight requests between retrieve and full-score in perdocdata. So no extra storage is needed on muppet side.
type GoogleApi.ContentWarehouse.V1.Model.SpamMuppetjoinsMuppetSignals.t · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.trendspamScore`
Rule: How many trending-spam queries the document matches.
Google's description: For now, the count of matching trendspam queries.
type integer() · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PornFlagData.finalOffensiveScore`
Rule: The final offensive-content score for an image, built from its text and OCR scores.
Google's description: Final offensive score based on image salient terms and image OCR vulgar and offensive scores.
type number() · confidence medium

`PornFlagData.finalViolenceScore`
Rule: The final violence score for an image, combined from several image signals.
Google's description: Final violence score based on some image signals (brain pixel score, co-clicked images violence score, navboost queries score, etc.).
type number() · confidence medium

`PornFlagData.offensiveSymbolDetection`
Rule: Whether hate symbols such as a swastika were detected in the image.
Google's description: QuimbyCongas-based detection of offensive symbols in the image (currently swastika and Nazi yellow badge).
type GoogleApi.ContentWarehouse.V1.Model.ImageSafesearchContentOffensiveSymbolDetection.t · confidence medium

`PornFlagData.qbstOffensiveScore`
Rule: An offensive score for an image derived from the queries people used to reach it.
Google's description: QBST-based image offensive score, Navboost based
type number() · confidence medium

`PornFlagData.qbstSpoofScore`
Rule: A score for how misleading an image is, derived from query logs.
Google's description: QBST-based image spoof score, Navboost based, unrelated to the pixel-based score in PornAnnotation.
type number() · confidence medium

`PornFlagData.queryTextViolenceScore`
Rule: A violence score for an image summed up from the queries that led to it.
Google's description: Aggregated navboost query violence score.
type number() · confidence medium

`PornFlagData.referrerCounts`
Rule: Counts of pages linking to this image and how adult those pages are.
Google's description: Information about referrers and their porn classification. For details see classifier/porn/proto/image_porn_classifier_signals.proto.
type GoogleApi.ContentWarehouse.V1.Model.ClassifierPornReferrerCounts.t · confidence medium

`PornFlagData.semanticSexualizationScore`
Rule: A model score for how sexualized the image looks.
Google's description: Starburst-based score predicting sexualization level of the image.
type number() · confidence medium

`QualityShoppingShoppingAttachmentOffer.nonDisplayableOrganicScoreMillis`
Rule: A one-to-1000 riskiness rating for a merchant destination in organic results.
Google's description: Normalized riskiness score for Organic destinations. It's in range [1,1000] with 1 being the worst score and 1000 being the best.
type integer() · confidence medium

`SafesearchInternalImageSignals.starburstPornScore`
Rule: An image adult-content score that feeds the final SafeSearch scores.
Google's description: Additional SafeSearch signals that are used to compute final scores.
type number() · confidence medium

`SmartphonePerDocData.adsDensityInterstitialViolationStrength`
Rule: How badly the page breaks the rules on ad-heavy mobile interstitials.
Google's description: Indicates if the page is violating mobile ads density interstitial policy and the violation strength. See go/interstitials-for-ads and
type integer() · confidence medium · stages: 12. Personalization, Local & Context

`SmartphonePerDocData.isErrorPage`
Rule: Whether the page returns an error to the smartphone crawler.
Google's description: Indicates if the page serves error to smartphone crawler. go/ramsey-sp404demotion
type boolean() · confidence medium · stages: 12. Personalization, Local & Context

`SmartphonePerDocData.isN1Redirect`
Rule: Whether the mobile page redirects visitors away from the matching content.
Google's description: Indicates if the page has mobile N-1 redirection. go/ramsey-n1demotion
type boolean() · confidence medium · stages: 12. Personalization, Local & Context

`SpamBrainData.site`
Rule: The site grouping that a spam classifier result applies to.
Google's description: Sitechunker site granularity for this result
type String.t · confidence medium · stages: 8. Spam, Penalty & Trust Systems

`WWWDocInfo.additionalSafesearchStats`
Rule: Extra statistics produced by the adult-content classifier for this document.
Google's description: Additional stats output by SafeSearch. See classifier/porn/public/porn-attachments.h.
type list(integer()) · confidence medium

`WWWDocInfo.isPorn`
Rule: Whether the document was classified as pornographic.
Google's description: Doc porn classification.
type boolean() · confidence medium

`WWWDocInfo.isSitePorn`
Rule: Whether the whole site was classified as pornographic.
Google's description: Consider the page classification is_porn as an alternative for is_site_porn, and talk to safesearch@google.com for additional information if needed.
type boolean() · confidence medium

`WWWDocInfo.isSoftporn`
Rule: Whether the document was classified as soft pornography.
Google's description: Doc softporn classification.
type boolean() · confidence medium

`WWWResultInfoSubImageDocInfo.pornSignals`
Rule: Packed adult-content annotations for the image.
Google's description: Encoded Safe Search annotations of the image. See image/safesearch/overall/public/image_porn_attachments.h for decoding functions.
type integer() · confidence medium

`AnchorsAnchor.expired`
Rule: Flags that the linking domain is an expired one.
Google's description: true iff exp domain
type boolean() · confidence low · stages: 4. Link Analysis & Authority Computation

`ClassifierPornSiteData.avgPedoPageScore`
Rule: The site's average score for suspected child exploitation pages.
Google's description: The average pedo page score for the site.
type number() · confidence low

`ClassifierPornSiteData.numberOfPedoPages`
Rule: How many pages of the site were flagged under the child exploitation restriction.
Google's description: The number of pages with pedo restrict.
type String.t · confidence low

`ClassifierPornSiteDataVersionedScore.verticals4Score`
Rule: An internal safe search score from one model version.
Google's description: Please talk to safesearch@ before relying on any of these internal fields:
type number() · confidence low

`CompositeDocIndexingInfo.hostid`
Rule: The page's host id, used mainly to tell whether it sits on a parked domain.
Google's description: The host id of the document. Used chiefly to determine whether the document is part of a parked domain.
type String.t · confidence low · stages: 2. Document Processing & Indexing

`ImageData.hateLogoDetection`
Rule: Detections of hate-related logos found in the picture.
Google's description: Hate logo detections from the VSS logo_recognition module.
type GoogleApi.ContentWarehouse.V1.Model.ImageUnderstandingIndexingAnnotationGroup.t · confidence low

`ImageSafesearchContentOCRAnnotation.ocrProminenceScore`
Rule: How visible or prominent the detected text is within the image.
Google's description: The score produced by Aksara geometry and spoof score. Describes the 'visibility' or 'importance' of the text on the image [0, 1]
type number() · confidence low

`ImageSafesearchContentOffensiveSymbolMatch.score`
Rule: Confidence that an offensive symbol was matched in the image.
Google's description: Confidence score of the match. The higher, the more likely to match the symbol.
type number() · confidence low

`IndexingDocjoinerAnchorSpamInfo.anchorEnd`
Rule: Date of the most recent anchor pointing at the page.
Google's description: End date of the last anchor of the document.
type integer() · confidence low · stages: 4. Link Analysis & Authority Computation; 8. Spam, Penalty & Trust Systems

`IndexingDocjoinerAnchorSpamInfo.anchorStart`
Rule: Date of the earliest anchor pointing at the page.
Google's description: Start date of the first anchor of the document.
type integer() · confidence low · stages: 4. Link Analysis & Authority Computation; 8. Spam, Penalty & Trust Systems

`IndexingDocjoinerAnchorStatistics.pageFromExpiredTaggedAnchors`
Rule: Count of anchors coming from pages tagged as expired.
Google's description: Set in SignalPenalizer::FillInAnchorStatistics.
type integer() · confidence low · stages: 4. Link Analysis & Authority Computation

`PerDocData.QuarantineInfo`
Rule: A bitmask of quarantine flags recorded for the document, such as whitelisted or url-in-url.
Google's description: bitmask of QuarantineBits (or'd together) used to store quarantine related information. For example: QUARANTINE_WHITELIST | QUARANTINE_URLINURL.
type integer() · confidence low · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`WWWResultInfoSubImageDocInfo.safeForUniversal`
Rule: Whether the image is clear of the blocklist used for image results.
Google's description: Result is not on the Images Universal blacklist. For more comprehensive filtering of IU images, including this bit, see superroot/impls/images/quality/safesearch/iu_inappropriate_filter_lib.h
type boolean() · confidence low

Links & authority (84)

`AnchorsAnchor.isLocal`
Rule: Roughly whether the linking page and the target page share a domain.
Google's description: The bit ~roughly~ indicates whether an anchor's source and target pages are on the same domain. Note: this plays no role in determining whether an anchor is onsite, ondomain, or offdomain in mustang (i.e., the bit above).
type boolean() · confidence high · stages: 4. Link Analysis & Authority Computation

`AnchorsAnchor.locality`
Rule: A measure of how local a link is, used in judging link quality.
Google's description: For ranking purposes, the quality of an anchor is measured by its "locality" and "bucket". See quality/anchors/definitions.h for more information.
type integer() · confidence high · stages: 4. Link Analysis & Authority Computation

`AnchorsAnchor.pagerankWeight`
Rule: Weight given to this link when PageRank is computed.
Google's description: Weight to be stored in linkmaps for pageranker
type number() · confidence high · stages: 4. Link Analysis & Authority Computation

`AnchorsAnchor.sourceType`
Rule: Quality band of the page the link came from: high, medium or low.
Google's description: is to record the quality of the anchor's source page and is correlated with but not identical to the index tier of the source page. In the docjoins built by the indexing pipeline (Alexandria), - Anchors marked TYPE_HIGH_QUALITY are from base documents. - Anchors marked TYPE_MEDIUM_QUALITY are from documents of medium quality (roughly but not exactly supplemental tier documents). - Anchors marked TYPE_LOW_QUALITY are from documents of low quality (roughly but not exactly blackhole documents). Not
type integer() · confidence high · stages: 4. Link Analysis & Authority Computation

`AnchorsAnchor.text`
Rule: The words that make up the link text.
Google's description: Space-delimited anchor words. Text that needs segmentation (like CJK or Thai) is unsegmented, since we set FLAGS_segment_during_lexing to false in mr-linkextractor.cc .
type String.t · confidence high · stages: 4. Link Analysis & Authority Computation

`AnchorsAnchorSource.pagerank`
Rule: PageRank of the linking page, stored on a 16-bit scale.
Google's description: uint16 scale
type integer() · confidence high · stages: 4. Link Analysis & Authority Computation

`CompositeDoc.anchorStats`
Rule: Aggregated statistics about the links pointing at this page.
Google's description: Mark as non-personal since it's an aggregation of anchors. For more details of Search personal data, see go/dma52-search-cdoc-fields.
type GoogleApi.ContentWarehouse.V1.Model.IndexingDocjoinerAnchorStatistics.t · confidence high · stages: 2. Document Processing & Indexing

`CompositeDoc.anchors`
Rule: The incoming links and their anchor text for this page.
Google's description: Mark as non-personal since no personal fields will be populated in
type GoogleApi.ContentWarehouse.V1.Model.Anchors.t · confidence high · stages: 2. Document Processing & Indexing

`IndexingDocjoinerAnchorStatistics.anchorPhraseCount`
Rule: How many distinct anchor phrases point at the page, capped at 5000.
Google's description: The number of unique anchor phrases. Capped by the constant kMaxAnchorPhraseCountInStats (=5000) defined in indexing/docjoiner/anchors/anchor-manager.cc.
type integer() · confidence high · stages: 4. Link Analysis & Authority Computation

`IndexingDocjoinerAnchorStatistics.topPrOnsiteAnchorCount`
Rule: How many high-PageRank same-site anchors point at the page.
Google's description: According to anchor quality bucket, anchor with pagrank > 51000 is the best anchor. anchors with pagerank < 47000 are all same.
type integer() · confidence high · stages: 4. Link Analysis & Authority Computation

`IndexingDocjoinerAnchorStatistics.totalDomainsSeen`
Rule: How many distinct domains link to the page.
Google's description: Number of domains seen in total.
type integer() · confidence high · stages: 4. Link Analysis & Authority Computation

`IndexingDocjoinerAnchorStatisticsPerDupStats.offdomainAnchorCount`
Rule: How many anchors came from other domains.
Google's description: Count of offdomain anchors.
type integer() · confidence high

`IndexingSignalAggregatorAgeWeightedCoverageData.firstCoveragePagerankNs`
Rule: The URL's PageRank value at the time it was first served.
Google's description: The pagerank when the url was serving for the first time.
type integer() · confidence high · stages: 2. Document Processing & Indexing

`NlpSaftHyperlink.anchorText`
Rule: The plain visible text of a link, with markup stripped out.
Google's description: Clean anchor text (no HTML markup).
type String.t · confidence high

`QualityNsrPQData.numOffdomainAnchors`
Rule: How many links to this page come from other domains.
Google's description: The total number of offdomain anchors seen by the NSR pipeline for this page.
type number() · confidence high · stages: 3. Content Quality & Site-Level Scoring

`RepositoryWebrefDocumentMetadata.numIncomingAnchors`
Rule: The weighted count of links pointing at this page from other pages.
Google's description: The (weighted) number of incoming anchors (links from other documents).
type number() · confidence high · stages: 6. Entity & Knowledge Graph Integration

`RepositoryWebrefSimplifiedAnchor.anchorText`
Rule: The visible text of a link pointing at the page.
Google's description: The anchor text. Note that the normalized text is not populated.
type GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefLocalizedString.t · confidence high · stages: 6. Entity & Knowledge Graph Integration

`RepositoryWebrefSimplifiedAnchor.count`
Rule: How many times this exact anchor text was seen.
Google's description: The number of times we see this anchor text.
type String.t · confidence high · stages: 6. Entity & Knowledge Graph Integration

`RepositoryWebrefSimplifiedAnchor.countFromOffdomain`
Rule: Count of these anchors coming from other domains.
Google's description: Count, score, normalized score, and volume of offdomain anchors.
type String.t · confidence high · stages: 6. Entity & Knowledge Graph Integration

`RepositoryWebrefSimplifiedAnchor.countFromOnsite`
Rule: Count of these anchors coming from the same site.
Google's description: Count, score, normalized score, and volume of onsite anchors.
type String.t · confidence high · stages: 6. Entity & Knowledge Graph Integration

`RepositoryWebrefSimplifiedAnchor.score`
Rule: Combined score of all anchors sharing this text.
Google's description: The sum/aggregate of the anchor scores that have the same text.
type number() · confidence high · stages: 6. Entity & Knowledge Graph Integration

`RepositoryWebrefWebrefOutlinkInfo.isNofollow`
Rule: Whether an outgoing link is marked nofollow.
Google's description: Whether this is a nofollow link (
type boolean() · confidence high

`AnchorsAnchor.context2`
Rule: A hash of the words surrounding the link on the page.
Google's description: This is a hash of terms near the anchor. (This is a second-generation hash replacing the value stored in the 'context' field.)
type integer() · confidence medium · stages: 4. Link Analysis & Authority Computation

`AnchorsAnchor.linkAdditionalInfo`
Rule: Extra data about the link, such as further anchor text or scores.
Google's description: Additional information related to the anchor, such as additional anchor text or scores.
type GoogleApi.ContentWarehouse.V1.Model.Proto2BridgeMessageSet.t · confidence medium · stages: 4. Link Analysis & Authority Computation

`AnchorsAnchor.linkTags`
Rule: Tags describing the link type and the page it came from.
Google's description: Contains info on link type, source page, etc.
type list(integer()) · confidence medium · stages: 4. Link Analysis & Authority Computation

`AnchorsAnchor.origText`
Rule: The link text exactly as written, keeping capitalization and punctuation.
Google's description: Original text, including capitalization and punctuation. Runs of whitespace are collapsed into a single space.
type String.t · confidence medium · stages: 4. Link Analysis & Authority Computation

`AnchorsAnchor.parallelLinks`
Rule: How many other links on the same page point to the same target domain.
Google's description: The number of additional links from the same source page to the same target domain. Not populated if is_local is true.
type integer() · confidence medium · stages: 4. Link Analysis & Authority Computation

`AnchorsAnchor.weight`
Rule: The link's weight on a scale of 0 to 127.
Google's description: weights are 0-127
type integer() · confidence medium · stages: 4. Link Analysis & Authority Computation

`AnchorsAnchorSource.homePageInfo`
Rule: Whether the linking page is a homepage and how trusted that homepage is.
Google's description: Information about if the source page is a home page. It can be one of the enum values defined in PerDocData::HomePageInfo (NOT_HOMEPAGE, NOT_TRUSTED, PARTIALLY_TRUSTED, and FULLY_TRUSTED).
type integer() · confidence medium · stages: 4. Link Analysis & Authority Computation

`AnchorsAnchorSource.nsr`
Rule: Site quality score of the linking page, scaled to a 0 to 1000 range.
Google's description: This NSR value has range [0,1000] and is the original value [0.0,1.0] multiplied by 1000 rounded to an integer.
type integer() · confidence medium · stages: 4. Link Analysis & Authority Computation

`AnchorsAnchorSource.outsites`
Rule: Roughly how many different sites the linking page points out to.
Google's description: approx num of pointed-to sites
type integer() · confidence medium · stages: 4. Link Analysis & Authority Computation

`AnchorsAnchorSource.pagerankNs`
Rule: A variant PageRank value for the linking page.
Google's description: unit16 scale
type integer() · confidence medium · stages: 4. Link Analysis & Authority Computation

`CompositeDocForwardingDup.purposes`
Rule: Which data a duplicate passes to the main page, such as anchors or click boost.
Google's description: The purpose(s) of the forwarding dup indicating if it is used for forwarding signal/anchors generally, or only for forwarding some specific signal (e.g. navboost), or for some other purposes (e.g., not for forwarding any data but for making "info:" complete). See indexing/dups/public/dups.h for more details.
type integer() · confidence medium

`ImageExactBoostNavQuery.referrerDocid`
Rule: Identifiers of the pages that show this image, ordered by their rank.
Google's description: List of all referrers, sorted by their rank (stored in Moosedog).
type list(String.t) · confidence medium

`ImageExactBoostNavQuery.referrerRank`
Rule: Where the current page ranks among the pages that show this image.
Google's description: Rank of the current web doc referrer (stored in docjoins).
type integer() · confidence medium

`IndexingDocjoinerAnchorSpamInfo.trustedTarget`
Rule: Whether this URL sits on a trusted source.
Google's description: Following fields record details about trusted anchors True if is this URL is on trusted source.
type boolean() · confidence medium · stages: 4. Link Analysis & Authority Computation; 8. Spam, Penalty & Trust Systems

`IndexingDocjoinerAnchorSpamInfo.trustedTotal`
Rule: How many trusted sources link to this URL.
Google's description: Total number of trusted sources for this URL.
type integer() · confidence medium · stages: 4. Link Analysis & Authority Computation; 8. Spam, Penalty & Trust Systems

`IndexingDocjoinerAnchorStatistics.anchorsWithDedupedImprovanchors`
Rule: How many anchors had duplicate phrases removed within the linking organization.
Google's description: The number of anchors for which some ImprovAnchors phrases have been removed due to duplication within source org.
type integer() · confidence medium · stages: 4. Link Analysis & Authority Computation

`IndexingDocjoinerAnchorStatistics.droppedRedundantAnchorCount`
Rule: How many anchors were dropped as redundant.
Google's description: Sum of anchors_dropped in the repeated group RedundantAnchorInfo, but can go higher if the latter reaches the cap of kMaxRecordsToKeep. (indexing/docjoiner/anchors/anchor-loader.cc), currently 10,000
type integer() · confidence medium · stages: 4. Link Analysis & Authority Computation

`IndexingDocjoinerAnchorStatistics.minDomainHomePageLocalOutdegree`
Rule: Smallest outbound link count among same-domain home pages linking here.
Google's description: Minimum local outdegree of all anchor sources that are domain home pages as well as on the same domain as the current target URL.
type integer() · confidence medium · stages: 4. Link Analysis & Authority Computation

`IndexingDocjoinerAnchorStatistics.minHostHomePageLocalOutdegree`
Rule: Smallest outbound link count among same-host home pages linking here.
Google's description: Minimum local outdegree of all anchor sources that are host home pages as well as on the same host as the current target URL.
type integer() · confidence medium · stages: 4. Link Analysis & Authority Computation

`IndexingDocjoinerAnchorStatistics.redundantAnchorForPhraseCapCount`
Rule: How many anchors were dropped for exceeding the per-domain phrase cap.
Google's description: Total anchor dropped due to exceed per domain phrase cap. Equals to sum of anchors_dropped in the repeated group RedundantAnchorInfoForPhraseCap, but can go higher if the latter reaches the cap of kMaxDomainsToKeepForPhraseCap (indexing/docjoiner/anchors/anchor-loader.h), currently 1000.
type integer() · confidence medium · stages: 4. Link Analysis & Authority Computation

`IndexingDocjoinerAnchorStatistics.totalDomainPhrasePairsAboveLimit`
Rule: How many domain and phrase pairs exceeded the cap.
Google's description: The following should be equal to the size of the following repeated group, except that it can go higher than 10,000.
type integer() · confidence medium · stages: 4. Link Analysis & Authority Computation

`IndexingDocjoinerAnchorStatistics.totalDomainPhrasePairsSeenApprox`
Rule: Roughly how many domain and phrase pairs were seen before capping.
Google's description: Number of domain/phrase pairs in total -- i.e. how many anchors we would have if the domain/phrase cutoff was set to 1 instead of 200. This is "approx" for large anchor clusters because there can be double counting when the LRU cache forgets about rare domain/phrase pairs.
type integer() · confidence medium · stages: 4. Link Analysis & Authority Computation

`IndexingDocjoinerAnchorStatistics.totalDomainsAbovePhraseCap`
Rule: How many domains exceeded the per-domain phrase cap.
Google's description: Number of domains above per domain phrase cap. We see too many phrases in the domains.
type integer() · confidence medium · stages: 4. Link Analysis & Authority Computation

`IndexingDocjoinerAnchorStatisticsPerDupStats.anchorCount`
Rule: How many anchors were kept for this duplicate URL.
Google's description: Count of anchors kept from forwarding.
type integer() · confidence medium

`IndexingDocjoinerAnchorStatisticsPerDupStats.redundantAnchorCount`
Rule: How many anchors were considered redundant.
Google's description: Count of redundant anchors.
type integer() · confidence medium

`IndexingDocjoinerAnchorTrustedInfo.text`
Rule: The tokenized anchor text from that site.
Google's description: Tokenized text of all anchors from the site.
type list(String.t) · confidence medium

`IndexingDocjoinerAnchorTrustedInfo.trustedScore`
Rule: Fraction of the site's pages with newsy anchors, higher for trusted sites.
Google's description: Fraction of pages with newsy anchors on the site, >0 for trusted sites.
type number() · confidence medium

`KaltixPerDocData.KaltixRank`
Rule: A per-document rank value stored in compact form for the top billion documents.
Google's description: approx. 2 bytes for top 1B
type integer() · confidence medium · stages: 4. Link Analysis & Authority Computation

`NlpSaftDocument.hyperlink`
Rule: The list of links found in the document, in the order they appear.
Google's description: The hyperlinks in the document. Multiple hyperlinks are sorted in left-to-right order.
type list(GoogleApi.ContentWarehouse.V1.Model.NlpSaftHyperlink.t) · confidence medium · stages: 9. Query Understanding & Intent Classification

`NlpSaftHyperlink.url`
Rule: The full address that a link points to.
Google's description: (Absolute) URL that the links to.
type String.t · confidence medium

`PerDocData.toolbarPagerank`
Rule: A stored copy of the page's 0-10 toolbar pagerank value.
Google's description: A copy of the value stored in /namespace/indexing/wwwglobal//fakepr/* for this document. A value of quality_bakery::FakeprUtils::kUnknownToolbarPagerank indicates that we don't have toolbar pagerank for this document. A value between 0 and 10 (inclusive) means that this is the toolbar pagerank of the page. Finally, if this value is not set it means that the toolbar pagerank is equivalent to: quality_bakery::FakeprUtils::EstimatePreDemotionFromPagerankNearestSeeds(basic_info.pagerank_ns()) calle
type integer() · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityNsrNsrData.siteLinkIn`
Rule: The average incoming link score of the site's pages.
Google's description: Average value of the site_link_in for pages in the sitechunk.
type number() · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityNsrNsrData.siteLinkOut`
Rule: The aggregated outgoing link score of the site's pages.
Google's description: Aggregated value of url link out scores for this sitechunk.
type number() · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityTravelGoodSitesData.normalizationFactor`
Rule: How much credit from local links is scaled before being added to global links.
Google's description: Factor that determines how local anchor credit is scaled before being added to global anchors.
type number() · confidence medium

`RepositoryWebrefSimplifiedAnchor.normalizedScore`
Rule: The anchor score after being scaled against total anchor volume.
Google's description: The normalized score, which is computed from the score and the total_volume.
type number() · confidence medium · stages: 6. Entity & Knowledge Graph Integration

`RepositoryWebrefSimplifiedAnchor.scoreFromFragment`
Rule: Combined score of anchors with this text that point to a page fragment.
Google's description: The sum/aggregate of the anchor scores that direct to a fragment and have the same text.
type number() · confidence medium · stages: 6. Entity & Knowledge Graph Integration

`RepositoryWebrefSimplifiedAnchor.scoreFromRedirect`
Rule: Combined score of anchors with this text that arrive via a redirect.
Google's description: The sum/aggregate of the anchor scores that direct to a different wiki title and have the same text. NOTE: url direct to a fragment score is not included in this value.
type number() · confidence medium · stages: 6. Entity & Knowledge Graph Integration

`RepositoryWebrefSimplifiedAnchor.totalVolume`
Rule: Total anchor score volume used as the denominator when normalizing.
Google's description: The total score volume used for normalization.
type number() · confidence medium · stages: 6. Entity & Knowledge Graph Integration

`RepositoryWebrefSimplifiedCompositeDoc.anchors`
Rule: The page's anchors, trimmed down and grouped by identical text.
Google's description: The composite doc anchors trimmed and transformed in a smaller data structure and aggregated (if they have the exact same text).
type GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefSimplifiedAnchors.t · confidence medium

`RepositoryWebrefWebrefOutlinkInfo.topicalityWeight`
Rule: How much topical weight each outgoing link to this address carries.
Google's description: The topicality_weight for each link with this target URL.
type list(number()) · confidence medium

`RepositoryWebrefWebrefOutlinkInfos.outlinkInfo`
Rule: Details about every address the page links out to.
Google's description: Information about each target URL referred to in the document's outlinks. If a given URL has multiple links, they are grouped in a single WebrefLinkInfo.
type list(GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefWebrefOutlinkInfo.t) · confidence medium

`ResearchScienceSearchSourceUrlDocjoinInfo.pagerankNs`
Rule: The document's live PageRank value.
Google's description: The production pagerank value of the document.
type integer() · confidence medium

`ScienceCitationAnchor.weight`
Rule: A weight from 1 to 128 for the anchor based on where it came from.
Google's description: weight of the anchor by looking where we get this anchor text. It can be PR, court level, year, or the combination of differerent aspects. weights are 1 - 128 defined as "Weights"
type integer() · confidence medium

`ScienceIndexSignal.NumBackwardLinks`
Rule: How many links point to the document.
Google's description: Summary statistics.
type integer() · confidence medium

`Anchors.homepageAnchorsDropped`
Rule: How many links from other homepages on the same site were discarded during anchor processing.
Google's description: The total # of local homepage anchors dropped in AnchorAccumulator.
type String.t · confidence low · stages: 4. Link Analysis & Authority Computation

`Anchors.localAnchorsDropped`
Rule: How many same-site non-homepage links were discarded during anchor processing.
Google's description: The total # of local non-homepage anchors dropped in AnchorAccumulator.
type String.t · confidence low · stages: 4. Link Analysis & Authority Computation

`Anchors.nonlocalAnchorsDropped`
Rule: How many links from other sites were discarded during anchor processing.
Google's description: The total # of non-local anchors dropped in AnchorAccumulator.
type String.t · confidence low · stages: 4. Link Analysis & Authority Computation

`AnchorsAnchor.catfishTags`
Rule: Tags attached to a link that are generated on the fly rather than stored.
Google's description: CATfish tags attached to a link. These are similar to link tags, except the values are created on the fly within Cookbook. See:
type list(integer()) · confidence low · stages: 4. Link Analysis & Authority Computation

`AnchorsAnchor.fullLeftContext`
Rule: The full text appearing before the link on the page.
Google's description: The full context. These are not written out in the linklogs.
type list(String.t) · confidence low · stages: 4. Link Analysis & Authority Computation

`AnchorsAnchor.setiPagerankWeight`
Rule: A second, temporary link weight for a PageRank variant.
Google's description: TEMPORARY
type number() · confidence low · stages: 4. Link Analysis & Authority Computation

`AnchorsAnchorSource.indyrank`
Rule: A stored rank value for the linking page.
Google's description: uint16 scale
type integer() · confidence low · stages: 4. Link Analysis & Authority Computation

`CompositeDoc.scaledIndyRank`
Rule: A scaled independence rank value copied into the per document data.
Google's description: to copy to per-doc
type integer() · confidence low · stages: 2. Document Processing & Indexing

`FatcatCompactDocClassification.weight`
Rule: How important this page is within its site, based on PageRank or impressions.
Google's description: The relative weight of this doc within a site, typically something like pagerank or navboost impressions. May be a large number (like an actual pageviews estimate), not limited to a small range.
type String.t · confidence low

`IndexingDocjoinerAnchorStatistics.globalAnchorDelta`
Rule: How much the set of links pointing at the page changed.
Google's description: Metric of number of changed global anchors computed as, size(union(previous, new) - intersection(previous, new)).
type integer() · confidence low · stages: 4. Link Analysis & Authority Computation

`OceanDocTag.pagerank`
Rule: PageRank value stored for a book page.
Google's description: the page rank value of the book page
type float() · confidence low

`PerDocData.ScaledIndyRank`
Rule: An independence rank stored as a 16-bit number that is scaled back to its real value.
Google's description: The independence rank is represented as a 16-bit integer, which is multiplied by (max_indy_rank / 65536) to produce actual independence rank values. max_indy_rank is typically 0.84.
type integer() · confidence low · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityShoppingShoppingAttachmentProduct.outlinkDomainRelationship`
Rule: Whether an offer link points to the same domain or off to another one.
Google's description: Whether an outlink points to the same domain or off-domain. Only added if the relationship is known, and the Offer has ref_type of OUTLINK.
type String.t · confidence low

`RepositoryWebrefWebrefOutlinkInfo.byteLength`
Rule: Length in bytes of the link text for an outgoing link.
Google's description: The length in bytes of such a link (including internal spaces); e.g. if the link text is "click here" then the length is 10.
type list(String.t) · confidence low

`RepositoryWebrefWebrefOutlinkInfo.url`
Rule: The address an outgoing link points to.
Google's description: The target URL of the link.
type String.t · confidence low

`ScienceCitation.Anchors`
Rule: The anchor text around a citation link on the referring page.
Google's description: All the anchor text (before, after, formal, etc) for this citation in the referring page.
type list(GoogleApi.ContentWarehouse.V1.Model.ScienceCitationAnchor.t) · confidence low

`ScienceCitationAnchor.count`
Rule: How many times this anchor text appears.
Google's description: Number of times this anchor text appears, only consider the text itself
type integer() · confidence low

`ScienceCitationAnchor.text`
Rule: The words used in the anchor text pointing at the document.
Google's description: Space-delimited anchor words. Text that needs segmentation (like CJK or Thai) is unsegmented. Generated by ScienceParseUtils::AppendTokenSeqToString()
type String.t · confidence low

Images & video (82)

`ClassifierPornQueryStats.queryTextPornScore`
Rule: A click weighted average of how adult the queries that click this image are.
Google's description: A query text porn score for the queries which have clicks to the image: query_text_porn_score := sum(clicks(query) * text_porn_score(query)) / sum(clicks(query))
type number() · confidence medium

`CompositeDoc.docImages`
Rule: Data about the images chosen to represent this page.
Google's description: Info about "selected" images associated with the document for which we (already) have ImageData. For each image URL, some fixed number of documents are selected as web referrers for the image URL, and within those selected documents, we say the image is "selected". Within the remaining documents, we say the image is "rejected". Note that this distinction is slightly different from selected for indexing. Only images within doc_images where is_indexed_by_imagesearch i
type list(GoogleApi.ContentWarehouse.V1.Model.ImageData.t) · confidence medium · stages: 2. Document Processing & Indexing

`CompositeDoc.docVideos`
Rule: Data about videos embedded in the page.
Google's description: Info about videos embedded in the document.
type list(GoogleApi.ContentWarehouse.V1.Model.ImageRepositoryVideoProperties.t) · confidence medium · stages: 2. Document Processing & Indexing

`ImageContentQueryBoostQueryBoost.query`
Rule: The cleaned-up query text that a per-query image boost applies to.
Google's description: Canonicalized query string.
type String.t · confidence medium

`ImageContentStarburstVersionGroup.descriptorFloat`
Rule: A list of numbers describing what an image looks like.
Google's description: Raw dense float feature vector.
type list(number()) · confidence medium

`ImageContentStarburstVersionGroup.descriptorShort`
Rule: A compact compressed version of the image's feature numbers.
Google's description: Short descriptor for image content features, e.g. compressed bytes. This is the compressed version of descriptor_float below. It can be can be decompressed to descriptor_float with a tiny bit of compression error (in most cases it should be totally fine).
type String.t · confidence medium

`ImageContentStarburstVersionGroup.starburstTokens`
Rule: Short codes standing for the visual clusters an image belongs to.
Google's description: Starburst tokens.
type list(integer()) · confidence medium

`ImageContentStarburstVersionGroup.starburstTokensDistances`
Rule: How far the image sits from each of the visual clusters assigned to it.
Google's description: Squared distance of the current embedding to each cluster center. Elements are aligned with starburst_tokens.
type list(number()) · confidence medium

`ImageData.clipartDetectorScore`
Rule: A classifier score for whether the picture looks like clipart.
Google's description: Image content based classifier scores.
type number() · confidence medium

`ImageData.deepImageEngagingness`
Rule: A model score for how engaging the picture is likely to be.
Google's description: Productionized Deep Image Engagingness score.
type GoogleApi.ContentWarehouse.V1.Model.ImageRepositoryDeepImageEngagingnessOutput.t · confidence medium

`ImageData.flowOutput`
Rule: Bundled image-content features used to compare how similar pictures are.
Google's description: The output of various features generated by the Flow framework, most importantly data from Starburst (go/starburst). Do NOT interact with the internals of this proto since they may change over time. Instead, use the existing interfaces that consume FlowProtoProd's directly, e.g., image/mustang/content/image_content_distance.h For more info, please contact image-content-core@.
type GoogleApi.ContentWarehouse.V1.Model.ImageContentFlowProtoProd.t · confidence medium

`ImageData.imageContentQueryBoost`
Rule: Per-query score boosts for the image produced by an image content model.
Google's description: Image Content Scored per query boosts. Currently this is filled by the pamir algorithm and populates the pamir_section.
type GoogleApi.ContentWarehouse.V1.Model.ImageContentQueryBoost.t · confidence medium

`ImageData.imageExactBoost`
Rule: Queries the image is believed to match, with a confidence score for each.
Google's description: A set of query fingerprints and confidence scores. There queries are supposed to be relevant to the image with high confidence.
type GoogleApi.ContentWarehouse.V1.Model.ImageExactBoost.t · confidence medium

`ImageData.isVisible`
Rule: Whether the image is shown on the page or only linked from it.
Google's description: True if the image is inlined on the page (typicially via) or false if the image is linked to (via an href).
type boolean() · confidence medium

`ImageData.nimaAva`
Rule: A model score for the aesthetic appeal of the picture.
Google's description: Productionized Nima AVA score. Both this field and nima_vq were added on the top of nima_ava_score and nima_vq_score because the signals are already integrated with Batch Amarna in image/repository/schema/global_output_tags.h using NimaOutput.
type GoogleApi.ContentWarehouse.V1.Model.ImageRepositoryNimaOutput.t · confidence medium

`ImageData.nimaVq`
Rule: A model score for the technical visual quality of the picture.
Google's description: Productionized Nima VQ score.
type GoogleApi.ContentWarehouse.V1.Model.ImageRepositoryNimaOutput.t · confidence medium

`ImageData.ocrGoodoc`
Rule: A trimmed record of text read out of the image by character recognition.
Google's description: Pruned OCR Goodoc see vss_aksara_ocr_util.h for the fields copied.
type GoogleApi.ContentWarehouse.V1.Model.GoodocDocument.t · confidence medium

`ImageData.ocrTaser`
Rule: Text read out of the image by a second character recognition system.
Google's description: Ocr detected by ocr_taser module.
type GoogleApi.ContentWarehouse.V1.Model.GoodocDocument.t · confidence medium

`ImageData.ocrTextboxes`
Rule: The individual lines of text found inside the image.
Google's description: Text lines detected by OCR engine.
type list(GoogleApi.ContentWarehouse.V1.Model.OcrPhotoTextBox.t) · confidence medium

`ImageData.rankInNeardupCluster`
Rule: Where this image places among its near duplicates, with one being best.
Google's description: Rank in near-dup cluster (go/image-rank-in-neardup-cluster). The rank is 1-indexed: rank 1 is the best.
type integer() · confidence medium

`ImageData.smearedTopWebLandingPageDocids`
Rule: Identifiers of top-ranked web pages that this image leads to.
Google's description: Web docids that correspond to high ranked smeared landing pages for this image. Used for conditional retrieval of actionable landing pages for image search.
type list(String.t) · confidence medium

`ImageMustangImageLinkSelectionInfo.webRelevanceScore`
Rule: How well an image matches the page it was picked from.
Google's description: score calculated in image selection phase, higher imagelink_selection_score more relevant the link is related to the web page
type float() · confidence medium

`ImageRegionsImageRegion.knnScore`
Rule: How visually similar this region is to its closest look-alikes.
Google's description: The score for this region based on how visually similar its neighbors are.
type float() · confidence medium

`ImageRegionsImageRegion.primaryLabel`
Rule: The best label describing what is in this part of the image.
Google's description: The primary label associated with the region. Specifically, the highest-scored whitelisted label associated with the region. See ImageRegionsConfig for details on the whitelist.
type GoogleApi.ContentWarehouse.V1.Model.ImageUnderstandingIndexingLabel.t · confidence medium

`ImageRepositoryAmarnaCloudSpeechSignals.transcriptAsr`
Rule: The full stitched speech transcript of the video with word timing offsets.
Google's description: This field contains full (stitched) transcription, word-level time offset , and word-level byte offset. The value of this field is derived from the SpeechRecognitionResult field above.
type GoogleApi.ContentWarehouse.V1.Model.PseudoVideoData.t · confidence medium

`ImageRepositoryContentBasedVideoMetadata.thumbnailQualityScore`
Rule: Model score for how visually appealing a video thumbnail is.
Google's description: Thumbnail quality score predict how visual pleasing a thumbnail is, based on the model trained with deep neural networks.(go/thumb_features_dd) Note the signal currently only available for Youtube videos.
type GoogleApi.ContentWarehouse.V1.Model.VideoThumbnailsThumbnailScore.t · confidence medium

`ImageRepositoryContentBasedVideoMetadata.transcriptAsr`
Rule: Speech transcript of the video generated from the caption speech recognizer.
Google's description: Speech related metadata The transcript_asr field is generated from the YT caption's SPEECH_RECOGNIZER asset.
type GoogleApi.ContentWarehouse.V1.Model.PseudoVideoData.t · confidence medium

`ImageRepositoryDeepImageEngagingnessOutput.score`
Rule: Model score predicting how engaging an image is.
Google's description: DeepImageEngagingness score.
type number() · confidence medium

`IndexingSpeechSpeechPropertiesProto.languageCode`
Rule: The language spoken in the audio, which may differ from the page's language.
Google's description: The spoken language, see i18n/identifiers/languagecode.h and go/gl2014. This may or may not match the language of the written page. (Examples: "en", "sv", "zh-CN").
type String.t · confidence medium

`IndexingSpeechSpeechPropertiesProto.numWords`
Rule: How many words the speech recogniser picked out of the audio.
Google's description: Total number of recognized words in processed fragment.
type integer() · confidence medium

`IndexingSpeechSpeechPropertiesProto.recognizerAccuracy`
Rule: The estimated accuracy of the speech recognition output, from zero to one.
Google's description: Measure of the estimated output accuracy from the speech recognition code, from 0 to 1. Based on word-level confidence and possibly other factors.
type number() · confidence medium

`MediaIndexRegion.labels`
Rule: Labels with confidence scores for an image region, stored to make sorting fast.
Google's description: The labels associated with the region encoded as a SparseFloatVector to facilitate dot product computation during sorting. The columns are the fingerprints of the labels and the values are the corresponding confidence scores. The vector is L2 normalized.
type GoogleApi.ContentWarehouse.V1.Model.MediaIndexSparseFloatVector.t · confidence medium

`PornFlagData.ocrAnnotation`
Rule: Text read out of the image by OCR, along with notes about it.
Google's description: Information about image OCR text. For details see image/safesearch/content/public/ocr_annotation.proto.
type GoogleApi.ContentWarehouse.V1.Model.ImageSafesearchContentOCRAnnotation.t · confidence medium

`PseudoVideoDataTranscript.Text`
Rule: The full spoken-word transcript of the video or audio.
Google's description: The complete transcription text.
type String.t · confidence medium

`QualityNsrNsrData.isVideoFocusedSite`
Rule: Whether most of the site's pages are video watch pages.
Google's description: Bit to determine whether the site has mostly video content, but is not hosted on any known video-hosting domains. Site is considered to be video-focused, if it has > 50% of the URLs with watch pages (with smoothing prior). ariane/4045246
type boolean() · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityOrbitAsteroidBeltDocumentIntentScores.imageIntentScores`
Rule: Intent scores for each image on the page, keyed by image id.
Google's description: Map of imageid key to ImageIntentScores, for images on cdoc.doc_images
type %{optional(String.t) => GoogleApi.ContentWarehouse.V1.Model.QualityOrbitAsteroidBeltImageIntentScores.t} · confidence medium · stages: 9. Query Understanding & Intent Classification

`QualityOrbitAsteroidBeltImageIntentScores.intents`
Rule: The list of intents assigned to an image.
Google's description: The unique identifier for an Asteroid Belt document intent is being reused here
type list(String.t) · confidence medium

`QualityOrbitAsteroidBeltImageIntentScores.scores`
Rule: The strength of each image intent, on a 0 to 100 scale.
Google's description: The intent scores, scaled to integers between 0 and 100 for compactness.
type list(integer()) · confidence medium

`QualityOrbitOrbitImageIntent.imageIntent`
Rule: Which intent this image classification entry is about.
Google's description: Intent of this proto instance.
type String.t · confidence medium

`QualityOrbitOrbitImageIntent.score`
Rule: How strongly the image matches this intent.
Google's description: Score of the intent.
type number() · confidence medium

`QualityOrbitOrbitImageIntent.trigger`
Rule: Whether the image intent passed its default threshold.
Google's description: Whether the intent triggered with the default threshold or not.
type boolean() · confidence medium

`QualityOrbitOrbitImageIntents.intents`
Rule: The image intents that scored high enough to keep.
Google's description: Intents with non-trivial scores.
type list(GoogleApi.ContentWarehouse.V1.Model.QualityOrbitOrbitImageIntent.t) · confidence medium

`WWWResultInfoSubImageDocInfo.salience`
Rule: How important the image is to the page it appears on.
Google's description: Salient score, indicating how important an image is to the page it's on. Check go/salient-images-design-doc for details.
type number() · confidence medium

`CorpusSelectionInfo.corpusScore`
Rule: A per-corpus score for an image.
Google's description: Corpus specific score for an image
type number() · confidence low

`ImageContentStarburstVersionGroup.descriptorFloatBeforeProjection`
Rule: The image feature numbers before they are compressed into an embedding.
Google's description: Raw dense float feature vector prior to embedding. Only for Starburst V6.
type list(number()) · confidence low

`ImageData.colorScore`
Rule: How much of each of twelve basic colors appears in the image.
Google's description: Colorness scores for the image. Each score represents the amount of a particular color in the image. At the current time, there are 12 colors, so there should always be 0 or 12 values in this array. The 12 colors are black, blue, brown, gray, green, orange, pink, purple, red, teal, white, yellow. The convention is that the scores are stored in alphabetical order, so the first score is black, and the last score is yellow.
type list(number()) · confidence low

`ImageData.coloredPixelsFrac`
Rule: The share of pixels with real color, used to tell color from grayscale.
Google's description: Fraction of the image that contains pixels over a certain saturation threshold: can be used to determine if the image is grayscale or not.
type number() · confidence low

`ImageData.imageRegions`
Rule: Boxes around objects found inside the picture, each with a label.
Google's description: Regions detected within the image (go/images-region-search-edd). Regions contain bounding boxes circumscribing objects of interest in the image, along with object labels. Regions may overlap.
type GoogleApi.ContentWarehouse.V1.Model.ImageRegionsImageRegions.t · confidence low

`ImageData.largestFaceFrac`
Rule: How much of the picture is taken up by the biggest face in it.
Google's description: Fraction of image covered by the largest face (should match largestFaceFraction, but without scaling). In perdocs, is set only if numberFaces > 0.
type number() · confidence low

`ImageData.numberFaces`
Rule: How many faces were found in the picture.
Google's description: Number of faces detected in the image
type integer() · confidence low

`ImageData.packedFullFaceInfo`
Rule: Face count and largest-face size squeezed into a compact index value.
Google's description: Encodes face number and largest face frac into a small package for storage in mustang. This is calculated directly from FaceDetectionResult.
type GoogleApi.ContentWarehouse.V1.Model.FaceIndexing.t · confidence low

`ImageData.whiteBackgroundScoreVersion`
Rule: How likely the picture is a single object shot on a white background.
Google's description: Image is likely an object on a white background (value on [0,1]).
type integer() · confidence low

`ImageRegionsImageRegion.boundingBoxScore`
Rule: How confident the system is in that detected rectangle.
Google's description: The confidence score associated with the bounding box.
type number() · confidence low

`ImageRegionsImageRegion.labelGroup`
Rule: The set of labels attached to this part of the image.
Google's description: The label group corresponding to the first LabelParams listed in ImageRegionsConfig.
type GoogleApi.ContentWarehouse.V1.Model.ImageUnderstandingIndexingLabelGroup.t · confidence low

`ImageRepositoryAmarnaCloudSpeechSignals.results`
Rule: Raw speech-to-text output returned by the Cloud Speech API for the video.
Google's description: Raw results from Cloud Speech API
type list(GoogleApi.ContentWarehouse.V1.Model.ImageRepositorySpeechRecognitionResult.t) · confidence low

`ImageRepositoryContentBasedVideoMetadata.cloudSpeechSignals`
Rule: Speech transcript generated for the video by the Cloud Speech API.
Google's description: Transcript generated from Cloud Speech API
type GoogleApi.ContentWarehouse.V1.Model.ImageRepositoryAmarnaCloudSpeechSignals.t · confidence low

`ImageRepositoryContentBasedVideoMetadata.featureSetData`
Rule: Stored set of video understanding features describing the video's content.
Google's description: Video Understanding Golden features. (go/amarna-video-signals#golden-signals) Note: Golden6 features (names matching "video_
type GoogleApi.ContentWarehouse.V1.Model.DrishtiFeatureSetData.t · confidence low

`ImageRepositoryContentBasedVideoMetadata.s3Asr`
Rule: Speech transcript generated for the video by the internal speech engine.
Google's description: Trnascript generated through AMARNA_CLOUD_SPEECH asset in Venom. Note that AMARNA_CLOUD_SPEECH uses S3 as the speech engine backend, similar to YT caption's SPEECH_RECOGNIZER asset. However, they may use different S3 models.
type GoogleApi.ContentWarehouse.V1.Model.ImageRepositoryAmarnaCloudSpeechSignals.t · confidence low

`ImageRepositoryNimaOutput.score`
Rule: A model score rating the quality of the image.
Google's description: NIMA score.
type number() · confidence low

`ImageRepositorySpeechRecognitionAlternative.transcript`
Rule: The recognized spoken text from the audio.
Google's description: Transcript text representing the words that the user spoke.
type String.t · confidence low

`ImageUnderstandingIndexingAnnotation.labelGroup`
Rule: Label annotations describing what is inside a detected box in an image.
Google's description: Multiple label annotations for this bounding box.
type list(GoogleApi.ContentWarehouse.V1.Model.ImageUnderstandingIndexingLabelGroup.t) · confidence low

`ImageUnderstandingIndexingLabel.canonicalText`
Rule: Human readable text of the label attached to image content.
Google's description: Human readable text.
type String.t · confidence low

`ImageUnderstandingIndexingLabel.metaData`
Rule: Extra label details covering topicality, visibility and attributes.
Google's description: Meta data for topicality, visible labels, attribute, etc.
type list(GoogleApi.ContentWarehouse.V1.Model.ImageUnderstandingIndexingMetaData.t) · confidence low

`ImageUnderstandingIndexingLabel.score`
Rule: Confidence score for an image label.
Google's description: Confidence score.
type number() · confidence low

`IndexingSpeechSpeechPropertiesProto.audioDuration`
Rule: How many seconds of audio were processed in this media fragment.
Google's description: Duration of audio in processed fragment (including non-speech), in seconds.
type integer() · confidence low

`IndexingSpeechSpeechPropertiesProto.audioOnly`
Rule: Whether the media file has only audio and no video track.
Google's description: If true, the media file is audio-only. If false, also has video track(s).
type boolean() · confidence low

`IndexingSpeechSpeechPropertiesProto.estimatedAudioDuration`
Rule: The estimated total audio length of the whole media file, in seconds.
Google's description: Estimated duration of audio in the whole file (including non-speech), in seconds. If this is greater than 0, then it will either be equal to audio_duration (when truncated_file is false), or to the length of the content (audio or video) according to the file header (when truncated_file is true).
type integer() · confidence low

`IndexingSpeechSpeechPropertiesProto.speechDuration`
Rule: How many seconds of actual speech were in the processed audio.
Google's description: Duration of speech in processed fragment, in seconds.
type integer() · confidence low

`IndexingVideosAsrTranscriptRepairAnnotation.numGibberishSegments`
Rule: How many transcript segments were judged to be gibberish.
Google's description: The number of the above segments that were classified as gibberish.
type integer() · confidence low

`IndexingVideosAsrTranscriptRepairAnnotation.numSegmentsClassified`
Rule: How many transcript segments were sent off for gibberish classification.
Google's description: The number of ASR segments that were sent for classification.
type integer() · confidence low

`MediaIndexRegion.entityFields`
Rule: The entities recognized inside a marked region of an image.
Google's description: Detected Entities found within this region.
type list(GoogleApi.ContentWarehouse.V1.Model.MediaIndexEntityField.t) · confidence low

`MediaIndexVideoCentroidDomainScore.numDocs`
Rule: How many pages from that domain were used to compute the video domain score.
Google's description: Number of pages from the domain used to generate this DomainScore.
type integer() · confidence low

`MediaIndexVideoCentroidDomainScore.score`
Rule: A score that is lower when a video appears across more varied pages of a domain.
Google's description: In general, lower scores indicate the video is appearing on more diverse pages.
type number() · confidence low

`PerDocData.s3AudioLanguage`
Rule: The spoken language of a watch page's main video, detected from the audio.
Google's description: Primary video's audio language classified by S3 based Automatic Language Identification (only for watch pages).
type GoogleApi.ContentWarehouse.V1.Model.S3AudioLanguageS3AudioLanguage.t · confidence low · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.videoLanguage`
Rule: The spoken language of a watch page's video, detected from the audio.
Google's description: Audio-based language classified by Automatic Language Identification (only for watch pages).
type GoogleApi.ContentWarehouse.V1.Model.QualityVidyaVideoLanguageVideoLanguage.t · confidence low · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.watchpageLanguageResult`
Rule: The language of a watch page as judged by a dedicated model.
Google's description: Language classified by the WatchPageLanguage Model (go/watchpage-language). Only present for watch pages.
type GoogleApi.ContentWarehouse.V1.Model.WatchpageLanguageWatchPageLanguageResult.t · confidence low · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PornFlagData.numberFaces`
Rule: How many faces were detected in the image.
Google's description: number of faces
type integer() · confidence low

`QualityVidyaVideoLanguageVideoLanguage.language`
Rule: The spoken language detected in the video's audio.
Google's description: Audio language of video classified by Automatic Language Identification. It corresponds to the first language (the highest confidence) in ALIResults.lang_results.
type String.t · confidence low

`QualityWebanswersVideoTranscriptAnnotations.asrRepair`
Rule: The outcome of checking how good the video's automatic transcript is.
Google's description: The results of ASR transcript quality analysis.
type GoogleApi.ContentWarehouse.V1.Model.IndexingVideosAsrTranscriptRepairAnnotation.t · confidence low

`SafesearchVideoContentSignals.videoClassifierOutput`
Rule: The combined output of the SafeSearch classifiers run on a video.
Google's description: Output of all SafeSearch video classifiers in Amarna.
type GoogleApi.ContentWarehouse.V1.Model.SafesearchVideoClassifierOutput.t · confidence low

`WWWResultInfoSubImageDocInfo.estSigmaU`
Rule: An uncertainty estimate derived from the image usefulness score.
Google's description: Estimated Sigma U calculated from U*.
type number() · confidence low

`WWWResultInfoSubImageDocInfo.humanModelScore`
Rule: A model score about people appearing in the image, from visual shopping tagging.
Google's description: DeepTags human model score. go/VisualShoppingImageAttributes
type number() · confidence low

Freshness (66)

`AnchorsAnchor.firstseenDate`
Rule: Day the link was first seen, counted from a fixed 1994 date.
Google's description: # days past Dec 31, 1994, 23:00:00 UTC (Unix time @788914800) that this link was first seen. Should never occupy more than 15 bits. NOTE: this is NOT the same as creation_date; firstseen_date is filled during link extraction
type integer() · confidence high · stages: 4. Link Analysis & Authority Computation

`CompositeDoc.urldate`
Rule: A date parsed out of the page's web address.
Google's description: Date in the url extracted by quality/snippets/urldate/date-in-url.cc This is given as midnight GMT on the date in question.
type String.t · confidence high · stages: 2. Document Processing & Indexing

`NlpSaftDocument.contentFirstseen`
Rule: The earliest time Google crawled or indexed the page with content, as a timestamp.
Google's description: Stores minimum of first time google successfully crawled a document, or indexed the document with contents (i.e, not roboted). It is stored as the number of seconds since epoch. See quality/historical/signals/firstseen/firstseen.proto
type String.t · confidence high · stages: 9. Query Understanding & Intent Classification

`NlpSaftDocument.contentage`
Rule: How old the content of the page is, stored as a timestamp.
Google's description: Age of the content of the document. For details, see: quality/historical/shingle/signals/contentage.proto The format has been translated to a canonical timestamp (seconds since epoch).
type String.t · confidence high · stages: 9. Query Understanding & Intent Classification

`NlpSaftDocument.lastSignificantUpdate`
Rule: The time of the last meaningful change to the page content.
Google's description: Last significant update of the page content, in the same format as the contentage field, and also derived from ContentAge.last_significant_update in quality/historical/shingle/signals/contentage.proto.
type String.t · confidence high · stages: 9. Query Understanding & Intent Classification

`NlpSaftDocument.syntacticDate`
Rule: A date taken from the page URL or title, stored as a timestamp.
Google's description: Document's syntactic date (e.g. date explicitly mentioned in the URL of the document or in the document title). It is stored as the number of seconds since epoch. See quality/timebased/syntacticdate/proto/syntactic-date.proto
type String.t · confidence high · stages: 9. Query Understanding & Intent Classification

`PerDocData.datesInfo`
Rule: Encoded information about the page's dates, such as whether its dates look old.
Google's description: Stores dates-related info (e.g. page is old based on its date annotations). Used in FreshnessTwiddler. Use encode/decode functions from quality/timebased/utils/dates-info-helper-inl.h
type String.t · confidence high · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.freshboxArticleScores`
Rule: Packed scores from freshness classifiers, including news article and live blog scores.
Google's description: Stores scores of freshness-related classifiers: freshbox article score, live blog score and host-level article score. The encoding/decoding API is in quality/freshness/freshbox/goldmine/freshbox_annotation_encoder.h. To use this field, you MUST join g/pq-classifiers-announce and add your use case at
type integer() · confidence high · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.lastSignificantUpdate`
Rule: The timestamp of the last meaningful change to the document.
Google's description: Last significant update of the document. This is sourced from the quality_timebased.LastSignificantUpdate proto as computed by the LSUSelector from various signals. The value is a UNIX timestamp in seconds.
type String.t · confidence high · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.semanticDate`
Rule: An estimated date for the page's content, derived from the page, its anchors and related pages.
Google's description: SemanticDate, estimated date of the content of a document based on the contents of the document (via parsing), anchors and related documents. Date is encoded as a 32-bits UNIX date (1970 Jan 1 epoch). Confidence is encoded using a SemanticDate specific format. For details of encoding, please refer to quality/freshness/docclassifier/semanticdate/public/semantic_date.proto
type integer() · confidence high · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.semanticDateInfo`
Rule: Packed confidence and metadata about the page's estimated date, used by freshness reranking.
Google's description: Info is encoded using a SemanticDate specific format. Contains confidence scores for day/month/year components as well as various meta data required by the freshness twiddlers.
type integer() · confidence high · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityTimebasedLastSignificantUpdate.date`
Rule: The time the page was judged to have last meaningfully changed.
Google's description: LastSignificantUpdate as UNIX timestamp in seconds. This is the new signal (go/lsu-dd) from LSU Selector V2 (once that is enabled, see b/171879888 for status), falling back to the legacy V1 signal if the HIGH_PRECISION signal does not exist. Please use the 'source' field to determine where the value comes from.
type String.t · confidence high · stages: 7. Freshness & Temporal Signals

`QualityTimebasedSyntacticDate.date`
Rule: The page's publication date as seconds since 1970.
Google's description: The number of seconds since epoch (Jan 1, 1970). This can be negative to indicate a publication date that is before 1970. For example, the ones from NY Times archive: "
type String.t · confidence high · stages: 7. Freshness & Temporal Signals

`WeboftrustLiveResultDocBoostData.hotTimes`
Rule: The time window during which the live result counts as timely.
Google's description: Specifies the time range within which this LiveResult is relevant. Used for deciding whether the rank of the corresponding page should be boosted. For example, this range can be set to encompass a few days before and after a sports game to which the page refers. If this field is missing, no boosting is performed. Specifying that a page should always be boosted is not recommended, but can be accomplished by setting hot_times.start_unix_time=0 and hot_time.end_unix_time=0x7fffffff. Note that multi
type GoogleApi.ContentWarehouse.V1.Model.WeboftrustTimeRange.t · confidence high · stages: 8. Spam, Penalty & Trust Systems

`AnchorsAnchor.creationDate`
Rule: When the link was first seen, also used as retweet time in fresh indexing.
Google's description: used for history - the first and last time we have seen this anchor. creation_date also used for Freshdocs Twitter indexing, a retweet is an anchor of the original tweet. This field records the time when a retweet is created.
type integer() · confidence medium · stages: 4. Link Analysis & Authority Computation

`AnchorsAnchor.encodedNewsAnchorData`
Rule: Packed data marking links that come from newsy, high quality sites.
Google's description: Encoded data containing information about newsiness of anchor. Populated only if anchor is classified as coming from a newsy, high quality site. Encoded data for anchor sources are being stored in googledata/quality/freshness/news_anchors/encoded_news_anchors_data.txt Scores are being computed with quality/freshness/news_anchors/ routines.
type integer() · confidence medium · stages: 4. Link Analysis & Authority Computation

`AnchorsAnchor.firstseenNearCreation`
Rule: Whether the first-seen date is close to when the link was actually added.
Google's description: true if we think 'firstseen_date' is an accurate estimate of when the link was actually added to the source page. false if it may have existed for some time before we saw it.
type boolean() · confidence medium · stages: 4. Link Analysis & Authority Computation

`AnchorsAnchor.lastUpdateTimestamp`
Rule: When this link record was last refreshed during indexing.
Google's description: Used for history and freshness tracking - the timestamp this anchor is updated in indexing.
type integer() · confidence medium · stages: 4. Link Analysis & Authority Computation

`AnchorsAnchorSource.crawlTimestamp`
Rule: When the linking page was crawled.
Google's description: Source page crawl timestamp.
type String.t · confidence medium · stages: 4. Link Analysis & Authority Computation

`CompositeDoc.registrationinfo`
Rule: When this domain was most recently registered and when it expires.
Google's description: Information about the most recent creation and expiration of this domain. It's extracted from domainedge signal.
type GoogleApi.ContentWarehouse.V1.Model.RegistrationInfo.t · confidence medium · stages: 2. Document Processing & Indexing

`CompositeDocIndexingInfo.urlChangerate`
Rule: How often the page's content changes.
Google's description: Changerate information for this doc (see crawler/changerate/changerate.proto for details).
type GoogleApi.ContentWarehouse.V1.Model.CrawlerChangerateUrlChangerate.t · confidence medium · stages: 2. Document Processing & Indexing

`CompositeDocIndexingInfo.urlHistory`
Rule: A record of the page's recent content changes.
Google's description: Url change history for this doc (see crawler/changerate/changerate.proto for details). Note if a doc has more than 20 changes, we only keep the last 20 changes here to avoid adding to much data in its docjoin.
type GoogleApi.ContentWarehouse.V1.Model.CrawlerChangerateUrlHistory.t · confidence medium · stages: 2. Document Processing & Indexing

`GDocumentBaseContent.CrawlTime`
Rule: When the page was crawled, in seconds since 1970.
Google's description: Seconds since Unix epoch.
type String.t · confidence medium

`ImageData.contentFirstCrawlTime`
Rule: The earliest crawl time seen for any near duplicate of this image.
Google's description: Earliest known crawl time among all neardups of this image (go/imagecontentage).
type integer() · confidence medium

`ImageData.firstCrawlTime`
Rule: The earliest time this image is known to have existed.
Google's description: Earliest known timestamp about this image. Today, this is the timestamp when the content key was generated for this image. The time is in seconds.
type integer() · confidence medium

`ImageData.firstTimeSeenOnDocSec`
Rule: The first time this image address was seen on the page that shows it.
Google's description: The first time this image URL was seen on the containing web page. Only set during web indexing.
type integer() · confidence medium

`IndexingSignalAggregatorAgeWeightedCoverageData.firstCoverageTimestamp`
Rule: When this URL first got any coverage data.
Google's description: Epoch seconds at which this url first gets coverage data.
type String.t · confidence medium · stages: 2. Document Processing & Indexing

`IndexingSignalAggregatorUrlScore.firstServedTimestamp`
Rule: The first time this document was ever served anywhere.
Google's description: The timestamp of the first time this document is served anywhere.
type String.t · confidence medium

`NlpSaftDocument.bylineDate`
Rule: The byline date of the page, stored as a timestamp and shown in search snippets.
Google's description: Document's byline date, if available: this is the date that will be shown in the snippets in web search results. It is stored as the number of seconds since epoch. See segindexer/compositedoc.proto
type String.t · confidence medium · stages: 9. Query Understanding & Intent Classification

`NlpSaftDocument.date`
Rule: The document's anchor date written as a full year-month-day timestamp.
Google's description: Document anchor date in YYYYMMDDhhmmss format.
type String.t · confidence medium · stages: 9. Query Understanding & Intent Classification

`PerDocData.lastSignificantUpdateInfo`
Rule: Extra detail recording where the last-significant-update timestamp came from.
Google's description: Metadata about last significant update. Currently this only encodes the quality_timebased.LastSignificantUpdate.source field which contains the info on the source of the signal. NOTE: Please do not read the value directly. Use helpers from quality/timebased/lastsignificantupdate/lsu-helper.h instead.
type String.t · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.timeSensitivity`
Rule: An encoded measure of how time-sensitive the document is.
Google's description: Encoded Document Time Sensitivity signal.
type integer() · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityCopiaFireflySiteSignal.numOfArticlesByPeriods`
Rule: How many news-quality articles a site published in each of the last 30-day windows.
Google's description: number of articles (lattice article score is 0.8 or more) sliced by 30 days (num_of_articles_by_periods[0] is the newest).
type list(String.t) · confidence medium · stages: 3. Content Quality & Site-Level Scoring

`QualityCopiaFireflySiteSignal.numOfUrlsByPeriods`
Rule: How many pages a site published in each of the last 30-day windows.
Google's description: number of urls sliced by 30 days (num_of_urls_by_periods[0] is the newest).
type list(String.t) · confidence medium · stages: 3. Content Quality & Site-Level Scoring

`QualityTimebasedDateUnreliability.unreliableDatesScore`
Rule: A score for how untrustworthy the dates on this page look.
Google's description: If scores are missing, the date is reliable by default. max(petacat.unreliable_dates_score)
type number() · confidence medium · stages: 7. Freshness & Temporal Signals

`QualityTimebasedDateUnreliability.unreliableDatesScoreExposureAdjusted`
Rule: The date-unreliability score adjusted by how prominent dates are on the page.
Google's description: max(petacat.unreliable_dates_score * petacat.date_exposure_score)
type number() · confidence medium · stages: 7. Freshness & Temporal Signals

`QualityTimebasedLastSignificantUpdateAdjustments.isUpperboundTimestampPrecise`
Rule: Whether the upper-bound timestamp came from a precise passage date.
Google's description: The timestamp is precise when it's derived from existing (>March 2022) passage timestamp.
type boolean() · confidence medium · stages: 7. Freshness & Temporal Signals

`QualityTimebasedLastSignificantUpdateAdjustments.upperboundTimestampInSeconds`
Rule: The latest time the page's last-update date is allowed to be.
Google's description: The upperbound value derived from passage timestamps. If present, the LSU date should never exceed this value. Design doc: go/lsu-max-passage-timestamp
type String.t · confidence medium · stages: 7. Freshness & Temporal Signals

`QualityTimebasedPageType.isPageWithFreshRepeatedDates`
Rule: Whether the page shows a run of recent repeated dates.
Google's description: Set to true if this page has a fresh repeated date sequence.
type boolean() · confidence medium · stages: 7. Freshness & Temporal Signals

`QualityTimebasedPetacatDateUnreliability.dateExposureScore`
Rule: How prominently dates appear on pages of this kind, smoothed to a score.
Google's description: This is date_exposure after sigmoid transformation.
type number() · confidence medium

`QualityTimebasedPetacatDateUnreliability.dateVsContentageDistributionSkew`
Rule: A component score comparing stated dates against page age.
Google's description: Auxiliary scores. The unreliable_dates_score is a product of these scores excl. date_exposure. The definition for each of the scores you can find at go/unreliable-dates-dd
type number() · confidence medium

`QualityTimebasedPetacatDateUnreliability.unreliableDatesScore`
Rule: The final untrustworthy-dates score before the date-prominence part.
Google's description: This is a final score excl. date exposure component.
type number() · confidence medium

`QualityTimebasedSyntacticDate.bylineDate`
Rule: The byline date, stored when it differs from the page's main date.
Google's description: The following field is set only when the byline date is different from the "date" field above. Currently this happens when the byline date is within the 24 hours of the crawl time, or close but not exactly the same as blog post date due to time zone. The syntactic date is never later than the crawl time. NOTE: If this field is set, use_as_byline_date will be meaningless, and better to be cleared.
type String.t · confidence medium · stages: 7. Freshness & Temporal Signals

`QualityTimebasedSyntacticDate.info`
Rule: Flag bits saying whether the byline date is considered high confidence.
Google's description: Used to store extra information about the syntactic date. For now only two bits are set. Please refer to the encoding/decoding functions provided in: quality/timebased/syntacticdate/util.h Bit 1 = High confidence byline. This bit is set if the syntactic date has a byline date and this date is considered to be high confidence. Bit 2 = High confidence byline without content age. This bit is set if the syntactic date has a byline date and this date is considered to be high confidence without suppor
type integer() · confidence medium · stages: 7. Freshness & Temporal Signals

`QualityTimebasedSyntacticDate.precisionMark`
Rule: How precise the extracted date is, for example day, month or year.
Google's description: The precision mark should be of type PRECISION_MARK.
type integer() · confidence medium · stages: 7. Freshness & Temporal Signals

`RepositoryWebrefDocumentMetadata.crawlTime`
Rule: When the page was crawled.
Google's description: The timestamp of when the document was crawled (if known). Copied from CompositeDoc.Content.CrawlTime.
type String.t · confidence medium · stages: 6. Entity & Knowledge Graph Integration

`RepositoryWebrefWebrefEntities.dateRange`
Rule: Date ranges judged most relevant to the page, such as a film's release date.
Google's description: Dates ranges that are most relevant to the document. E.g. on a document about Dune the 2021 movie, this might hold the release date of that movie.
type list(GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefSemanticDateRange.t) · confidence medium · stages: 6. Entity & Knowledge Graph Integration

`ResearchScienceSearchSourceUrlDocjoinInfo.latestPageUpdateDate`
Rule: The date on the page that reflects when its content was published.
Google's description: The syntactic date of a dataset document that reflects the publication date of the content.
type String.t · confidence medium

`ScienceIndexSignal.PublicationYear`
Rule: The year the article was published.
Google's description: Publication date.
type integer() · confidence medium

`WeboftrustLiveResultsDocAttachments.docBoost`
Rule: The bundle of live-result boost information attached to a document.
Google's description: Information about potential rank boosting for the document by virtue of its Live Result feed.
type list(GoogleApi.ContentWarehouse.V1.Model.WeboftrustLiveResultDocBoostData.t) · confidence medium

`ImageExifIPTCMetadata.dateCreated`
Rule: The creation date recorded in the image's embedded metadata.
Google's description: Time (in seconds)
type String.t · confidence low

`ImageExifImageEmbeddedMetadata.captureTime`
Rule: The time the photograph was taken, from the image file's metadata.
Google's description: Capturing settings in time_t
type String.t · confidence low

`ImageExifImageEmbeddedMetadata.modificationTime`
Rule: The time the image file was last modified, from its metadata.
Google's description: in time_t
type String.t · confidence low

`PerDocData.domainAge`
Rule: The age of the domain, stored as a 16-bit value.
Google's description: 16-bit
type integer() · confidence low · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.eventsDate`
Rule: The start dates of events listed on the page, one date per event.
Google's description: Date for Events. A web page might list multiple events with different dates. We only take one date (start date) per event.
type list(String.t) · confidence low · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.isHotdoc`
Rule: Flag set by the fresh-document pipeline marking this as a currently hot document.
Google's description: Set by the FreshDocs instant doc joiner. See //indexing/instant/hotdocs/README and
type boolean() · confidence low · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityTimebasedSyntacticDate.timeZoneOffsetSeconds`
Rule: The time-zone shift in seconds applied when working out the date.
Google's description: Indicates the time zone offset in seconds applied to derive
type String.t · confidence low · stages: 7. Freshness & Temporal Signals

`RepositoryWebrefSemanticDateRange.confidence`
Rule: How sure the system is that an extracted date range actually applies to the document.
Google's description: Indicates how confident we are this extracted range is relevant to a document (document to be infered from context).
type number() · confidence low

`RepositoryWebrefSemanticDateRange.end`
Rule: The end date pulled from the entity's knowledge graph data.
Google's description: End date extracted from the entity along end_source_property.
type String.t · confidence low

`RepositoryWebrefSemanticDateRange.start`
Rule: The start date pulled from the entity's knowledge graph data.
Google's description: Start date extracted from the entity along start_source_property.
type String.t · confidence low

`ShingleSource.timestamp`
Rule: When this source was first seen.
Google's description: First-seen timestamp of the source.
type integer() · confidence low

`SnapshotSnapshotMetadata.crawlTimestamp`
Rule: The time the page's main file was fetched, in seconds since epoch.
Google's description: The time at which the main resource of the Snapshot was fetched, in seconds since epoch. Note that the various page dependencies may have been fetched at much earlier points in time (hours, maybe days) and that this could be off from the actual rendering time.
type String.t · confidence low

`TrawlerFetchReplyDataCrawlDates.NotChangedDate`
Rule: Timestamp for when the page was last actually fetched from the web.
Google's description: fetched from the web. Timestamp indicates when
type integer() · confidence low

`TrawlerFetchReplyDataCrawlDates.OriginalCrawlDate`
Rule: Timestamp for when the page was originally crawled.
Google's description: Timestamp when the page was
type integer() · confidence low

`TrawlerFetchReplyDataCrawlDates.ReuseDate`
Rule: Timestamp for when the page was last checked and found unchanged.
Google's description: the page was last checked but found to be the same as before. If set, timestamp to indicate
type integer() · confidence low

`WeboftrustTimeRange.startUnixTime`
Rule: The start time of a live result's relevant time window.
Google's description: Start and end times should always appear. Marked as optional to avoid breaking code.
type String.t · confidence low

Clicks & engagement (54)

`CompositeDocIndexingInfo.rawNavboost`
Rule: The page's own click boost count before duplicates are merged in.
Google's description: The raw navboost count for the canonical url without aggregating the navboost from dup urls. This field is used when building forwarding map.
type integer() · confidence high · stages: 2. Document Processing & Indexing

`CompressedQualitySignals.crapsAbsoluteHostSignals`
Rule: Raw host-level impression counts used by the Craps click and impression system.
Google's description: Impressions, unsquashed, host level, not to be used with compressed ratios. Not to be used in Pattern Data.
type integer() · confidence high · stages: 3. Content Quality & Site-Level Scoring; 8. Spam, Penalty & Trust Systems; 11. Re-Ranking & Twiddler System

`CountryClickDistribution.total`
Rule: The total number of clicks counted for the page or domain.
Google's description: To store total clicks on this page/domain.
type float() · confidence high

`ImageData.h2i`
Rule: How often people hover over this image compared with seeing or clicking it.
Google's description: 'Hovers to Impressions' and 'Hovers to Clicks' ratios for an image. These are considered Search CPS Personal Data due to concerns that they may be used to reidentify or confirm the presence of specific singleton (unique) queries.
type number() · confidence high

`ImageData.imageQualityClickSignals`
Rule: Click behavior used to judge how useful and appealing an image is.
Google's description: Click signals for measuring image quality (e.g., usefulness, presentation, appealingness, and engagingness). Click signals are considered Search CPS Personal Data.
type GoogleApi.ContentWarehouse.V1.Model.ImageQualityNavboostImageQualityClickSignals.t · confidence high

`ImageExactBoostNavQuery.imageClickRank`
Rule: Where the image ranks for this query based on clicks it received.
Google's description: Click-based rank of the image for this query.
type integer() · confidence high

`IndexingSignalAggregatorAgeWeightedCoverageData.chances`
Rule: The time-decayed count of chances the URL had to be seen, used as the denominator for click and impression rates.
Google's description: Numbers below are all total in the decayed manner. To get rate of impression/clicks, divide by chances.
type float() · confidence high · stages: 2. Document Processing & Indexing

`IndexingSignalAggregatorAgeWeightedCoverageData.dwells`
Rule: Count of time-decayed dwells the page got from Knowledge Panel and Web Answers.
Google's description: Dwells from KnowledgePanel and WebAnswers.
type float() · confidence high · stages: 2. Document Processing & Indexing

`IndexingSignalAggregatorAgeWeightedCoverageData.intervalData`
Rule: Tracks the average time between clicks, good clicks, and weighted impressions for the URL.
Google's description: Interval Data to track the average time between clicks_total, clicks_good, and ctr_weighted_impression.
type GoogleApi.ContentWarehouse.V1.Model.IndexingSignalAggregatorAdaptiveIntervalData.t · confidence high · stages: 2. Document Processing & Indexing

`IndexingSignalAggregatorAgeWeightedCoverageData.lastDwellDateInDays`
Rule: The most recent day the page received a Knowledge Panel or Web Answer dwell.
Google's description: Indicates the date when this document received the last KnowledgePanel or WebAnswer dwell. Note: The date is identified in terms of number of days since Epoch.
type integer() · confidence high · stages: 2. Document Processing & Indexing

`IndexingSignalAggregatorAgeWeightedCoverageData.lastGoodClickDateInDays`
Rule: The most recent day the page received a good click.
Google's description: Indicates the date when this document received the last good click. Note: The date is identified in terms of number of days since Epoch.
type integer() · confidence high · stages: 2. Document Processing & Indexing

`IndexingSignalAggregatorAgeWeightedCoverageData.lastImpressionDateInDays`
Rule: The most recent day the page was shown in results.
Google's description: Indicates the date when this document received the last impression. Note: The date is identified in terms of number of days since Epoch.
type integer() · confidence high · stages: 2. Document Processing & Indexing

`QualityNavboostCrapsCrapsData.lastLongestClicks`
Rule: How many clicks were the final and longest click in a related search session.
Google's description: The number of clicks that were last and longest in related user queries.
type float() · confidence high · stages: 5. User Engagement & NavBoost (Click Signals)

`RepositoryWebrefDocumentMetadata.totalClicks`
Rule: The total number of clicks on this page taken from navboost data.
Google's description: The total clicks on this document, taken from navboost data.
type number() · confidence high · stages: 6. Entity & Knowledge Graph Integration

`ResearchScienceSearchNavboostQueryInfo.impCount`
Rule: Estimated number of times this page was shown for this query.
Google's description: imp_count stores an estimate of the number of impressions for this tuple.
type number() · confidence high

`ResearchScienceSearchNavboostQueryInfo.lccCount`
Rule: Estimated number of long clicks recorded for this query and page.
Google's description: lcc_count stores an estimate of the number of long clicks for this tuple. NOTE: It is similar to query_doc_count, but calculated in different manner.
type number() · confidence high

`ResearchScienceSearchNavboostQueryInfo.query`
Rule: The query text that these click counts belong to.
Google's description: The query string.
type String.t · confidence high

`ResearchScienceSearchNavboostQueryInfo.queryDocCount`
Rule: Number of long clicks on this particular query and page pair.
Google's description: The query_doc_count stores the number of long-clicks on this pair.
type number() · confidence high

`CompositeDocIndexingInfo.normalizedClickScore`
Rule: A click score rescaled to match the score of a similarly ranked normal page.
Google's description: This score is calculated by re-mapping the back onto the partition's score distribution, such that the score represents the score of the equivalently ranked organically-selected document.
type number() · confidence medium · stages: 2. Document Processing & Indexing

`CompressedQualitySignals.crapsNewUrlSignals`
Rule: URL-level click and impression data for the Craps system.
Google's description: For craps
type String.t · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 8. Spam, Penalty & Trust Systems; 11. Re-Ranking & Twiddler System

`ImageExactBoostNavQuery.confidence`
Rule: How confident the system is that this image matches the stored query.
Google's description: Associated confidence scores for the image for the query.
type integer() · confidence medium

`ImageExactBoostNavQuery.navQuery`
Rule: The tidied-up text of a query the image is strongly associated with.
Google's description: The normalized raw query text.
type String.t · confidence medium

`ImageRegionsImageRegions.hasNavboost`
Rule: Whether the image has Navboost click data attached to it.
Google's description: True if the image has navboost.
type boolean() · confidence medium

`IndexingSignalAggregatorAgeWeightedCoverageData.averageChanceTime`
Rule: The average time of the time-decayed chances a URL had to be shown.
Google's description: Weighted averged timestamps of the decayed chances.
type float() · confidence medium · stages: 2. Document Processing & Indexing

`IndexingSignalAggregatorAgeWeightedCoverageData.lastLuDwellDateInDays`
Rule: The most recent day the page received a Local Universal dwell.
Google's description: Indicates the date when this document received the last LocalUniversal dwell. Note: The date is identified in terms of number of days since Epoch.
type integer() · confidence medium · stages: 2. Document Processing & Indexing

`IndexingSignalAggregatorAgeWeightedCoverageData.lastPseudoImpressionsDateInDays`
Rule: The last day the page was retrieved as a result but withheld because of its age.
Google's description: Indicates the date when this document received the last pseudo-impression. I.e., when it was retrieved as a result but GWS would not show it because of the document's age in the index. Note: The date is identified in terms of number of days since Epoch.
type integer() · confidence medium · stages: 2. Document Processing & Indexing

`IndexingSignalAggregatorAgeWeightedCoverageData.luDwells`
Rule: Count of time-decayed dwells the page got from Local Universal results.
Google's description: Dwells from LocalUniversal.
type float() · confidence medium · stages: 2. Document Processing & Indexing

`IndexingSignalAggregatorAgeWeightedCoverageData.totalChances`
Rule: The raw, undecayed number of chances this URL had to be shown.
Google's description: Total number of chances on this urls (not decayed).
type String.t · confidence medium · stages: 2. Document Processing & Indexing

`IndexingSignalAggregatorSccSignal.numImageUrls`
Rule: How many of the URL's clicks were image clicks, as a count or a ratio.
Google's description: This represents the number of urls with image clicks. A url can have both image and non-image clicks, in which case we set num_image_urls to be the ratio of image_clicks vs total clicks. For example, if a url has 10 total clicks and 7 image clicks, num_image_urls will be set to 0.7.
type float() · confidence medium

`IndexingSignalAggregatorUrlScore.isImportant`
Rule: Marks a URL as having a notable signal, such as good clicks.
Google's description: Whether this url has important signal. Used for keeping patterns that match too few URLs but some of them have good clicks.
type boolean() · confidence medium

`PornFlagData.queryStats`
Rule: Statistics about the queries that led to this image, taken from click logs.
Google's description: Query statistics from Navboost logs. For more details see classifier/porn/proto/image_porn_classifier_signals.proto.
type GoogleApi.ContentWarehouse.V1.Model.ClassifierPornQueryStats.t · confidence medium

`QualityNavboostCrapsCrapsClickSignals.absoluteImpressions`
Rule: How many times results from this host were shown, as a raw count.
Google's description: Thus far this field is only used for host level unsquashed impressions. When compressed (e.g., in perdocdata.proto, CompressedQualitySignals), this value is represented individually and thus is generally incompatible with the other values which are compressed as click-ratios.
type float() · confidence medium · stages: 5. User Engagement & NavBoost (Click Signals)

`QualityNavboostCrapsCrapsData.features`
Rule: Click statistics broken out for particular slices such as mobile users in one city.
Google's description: Contains CrapsClickSignals for specific features. (i.e. for mobile, US, metro id -123")
type list(GoogleApi.ContentWarehouse.V1.Model.QualityNavboostCrapsFeatureCrapsData.t) · confidence medium · stages: 5. User Engagement & NavBoost (Click Signals)

`QualityNavboostCrapsCrapsData.impressions`
Rule: How many times the result was shown to users.
Google's description: These fields may become legacy fields; we may retire them and use the squashed field (below) instead, to allow for some nesting.
type float() · confidence medium · stages: 5. User Engagement & NavBoost (Click Signals)

`QualityNavboostCrapsCrapsData.mobileSignals`
Rule: Click statistics gathered only from mobile searches.
Google's description: The portion of this CrapsData aggregated on data from tier 1/2 mobile interfaces in QSessions.
type GoogleApi.ContentWarehouse.V1.Model.QualityNavboostCrapsCrapsClickSignals.t · confidence medium · stages: 5. User Engagement & NavBoost (Click Signals)

`QualityNavboostCrapsCrapsData.unscaledIpPriorBadFraction`
Rule: The raw share of bad activity expected from this IP address, before scaling.
Google's description: Used to assign a prior based on IP address. See quality/navboost/craps/craps-ip-prior.h. This value is prior to the linear transformation (scaling / offset / min / max) that's applied in craps-penalty.cc.
type float() · confidence medium · stages: 5. User Engagement & NavBoost (Click Signals)

`QualityNavboostCrapsCrapsData.unsquashed`
Rule: Uncompressed click statistics for this record.
Google's description: We will start using this one for the retuning rollout.
type GoogleApi.ContentWarehouse.V1.Model.QualityNavboostCrapsCrapsClickSignals.t · confidence medium · stages: 5. User Engagement & NavBoost (Click Signals)

`QualityNavboostCrapsFeatureCrapsData.signals`
Rule: The click statistics for this particular locale slice.
Google's description: CRAPS Signals for the locale.
type GoogleApi.ContentWarehouse.V1.Model.QualityNavboostCrapsCrapsClickSignals.t · confidence medium · stages: 5. User Engagement & NavBoost (Click Signals)

`QualityNsrNsrData.chromeInTotal`
Rule: How much Chrome viewing traffic the site gets overall.
Google's description: Site-level Chrome views.
type number() · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityNsrNsrData.impressions`
Rule: How often the site's pages were shown in results.
Google's description: Site-level impressions.
type number() · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityNsrNsrData.pnavClicks`
Rule: The click count used as the denominator of the pnav fraction.
Google's description: denominator for the pnav computation
type number() · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualitySitemapTargetGroup.topUrl`
Rule: The links with the most browser transition clicks in this group.
Google's description: A list of top urls with highest two_level_score, i.e., chrome_trans_clicks.
type list(GoogleApi.ContentWarehouse.V1.Model.QualitySitemapTopURL.t) · confidence medium

`RepositoryWebrefRefconQueryStats.aggregatedQueryCount`
Rule: Total query count leading to the document, usable to estimate its popularity.
Google's description: Total query count for the document from all the query terms. Can be used to estimate the popularity of the document.
type float() · confidence medium

`ResearchScienceSearchNavboostQueryInfo.queryCount`
Rule: How many times this query was seen overall.
Google's description: The query_count stores the counts on this query.
type number() · confidence medium

`ResearchScienceSearchSourceUrlDocjoinInfo.navboostQuery`
Rule: The list of queries and click counts recorded for the page.
Google's description: A sequence of Navboost queries for the dataset source_url.
type list(GoogleApi.ContentWarehouse.V1.Model.ResearchScienceSearchNavboostQueryInfo.t) · confidence medium

`ToolBarPerDocData.VisitorsPastDay`
Rule: Count of distinct Google Toolbar visitors this page had in the past day.
Google's description: Indicates how many distinct toolbar visitors this page had in the past day. Will only be present if the number is reasonably large.
type integer() · confidence medium

`VideoContentSearchVideoInfo.crapsData`
Rule: Stored click and impression data for the video page.
Google's description: Craps data from the video cdoc.
type GoogleApi.ContentWarehouse.V1.Model.QualityNavboostCrapsCrapsData.t · confidence medium

`VideoContentSearchVideoInfo.uniqueChromeViews`
Rule: How many unique Chrome visits the page received.
Google's description: Number of unique visits in Chrome.
type integer() · confidence medium

`QualityNavboostCrapsCrapsClickSignals.unicornClicks`
Rule: The share of clicks tied to a particular internal class of user sessions.
Google's description: The subset of clicks that are associated with an event from a Unicorn user.
type float() · confidence low · stages: 5. User Engagement & NavBoost (Click Signals)

`QualityNavboostCrapsCrapsData.packedIpAddress`
Rule: The packed IP address used to look up a click-trust prior.
Google's description: Contains a packed string in network byte order, as expected by CrapsIpPrior. Only populated if we looked up the ip_prior_bad_fraction at retrieval time.
type String.t · confidence low · stages: 5. User Engagement & NavBoost (Click Signals)

`QualityNavboostCrapsCrapsData.patternSccStats`
Rule: Statistics about the individual URLs that contributed to a pattern's click data.
Google's description: For pattern data, this will contain stats of the SCC's of the individual urls contributing to the pattern.
type GoogleApi.ContentWarehouse.V1.Model.QualityNavboostCrapsStatsWithWeightsProto.t · confidence low · stages: 5. User Engagement & NavBoost (Click Signals)

`QualityNavboostCrapsCrapsData.voterTokenCount`
Rule: How many distinct user tokens contributed to this click record.
Google's description: The number of distinct voter tokens (a lower bound on the number of distinct users that contributed to the entry, used for privacy-related filtering).
type integer() · confidence low · stages: 5. User Engagement & NavBoost (Click Signals)

`QualityNsrNsrData.pnav`
Rule: A site-level fraction describing navigational click behavior.
Google's description: Fractional signals.
type number() · confidence low · stages: 3. Content Quality & Site-Level Scoring; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`VideoContentSearchVideoInfo.numViews`
Rule: How many times the video has been viewed.
Google's description: Number of views.
type String.t · confidence low

Personalization & local (44)

`RepositoryAnnotationsGeoTopic.normalizedScore`
Rule: How likely it is that the page is about this place, from zero to one.
Google's description: A score [0, 1] indicating the likelihood of the location being the GeoTopicality.
type number() · confidence high

`AnchorsAnchorSource.localCountryCodes`
Rule: Countries the linking page is considered most relevant to.
Google's description: Countries to which the source page is local/most relevant; stored as III identifiers for country/region codes (see
type list(integer()) · confidence medium · stages: 4. Link Analysis & Authority Computation

`CountryClickDistribution.confidence`
Rule: How trustworthy the recorded spread of clicks by country is.
Google's description: To store confidence in the distribution in cases when total is not set.
type float() · confidence medium

`CountryCountryAttachment.clickDistribution`
Rule: How a page's clicks spread across countries, used to guess its country.
Google's description: Store weighted click distribution for page level country-id classification.
type GoogleApi.ContentWarehouse.V1.Model.CountryClickDistribution.t · confidence medium · stages: 12. Personalization, Local & Context

`CountryCountryAttachment.geoLocations`
Rule: The specific places, such as cities or states, tied to the page.
Google's description: New MetroID: Now called GeoLocations since the locations could be sublocalities, cities or states. GeoLocations are always more fine grained than country. TODO (jayeshv): Once new MetroID/GeoLocations is launched everywhere, deleted old MetroID related fields.
type GoogleApi.ContentWarehouse.V1.Model.CountryGeoLocations.t · confidence medium · stages: 12. Personalization, Local & Context

`CountryCountryAttachment.localCountries`
Rule: The countries the page is considered local to.
Google's description: two-letter(lower-case) countrycode, e.g. us countries that is local to
type list(String.t) · confidence medium · stages: 12. Personalization, Local & Context

`CountryCountryAttachment.localCountryCodes`
Rule: The stored numeric country codes for the page.
Google's description: Fields that actually store the country id in docjoins. The format of this data is defined in //i18n/identifiers/stableinternalregionconverter.cc. Converter defined there can be used to convert it to RegionCode format.
type list(integer()) · confidence medium · stages: 12. Personalization, Local & Context

`CountryCountryAttachment.metroIdList`
Rule: Metro-area features tied to the page, with the surrounding province.
Google's description: Metro locations: list of NavBoost feature V2 associated with a doc, along with the enclosing province. Metro locations with new tags.
type list(GoogleApi.ContentWarehouse.V1.Model.CountryMetroNBFeature.t) · confidence medium · stages: 12. Personalization, Local & Context

`CountryCountryAttachment.relatedCountries`
Rule: Countries the page relates to without being local to them.
Google's description: two-letter(lower-case) countrycode, e.g. us countries that is related to, but not local to
type list(String.t) · confidence medium · stages: 12. Personalization, Local & Context

`CountryCountryAttachment.superGlobal`
Rule: A flag marking a page as super global, which gets a smaller country demotion.
Google's description: Super global pages get lesser demotion than global pages. A document can only be either global or super_global but not both.
type boolean() · confidence medium · stages: 12. Personalization, Local & Context

`CountryCountryAttachment.urlPatternBasedLanguage`
Rule: Language and country guessed from the shape of the URL.
Google's description: Language and country extracted using the URL pattern map.
type integer() · confidence medium · stages: 12. Personalization, Local & Context

`CountryCountryAttachment.wmxCountry`
Rule: The country a site owner declared in the webmaster console.
Google's description: Country specified for a web-site through webmaster console.
type String.t · confidence medium · stages: 12. Personalization, Local & Context

`CountryGeoLocation.clickRadius50Percent`
Rule: The distance around the page's location that covers half of its clicks.
Google's description: The radius (in miles) around the assigned location that the document gets 50% of its clicks.
type integer() · confidence medium

`CountryGeoLocation.confidencePercent`
Rule: How sure the system is about the assigned location, on a 0 to 100 scale.
Google's description: Confidence mapped to [0, 100]. Converted to integer for efficient storage. Populated during index creation.
type integer() · confidence medium

`CountryGeoLocations.isNonLocationSpecific`
Rule: A flag for pages that get many clicks but belong to no particular place.
Google's description: This will be set to true for documents which receive several clicks but are not assigned any location because the click distribution is flat. Typical examples are global sites like facebook.com, chains like walmart.com, informational sites like wikipedia.org etc. This flag is not propagated to deeper pages since this signal is meant to indicate that a website or a part of website is conclusively non-local, so propagating this information to deeper pages does not make sense. If this flag is set,
type boolean() · confidence medium

`CountryMetroNBFeature.id`
Rule: A packed identifier combining country, language and metro area.
Google's description: A 32 bit navboost v2 feature id encoding (country, language, metro). NavBoosterUtils class (google3/quality/navboost/nav_booster_utils.h) provides functions to decode this feature.
type integer() · confidence medium

`DocProperties.languages`
Rule: The languages the document is written in.
Google's description: A Language enum value. See: go/language-enum
type list(integer()) · confidence medium

`IndexingDupsLocalizedLocalizedCluster.documentLanguage`
Rule: The language detected from the page's own text.
Google's description: The language of this document (as detected by on-page language detection, not influenced by external anchor signals or other indirect conclusions).
type String.t · confidence medium

`LocalWWWInfo.geotopicality`
Rule: Annotations about the geographic places a page is about, not specific businesses.
Google's description: Information about geo locations, rather than individual businesses.
type GoogleApi.ContentWarehouse.V1.Model.RepositoryAnnotationsGeoTopicality.t · confidence medium · stages: 12. Personalization, Local & Context

`LocalWWWInfoCluster.annotationConfidence`
Rule: Confidence that a business mention found on the page is really that business.
Google's description: Confidence score for business mention annotations which is copied from LocalEntityAnnotations::location_confidence.
type number() · confidence medium · stages: 12. Personalization, Local & Context

`LocalWWWInfoCluster.confidence`
Rule: Probability that this page is the business's own main page.
Google's description: Probability that this is the authority page of the business. Same as LocalListing.authority_page_probability, only set for pages with page_type_flags & AUTHORITY.
type number() · confidence medium · stages: 12. Personalization, Local & Context

`PerDocData.brainloc`
Rule: Location information attached to the document.
Google's description: Brainloc contains location information for the document. See ariane/273189 for details.
type GoogleApi.ContentWarehouse.V1.Model.QualityGeoBrainlocBrainlocAttachment.t · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.countryInfo`
Rule: The country information recorded for the document.
Google's description: This field stores the country information for the document in the form of CountryAttachment.
type GoogleApi.ContentWarehouse.V1.Model.CountryCountryAttachment.t · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityNsrNsrData.localityScore`
Rule: How local the site is, as the locality part of a local authority signal.
Google's description: Locality score of the site, i.e. the locality component of the LocalAuthority signal (see go/pq-localauthority).
type number() · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualitySalientCountriesSalientCountry.country`
Rule: The two-letter code of a country this document is about.
Google's description: 2-letter country format.
type String.t · confidence medium

`QualitySalientCountriesSalientCountry.salience`
Rule: How strongly the document relates to that country, from zero to one.
Google's description: How salient this country is for the document. [0,1] range.
type number() · confidence medium

`RepositoryAnnotationsGeoTopic.address`
Rule: The city, region and country containing the place the page is about.
Google's description: Stores parent/container information containing city, province & country.
type GoogleApi.ContentWarehouse.V1.Model.GeostoreAddressProto.t · confidence medium

`RepositoryAnnotationsGeoTopic.componentScores`
Rule: The raw scores that were combined into the final place score.
Google's description: The raw scores used to calculate the normalized_score. Note that not all these scores may be exposed to the users.
type list(GoogleApi.ContentWarehouse.V1.Model.RepositoryAnnotationsGeoTopicalityScore.t) · confidence medium

`RepositoryAnnotationsGeoTopic.confidence`
Rule: How confident the system is that the page is about this place.
Google's description: A score [0, 1] indicating the confidence.
type number() · confidence medium

`RepositoryAnnotationsGeoTopic.locationName`
Rule: The standardized name of the place.
Google's description: Name of the Geographic location. This is the normalized name.
type String.t · confidence medium

`RepositoryAnnotationsGeoTopic.sumContainedPoiNormalizedScores`
Rule: The added-up scores of points of interest inside this area.
Google's description: The sum of the normalized scores of POIs contained within a particular locality.
type number() · confidence medium

`RepositoryAnnotationsGeoTopicality.geotopics`
Rule: The list of places the page is about, strongest first.
Google's description: The geotopics are ordered by normalized_score in descending order.
type list(GoogleApi.ContentWarehouse.V1.Model.RepositoryAnnotationsGeoTopic.t) · confidence medium · stages: 12. Personalization, Local & Context

`WWWDocInfo.languageTag`
Rule: The single most likely language of the document.
Google's description: This returns the most probable language for the document. The complete set of languages is in the GenericSearchResponse. (If some future use requires all languages from the doc request, note that fetching that will require decoding the entire per-doc data attachment, which is a performance hit) Use docinfo-util.h to set & read language fields. Language tag as defined by
type String.t · confidence medium

`CountryCountryAttachment.salientCountries`
Rule: The countries a document is most relevant to.
Google's description: [Experimental]: Top salient countries for a doc. If a country can not be found on this field it can be considered that this doc is not relevant to it.
type list(GoogleApi.ContentWarehouse.V1.Model.CountrySalientCountry.t) · confidence low · stages: 12. Personalization, Local & Context

`CountryGeoLocation.propagatedFromASubpage`
Rule: Whether the location came from a subpage rather than the page itself.
Google's description: True if this location is assigned to one of the subpages, and not to the page itself. If the total number of locations assigned to all the subpages of a page is small (usually up to 5), then that page also gets assigned those locations, and this flag is set for those locations.
type boolean() · confidence low

`CountryGeoLocations.propagationDepthFromParent`
Rule: How many URL levels away the parent page that supplied the location is.
Google's description: Depth of the URL from it's nearest parent in GeoLocation data. Webpages inherhit locations from their parent pages. For example, if foo.com/a is assigned location L1, and foo.com/a/b is not assigned any location, then
type integer() · confidence low

`CountryMetroNBFeature.enclosingProvinceGeotoken`
Rule: A fingerprint of the state around a metro area, used to tell metros apart.
Google's description: The enclosing_province_geotoken is a 32 bit fingerprint of the state encosing the (metro) id. MetroId's can span multiple states. Enclosing geotoken is filled in with the state name for disambiguation. ProvinceGeotoken field is different as it indicates an "interest". Format: 32 bit fingerprint(
type integer() · confidence low

`PerDocData.geodata`
Rule: A compact blob of geographic data about the page.
Google's description: geo data; approx 24 bytes for 23M U.S. pages
type String.t · confidence low · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityGeoBrainlocBrainlocAttachment.topCitiesVocabIds`
Rule: The top locations linked to the document, with their scores, stored compactly.
Google's description: Compressed top locations and their scores.
type list(integer()) · confidence low

`QualitySalientCountriesSalientCountrySet.packedCountry`
Rule: Country codes and their salience values packed together for storage.
Google's description: Packed Country and salience optimized for index storage
type list(integer()) · confidence low

`QualityShoppingShoppingAttachmentLocale.languageId`
Rule: A numeric language and region code kept for fast scoring.
Google's description: Use integers for fast scoring. Note: 26 is UNKNOWN_LANGUAGE_ID, 0 is UNKNOWN region, see i18n::languages::Language and StableInternalRegionconverter Use -1 as default for both.
type integer() · confidence low

`RepositoryAnnotationsGeoTopic.denseCity`
Rule: Whether the place is a densely populated city.
Google's description: Is this a dense city (e.g., population > 100k)?
type boolean() · confidence low

`RepositoryAnnotationsGeoTopic.latE7`
Rule: The latitude and longitude of the place.
Google's description: Latitude and Longitude of the location.
type integer() · confidence low

`ResearchScienceSearchSourceUrlDocjoinInfo.languageCode`
Rule: The detected language of the document.
Google's description: The language of the document in the string representation of LanguageCode. Converts from Language Enum to LanguageCode through i18n/identifiers/langenclanguagecodeconverter.h Please use i18n/identifiers/languagecodeconverter.h for converting between LanguageCode and string representation.
type String.t · confidence low

Site structure (23)

`CompositeDocIndexingInfo.urlPatternSignals`
Rule: Signals drawn from the address pattern that feed the document's score.
Google's description: UrlPatternSignals for this doc, used to compute document score in LTG (see indexing/signal_aggregator/proto/signal-aggregator.proto for details).
type GoogleApi.ContentWarehouse.V1.Model.IndexingSignalAggregatorUrlPatternSignals.t · confidence medium · stages: 2. Document Processing & Indexing

`ImageData.codomainStrength`
Rule: Confidence that the image is hosted on a domain related to the page showing it.
Google's description: Like is_visible, this is a property of the (web-doc, img_url) pair not just the image. A high codomain_strength indicates high confidence based on collected stats that the image is hosted on a companion domain. If not enough stats are available for codomain strength, this field may be absent in ImageData, and hence the CompositeDoc. Do not place negative values here. Permitted values range between 0 and image_quality_codomain::kMaxCodomainStrength defined in //image/quality/codomain/codomain-sta
type integer() · confidence medium

`IndexingSignalAggregatorAggregatedScore.deviation`
Rule: How spread out the individual URL scores are within the group.
Google's description: A number reflecting the deviation of Url scores.
type float() · confidence medium · stages: 2. Document Processing & Indexing; 11. Re-Ranking & Twiddler System

`IndexingSignalAggregatorAggregatedScore.meanScore`
Rule: The averaged score across all URLs in this group.
Google's description: Overall stats that are only available in final aggregation results. The aggregated score.
type float() · confidence medium · stages: 2. Document Processing & Indexing; 11. Re-Ranking & Twiddler System

`IndexingSignalAggregatorAggregatedScore.numUrlsMatched`
Rule: How many URLs matched this pattern or class.
Google's description: Final Stats that are also available in intermediate output. Number of Urls matching the class.
type String.t · confidence medium · stages: 2. Document Processing & Indexing; 11. Re-Ranking & Twiddler System

`IndexingSignalAggregatorAggregatedScore.numUrlsWithSignal`
Rule: How many of the matching URLs actually had a signal value.
Google's description: Not every matching url has a signal.
type String.t · confidence medium · stages: 2. Document Processing & Indexing; 11. Re-Ranking & Twiddler System

`IndexingSignalAggregatorAggregatedScore.patternsUsedInMediation`
Rule: The list of URL patterns that actually contributed to the combined signal.
Google's description: Optionally populated in mediators. A list of patterns that actually contributed to the final mediated signal.
type list(String.t) · confidence medium · stages: 2. Document Processing & Indexing; 11. Re-Ranking & Twiddler System

`IndexingSignalAggregatorAggregatedScore.percentile`
Rule: Score cut-offs that split the matching URLs into equal-sized buckets.
Google's description: Score percentile of matching urls. If present, it has N entries for buckets of roughly equal number of urls. N is specified the aggregation. The value is the min score in that bucket.
type list(float()) · confidence medium · stages: 2. Document Processing & Indexing; 11. Re-Ranking & Twiddler System

`LocalWWWInfo.siteSiblings`
Rule: Per document counts of sibling pages on the same site, independent of any address.
Google's description: These are per-document signals independent of any particular address.
type integer() · confidence medium · stages: 12. Personalization, Local & Context

`PerDocData.onsiteProminence`
Rule: A measure of how important the page is within its own site, from simulated traffic.
Google's description: Onsite prominence measures the importance of the document within its site. It is computed by propagating simulated traffic from the homepage and high craps click pages. It is a 13-bit int.
type integer() · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityAuthorityTopicEmbeddingsVersionedItem.siteEmbedding`
Rule: A compact numeric summary of what a whole site or page is about.
Google's description: Compressed site/page embeddings.
type String.t · confidence medium

`QualityAuthorityTopicEmbeddingsVersionedItem.siteFocusScore`
Rule: A number for how tightly a site sticks to a single topic.
Google's description: Number denoting how much a site is focused on one topic.
type number() · confidence medium

`QualityAuthorityTopicEmbeddingsVersionedItem.siteRadius`
Rule: How far a site's individual pages stray from the site's main topic.
Google's description: The measure of how far page_embeddings deviate from the site_embedding.
type number() · confidence medium

`QualitySitemapBreadcrumbTargetDoc.count`
Rule: How many web pages list this address in their breadcrumb trails.
Google's description: The number of web pages that contains the url in their breadcrumbs.
type integer() · confidence medium

`WWWDocInfo.nsrSitechunk`
Rule: The site grouping key used by the site-level quality system, sometimes at channel level.
Google's description: Sitechunk used by NSR. For most pages this is equivalent HOST_LEVEL_V3 sitechunk. Main difference is that, for sites like youtube.com and vimeo.com, nsr_sitechunks are channel level (based on schema.org markup, not url alone). See go/nsr-chunks for more details.
type String.t · confidence medium

`IndexingSignalAggregatorAggregatedScore.maxScoreUrl`
Rule: The individual URL that had the highest score in this group.
Google's description: Input UrlScore with max score.
type GoogleApi.ContentWarehouse.V1.Model.IndexingSignalAggregatorUrlScore.t · confidence low · stages: 2. Document Processing & Indexing; 11. Re-Ranking & Twiddler System

`IndexingSignalAggregatorAggregatedScore.minScoreUrl`
Rule: The individual URL that had the lowest score in this group.
Google's description: Input UrlScore with min score.
type GoogleApi.ContentWarehouse.V1.Model.IndexingSignalAggregatorUrlScore.t · confidence low · stages: 2. Document Processing & Indexing; 11. Re-Ranking & Twiddler System

`IndexingSignalAggregatorAggregatedScore.scores`
Rule: The collected list of scores used to work out percentiles.
Google's description: for calculating percentile
type list(float()) · confidence low · stages: 2. Document Processing & Indexing; 11. Re-Ranking & Twiddler System

`IndexingSignalAggregatorAggregatedScore.singleUrlScore`
Rule: Holds the score when this record covers just one URL.
Google's description: If this field presents, it is for a single url. No other field should appear.
type GoogleApi.ContentWarehouse.V1.Model.IndexingSignalAggregatorUrlScore.t · confidence low · stages: 2. Document Processing & Indexing; 11. Re-Ranking & Twiddler System

`IndexingSignalAggregatorAggregatedScore.totalScore`
Rule: A running sum of scores used to compute the mean.
Google's description: summation varaible used to get mean
type float() · confidence low · stages: 2. Document Processing & Indexing; 11. Re-Ranking & Twiddler System

`IndexingSignalAggregatorAggregatedScore.totalWeight`
Rule: A running sum of weights used for weighted averages.
Google's description: for calculating weighted mean/dev
type float() · confidence low · stages: 2. Document Processing & Indexing; 11. Re-Ranking & Twiddler System

`IndexingSignalAggregatorSccData.pattern`
Rule: The closest matching URL-pattern record for this data.
Google's description: The most immediate pattern data.
type GoogleApi.ContentWarehouse.V1.Model.IndexingSignalAggregatorSccSignal.t · confidence low

`IndexingSignalAggregatorUrlScore.weight`
Rule: How much this URL counts when scores are combined.
Google's description: weight for this url.
type float() · confidence low

Crawling & indexing (18)

`CompositeDocIndexingInfo.selectionTierRank`
Rule: A zero to one score placing the document within its serving index tier.
Google's description: Selection tier rank is a language normalized score ranging from 0-1 over the serving tier (Base, Zeppelins, Landfills) for this document.
type number() · confidence medium · stages: 2. Document Processing & Indexing

`GDocumentBase.NoIndexReason`
Rule: Flags recording why a page must not be indexed, snippeted, or cached.
Google's description: When these reasons are set to a non zero value, the document should not be indexed, or show a snippet, or show a cache, etc. These reasons are bit maps of indexing.converter.RobotsInfo.RobotedReasons enum values reflecting the places where the restriction was found: //depot/google3/indexing/converter/proto/converter.proto
type integer() · confidence medium · stages: 2. Document Processing & Indexing

`HtmlrenderWebkitHeadlessProtoDocument.title`
Rule: The title of the rendered page.
Google's description: Document's title.
type String.t · confidence medium

`HtmlrenderWebkitHeadlessProtoRenderResponse.title`
Rule: The page title produced by the renderer.
Google's description: Contains the page title produced by webkit, in the UTF-8 encoding.
type String.t · confidence medium

`HtmlrenderWebkitHeadlessProtoRenderTreeNode.renderedText`
Rule: The actual visible text that was rendered for a text node.
Google's description: The actual text that was rendered. This is applicable only for text nodes.
type String.t · confidence medium · stages: 2. Document Processing & Indexing

`ImageData.corpusSelectionInfo`
Rule: Scoring information about which image collections this image was selected into.
Google's description: Corpus scoring info for images indexed through Amarna.
type list(GoogleApi.ContentWarehouse.V1.Model.CorpusSelectionInfo.t) · confidence medium

`ImageData.lastCrawlTime`
Rule: The most recent time this image address was fetched.
Google's description: Most recent timestamp in seconds when this URL was crawled.
type integer() · confidence medium

`IndexingSignalAggregatorAgeWeightedCoverageData.firstBaseCoverageTimestamp`
Rule: When this URL first appeared in the BASE index coverage.
Google's description: Epoch seconds at which this url first gets coverage in BASE.
type String.t · confidence medium · stages: 2. Document Processing & Indexing

`PerDocData.scaledSelectionTierRank`
Rule: A 0-32767 value giving the document's position within its index serving tier.
Google's description: Selection tier rank is a language normalized score ranging from 0-32767 over the serving tier (Base, Zeppelins, Landfills) for this document. This is converted back to fractional position within the index tier by scaled_selection_tier_rank/32767.
type integer() · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`ScienceIndexSignal.IndexSelectionScore`
Rule: A score for whether the page belongs in the main index or the supplemental one.
Google's description: Index selection score for websearch, bigger is better: (0.5,1.0] - prefer selection into the base index, (0.0,0.5] - prefer selection into the supplemental index.
type number() · confidence medium

`SentenceBoundaryAnnotationsInstance.toIndex`
Rule: Marks which annotations were picked to be indexed.
Google's description: Used to mark the annotations selected to be indexed.
type boolean() · confidence medium

`SmartphonePerDocData.isWebErrorMobileContent`
Rule: Whether the URL errors for the desktop crawler but works for the smartphone crawler.
Google's description: Indicates if the current URL serves error page to desktop crawler and non error page to smartphone crawler.
type boolean() · confidence medium · stages: 12. Personalization, Local & Context

`Anchors.indexTier`
Rule: Which index tier the anchors were pulled from.
Google's description: The index tier from which the anchors were extracted. Note that this is only valid in the anchor record written by linkextractor. The value can be one of the enum values defined in segindexer/types.h.
type integer() · confidence low · stages: 4. Link Analysis & Authority Computation

`CompositeDoc.embeddedContentInfo`
Rule: Information about content and links pulled in from embedded resources.
Google's description: Data produced by the embedded-content system. This is a thin message, containing only embedded_links_info data for the embedder and JavaScript/CSS embedded links (the embedded-content bigtable also contains snapshots, compressed document trees and all embedded link types). Provided using the index signal API.
type GoogleApi.ContentWarehouse.V1.Model.IndexingEmbeddedContentEmbeddedContentInfo.t · confidence low · stages: 2. Document Processing & Indexing

`HtmlrenderWebkitHeadlessProtoDocument.contentLanguage`
Rule: The language the rendered page is written in.
Google's description: Document's language.
type String.t · confidence low

`RepositoryWebrefDocumentMetadata.forwardingUrls`
Rule: Other addresses that redirect to this page.
Google's description: Urls that forward to this url. Needed for url -> topical entity entries.
type GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefForwardingUrls.t · confidence low · stages: 6. Entity & Knowledge Graph Integration

`RepositoryWebrefForwardingUrls.forwardingUrl`
Rule: Addresses that redirect to this page.
Google's description: Urls that forward to this url. Used for url -> topical entity entries.
type list(String.t) · confidence low

`ResearchScienceSearchSourceUrlDocjoinInfo.indexTier`
Rule: Which index tiers the document is stored in.
Google's description: Index tiers (BASE, UNIFIED_ZEPPELIN, etc) that the document belongs to. NOTE: Each document may belong to multiple tiers. NOTE: The original data type is an enum CompositeDoc::SubIndexType. However we don't want to depend on segindexer/compositedoc.proto because the proto is too large. Instead, we use CompositeDoc::SubIndexType_Name(subindexid) to convert into a string representation. To convert string back to CompositeDoc::SubIndexType, use CompositeDoc::SubIndexType_Parse.
type list(String.t) · confidence low

Other (18)

`GDocumentBaseContent.Language`
Rule: The language the page content is written in.
Google's description: A Language enum value. See: go/language-enum Default is english
type integer() · confidence medium

`IndexingMlVerticalVerticalItem.probability`
Rule: How confident the model is that the page belongs to this vertical.
Google's description: The probability of the vertical, whose value is in [0.0, 1.0].
type number() · confidence medium

`IndexingSignalAggregatorUrlScore.score`
Rule: The signal score for this URL, absent when the URL has no signal.
Google's description: score might be missing if the url does not have signal.
type float() · confidence medium

`PerDocData.languages`
Rule: The likely languages of the page, listed most likely first.
Google's description: Plausible languages in order of decreasing plausibility. Language values are small, IE < 127 so this should compress to one byte each.
type list(integer()) · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.rosettaLanguages`
Rule: The page's top two language codes from the Rosetta language annotator.
Google's description: Top two document language BCP-47 codes as generated by the RosettaLanguageAnnotator in the decreasing order of probability.
type list(String.t) · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.saftLanguageInt`
Rule: The page's top language from a second language identifier, stored when it differs.
Google's description: Top document language as generated by SAFT LangID. For now we store bare minimum: just the top 1 language value, converted to the language enum, and only when different from the first value in 'languages'.
type list(integer()) · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`AnchorsAnchorSource.language`
Rule: Language of the linking page.
Google's description: default -> English
type integer() · confidence low · stages: 4. Link Analysis & Authority Computation

`BiasingPerDocData2BiasingField.value`
Rule: The document's biasing value, stored as a decimal number.
Google's description: The value, under various representations to get maximum compression. Exactly one of them is guaranteed to be filled. value as a double.
type float() · confidence low

`BiasingPerDocData2BiasingField.valueFloat`
Rule: The document's biasing value stored as an integer scaled by one thousand.
Google's description: a floating value, represented as an integer by converting using floating_value * 1000. Useable for all floating values that need 3 digits of precision, and are small enough.
type integer() · confidence low

`BiasingPerDocData2BiasingField.valueInt`
Rule: The document's biasing value when it fits in a plain integer.
Google's description: value as an int32. When the value is encode-able as an integer.
type integer() · confidence low

`BiasingPerDocDataBiasingField.Value`
Rule: The document's biasing value converted to a decimal so values compare uniformly.
Google's description: Biasing value translated into a double for uniform comparison
type float() · confidence low

`IndexingMlVerticalVerticalItem.name`
Rule: The name of the predicted vertical.
Google's description: The description name.
type String.t · confidence low

`PerDocData.biasingdata2`
Rule: A more space-efficient container of per-document biasing data.
Google's description: A replacement for BiasingPerDocData that is more space efficient. Once this is live everywhere, biasingdata will be deprecated.
type GoogleApi.ContentWarehouse.V1.Model.BiasingPerDocData2.t · confidence low · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityNsrVersionedFloatSignal.value`
Rule: The decimal value of a signal for one version.
Google's description: The corresponding float value.
type number() · confidence low

`QualityNsrVersionedIntSignal.value`
Rule: The whole-number value of a signal for one version.
Google's description: The corresponding int value.
type String.t · confidence low

`RegistrationInfo.createdDate`
Rule: The day the domain was last registered, counted from January 1995.
Google's description: This is the number of days since January 1st 1995 that this domain was last created. This should always fit in 15 bits.
type integer() · confidence low

`RegistrationInfo.expiredDate`
Rule: The day the domain last expired, counted from January 1995.
Google's description: This is the number of days since January 1st 1995 that this domain last expired. This should always fit in 15 bits. Jan 1st 1995 was chosen by the history project as a special epoch date. Both the registrationinfo dates and the linkage dates are measured in days since this epoch.
type integer() · confidence low

`ScienceCitationUnionCatalog.NumLibraries`
Rule: How many libraries hold this work, noted as useful for ranking.
Google's description: Information about the number of libraries the citation appears. It should be useful for ranking.
type integer() · confidence low

Query intent (17)

`PerDocData.asteroidBeltIntents`
Rule: Stored intent scores for the document from the Asteroid Belt project.
Google's description: For indexing Asteroid Belt intent scores. See go/asteroid-belt for details.
type GoogleApi.ContentWarehouse.V1.Model.QualityOrbitAsteroidBeltDocumentIntentScores.t · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.topPetacatTaxId`
Rule: The site's main topic category, used to check how well a result matches a query.
Google's description: Top petacat of the site. Used in SiteboostTwiddler to determine result/query matching.
type integer() · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityFringeFringeQueryPriorPerDocData.encodedEntityPriorScore`
Rule: The highest chance a query is fringe based on the main entities on the page.
Google's description: Highest entity prior seen for document's Headline and SingleTopic entities (see go/topicality-score for definitions of entity topicalities). Represents probability that a query is fringe, given that the entity is in the result set with topicality >= Headline. Scores scaled to integers between 0 and 1000 for compactness. Scores must be interpreted through FringeQueryPriorEncoding::Decode API.
type integer() · confidence medium

`QualityFringeFringeQueryPriorPerDocData.encodedFringePriorScore`
Rule: The chance that a query is fringe when this page appears in results.
Google's description: Probability that a query is fringe, given this document is in the result set. Scores scaled to integers between 0 and 1000 for compactness. Scores must be interpreted through FringeQueryPriorEncoding::Decode API.
type integer() · confidence medium

`QualityFringeFringeQueryPriorPerDocData.encodedFringeSitePriorScore`
Rule: The chance a query is fringe when this page's site appears in results.
Google's description: Probability that a query is fringe, given this document's site is in the result set. Scores scaled to integers between 0 and 1000 for compactness. Scores must be interpreted through FringeQueryPriorEncoding::Decode API.
type integer() · confidence medium

`QualityOrbitAsteroidBeltDocumentIntentScores.belowThresholdIntents`
Rule: Topic intents found on the page that scored below the triggering threshold.
Google's description: Same as above, but for intents below triggering threshold. It can be assumed that any intent in this list has trigger=false. Most intents do not have below-threshold annotations.
type list(String.t) · confidence medium · stages: 9. Query Understanding & Intent Classification

`QualityOrbitAsteroidBeltDocumentIntentScores.intents`
Rule: The list of intents the document was classified into.
Google's description: The 'intents' and 'scores' fields are stored as parallel lists for compactness. The 'scores' field should not be accessed directly, but instead through the functions in document_intent_scores_utils.
type list(String.t) · confidence medium · stages: 9. Query Understanding & Intent Classification

`QualityOrbitAsteroidBeltDocumentIntentScores.scores`
Rule: The strength of each detected document intent, on a 0 to 100 scale.
Google's description: The intent scores, scaled to integers between 0 and 100 for compactness.
type list(integer()) · confidence medium · stages: 9. Query Understanding & Intent Classification

`RepositoryWebrefUniversalNgramData.idf`
Rule: How rare a word or phrase is across documents.
Google's description: IDF of the n-gram.
type number() · confidence medium

`ImageData.orbitIntents`
Rule: Predicted user intents associated with the image.
Google's description: Orbit intents. More details on go/orbit-in-image-indexing.
type GoogleApi.ContentWarehouse.V1.Model.QualityOrbitOrbitImageIntents.t · confidence low

`NlpSaftDocument.language`
Rule: The language the document is written in.
Google's description: Document language (default is English). This field's value maps cleanly to the i18n.languages.Language proto enum (i18n::languages::Language in C++).
type integer() · confidence low · stages: 9. Query Understanding & Intent Classification

`PerDocData.fringeQueryPrior`
Rule: Encoded prior about fringe queries for this document, owned by the fringe ranking team.
Google's description: Contains encoded FringeQueryPrior information. Unlikely to be meaningful for anyone other than fringe-ranking team. Contact fringe-ranking team if any questions, but do NOT use directly without consulting them.
type GoogleApi.ContentWarehouse.V1.Model.QualityFringeFringeQueryPriorPerDocData.t · confidence low · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityFringeFringeQueryPriorPerDocData.encodedFringeSitePriorScoreForQfsTraining`
Rule: A training-only version of the site fringe prior that leaves out some inputs.
Google's description: Probability that a query is fringe, given this document's site is in the result set. Does not use signals with a dependency on the QueryFringeScore of a document. Scores scaled to integers between 0 and 1000 for compactness. Scores must be interpreted through FringeQueryPriorEncoding::Decode API. Will NOT be present if the fringe_site_prior_score_for_qfs_training is not significantly different from the site_prior_score.
type integer() · confidence low

`QualityViewsExtractionClusterInfo.clusterSetScore`
Rule: How confident the system is in the entity group interpretation.
Google's description: Cluster set qref confidence score.
type number() · confidence low

`QualityViewsExtractionClusterInfo.score`
Rule: How strong this entity is within its group.
Google's description: The score represents the score of the entity within the cluster.
type number() · confidence low

`RepositoryWebrefUniversalNgramData.pluralProb`
Rule: Chance that a phrase is the plural form of a word.
Google's description: Probability that the n-gram is a plural form of a word. This information is extracted from SAFT annotations of queries. See HasPluralProperty().
type number() · confidence low

`VideoContentSearchVideoInfo.navqueries`
Rule: Queries treated as navigational lookups for this video.
Google's description: Navqueries for the video.
type list(String.t) · confidence low

Part III. Contextual ranking signals (1,047)

Fields that shape ranking only in specific contexts: verticals, result types, devices, locales, or special features.

Snippets & display (354)

`CompositeDoc.docinfoPassthroughAttachments`
Rule: Extra data passed through for display only, deliberately kept out of scoring.
Google's description: This message set is used for data pushed into the index using the signals framework that is never to be used in Mustang or TG Continuum scoring/snippeting code. Any protocol buffer stored in this message set is automatically returned in a docinfo response - it ends up in the "info" message set in the WWWSnippetResponse, so it can be used in post-doc twiddlers and for display in GWS with no code changes in Mustang or Teragoogle.
type GoogleApi.ContentWarehouse.V1.Model.Proto2BridgeMessageSet.t · confidence high · stages: 2. Document Processing & Indexing

`ExtraSnippetInfoResponseTidbit.begin`
Rule: Where in the document the snippet fragment starts.
Google's description: For tidbits only: position of tidbit in the document. More specifically, tidbit is found at [begin, end) in the document's tokens.
type integer() · confidence high

`ExtraSnippetInfoResponseTidbit.text`
Rule: The snippet fragment text with search terms already highlighted.
Google's description: the tidbit text, with search terms already highlighted
type String.t · confidence high

`GDocumentBase.DisplayUrl`
Rule: The URL to show in search results when it should differ from the indexed one.
Google's description: Sometimes the URL displayed in search results should be different from what gets indexed (e.g. in enterprise, content management systems). If this value is not set, we default to the regular URL.
type String.t · confidence high · stages: 2. Document Processing & Indexing

`GenericSnippetResponse.snippet`
Rule: The lines of snippet HTML shown beneath the result.
Google's description: Lines of the snippet HTML. Typically gws concatenates these and lets the browser wrap. The values include trailing spaces, so inserting additional spaces is not necessary. However, for very old browsers, gws may insert break tags after each snippet line. This field is confusing and poorly named; "snippet_line" would be better. In particular, note that this does not return multiple snippets for a result. Nor are these fields the individual tidbits of the snippet.
type list(String.t) · confidence high

`GenericSnippetResponse.title`
Rule: The result title HTML, possibly shortened and with query matches marked.
Google's description: The title HTML. It may contain tags to denote query term matches. It may be already truncated and "..." is put instead (note that truncation does not always happen at the very end of the title text). However the existence of "..." does not guarantee that the snippet generation algorithm truncated it; e.g. webmasters themselves can write "...".
type String.t · confidence high

`IndexingConverterRobotsInfo.maxImagePreview`
Rule: The largest image preview size the page allows.
Google's description: Max image preview restriction applied to this data. A value of THUMBNAIL_UNSPECIFIED can be treated as though there is no restriction.
type String.t · confidence high

`IndexingConverterRobotsInfo.maxSnippetLength`
Rule: The maximum snippet length the page allows.
Google's description: Max snippet preview restriction applied to this data. If this field is zero, it indicates that no snippet data can be displayed, therefore this field should be checked using has_max_snippet_length to determine if it was set.
type String.t · confidence high

`ListSnippetResponse.isTable`
Rule: Whether a list snippet should be laid out as a table.
Google's description: Should list be formatted as a table?
type boolean() · confidence high

`ListSnippetResponse.totalRows`
Rule: How many rows were found in the page's list, of which the shown rows are a subset.
Google's description: The number of rows annotated in the doc, of which 'row' is a subset.
type integer() · confidence high

`LongStructuredSnippetEntry.header`
Rule: Whether this piece of a structured snippet is a heading or a paragraph.
Google's description: Is this a header or normal paragraph?
type boolean() · confidence high

`LongStructuredSnippetEntry.text`
Rule: The wording shown in this part of a structured snippet.
Google's description: The text of the header or paragraph.
type String.t · confidence high

`MustangReposWwwSnippetsOrganicListSnippetResponse.header`
Rule: The header text and list item text pulled from a page for a list-style snippet.
Google's description: The texts of header and listing items.
type String.t · confidence high

`MustangReposWwwSnippetsOrganicListSnippetResponse.headerUsedInSnippet`
Rule: Whether the extracted header was actually used in the shown snippet.
Google's description: If the header being used in organic snippet.
type boolean() · confidence high

`MustangReposWwwSnippetsSnippetsRanklabFeatures.browserWidth`
Rule: The width of the user's browser, recorded with snippet features.
Google's description: Browser width.
type integer() · confidence high

`MustangReposWwwSnippetsSnippetsRanklabFeatures.candidates`
Rule: Recorded features for the snippet candidates produced by the snippet scorers.
Google's description: Features for snippets candidates, generated by both old and new scorer. Currently only features for chosen candidate is generated.
type list(GoogleApi.ContentWarehouse.V1.Model.MustangReposWwwSnippetsSnippetCandidate.t) · confidence high

`MustangReposWwwSnippetsSnippetsRanklabFeatures.displaySnippet`
Rule: Recorded features for the snippet that was finally displayed.
Google's description: Snippet features for the final chosen snippet. This field is firstly populated by Muppet, and then overwriten by Superroot if SnippetBrain is triggered.
type GoogleApi.ContentWarehouse.V1.Model.QualityPreviewRanklabSnippet.t · confidence high

`MustangReposWwwSnippetsSnippetsRanklabFeatures.snippets`
Rule: Recorded features for each snippet candidate considered.
Google's description: Snippet features for Muppet snippet candidates. In production, only the data for chosen snippet will be recorded.
type list(GoogleApi.ContentWarehouse.V1.Model.QualityPreviewRanklabSnippet.t) · confidence high

`MustangReposWwwSnippetsSnippetsRanklabFeatures.titles`
Rule: Recorded features for each title candidate, best first, with the chosen one first.
Google's description: Per-candidate title features for ranklab models, sorted from the best candidate to the worst candidate (i.e., the first element is the actually selected title).
type list(GoogleApi.ContentWarehouse.V1.Model.QualityPreviewRanklabTitle.t) · confidence high

`MustangSnippetsRenderedToken.bolded`
Rule: Whether this word is shown in bold in the rendered snippet.
Google's description: Is the rendered token bolded (insided)
type boolean() · confidence high

`PerDocData.noimageframeoverlayreason`
Rule: A reason code saying the image should not be shown in overlay mode in image snippets.
Google's description: If not 0, we should not show the image in overlay mode in image snippets
type integer() · confidence high · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityDniDocPreviewRestrictions.faviconDisplay`
Rule: Whether the site's favicon may be shown next to the result.
Google's description: Whether the favicon for a given domain should be displayed. FAVICON_DISPLAY_UNSPECIFIED - Display the favicon DISABLE_FAVICON - the favicon should not be rendered by the feature
type String.t · confidence high

`QualityDniDocPreviewRestrictions.maxSnippetLength`
Rule: The longest snippet allowed for this page, based on its markup.
Google's description: The max number of snippet characters allowed. Based on document markup. No limit if value is less than 0, Google could use any length of snippets. Default value 0 is the strictest restriction, to avoid violating mistakenly If not set, there is no snippet length policy to enforce. Features must first check has_max_snippet_length to avoid applying an overly strict policy.
type integer() · confidence high

`QualityDniDocPreviewRestrictions.maxSnippetLengthFromPublisher`
Rule: Preview size limits taken from the publisher's Search Console settings.
Google's description: Same as the max_snippet_length, max_thumbnail_size, max_video_preview_secs listed above. But values are based on publisher's preferences from Search Console's robots meta tag tool.
type integer() · confidence high

`QualityDniDocPreviewRestrictions.maxThumbnailSize`
Rule: The largest thumbnail allowed for this page.
Google's description: The max thumbnail size allowed. Based on document markup Default value NONE is the strictest restriction, to avoid violating mistakenly. If not set, there is no thumbnail policy to enforce.
type String.t · confidence high

`QualityDniDocPreviewRestrictions.maxVideoPreviewSecs`
Rule: The longest video preview allowed for this page.
Google's description: The max seconds of video preview allowed. Based on document markup. No limit if value is less than 0, Google could show any seconds of video. Default value 0 is the strictest restriction, to avoid violating mistakenly If not set, there is no preview length policy to enforce. Features must first check has_max_video_preview_secs to avoid applying an overly strict policy.
type integer() · confidence high

`QualityPreviewChosenSnippetInfo.snippetHtml`
Rule: The finished HTML of the snippet shown under the result.
Google's description: The rendered snippet html.
type String.t · confidence high

`QualityPreviewChosenSnippetInfo.source`
Rule: Which snippet generator produced the chosen snippet.
Google's description: Source of the chosen snippet, decided in PORC. String value of quality.porc.TextSnippetCandidate.TextSnippetSource defined at google3/quality/porc/proto/text_snippet.proto
type String.t · confidence high

`QualityPreviewChosenSnippetInfo.trailingEllipsis`
Rule: Whether the snippet ends with an ellipsis.
Google's description: Whether this snippet has trailing ellipsis.
type boolean() · confidence high

`QualityPreviewChosenSnippetInfoTidbitInfo.separator`
Rule: The characters placed before this snippet fragment.
Google's description: Separator to put before this tidbit.
type String.t · confidence high

`QualityPreviewRanklabTitle.dataSourceType`
Rule: Where the title candidate came from.
Google's description: Title source type.
type String.t · confidence high · stages: 10. Snippet & Answer Extraction

`QualityPreviewRanklabTitle.dupTokens`
Rule: How many repeated words the title contains.
Google's description: Numbers of duplicated tokens. For example, duplicated tokens for a title "dog cat cat cat" is 2 (for 2 extra "cat").
type integer() · confidence high · stages: 10. Snippet & Answer Extraction

`QualityPreviewRanklabTitle.hasSiteInfo`
Rule: Whether the title includes the site's name.
Google's description: Whether a title contains site information.
type boolean() · confidence high · stages: 10. Snippet & Answer Extraction

`QualityPreviewRanklabTitle.isTruncated`
Rule: Whether the title candidate was cut short.
Google's description: Whether this title candidate is truncated or not.
type boolean() · confidence high · stages: 10. Snippet & Answer Extraction

`QualityPreviewRanklabTitle.isValid`
Rule: Whether the title candidate is non-empty and usable.
Google's description: Whether a title is valid (i.e., not empty).
type boolean() · confidence high · stages: 10. Snippet & Answer Extraction

`QualityPreviewRanklabTitle.widthFraction`
Rule: How wide the rendered title is compared to the maximum width allowed for a title.
Google's description: A rendered width of this title divided by the max allowed width for title. Takes values in [0, 1].
type number() · confidence high · stages: 10. Snippet & Answer Extraction

`QualityPreviewSnippetBrainFeatures.isSnippetBrainBoldingTriggered`
Rule: Whether bolding was applied inside the generated snippet.
Google's description: Is the bolding triggered.
type boolean() · confidence high

`QualityPreviewSnippetBrainFeatures.modelScore`
Rule: The score a snippet-generation model gave to a candidate snippet.
Google's description: The score by SnippetBrain model.
type number() · confidence high

`QualityPreviewSnippetRadishFeatures.passageCoverage`
Rule: Share of words shared between the source passage and the candidate snippet.
Google's description: The ratio of overlapping tokens between the radish passage and snippet candidate.
type number() · confidence high · stages: 10. Snippet & Answer Extraction

`QualitySnippetsTruncationSnippetBoldedRange.begin`
Rule: Where the bolded stretch of snippet text starts.
Google's description: Bolded range [begin, end)
type GoogleApi.ContentWarehouse.V1.Model.QualitySnippetsTruncationSnippetBoldedRangePosition.t · confidence high

`ResearchScienceSearchCitation.safeHtmlCitation`
Rule: A cleaned-up citation string prepared only for showing in the interface.
Google's description: The citation after HTML sanitation. Used only by the front-end.
type GoogleApi.ContentWarehouse.V1.Model.WebutilHtmlTypesSafeHtmlProto.t · confidence high

`ResearchScienceSearchSourceUrlDocjoinInfo.displayUrl`
Rule: The web address shown to users in the search result.
Google's description: The url used to display in the google search results.
type String.t · confidence high

`ScienceCitationDownloadURL.DisplayOrg`
Rule: The publisher name to show for this URL.
Google's description: publisher display name
type String.t · confidence high

`ScienceCitationDownloadURL.NoArchive`
Rule: A page request not to show a cached copy.
Google's description: metatag: don't show cached version
type boolean() · confidence high

`ScienceCitationDownloadURL.NoSnippet`
Rule: A page request not to show a snippet.
Google's description: metatag: don't show snippet
type boolean() · confidence high

`SdrPageAnchorsSitelink.text`
Rule: The heading or reworded text shown for the jump link.
Google's description: Heading/Reformulated text is needed to display.
type String.t · confidence high

`SdrScrollTo.prefix`
Rule: Text just before the target passage, used to pick the right match on the page.
Google's description: Prefix to help with disambiguating between multiple text matches on page. Optional.
type String.t · confidence high

`SdrScrollTo.suffix`
Rule: Text just after the target passage, used to pick the right match on the page.
Google's description: Suffix to help with disambiguating between multiple text matches on page. Optional.
type String.t · confidence high

`SdrScrollTo.textEnd`
Rule: Where the highlighted passage ends.
Google's description: End of the text span to be highlighted. Optional.
type String.t · confidence high

`SdrScrollTo.textStart`
Rule: Where the highlighted passage starts.
Google's description: Start of the text span to be highlighted.
type String.t · confidence high

`Sitemap.TargetGroups`
Rule: Groups of sitelink targets, of which only one is picked to show the user.
Google's description: One Sitemap can contain multiple TargetGroups, but only one of them will be displayed to the user - this decision will be made at displaying time and can take into account various factors, such as the users' language and country, currently running experiments, etc.
type list(GoogleApi.ContentWarehouse.V1.Model.QualitySitemapTargetGroup.t) · confidence high

`SnippetExtraInfo.candidateInfo`
Rule: The list of possible snippets considered for a result, ordered by their id.
Google's description: Candidates are ordered by their id.
type list(GoogleApi.ContentWarehouse.V1.Model.SnippetExtraInfoSnippetCandidateInfo.t) · confidence high · stages: 10. Snippet & Answer Extraction

`SnippetExtraInfo.snippetBrainSelectedCandidateIndex`
Rule: Which candidate snippet the SnippetBrain model picked.
Google's description: Snippet candidate index selected by snippet brain model. This field will get populated in SnippetFlow in superroot. go/snippets-brain
type integer() · confidence high · stages: 10. Snippet & Answer Extraction

`SnippetExtraInfoSnippetCandidateInfo.boldedRanges`
Rule: Which parts of the snippet text get shown in bold.
Google's description: Bolded ranges in the printed snippet lines.
type list(GoogleApi.ContentWarehouse.V1.Model.QualitySnippetsTruncationSnippetBoldedRange.t) · confidence high · stages: 10. Snippet & Answer Extraction

`SnippetExtraInfoSnippetCandidateInfo.sentenceStarts`
Rule: Where each sentence begins inside the snippet text.
Google's description: Sentence starting positions in the printed snippet lines.
type list(GoogleApi.ContentWarehouse.V1.Model.QualitySnippetsTruncationSnippetBoldedRangePosition.t) · confidence high · stages: 10. Snippet & Answer Extraction

`SnippetExtraInfoSnippetCandidateInfo.snippet`
Rule: The actual lines of snippet text for this candidate.
Google's description: Muppet fills snippet lines in
type list(String.t) · confidence high · stages: 10. Snippet & Answer Extraction

`TitleSizeParams.muppetTitleLengthInDeciems`
Rule: Maximum length allowed for the title shown in results.
Google's description: Total max length of title in deciems
type integer() · confidence high

`TitleSizeParams.muppetTitleNumLines`
Rule: How many lines the displayed result title may use.
Google's description: Number of lines for title
type integer() · confidence high

`VideoContentSearchVideoAnchor.destinationUrl`
Rule: The link used when someone clicks a key moment, jumping to that point in the video.
Google's description: When set, this is the link that should be used when clicking on a video anchor. This should jump to the given time in the video.
type String.t · confidence high

`VideoContentSearchVideoAnchor.label`
Rule: The text shown for a video key moment.
Google's description: The text label of the video anchor.
type String.t · confidence high

`VideoContentSearchVideoAnchor.thumbnail`
Rule: Details of the thumbnail image to show for this key moment.
Google's description: Data about the thumbnail to display for this anchor.
type GoogleApi.ContentWarehouse.V1.Model.VideoContentSearchAnchorThumbnail.t · confidence high

`VideoContentSearchVideoAnchor.thumbnailUrl`
Rule: Address of the video frame image shown for this key moment.
Google's description: The url for a frame to display for this anchor.
type String.t · confidence high

`VideoContentSearchVideoAnchors.shouldServeThumbnails`
Rule: Whether thumbnails should be displayed alongside the key moments.
Google's description: Whether or not thumbnails should be displayed when serving anchors.
type boolean() · confidence high

`WWWDocInfo.cropData`
Rule: Instructions for how to crop the thumbnail image.
Google's description: Thumbnail cropping information.
type integer() · confidence high

`WWWDocInfo.noimageframeoverlayreason`
Rule: A reason code saying the image should not be shown in overlay mode.
Google's description: If not 0, we should not show the image in overlay mode in image snippets.
type integer() · confidence high

`WWWDocInfo.seenNopreview`
Rule: Whether the page carries a nopreview robots tag.
Google's description: Has nopreview meta robots flag
type boolean() · confidence high

`WWWDocInfo.seenNosnippet`
Rule: Whether the page carries a nosnippet robots tag.
Google's description: Has nosnippet meta robots flag
type boolean() · confidence high

`WWWDocInfo.seenNotranslate`
Rule: Whether the page carries a notranslate robots tag.
Google's description: Has notranslate meta robots flag
type boolean() · confidence high

`WWWResultInfoSubImageDocInfo.deepCropBytes`
Rule: Hints for how to crop the thumbnail using deep cropping.
Google's description: Deepcrop thumbnail cropping hints.
type String.t · confidence high

`WWWSnippetResponse.chosenBodyTidbits`
Rule: The text ranges from the page body picked to build the snippet.
Google's description: Tidbits chosen from the document body. Consists of repeated [begin, end) half-open ranges in token offsets from the beginning of the document.
type list(integer()) · confidence high · stages: 10. Snippet & Answer Extraction

`WWWSnippetResponse.docPreviewRestrictions`
Rule: Rules limiting how much of the page can be previewed.
Google's description: DocPreviewRestrictions for canonical url.
type GoogleApi.ContentWarehouse.V1.Model.QualityDniDocPreviewRestrictions.t · confidence high · stages: 10. Snippet & Answer Extraction

`WWWSnippetResponse.docPreviewRestrictionsForAmp`
Rule: Rules limiting how much of the AMP version of the page can be previewed.
Google's description: DocPreviewRestrictions for amp result.
type GoogleApi.ContentWarehouse.V1.Model.QualityDniDocPreviewRestrictions.t · confidence high · stages: 10. Snippet & Answer Extraction

`WWWSnippetResponse.info`
Rule: A bundle of extra result data such as sitelinks, rich snippets, and breadcrumbs.
Google's description: Additional data. Currently, this is used for sitelinks, localinfo, manybox, discussion metadata, richsnippets, similarpages and breadcrumbs.
type GoogleApi.ContentWarehouse.V1.Model.Proto2BridgeMessageSet.t · confidence high · stages: 10. Snippet & Answer Extraction

`WWWSnippetResponse.listSnippet`
Rule: Snippet data formatted as a list.
Google's description: List snippet data.
type GoogleApi.ContentWarehouse.V1.Model.ListSnippetResponse.t · confidence high · stages: 10. Snippet & Answer Extraction

`WWWSnippetResponse.listSummary`
Rule: A summary phrase used for pages that are lists.
Google's description: List summary phrase for list pages.
type String.t · confidence high · stages: 10. Snippet & Answer Extraction

`WWWSnippetResponse.longStructuredSnippet`
Rule: A longer structured snippet that replaces the normal snippet when present.
Google's description: Returned if want_long_structured_snippets. If present, caller should ignore the normal snippet.
type GoogleApi.ContentWarehouse.V1.Model.LongStructuredSnippet.t · confidence high · stages: 10. Snippet & Answer Extraction

`WWWSnippetResponse.sectionHeadingAnchorName`
Rule: The matched table-of-contents entry and the on-page anchor it links to.
Google's description: If requested, the snippet generator may take note of query items present in an entry in an on-page table-of-contents (i.e. a series of on-page links to named anchors.) If so, these two fields contain the formatted and highlighted entry and the name of the on-page anchor it links to, respectively. This may be used by GWS to show a direct link to that named anchor on the page.
type String.t · confidence high · stages: 10. Snippet & Answer Extraction

`WWWSnippetResponse.snippetHighlightMatchesBitmap`
Rule: Bitmaps marking query term matches inside each highlighted snippet fragment.
Google's description: List of bitmaps representing matches to leaf query terms within each of the highlighted snippet text fragments. Consecutive bitmaps correspond to consecutive text fragments. It gets populated iff return_query_snippet_highlight_matches = true. Example: document body section: "This cafe has pet friendly patio." squery: (a (o dog :o pet :syn:general) friendly :o (o restaurant :o cafe :syn:general)) Let's assume the returned snippet text contains the whole document body section where two f
type list(GoogleApi.ContentWarehouse.V1.Model.WWWSnippetResponseBitmapPB.t) · confidence high · stages: 10. Snippet & Answer Extraction

`WWWSnippetResponse.snippetPageNumber`
Rule: The page number where the snippet starts, for paged documents like PDFs.
Google's description: If requested the page number on which the snippet begins. (Only for documents such as PDFs where page numbers are well-defined.)
type integer() · confidence high · stages: 10. Snippet & Answer Extraction

`WWWSnippetResponse.snippetPrefixCharCount`
Rule: How many characters the snippet prefix takes up.
Google's description: Character counts of snippet prefix, if any. E.g. section heading, list summary, byline date.
type integer() · confidence high · stages: 10. Snippet & Answer Extraction

`WWWSnippetResponse.snippetRenderedToken`
Rule: How each token was rendered when building the snippet.
Google's description: How tokens are rendered in generating snippet.
type list(GoogleApi.ContentWarehouse.V1.Model.MustangSnippetsRenderedToken.t) · confidence high · stages: 10. Snippet & Answer Extraction

`WWWSnippetResponse.titleLengthAdjustedForBrowserWidth`
Rule: Whether the title was already shortened to fit the browser width.
Google's description: True if the title length is already adjusted for the browser width. If it is true, GWS needs not truncate the title.
type boolean() · confidence high · stages: 10. Snippet & Answer Extraction

`WWWSnippetResponse.titleRenderedToken`
Rule: How each token was rendered when building the title.
Google's description: How tokens are rendered in generating title. Note: In rendering a title, the page title part and the site/host/domain title part can be flipped after initial rendering. The flip, if happened, may not be reflected in this field. That is, this field may contain the tokens in the original, pre-flip, order.
type list(GoogleApi.ContentWarehouse.V1.Model.MustangSnippetsRenderedToken.t) · confidence high · stages: 10. Snippet & Answer Extraction

`WWWSnippetResponse.truncatedTitle`
Rule: A shortened version of the title.
Google's description: Only for desktop web search. Please refer to Title.keep_original_title_and_populate_truncated_one for more details.
type String.t · confidence high · stages: 10. Snippet & Answer Extraction

`BlogPerDocData.convTree`
Rule: The reply structure used to show a conversation view of posts.
Google's description: For the threaded conversation view. Only populated in docs with provider type SYNTHETIC_CONVERSATION_DOC.
type GoogleApi.ContentWarehouse.V1.Model.BlogsearchConversationTree.t · confidence medium

`BlogsearchConversationNode.authorName`
Rule: The username of the author of a post in a conversation thread.
Google's description: The username of the author of the microblog post represented by this node.
type String.t · confidence medium

`BlogsearchConversationNode.children`
Rule: The document ids of replies hanging under this post.
Google's description: A list of docids of child nodes.
type list(String.t) · confidence medium

`BlogsearchConversationNode.parent`
Rule: The document id of the post this one is replying to.
Google's description: The docid of the parent node. The root of the tree will leave this empty.
type String.t · confidence medium

`BlogsearchConversationTree.nodes`
Rule: The set of posts that make up a conversation thread.
Google's description: The nodes in this conversation. No particular order is assumed.
type list(GoogleApi.ContentWarehouse.V1.Model.BlogsearchConversationNode.t) · confidence medium

`CompositeDoc.richsnippet`
Rule: Structured rich snippet data pulled out of the page's content.
Google's description: rich snippet extracted from the content of a document.
type GoogleApi.ContentWarehouse.V1.Model.RichsnippetsPageMap.t · confidence medium · stages: 2. Document Processing & Indexing

`CrowdingPerDocDataNewsCluster.ClusterSize`
Rule: How many related articles are in the cluster, shown in the related link.
Google's description: This is the X in the "and X related >>" link on headlines and search results
type integer() · confidence medium

`DocProperties.leadingtext`
Rule: Information about the text leading into the page's content, from snippet code.
Google's description: Leading text information generated by google3/quality/snippets/leadingtext/leadingtext-detector.cc
type GoogleApi.ContentWarehouse.V1.Model.SnippetsLeadingtextLeadingTextInfo.t · confidence medium

ExtraSnippetInfoResponseMatchInfo.titleMatches
Rule: Bit flags marking which query terms appear in the title.
Google's description: bitvector of query items matching the title
type String.t · confidence medium

ExtraSnippetInfoResponseMatchInfo.urlMatches
Rule: Bit flags marking which query terms appear in the URL.
Google's description: bitvector of query items matching the url
type String.t · confidence medium

ExtraSnippetInfoResponseMatchInfo.weightedItems
Rule: Bit flags marking which query terms the snippet chooser considered.
Google's description: bitvector of query items considered by chooser
type String.t · confidence medium

ExtraSnippetInfoResponseQuerySubitem.isHighlighted
Rule: Whether this query term was highlighted in the snippet.
Google's description: Additional information from the SnippetQuery.
type boolean() · confidence medium

ExtraSnippetInfoResponseQuerySubitem.isOriginal
Rule: Whether this piece was a real query term rather than an added variant.
Google's description: true iff this subitem was an original query term or phrase. Can only be false if want_all_query_subitems == true in the request.
type boolean() · confidence medium

ExtraSnippetInfoResponseQuerySubitem.items
Rule: Bit flags identifying which query terms this piece covers.
Google's description: a bitvector of the query items corresponding to this subitem. Typically only one bit is set, but see comment above.
type integer() · confidence medium

ExtraSnippetInfoResponseQuerySubitem.text
Rule: The text of this query term.
Google's description: text associated with this query item
type String.t · confidence medium

ExtraSnippetInfoResponseQuerySubitem.weight
Rule: How much weight the snippet code gave this query term.
Google's description: the weight of this query item, as calculated by SubitemWeight():
type integer() · confidence medium

ExtraSnippetInfoResponseTidbit.items
Rule: Bit flags for which query terms appear in this snippet fragment.
Google's description: a bitvector of each query term within this tidbit
type String.t · confidence medium

ExtraSnippetInfoResponseTidbit.score
Rule: A score for this snippet fragment, comparable only to similar fragments.
Google's description: the score for this tidbit if there was one this is returned for Snippets and Tidbits and is only meaningful for comparing between objects of the same type (snippet to snippet, tidbit to tidbit)
type number() · confidence medium

ImageBaseThumbnailMetadata.byteSize
Rule: How many bytes the stored thumbnail image takes up.
Google's description: the size of the stored thumbnail
type integer() · confidence medium

ImageBaseThumbnailMetadata.expirationTimestampMicros
Rule: When the stored thumbnail is allowed to be dropped from serving.
Google's description: expiration timestamp of thumbnail
type String.t · confidence medium

ImageBaseThumbnailMetadata.height
Rule: The height in pixels of the stored thumbnail.
Google's description: the height of the stored thumbnail
type integer() · confidence medium

ImageBaseThumbnailMetadata.mimeType
Rule: The file type of the stored thumbnail, such as JPEG or PNG.
Google's description: The mime_type of the thumbnail ("image/jpeg", "image/png", etc.).
type String.t · confidence medium

ImageBaseThumbnailMetadata.width
Rule: The width in pixels of the stored thumbnail.
Google's description: the width of the stored thumbnail
type integer() · confidence medium

ImageBaseVideoPreviewMetadata.byteSize
Rule: How many bytes the stored video preview clip takes up.
Google's description: Size of the stored preview.
type integer() · confidence medium

ImageBaseVideoPreviewMetadata.duration
Rule: How long the video preview clip runs, in milliseconds.
Google's description: TODO (yzliu): consider using duration_ms as field name since it is number of milliseconds. Duration of the preview in ms.
type integer() · confidence medium

ImageBaseVideoPreviewMetadata.expirationTimestampMicros
Rule: When the stored video preview is allowed to be dropped from serving.
Google's description: Expiration timestamp of preview in microseconds since epoch.
type String.t · confidence medium

ImageBaseVideoPreviewMetadata.height
Rule: The height in pixels of the stored video preview.
Google's description: Height of the stored preview.
type integer() · confidence medium

ImageBaseVideoPreviewMetadata.mimeType
Rule: The file type of the stored video preview, such as MP4.
Google's description: Mime type of the preview ("video/mp4").
type String.t · confidence medium

ImageBaseVideoPreviewMetadata.timestamp
Rule: Where in the source video the preview clip starts, in milliseconds.
Google's description: Timestamp of start of preview in ms.
type integer() · confidence medium

ImageBaseVideoPreviewMetadata.videoSegmentEndUs
Rule: The end point in the source video of the segment this preview covers.
Google's description: The end timestamp of the video segment in microseconds that this preview is generated from. Used for segmented video previews.
type String.t · confidence medium

ImageBaseVideoPreviewMetadata.videoSegmentStartUs
Rule: The start point in the source video of the segment this preview covers.
Google's description: The start timestamp of the video segment in microseconds that this preview is generated from. Used for segmented video previews.
type String.t · confidence medium

ImageBaseVideoPreviewMetadata.width
Rule: The width in pixels of the stored video preview.
Google's description: Width of the stored preview.
type integer() · confidence medium

ImageData.crops
Rule: Suggested crop areas so the image can be shown well at small sizes.
Google's description: The content-aware cropping information.
type GoogleApi.ContentWarehouse.V1.Model.ContentAwareCropsIndexing.t · confidence medium

ImageData.deepCrop
Rule: Model-based suggested crop areas for showing the image.
Google's description: DeepCrop based cropping information. See go/creatism/deepcrop for details.
type GoogleApi.ContentWarehouse.V1.Model.DeepCropIndexing.t · confidence medium

ImageData.expirationTimestamp
Rule: How long the thumbnail is guaranteed to stay available for serving.
Google's description: The thumbnail is guaranteed to be kept in the serving system until the expiration timestamp has passed, in microseconds.
type String.t · confidence medium

ImageData.imageLicenseInfo
Rule: Licensing details telling users how they may legally reuse the image.
Google's description: Indicates license info of this image, which will tell image search users how to use this image legally.
type GoogleApi.ContentWarehouse.V1.Model.ImageSearchImageLicenseInfo.t · confidence medium

ImageData.licensedWebImagesOptInState
Rule: Whether the site owner opted this image into uses such as large previews.
Google's description: Indicates the web-master opt-in state of this image, and will be used for Google products to decide usage rights like showing large previews.
type String.t · confidence medium

`ImageData.thumbWidth`
Rule: The width of the thumbnail shown for this image.
Google's description: Thumbnail width.
type integer() · confidence medium

`ImageDataThumbnail.expirationTimestampMicros`
Rule: How long the thumbnail is guaranteed to stay available for serving.
Google's description: The thumbnail is guaranteed to be kept in the serving system until the expiration timestamp has passed, in microseconds.
type String.t · confidence medium

`ImageExifIPTCMetadata.acquireLicensePage`
Rule: A link to the page where a license for the image can be bought.
Google's description: Page URL about how to acquire this licensable image.
type String.t · confidence medium

`ImageExifIPTCMetadata.licenseUrl`
Rule: A link to the terms describing how the image may be distributed.
Google's description: License URL about how to distribute the image.
type String.t · confidence medium

`ImageRegionsImageRegions.has300kThumb`
Rule: Whether a 300k-size thumbnail exists for the image.
Google's description: True if the image has a 300k thumb.
type boolean() · confidence medium

`ImageRepositoryContentBasedVideoMetadata.videoTranscriptAnnotations`
Rule: Sentence-split video transcript with timing, used to build video answers.
Google's description: VideoTranscriptAnnotations holds sentence segmented text and timing information to be used for VideoAnswers (go/video-answers). Note that only punctuated_transcript, timing_info, and lang field are filled, and other fields will be filled in the later stage.
type GoogleApi.ContentWarehouse.V1.Model.QualityWebanswersVideoTranscriptAnnotations.t · confidence medium

`ImageRepositoryVideoPreviewsVideoPreview.content`
Rule: The actual bytes of the short preview clip generated for the video.
Google's description: The actual video preview bytes generated for the video.
type String.t · confidence medium

`ImageRepositoryVideoPreviewsVideoPreview.metadata`
Rule: Metadata describing the type and size of a video preview.
Google's description: The metadata associated with the preview (i.e. the type: 8k, 90k, etc.)
type GoogleApi.ContentWarehouse.V1.Model.ImageBaseVideoPreviewMetadata.t · confidence medium

`ImageRepositoryVideoPreviewsVideoPreview.previewFrameZero`
Rule: The first still frame taken from the video preview.
Google's description: 0th frame image of the preview. This frame has the same resolution as the associated preview video bytes, as it is taken directly from the preview bytes in Venom/Viper processing. Right now, we only populate preview_frame_zero only for the VPREVIEW_TYPE_540K_ORIGINAL_HQ_LICENSED preview type. Note that preview_frame_zero.thumbnails(0).timestamp_ms() is the timestamp from the full video, not from the preview.
type GoogleApi.ContentWarehouse.V1.Model.DrishtiVesperVideoThumbnail.t · confidence medium

`ImageSearchImageLicenseInfo.acquireLicensePage`
Rule: Page address explaining how to license the image.
Google's description: Records web page url about how to use the licensed image.
type String.t · confidence medium

`ImageSearchImageLicenseInfo.copyrightNotice`
Rule: Copyright notice text recorded for the image.
Google's description: Provides copyright info.
type String.t · confidence medium

`ImageSearchImageLicenseInfo.creditText`
Rule: Text used to credit the people or organizations behind the image.
Google's description: Text for crediting persons or organizations.
type String.t · confidence medium

`IndexingEmbeddedContentEmbeddedContentInfo.isAlternateSnapshot`
Rule: Whether the screenshot came from an alternate snapshot.
Google's description: Indicate if the snapshot is generated from alternate snapshot. If true, the snapshot will be exported even if the snapshot quality score is low.
type boolean() · confidence medium

`IndexingEmbeddedContentEmbeddedContentInfo.renderedSnapshotImage`
Rule: The screenshot image of the rendered page.
Google's description: Snapshot image of a rendered html document (possibly encoded as png, jpeg, or webp).
type String.t · confidence medium

`IndexingEmbeddedContentEmbeddedContentInfo.renderedSnapshotQualityScore`
Rule: How complete the rendered screenshot is, from 0 to 1.
Google's description: The quality of the image, 0.0 is the worst, 1.0 is the best. If all dependencies are successfully crawled, the quality should be 1.0. If one or more of the dependencies are unknown, the quality will be lower.
type float() · confidence medium

`LocalWWWInfoCluster.makePlusboxVisible`
Rule: A hint telling the front end whether to show the expandable local box.
Google's description: A hint for frontend to decide whether this plusbox should be visible or not.
type boolean() · confidence medium · stages: 12. Personalization, Local & Context

`MustangReposWwwSnippetsOrganicListSnippetResponse.headerTitleRedundancy`
Rule: How much of the snippet header text is already covered by the page title.
Google's description: The ratio of header tokens covered by title.
type number() · confidence medium

`MustangReposWwwSnippetsOrganicListSnippetResponse.originalTotalItems`
Rule: How many items the original list on the page contained.
Google's description: The number of items in the original list.
type integer() · confidence medium

`MustangReposWwwSnippetsSnippetCandidate.dataSourceType`
Rule: Which source the snippet candidate text was taken from.
Google's description: data_source_type corresponds to the ChosenSnippet::SnippetType enum.
type integer() · confidence medium

`MustangReposWwwSnippetsSnippetsRanklabFeatures.documentLanguage`
Rule: The language or locale of the document, logged alongside snippet features.
Google's description: locale of the document.
type String.t · confidence medium

`MustangReposWwwSnippetsSnippetsRanklabFeatures.originalQueryTermCoverages`
Rule: How much of the original query wording appears in the title and snippet.
Google's description: Original query term coverage in titles and / or snippets.
type GoogleApi.ContentWarehouse.V1.Model.QualityPreviewSnippetQueryTermCoverageFeatures.t · confidence medium

`MustangReposWwwSnippetsSnippetsRanklabFeatures.queryLanguage`
Rule: The language or locale of the query, logged alongside snippet features.
Google's description: locale of the query,
type String.t · confidence medium

`MustangReposWwwSnippetsSnippetsRanklabFeatures.snippetDataSourceType`
Rule: Which source the displayed snippet came from.
Google's description: Snippet data source.
type integer() · confidence medium

`MustangReposWwwSnippetsSnippetsRanklabFeatures.snippetQueryTermCoverage`
Rule: How much of the query wording appears in the snippet text.
Google's description: Query term coverage in snippets.
type number() · confidence medium

`MustangReposWwwSnippetsSnippetsRanklabFeatures.titleDataSourceType`
Rule: Which source the displayed title came from.
Google's description: Title data source.
type integer() · confidence medium

`MustangReposWwwSnippetsSnippetsRanklabFeatures.titleQueryTermCoverage`
Rule: How much of the query wording appears in the title.
Google's description: Query term coverage in titles.
type number() · confidence medium

`MustangReposWwwSnippetsSnippetsRanklabFeatures.titleSnippetQueryTermCoverage`
Rule: How much of the query wording appears in the title and snippet combined.
Google's description: Query term coverage in titles and snippets.
type number() · confidence medium

`MustangSnippetsRenderedToken.byteOffsetBegin`
Rule: Where this word starts in the rendered snippet text.
Google's description: Byte offset range in the rendered text that corresponds to this token. [byte_offset_begin, byte_offset_end) inclusive
type integer() · confidence medium

`MustangSnippetsRenderedToken.byteOffsetEnd`
Rule: Where this word ends in the rendered snippet text.
Google's description: exclusive
type integer() · confidence medium

`MustangSnippetsRenderedToken.section`
Rule: Which part of the page the rendered snippet word came from.
Google's description: Section and TokenPos of the token.
type String.t · confidence medium

`OceanDocInfo.docTag`
Rule: Bundle of book metadata returned with search results for building snippets.
Google's description: data returned with search docresults (snippets)
type GoogleApi.ContentWarehouse.V1.Model.OceanDocTag.t · confidence medium

`OceanDocTag.coverPage`
Rule: Cover page reference used to build result thumbnail images.
Google's description: cover page (PrintedAsSeen string), to generate results snippet thumbnail image urls
type String.t · confidence medium

`OceanDocTag.coverPageSize`
Rule: Pixel width and height of the cover page image.
Google's description: The size (in pixels) of the full-resolution clean images used for the cover page. The width and height will be zero if no image for that page.
type GoogleApi.ContentWarehouse.V1.Model.OceanImageSize.t · confidence medium

`OceanDocTag.editors`
Rule: Editor names shown in the book result display.
Google's description: Editors string for front end.
type String.t · confidence medium

`OceanDocTag.metadataCoverExists`
Rule: Whether a publisher-supplied cover thumbnail exists.
Google's description: metadata_cover_exists will be set if there's a metadata-provided cover thumbnail. the thumbnail will be used for scanless books or when a scanned book is in metadata-only view.
type boolean() · confidence medium

`OceanDocTag.metadataCoverSize`
Rule: Pixel size of the metadata cover image.
Google's description: The size (in pixels) of the metadata cover image.
type GoogleApi.ContentWarehouse.V1.Model.OceanImageSize.t · confidence medium

`OceanDocTag.printedPageNumber`
Rule: The page number as printed in the physical book.
Google's description: printed page number (OceanPrintedPageNumber; from OceanPageInfoMap::Page::printed_page_number)
type String.t · confidence medium

`OceanDocTag.refPageUrl`
Rule: URL of the 'About this book' reference page.
Google's description: the URL of the reference page (About this book)
type String.t · confidence medium

`OceanDocTag.searchInBookUrl`
Rule: URL for searching inside the book.
Google's description: the URL for the "search in book"
type String.t · confidence medium

`OceanDocTag.structuredPageNumber`
Rule: Structured version of the printed page number.
Google's description: In CAv2 only: structured page number (printed number as we understand it) (result of ocean::StructuredPageNumberProto::AppendToString)
type String.t · confidence medium

`OceanDocTag.subTitle`
Rule: Book subtitle shown in the result display.
Google's description: Sub title string for front end
type String.t · confidence medium

`OceanDocTag.thumbnailUrl`
Rule: URL of the book's cover image.
Google's description: the URL of the cover page.
type String.t · confidence medium

`OceanDocTag.title`
Rule: Book title shown in the result display.
Google's description: Title string for front end.
type String.t · confidence medium

`OceanDocTag.usingActualCover`
Rule: Whether the shown cover is the book's real cover rather than a generated one.
Google's description: using_actual_cover will be set if we are using the actual cover of the book (instead of the table of content, etc.). This is particularly useful to identify books where we inserted a generated cover via Coverups.
type boolean() · confidence medium

`OceanDocTagCatalogSpecific.publicationTimeToDisplay`
Rule: Human-readable publication period shown to users, such as 'Spring 2002'.
Google's description: String to be displayed as catalog publication time, e.g. "Spring 2002".
type String.t · confidence medium

`OceanDocTagMagazineSpecific.displayDate`
Rule: Human-readable issue date shown in the interface.
Google's description: A human-readable date for display in the UI. Unlike "publication
type String.t · confidence medium

`OceanDocTagMagazineSpecific.issueDescription`
Rule: Description of a magazine issue's featured articles.
Google's description: Description specific to a magazine issue, such as featured articles and article summaries.
type String.t · confidence medium

`OceanDocTagMagazineSpecific.serialTitle`
Rule: Series title used in the metadata line of a magazine snippet.
Google's description: This is used to render the metadata line of the snippet and should be present in all magazine documents.
type String.t · confidence medium

`OceanDocTagPatentSpecific.relativeThumbnailPath`
Rule: Path to the image used as the patent's thumbnail.
Google's description: Path identifying the image used for the thumbnail of this patent. e.g. "EP1234567B1/imgf0001.png" The client is expected to fill in the rest of the url such as:
type String.t · confidence medium

`PerDocData.rsApplication`
Rule: Rich snippet application information found on the page.
Google's description: Application information associated to the document.
type GoogleApi.ContentWarehouse.V1.Model.RepositoryAnnotationsRdfaRdfaRichSnippetsApplication.t · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityDniDocPreviewRestrictions.extendedNewsPreviewsDomain`
Rule: Whether the page's domain falls under the extended news preview rules, and its status.
Google's description: Whether this document comes from a domain that is affected by Extended News Previews (ENP) and its status (approved/rejected).
type GoogleApi.ContentWarehouse.V1.Model.QualityDniExtendedNewsPreviews.t · confidence medium

`QualityDniDocPreviewRestrictions.isAmp`
Rule: Whether the page itself is an AMP page.
Google's description: Only be true when the page itself is an AMP page. For paired AMP, the canonical page will have this bit as false.
type boolean() · confidence medium

`QualityDniDocPreviewRestrictions.snippetLengthSource`
Rule: Where the snippet length limit came from.
Google's description: Source of snippet length.
type String.t · confidence medium

`QualityDniDocPreviewRestrictions.thumbnailSizeSource`
Rule: Where the thumbnail size limit came from.
Google's description: Source of thumbnail size.
type String.t · confidence medium

`QualityDniDocPreviewRestrictions.videoPreviewSecsSource`
Rule: Where the video preview length limit came from.
Google's description: Source of video preview seconds.
type String.t · confidence medium

`QualityDniExtendedNewsPreviews.desnippetedCountryCode`
Rule: Countries where this publisher's snippets are withheld.
Google's description: List of countries that desnippet the publisher. ISO 3166-1-alpha-2 country code (such as FR). See go/iiuse#region-identifiers. Used in V1.
type list(String.t) · confidence medium

`QualityDniExtendedNewsPreviews.status`
Rule: Whether the publisher is approved or rejected for extended news previews.
Google's description: ENP status.
type String.t · confidence medium

`QualityPreviewChosenSnippetInfo.isVulgar`
Rule: Whether the snippet was flagged as vulgar.
Google's description: Whether this snippet is a vulgar candidate.
type boolean() · confidence medium

`QualityPreviewChosenSnippetInfoTidbitInfo.sectionName`
Rule: The page section the snippet fragment came from.
Google's description: Section name of current snippet.
type String.t · confidence medium

`QualityPreviewChosenSnippetInfoTidbitInfo.tidbitText`
Rule: The text of a snippet fragment, kept for validation.
Google's description: Tidbit text for validation.
type String.t · confidence medium

`QualityPreviewChosenSnippetInfoTidbitInfo.tokenBegin`
Rule: Where the snippet fragment starts within its section.
Google's description: Tidbit token range in the section.
type String.t · confidence medium

`QualityPreviewRanklabSnippet.brainFeatures`
Rule: Model features used when choosing which snippet to show.
Google's description: Features from SnippetFlow in Superroot.
type GoogleApi.ContentWarehouse.V1.Model.QualityPreviewSnippetBrainFeatures.t · confidence medium · stages: 10. Snippet & Answer Extraction

`QualityPreviewRanklabSnippet.finalScore`
Rule: The final score of this snippet candidate.
Google's description: The final score of this candidate.
type number() · confidence medium · stages: 10. Snippet & Answer Extraction

`QualityPreviewRanklabSnippet.isMuppetSelectedSnippet`
Rule: Whether this candidate was the snippet a particular scorer picked.
Google's description: Is this the candidate chosen by Muppet scorer.
type boolean() · confidence medium · stages: 10. Snippet & Answer Extraction

`QualityPreviewRanklabSnippet.originalQueryTermCoverageFeatures`
Rule: How much of the raw query wording the snippet covers.
Google's description: Query term (original terms only) coverage features.
type GoogleApi.ContentWarehouse.V1.Model.QualityPreviewSnippetQueryTermCoverageFeatures.t · confidence medium · stages: 10. Snippet & Answer Extraction

`QualityPreviewRanklabSnippet.queryFeatures`
Rule: The main query-related features of the snippet candidate.
Google's description: Core set of snippet features.
type GoogleApi.ContentWarehouse.V1.Model.QualityPreviewSnippetQueryFeatures.t · confidence medium · stages: 10. Snippet & Answer Extraction

`QualityPreviewRanklabSnippet.queryTermCoverageFeatures`
Rule: How much of the query, including synonyms, the snippet covers.
Google's description: Query term (including synonyms) coverage features.
type GoogleApi.ContentWarehouse.V1.Model.QualityPreviewSnippetQueryTermCoverageFeatures.t · confidence medium · stages: 10. Snippet & Answer Extraction

`QualityPreviewRanklabSnippet.snippetInfo`
Rule: Details identifying the snippet that was chosen.
Google's description: Information to identify current chosen snippet.
type GoogleApi.ContentWarehouse.V1.Model.QualityPreviewChosenSnippetInfo.t · confidence medium · stages: 10. Snippet & Answer Extraction

`QualityPreviewRanklabTitle.baseRank`
Rule: Where this title candidate ranked in the baseline run.
Google's description: The ranking index of this candidate (starting from 0) in base.
type integer() · confidence medium · stages: 10. Snippet & Answer Extraction

`QualityPreviewRanklabTitle.docLang`
Rule: The page language used when scoring this title candidate.
Google's description: Document language for this title. It is used for model inference and hence flattened into RanklabTitle instead of RanklabDoc.
type String.t · confidence medium · stages: 10. Snippet & Answer Extraction

`QualityPreviewRanklabTitle.forcedExperimentScore`
Rule: A score given to title candidates forced in by an experiment.
Google's description: A score assigned for candidates forced by experiments.
type integer() · confidence medium · stages: 10. Snippet & Answer Extraction

QualityPreviewRanklabTitle.goldmineAnchorFactor
Rule: How much anchor text contributed to the title candidate's boost score.
Google's description: == Internal boost feature signals used to compute
type number() · confidence medium · stages: 10. Snippet & Answer Extraction

QualityPreviewRanklabTitle.goldmineOtherBoostFeatureCount
Rule: How many other title boost features were present but not listed separately.
Google's description: The number of BoostFeatures present in AlternativeTitlesGeneator but not populated above.
type integer() · confidence medium · stages: 10. Snippet & Answer Extraction

QualityPreviewRanklabTitle.perTypeQuality
Rule: How good the title is judged to be for its type.
Google's description: How good or bad this title is as a
type String.t · confidence medium · stages: 10. Snippet & Answer Extraction

QualityPreviewRanklabTitle.perTypeRank
Rule: Where the title ranks among candidates of the same type.
Google's description: Rank of this title among titles of the same
type integer() · confidence medium · stages: 10. Snippet & Answer Extraction

QualityPreviewRanklabTitle.percentBodyTitleTokensCovered
Rule: What share of the on-page heading's words this title covers.
Google's description: Numbers of body title tokens covered by this title, in range of [0, 1]. Not set if body title is considered "bad".
type number() · confidence medium · stages: 10. Snippet & Answer Extraction

QualityPreviewRanklabTitle.percentTokensCoveredByBodyTitle
Rule: What share of this title's words also appear in the on-page heading.
Google's description: Numbers of tokens covered by body title, in range of [0, 1]. Not set if body title is considered "bad".
type number() · confidence medium · stages: 10. Snippet & Answer Extraction

QualityPreviewRanklabTitle.queryMatch
Rule: How many query terms, including synonyms, the title matches.
Google's description: The number of (different) terms with a query match. It may include the match with any SQuery node (e.g., synonyms).
type integer() · confidence medium · stages: 10. Snippet & Answer Extraction

QualityPreviewRanklabTitle.queryMatchFraction
Rule: The share of the user's query words matched by the title.
Google's description: A number of matched query terms divided by the number of all terms in query. Synonyms or other terms that appear in squery but not in the raw query are excluded. Takes values in [0, 1].
type number() · confidence medium · stages: 10. Snippet & Answer Extraction

QualityPreviewRanklabTitle.testRank
Rule: Where this title candidate ranked in the experiment run.
Google's description: The ranking index of this candidate (starting from 0) in test.
type integer() · confidence medium · stages: 10. Snippet & Answer Extraction

QualityPreviewSnippetRadishFeatures.answerScore
Rule: How well a passage answers the query, used when picking a snippet.
Google's description: Answer score of the passage for this
type number() · confidence medium · stages: 10. Snippet & Answer Extraction

QualityPreviewSnippetRadishFeatures.similarityScore
Rule: How similar this passage is to the snippet candidate.
Google's description: Similarity score between this
type number() · confidence medium · stages: 10. Snippet & Answer Extraction

QualitySitemapSporcSignals.oslScore
Rule: A score for showing a plain one-line sitelink.
Google's description: Score for normal oneline sitelink
type number() · confidence medium

QualitySitemapSporcSignals.scrolltoScore
Rule: A score for showing a sitelink that scrolls to a spot on the page.
Google's description: Score for scroll-to sitelink
type number() · confidence medium

QualitySitemapSporcSignals.tocScore
Rule: A score for showing a table-of-contents style sitelink.
Google's description: Score for table of contents sitelink
type number() · confidence medium

`QualitySitemapTarget.sectionTexts`
Rule: Text of page sections used to preview page anchors.
Google's description: Section texts used for Page Anchors Preview (go/page-anchor-preview-dd).
type list(String.t) · confidence medium

`QualitySitemapTarget.snippetResponse`
Rule: The snippet generated for the sitelink target page.
Google's description: The snippet response for the target document for an empty query.
type GoogleApi.ContentWarehouse.V1.Model.GenericSnippetResponse.t · confidence medium

`QualitySitemapTarget.sporcSignals`
Rule: Sitelink scores filled in only at serving time, never stored in the index.
Google's description: FIELD SHOULD ONLY BE POPULATED DURING ONLINE/SERVING TIME. Thus it should have no impact on cdoc/mdu.
type GoogleApi.ContentWarehouse.V1.Model.QualitySitemapSporcSignals.t · confidence medium

`QualitySitemapTargetGroup.allTargetsNamedAnchors`
Rule: Whether every link in this group points to a named anchor on the source page.
Google's description: If all the targets in this group are named anchors on the source page.
type boolean() · confidence medium

`QualitySitemapTargetGroup.allTargetsNamedTopictagsScrollto`
Rule: Whether every link in this group points to a named scroll-to tag.
Google's description: If all the targets in this group are named topictags_scrollto on the source page.
type boolean() · confidence medium

`RepositoryAnnotationsMustangSentimentSnippetAnnotations.snippetText`
Rule: The actual text of a sentiment snippet, filled in only when a snippet is returned for display.
Google's description: This protobuffer is serving double duty as both a Mustang attachment and the response proto that gets returned by Mustang in the WWWSnippetResponse's info MessageSet. When stored as an attachment, this field will always be empty. However, when returned with the WWWSnippetResponse, Mustang will print and store the actual sentiment snippet's text here.
type String.t · confidence medium

`RepositoryAnnotationsMustangSentimentSnippetAnnotations.start`
Rule: The token position where the snippet span begins.
Google's description: begin and end are token offsets.
type integer() · confidence medium

`RepositoryAnnotationsRdfaBreadcrumbs.crumb`
Rule: The list of links making up a page's breadcrumb trail.
Google's description: Each crumb represents one link of the breadcrumb chain.
type list(GoogleApi.ContentWarehouse.V1.Model.RepositoryAnnotationsRdfaCrumb.t) · confidence medium

`RepositoryAnnotationsRdfaBreadcrumbs.url`
Rule: The address of the page the breadcrumb trail was taken from.
Google's description: The URL of the document from which this breadcrumb trail was extracted.
type String.t · confidence medium

`RepositoryAnnotationsRdfaCrumb.title`
Rule: The visible text of one step in a breadcrumb trail.
Google's description: The text that represented this crumb in the document.
type String.t · confidence medium

`RepositoryAnnotationsRdfaCrumb.url`
Rule: The address that one breadcrumb step links to.
Google's description: The URL linked from this crumb.
type String.t · confidence medium

`RepositoryAnnotationsRdfaRdfaRichSnippetsApplication.category`
Rule: The category labels listed for a mobile app.
Google's description: Application information.
type list(String.t) · confidence medium

`RepositoryAnnotationsRdfaRdfaRichSnippetsApplication.extractedIconColor`
Rule: The single main color pulled from an app icon, kept for older code.
Google's description: Top 1 of extracted icon colors. We keep this field for backward compatibility.
type integer() · confidence medium

`RepositoryAnnotationsRdfaRdfaRichSnippetsApplication.extractedIconColors`
Rule: The ten most common colors pulled from an app icon.
Google's description: Top 10 of extracted icon colors. r = (rgb >> 16) & 0xff; g = (rgb >> 8) & 0xff; b = rgb & 0xff;
type list(integer()) · confidence medium

RepositoryAnnotationsRdfaRdfaRichSnippetsApplication.hasEditorsChoiceBadge
Rule: Whether the app carries an editors choice badge.
Google's description: Whether the app has editors choice tag
type boolean() · confidence medium

RepositoryAnnotationsRdfaRdfaRichSnippetsApplication.iconUrlHref
Rule: The address of the app's icon image.
Google's description: Icon and Screenshots
type String.t · confidence medium

RepositoryAnnotationsRdfaRdfaRichSnippetsApplication.inAppPurchase
Rule: Whether the app sells things inside the app.
Google's description: Whether the app offers in-app purchase.
type boolean() · confidence medium

RepositoryAnnotationsRdfaRdfaRichSnippetsApplication.optionalResult
Rule: Whether this app is an optional extra result in a grid layout.
Google's description: Whether this App is optional result for Grid UI.
type boolean() · confidence medium

RepositoryAnnotationsRdfaRdfaRichSnippetsApplication.price
Rule: The app's price as shown.
Google's description: Price
type String.t · confidence medium

RepositoryAnnotationsRdfaRdfaRichSnippetsApplication.rating
Rule: The app's star rating to display.
Google's description: Ratings and reviews Either for this version only or for all versions to be displayed.
type String.t · confidence medium

RepositoryAnnotationsRdfaRdfaRichSnippetsApplication.size
Rule: How much space the app takes up.
Google's description: bytes or numeric with MB or GB
type String.t · confidence medium

RepositoryAnnotationsRdfaRdfaRichSnippetsApplication.supportsAndroidTv
Rule: Whether the app works on Android TV.
Google's description: Whether this App supports Android TV. Note that some App supports more than one platforms. So we would use boolean for a platform.
type boolean() · confidence medium

RepositoryAnnotationsRdfaRdfaRichSnippetsApplication.supportsChromecast
Rule: Whether the app works with Google Cast.
Google's description: Whether this App supports Google Cast.
type boolean() · confidence medium

RepositoryAnnotationsRdfaRdfaRichSnippetsApplication.totalRatingCount
Rule: How many ratings the app has across all versions.
Google's description: Rating_count including all versions of this application.
type integer() · confidence medium

RepositoryAnnotationsRdfaRdfaRichSnippetsApplication.vendor
Rule: The company or person that makes the app.
Google's description: Vendor
type String.t · confidence medium

RepositoryAnnotationsRdfaRdfaRichSnippetsApplicationCountryPrice.currencyCode
Rule: The currency an app price is quoted in.
Google's description: ISO 4217 currency code.
type String.t · confidence medium

RepositoryAnnotationsRdfaRdfaRichSnippetsApplicationCountryPrice.price
Rule: The app's price for one country as a plain number string.
Google's description: Price string converted from double value in a standard currency unit, like '199.35' or '1400'.
type String.t · confidence medium

RepositoryAnnotationsRdfaRdfaRichSnippetsApplicationRankData.playwrightCategoryId
Rule: The Play category used to decide which app ranking list to show.
Google's description: Copied from the category_id field from Playwright docs. It helps decide which category to show in app ranking info.
type list(String.t) · confidence medium

RepositoryWebrefDisplayInfo.displayName
Rule: Names for showing an entity, one per language, taken from trusted sources.
Google's description: Per language display name from reliable sources.
type list(GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefDisplayName.t) · confidence medium

`RepositoryWebrefDisplayName.canonicalName`
Rule: The encyclopedia-style name of a topic.
Google's description: An encylopedia style topic name (e.g. "Dog", not "Dogs"). Corresponds to Freebase /type/object/name and similar sources.
type String.t · confidence medium

`RepositoryWebrefDisplayName.subjectName`
Rule: The name used when the entity appears as a topic or interest in a list.
Google's description: A name as it would be used for a news topic, an interest, the subject of a story. E.g. in list of things ("Related topics: Smartphones, computers, accidents". "Interests: Dogs").
type String.t · confidence medium

`RepositoryWebrefExtraMetadata.displayInfo`
Rule: Information used to show the entity in products and interfaces.
Google's description: Information for displaying the entity in applications.
type GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefDisplayInfo.t · confidence medium

`RichsnippetsDataObject.type`
Rule: The type of a structured-data object found on the page.
Google's description: The object type.
type String.t · confidence medium

`RichsnippetsDataObjectAttribute.idata`
Rule: Whole-number data stored under a structured-data attribute, such as a thumbnail reference.
Google's description: idata holds integer data under the attribute name, and could be interpreted differently according to the attribute name. Example: stores the ImadeData.docid used to generate thumbnails. idata will not be automatically converted into xml (the default behavior), but the behavior can be overwritten if necessary.
type String.t · confidence medium

`RichsnippetsDataObjectAttribute.subobject`
Rule: A structured-data object nested inside another one.
Google's description: A data object can have other data objects nested inside it. This is needed to represent Microformats and RDFa which have nestings e.g., a review with a business with an address, or a review with a rating object. See the Webmaster Central 2009/05 blog on "Introducing Rich Snippets".
type GoogleApi.ContentWarehouse.V1.Model.Proto2BridgeMessageSet.t · confidence medium

`RichsnippetsDataObjectAttribute.value`
Rule: The value held by a structured-data attribute.
Google's description: Either of cdata or value should be present.
type String.t · confidence medium

`ScienceCitation.AbstractHtml`
Rule: The abstract kept in HTML form for showing to users.
Google's description: Version of abstract field for display. Contains unsanitized XML/HTML.
type String.t · confidence medium

`ScienceCitation.NumKeyQuotes`
Rule: How many key quotes are available, kept for showing to users.
Google's description: for display in gws
type integer() · confidence medium

`ScienceCitation.NumRelated`
Rule: How many related articles exist, kept for showing to users.
Google's description: for display in gws
type integer() · confidence medium

`ScienceCitation.NumVersions`
Rule: How many versions of the article exist, kept for showing to users.
Google's description: for display in gws
type integer() · confidence medium

`ScienceCitation.TitleHtml`
Rule: The article title kept with its original HTML markup so it can be displayed.
Google's description: Version of title for display. Contains unsanitized HTML/XML.
type String.t · confidence medium

`ScienceCitationAlternateAbstract.AbstractHtml`
Rule: The abstract kept with its original HTML tags so it can be displayed.
Google's description: Version of abstract field for display. This may contain XML/HTML tags.
type String.t · confidence medium

`ScienceCitationAlternateTitle.TitleHtml`
Rule: An alternate article title kept with its original HTML markup for display.
Google's description: Version of title for display. Contains unsanitized HTML/XML.
type String.t · confidence medium

`ScienceCitationDownloadURL.ContentType`
Rule: The content type of the URL, used to make the result display better.
Google's description: makes gws display nicer :)
type integer() · confidence medium

`ScienceCitationDownloadURL.DisplayPriority`
Rule: A score for how much this URL is preferred when showing the article.
Google's description: display preference score
type integer() · confidence medium

`ScienceCitationDownloadURL.ExcerptContent`
Rule: The first few lines of an abstract-like excerpt from the page.
Google's description: first few lines of abstract'ish excerpt
type String.t · confidence medium

`ScienceCitationUnionCatalog.Url`
Rule: The catalog link shown to users.
Google's description: UnionCatalog url to display to users
type String.t · confidence medium

`ScienceIndexSignal.Title`
Rule: The article title, stored only when the page's own HTML title is poor.
Google's description: Title of the article. Its only filled in when the html title of the page isn't good.
type String.t · confidence medium

`SdrPageAnchorsSitelink.embedding`
Rule: A vector representation of a page section used when picking jump links.
Google's description: Needed for relevance scoring.
type GoogleApi.ContentWarehouse.V1.Model.SdrEmbedding.t · confidence medium

`SdrPageAnchorsSitelink.geometryScore`
Rule: A combined score describing the shape and layout of the page section.
Google's description: aggregate score from Section Geometry.
type number() · confidence medium

`SdrPageAnchorsSitelink.headingAbbrvScore`
Rule: A score for how well the heading works as a shortened label.
Google's description: Heading Abbreviation score.
type number() · confidence medium

`SdrPageAnchorsSitelink.hpScore`
Rule: A score used to filter which headings and passages qualify as jump links.
Google's description: Needed for heading/passage filtering.
type number() · confidence medium

`SdrPageAnchorsSitelink.sectionHeight`
Rule: How tall the page section is.
Google's description: Needed for Geometry Scoring and backoffs. from Section Geometry.
type integer() · confidence medium

`Sitemap.pageAnchorsDocInfo`
Rule: Stored page anchor details used to build jump links for a result.
Google's description: This field is populated in the Sitemap MDU subpopulator from cdoc data. This is used to store page anchors information for TopicTagsScrolltoFlow.
type GoogleApi.ContentWarehouse.V1.Model.SdrPageAnchorsDocInfo.t · confidence medium

`Sitemap.searchInSite`
Rule: Turns on the search-this-site box for the result.
Google's description: Enable site search.
type boolean() · confidence medium

`Sitemap.subresultList`
Rule: The list of sub-results shown underneath the main result.
Google's description: This field is populated in the Sitemap MDU subpopulator from cdoc data. It's not set in the cdoc Sitemap.
type GoogleApi.ContentWarehouse.V1.Model.QualitySitemapSubresultList.t · confidence medium

`SnapshotSnapshotDocument.metaNoSnippet`
Rule: Records that the page asked, via a tag, not to have a snippet shown for it.
Google's description: These are set from tags in the web page:
type boolean() · confidence medium

`SnapshotTextNode.maxSplit`
Rule: A 0-7 rating of the strongest text-breaking tag seen before this chunk of text.
Google's description: A value in the range [0,7] (zero if not present) indicating the most "powerful" splitting tag since the last text node. See "enum Category" in mustang/snippets/taginfo.h.
type integer() · confidence medium

`SnippetExtraInfo.containUserQuotes`
Rule: Notes that every candidate snippet contains text the user quoted in the query.
Google's description: Indicates that the snippet candidates all contain uesr quotes.
type boolean() · confidence medium · stages: 10. Snippet & Answer Extraction

`SnippetExtraInfo.containVulgarCandidates`
Rule: Notes that some of the candidate snippets contain vulgar wording.
Google's description: Indicates if there are any vulgar snippet candidates.
type boolean() · confidence medium · stages: 10. Snippet & Answer Extraction

`SnippetExtraInfo.disableNg3Scoring`
Rule: A switch telling the system to pick the snippet using only the SnippetBrain model.
Google's description: Indicates that SR side should disable the ng3 scoring and soly rely on the SnippetBrain scoring for selecting the final candidate. E.g. this can happen when the offline snippet generation routine is triggered.
type boolean() · confidence medium · stages: 10. Snippet & Answer Extraction

`SnippetExtraInfo.disableQueryFeatures`
Rule: A switch turning off query-relevance features when scoring snippets.
Google's description: Indicates whether the query relevance features is disabled or not in Muppet scoring.
type boolean() · confidence medium · stages: 10. Snippet & Answer Extraction

`SnippetExtraInfo.forceLeadingTextOrMeta`
Rule: A switch preventing new snippet candidates from being added later in the flow.
Google's description: Indicates to not add any new candidates in SnippetFlow.
type boolean() · confidence medium · stages: 10. Snippet & Answer Extraction

`SnippetExtraInfoSnippetCandidateInfo.id`
Rule: A number identifying each candidate snippet for a document within one request.
Google's description: Candidate identifier number, unique among all snippet candidates under each document in each request. What does this number mean: - Muppet candidates: This equals to the candidate's rank by Muppet snippets scorer. - Superroot candidates: No specific meaning, this number should be larger than that of Muppet candidates. This field is used to: - Verify whether snippet brain chooses a different snippet from Muppet (the one chosen by Muppet is always in id 0). - Print debugging information and sort c
type integer() · confidence medium · stages: 10. Snippet & Answer Extraction

`SnippetExtraInfoSnippetCandidateInfo.isMuppetSelectedSnippet`
Rule: Marks whether this candidate is the one the earlier snippet picker chose.
Google's description: If this snippet is chosen by Muppet.
type boolean() · confidence medium · stages: 10. Snippet & Answer Extraction

`SnippetExtraInfoSnippetCandidateInfo.isSnippetBrainBoldingTriggered`
Rule: Marks whether the model generated a bolding span for this snippet.
Google's description: If SnippetsBrain bolding model triggered and a bolding span is generated.
type boolean() · confidence medium · stages: 10. Snippet & Answer Extraction

`SnippetExtraInfoSnippetCandidateInfo.listInfo`
Rule: Extra list layout details, filled in only for snippets shown as lists.
Google's description: List information for this candidate, only populated for RADISH_LIST snippets.
type GoogleApi.ContentWarehouse.V1.Model.MustangReposWwwSnippetsOrganicListSnippetResponse.t · confidence medium · stages: 10. Snippet & Answer Extraction

`SnippetExtraInfoSnippetScoringInfo.brainScore`
Rule: The score a machine learning model gave this candidate snippet.
Google's description: The below fields are populated by SnippetFlow in superroot. Score generated from snippet brain model. go/snippets-brain
type number() · confidence medium

`SnippetExtraInfoSnippetScoringInfo.features`
Rule: The set of measured features used by the snippet scorer.
Google's description: Snippets ranklab features generated by scorer V2.
type GoogleApi.ContentWarehouse.V1.Model.QualityPreviewRanklabSnippet.t · confidence medium

`SnippetExtraInfoSnippetScoringInfo.finalScore`
Rule: The final score used to choose among candidate snippets.
Google's description: Final snippet score by chooser.
type number() · confidence medium

`SnippetExtraInfoSnippetScoringInfo.rankBySnippetFlow`
Rule: The final ordering position this snippet candidate was given.
Google's description: Final rank given by SnippetFlow.
type integer() · confidence medium

`SnippetsLeadingtextLeadingTextAnnotation.type`
Rule: Which kind of leading text this annotation marks.
Google's description: Type of this leading text. Should be an enum of LeadingTextInfo.LeadingTextType
type integer() · confidence medium

`SnippetsLeadingtextLeadingTextAnnotationPiece.begin`
Rule: The starting byte position of a piece of the page's leading text.
Google's description: A piece of leading text is text within [begin, end). For example, a document is "ABCDEF". If we want to set leading text as 'CD', the value of begin is byte offset of 'C', the value of end is byte offset of 'E'. end == -1 means to the end of document.
type integer() · confidence medium

`SnippetsLeadingtextLeadingTextAnnotationPiece.beginText`
Rule: A copy of the starting text, kept so positions can be re-aligned later.
Google's description: UTF8 text, for alignment when using reusableinfo. Those text are not available in docjoins.
type String.t · confidence medium

`SnippetsLeadingtextLeadingTextInfo.beginPos`
Rule: The byte position in the page where the leading text starts.
Google's description: Leading text start position, byte offset of page content. The offset is got in ParseMaster. So it is the offset after the content is converted to UTF8.
type integer() · confidence medium

`SnippetsLeadingtextLeadingTextInfo.leadingtext`
Rule: One or more candidate stretches of opening text taken from the page.
Google's description: Note: You can also use it to save multiple leading text candidates.
type list(GoogleApi.ContentWarehouse.V1.Model.SnippetsLeadingtextLeadingTextAnnotation.t) · confidence medium

`SnippetsLeadingtextLeadingTextInfo.text`
Rule: A copy of the leading text itself, used to line positions back up.
Google's description: UTF8 text, for alignment when using reusableinfo. The text is not available in docjoins.
type String.t · confidence medium

`SnippetsLeadingtextLeadingTextInfo.type`
Rule: Which style of leading text was chosen for this kind of document.
Google's description: Type of leading text which is optimized for this type of document.
type String.t · confidence medium

`UniversalsearchNewPackerKnowledgeResultSupport.rank`
Rule: The zero based position of this result, used to place it against web results.
Google's description: The 0 based rank for this result. When the source is MAIN_GSR this corresponds to the index of the result in the generic search response specific in AddKnowledgePackerPreprocessors. Otherwise, the rank should be estimated so the result is closest in importance to the web result at the same rank.
type integer() · confidence medium

`VideoContentSearchTextMatchInfo.matchedDescriptionText`
Rule: The video description text used as the label for a key moment anchor.
Google's description: The video description text matched with the ASR that's used as the anchor label.
type String.t · confidence medium

`VideoContentSearchVideoAnchors.thumbnailForced`
Rule: Whether a hidden thumbnail is being shown anyway.
Google's description: This field indicates that the thumbnail should be hidden but is forced to show.
type boolean() · confidence medium

`VideoContentSearchVideoAnchors.thumbnailSetInfo`
Rule: Details about the group of thumbnails shown for the key moments.
Google's description: Information about the set of thumbnails.
type GoogleApi.ContentWarehouse.V1.Model.VideoContentSearchAnchorsThumbnailInfo.t · confidence medium

`WWWDocInfo.foreignMetadescription`
Rule: Marks that the meta description is written in a different language than the page.
Google's description: Indicate if the meta description in a different language than its page.
type boolean() · confidence medium

`WWWDocInfo.metaDescriptionLanguages`
Rule: The languages detected for the meta description or body title when they differ from the page.
Google's description: If meta description/body title were detected to be in a different language from the document language (the 'language' field above) in RosettaLanguageAnnotator, the detected languages are populated here. Note: as of ariane/154728, no more than one language is populated for each field.
type list(String.t) · confidence medium

`WWWDocInfo.title`
Rule: The title of the landing page.
Google's description: Landing page title.
type String.t · confidence medium

`WWWDocInfo.visualType`
Rule: Whether the document is laid out right-to-left or otherwise visual.
Google's description: Is this doc visual RTL? See enum VisualType in visualtype.h. Default is NOT_VISUAL_DOCUMENT.
type String.t · confidence medium

`WWWSnippetResponse.answerDocMatches`
Rule: Which prepared answers had a match inside the document.
Google's description: A list of answers that had at least one hit in the document. Answers are identified by their index into the QRewriteQueryParams_AnswerSnippetInfo array (see //query/proto/query-params.proto).
type list(integer()) · confidence medium · stages: 10. Snippet & Answer Extraction

`WWWSnippetResponse.events`
Rule: A packed set of flags recording what happened while building the snippet.
Google's description: Bitfield of snippet events and the various events. See SnippetEvents in ./snippets/defines.h for details on the contents.
type String.t · confidence medium · stages: 10. Snippet & Answer Extraction

`WWWSnippetResponse.extraInfo`
Rule: Extra snippet details returned when the caller asks for them.
Google's description: If requested, the extra snippet info
type GoogleApi.ContentWarehouse.V1.Model.ExtraSnippetInfoResponse.t · confidence medium · stages: 10. Snippet & Answer Extraction

`WWWSnippetResponse.findyTidbits`
Rule: Marks which snippet pieces suit the Quick Scroll browser extension.
Google's description: A bitvector of the tidbits in the snippet that are appropriate for the Quick Scroll (Findy) Chrome extension. Typically these contain "extra body matches", i.e., important query items not in the title.
type integer() · confidence medium · stages: 10. Snippet & Answer Extraction

`WWWSnippetResponse.matchesBitmapEncoded`
Rule: A bitmap marking where query terms matched in the body and URL.
Google's description: Bitmap representing matches to leaf query terms within document (body section and url). It gets populated if query_matches_info = true.
type String.t · confidence medium · stages: 10. Snippet & Answer Extraction

`WWWSnippetResponse.seenNotTerm`
Rule: Whether a negative query term matched the meta description.
Google's description: Did a negative query term match the meta description?
type boolean() · confidence medium · stages: 10. Snippet & Answer Extraction

`WWWSnippetResponse.snippetBylineDate`
Rule: The date shown alongside time-sensitive snippets.
Google's description: Byline date for time sensitive snippets. Most of the time it originates from quality_timebased::SyntacticDate and it is floored to PT midnight.
type String.t · confidence medium · stages: 10. Snippet & Answer Extraction

`WWWSnippetResponse.snippethash`
Rule: A hash of the snippet body used to filter duplicate results.
Google's description: A hash for duplicate detection. Two results with the same content can return different snippets if, for example, one has an ODP entry and the other does not. Gws can use this value reliably to filter duplicates. It is a hash of body only tidbits.
type String.t · confidence medium · stages: 10. Snippet & Answer Extraction

`WWWSnippetResponse.titleSizeParams`
Rule: Sizing parameters used when laying out the title.
Google's description: Will only be set when
type GoogleApi.ContentWarehouse.V1.Model.TitleSizeParams.t · confidence medium · stages: 10. Snippet & Answer Extraction

`WeboftrustLiveResultsDocAttachments.providerAttachment`
Rule: The live result content attached to the document.
Google's description: Identifies a Live Result which is to be attached to the document.
type list(GoogleApi.ContentWarehouse.V1.Model.WeboftrustLiveResultProviderDocAttachment.t) · confidence medium

`BlogsearchConversationNode.date`
Rule: The date the post in the conversation thread was created.
Google's description: The creation date of the doc.
type String.t · confidence low

`GDocumentBaseDirectory.Description`
Rule: The web directory's written description of the page.
Google's description: encoded in UTF8
type String.t · confidence low

`GDocumentBaseDirectory.Title`
Rule: The title the web directory gave the page.
Google's description: encoded in UTF8
type String.t · confidence low

`GDocumentBaseDirectory.TitleScore`
Rule: A deprecated score for the directory title that nothing populates.
Google's description: Deprecated; do not use. There is no code populating these fields as of Oct 2017.
type number() · confidence low

`GenericSnippetResponse.rankingSnippet`
Rule: An experimental snippet field that is only filled in some of the time.
Google's description: The experimental ranking snippet field. This will be populated only when
type String.t · confidence low

`ImageBaseThumbnailMetadata.crops`
Rule: Suggested crop areas for the thumbnail, normally left empty.
Google's description: SmartCrop crop-hints By default, this field is not populated.
type GoogleApi.ContentWarehouse.V1.Model.ContentAwareCropsIndexing.t · confidence low

`ImageBaseThumbnailMetadata.deepCrop`
Rule: An older style of suggested crop area for the thumbnail, normally left empty.
Google's description: DeepCrop crop-hints. Usage in thumbnails could be deprecated in favor or deep_crop_pixels (below). By default, this field is not populated.
type GoogleApi.ContentWarehouse.V1.Model.DeepCropIndexing.t · confidence low

`ImageBaseThumbnailMetadata.deepCropPixels`
Rule: Suggested crop area for the thumbnail expressed in pixels, normally left empty.
Google's description: DeepCrop signal in pixels, equivalent to deep_crop (above) but with pixels instead of percentages. By default, this field is not populated.
type GoogleApi.ContentWarehouse.V1.Model.DeepCropPixels.t · confidence low

`ImageRepositoryContentBasedVideoMetadata.videoAnchorSet`
Rule: Sets of anchor points in the video with thumbnails and metadata for key moments.
Google's description: Video anchor sets hold set of anchors with multiple anchor types and sequence of VideoAnchor which contains metadata about the anchor, such as thumbnail, perdoc data.
type GoogleApi.ContentWarehouse.V1.Model.VideoContentSearchVideoAnchorSets.t · confidence low

`IndexingEmbeddedContentRenderingOutputMetadata.snapshotQualityScore`
Rule: How good the rendered screenshot is, copied from the renderer.
Google's description: These scores are copied from htmlrender_webkit_headless_proto.Document.
type float() · confidence low

`KnowledgeAnswersIntentQueryArgumentProvenanceSearchAnswerValue.text`
Rule: The text shown to the user for this answer value.
Google's description: The display text of this answer value. It's taken from the search answer value display text if present, or the canonical name if it's an entity.
type String.t · confidence low

`KnowledgeAnswersIntentQueryArgumentSignals.resultSupport`
Rule: List of results that back up this query argument when assembling the answer.
Google's description: The list of result supports for this Argument.
type list(GoogleApi.ContentWarehouse.V1.Model.UniversalsearchNewPackerKnowledgeResultSupport.t) · confidence low

`KnowledgeAnswersIntentQueryFunctionCallSignals.resultSupport`
Rule: List of results that back up this parsed request when assembling the answer.
Google's description: The list of result supports for this FunctionCall.
type list(GoogleApi.ContentWarehouse.V1.Model.UniversalsearchNewPackerKnowledgeResultSupport.t) · confidence low

`KnowledgeAnswersIntentQueryResponseMeaningSignalsResponseMeaningSignals.answersHeaderSignals`
Rule: Data used to build the header shown above an answer.
Google's description: Signals that are used to generate answers header.
type GoogleApi.ContentWarehouse.V1.Model.KnowledgeAnswersIntentQueryResponseMeaningSignalsAnswersHeaderSignals.t · confidence low

`LocalWWWInfoCluster.isPlusbox`
Rule: An old flag for showing an expandable local result box.
Google's description: TODO(local-universal) Consider deleting is_plusbox once the new scheme that uses make_plusbox_visible rolled out.
type boolean() · confidence low · stages: 12. Personalization, Local & Context

`MustangReposWwwSnippetsCandidateFeature.name`
Rule: The name of a feature used by the snippet chooser scorer.
Google's description: Name corresponds to the names in WebChooserScorer::FeatureNames.
type String.t · confidence low

`MustangReposWwwSnippetsOrganicListSnippetResponse.radishScore`
Rule: A score for the list-snippet detector applied to this page.
Google's description: The score of the radish signal.
type number() · confidence low

`OceanDocTag.authors`
Rule: Author names shown in the book result display; slated for removal.
Google's description: TODO(leonid) Deprecate these Authors string for front end.
type String.t · confidence low

`QualityDniExtendedNewsPreviews.policyCriteriaBase64`
Rule: Encoded Search Console settings that change how preview policy is computed.
Google's description: Search Console Signals that modifies how policy are calculated. Didn't add
type String.t · confidence low

`QualityPreviewRanklabSnippet.experimentalFeatures`
Rule: A placeholder for experimental snippet features that is not filled in.
Google's description: For experimental usage, not populated yet.
type GoogleApi.ContentWarehouse.V1.Model.QualityPreviewSnippetExperimentalFeatures.t · confidence low · stages: 10. Snippet & Answer Extraction

`QualityPreviewRanklabSnippet.radishFeatures`
Rule: Features from the Radish system about this snippet.
Google's description: Radish related information.
type GoogleApi.ContentWarehouse.V1.Model.QualityPreviewSnippetRadishFeatures.t · confidence low · stages: 10. Snippet & Answer Extraction

`QualityPreviewRanklabTitle.goldmineAdjustedScore`
Rule: An adjusted score for the title candidate.
Google's description: The score for
type number() · confidence low · stages: 10. Snippet & Answer Extraction

`QualityPreviewRanklabTitle.goldmineFinalScore`
Rule: A deprecated final score for the title candidate.
Google's description: Deprecated: use
type number() · confidence low · stages: 10. Snippet & Answer Extraction

`QualityPreviewRanklabTitle.goldminePageScore`
Rule: A page-based score for the title candidate.
Google's description: The score for the
type number() · confidence low · stages: 10. Snippet & Answer Extraction

`QualityPreviewRanklabTitle.queryRelevance`
Rule: A deprecated experimental relevance value for the title.
Google's description: Deprecated experimental features.
type number() · confidence low · stages: 10. Snippet & Answer Extraction

`QualityPreviewSnippetRadishFeatures.navboostQuery`
Rule: The query string associated with this snippet passage signal.
Google's description: Navboost query for this radish signal.
type String.t · confidence low · stages: 10. Snippet & Answer Extraction

`QualitySitemapTarget.languages`
Rule: The languages of the page a sitelink points to.
Google's description: The languages of the document, taken from its cdoc.properties().languages()
type list(integer()) · confidence low

`QualitySitemapTarget.salientImage`
Rule: An image for the sitelink target, fetched live and used for experiments.
Google's description: The image data will be copied from the DocInfo response, and will be retrieved online, so this field should not be populated during indexing. This is a temporary field for experimentation.
type GoogleApi.ContentWarehouse.V1.Model.WWWResultInfoSubImageDocInfo.t · confidence low

`RepositoryAnnotationsMustangSentimentSnippetAnnotations.deprecatedMagnitude`
Rule: An old sentiment strength value, replaced by a snippet score.
Google's description: Deprecated: use snippet_score instead
type number() · confidence low

`RepositoryAnnotationsMustangSentimentSnippetAnnotations.deprecatedPolarity`
Rule: An old sentiment polarity number that has been replaced by a snippet score.
Google's description: Deprecated: use snippet_score instead
type number() · confidence low

`RepositoryAnnotationsRdfaRdfaRichSnippetsApplication.operatingSystems`
Rule: The operating systems an app runs on.
Google's description: See google3/quality/richsnippets/schema/data/operating_systems_rules.txt for possible values.
type list(String.t) · confidence low

`RepositoryAnnotationsRdfaRdfaRichSnippetsApplication.screenUrlHref`
Rule: Screenshot links for an app, being replaced by a newer image field.
Google's description: TODO(b/260128276) deprecate this field in favor of image_data.
type list(String.t) · confidence low

`RepositoryWebrefExtraMetadata.kcAttributeMetadata`
Rule: Details tied to knowledge attributes and question-and-answer triggering.
Google's description: Metadata related to KC attributes and Question & Answer triggering.
type GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefKCAttributeMetadata.t · confidence low

`ScienceCitationDownloadURL.HtmlTitle`
Rule: The HTML title of the page.
Google's description: html title of the page
type String.t · confidence low

`ShoppingWebentityShoppingAnnotationProductRating.count`
Rule: How many ratings or reviews were combined into this product rating.
Google's description: Number of ratings/reviews aggregated to create this product rating. If there are no ratings yet, this field will be explicitly set to zero, so whether this field is set should be checked using has_count.
type String.t · confidence low

`ShoppingWebentityShoppingAnnotationProductRating.minValueMillis`
Rule: The lowest and highest rating values that could possibly be given for a product.
Google's description: The lower and upper bounds of the rating values that could be submitted for the product. (Note that it is not the min/max ratings submitted for the product, it is the min/max that can hypothetically be submitted.)
type String.t · confidence low

`ShoppingWebentityShoppingAnnotationProductRating.value`
Rule: The product's average rating rescaled onto a zero to five scale.
Google's description: The value of this rating normalized between 0 and 5. This will not be set if count is set to 0.
type float() · confidence low

`ShoppingWebentityShoppingAnnotationProductRating.valueMillis`
Rule: The product's combined rating value before it is rescaled.
Google's description: The non-normalized aggregated value of the ratings for this product.
type String.t · confidence low

`SnapshotSnapshotMetadata.snapshotQualityScore`
Rule: A zero-to-one rating of how good the stored page snapshot is.
Google's description: The score here corresponds to the score in Snapshot, a number between 0.0 and 1.0 (higher the better).
type number() · confidence low

`SnapshotTextNode.inLink`
Rule: Marks whether this chunk of page text sits inside a link.
Google's description: One if the current text node is within a link; otherwise zero/not present.
type integer() · confidence low

`VideoContentSearchVideoActions.skipPatternStartOffsetInput`
Rule: Name of the input used when building a link that jumps to a point in a video.
Google's description: startOffset_input name as defined in
type String.t · confidence low

`VideoContentSearchVideoActions.skipToTimePattern`
Rule: Pattern used to build a link that skips to a specific time in a video.
Google's description: Skip to time pattern as defined in
type String.t · confidence low

`VideoContentSearchVideoAnchorScoreInfo.thumbnailInfo`
Rule: Details about the thumbnail chosen for this key moment.
Google's description: Information about the thumbnail anchor.
type GoogleApi.ContentWarehouse.V1.Model.VideoContentSearchAnchorThumbnailInfo.t · confidence low

`WWWDocInfo.badMetadescription`
Rule: A flag marking the page's meta description as bad.
Google's description: Bad meta flag
type boolean() · confidence low

`WWWDocInfo.googleLabelData`
Rule: Extra label data attached to an individual search result.
Google's description: Addition to support google label per-search-result annotation.
type String.t · confidence low

`WWWDocInfo.rootpageDuplicateMetadescription`
Rule: Marks a page as the main page among many sharing the same duplicated meta description.
Google's description: True if the meta-description is duplicated on many other pages and this page is the rootpage of such pages which have the same meta-description.
type boolean() · confidence low

`WWWDocInfo.seenNoodp`
Rule: A flag for an open-directory snippet feature that is no longer used.
Google's description: NOTE(kinoue): ODP/GWD snippet is unlaunched as of June 2017. This is no longer used.
type boolean() · confidence low

`WWWDocInfo.shoppingOffers`
Rule: Product offer details attached to the document from an inventory service.
Google's description: Shopping offer info from Inventory & Policy Service.
type list(GoogleApi.ContentWarehouse.V1.Model.ImageMustangShoppingOffer.t) · confidence low

`WWWSnippetResponse.metaTags`
Rule: The page's meta tags.
Google's description: Meta tags
type list(GoogleApi.ContentWarehouse.V1.Model.WWWMetaTag.t) · confidence low · stages: 10. Snippet & Answer Extraction

`WWWSnippetResponse.odp`
Rule: Whether open-directory text was used in the snippet, no longer populated.
Google's description: Was odp used in the snippets? DEPRECATED - this is no longer populated as of June 2017.
type boolean() · confidence low · stages: 10. Snippet & Answer Extraction

`WWWSnippetResponse.sentimentSnippets`
Rule: Deprecated sentiment-based snippets.
Google's description: DEPRECATED Sentiment snippets
type list(GoogleApi.ContentWarehouse.V1.Model.RepositoryAnnotationsMustangSentimentSnippetAnnotations.t) · confidence low · stages: 10. Snippet & Answer Extraction

`WWWSnippetResponse.siteDisplayName`
Rule: The friendly site name shown for the result, now being replaced.
Google's description: The domain-level display name of the website, such as "Google" for google.com. See go/site-display-name for more details. As of Aug 2023, this field is being deprecated in favor of
type String.t · confidence low · stages: 10. Snippet & Answer Extraction

Images & video (212)

`VideoContentSearchVideoAnchor.isFiltered`
Rule: Whether this key moment is held back and not shown in results.
Google's description: If true, the anchor is filtered and not served online.
type boolean() · confidence high

`VideoContentSearchVideoAnchor.time`
Rule: The point in the video, in milliseconds, that this key moment marks.
Google's description: The time stamp of the video anchor in milliseconds.
type String.t · confidence high

`VideoContentSearchVideoAnchors.isFiltered`
Rule: Whether the whole set of key moments is held back and not shown.
Google's description: If true, the anchor set is filtered and not served online.
type boolean() · confidence high

`WWWDocInfo.imageHeight`
Rule: The height of the image in pixels.
Google's description: image height
type integer() · confidence high

`WWWDocInfo.imageLicenseInfo`
Rule: Licensing details for the image, such as where to get the license.
Google's description: Image license info such as license url and how to acquire the license.
type GoogleApi.ContentWarehouse.V1.Model.ImageSearchImageLicenseInfo.t · confidence high

`WWWDocInfo.imageSize`
Rule: The image's file size in bytes.
Google's description: size in bytes;
type integer() · confidence high

`WWWDocInfo.imageWidth`
Rule: The width of the image in pixels.
Google's description: image width
type integer() · confidence high

`WWWDocInfo.isAnimated`
Rule: Whether the image is animated.
Google's description: Is this image animated?
type boolean() · confidence high

`WWWDocInfo.thumbHeight`
Rule: The height of the thumbnail image.
Google's description: thumbnail height
type integer() · confidence high

`WWWDocInfo.thumbWidth`
Rule: The width of the thumbnail image.
Google's description: Additions for image search.
type integer() · confidence high

`WWWDocInfoThumbnail.type`
Rule: Which kind of thumbnail this entry is.
Google's description: The type here corresponds to image_base::ThumbnailType defined in image/base/thumbnail-type.proto.
type integer() · confidence high

`WWWResultInfoSubImageDocInfo.bestThumbnailType`
Rule: Which of the two thumbnail sizes is the better one for this image.
Google's description: The best thumbnail type is either 300K or 50K.
type String.t · confidence high

`WWWResultInfoSubImageDocInfo.width50k`
Rule: The size of the small thumbnail generated for this image.
Google's description: Width and height of the AREA_50K thumbnail for this image.
type integer() · confidence high

`CorpusSelectionInfo.referrerDocid`
Rule: Identifiers of the pages indexed as carrying this image.
Google's description: Set of referrers indexed with the image.
type list(String.t) · confidence medium

`CorpusSelectionInfo.referrerUrls`
Rule: URLs of the pages indexed as carrying this image.
Google's description: Set of referrer urls indexed with the image.
type list(String.t) · confidence medium

`DeepCropPixels.x0`
Rule: A pixel edge of the crop box chosen for an image.
Google's description: Pixels version of the DeepCropIndexing bytes, this corresponds to the crop box for a given image (based input image size and desired aspect ratio).
type integer() · confidence medium

`ImageData.fileFormat`
Rule: The image's actual file format, regardless of its file extension.
Google's description: True file format (not extension).
type String.t · confidence medium

`ImageData.height`
Rule: The height of the image in pixels.
Google's description: Height
type integer() · confidence medium

`ImageData.isMultiframe`
Rule: Whether the original file contains more than one frame.
Google's description: True if the original image contains multiple frames (e.g., for animated or stereoscopic images).
type boolean() · confidence medium

`ImageData.size`
Rule: The byte size of the original full-size image file.
Google's description: Size in bytes of original (non-thumbnail)
type integer() · confidence medium

`ImageRegionsImageRegion.boundingBox`
Rule: The rectangle marking where a detected object sits in the image.
Google's description: The bounding box of the region.
type GoogleApi.ContentWarehouse.V1.Model.PhotosVisionGroundtruthdbNormalizedBoundingBox.t · confidence medium

`ImageRepositoryContentBasedVideoMetadata.videoPreviewBytes`
Rule: The raw bytes of a generated video preview, passed along for serving only.
Google's description: video_preview_bytes is only exported as virtual dataset by IE VideoUnderstanding and should not be persisted. It will be used by downstream IE functions to push for serving.
type list(GoogleApi.ContentWarehouse.V1.Model.ImageRepositoryVideoPreviewsVideoPreview.t) · confidence medium

`ImageRepositoryContentBasedVideoMetadata.videoPreviews`
Rule: Metadata describing available video previews without the preview bytes.
Google's description: video_previews contain the preview metadata but no bytes. It exits for IE and non-IE cases.
type list(GoogleApi.ContentWarehouse.V1.Model.ImageBaseVideoPreviewMetadata.t) · confidence medium

`ImageSearchImageLicenseInfo.licenseUrl`
Rule: The address of the license that applies to the image.
Google's description: Records license URL.
type String.t · confidence medium

`PerDocData.mediaOrPeopleEntities`
Rule: The five most topical person or media entities on the page, used by Image Search.
Google's description: Contains the mids of the 5 most topical entities annotated with selected KG collections. This information is currently used on Image Search to detect cases where results converged to mostly a single person or media entity. More details: go/result-set-convergence.
type GoogleApi.ContentWarehouse.V1.Model.ImageQualitySensitiveMediaOrPeopleEntities.t · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityWebanswersVideoTranscriptAnnotations.timingInfo`
Rule: Start and end times for each sentence in the video transcript.
Google's description: Timing information that maps sentence boundaries in the punctuated transcript with timing offsets for the start and end of those sentences.
type GoogleApi.ContentWarehouse.V1.Model.QualityWebanswersVideoYouTubeCaptionTimingInfoAnnotations.t · confidence medium

`RepositoryAnnotationsRdfaRdfaRichSnippetsApplication.imageData`
Rule: Image data for a mobile app listing.
Google's description: Copied from google3/contentads/shared/boulder/mobile-app-data-image-data.proto.
type GoogleApi.ContentWarehouse.V1.Model.QualityCalypsoAppsUniversalImageData.t · confidence medium

`SnapshotImageNode.url`
Rule: The full web address of an image exactly as it appears in the page.
Google's description: The absolute url of the image as present in the page.
type String.t · confidence medium

`VideoContentSearchAnchorCommonFeatureSet.anchorQbstDistance`
Rule: How far the key moment text is from the video's top query or its title.
Google's description: QBST distance between the anchor and the top navboost query of the video if exists, or the video title otherwise.
type number() · confidence medium

`VideoContentSearchAnchorCommonFeatureSet.asrAverageBabelSimilarityScore`
Rule: Average text similarity between the key moment label and all spoken sentences.
Google's description: Average of babel similarity between the anchor and all asr sentences.
type number() · confidence medium

`VideoContentSearchAnchorCommonFeatureSet.asrMaximumBabelSimilarityScore`
Rule: Best text similarity between the key moment label and any spoken sentence.
Google's description: Maximum babel similarity between the anchor and the asr sentences.
type number() · confidence medium

`VideoContentSearchAnchorCommonFeatureSet.bleurtFeatures`
Rule: The inputs collected for running the Bleurt text model on a key moment.
Google's description: Features needed for Bleurt inference.
type GoogleApi.ContentWarehouse.V1.Model.VideoContentSearchBleurtFeatures.t · confidence medium

`VideoContentSearchAnchorCommonFeatureSet.bleurtScore`
Rule: The Bleurt model score for a video key moment label.
Google's description: The Bleurt inference score generated using the bleurt_features.
type number() · confidence medium

`VideoContentSearchAnchorCommonFeatureSet.descartesScoreWithTitle`
Rule: Similarity score between the video title and the key moment label.
Google's description: Descartes similarity score between video title and anchor label.
type number() · confidence medium

`VideoContentSearchAnchorCommonFeatureSet.descriptionAverageBabelSimilarityScore`
Rule: Average similarity between the key moment label and the description sentences.
Google's description: Average of babel similarity between the anchor and all description sentences.
type number() · confidence medium

`VideoContentSearchAnchorCommonFeatureSet.descriptionMaximumBabelSimilarityScore`
Rule: Best similarity between the key moment label and any description sentence.
Google's description: Maximum babel similarity between the anchor and the description sentences.
type number() · confidence medium

`VideoContentSearchAnchorCommonFeatureSet.dolphinDescriptivenessScore`
Rule: Predicted rating of how descriptive and useful a video key moment is.
Google's description: The predicted descriptiveness and usefulness rating scores generated by the Unified Dolphin model. Rating template: experimental/video/video_anchors_oneside_without_thumbnail/template.jhtml
type number() · confidence medium

`VideoContentSearchAnchorCommonFeatureSet.dolphinEnsembleScore`
Rule: Individual model scores when the key moment scorer is an ensemble.
Google's description: If the dolphin model is an ensemble model, this contains the scores associated to each individual ensemble model.
type list(GoogleApi.ContentWarehouse.V1.Model.VideoContentSearchDolphinEnsembleScore.t) · confidence medium

`VideoContentSearchAnchorCommonFeatureSet.dolphinFeatures`
Rule: The inputs used to produce the key moment's Dolphin score.
Google's description: The features used to generate the Dolphin score.
type GoogleApi.ContentWarehouse.V1.Model.VideoContentSearchDolphinFeatures.t · confidence medium

`VideoContentSearchAnchorCommonFeatureSet.dolphinScore`
Rule: The Dolphin model score for a video key moment.
Google's description: The score generated by the Dolphin callout model.
type number() · confidence medium

`VideoContentSearchAnchorCommonFeatureSet.labelPhraseEmbedding`
Rule: A numeric embedding of the key moment label text.
Google's description: A phrase embedding for the anchor label. The model used to generate the embedding can be found in VideoAnchorSets: video_score_info.common_features.label_phrase_embedding_model
type list(number()) · confidence medium

`VideoContentSearchAnchorCommonFeatureSet.mumDescriptivenessScore`
Rule: Predicted descriptiveness of a key moment from the MUM model.
Google's description: The predicted descriptiveness of the anchor using the MUM unified scoring model.
type number() · confidence medium

`VideoContentSearchAnchorCommonFeatureSet.mumUsefulnessScore`
Rule: Predicted usefulness of a key moment from the MUM model.
Google's description: The predicted usefulness of the anchor using the MUM unified scoring model.
type number() · confidence medium

`VideoContentSearchAnchorCommonFeatureSet.saftDocument`
Rule: A linguistic annotation built from the key moment label text.
Google's description: A saft document generated from the anchor label.
type GoogleApi.ContentWarehouse.V1.Model.NlpSaftDocument.t · confidence medium

`VideoContentSearchAnchorCommonFeatureSet.timedLabelFeatures`
Rule: Labels found in a passage with their timing and surrounding context.
Google's description: For annotating labels and their timing and context info. For example, this is used for anchor labels within a passage.
type list(GoogleApi.ContentWarehouse.V1.Model.VideoContentSearchCaptionLabelFeatures.t) · confidence medium

`VideoContentSearchAnchorCommonFeatureSet.titleAnchorBabelMatchScore`
Rule: Text similarity between the key moment label and the video title.
Google's description: Babel similarity between the anchor and the video title.
type number() · confidence medium

`VideoContentSearchAnchorCommonFeatureSetLabelSpanTimestamp.asrConfidence`
Rule: How confident speech recognition was about this piece of text.
Google's description: The ASR confidence for the label span, if available.
type number() · confidence medium

`VideoContentSearchAnchorThumbnail.isThumbnailMissing`
Rule: Whether no thumbnail could be made for this key moment.
Google's description: Set to true when no thumbnail could be generated for this anchor.
type boolean() · confidence medium

`VideoContentSearchAnchorThumbnail.servingMetadata`
Rule: Size and dimension details for the key moment thumbnail.
Google's description: Metadata about the anchor thumbnail computed by Amarna, including dimensions and the size in bytes.
type GoogleApi.ContentWarehouse.V1.Model.ImageBaseThumbnailMetadata.t · confidence medium

`VideoContentSearchAnchorThumbnail.timestampMs`
Rule: The moment in the video that the thumbnail frame was taken from.
Google's description: Millisecond timestamp of the frame used for the thumbnail.
type integer() · confidence medium

`VideoContentSearchAnchorThumbnailInfo.colorEntropy`
Rule: How varied the colours are in the thumbnail image.
Google's description: Entropy of the clustered color distribution.
type number() · confidence medium

`VideoContentSearchAnchorThumbnailInfo.imageData`
Rule: The thumbnail image itself, kept for safety classification.
Google's description: Thumbnail image data for SafeSearch classification.
type GoogleApi.ContentWarehouse.V1.Model.ImageData.t · confidence medium

`VideoContentSearchAnchorThumbnailInfo.isUnsafe`
Rule: A single flag saying whether the thumbnail is considered unsafe.
Google's description: Convenience field that consolidates signals for whether this thumbnail is safe.
type boolean() · confidence medium

`VideoContentSearchAnchorThumbnailInfo.starburstV4Embedding`
Rule: A 64 number visual embedding of the thumbnail.
Google's description: 64d float vector of starburst v4 embedings.
type GoogleApi.ContentWarehouse.V1.Model.DrishtiDenseFeatureData.t · confidence medium

`VideoContentSearchAnchorsCommonFeatureSet.dolphinDescriptivenessStats`
Rule: Summary of descriptiveness scores across a video's key moments.
Google's description: A summary of the Dolphin descriptiveness scores of the anchors in the set.
type GoogleApi.ContentWarehouse.V1.Model.VideoContentSearchMetricStats.t · confidence medium

`VideoContentSearchAnchorsCommonFeatureSet.dolphinUsefulnessStats`
Rule: Summary of usefulness scores across a video's key moments.
Google's description: A summary of the Dolphin usefulness scores of the anchors in the set.
type GoogleApi.ContentWarehouse.V1.Model.VideoContentSearchMetricStats.t · confidence medium

`VideoContentSearchAnchorsCommonFeatureSet.mumDescriptivenessStats`
Rule: Summary of MUM descriptiveness scores across a video's key moments.
Google's description: A summary of the MUM descriptiveness scores of the anchors in the set.
type GoogleApi.ContentWarehouse.V1.Model.VideoContentSearchMetricStats.t · confidence medium

`VideoContentSearchAnchorsCommonFeatureSet.mumUsefulnessStats`
Rule: Summary of MUM usefulness scores across a video's key moments.
Google's description: A summary of the MUM usefulness scores of the anchors in the set.
type GoogleApi.ContentWarehouse.V1.Model.VideoContentSearchMetricStats.t · confidence medium

`VideoContentSearchAnchorsThumbnailInfo.thumbnailDiversity`
Rule: How visually different a video's key moment thumbnails are from each other.
Google's description: A score representing how diverse a set of thumbnails is. This is currently defined as one minus the median pairwise cosine similarity between thumbnail Starburst embeddings.
type number() · confidence medium

`VideoContentSearchAsrCaption.confidence`
Rule: How confident speech recognition is about a transcribed word.
Google's description: The confidence score of the token: between 0 and 1.
type number() · confidence medium

`VideoContentSearchAsrCaption.durationMs`
Rule: How long the spoken word lasts.
Google's description: The duration that the token is spoken for.
type number() · confidence medium

`VideoContentSearchAsrCaption.startTimeMs`
Rule: When in the video the word starts being spoken.
Google's description: The time in the video at which the token starts being spoken.
type integer() · confidence medium

`VideoContentSearchAsrCaption.text`
Rule: The transcribed spoken word itself.
Google's description: The speech token.
type String.t · confidence medium

`VideoContentSearchCaptionEntityAnchorFeatures.asrSentence`
Rule: The closest sentence from the spoken transcript.
Google's description: The nearest ASR sentence.
type String.t · confidence medium

`VideoContentSearchCaptionEntityAnchorFeatures.asrStartTime`
Rule: When the matched transcript sentence begins.
Google's description: The begin time in ms of the ASR sentence.
type integer() · confidence medium

`VideoContentSearchCaptionEntityAnchorFeatures.bertScores`
Rule: BERT outputs computed from the title, transcript sentence and mention text.
Google's description: The result of the BERT inference using the title, ASR sentence and entity mention text.
type list(number()) · confidence medium

`VideoContentSearchCaptionEntityAnchorFeatures.durationCoverage`
Rule: What share of the video lies between the first and last mention of the topic.
Google's description: The percentage of the video covered by the span of the first mention to the last mention.
type number() · confidence medium

`VideoContentSearchCaptionEntityAnchorFeatures.estimatedMentionTime`
Rule: The estimated time in the video when the topic is mentioned.
Google's description: The estimated begin time in ms of the entity mention using the text offset divided by the ASR duration.
type integer() · confidence medium

`VideoContentSearchCaptionEntityAnchorFeatures.groupCohesion`
Rule: How similar this key moment is to the others in its group.
Google's description: Average similarity between this anchor and other anchors in the set.
type number() · confidence medium

`VideoContentSearchCaptionEntityAnchorFeatures.msFromLastAnchor`
Rule: Time since the previous key moment, or since the video started.
Google's description: The duration in ms between this anchor and the previous anchor or the beginning of the video if this is the first anchor.
type integer() · confidence medium

`VideoContentSearchCaptionEntityAnchorFeatures.nextAsrSentence`
Rule: The transcript sentence that comes after the matched one.
Google's description: The ASR sentence after asr_sentence or "" if it is the last sentence.
type String.t · confidence medium

`VideoContentSearchCaptionEntityAnchorFeatures.previousAsrSentence`
Rule: The transcript sentence that comes before the matched one.
Google's description: The ASR sentence before asr_sentence or "" if it is the first sentence.
type String.t · confidence medium

`VideoContentSearchCaptionLabelFeatures.alignedOcrTexts`
Rule: On screen text detected during the same stretch of video as this key moment.
Google's description: OCR anchors with overlapping time-window with this anchor
type list(GoogleApi.ContentWarehouse.V1.Model.VideoContentSearchOCRText.t) · confidence medium

`VideoContentSearchCaptionLabelFeatures.alignedTime`
Rule: The video timestamp that the reference text points at.
Google's description: The time stamp in milliseconds for the reference text (e.g. description anchor time).
type String.t · confidence medium

`VideoContentSearchCaptionLabelFeatures.contextText`
Rule: Transcript text from a wide window around the aligned time.
Google's description: Text around the aligned_time of a long duration, say [-15 minutes, +15 minutes]
type String.t · confidence medium

`VideoContentSearchCaptionLabelFeatures.labelText`
Rule: The main label text for the key moment.
Google's description: The main label text for the feature.
type String.t · confidence medium

`VideoContentSearchCaptionLabelFeatures.textSimilarityFeatures`
Rule: Matching text found by similarity comparison.
Google's description: Identified matching text by similarity.
type GoogleApi.ContentWarehouse.V1.Model.VideoContentSearchTextSimilarityFeatures.t · confidence medium

`VideoContentSearchCaptionLabelFeatures.textSpanAtAlignedTime`
Rule: The stretch of transcript text starting at the aligned time.
Google's description: The text span in the passage starting from the aligned time.
type String.t · confidence medium

`VideoContentSearchCaptionSpanAnchorFeatures.dolphinFeatures`
Rule: The inputs assembled for the Dolphin span model.
Google's description: The features used to construct the inference example.
type GoogleApi.ContentWarehouse.V1.Model.VideoContentSearchSpanDolphinFeatures.t · confidence medium

`VideoContentSearchCaptionSpanAnchorFeatures.dolphinScores`
Rule: Scores returned by the Dolphin span model.
Google's description: The inference result from the Dolphin span model.
type GoogleApi.ContentWarehouse.V1.Model.VideoContentSearchSpanDolphinScores.t · confidence medium

`VideoContentSearchCaptionSpanAnchorFeatures.embeddingDistance`
Rule: How far this key moment's embedding sits from the video's other key moments.
Google's description: Embedding distances (e.g. cosine distance) to the other anchors of the same video.
type list(number()) · confidence medium

`VideoContentSearchCaptionSpanAnchorFeatures.postGapInMs`
Rule: Time until the next key moment.
Google's description: Time gap in ms to the next anchor. Always positive number.
type integer() · confidence medium

`VideoContentSearchCaptionSpanAnchorFeatures.preGapInMs`
Rule: Time since the previous key moment.
Google's description: Time gap in ms to the previous anchor. Always a positive number.
type integer() · confidence medium

`VideoContentSearchCaptionSpanAnchorFeatures.spanAsrConfidenceStats`
Rule: Summary of speech recognition confidence for the chosen text span.
Google's description: A summary of the ASR confidence for the selected candidate.
type GoogleApi.ContentWarehouse.V1.Model.VideoContentSearchMetricStats.t · confidence medium

`VideoContentSearchCaptionSpanAnchorFeatures.spanDolphinScore`
Rule: Summary of Dolphin word scores for the chosen text span.
Google's description: A summary of the Dolphin span token scores for the selected candidate.
type GoogleApi.ContentWarehouse.V1.Model.VideoContentSearchMetricStats.t · confidence medium

`VideoContentSearchCaptionSpanAnchorFeatures.wordCount`
Rule: How many words the text span contains.
Google's description: Word count of the span text, tokenized with SAFT.
type integer() · confidence medium

`VideoContentSearchCaptionSpanAnchorSetFeatures.anchorSetDolphinScoreStats`
Rule: Summary of Dolphin scores across a video's set of key moments.
Google's description: A summary of the dolphin scores over the anchor set.
type GoogleApi.ContentWarehouse.V1.Model.VideoContentSearchMetricStats.t · confidence medium

`VideoContentSearchDescriptionAnchorFeatures.inAsr`
Rule: Whether the description key moment is also mentioned in the spoken transcript.
Google's description: Whether or not a mention of the description anchor exists in the ASR.
type boolean() · confidence medium

`VideoContentSearchDescriptionAnchorFeatures.spanToAsrTime`
Rule: How far the transcript sentence sits in time from the description key moment.
Google's description: The distance from the ASR sentence to the description anchor time in ms.
type integer() · confidence medium

`VideoContentSearchDescriptionAnchorSetFeatures.asrAnchorCount`
Rule: How many description key moments were matched to the transcript.
Google's description: The number of description anchors that were matched to captions in the ASR.
type integer() · confidence medium

`VideoContentSearchDescriptionAnchorSetFeatures.asrAnchorFraction`
Rule: The share of key moments that were matched to the transcript.
Google's description: The fraction of anchors that were matched to captions in the ASR.
type number() · confidence medium

`VideoContentSearchDescriptionSpanInfo.contextTokenCount`
Rule: How many words are in the sentence the description span came from.
Google's description: The number of tokens in the context (sentence) where the description span is extracted from.
type integer() · confidence medium

`VideoContentSearchDescriptionSpanInfo.dolphinScores`
Rule: Dolphin span results when the key moment comes from a description span.
Google's description: The inference result from the Dolphin span model if the anchor's source is description span.
type GoogleApi.ContentWarehouse.V1.Model.VideoContentSearchSpanDolphinScores.t · confidence medium

`VideoContentSearchDescriptionSpanInfo.spanDolphinScoreStats`
Rule: Summary of Dolphin word scores for the description span.
Google's description: A summary of the Dolphin span token scores for the selected candidate. Currently, dolphin_scores in DescriptionSpanInfo would only contain a single span candidate so this field is essentially the copy of the score_stats for that span candidate.
type GoogleApi.ContentWarehouse.V1.Model.VideoContentSearchMetricStats.t · confidence medium

`VideoContentSearchDescriptionSpanInfo.spanTokenCount`
Rule: How many words the description span contains.
Google's description: The number of tokens in the description span. The description span is formed from non-contiguous segment spans of a sentence (context), where each segment span's score satifsies the min span thresholds.
type integer() · confidence medium

`VideoContentSearchDescriptionSpanInfo.spanTokenCountRatio`
Rule: The span's word count as a fraction of its sentence's word count.
Google's description: The ratio of span_token_count / context_token_count.
type number() · confidence medium

`VideoContentSearchDolphinEnsembleScore.dolphinScore`
Rule: A score from one model inside the Dolphin ensemble.
Google's description: The score generated by the Dolphin callout model.
type number() · confidence medium

`VideoContentSearchDolphinFeatures.altQuery`
Rule: The alternative query text used to build the model input.
Google's description: The alt query used for building the Dolphin example.
type String.t · confidence medium

`VideoContentSearchDolphinFeatures.answer`
Rule: The answer text used to build the model input.
Google's description: The answer used for building the Dolphin example.
type String.t · confidence medium

`VideoContentSearchDolphinFeatures.query`
Rule: The query text used to build the model input.
Google's description: The query used for building the Dolphin example.
type String.t · confidence medium

`VideoContentSearchDolphinFeatures.timeMs`
Rule: The video timestamp of the key moment used in the model input.
Google's description: The time stamp of the video anchor in milliseconds.
type String.t · confidence medium

`VideoContentSearchDolphinFeatures.title`
Rule: The video title used to build the model input.
Google's description: The title used for building the Dolphin example.
type String.t · confidence medium

`VideoContentSearchEntityGroupInfo.label`
Rule: The label shown for a group of key moments.
Google's description: Label for this anchor group.
type String.t · confidence medium

`VideoContentSearchFrameSimilarityInterval.frameSimilarity`
Rule: How closely each video frame matches the topic.
Google's description: The similarity between this topic and starburst features for frames in [frames[frame_level_starburst_start_index], frames[frame_level_starburst_start_index + len(frame_similarity)].
type list(number()) · confidence medium

`VideoContentSearchFrameSimilarityInterval.framesEndTimestampMs`
Rule: When the last frame in this stretch of video occurs.
Google's description: Timestamp in milliseconds for the last frame in this frame interval.
type String.t · confidence medium

`VideoContentSearchFrameSimilarityInterval.framesStartTimestampMs`
Rule: When the first frame in this stretch of video occurs.
Google's description: Timestamp in milliseconds for the first frame in this frame interval.
type String.t · confidence medium

`VideoContentSearchFrameStarburstData.denseVector`
Rule: The numeric visual embedding of a video frame.
Google's description: Raw float feature vector of the starburst representation.
type list(number()) · confidence medium

`VideoContentSearchFrameStarburstData.timestampMs`
Rule: When this frame occurs in the video.
Google's description: Timestamp in milliseconds for this frame.
type String.t · confidence medium

`VideoContentSearchGenerativePredictionFeatures.passage`
Rule: The passage of text fed to a generative model.
Google's description: Features for inferences from generative models.
type String.t · confidence medium

`VideoContentSearchGenerativePredictionFeatures.predictions`
Rule: The text that a generative model produced.
Google's description: Inference results.
type list(String.t) · confidence medium

`VideoContentSearchGenerativeTopicPredictionFeatures.predictions`
Rule: The candidate topics a generative model produced for the video.
Google's description: Inference results from the prediction service. Since we generally use beam search with beam_size > 1, this field is repeated to capture all the generated topic beams.
type list(String.t) · confidence medium

`VideoContentSearchListAnchorFeatures.babelMatch`
Rule: How well a list item from the description matched the transcript text.
Google's description: The babel match info of the list anchor with its matched ASR text.
type GoogleApi.ContentWarehouse.V1.Model.VideoContentSearchTextMatchInfo.t · confidence medium

`VideoContentSearchListAnchorFeatures.descriptionSpanInfo`
Rule: Details about the description span a list key moment came from.
Google's description: The description span metadata about list anchor when the anchor source is DESCRIPTION_SPANS.
type GoogleApi.ContentWarehouse.V1.Model.VideoContentSearchDescriptionSpanInfo.t · confidence medium

`VideoContentSearchListAnchorFeatures.matchScores`
Rule: How well this list item matched various pieces of the transcript.
Google's description: The metadata about this list item's matches with different ASR snippets. This is currently used in the base model (DTW) to generate candidate anchors.
type list(GoogleApi.ContentWarehouse.V1.Model.VideoContentSearchMatchScores.t) · confidence medium

`VideoContentSearchListAnchorFeatures.pretriggerScore`
Rule: The score from the early filtering model.
Google's description: The score from the pretrigger model.
type number() · confidence medium

`VideoContentSearchListAnchorFeatures.titleAnchorBabelMatchScore`
Rule: Text similarity between the key moment and the video title.
Google's description: Babel similarity between the anchor and the video title.
type number() · confidence medium

`VideoContentSearchListAnchorSetFeatures.aggregatedSpanText`
Rule: The combined span text across all candidates in the key moment set.
Google's description: The following fields are used for description span anchors, The aggregated span token texts over all the span candidates of the anchor set.
type String.t · confidence medium

`VideoContentSearchListAnchorSetFeatures.anchorSetSpanScoreStats`
Rule: Summary of span scores across the key moment set.
Google's description: A summary of the span scores over the anchor set. This summary is calculated over the aggregation of the individual token spans belonging to the span candidates of anchors.
type GoogleApi.ContentWarehouse.V1.Model.VideoContentSearchMetricStats.t · confidence medium

`VideoContentSearchListAnchorSetFeatures.babelMatchScoreStats`
Rule: Median, average and spread of text match scores across the key moment set.
Google's description: Median, average and standard deviation of babel_match_score among anchors in the same VideoAnchors cluster.
type GoogleApi.ContentWarehouse.V1.Model.VideoContentSearchMetricStats.t · confidence medium

`VideoContentSearchListAnchorSetFeatures.contextTokenCountStats`
Rule: Summary of sentence word counts across the key moment set.
Google's description: A summary of the context token counts over the anchor set.
type GoogleApi.ContentWarehouse.V1.Model.VideoContentSearchMetricStats.t · confidence medium

`VideoContentSearchListAnchorSetFeatures.durationSpanRatio`
Rule: The share of the video's length from the first key moment to the last.
Google's description: The ratio of anchors timespan duration over the total duration of the video. Anchors timespan duration is defined as the time span from the first anchor to the last anchor in VideoAnchors.
type number() · confidence medium

`VideoContentSearchListAnchorSetFeatures.durationToPredictedTimeMsStats`
Rule: Median, average and spread of the gap to predicted times across key moments.
Google's description: Median, average and standard deviation of duration_to_predicted_time_ms among anchors in the same VideoAnchors cluster.
type GoogleApi.ContentWarehouse.V1.Model.VideoContentSearchMetricStats.t · confidence medium

`VideoContentSearchListAnchorSetFeatures.listAnchorSource`
Rule: Where the list key moments were taken from in the video description.
Google's description: The source of anchors extracted from the video descriptions.
type String.t · confidence medium

`VideoContentSearchListAnchorSetFeatures.listDescriptionItemsSize`
Rule: How many list items the video description contains.
Google's description: The total number of list items mentioned in the video description. Not all these list items are necessarily found as list anchors.
type integer() · confidence medium

VideoContentSearchListAnchorSetFeatures.matchedListDescriptionAnchorsRatio
Rule: The share of eligible description list items that matched the transcript.
Google's description: The number of matched anchors in the list anchors over the total number of post-filtering list items in the video description, i.e. matched_list_description_anchors_ratio = matched_list_description_anchors_size / post_filtering_list_description_items_size.
type number() · confidence medium

VideoContentSearchListAnchorSetFeatures.matchedListDescriptionAnchorsSize
Rule: How many description list items were matched in the transcript.
Google's description: The number of matched list anchors found in the ASR. The matched list anchors are a subset of the post-filtering list items in the video description, and as such matched_list_description_anchors_size <= post_filtering_list_description_items_size.
type integer() · confidence medium

VideoContentSearchListAnchorSetFeatures.postFilteringListDescriptionItemsSize
Rule: How many description list items survived filtering and were considered.
Google's description: The total number of list items in the video description that are actually considered for matching. This is a subset of list items in the video description that passed filterings such as language filtering, i.e. post_filtering_list_description_items_size <= list_description_items_size
type integer() · confidence medium

VideoContentSearchListAnchorSetFeatures.pretriggerScoreStats
Rule: Median, average and spread of early filter scores across key moments.
Google's description: Median, average and standard deviation of pretrigger_score among anchors in the same cluster.
type GoogleApi.ContentWarehouse.V1.Model.VideoContentSearchMetricStats.t · confidence medium

VideoContentSearchListAnchorSetFeatures.spanTokenCountRatioStats
Rule: Summary of span to sentence word ratios across the key moment set.
Google's description: A summary of the span token count ratios over the anchor set.
type GoogleApi.ContentWarehouse.V1.Model.VideoContentSearchMetricStats.t · confidence medium

VideoContentSearchListAnchorSetFeatures.spanTokenCountStats
Rule: Summary of span word counts across the key moment set.
Google's description: A summary of the span token counts over the anchor set.
type GoogleApi.ContentWarehouse.V1.Model.VideoContentSearchMetricStats.t · confidence medium

VideoContentSearchMultimodalTopicFeatures.topic
Rule: The text of a topic generated for a section of the video.
Google's description: The text of the generated topic.
type String.t · confidence medium

VideoContentSearchMultimodalTopicFeatures.topicEndMs
Rule: The end time of the video topic.
Google's description: End time of the topic.
type String.t · confidence medium

VideoContentSearchMultimodalTopicFeatures.topicStartMs
Rule: The start time of the video topic.
Google's description: Start time of the topic.
type String.t · confidence medium

VideoContentSearchQnaAnchorFeatures.answer
Rule: A passage of the video's spoken text used as the answer.
Google's description: Segment of text from the ASR.
type String.t · confidence medium

VideoContentSearchQnaAnchorFeatures.endMs
Rule: The end time of the answer segment in the video.
Google's description: End time in milliseconds relative to the beginning of the video.
type String.t · confidence medium

VideoContentSearchQnaAnchorFeatures.question
Rule: The question text attached to this point in the video.
Google's description: Question from Related Questions SSTable or NavBoost.
type String.t · confidence medium

VideoContentSearchQnaAnchorFeatures.startMs
Rule: The start time of the answer segment in the video.
Google's description: Start time in milliseconds relative to the beginning of the video.
type String.t · confidence medium

VideoContentSearchShoppingOpinionsAnchorFeatures.anchorLabel
Rule: The label shown for this video opinion anchor.
Google's description: The anchor label.
type String.t · confidence medium

`VideoContentSearchTextSimilarityFeatures.similarityScore`
Rule: How closely a candidate anchor label matches the video's spoken text.
Google's description: The similarity score given by the scoring method specified by the message scoring_method_name.
type number() · confidence medium

`VideoContentSearchVideoAnchor.anchorScore`
Rule: How confident the system is in a key moment marker for a video.
Google's description: The score indicating anchor confidence.
type number() · confidence medium

`VideoContentSearchVideoAnchor.duration`
Rule: How long the key moment segment lasts, in milliseconds.
Google's description: The duration of the video anchor in milliseconds.
type String.t · confidence medium

`VideoContentSearchVideoAnchor.isSafe`
Rule: A combined flag for whether the key moment label is safe to show.
Google's description: Convenience field that consolidates signals for whether this label is safe.
type boolean() · confidence medium

`VideoContentSearchVideoAnchor.labelScore`
Rule: How confident the system is in the text chosen for a key moment.
Google's description: The score indicating label confidence.
type number() · confidence medium

`VideoContentSearchVideoAnchor.precisionScore`
Rule: The precision level a key moment must clear before it is allowed to show.
Google's description: The precision for which the anchor should trigger. For example, if the desired precision is 95%, anchors with precision_score < 0.95 should be removed.
type number() · confidence medium

`VideoContentSearchVideoAnchors.predictedQuerylessTocUsefulness`
Rule: Predicted usefulness of this key moment set as a table of contents.
Google's description: The score that predicts the usefulness of this anchor set on the Huh table of contents eval without considering the query.
type number() · confidence medium

`VideoContentSearchVideoAnchors.score`
Rule: An overall quality score for the set of key moments.
Google's description: The quality of the anchor set.
type number() · confidence medium

`VideoContentSearchVideoAnchors.videoIntroduction`
Rule: Information about the video's opening segment.
Google's description: Information about the video's introduction segment.
type GoogleApi.ContentWarehouse.V1.Model.VideoContentSearchVideoIntroduction.t · confidence medium

`VideoContentSearchVideoInfo.description`
Rule: The video's description text.
Google's description: Video description.
type String.t · confidence medium

`VideoContentSearchVideoInfo.durationMs`
Rule: How long the video runs, in milliseconds.
Google's description: Video duration in ms.
type number() · confidence medium

`VideoContentSearchVideoInfo.hasAsr`
Rule: Whether speech recognition has produced a transcript for this video.
Google's description: Whether or not automatic speech recognition has been generated for this video.
type boolean() · confidence medium

`VideoContentSearchVideoInfo.hasDescriptionAnchors`
Rule: Whether the video has chapter markers written in its description.
Google's description: Whether or not the video has description anchors.
type boolean() · confidence medium

`VideoContentSearchVideoInfo.isSafe`
Rule: Whether all of the video's key moments were judged safe.
Google's description: Convenience field that is false if any of the video's anchors have their is_safe field set to false.
type boolean() · confidence medium

`VideoContentSearchVideoInfo.title`
Rule: The video's title.
Google's description: Video title.
type String.t · confidence medium

`VideoContentSearchVideoIntroduction.hasIntro`
Rule: Whether the video has an intro segment.
Google's description: If set to true, it means the video has an introduction spanning from intro_start_ms to intro_end_ms.
type boolean() · confidence medium

`VideoContentSearchVideoIntroduction.introEndMs`
Rule: The time the video's intro ends, so viewers can skip to it.
Google's description: Timestamp of the end of an introduction. Will only be set if has_intro is true. Indicates video may be skipped to this timestamp with minimal impact on understanding the overall video contents.
type String.t · confidence medium

`VideoContentSearchVideoIntroduction.introStartMs`
Rule: The time the video's intro begins.
Google's description: Timestamp of the beginning of an introduction. Will only be set if has_intro is true. This value may be nonzero.
type String.t · confidence medium

`WWWDocInfo.colorDetectionResult`
Rule: The dominant color detected in the image, packed as a number.
Google's description: Detected color in the image in RGB565 format in the lower 16 bits.
type integer() · confidence medium

`WWWDocInfo.licensedWebImagesOptInState`
Rule: Whether the site owner opted this image into the licensed-images program.
Google's description: Indicates the web-master opt-in state of this image. This project is still in MVP stage, please contact us licensed-media-team@ before use.
type String.t · confidence medium

`WWWDocInfo.nearbyText`
Rule: Text surrounding the image on its page, used to build scroll-to-image links.
Google's description: Nearby text of the image on landing page. Used to construct Scroll to Image urls.
type String.t · confidence medium

`WWWResultInfoSubImageDocInfo.salientColorInfo`
Rule: The main color of the image and where that color reading came from.
Google's description: 4-bytes: (low order on the left) RRRRRRRR GGGGGGGG BBBBBBBB SS where R: 8-bits encoding color 'r' G: 8-bits encoding color 'g' B: 8-bits encoding color 'b' S: 2-bits encoding the color source - 00 = from color detection result - 01 = from cairo This field has the salient color information.
type integer() · confidence medium

`ImageData.embeddedMetadata`
Rule: The camera and authoring metadata embedded in the image file.
Google's description: the EXIF/IPTC metadata
type GoogleApi.ContentWarehouse.V1.Model.ImageExifImageEmbeddedMetadata.t · confidence low

`ImageData.extendedExif`
Rule: A richer set of embedded image metadata produced by the photos backend.
Google's description: The EXIF generated by photos backend team's (more specifically FIFE's) thumbnailer library. This exif model is more comprehensive since a dedicated team is constantly improving it and adding new fields over time. This is currently populated by moonshine for selected corpora.
type GoogleApi.ContentWarehouse.V1.Model.PhotosImageMetadata.t · confidence low

`ImageData.generatedImageLikelihoodSignal`
Rule: How likely it is that the picture was made by Google's image generation.
Google's description: The signal about the likelihood of whether the image is generated by Google.
type GoogleApi.ContentWarehouse.V1.Model.ImageRepositoryGeneratedImageLikelihoodSignal.t · confidence low

`ImageData.largestFaceFraction`
Rule: An older scaled version of the largest-face fraction that should not be used.
Google's description: Fraction of image covered by the largest face, scaled by 1000. Warning: Is DEPRECATED in favor of largest_face_frac. Do not use.
type integer() · confidence low

`ImageExifIPTCMetadata.digitalSourceType`
Rule: A tag in the image metadata saying how the image was originally produced.
Google's description: See go/gi-annotations-indexing for more information.
type String.t · confidence low

`ImageExifIPTCMetadata.title`
Rule: The title given to the image in its embedded metadata.
Google's description: IPTC description related fields.
type String.t · confidence low

`ImageExifIPTCMetadataArtwork.title`
Rule: The title of the artwork shown in the image.
Google's description: Other fields omitted.
type String.t · confidence low

`ImageExifImageEmbeddedMetadata.comments`
Rule: A free-text comment stored inside the image file.
Google's description: EXIF_TAG_USER_COMMENT
type String.t · confidence low

`ImageExifImageEmbeddedMetadata.comments2`
Rule: A second free-text comment field stored inside the image file.
Google's description: EXIF_TAG_XP_COMMENT
type String.t · confidence low

`ImageExifImageEmbeddedMetadata.description`
Rule: A description of the picture stored inside the image file.
Google's description: EXIF_TAG_IMAGE_DESCRIPTION
type String.t · confidence low

`ImageExifImageEmbeddedMetadata.keywords`
Rule: Keywords the author stored inside the image file.
Google's description: EXIF_TAG_XP_KEYWORDS
type String.t · confidence low

`ImageExifImageEmbeddedMetadata.subject`
Rule: A subject line stored inside the image file.
Google's description: EXIF_TAG_XP_SUBJECT
type String.t · confidence low

`ImageExifImageEmbeddedMetadata.title`
Rule: A title stored inside the image file by the author.
Google's description: EXIF_TAG_XP_TITLE
type String.t · confidence low

`ImagePerDocData.DEPRECATEDEntropyColor`
Rule: An old packed color and detail summary of the thumbnail, no longer used.
Google's description: entropy and color values for thumbnail (4 bytes consisting of R, G, B and entropy values)
type integer() · confidence low

`ImagePerDocData.filename`
Rule: The file name of the image stored with the page record.
Google's description: about 10 bytes
type String.t · confidence low

`ImageRegionsImageRegion.isProduct`
Rule: Whether the detected region is considered a product.
Google's description: Set true if the region represents a product, i.e., if any of its labels are on a product labels whitelist. See ImageRegionsConfig for details on the product whitelist.
type boolean() · confidence low

`ImageRegionsImageRegions.precomputedRestricts`
Rule: Precomputed restriction flags attached to the image for filtering at serving time.
Google's description: The precomputed restricts for the image.
type GoogleApi.ContentWarehouse.V1.Model.PrecomputedRestricts.t · confidence low

`ImageRepositoryAnimatedImagePerdocData.durationMs`
Rule: Total length of the animation in milliseconds.
Google's description: Total duration of animation, in ms.
type integer() · confidence low

`ImageRepositoryContentBasedVideoMetadata.mediaInfo`
Rule: Basic facts about the media file such as duration, resolution and stream details.
Google's description: Information about the media file, such as duration, resolution, and detail about each audio/video stream. Note that it contains no PII.
type GoogleApi.ContentWarehouse.V1.Model.VideoMediaInfo.t · confidence low

`ImageRepositoryContentBasedVideoMetadata.videoDurationSec`
Rule: Length of the video in seconds.
Google's description: Set from the video header if truncated, or is the verified length if completely crawled.
type float() · confidence low

`ImageSearchImageLicenseInfo.isRetiredCcUrl`
Rule: Whether the image's license link is on the retired licenses list.
Google's description: Whether this license url is in retired license list, which is from:
type boolean() · confidence low

`IndexingSpeechSpeechPropertiesProto.estimatedAudioDurationConfidence`
Rule: How much to trust the audio length estimate, on a zero to one scale.
Google's description: Our confidence in the duration estimate, on a scale from 0 (not confident) to 1 (very confident). An estimate should have a confidence of at least 0.5 if it is to be shown to users.
type number() · confidence low

`MediaIndexVideoFrame.thumbnailType`
Rule: Which thumbnail formats are available for this video frame.
Google's description: Set of available thumbnail types for this frame. Should be valid image_base.ThumbnailType values (enumerated at
type list(String.t) · confidence low

`PerDocData.nsrIsVideoFocusedSite`
Rule: Deprecated flag marking the site as video-focused but not a major video host.
Google's description: This field is propagated to shards. It will also be populated at serving time by go/web-signal-joins (see b/170607253). Bit indicating whether this site is video-focused, but not hosted on any major known video hosting domains. This field is deprecated - used the equivalent field inside nsr_data_proto instead.
type boolean() · confidence low · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityShoppingShoppingAttachmentOffer.inferredImages`
Rule: Images guessed to belong to an offer, sorted for fast lookup at serving.
Google's description: inferred_images are sorted by inferred_image_id for efficient search during serving.
type list(GoogleApi.ContentWarehouse.V1.Model.ShoppingWebentityShoppingAnnotationInferredImage.t) · confidence low

`QualityVidyaVideoLanguageVideoLanguage.speechClass`
Rule: The kind of speech detected in the video.
Google's description: Type of detected speech.
type String.t · confidence low

`QualityWebanswersVideoTranscriptAnnotations.lang`
Rule: The language the video transcript is written in.
Google's description: The language of the transcript as recorded in Amarna.
type String.t · confidence low

`RepositoryWebrefEntityAnnotations.imageMention`
Rule: Experimental record of the entity appearing in an image.
Google's description: This is an experimental output for go/multiref. Don't use it without consulting the Webref team
type list(GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefImageMention.t) · confidence low

`SnapshotImageNode.isExternal`
Rule: Flags whether an image on a page is hosted or linked from a different site than the page itself.
Google's description: An image is considered external iff both: 1. The image appears in a link that is not in the same org as the document, or the target URL is in a different org. 2. The image src is not in the same org as the document.
type boolean() · confidence low

`VideoContentSearchAnchorCommonFeatureSetLabelSpanTimestamp.isSentenceStart`
Rule: Whether this word is the first word of a sentence.
Google's description: Whether or not this token is the first token in a sentence.
type boolean() · confidence low

`VideoContentSearchAnchorCommonFeatureSetLabelSpanTimestamp.labelBeginCharIndex`
Rule: Where the label text starts within the surrounding text.
Google's description: The character index range for the span. The end index is exclusive.
type String.t · confidence low

`VideoContentSearchAnchorCommonFeatureSetLabelSpanTimestamp.timeMs`
Rule: The point in the video where this piece of text occurs.
Google's description: The time of the span.
type String.t · confidence low

`VideoContentSearchAnchorThumbnail.thumbnailInfo`
Rule: Details about the thumbnail chosen for a key moment.
Google's description: Information about the thumbnail anchor.
type GoogleApi.ContentWarehouse.V1.Model.VideoContentSearchAnchorThumbnailInfo.t · confidence low

`VideoContentSearchAnchorsThumbnailInfo.hasMissingStarburst`
Rule: Whether any key moment thumbnail is missing its visual embedding.
Google's description: Whether or not any of the anchor thumbnails have missing Starburst embeddings.
type boolean() · confidence low

`VideoContentSearchAnchorsThumbnailInfo.hasMissingThumbnails`
Rule: Whether any key moment is missing a thumbnail.
Google's description: Whether or not any of the anchors have missing thumbnails.
type boolean() · confidence low

`VideoContentSearchAspect.fallbackAspect`
Rule: A backup product aspect used when no other aspect is available.
Google's description: Product aspect to be used if non of the other aspects exist.
type String.t · confidence low

`VideoContentSearchAspect.saftAspect`
Rule: A product aspect found by text annotation.
Google's description: Product aspect produced by running SAFT annotation.
type String.t · confidence low

`VideoContentSearchDolphinFeatures.url`
Rule: The address of the video used to build the model input.
Google's description: The url of the video.
type String.t · confidence low

`VideoContentSearchListAnchorFeatures.listItemIndex`
Rule: The position of this item in the description's list.
Google's description: The list item index of this anchor in the video description.
type integer() · confidence low

`VideoContentSearchTextMatchInfo.matchedDescriptionItemIndexRatio`
Rule: Which item in the video's description list this match came from, as a fraction.
Google's description: The ratio of the video description item index this match corresponds to over the total number of list description items for the video.
type number() · confidence low

`VideoContentSearchVideoAnchor.anchorType`
Rule: What kind of key moment marker this is.
Google's description: Specifies the type of the anchor.
type String.t · confidence low

`VideoContentSearchVideoAnchor.isScutiBad`
Rule: Whether an internal review tool flagged this label as bad.
Google's description: whether this label is bad by go/scuti
type boolean() · confidence low

`VideoContentSearchVideoAnchors.experimentalPredictedQuerylessTocUsefulness`
Rule: An experimental version of the predicted usefulness score for a key moment set.
Google's description: Same as above, but used for experimenting with new models.
type number() · confidence low

`VideoContentSearchVideoCommonFeatures.captionInfo`
Rule: The caption or transcript data for the video.
Google's description: The caption data for the video transcript. The models used for unified scorer. Should be a filepath that contains saved_model.pb and a variables/ folder
type GoogleApi.ContentWarehouse.V1.Model.VideoContentSearchCaptionInfo.t · confidence low

`VideoContentSearchVideoInfo.isWatchpage`
Rule: Whether this page is a video watch page.
Google's description: Whether or not this is watchpage.
type boolean() · confidence low

`VideoContentSearchVideoInfo.pseudoVideoData`
Rule: The video transcript with a timestamp for every word.
Google's description: ASR with timing info for each token copied from doc_videos.content_based_metadata.transcript_asr.
type GoogleApi.ContentWarehouse.V1.Model.PseudoVideoData.t · confidence low

`VideoContentSearchVideoInfo.transcriptAnnotations`
Rule: Annotations over the transcript covering timing and mentioned topics.
Google's description: Transcript annotations that include information about the ASR including timing and entity mentions.
type GoogleApi.ContentWarehouse.V1.Model.QualityWebanswersTranscriptAnnotations.t · confidence low

`VideoContentSearchVideoInfo.verticalItem`
Rule: The top subject categories assigned to the video, most confident first.
Google's description: Top petacat verticals of the video produced by indexing/ml/vertical, sorted in descending order by vertical confidence.
type list(GoogleApi.ContentWarehouse.V1.Model.IndexingMlVerticalVerticalItem.t) · confidence low

`VideoContentSearchVideoInfo.videoGenre`
Rule: The genre listed for the video on the page.
Google's description: Genre of the video from the page metadata. Concatenate all with a comma separator if there are multiple genres.
type String.t · confidence low

`WWWDocInfo.coupledUrl`
Rule: The address of a related document, such as the image paired with this page.
Google's description: Url of coupled doc (e.g. image)
type String.t · confidence low

`WWWDocInfo.isHostedImage`
Rule: Whether this is a Google-hosted image.
Google's description: Hosted Images related fields.
type boolean() · confidence low

`WWWDocInfo.visibleImage`
Rule: Whether the paired image was actually visible on the page.
Google's description: If an image request, was the coupled image visible on the page?
type boolean() · confidence low

Crawling & indexing (156)

`IndexingConverterRobotsInfo.noindexReason`
Rule: Bit flags saying why a page is marked noindex.
Google's description: Bit map of RobotedReasons values. When set to a non-zero value, the document should not be indexed or archived etc. based on the name of the tag.
type integer() · confidence high

`ScienceCitationDownloadURL.NoIndex`
Rule: A page request not to display this URL.
Google's description: metatag: don't display this url
type boolean() · confidence high

`WWWDocInfo.seenNoarchive`
Rule: Whether the page carries a noarchive robots tag.
Google's description: Has noarchive meta robots flag
type boolean() · confidence high

`WWWDocInfo.seenNoindex`
Rule: Whether the page carries a noindex robots tag.
Google's description: Has noindex meta robots flag
type boolean() · confidence high

`WWWDocInfo.url`
Rule: The document's address.
Google's description: Url
type String.t · confidence high

`CompositeDocIndexingInfo.contentProtected`
Rule: Whether the page's last good content is kept even though crawling hit an error.
Google's description: Whether current page is under content protection, i.e. a page has been crawled as an error page, but we preserve its last known good content and keep its crawl_status as converter.CrawlStatus::CONTENT.
type boolean() · confidence medium · stages: 2. Document Processing & Indexing

`CompositeDocIndexingInfo.convertToRobotedReason`
Rule: Why the page was treated as blocked by robots rules.
Google's description: If set, indicates that the crawl status was converted to ROBOTED for the reason specified by the enum value in converter.RobotedReasons.ConvertToRobotedReasons. See indexing/converter/proto/converter.proto for details. If unset, then the document was not converted to roboted, and if the document crawl status is ROBOTED, then the document is disallowed (at least to Google) in robots.txt.
type integer() · confidence medium · stages: 2. Document Processing & Indexing

`CompositeDocIndexingInfo.crawlStatus`
Rule: The crawl state of the page, such as having content or being blocked.
Google's description: One of the enum values in converter.CrawlStatus.State (see indexing/converter/proto/converter.proto for details). Default is converter.CrawlStatus::CONTENT. The document is roboted if the value is converter.CrawlStatus::ROBOTED.
type integer() · confidence medium · stages: 2. Document Processing & Indexing

`CompositeDocIndexingInfo.errorType`
Rule: Which kind of error page this document was detected as, if any.
Google's description: One of the enum values in converter.ErrorPageType (see indexing/converter/proto/error-page-detector-enum.proto for detail). Default is converter::ERROR_PAGE_NONE.
type integer() · confidence medium · stages: 2. Document Processing & Indexing

`CorpusSelectionInfo.isSelectedForIndexing`
Rule: Whether the image was chosen to be indexed.
Google's description: Whether an image was selected for indexing.
type boolean() · confidence medium

`CrawlerChangerateUrlChange.interval`
Rule: How many seconds the observed change spanned.
Google's description: The length in seconds of the change.
type integer() · confidence medium · stages: 1. Crawling & Discovery

`CrawlerChangerateUrlChange.offDomainLinksChange`
Rule: Whether the page's links to other domains changed.
Google's description: Whether the content of the off-domain links changed.
type boolean() · confidence medium · stages: 1. Crawling & Discovery

`CrawlerChangerateUrlChange.offDomainLinksCount`
Rule: The new number of links pointing off the page's own domain.
Google's description: The new count of off-domain links, if they changed.
type integer() · confidence medium · stages: 1. Crawling & Discovery

`CrawlerChangerateUrlChange.onDomainLinksCount`
Rule: The new number of links pointing within the page's own domain.
Google's description: The new count of on-domain links, if the count changed.
type integer() · confidence medium · stages: 1. Crawling & Discovery

`CrawlerChangerateUrlChange.onDomainLinksCountChange`
Rule: Whether the number of same-domain links changed.
Google's description: Whether the number of on-domain links changed.
type boolean() · confidence medium · stages: 1. Crawling & Discovery

`CrawlerChangerateUrlChangerate.averageChangeSignificance`
Rule: How big the page's typical change is, used to order crawling.
Google's description: The "significance" of the average change we saw of this document (from 0 to 1). Influenced by content changes. This can be used for prioritizing the crawl (higher significance first).
type float() · confidence medium · stages: 1. Crawling & Discovery; 7. Freshness & Temporal Signals

`CrawlerChangerateUrlChangerate.globalBasedChangePeriod`
Rule: Estimated seconds between page changes, using a global starting assumption.
Google's description: // The changerate estimation based on the global prior. // The global-based changeperiod. This is our estimate (in seconds) for the average time between changes. It is computed using the new prior method based on global_based_prior_period and the global_based_prior_strength specified below. This is used for computing pattern priors. Use pattern_based
type integer() · confidence medium · stages: 1. Crawling & Discovery; 7. Freshness & Temporal Signals

`CrawlerChangerateUrlChangerate.lastChangeSignificance`
Rule: How big the page's most recent change was, used to order crawling.
Google's description: The "significance" of the last change we saw of this document (from 0 to 1). Influenced by content changes, etc. This can be used for prioritizing the crawl (higher significance first).
type float() · confidence medium · stages: 1. Crawling & Discovery; 7. Freshness & Temporal Signals

`CrawlerChangerateUrlChangerate.lastFetched`
Rule: The last time the crawler fetched the page.
Google's description: The last time (unix timestamp) we saw a fetched copy of the document.
type integer() · confidence medium · stages: 1. Crawling & Discovery; 7. Freshness & Temporal Signals

`CrawlerChangerateUrlChangerate.patternBasedChangePeriod`
Rule: Estimated seconds between page changes, using a URL-pattern assumption.
Google's description: // The changerate estimation based on the pattern prior. // The pattern-based changeperiod. This is our estimate (in seconds) for the average time between changes. It is calculated based on the pattern_based_prior_period and pattern_based_prior_strength below. This quantity will eventually replace the old changeperiod calculation.
type integer() · confidence medium · stages: 1. Crawling & Discovery; 7. Freshness & Temporal Signals

`CrawlerChangerateUrlChangerate.ugcChangePeriod`
Rule: Estimated time between changes worked out from user-generated content.
Google's description: // The UGC changerate estimation. // Information on change period generated from user generated content (UGC) change history.
type integer() · confidence medium · stages: 1. Crawling & Discovery; 7. Freshness & Temporal Signals

`CrawlerChangerateUrlHistory.change`
Rule: The full list of changes seen for this URL.
Google's description: All the changes we've seen for this URL.
type list(GoogleApi.ContentWarehouse.V1.Model.CrawlerChangerateUrlChange.t) · confidence medium · stages: 1. Crawling & Discovery

`CrawlerChangerateUrlHistory.latestVersion`
Rule: The most recent recorded version of the URL.
Google's description: The latest version we've seen.
type GoogleApi.ContentWarehouse.V1.Model.CrawlerChangerateUrlVersion.t · confidence medium · stages: 1. Crawling & Discovery

`CrawlerChangerateUrlVersion.timestamp`
Rule: When the crawler fetched this version of the page.
Google's description: The timestamp we crawled the page.
type integer() · confidence medium · stages: 1. Crawling & Discovery

`GDocumentBase.userAgentName`
Rule: Which crawler user agent fetched the URL, such as desktop or smartphone.
Google's description: The user agent name used to crawl the URL. See //crawler/engine/webmirror_user_agents.h for the list of user-agents (e.g. crawler::WebmirrorUserAgents::kGoogleBot). NOTE: This field is copied from the first WEBMIRROR FetchReplyClientInfo in trawler_fetch_info column. We leave this field unpopulated if no WEBMIRROR FecthReplyClientInfo is found. As the submission of cl/51488336, Alexandria starts to populate this field. However, docjoins from freshdocs (or any other source), won't have this field
type String.t · confidence medium · stages: 2. Document Processing & Indexing

`GDocumentBaseContent.ContentType`
Rule: The content type of the fetched document, such as HTML or PDF.
Google's description: See enum ContentType in //depot/google3/webutil/http/content-type.proto.
type integer() · confidence medium

`ImageData.isIndexedByImagesearch`
Rule: Whether image search chose to index this image, and why not if it did not.
Google's description: If this image was not selected for indexing by imagesearch, these fields say so and explain why.
type boolean() · confidence medium

`ImageData.precomputedRestricts`
Rule: Filters worked out ahead of time and stored with the index entry.
Google's description: Restricts computed before building a Mustang index.
type GoogleApi.ContentWarehouse.V1.Model.PrecomputedRestricts.t · confidence medium

`ImageData.restrictStrings`
Rule: A written-out list of all the filters attached to this image.
Google's description: A string representation of all the restricts associated with this image.
type list(String.t) · confidence medium

`ImageData.robotedAgents`
Rule: Which crawler user agents are blocked from this image by robots rules.
Google's description: CSV list of user agents for which this image should be considered roboted. Note: All images are crawled using googlebot-images, this exists for clients that require additional restrictions beyond googlebot-images such as news.
type String.t · confidence medium

`ImageData.unavailableAfterSecs`
Rule: The time after which this image should be treated as no longer available.
Google's description: Time in seconds since epoch after which this image should be considered unavailable.
type String.t · confidence medium

`ImageMoosedogCrawlState.noIndexAfterTimestamp`
Rule: The time after which this image should be treated as no longer available.
Google's description: Time in seconds since epoch after which this image should be considered unavailable.
type String.t · confidence medium

`ImageMoosedogCrawlState.urlDeleted`
Rule: Whether the address was taken down so its fetch result can be removed quickly.
Google's description: Set to true if the url is taken down by clients. This indicates that this crawl state is used to fast remove the crawl result of the url instead of waiting for Multiverse crawl results. For more information, please refer to go/amarna-url-deletion.
type boolean() · confidence medium

`ImageRepositoryContentBasedVideoMetadata.unwantedContent`
Rule: Marks that the video was removed through the media deletion service.
Google's description: If set, video has been deleted using the deletion service (MediaDeletionService).
type GoogleApi.ContentWarehouse.V1.Model.ImageRepositoryUnwantedContent.t · confidence medium

`ImageRepositoryVideoProperties.indexedVerticals`
Rule: Which search verticals the video has been indexed into.
Google's description: Proto to store vertical indexing information for the document. The video can be indexed to multiple verticals, including video search. It can also be indexed only in verticals, not in video search.
type list(GoogleApi.ContentWarehouse.V1.Model.ImageRepositoryVideoPropertiesVerticalIndexingInfoVideo.t) · confidence medium

`ImageSearchImageIndexingInfo.imageLinkSelectionInfo`
Rule: Details about which images on the page were picked for image indexing.
Google's description: Image Selection Info
type list(GoogleApi.ContentWarehouse.V1.Model.ImageSearchImageSelectionInfo.t) · confidence medium

`ImageSearchImageIndexingInfo.rejectedNotIndexedImageLink`
Rule: Images on the page that were neither ready nor selected for image indexing.
Google's description: URLs and Amarna status of images on the page for which image data is not yet available and weren't selected for indexing in image search. Used by consumers of docjoins that need a complete view of image urls on the page (i.e. Digdug).
type list(GoogleApi.ContentWarehouse.V1.Model.ImageSearchUnindexedImageLink.t) · confidence medium

`ImageSearchImageIndexingInfo.selectedNotIndexedImageLink`
Rule: Images on the page selected for indexing but whose data is not ready yet.
Google's description: URLs and Amarna status of images on the page for which image data is not yet available and were otherwise selected for indexing in image search. Used by consumers of docjoins that need a complete view of selected image urls on the page (i.e. Hearse, the index selection testbed).
type list(GoogleApi.ContentWarehouse.V1.Model.ImageSearchUnindexedImageLink.t) · confidence medium

`IndexingConverterRawRedirectInfo.rawFinalTargetFromRendering`
Rule: The last URL a redirect chain landed on when the page was rendered.
Google's description: Final redirect target found from rendering. It is the same as the last element of raw_redirect_chain_from_rendering. It is used as an input source for the indexable fragment detection pipeline and also downstream phases.
type String.t · confidence medium

`IndexingConverterRawRedirectInfo.rawFinalTargetFromTrawler`
Rule: The redirect target including its fragment, used to serve smartphone versions of pages.
Google's description: This is with-fragment version of redirect_with_contents. This field is populated only if there was a fragment. This field is used by indexing::mobile::GetRedirectTarget() defined in indexing/mobile/internal/smartphone-util.cc, which extracts the redirect target for smartphone optimized pages. The extracted target in turn is served in search results for smartphone users. We need with-fragment version because with-fragment url can return different content than fragment-stripped url. For example,
type String.t · confidence medium

`IndexingConverterRobotsInfo.contentExpiry`
Rule: The time after which the page should no longer be shown in results.
Google's description: time in unix time format after which this content should not be shown in the results. This in inferred from the X-Robots-Tag HTTP header with unavailable_after: Do not use this field directly. There is a column called content_expiration in Alexandria that includes this and other signals.
type integer() · confidence medium

`IndexingDupsLocalizedLocalizedClusterTargetLinkTargetDocData.isCanonical`
Rule: Whether the URL was the canonical one when checked.
Google's description: Whether the URL being validated is canonical at the time of processing.
type boolean() · confidence medium

`IndexingEmbeddedContentEmbeddedContentInfo.embeddedLinksInfo`
Rule: Details of the external files the page needs to render.
Google's description: Information about all external resources needed to render this page, a.k.a. embedded links. This includes .css files, images embedded in a page, external javascripts, iframes etc.
type GoogleApi.ContentWarehouse.V1.Model.IndexingEmbeddedContentEmbeddedLinksInfo.t · confidence medium

`IndexingEmbeddedContentEmbeddedLinksInfo.pageSizeInfo`
Rule: How much data the page and its resources weigh.
Google's description: Page download size.
type GoogleApi.ContentWarehouse.V1.Model.IndexingEmbeddedContentPageSizeInfo.t · confidence medium

`IndexingEmbeddedContentEmbeddedLinksInfo.sumHttpResponseLength`
Rule: Combined byte size of the page and all resources it loads.
Google's description: This field is the sum of http_response_length for the embedder and all embedded resources. This is expected to be set only in the docjoins, not in the pinax tables or the exported bigtable.
type integer() · confidence medium

`IndexingEmbeddedContentFetchUrlResponseMetadata.crawlStatus`
Rule: Whether the fetch returned content, was blocked by robots, or errored.
Google's description: The field always exists, and has four options: UNKNOWN, CONTENT, ROBOTED and ERROR, which are defined in indexing.converter.CrawlStatus.
type integer() · confidence medium

`IndexingEmbeddedContentFetchUrlResponseMetadata.fetchWithSmartphoneUa`
Rule: Whether the fetch used the smartphone crawler.
Google's description: True if the response is fetched with SMARTPHONE user agent.
type boolean() · confidence medium

`IndexingEmbeddedContentFetchUrlResponseMetadata.robotsInfo`
Rule: Robots directives from the fetch, used only for the noindex reason.
Google's description: Note that this robots_info should only be used for noindex_reason and will only be present for TARGET_MAIN_FRAME / TARGET_SUBFRAME fetches.
type GoogleApi.ContentWarehouse.V1.Model.IndexingConverterRobotsInfo.t · confidence medium

`IndexingEmbeddedContentLinkInfo.contentLength`
Rule: Byte size of the response body before decompression.
Google's description: Size of the HTTP body (payload of the HTTP response, excluding headers), pre-decompression. Equal to the value of the Content-Length header if any. NOTE: if this proto is converted to from ReferencedResource, we have to use the size of the full HTTP response (i.e. http_response_length) as an approximation, as we could not get the size of HTTP headers.
type integer() · confidence medium

`IndexingEmbeddedContentLinkInfo.crawlDuration`
Rule: How many milliseconds the resource took to download.
Google's description: Time spent downloading this resource, in milliseconds. Not a timestamp!
type integer() · confidence medium

`IndexingEmbeddedContentLinkInfo.crawlStatus`
Rule: Whether the resource fetch returned content, was blocked, or errored.
Google's description: Enum values for crawl_status are defined in indexing/converter/proto/converter.proto
type integer() · confidence medium

`IndexingEmbeddedContentLinkInfo.fetchStatus`
Rule: The crawler's fetch status for the resource.
Google's description: Fetch status from trawler.
type GoogleApi.ContentWarehouse.V1.Model.TrawlerFetchStatus.t · confidence medium

`IndexingEmbeddedContentLinkInfo.httpResponseLength`
Rule: Byte size of the whole HTTP response including headers.
Google's description: Size of the full HTTP response (headers and body pre-decompression). Semantically equal to content_length plus size of the HTTP headers.
type integer() · confidence medium

`IndexingEmbeddedContentLinkInfo.uncompressedContentLength`
Rule: Byte size of the response body after decompression.
Google's description: Size of the HTTP body (payload of the HTTP response, excluding headers), post-decompression. Equal to content_length if the body was not compressed to begin with. NOTE: if this proto is converted to from ReferencedResource, we have to use the size of the full HTTP response as an approximation, as we could not get the size of HTTP headers.
type integer() · confidence medium

`IndexingEmbeddedContentPageSizeInfo.numImages`
Rule: How many images the page loads.
Google's description: Images are also resources. num_images <= num_resources.
type integer() · confidence medium · stages: 2. Document Processing & Indexing

`IndexingEmbeddedContentPageSizeInfo.numImagesWithContent`
Rule: How many of the page's images were fetched successfully.
Google's description: Number of images whose crawl status is CONTENT.
type integer() · confidence medium · stages: 2. Document Processing & Indexing

`IndexingEmbeddedContentPageSizeInfo.numResources`
Rule: How many external resources the page loads.
Google's description: Numbers below don't include embedder.
type integer() · confidence medium · stages: 2. Document Processing & Indexing

`IndexingEmbeddedContentPageSizeInfo.numResourcesWithContent`
Rule: How many of those resources were fetched successfully.
Google's description: Number of resources whose crawl status is CONTENT.
type integer() · confidence medium · stages: 2. Document Processing & Indexing

`IndexingEmbeddedContentPageSizeInfo.sumHttpResponseLength`
Rule: Combined byte size of the page and its resources.
Google's description: Sum of embedder and all referenced resources.
type integer() · confidence medium · stages: 2. Document Processing & Indexing

`IndexingEmbeddedContentRenderingOutputMetadata.withMissingResources`
Rule: Whether some resources were missing when the page was rendered.
Google's description: True if there were any missing resources during the rendering.
type boolean() · confidence medium

`IndexingSpeechSpeechPropertiesProto.truncatedFile`
Rule: Whether only part of the media file was processed, usually due to a crawl size limit.
Google's description: If true, we may have processed a truncated file (most likely due to a size-cutoff when crawling). As a result, the audio duration is a lower bound and the other fields reflect only the processed prefix of the file.
type boolean() · confidence medium

`LogsProtoIndexingCrawlerIdCrawlerIdProto.country`
Rule: The country whose crawling machines fetched the page.
Google's description: The country to crawl the country from, defaults to the default non-specified crawling node (which is interpreted by most web-servers as USA). When specified, the crawling will fetch the document from a node in that country instead.
type String.t · confidence medium

`LogsProtoIndexingCrawlerIdCrawlerIdProto.deviceType`
Rule: Whether the page was fetched as desktop or smartphone Googlebot.
Google's description: The device type, which maps into the useragent to be set when initiating the fetch-request, e.g. desktop-googlebot vs. smartphone-googlebot.
type String.t · confidence medium

`LogsProtoIndexingCrawlerIdCrawlerIdProto.indexGrowthExptType`
Rule: Marks a page as a duplicate copy created for an index size experiment.
Google's description: Specifies whether the document is a duplicated document from the index growth experiment, detailed at go/indexsize_exp, defaults to not in any experiment.
type String.t · confidence medium

`LogsProtoIndexingCrawlerIdCrawlerIdProto.language`
Rule: The language header the crawler sent when fetching the page.
Google's description: The language being set by the crawler. Defaults to UNKNOWN_LANGUAGE which indicates to not apply an accept-language header on the FetchRequest. When a language is specified, on crawling this language is converted into an accept-language header (e.g. GERMAN -> "Accept-language: de"). Script variations, e.g. ZH-HANS vs. ZH-HANT, are handled as different enum values (e.g. CHINESE vs. CHINESE_T).
type String.t · confidence medium

`LogsProtoIndexingCrawlerIdCrawlerIdProto.languageCode`
Rule: A language label applied to manually translated pages such as feeds.
Google's description: Language-code used for identifying the locale of the document. 'language' and 'country' above are used for web-based documents, representing the detected language of the document and the country it was crawled from. The language code here, however, rather represents an artifical language_code applied to manually translated webpages (e.g. feeds), for instance for the pidgin-usecase. They are limited to the set of III-codes being supported by the client, yet are beyond the enum in 'language', e.g.
type String.t · confidence medium

`PerDocData.inNewsstand`
Rule: Flag saying the document belongs to the newsstand collection.
Google's description: This field indicates whether the document is in the newsstand corpus.
type boolean() · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.localizedCluster`
Rule: Grouping information linking translated or localized versions of a page.
Google's description: Information on localized clusters, which is the relationship of translated and/or localized pages.
type GoogleApi.ContentWarehouse.V1.Model.IndexingDupsLocalizedLocalizedCluster.t · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.servingTimeClusterIds`
Rule: Cluster identifiers used to remove near-duplicate results when serving.
Google's description: A set of cluster ids which are generated in Alexandria and used to de-dup results at serving time.
type GoogleApi.ContentWarehouse.V1.Model.IndexingDocjoinerServingTimeClusterIds.t · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`ScienceCitationDownloadURL.BrokenLandingPage`
Rule: A flag set when the landing page for this article is known to be broken.
Google's description: set if we know the landing page is broken
type boolean() · confidence medium

`ScienceCitationDownloadURL.CrawlTimestamp`
Rule: The time the page was crawled, in seconds since the epoch.
Google's description: seconds since the epoch
type String.t · confidence medium

`ScienceCitationDownloadURL.FirstDiscovered`
Rule: The time this URL was first discovered, in seconds since the epoch.
Google's description: seconds since the epoch
type String.t · confidence medium

`ScienceCitationDownloadURL.InPrevIndex`
Rule: Whether this URL was part of a previous index.
Google's description: is url included in a previous index
type boolean() · confidence medium

`ScienceCitationDownloadURL.IndexPriority`
Rule: A score for how strongly this URL is preferred for indexing.
Google's description: indexing preference score
type integer() · confidence medium

`ScienceCitationDownloadURL.LegalMustInclude`
Rule: Marks the URL as one to keep because it matches a law-articles pattern.
Google's description: e.g., in law_articles.pat
type boolean() · confidence medium

`ScienceCitationDownloadURL.LikelyNoCache`
Rule: Whether the URL was on the do-not-cache bad-URL list when indexed.
Google's description: badurls_nocache at indexing time
type boolean() · confidence medium

`ScienceCitationDownloadURL.LikelyNoIndex`
Rule: Whether the URL was on the do-not-return bad-URL list when indexed.
Google's description: badurls_noreturngws at indexing time
type boolean() · confidence medium

`ScienceCitationDownloadURL.MaybeNoIndexReparse`
Rule: Marks a reparsed page as not indexed when the original version was not indexed.
Google's description: Incremental only: mark as NoIndexed if this is a reparse and the base version is NoIndexed.
type boolean() · confidence medium

`ScienceCitationDownloadURL.MustInclude`
Rule: Marks the URL as one to keep because it matches a science-articles pattern.
Google's description: e.g., in science_articles.pat
type boolean() · confidence medium

`TrawlerFetchReplyData.LastUrlStatus`
Rule: The crawl outcome for the final URL in a redirect chain.
Google's description: Crawl status of the last url on chain
type GoogleApi.ContentWarehouse.V1.Model.TrawlerFetchStatus.t · confidence medium · stages: 1. Crawling & Discovery

`TrawlerFetchReplyData.RobotsInfo`
Rule: Robots directives found for the page, such as noindex, nofollow, nosnippet or noarchive.
Google's description: Extra information in robots.txt for this page (ORed together bits from trawler::RobotsInfo). e.g. nosnippet vs. noarchive vs nofollow vs noindex vs disallow Consider using HopRobotsInfo instead, which has per-redirect hop detail.
type integer() · confidence medium · stages: 1. Crawling & Discovery

`TrawlerFetchReplyData.Status`
Rule: The final crawl outcome for this URL after any redirects were followed.
Google's description: Status of the fetch - refers to the final status at the end of the redirect chain.
type GoogleApi.ContentWarehouse.V1.Model.TrawlerFetchStatus.t · confidence medium · stages: 1. Crawling & Discovery

`TrawlerFetchReplyDataRedirects.HopPageNoIndexInfo`
Rule: Noindex instructions found in the headers or content of this hop's source URL.
Google's description: Extra trawler::PageNoIndexInfo for this hop. Integer: ORed together bits from trawler::PageNoIndexInfo. The information specified by this field comes from the http header or content of the source url, not the "TargetUrl" in this Redirects group.
type integer() · confidence medium · stages: 1. Crawling & Discovery

`TrawlerFetchReplyDataRedirects.HopRobotsInfo`
Rule: Robots directives that applied to this redirect hop.
Google's description: Extra trawler::RobotsInfo for this hop. Integer: ORed together bits from trawler::RobotsInfo
type integer() · confidence medium · stages: 1. Crawling & Discovery

`TrawlerFetchReplyDataRedirects.TargetUrl`
Rule: The cleaned up URL the redirect pointed to and that the crawler followed.
Google's description: Difference between the following two fields: TargetUrl is set when we have followed the redirect target, and the url is canonicalized. RawTargetUrl is set in either of the following two cases: (1) The url has not be been followed. For example, the redirect is intended to be handled by the client. In the fetch reply response, you will see the url's status as URL_NOT_FOLLOWED-NOT_FOLLOWED
type String.t · confidence medium · stages: 1. Crawling & Discovery

`TrawlerFetchStatus.State`
Rule: The basic outcome of a fetch, such as crawled, blocked by robots, or error.
Google's description: The State field describes the basic outcome of a fetch (URL_CRAWLED, URL_ROBOTED, URL_ERROR, etc). The value is one of the UrlStatusType enum values from crawler/trawler/trawler_enums.proto Note, there are several combinations of this Status/Reason tuple that could mean that your content is crawled or can be bucketed in a particular type of error. So instead of comparing the enumeration values manually, we suggest to use the predicate functions such as IsContentCrawled() provided in crawler/traw
type String.t · confidence medium

`WWWDocInfo.crawlTime`
Rule: The last time this document was crawled.
Google's description: Last time this doc crawled
type String.t · confidence medium

`WWWDocInfo.fileTypeId`
Rule: An identifier for the original file type before conversion to text or HTML.
Google's description: If converted to TEXT or HTML
type String.t · confidence medium

`WWWDocInfo.isRoboted`
Rule: Whether the host's robots.txt disallows crawling this page.
Google's description: Is disallowed for crawling according to host's robots.txt.
type boolean() · confidence medium

`WWWDocInfo.urlAfterRedirects`
Rule: The final address after following redirects, empty when there were none.
Google's description: empty => same as url
type String.t · confidence medium

`CompositeDoc.csePagerankCutoff`
Rule: The pagerank a page must beat to be picked up for the custom search index.
Google's description: URL should only be selected for CSE Index if it's pagerank is higher than cse_pagerank_cutoff.
type integer() · confidence low · stages: 2. Document Processing & Indexing

`CompositeDocIndexingInfo.imageIndexingInfo`
Rule: Notes on how the page's images were indexed, such as missing image data.
Google's description: Indexing info about images (i.e. image links missing image data, etc).
type GoogleApi.ContentWarehouse.V1.Model.ImageSearchImageIndexingInfo.t · confidence low · stages: 2. Document Processing & Indexing

`CompositeDocIndexingInfo.videoIndexingInfo`
Rule: Notes on how the page's videos were indexed.
Google's description: Indexing info about videos.
type GoogleApi.ContentWarehouse.V1.Model.ImageRepositoryVideoIndexingInfo.t · confidence low · stages: 2. Document Processing & Indexing

`CrawlerChangerateUrlChange.fractionalTileChange`
Rule: A deprecated measure of what share of the page's tiles changed.
Google's description: Deprecated fields. The fraction of tiles (0 to 1) that changed.
type float() · confidence low · stages: 1. Crawling & Discovery

`CrawlerChangerateUrlChangerate.changeperiod`
Rule: An old estimate of seconds between page changes that is generally not used.
Google's description: // The classic changerate estimation. // The classic estimate of change period (in seconds). It is computed by inserted a "fake" change and no-change interval as a prior distribution. This field is generally not used and should NOT be accessed directly. See above for correct method for determining the change period estimate.
type integer() · confidence low · stages: 1. Crawling & Discovery; 7. Freshness & Temporal Signals

`CrawlerChangerateUrlChangerate.confidence`
Rule: How sure the system is about the estimated time between changes.
Google's description: The confidence (between 0 and 1) in the changeperiod guess.
type float() · confidence low · stages: 1. Crawling & Discovery; 7. Freshness & Temporal Signals

`CrawlerChangerateUrlChangerate.globalBasedChangePeriodConfidence`
Rule: How sure the system is about the global-prior change estimate.
Google's description: The 'confidence' of the global-based changeperiod. This is the n-th root of the posterior evaluated at MAP point, where n is the number of history intervals. For now, it is hard to interpret the meaning of the absolute values of 'average' posterior cross different sets of data.
type float() · confidence low · stages: 1. Crawling & Discovery; 7. Freshness & Temporal Signals

`CrawlerChangerateUrlChangerate.numIntervals`
Rule: How many pairs of page versions have been observed.
Google's description: The number of intervals we've seen for this document (where an interval is two different versions).
type integer() · confidence low · stages: 1. Crawling & Discovery; 7. Freshness & Temporal Signals

`CrawlerChangerateUrlChangerate.patternBasedChangePeriodConfidence`
Rule: How sure the system is about the pattern-based change estimate.
Google's description: The same as global_based_change_period_confidence, except it is computed using pattern based priors.
type float() · confidence low · stages: 1. Crawling & Discovery; 7. Freshness & Temporal Signals

`CrawlerChangerateUrlChangerate.patternBasedLowerChangePeriod`
Rule: The low end of the range for the pattern-based change estimate.
Google's description: The lower edge of a confidence interval for the pattern-based change period.
type integer() · confidence low · stages: 1. Crawling & Discovery; 7. Freshness & Temporal Signals

`CrawlerChangerateUrlChangerate.type`
Rule: What kind of document the crawl history suggests this is.
Google's description: // Basic information of a document. // The type of the document determined by crawl histories, refer to TYPE for more information.
type integer() · confidence low · stages: 1. Crawling & Discovery; 7. Freshness & Temporal Signals

`CrawlerChangerateUrlVersion.contentType`
Rule: The content type of the fetched page.
Google's description: The content type of the page.
type integer() · confidence low · stages: 1. Crawling & Discovery

`GDocumentBase.ContentExpiryTime`
Rule: The time, in seconds since 1970, when the page's content expires.
Google's description: unix secs from epoch
type integer() · confidence low · stages: 2. Document Processing & Indexing

`GDocumentBase.IPAddr`
Rule: The IP address of the server the page was fetched from.
Google's description: IP addr in binary (allows for IPv6)
type String.t · confidence low · stages: 2. Document Processing & Indexing

`GDocumentBase.Repid`
Rule: The id of the duplicate cluster that this page's canonical URL belongs to.
Google's description: is the webmirror representative id of the canonical url. Urls with the same repid are considered as dups in webmirror. WARNING: use this field with caution! The webmirror duprules change frequently, so this value only reflects the duprules at the time when the canonical's docjoin is built.
type String.t · confidence low · stages: 2. Document Processing & Indexing

`GDocumentBaseContent.ContentLength`
Rule: The length of the stored page content.
Google's description: The actual length of the content: If Representation is compressed, this equals to Content.UncompressedLength; otherwise it is the length of the representation string.
type integer() · confidence low

`GDocumentBaseContent.Representation`
Rule: The stored copy of the crawled page content.
Google's description: Possibly compressed for old documents. It is not compressed for docjoins produced by Raffia after ~2012.
type String.t · confidence low

`GDocumentBaseContent.crawledFileSize`
Rule: The file size of the document as it was crawled.
Google's description: Crawled file size of the original document.
type integer() · confidence low

`GDocumentBaseContent.encodedGeometryAnnotations`
Rule: Page layout geometry stored in a compressed form.
Google's description: GeometryAnnotations, encoded with GeometryUtil::DeltaEncode() to reduce disk space usage. Use GeometryUtil::DeltaDecode() to decode this field.
type String.t · confidence low

`HtmlrenderWebkitHeadlessProtoDocument.renderTreeQualityScore`
Rule: Old score for how completely the page's layout could be rendered, now removed.
Google's description: DEPRECATED - This field to be removed mid-2011. If you need this, use the library directly: //google3/htmlrender/webkit_headless/snapshot_quality Indicates how good or bad the rendering is from the perspective of the render tree. This is different from snapshot_quality_score in that the quality analysis examines everything that can be rendered, not just the portion within the document's viewport. It also ignores missing resouces with fixed width/height specified in the tag. A score of 1 (100%) i
type float() · confidence low

`HtmlrenderWebkitHeadlessProtoDocument.snapshotQualityScore`
Rule: Old score for how usable the rendered page screenshot was, now removed.
Google's description: DEPRECATED - This field to be removed mid-2011. If you need this, use the library directly: //google3/htmlrender/webkit_headless/snapshot_quality Indicates how good or bad the rendered snapshot is within the rendered content area within the document's viewport. A score of 1 (100%) implies the snapshot is of best quality and a score of 0 implies the snapshot is unusable.
type float() · confidence low

`ImageData.indexedVerticals`
Rule: Which specialized indexes this image and its links were placed into.
Google's description: Vertical indexing information for the document and associated image outlinks.
type list(GoogleApi.ContentWarehouse.V1.Model.ImageDataVerticalIndexingInfoImage.t) · confidence low

`ImageDataVerticalIndexingInfoImage.name`
Rule: The name of a specialized index this image was placed into.
Google's description: List the string representation of the vertical that this image is indexed into.
type String.t · confidence low

`ImageMoosedogCrawlState.detailedReason`
Rule: A more specific code explaining why an image was not fetched.
Google's description: Each of the above not_crawled_reason will have a set of detailed reason defined in crawler/trawler/trawler_enums.proto.
type integer() · confidence low

`ImageMoosedogCrawlState.robotedAgents`
Rule: Crawler user agents blocked from this image, recorded for information only.
Google's description: A comma separated list of user agents for which this image should be considered roboted. All images are crawled using googlebot-images and this exists here purely for informative reasons.
type String.t · confidence low

`ImageRepositoryVideoProperties.crawlState`
Rule: The raw record of how the video was crawled.
Google's description: The raw crawl state.
type GoogleApi.ContentWarehouse.V1.Model.ImageMoosedogCrawlState.t · confidence low

`ImageRepositoryVideoProperties.firstCrawlTimestampSec`
Rule: When the video was first successfully fetched.
Google's description: Timestamp of the first time that the video was successfully crawled.
type String.t · confidence low

`ImageRepositoryVideoPropertiesVerticalIndexingInfoVideo.name`
Rule: Name of a vertical the item is indexed into.
Google's description: List the string representation of the vertical that this image is indexed into.
type String.t · confidence low

`ImageSearchImageSelectionInfo.imageLinkSelectionInfo`
Rule: Details about how an image on the page was selected for indexing.
Google's description: Image Selection Info.
type GoogleApi.ContentWarehouse.V1.Model.ImageMustangImageLinkSelectionInfo.t · confidence low

`IndexingBadSSLCertificate.urlWithBadSslCertificate`
Rule: The address that the bad SSL certificate actually came from.
Google's description: The URL where the bad SSL certificate really comes from. Present iff it is different from the source URL, i.e. a redirect target of the source URL).
type String.t · confidence low

`IndexingConverterLocalizedAlternateName.feedUrl`
Rule: The sitemap address that supplied this alternate URL, when applicable.
Google's description: Populated if annotation_source is SITEMAP.
type String.t · confidence low

`IndexingConverterRawRedirectInfo.rawRedirectChainFromRendering`
Rule: The ordered list of redirects seen while rendering the page.
Google's description: Redirect chain generated from redirect events in rendering. At the beginning of it, there could be some redirects from trawler (i.e. could be partial or entire trawler redirect chain), other redirects have their RedirectParams::is_redirect_from_rendering fields set to true. Redirects here have no RedirectChain::Hop::raw_target fields populated, and targets stored in RedirectChain::Hop::target fields are likely cleaned while keeping fragments (also sometimes they could be uncleaned ones because o
type GoogleApi.ContentWarehouse.V1.Model.IndexingConverterRedirectChain.t · confidence low

`IndexingConverterRedirectChainHop.rawTarget`
Rule: A redirect hop's target URL including the fragment.
Google's description: Redirect target with fragment. This field is populated only if there was a fragment.
type String.t · confidence low

`IndexingConverterRedirectChainHop.target`
Rule: The target URL of one hop in a redirect chain.
Google's description: Redirect target URL and params of the current hop in the redirect chain.
type String.t · confidence low

`IndexingConverterRedirectParams.delay`
Rule: How many seconds after page load the redirect happened.
Google's description: The time difference between page loading and redirect occurrence. When missing, it means the redirect happens immediately (i.e. delay = 0). In seconds.
type integer() · confidence low

`IndexingConverterRedirectParams.frameTargetDeniesFraming`
Rule: Whether the redirect target refuses to be shown inside a frame.
Google's description: Populated for SINGLE_FRAME and SINGLE_IFRAME redirects only and indicates that the target url requested not to be framed, by virtue of using the "X-Frame-Options" HTTP header.
type boolean() · confidence low

`IndexingConverterRedirectParams.isDownload`
Rule: Whether the redirect resulted in a file download.
Google's description: Indicates corresponding redirect is a download. This field is only set when rendering redirect chain is used. This field represents the value of corresponding "RedirectEvent.target_content_downloaded" field.
type boolean() · confidence low

`IndexingConverterRobotsInfo.indexifembeddedReason`
Rule: Code recording why the indexifembedded robots rule applied.
Google's description: IMPORTANT: if you add a new field here, update the MergeRobotsInfo() function to merge the new field.
type integer() · confidence low

`IndexingDupsLocalizedLocalizedClusterLinkBasedClusterInfoLinkData.crawlTimestamp`
Rule: When the linked page was last crawled.
Google's description: If set, represents the crawl timestamp. If not set, there is no known crawl timestamp for that url.
type integer() · confidence low

`IndexingDupsLocalizedLocalizedClusterTargetLinkLinkAnnotationSourceInfo.sourceFeedUrl`
Rule: The sitemap feed URL where the annotation was found.
Google's description: Optional field that stores the feed URL where a Sitemap annotation was discovered. Only populated if annotation_source is SITEMAP.
type String.t · confidence low

`IndexingDupsLocalizedLocalizedClusterTargetLinkMetadata.firstSeenMs`
Rule: When the link was first seen.
Google's description: When was the first time a link seen. Defaults to last crawled timestamp.
type String.t · confidence low

`IndexingDupsLocalizedLocalizedClusterTargetLinkMetadata.lastVerifiedMs`
Rule: When the link was last re-checked.
Google's description: When was the last time a link validated. Validation is the process of (re)reading the relevant information for a linked document from its respective row in the document table. Data needed for understanding the correctness of the cluster is copied over to have it available locally.
type String.t · confidence low

`IndexingDupsLocalizedLocalizedClusterTargetLinkTargetDocData.crawlTimestampSeconds`
Rule: When the linked page was last crawled.
Google's description: The timestamp of the last crawl attempt from crawl_timestamp column.
type integer() · confidence low

`IndexingEmbeddedContentEmbeddedContentInfo.compressedDocumentTrees`
Rule: The stored DOM and render tree produced when the page was rendered.
Google's description: The document's DOM and render tree produced by WebKit as a side effect of rendering the page. It might be compressed or not. Thus, use indexing::embedded_content::UncompressWebkitDocument to decode it.
type String.t · confidence low

`IndexingEmbeddedContentEmbeddedContentInfo.convertedContents`
Rule: The converted page contents produced alongside the render tree.
Google's description: The converted contents, as produced by the same DocumentUpdater transaction that generated the render tree. Useful whenever one of our users wants to experiment with deriving an annotation from the render tree.
type String.t · confidence low

`IndexingEmbeddedContentEmbeddedContentInfo.referencedResource`
Rule: List of external resources used while rendering the page.
Google's description: Information about all external resources used to render this page, a.k.a. embedded links. This includes .css files, images embedded in a page, external javascripts, iframes etc.
type list(GoogleApi.ContentWarehouse.V1.Model.HtmlrenderWebkitHeadlessProtoReferencedResource.t) · confidence low

`IndexingEmbeddedContentEmbeddedContentInfo.richcontentData`
Rule: Stored data allowing the original page content to be reconstructed.
Google's description: The rich content data to recover the original contents from the converted_contents. Useful for offline content analysis.
type GoogleApi.ContentWarehouse.V1.Model.IndexingConverterRichContentData.t · confidence low

`IndexingEmbeddedContentRenderingOutputMetadata.importance`
Rule: An importance value assigned to the rendered page.
Google's description: The importance value of the rendered document.
type integer() · confidence low

`IndexingEmbeddedContentRenderingOutputMetadata.newTokensPercentageAfterRendering`
Rule: Share of the page's words that only appeared after rendering.
Google's description: Percentage of new tokens in the rendered the document content.
type float() · confidence low

`MobilePerDocData.mobileurl`
Rule: A retired field holding the address of the mobile version of a page.
Google's description: DEPRECATED: Url of the mobile version of the document. This is set during canonicalization if we do not know that the Web url also serves the mobile version.
type String.t · confidence low

`OceanDocTag.isLandingPage`
Rule: Whether this page was chosen as the book's landing page at indexing time.
Google's description: whether this a landing page chosen at indexing time.
type boolean() · confidence low

`PerDocData.MobileData`
Rule: Extra index metadata kept for low-end mobile versions of documents.
Google's description: Additional metadata for lowend mobile documents in the Google index.
type GoogleApi.ContentWarehouse.V1.Model.MobilePerDocData.t · confidence low · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.PremiumData`
Rule: Extra index metadata kept for premium documents.
Google's description: Additional metadata for Premium document in the Google index.
type GoogleApi.ContentWarehouse.V1.Model.PremiumPerDocData.t · confidence low · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.smartphoneData`
Rule: Extra index metadata kept for smartphone versions of documents.
Google's description: Additional metadata for smartphone documents in the Google index.
type GoogleApi.ContentWarehouse.V1.Model.SmartphonePerDocData.t · confidence low · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`SnapshotSnapshotMetadata.countDistinctResources`
Rule: How many separate files had to be fetched to render the page.
Google's description: The number of distinct resources fetched to render the content. This may aid the calculation of total page load time for user experience. For example, if total_content_length is only a few dozen kilobytes, but that is from fetching 100 distinct resources, total page load time might be much higher than the total_content_length would otherwise infer.
type integer() · confidence low

`SnapshotSnapshotMetadata.totalContentSize`
Rule: Total bytes downloaded to render the page, including images and stylesheets.
Google's description: Number of bytes fetched to render the content. For example, to render a web page, this value would include the HTML, stylesheets, images, and all other dependencies. This can be used to calculate a coarse estimate of the total page load time a user might experience.
type String.t · confidence low

`TrawlerFetchReplyDataRedirects.HTTPResponseCode`
Rule: The HTTP response code returned for this redirect hop.
Google's description: The HTTP response code for this hop. We need this since multiple response codes may have the same redirect type (e.g., 302 and 307 are both REDIRECT_TEMPORARILY), but clients may want to know which one was received. Note this is set only for the hops that are followed (i.e., TargetUrl is present). If the last redirect hop was not followed the fetch status will be URL_NOT_FOLLOWED, and the response code will be in the top level ProtocolResponse field.
type integer() · confidence low · stages: 1. Crawling & Discovery

`TrawlerFetchReplyDataRedirects.Type`
Rule: The kind of redirect that was seen on this hop.
Google's description: URL and redirect type
type String.t · confidence low · stages: 1. Crawling & Discovery

`UrlPoisoningData.NotChangedTimeMs`
Rule: Timestamp for when the page was last fetched from the web.
Google's description: fetched from the web. Time when the page was last
type String.t · confidence low

`UrlPoisoningData.OriginalCrawlTimeMs`
Rule: Timestamp for when the page was originally crawled.
Google's description: Time when the page was
type String.t · confidence low

`UrlPoisoningData.ReuseTimeMs`
Rule: Timestamp for when the page was last checked and found unchanged.
Google's description: checked but found to be the same as before. If set, timestamp to indicate
type String.t · confidence low

`VideoContentSearchVideoInfo.subindexid`
Rule: A copy of which index tier the document sits in.
Google's description: The subindexid from the cdoc. Stored as an int to avoid a cyclical dependency. Should be convertible to CompositeDoc.SubIndexType.
type list(integer()) · confidence low

`WWWDocInfo.bodySize`
Rule: The size of the document.
Google's description: Size of document
type integer() · confidence low

`WWWDocInfo.contentType`
Rule: The content type of the document.
Google's description: If not present, then the type
type String.t · confidence low

`WWWDocInfo.hasBadSslCertificate`
Rule: Whether the original document was served with a bad SSL certificate.
Google's description: If true, the original document has a bad SSL certificate.
type boolean() · confidence low

`WWWDocInfo.referrerUrl`
Rule: The address of the page that referred to this document.
Google's description: Url of referring doc
type String.t · confidence low

Entities & authors (93)

`ImageSearchImageLicenseInfo.creator`
Rule: Names of the creators or authors of the image.
Google's description: creator, authors.
type list(String.t) · confidence medium

`KnowledgeAnswersIntentQueryArgumentSignals.relatedEntity`
Rule: Entities related to this query argument and how they are related.
Google's description: List of entities that are semantically related to the argument as well as details of the relationship.
type list(GoogleApi.ContentWarehouse.V1.Model.NlpSemanticParsingRelatedEntity.t) · confidence medium

`KnowledgeAnswersIntentQueryCollectionMembership.collectionScore`
Rule: How strongly an entity belongs to a given group of entities.
Google's description: The collection score for a entity.
type number() · confidence medium

`KnowledgeAnswersIntentQueryImpliedEntity.mid`
Rule: Knowledge Graph id of an entity implied by the query.
Google's description: The KG mid of the implied entity.
type String.t · confidence medium

`NlpSemanticParsingQRefAnnotation.displayName`
Rule: A canonical display name for the entity, so different phrasings map to one name.
Google's description: Copy the display info. This can be used by annotators to give grammars a canonical name for an entity. For instance, the media grammar could use it to output the same canonical name for "rock music" and "rock".
type String.t · confidence medium · stages: 9. Query Understanding & Intent Classification

`NlpSemanticParsingQRefAnnotation.freebaseMid`
Rule: The Knowledge Graph ID of the entity matched to this piece of text.
Google's description: The mid of the entity in freebase associated with this span.
type String.t · confidence medium · stages: 9. Query Understanding & Intent Classification

`VideoContentSearchCaptionEntityAnchorFeatures.asrMentionText`
Rule: The wording used in the spoken transcript when the topic is mentioned.
Google's description: If the description anchor has been recognized as an entity and that entity has also been mention in the ASR, this is the mention text from the ASR.
type String.t · confidence medium

`VideoContentSearchCaptionEntityAnchorFeatures.broadness`
Rule: How broad or general the topic is.
Google's description: A score to help determine how specific this entity is.
type number() · confidence medium

`VideoContentSearchCaptionEntityAnchorFeatures.entityConnectedness`
Rule: How well connected the topic is in Google's entity graph.
Google's description: The webref connectedness score of the entity.
type number() · confidence medium

`VideoContentSearchCaptionEntityAnchorFeatures.entityDescription`
Rule: A short description of the topic.
Google's description: A short text describing the entity.
type String.t · confidence medium

`VideoContentSearchCaptionEntityAnchorFeatures.entityInfo`
Rule: Counts of how often and in how many videos the topic appears.
Google's description: Information about how many documents the entity occurred in and how many total mentions the entity has across the entire video corpus.
type GoogleApi.ContentWarehouse.V1.Model.VideoContentSearchCaptionEntityDocInfo.t · confidence medium

`VideoContentSearchCaptionEntityAnchorFeatures.entityMentionInDescription`
Rule: Whether the topic's wording also appears in the video description.
Google's description: Whether or not the entity mention text appears in the description of the video.
type boolean() · confidence medium

`VideoContentSearchCaptionEntityAnchorFeatures.hypernymConfidence`
Rule: Confidence in the broader category chosen as the group label.
Google's description: The confidence of the hypernym used as the set label.
type number() · confidence medium

`VideoContentSearchCaptionEntityAnchorFeatures.hypernymCount`
Rule: How many broader categories were used when measuring similarity.
Google's description: Number of hypernyms used for calculating similarity.
type number() · confidence medium

`VideoContentSearchCaptionEntityAnchorFeatures.hyperpediaSalientTermsSimilarity`
Rule: How close the topic's broader categories are to the video's main terms.
Google's description: The cosine similarity between the document salient terms and the hyperpedia hypernyms for a given entity.
type number() · confidence medium

`VideoContentSearchCaptionEntityAnchorFeatures.inWebrefEntities`
Rule: Whether this topic is among the video's recognised entities.
Google's description: Whether or not this entity is in the webref entities.
type boolean() · confidence medium

`VideoContentSearchCaptionEntityAnchorFeatures.isProduct`
Rule: Whether the topic counts as a product rather than a related topic.
Google's description: Whether this entity counts as a 'product' for the purpose of dividing entities between the 'related topics' and 'products in this video' features.
type boolean() · confidence medium

`VideoContentSearchCaptionEntityAnchorFeatures.maxMentionConfidence`
Rule: The highest confidence among all mentions of the topic in the transcript.
Google's description: The maximum confidence of all of the entity mentions in the transcript.
type number() · confidence medium

`VideoContentSearchCaptionEntityAnchorFeatures.mentionConfidence`
Rule: Confidence that the spoken mention refers to this particular topic.
Google's description: The confidence that the ASR mention matches the given mid.
type number() · confidence medium

`VideoContentSearchCaptionEntityAnchorFeatures.mentions`
Rule: How many times the topic is mentioned in the spoken transcript.
Google's description: Number of times an entity is mentioned in the ASR transcript.
type integer() · confidence medium

`VideoContentSearchCaptionEntityAnchorFeatures.otherAsrMentionText`
Rule: The transcript wording of every mention of the topic.
Google's description: The ASR text of each mention of the entity.
type list(String.t) · confidence medium

`VideoContentSearchCaptionEntityAnchorFeatures.otherEstimatedMentionTimes`
Rule: Every estimated time at which the topic was mentioned.
Google's description: Each time the entity was mentioned.
type list(integer()) · confidence medium

`VideoContentSearchCaptionEntityAnchorFeatures.trustedNameConfidence`
Rule: Confidence that the spoken mention is a reliable name for the topic.
Google's description: The confidence that the ASR mention is a trusted name.
type number() · confidence medium

`VideoContentSearchCaptionEntityAnchorFeatures.webrefEntityTopicality`
Rule: How on topic the recognised entity is, or zero if it is not one.
Google's description: The webref entity topicality score if the entity is a webref entity and 0 if not.
type number() · confidence medium

`VideoContentSearchCaptionEntityAnchorSetFeatures.aggregateScore`
Rule: The combined score used to filter and pick topic groups.
Google's description: The total score used for filtering and selecting entity sets.
type number() · confidence medium

`VideoContentSearchCaptionEntityAnchorSetFeatures.clusterSize`
Rule: How many topics were in the group before filtering.
Google's description: The prefiltered size of the entity set.
type integer() · confidence medium

`VideoContentSearchCaptionEntityAnchorSetFeatures.entitiesInWebrefEntities`
Rule: How many topics in the group are recognised entities.
Google's description: The number of entities in the anchor set that are in the webref entities.
type integer() · confidence medium

`VideoContentSearchCaptionEntityAnchorSetFeatures.entityMentionInDescriptionCount`
Rule: How many key moments use wording that also appears in the description.
Google's description: The number of anchors where the entity mention text appears in the description of the video.
type boolean() · confidence medium

`VideoContentSearchCaptionEntityAnchorSetFeatures.groupCohesion`
Rule: Average similarity between the broader categories of group members.
Google's description: The average cosine similarity between hypernyms of members of the set.
type number() · confidence medium

`VideoContentSearchCaptionEntityAnchorSetFeatures.hypernym`
Rule: The most common broader category across topics in the group.
Google's description: The most prominent hypernym across the entities in the set.
type String.t · confidence medium

`VideoContentSearchCaptionEntityAnchorSetFeatures.hypernymSalience`
Rule: How prominent the group's chosen broader category is.
Google's description: The salience of the best hypernym for the set.
type number() · confidence medium

`VideoContentSearchCaptionEntityAnchorSetFeatures.medianMentions`
Rule: The median number of transcript mentions among topics in the group.
Google's description: Median number of times any member of the set was mentioned in the ASR transcript.
type integer() · confidence medium

`VideoContentSearchCaptionEntityAnchorSetFeatures.mentionSalience`
Rule: The share of all topic mentions in the video that belong to this one.
Google's description: Mentions divided by the total number of entity mentions in the video.
type number() · confidence medium

`VideoContentSearchCaptionEntityAnchorSetFeatures.salience`
Rule: How prominent the topic group is relative to the video's main terms.
Google's description: Salience of the set computed by aggregating the hypernyms from each member and calculating the cosine similarity with the salient terms.
type number() · confidence medium

`VideoContentSearchCaptionEntityAnchorSetFeatures.topHypernym`
Rule: The leading broader categories for the topics in the group.
Google's description: The top N hypernyms for the entities in the set.
type list(String.t) · confidence medium

`VideoContentSearchCaptionEntityAnchorSetFeatures.totalMentions`
Rule: Total transcript mentions for all topics in the group.
Google's description: Number of times any member of the group was mentioned in the ASR transcript.
type integer() · confidence medium

`VideoContentSearchCaptionEntityDocInfo.entityDocCount`
Rule: How many videos mention this topic at least once.
Google's description: The number of documents where this entity was mentioned at least once.
type String.t · confidence medium

`VideoContentSearchCaptionEntityDocInfo.entityMentionCount`
Rule: How many times this topic is mentioned across the whole collection.
Google's description: The number of times the entity was mentioned across the entire corpus.
type String.t · confidence medium

`VideoContentSearchDescriptionAnchorFeatures.entityTextCoverage`
Rule: How much of the description key moment text a recognised topic covers.
Google's description: When the description anchor text has been recognized as an entity, how much of the description anchor text is covered by the entity mention.
type number() · confidence medium

`VideoContentSearchDescriptionAnchorSetFeatures.uniqueAsrMidCount`
Rule: How many distinct topics were matched to description key moments.
Google's description: The number of unique mids which where matched to description anchors.
type integer() · confidence medium

`VideoContentSearchEntityAnnotations.category`
Rule: Which category the recognised topic belongs to.
Google's description: The Webref category that this entity belongs to e.g. "/moka/software".
type String.t · confidence medium

`VideoContentSearchEntityAnnotations.confidence`
Rule: Confidence that the topic appears somewhere in the label.
Google's description: The overall confidence that this entity is annotated somewhere in the label.
type number() · confidence medium

`VideoContentSearchEntityAnnotations.isRestricted`
Rule: Whether the topic falls into a blocked category.
Google's description: Whether or not this entity belongs to a set of blocklisted categories.
type boolean() · confidence medium

`VideoContentSearchVideoAnchor.entityScore`
Rule: How useful the topic identified for this key moment is judged to be.
Google's description: The score indicating the usefulness of the entity identified by 'mid'.
type number() · confidence medium

`AppsPeopleOzExternalMergedpeopleapiPerson.searchProfile`
Rule: The person profile used by Search and Janata surfaces.
Google's description: Profile for Janata and Search. go/janata-profile-in-sgbe
type list(GoogleApi.ContentWarehouse.V1.Model.AppsPeopleOzExternalMergedpeopleapiSearchProfile.t) · confidence low

`ImageExifIPTCMetadata.creator`
Rule: The credited creator of the image from its embedded metadata.
Google's description: IPTC authorship & copyright related fields.
type list(String.t) · confidence low

`ImageExifImageEmbeddedMetadata.author`
Rule: The artist credited inside the image file's metadata.
Google's description: Text fields EXIF_TAG_ARTIST
type String.t · confidence low

`ImageExifImageEmbeddedMetadata.author2`
Rule: A second author field taken from the image file's metadata.
Google's description: EXIF_TAG_XP_AUTHOR
type String.t · confidence low

`ImageExifImageEmbeddedMetadata.copyright`
Rule: The copyright notice stored inside the image file.
Google's description: EXIF_TAG_COPYRIGHT
type String.t · confidence low

`KnowledgeAnswersIntentQueryCollectionMembership.score`
Rule: The different kinds of membership scores an entity has for a group.
Google's description: Different types of scores for the collection. Each score type has at most one score.
type list(GoogleApi.ContentWarehouse.V1.Model.KnowledgeAnswersIntentQueryCollectionScore.t) · confidence low

`KnowledgeAnswersIntentQueryQrefAnnotationSignals.trustedNameConfidence`
Rule: Confidence that a name in the query is a trusted name for the entity.
Google's description: Trusted name confidence signal
type float() · confidence low

`KnowledgeAnswersIntentQuerySupportTransferSignals.supportTransferSource`
Rule: Entities that passed supporting evidence to this entity.
Google's description: Entities that transferred support to this entity (mids).
type list(String.t) · confidence low

`KnowledgeAnswersIntentQuerySupportTransferSignals.supportTransferTarget`
Rule: Entities that received supporting evidence from this entity.
Google's description: Entities that received support from this entity (mids).
type list(String.t) · confidence low

`NlpSemanticParsingQRefAnnotation.clusterSiblingMid`
Rule: The other entity IDs that belong to the same cluster as this one.
Google's description: The set of mids that are members of the same cluster.
type list(String.t) · confidence low · stages: 9. Query Understanding & Intent Classification

`NlpSemanticParsingQRefAnnotation.entityRelationship`
Rule: Recorded relationships between this entity and other entities in the query.
Google's description: The relationship information from QRef. Only included if the QRefAnnotator is initialised with include_annotated_relationships.
type list(GoogleApi.ContentWarehouse.V1.Model.NlpSemanticParsingQRefAnnotationEntityRelationship.t) · confidence low · stages: 9. Query Understanding & Intent Classification

`NlpSemanticParsingQRefAnnotation.relatedEntity`
Rule: Other entity IDs related to the matched entity.
Google's description: Mids related to the given entity
type list(GoogleApi.ContentWarehouse.V1.Model.NlpSemanticParsingRelatedEntity.t) · confidence low · stages: 9. Query Understanding & Intent Classification

`NlpSemanticParsingQRefAnnotationCollectionMembership.collectionId`
Rule: The identifier of an entity collection, such as US states, that the entity belongs to.
Google's description: Identifier of the collection. Usually something like "/collection/us_states".
type String.t · confidence low

`NlpSemanticParsingQRefAnnotationEntityRelationship.impliedBy`
Rule: Flag saying this entity is implied by another entity in the query.
Google's description: True if this entity is implied by the other (includes geo contains).
type boolean() · confidence low

`NlpSemanticParsingQRefAnnotationEntityRelationship.implies`
Rule: Flag saying this entity implies another entity in the query.
Google's description: True if this entity implies the other (includes geo contained by).
type boolean() · confidence low

`NlpSemanticParsingQRefAnnotationEntityRelationship.linkPropertyName`
Rule: The names of the relationship links between two entities.
Google's description: Names of the relationship links.
type list(String.t) · confidence low

`NlpSemanticParsingRelatedEntity.clusterSupportTransferRelation`
Rule: Whether the related entity came from transferring support across an entity cluster.
Google's description: Denotes whether or not the related entity is derived from cluster support transfer.
type String.t · confidence low

`NlpSemanticParsingRelatedEntity.composedFromRelation`
Rule: Whether the related entity is one part of a larger compound entity.
Google's description: Denotes whether or not the related entity composes a compound entity together with other related entities.
type String.t · confidence low

`NlpSemanticParsingRelatedEntity.equivalentRelation`
Rule: Whether the two entity IDs are treated as the same thing.
Google's description: Whether or not the given mid is related to the other mid. Equivalent mids are usually mutually exclusive with other kinds of relations.
type String.t · confidence low

`NlpSemanticParsingRelatedEntity.mdvcRelation`
Rule: The dimension style relationship between this entity and a related one.
Google's description: The mdvc relation with the related mid.
type String.t · confidence low

`NlpSemanticParsingRelatedEntity.mid`
Rule: The ID of the related entity.
Google's description: Mid that is related.
type String.t · confidence low

`NlpSemanticParsingRelatedEntity.supportTransferRelation`
Rule: Whether support was passed between the two entities.
Google's description: Denotes whether or not there was support transfer between the two entities.
type String.t · confidence low

`NlpSemanticParsingSaftMentionAnnotation.entity`
Rule: Annotations for query phrases that name an entity, such as a sports team.
Google's description: Annotations for spans like "san francisco giants".
type GoogleApi.ContentWarehouse.V1.Model.NlpSemanticParsingSaftSpan.t · confidence low

`NlpSemanticParsingSaftMentionAnnotation.title`
Rule: Annotations for descriptive title phrases such as "the president of the United States".
Google's description: Annotations for spans like "the president of the United States".
type GoogleApi.ContentWarehouse.V1.Model.NlpSemanticParsingSaftSpan.t · confidence low

`PerDocData.DEPRECATEDAuthorObfuscatedGaia`
Rule: Old obfuscated Google profile IDs for the page's authors; replaced by a string version.
Google's description: The obfuscated google profile gaia id(s) of the author(s) of the document. This field is deprecated, use the string version.
type list(String.t) · confidence low · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

QualityFringeFringeQueryPriorPerDocData.sensitiveEntitiesIndices
Rule: Positions of entities on the page that refer to people or groups; being replaced.
Google's description: Indices on the repository_webref::WebrefEntities::entity field of entities that represent a person or a group of people (aka sensitive entities). Will be deprecated once sensitive_entities_mids is fully launched (see b/290268614).
type list(integer()) · confidence low

RepositoryWebrefCategoryAnnotation.mid
Rule: The category's knowledge graph id, which is not filled in for some outputs.
Google's description: Mid representation of the category. Eg "/m/05qt0". WARNING: In UDR this field is not populated, use document_entity.entity.mid instead of document_entity.category.mid.
type String.t · confidence low

RepositoryWebrefCategoryAnnotationHitCatSource.experimentalConfidence
Rule: A trial version of the category confidence that can change at any time.
Google's description: Note: For testing the next version. May change at any time. Experimental confidence of the category in the range (0.0, 1.0).
type number() · confidence low

RepositoryWebrefDetailedEntityScores.referencePageScores
Rule: Scores for finding new reference pages, normally not filled in.
Google's description: Signals used for mining new reference pages, set by the reference-page-scorer processor (that is turned off by default). This field is not populated, except for special reference page extraction runs.
type GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefReferencePageScores.t · confidence low

RepositoryWebrefDetailedMentionScores.resultEntityScore
Rule: How much search results backed up this mention, filled in only on request.
Google's description: How much support this mention received from the results for PostRef. This is populated only if explicitly requested and different from 0.
type number() · confidence low

RepositoryWebrefEntityScores.nameCapitalizationProb
Rule: An outdated estimate of how often a name is capitalized.
Google's description: The following field is deprecated and should eventually be removed. It uses data and rules that have not been refreshed for ~10y and KG has changed a lot in the meantime.
type number() · confidence low

RepositoryWebrefMdvcMetadata.resolutionPriority
Rule: Which entity resolutions win when a query has several, marked as being migrated away.
Google's description: Undergoing migration into the PerVertical message. Avoid using it. Resolution priority for this entity. In case a query has many possible resolutions, only the ones with the highest resolution priority are annotated.
type integer() · confidence low

RepositoryWebrefTripleAnnotation.isImplied
Rule: Flag saying this extracted fact is implied by other facts on the page.
Google's description: The information in this triple is implied by other triple(s) in the document.
type boolean() · confidence low

RepositoryWebrefTripleAnnotation.kgVerified
Rule: Flag saying the extracted fact already exists in the knowledge graph.
Google's description: Set to true if this triple is present in the webref model as either a link or property value. This implies that the information is in the Knowledge Graph. Note that it can happen that a triple is in KG but not present in the webref model.
type boolean() · confidence low

RepositoryWebrefTripleAnnotation.mentions
Rule: Where on the page the extracted fact appears.
Google's description: Occurrences of the triple on the document
type list(GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefTripleMention.t) · confidence low

RepositoryWebrefTripleMention.predMention
Rule: Where the relationship part of an extracted fact appears on the page.
Google's description: Document mention of the predicate
type GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefSegmentMention.t · confidence low

RepositoryWebrefTripleMention.subMention
Rule: Where the subject of an extracted fact appears on the page.
Google's description: Document mention of the subject
type GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefSegmentMention.t · confidence low

RepositoryWebrefTripleMention.valueMention
Rule: Where the value of an extracted fact appears on the page.
Google's description: Document mention of the value
type GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefSegmentMention.t · confidence low

`RepositoryWebrefWebrefEntities.tripleAnnotations`
Rule: Facts the system inferred from the page's own text, tables and lists.
Google's description: Triples inferred from the document When the annotator recognizes phrases, lists or tables associated with a property or relationship for an entity it generates triples that encode that information. This generated data is only substantiated by the document vs KG data which has been verified from multiple sources and/or human curators.
type GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefTripleAnnotations.t · confidence low · stages: 6. Entity & Knowledge Graph Integration

`SocialPersonalizationKnexAnnotationItem.generality`
Rule: An experimental, rarely filled score for how broad or narrow a topic is.
Google's description: A score that measures how broad / narrow a topic is, independent of the document and/or user profile. This is an experimental score and is not populated by default. Currently, it is in the range of [0, 1], where the higher the score is, the more general a topic is. Please talk to sherlock-dev@ before using this score. Populated when debug_level > 0.
type number() · confidence low

`UniversalsearchNewPackerKnowledgeResultSupportProvenance.entityGroupType`
Rule: The entity group type of the entity that this result supports.
Google's description: Entity group type of the supported entity. This field should only be set for support on entities.
type String.t · confidence low

`VideoContentSearchCaptionEntityDocInfo.mid`
Rule: The identifier of the topic these counts belong to.
Google's description: The entity id.
type String.t · confidence low

`VideoContentSearchCaptionEntityDocInfo.totalDocCount`
Rule: How many documents are in the collection, used for comparison.
Google's description: The number of documents in the corpus.
type String.t · confidence low

`VideoContentSearchCaptionEntityDocInfo.totalMentionCount`
Rule: Total mentions of any topic across the whole collection.
Google's description: The number of mentions of any entity across the entire corpus.
type String.t · confidence low

`VideoContentSearchEntityAnnotations.mid`
Rule: The identifier of the recognised topic.
Google's description: The Webref entity mid.
type String.t · confidence low

`VideoContentSearchVideoAnchor.mid`
Rule: Knowledge graph identifier for the topic of the key moment.
Google's description: The mid of the video anchor.
type String.t · confidence low

`VideoContentSearchVideoAnchor.namedEntity`
Rule: Named people, places or things mentioned in the key moment label.
Google's description: Specifies named enitities the label has.
type list(GoogleApi.ContentWarehouse.V1.Model.VideoContentSearchNamedEntity.t) · confidence low

`WWWSnippetResponse.orionEntities`
Rule: A deprecated set of entities found in the document.
Google's description: DEPRECATED If requested, the orion entities
type GoogleApi.ContentWarehouse.V1.Model.OrionDocEntitiesProto.t · confidence low · stages: 10. Snippet & Answer Extraction

`WWWSnippetResponse.scienceInfo`
Rule: Short bibliographic details for the document from Google Scholar.
Google's description: Abbreviated bibliographic data from Google Scholar.
type GoogleApi.ContentWarehouse.V1.Model.ScienceIndexSignal.t · confidence low · stages: 10. Snippet & Answer Extraction

Personalization & local (48)

`CountryCountryAttachment.userVisibleLocalCountry`
Rule: The country shown to users on the results page for this page.
Google's description: This is the country id we show to users on the result page. This is kept different from country demotion country id because we dont want to expose our backoff and url based detection algorithm - also we want to be ultra conservative in showing this.
type integer() · confidence high · stages: 12. Personalization, Local & Context

`LocalWWWInfoCluster.menuUrl`
Rule: Links to the business's menu.
Google's description: Menu link for the business. Currently only comes from Local attributes.
type list(String.t) · confidence high · stages: 12. Personalization, Local & Context

`CompositeDoc.localizedAlternateName`
Rule: Other language versions of the address, swapped in to match the user's locale.
Google's description: Localized alternate names are similar to alternate names, except that it is associated with a language different from its canonical. This is the subset of webmaster-provided localized alternate names being in the dup cluster of this document. Used during serving for swapping in the URL based on regional and language preferences of the user.
type list(GoogleApi.ContentWarehouse.V1.Model.IndexingConverterLocalizedAlternateName.t) · confidence medium · stages: 2. Document Processing & Indexing

`CompositeDocLocalizedVariations.dupsComputedAlternateNames`
Rule: Duplicate addresses in other languages that can be swapped in for the user.
Google's description: A subset of computed variations, only the members which are dups to the main url. Used during serving for swapping in the URL based on regional and language preferences of the user.
type list(GoogleApi.ContentWarehouse.V1.Model.IndexingDupsComputedLocalizedAlternateNamesLocaleEntry.t) · confidence medium

`CountryCountryAttachment.isValidForCountryRestrict`
Rule: Whether the page's country list can be used for country-restricted searches.
Google's description: Set to true if the local_countries field can be used for country restricts as well.
type boolean() · confidence medium · stages: 12. Personalization, Local & Context

`CountryCountryAttachment.metroLocationId`
Rule: Tokens that let searches be restricted to a metro area.
Google's description: Metro level data. metro_location_id stores geotokens for metro restricts.
type list(String.t) · confidence medium · stages: 12. Personalization, Local & Context

`CountryCountryAttachment.restrictCountries`
Rule: Country codes used to filter results by country.
Google's description: List of two-letter(lower-case) countrycodes(e.g. us) valid for restricts. Typically cloned out of local_countries if is_valid_for_country_restrict is set to true.
type list(String.t) · confidence medium · stages: 12. Personalization, Local & Context

`CountryCountryAttachment.userVisibleCountryFromLogs`
Rule: The country for the page worked out from log data.
Google's description: This is used to store the visible country id computed from logs data
type String.t · confidence medium · stages: 12. Personalization, Local & Context

`IndexingConverterLocalizedAlternateName.parsedLanguage`
Rule: Language and region code parsed out of an alternate URL's language field.
Google's description: Parsed language and region code from language field.
type String.t · confidence medium

`IndexingDupsComputedLocalizedAlternateNamesLocaleEntry.language`
Rule: The language and region this alternate URL serves.
Google's description: Language/Region code. E.g. "en-US" or "de". Allowed values are language-region codes based on the W3C recommendation
type String.t · confidence medium

`IndexingDupsComputedLocalizedAlternateNamesLocaleEntry.url`
Rule: The alternate URL for a particular language or region.
Google's description: The alternate url representing the content for a specific language and region (or language only).
type String.t · confidence medium

`IndexingDupsComputedLocalizedAlternateNamesLocaleEntry.urlRegionCode`
Rule: The region code taken from the URL itself.
Google's description: The region code that was extracted from the URL, either by the TLD or via a pattern (like 'en-ca' as a path element).. Always filled in if known, unlike the sometimes left out region part of the language field. Unknown Region Code
type integer() · confidence medium

`IndexingDupsLocalizedLocalizedClusterCluster.filteringEnabled`
Rule: Whether only one page from this cluster should be kept on a results page.
Google's description: Indicates that filtering can be applied on the category (if many results of one cluster show up on the SERP, only one should be kept).
type boolean() · confidence medium

`IndexingDupsLocalizedLocalizedClusterCluster.language`
Rule: The language the URL indicates.
Google's description: The language as represented by the URL, e.g. 'use this document on the cluster for German queries'.
type String.t · confidence medium

`IndexingDupsLocalizedLocalizedClusterCluster.regionCode`
Rule: The stable region code for the cluster.
Google's description: Same as language, except for the country. This is the Stable Region Code. This value may be UNKNOWN even though the URL region code is known, namely when the known region code was the main region for the language and for that language no other region is specified (e.g. de-DE being the only german variation). Unknown Region Code
type integer() · confidence medium

`IndexingDupsLocalizedLocalizedClusterCluster.urlRegionCode`
Rule: The region code taken from the URL.
Google's description: Similar to region_code, but always has the value filled in if known. Unknown Region Code
type integer() · confidence medium

`IndexingDupsLocalizedLocalizedClusterTargetLinkLinkAnnotationSourceInfo.languageCode`
Rule: The language code found for the linked URL.
Google's description: Language code extracted from the URL (hreflang or outlink). One URL can represent multiple language codes, like e.g. de-at and de-ch
type String.t · confidence medium

`LocalWWWInfoCluster.hours`
Rule: The business's opening hours as known from local data or the page.
Google's description: Opening hours for the business, from Local attributes and/or extracted annotations.
type GoogleApi.ContentWarehouse.V1.Model.GeostoreTimeScheduleProto.t · confidence medium · stages: 12. Personalization, Local & Context

`RepositoryWebrefSupportTransferRule.userCountry`
Rule: Country the entity transfer rule applies to.
Google's description: The user country this rule is for. Rules only take effect if the country is not set, set to an empty string or matches the country that is detected for the user - like the country user issued the query from. E.g. "US" for the United States.
type String.t · confidence medium

`RepositoryWebrefSupportTransferRule.userLanguage`
Rule: User language the entity transfer rule applies to.
Google's description: User language this rule is for. Rules only take effect if the language is not set, set to an empty string or matches the language that is detected for the user's query. E.g. set it to "en" if you want the STBR rule to only work for users working from computers with English being set as the main language. Keep it unset if you want the rule not to depend on local language settings of user's computer.
type String.t · confidence medium

`SearchPolicyRankableSensitivity.dasherUser`
Rule: Whether the query came from a managed Workspace account user.
Google's description: True iff the query is from a Dasher user.
type boolean() · confidence medium

`CountryCountryAttachment.documentLocationSource`
Rule: Where the document's location information came from.
Google's description: Specifies the origin of
type String.t · confidence low · stages: 12. Personalization, Local & Context

`CountryCountryAttachment.weightAboveIdealForLocalness`
Rule: A confidence number about how global a result is; cleared in production.
Google's description: If result is global, store weight above ideal, as a confidence signal. Used in query localness, cleared in production CountryAttachment.
type float() · confidence low · stages: 12. Personalization, Local & Context

`CountryGeoLocation.confidence`
Rule: How sure the system is about the location; removed when the index is built.
Google's description: Confidence on the location. Ranges in [0.0, 1.0]. Cleared during index creation.
type number() · confidence low

`CountryLocationInfo.type`
Rule: What kind of place this is, such as a city or a state.
Google's description: Type of the location (sub-locality, city, state etc).
type String.t · confidence low

`ImageExifIPTCMetadata.locationCreated`
Rule: Where the camera was when the picture was taken.
Google's description: Location of the camera
type GoogleApi.ContentWarehouse.V1.Model.ImageExifIPTCMetadataLocationInfo.t · confidence low

`ImageExifIPTCMetadata.locationShown`
Rule: The place or places shown in the picture.
Google's description: Location shown on the image
type list(GoogleApi.ContentWarehouse.V1.Model.ImageExifIPTCMetadataLocationInfo.t) · confidence low

`ImageExifImageEmbeddedMetadata.deprecatedCity`
Rule: An old city field from image metadata, replaced by a newer location field.
Google's description: Location from IPTC @deprecated: Use iptc.location instead.
type String.t · confidence low

`ImageExifImageEmbeddedMetadata.latitude`
Rule: The latitude where the picture was taken.
Google's description: GPS location +/- 90 inclusive
type float() · confidence low

`ImageExifImageEmbeddedMetadata.longitude`
Rule: The longitude where the picture was taken.
Google's description: +/- 180 inclusive
type float() · confidence low

`IndexingDupsLocalizedLocalizedClusterLinkBasedClusterInfo.linkMember`
Rule: The verified member pages of the cluster.
Google's description: All verified members of the cluster (including recursive inclusions).
type list(GoogleApi.ContentWarehouse.V1.Model.IndexingDupsLocalizedLocalizedClusterLinkBasedClusterInfoLinkMember.t) · confidence low

`IndexingDupsLocalizedLocalizedClusterTargetLinkLinkAnnotationSourceInfo.anchorText`
Rule: The anchor text the language code was read from.
Google's description: Optional field for storing the anchor text the language code was extracted from. Applies to outlinks only.
type String.t · confidence low

`IndexingDupsLocalizedLocalizedClusterTargetLinkLinkAnnotationSourceInfo.annotationSource`
Rule: Where the language code was found, such as hreflang or a sitemap.
Google's description: Information about where the language code was extracted from.
type String.t · confidence low

`IndexingDupsLocalizedLocalizedClusterTargetLinkTargetDocData.contentLanguage`
Rule: The language detected in the linked page's text.
Google's description: The detected on-page content language of the document.
type String.t · confidence low

`LocalWWWInfo.isLargeChain`
Rule: Whether the local business tied to this page is a widely spread chain.
Google's description: Does this LocalWWWInfo represent a widely-distributed chain?
type boolean() · confidence low · stages: 12. Personalization, Local & Context

`LocalWWWInfoCluster.featureType`
Rule: The category of place this listing is, mainly whether it is a point of interest.
Google's description: Feature type for this listing, from LocalListing::info::related_feature. A geostore::FeatureProto::TypeCategory. Intended primarily to indicate POI-ness (i.e., TYPE_ESTABLISHMENT_POI).
type integer() · confidence low · stages: 12. Personalization, Local & Context

`LocalWWWInfoCluster.pageTypeFlags`
Rule: Flags describing what kind of web reference this page is for the business.
Google's description: Type of the web reference.
type integer() · confidence low · stages: 12. Personalization, Local & Context

`NlpSemanticParsingQRefAnnotation.location`
Rule: The center point coordinates of a place entity.
Google's description: The center point of this location. This is either directly provided by the FeatureProto.center field or the centroid using the points of the polygon in the FeatureProto.
type GoogleApi.ContentWarehouse.V1.Model.GeostorePointProto.t · confidence low · stages: 9. Query Understanding & Intent Classification

`NlpSemanticParsingQRefAnnotation.locationType`
Rule: What kind of place the entity is, such as a city or a country, stored as a number.
Google's description: The location type of the entity, as an int32 representing a TypeCategory enum value. For example, this could be TYPE_LOCALITY (37) or TYPE_COUNTRY (33). We store this type as an int because including FeatureProto would cause java/com/google/ads/adh/pipeline/bigquery:ProtoCatalog to become too large, resulting in OOM errors.
type integer() · confidence low · stages: 9. Query Understanding & Intent Classification

`RepositoryAnnotationsGeoTopic.establishmentType`
Rule: A code for the kind of establishment the place is.
Google's description: Sub type for POI types like ESTABLISHMENT_POI, ESTABLISHMENT_GROUNDS & ESTABLISHMENT_BUILDING
type integer() · confidence low

`RepositoryWebrefMention.personalizationContextOutputs`
Rule: Personalization boost or penalty scores applied to this mention.
Google's description: Additional personalization output scores about this mention.
type GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefPersonalizationContextOutputs.t · confidence low · stages: 6. Entity & Knowledge Graph Integration

`RepositoryWebrefMention.referenceScore`
Rule: How likely the mention points at something from personal data.
Google's description: Whether the mention is a reference (e.g. it could be resolved to an entity coming from personal data) or not, currently only used for personal resolutions. Scores from 0 (not a reference) to 1 (reference). Use at your own risk as they are subject to change, advised to talk to refx-personal first.
type number() · confidence low · stages: 6. Entity & Knowledge Graph Integration

`RepositoryWebrefMention.resolutionScore`
Rule: How likely the mention resolves to a specific personal item.
Google's description: Whether the mention is a resolution or not, currently only used for personal resolutions. Scores from 0 (not a resolution) to 1 (resolution). Use at your own risk as they are subject to change, advised to talk to refx-personal first.
type number() · confidence low · stages: 6. Entity & Knowledge Graph Integration

`RepositoryWebrefPersonalizationContextOutput.score`
Rule: How strongly a personalization boost or penalty was applied.
Google's description: Score corresponding to some kind of biasing strength which was applied. The exact semantics of this score is subject to further changes. Don't make assumptions about specific values or ranges. Values > 0 represents that a boost was applied. Values < 0 represents that a penalty was appled.
type number() · confidence low

`RepositoryWebrefPersonalizationContextOutputs.outputs`
Rule: Personalization scores broken out by type.
Google's description: Detailed output scores per personalization type.
type list(GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefPersonalizationContextOutput.t) · confidence low · stages: 12. Personalization, Local & Context

`RepositoryWebrefWebrefAttachmentMetadata.latE7`
Rule: Latitude and longitude of a place linked to the entity.
Google's description: Latitude and longitude of the location. Same format as geostore.PointProto.
type integer() · confidence low

`RepositoryWebrefWikipediaGeocode.location`
Rule: Coordinates for a place taken from Wikipedia data.
Google's description: The location as extracted from the wikijoins.
type GoogleApi.ContentWarehouse.V1.Model.GeostorePointProto.t · confidence low

`WWWDocInfo.language`
Rule: The document's language, with no usable description given.
Google's description: go/iii-td b/130371355
type String.t · confidence low

Query intent (44)

`ClassifierPornQueryClassifierOutput.isPositive`
Rule: Whether the query counts as adult after applying the recommended cutoff.
Google's description: The bit that shows if this classifier outputs positive classification for the input query. Set by thresholding with a recommended threshold.
type boolean() · confidence medium

`ClassifierPornQueryClassifierOutput.score`
Rule: The adult content score the classifier gave the query.
Google's description: The score that the classifier assigned to the input query. This is filled by all verticals.
type number() · confidence medium

`KnowledgeAnswersIntentQueryArgumentSignals.qrefConfidenceScore`
Rule: Confidence that the entity matched to this part of the query is the right one.
Google's description: The QRef confidence score for an entity argument.
type number() · confidence medium

`KnowledgeAnswersIntentQueryArgumentSignals.rawQueryText`
Rule: Copy of the exact words in the query that this annotation covers.
Google's description: A copy of the span of canonical (raw) parser input text corresponding to this annotation.
type String.t · confidence medium

`KnowledgeAnswersIntentQueryImpliedEntity.annotatedSpan`
Rule: Copy of the query words in which this implied entity was found.
Google's description: A copy of the span of canonical (raw) parser input text corresponding to this annotation. Copied from QRefAnnotation.annotated_span.
type String.t · confidence medium

`KnowledgeAnswersIntentQueryImpliedEntity.qrefConfidenceScore`
Rule: Confidence between zero and one that the entity was correctly spotted.
Google's description: The QRef confidence (in [0, 1]) of the entity being correctly annotated.
type number() · confidence medium

`KnowledgeAnswersIntentQueryQrefAnnotationSignals.score`
Rule: Score attached to an entity annotation found in the query.
Google's description: Numeric value associated with each annotation within the Qref servlet output.
type float() · confidence medium

`NlpSemanticParsingQRefAnnotation.annotatedSpan`
Rule: A copy of the piece of query text this entity annotation covers.
Google's description: A copy of the span of canonical (raw) parser input text corresponding to this annotation.
type String.t · confidence medium · stages: 9. Query Understanding & Intent Classification

`NlpSemanticParsingQRefAnnotation.confidenceScore`
Rule: How confident the system is, from 0 to 1, that the entity was correctly identified.
Google's description: The confidence (in [0, 1]) of the entity being correctly annotated.
type float() · confidence medium · stages: 9. Query Understanding & Intent Classification

`RepositoryWebrefSupportTransferRule.allowWildcardIntents`
Rule: Switch letting an entity transfer rule trigger panel style results for any entity type.
Google's description: If set to true, allow STBR targets to trigger intents like ShowEntity that do not have explicit lists of allowed collections, accepting entities with any collections instead. This setting together with target_collection define what intents are allowed to be triggered by the STBR target. Be careful with setting this option to 'true', as in the case of our example the query [france] would result in the knowledge panel for the monarch. Louis XIV might've wanted such a behaviour, but you are probabl
type boolean() · confidence medium

`RepositoryWebrefSupportTransferRule.targetCollection`
Rule: Entity type given to the transferred annotation, limiting what results it can trigger.
Google's description: Collection that is going to be assigned to the target when the annotation for the target is created. This setting together with allow_wildcard_intents define what intents are allowed to be triggered by the STBR target. At the moment those annotations get created, we can not afford to fetch information about the STBR target from Topic Server, but the target mid by itself doesn't mean much for the Search stack. Setting up the collection for that annotated mid allows us to provide at least some inf
type GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefKGCollection.t · confidence medium

`SearchPolicyRankableSensitivityFollowOn.blockNonV2SearchBackends`
Rule: Whether parallel search backends should be blocked for this query.
Google's description: Should e2e search candidates running in parallel with QU (eg. GBot) be blocked.
type boolean() · confidence medium

`SearchPolicyRankableSensitivityFollowOn.ignoreQueryUnderstanding`
Rule: Whether this sensitivity takes precedence over ones found by query understanding.
Google's description: Iff true this follow-on sensitivity will rank above the ones determined by query understanding (QU).
type boolean() · confidence medium

`SearchPolicyRankableSensitivityQueryUnderstanding.intentOnlyNoPii`
Rule: Marks a query with a sensitive intent but no sensitive personal content.
Google's description: QU dectects a sensitive intent with no sensitive content (eg., [Send email]). This flag is only used to trigger a sensitive feature because as a precaution all sensitive features should check current sensitivity mode before triggering; but will NOT block any backends, which means all other features can compete fairly.
type boolean() · confidence medium

`SearchPolicyRankableSensitivityQueryUnderstanding.rewrittenQuery`
Rule: The rewritten version of the query this sensitivity was computed for.
Google's description: The rewritten query this sensitivity is for. Note different query rewrites could produce different sensitivities.
type String.t · confidence medium

`UniversalsearchNewPackerKnowledgeResultSupport.score`
Rule: How strongly this result supports a given query interpretation, from 0 to 1.
Google's description: A score from 0 to 1 inclusive that represents the strength of the result support. A score of one indicates this result is entirely about this interpretation.
type number() · confidence medium

`VideoContentSearchCaptionEntityAnchorFeatures.isOracleEntity`
Rule: Whether the topic also shows up as a follow up query.
Google's description: True if the given entity appears as an Oracle followup query.
type boolean() · confidence medium

`WWWSnippetResponse.isValidResult`
Rule: Whether the result actually matches the query and should be kept.
Google's description: Assume this is true unless we discover that the result doesn't match the the query, in which case this result is invalid despite having returned docinfo.
type boolean() · confidence medium · stages: 10. Snippet & Answer Extraction

`KnowledgeAnswersIntentQueryArgumentSignals.fromManualSymbolAnnotation`
Rule: Marks that the entity came from a hand-made graphic symbol annotation.
Google's description: Whether the argument entity comes from a manual graphic symbol annotation. This is later used as a heuristic for poor web result quality.
type boolean() · confidence low

`KnowledgeAnswersIntentQueryArgumentSignals.fromSymbolAnnotation`
Rule: Marks that the entity came from a graphic symbol annotation.
Google's description: Whether the argument entity comes from a graphic symbol annotation. This is later used as a heuristic for poor web result quality.
type boolean() · confidence low

`KnowledgeAnswersIntentQueryArgumentSignals.relatednessSignals`
Rule: Relatedness signals about the argument, such as how popular the query is.
Google's description: Relatedness Matrix signals about this argument, e.g., query_popularity.
type GoogleApi.ContentWarehouse.V1.Model.KnowledgeAnswersIntentQueryRelatednessSignals.t · confidence low

`KnowledgeAnswersIntentQueryLocalResultSignals.salientTermSet`
Rule: Notable terms tied to a local result based on its place categories.
Google's description: Salient terms associated with this Local result based on the result gcids.
type GoogleApi.ContentWarehouse.V1.Model.QualitySalientTermsSalientTermSet.t · confidence low

`KnowledgeAnswersIntentQueryLocalSignals.minSalientTermSimilarity`
Rule: Lowest similarity of notable terms between the local results shown.
Google's description: Minimum salient term similarity between Local results.
type number() · confidence low

`KnowledgeAnswersIntentQuerySimpleValue.ungroundedValue`
Rule: The part of the query that the parser could not make sense of.
Google's description: Ungrounded value contains the part of the query (or web snippet, etc.) that was not understood. For more information on this field, see the "Ungrounded value" item in the table of contents of go/iql-v1
type String.t · confidence low

`NlpSemanticParsingQRefAnnotation.clusterSetScore`
Rule: How confident the system is in the entity cluster it matched for the query.
Google's description: Cluster set qref confidence score.
type float() · confidence low · stages: 9. Query Understanding & Intent Classification

`NlpSemanticParsingQRefAnnotation.mdvcVerticals`
Rule: Which broad categories, or verticals, this compound entity belongs to.
Google's description: The set of verticals this summary node belongs to.
type list(String.t) · confidence low · stages: 9. Query Understanding & Intent Classification

`NlpSemanticParsingSaftCoreference.category`
Rule: Whether a referring phrase in the query is a pronoun or a noun phrase.
Google's description: Categories can be either a $PronounMention or $NominalMention.
type String.t · confidence low

`NlpSemanticParsingSaftCoreference.rawText`
Rule: The piece of the query text this coreference annotation covers.
Google's description: The substring of the raw query spanned by this annotation.
type String.t · confidence low

`NlpSemanticParsingSaftCoreference.referentText`
Rule: The name of the thing a referring phrase in the query points back to.
Google's description: The name of the entity this mentions refers to.
type String.t · confidence low

`NlpSemanticParsingSaftMeasure.category`
Rule: What kind of measurement the phrase expresses, such as mass.
Google's description: Defines the category of measure, like $Mass.
type String.t · confidence low

`NlpSemanticParsingSaftMeasure.rawText`
Rule: The piece of the query text this measurement annotation covers.
Google's description: The substring of the raw query spanned by this annotation.
type String.t · confidence low

`NlpSemanticParsingSaftMeasure.value`
Rule: The numeric amount of a measurement found in the query.
Google's description: The numerical value of the measure.
type float() · confidence low

`NlpSemanticParsingSaftMentionAnnotation.coreference`
Rule: Annotations for query phrases that refer back to something mentioned earlier.
Google's description: Annotations for spans that are resolved coreference mentions.
type GoogleApi.ContentWarehouse.V1.Model.NlpSemanticParsingSaftCoreference.t · confidence low

`NlpSemanticParsingSaftMentionAnnotation.measure`
Rule: Annotations for query phrases that state a measurement, such as "53 pounds".
Google's description: Annotations for spans "53 pounds".
type GoogleApi.ContentWarehouse.V1.Model.NlpSemanticParsingSaftMeasure.t · confidence low

`NlpSemanticParsingSaftSpan.category`
Rule: The grammatical or semantic label attached to a piece of query text.
Google's description: Categories can be either syntactic (NNS for fine-grained-POS) or semantics ($Mass for measures).
type String.t · confidence low

`NlpSemanticParsingSaftSpan.rawText`
Rule: The piece of the query text this annotation covers.
Google's description: The substring of the raw query spanned by this annotation.
type String.t · confidence low

`RepositoryWebrefEntityAnnotations.explainedRangeInfo`
Rule: The parts of a query that this entity explains.
Google's description: All ranges explained by the entity or any other entity it implies. Used in the context of partial query interpretation (go/partial-understanding).
type GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefExplainedRangeInfo.t · confidence low

`RepositoryWebrefEntityAnnotations.isResolution`
Rule: Whether the entity is a summary produced by combining other annotations.
Google's description: True if the entity is an MDVC summary entity, i.e. it might not be mentioned directly on the query, but it is the product of resolving a set of explicit annotations. E.g. "2014 FIFA World Cup" can be the summary for the query: [soccer world cup in brazil] even though none of the names of the entity is mentioned on the query. Summary nodes can also be synthetic, i.e. have a /t/ mid, as they represent the intersection between a set of regular annotations. For more information, see
type boolean() · confidence low

`RepositoryWebrefExplainedRangeInfo.explainedRange`
Rule: The stretches of text that an entity accounts for.
Google's description: All ranges explained by the entity.
type list(GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefExplainedRangeInfoExplainedRange.t) · confidence low

`RepositoryWebrefLexicalRange.direction`
Rule: How a measurable aspect relates to the facet mentioned in the query.
Google's description: Direction defines the relation between the measurable aspect and the facet.
type String.t · confidence low

`RepositoryWebrefWebrefMustangAttachment.iqlAttachment`
Rule: Encoded intent expressions attached to the page.
Google's description: A sub-proto to encode IQL expressions. To be used by Pianno page-level intents and Webref Compounds.
type GoogleApi.ContentWarehouse.V1.Model.KnowledgeAnswersIntentQueryIndexingIQLAttachment.t · confidence low

`RepositoryWebrefWebrefMustangAttachment.unboundIntentMid`
Rule: IDs of general intents detected on the page without specific arguments.
Google's description: The uint64-encoded MID of the unbound intents generated by Pianno. An unbound intent is the annotation of an intent without slots arguments (e.g. Age) declared in Intent Catalog. See go/pianno-asteroid-belt-migration for details. We only keep the top unbound intents with the highest orbit scores. Should have the same number of elements as unbound_intent_score_e2.
type list(String.t) · confidence low

`RepositoryWebrefWebrefMustangAttachment.unboundIntentScoreE2`
Rule: Confidence score for each detected page level intent.
Google's description: The confidence of the unbound intent. represented as a fixed-point integer with two decimal precision. In the range [0, 100]. Should have the same number of elements as unbound_intent_mid.
type list(integer()) · confidence low

`SearchPolicyRankableSensitivity.sensitivityMode`
Rule: A key naming which sensitivity category applies to the query.
Google's description: Key of a sensitivity.
type String.t · confidence low

Other (39)

`OceanDocTag.viewability`
Rule: Rules for how much of the book may be displayed in each country.
Google's description: Volume viewability, which defines how/if the volume should be displayed in various locales.
type GoogleApi.ContentWarehouse.V1.Model.OceanVolumeViewability.t · confidence medium

`OceanDocTagCatalogSpecific.latest`
Rule: Whether this is the newest issue of a catalog series, used for filtering.
Google's description: Is this the latest issue of this catalog series? This is required to filter results if the latest restrict is on.
type boolean() · confidence medium

`PerDocData.appsLink`
Rule: Android app IDs found in the page's outgoing links, used for app results.
Google's description: AppsLink contains Android application IDs in outlinks. It is used to improve results ranking within applications universal. See
type GoogleApi.ContentWarehouse.V1.Model.QualityCalypsoAppsLink.t · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.launchAppInfo`
Rule: Instructions for opening a mobile app to view this document's content.
Google's description: Info on how to launch a mobile app to consume this document's content, if applicable (see go/calypso).
type GoogleApi.ContentWarehouse.V1.Model.QualityRichsnippetsAppsProtosLaunchAppInfoPerDocData.t · confidence medium · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PremiumPerDocData.IsEntitled`
Rule: Whether the current user can read the paid content without paying.
Google's description: User is entitled to see the premium content for free.
type boolean() · confidence medium

`PremiumPerDocData.Price`
Rule: The price of a paid document, stored in cents.
Google's description: price * 100 if available
type integer() · confidence medium

`ScienceCitationDownloadURL.DMCANotice`
Rule: A notice URL recording that this result was taken down on copyright grounds.
Google's description: metatag: URL; result was taken down
type String.t · confidence medium

`VideoContentSearchVideoAnchorScoreInfo.safeSearchClassifierOutput`
Rule: Output of the adult content classifier for this key moment label.
Google's description: The output of Safe Search's MultiLabelClassifier.
type GoogleApi.ContentWarehouse.V1.Model.ClassifierPornQueryMultiLabelClassifierOutput.t · confidence medium

`VideoContentSearchVideoInfo.asrLanguage`
Rule: The language of the speech detected in the video.
Google's description: Language information, extracted from content_based_metadata.speech_properties.
type String.t · confidence medium

`VideoContentSearchVideoInfo.docLanguage`
Rule: The language of the page the video is on.
Google's description: Language information, extracted from DocProperties.
type String.t · confidence medium

`VideoContentSearchVideoInfo.titleLanguage`
Rule: The language detected in the video title.
Google's description: Video title language Language information, set automatically by the SAFT LangID.
type String.t · confidence medium

`VideoContentSearchVideoScoreInfo.safeSearchClassifierOutput`
Rule: Output of the adult content classifier for the video title.
Google's description: The output of Safe Search's MultiLabelClassifier for video title.
type GoogleApi.ContentWarehouse.V1.Model.ClassifierPornQueryMultiLabelClassifierOutput.t · confidence medium

`BlogPerDocData.isSyntacticReshare`
Rule: Marks a post that simply reshares another post.
Google's description: A syntactic reshare is a document that is
type boolean() · confidence low

`CompositeDocIndexingInfo.primaryVertical`
Rule: Which search vertical the document belongs to.
Google's description: Vertical membership of the document. -
type String.t · confidence low · stages: 2. Document Processing & Indexing

`DocProperties.proseRestrict`
Rule: Restriction tags used by custom structured search.
Google's description: The restricts for CSE structured search.
type list(String.t) · confidence low

`GDocumentBase.FilterForSafeSearch`
Rule: An old SafeSearch filter flag that has not been filled in since 2012.
Google's description: Deprecated, do not use, this field is not populated since 2012.
type integer() · confidence low · stages: 2. Document Processing & Indexing

`GDocumentBaseDirectory.Category`
Rule: The web directory category the page was listed under.
Google's description: encoded in UTF8
type String.t · confidence low

`ImageData.isIipInScope`
Rule: A flag marking the image as covered by a particular trust program's entities.
Google's description: IIP in scope signal (go/iip). Set to true if the image is annotated with any iip_in_scope entities (go/iukp-trust-v2).
type boolean() · confidence low

`NlxDataSchemaLanguageSpan.languageCode`
Rule: Language codes for the text in this span.
Google's description: A set of BCP-47 codes indicating the language(s) of this span of text.
type list(String.t) · confidence low

`OceanDocTag.subjectBitmap`
Rule: Bitmap of top-level subject categories for the book.
Google's description: Bitmap indicating top-level subjects associated with this document. See ocean/metadata/subjects/util.h for more detail.
type String.t · confidence low

`OceanDocTagBookSpecific.subject`
Rule: Subject category assigned to the book.
Google's description: Subject (from Bisac)
type String.t · confidence low

`OceanPerDocData.flags`
Rule: Per-document flags for rights, availability, and adult content.
Google's description: rights, mask-availability, porn, etc.
type String.t · confidence low

`PerDocData.scienceDoctype`
Rule: A code marking the document as a Scholar science document and how much of it is visible.
Google's description: Scholar/Science Document type: <0 == not a Science Document -- default 0 == Science doc fully visible >0 == Science doc but limited visibility, the number is the visible terms
type integer() · confidence low · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PremiumPerDocData.IsArchival`
Rule: A flag marking a free document as archival material.
Google's description: True if a free document is archival in nature.
type boolean() · confidence low

`QualityLabelsGoogleLabelDataLabel.globalLabelBucket`
Rule: A compact bucketed form of the label's value, used to save storage.
Google's description: A byte-size value representing 64 * (1 + global_label_value). Use this instead of global_label_value to save on label storage. See quality_prose::LabelValueToBucket() for more info.
type integer() · confidence low

`QualityLabelsGoogleLabelDataLabelProvider.labelBucket`
Rule: A compact bucketed form of the label's value, used to save storage.
Google's description: A byte-size value representing 64 * (1 + global_label_value). Use it instead of global_label_value to save on label storage.
type integer() · confidence low

`QualityNsrNsrDataClusterUplift.local`
Rule: An uplift score for the local-sites arm of a site experiment.
Google's description: Score for the local sites arm.
type number() · confidence low

`QualityNsrNsrDataClusterUplift.small`
Rule: An uplift score for the small-sites arm of a site experiment.
Google's description: Score for the small sites arm.
type number() · confidence low

`QualityShoppingShoppingAttachmentPBlock.fullTitle`
Rule: The longer product title pulled out of a product block on the page.
Google's description: Field full_title may contain duplicate info from title and list_title.
type String.t · confidence low

`QualityShoppingShoppingAttachmentPBlock.title`
Rule: The product title pulled out of a product block on the page.
Google's description: Product info extracted by Product Blocks go/sdu-shopping-page-intro and go/product-block-extraction. Here is an example of a page with a ## list_title (Shoes) and 3 blocks with their own titles: | Shoes | | ----------| |
type String.t · confidence low

`RepositoryAnnotationsRdfaRdfaRichSnippetsApplication.filteringTag`
Rule: Tags stored so app results can be filtered, such as a popularity threshold tag.
Google's description: Tags to be indexed for filtering, e.g. "ft_popular_score_gt_1m".
type list(String.t) · confidence low

`RepositoryWebrefDocumentMetadata.language`
Rule: The language the page is written in.
Google's description: The document language, as read from doc().content().language(). This is go/language-enum value.
type String.t · confidence low · stages: 6. Entity & Knowledge Graph Integration

`RepositoryWebrefWebrefEntities.rangeAnnotations`
Rule: Non-entity markings of text ranges such as parts of speech or stopwords.
Google's description: These messages contain non-entity annotations of ranges in the document. This might be used to hold part-of-speech annotations, stopword annotations, and other range based information. The exclude_field option is for Goldmine AnnotationsFinder to exclude ranged_annotations from retrieving annotation entities
type list(GoogleApi.ContentWarehouse.V1.Model.RepositoryWebrefRangeAnnotations.t) · confidence low · stages: 6. Entity & Knowledge Graph Integration

`ScienceCitationDownloadURL.DownloadMonth`
Rule: The month tied to this download URL, numbered from zero for January.
Google's description: DownloadMonth is a zero-indexed field (0 is January).
type integer() · confidence low

`ScienceCitationDownloadURL.DownloadYear`
Rule: The full year tied to this download URL.
Google's description: no abbrv
type integer() · confidence low

`SentenceBoundaryAnnotationsInstance.cleanText`
Rule: A tidied-up copy of the annotated text with extra whitespace removed.
Google's description: A clean version of .text() generated by using CleanText() and stripping unnecessary whitespace.
type String.t · confidence low

`SentenceBoundaryAnnotationsInstance.context`
Rule: The surrounding page text where the annotation was found.
Google's description: Plain text context from the page within which the annotation occurred.
type String.t · confidence low

`SentenceBoundaryAnnotationsInstance.text`
Rule: The original document text covered by this annotation.
Google's description: Original UTF-8 document text occurring in the range [begin, end).
type String.t · confidence low

`VideoContentSearchVideoAnchorScoreInfo.labelLanguage`
Rule: The language detected in the key moment label text.
Google's description: Detected language of label
type String.t · confidence low

Freshness (19)

`CompositeDocQualitySignals.lastSignificantUpdate`
Rule: A combined date of the page's last meaningful change, used for the last-modified search filter.
Google's description: Contains a date used for the "Date Last Modified" toolbelt restrict mode. Note: this date is a combined date and is different from the pure shingle-based signal stored in contentage.last_significant_update field.
type GoogleApi.ContentWarehouse.V1.Model.QualityTimebasedLastSignificantUpdate.t · confidence medium

`CrawlerChangerateUrlChangerate.lastChanged`
Rule: The last time the crawler saw the page's content differ.
Google's description: The last time (unix timestamp) we saw a changed copy of the document. Provided iff we have seen the page change.
type integer() · confidence medium · stages: 1. Crawling & Discovery; 7. Freshness & Temporal Signals

`CrawlerChangerateUrlVersion.lastModified`
Rule: The date the server reported in the page's last-modified header.
Google's description: The date from the LastModified header, if present.
type integer() · confidence medium · stages: 1. Crawling & Discovery

`OceanDocTagCatalogSpecific.publicationDate`
Rule: Catalog publication date used to decide which issue is newest.
Google's description: time_t date corresponding to the catalog publication date. Approximate when the catalog does not have an exact "date" of publication, e.g. For "Spring 2002", year, month and day are 2002, 03 and 21, respectivley. This value is used to compare catalog issues to determine the latest. The value is stored in seconds-since-epoch, 1/1/1970. This is not a problem for catalogs because we are not dealing with any catalogs from before the 70's.
type integer() · confidence medium

`QualityTimebasedSyntacticDate.syntacticDateNotForRestrict`
Rule: Whether this date may be used when a searcher filters by date.
Google's description: If this is true, do not use syntactic date in date restricts.
type boolean() · confidence medium · stages: 7. Freshness & Temporal Signals

`ScienceCitation.ClusterDiscoveryDate`
Rule: The earliest date the article group was discovered, used for logging and date sorting.
Google's description: Used for logging, recommendations, and sort-by-date. Contains the earliest discovery date of the cluster, adjusted for earlier publication dates. Stored in Universal time scale (100 ns ticks since 0001 AD) because Unix timestamp would lead to negative dates for pre-1970 docs.
type String.t · confidence medium

`AnchorsAnchor.possiblyOldFirstseenDate`
Rule: Old flag meaning the link may be older than recorded; deprecated.
Google's description: DEPRECATED. It used to be set if firstseen_date is not set. It's to indicate that the anchor is possibly old, but we don't have enough information to tell until the linkage map is updated. TODO(hxu) rename it to possibly_old_firstseen_date_DEPRECATED after clean up other dependencies.
type boolean() · confidence low · stages: 4. Link Analysis & Authority Computation

`OceanDocTagBookSpecific.publicationDate`
Rule: The book's publication date.
Google's description: In the format yyyy.mm.dd, or possibly just yyyy.
type String.t · confidence low

`OceanDocTagNewspaperSpecific.publicationDate`
Rule: Publication date of a scanned newspaper issue.
Google's description: These fields are only populated for Santorini (newspapers on goovols) formatted newspapers, not for Atlantis: YYYY.MM.DD format.
type String.t · confidence low

`PerDocData.freshnessEncodedSignals`
Rule: Deprecated packed freshness and aging data derived from URL-pattern signals.
Google's description: Stores freshness and aging related data, such as time-related quality metrics predicted from url-pattern level signals. Use the encoding decoding API in quality/freshness/docclassifier/aging/encoded-pattern-signals.h This field is deprecated.
type String.t · confidence low · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.semanticDateConfidence`
Rule: A deprecated confidence value for the semantic date, replaced by the newer info field.
Google's description: DEPRECATED: semantic_date_confidence replaced by semantic_date_info.
type integer() · confidence low · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualityDniDocPreviewRestrictions.bylineDateSecs`
Rule: The publication date the site owner declared for the page.
Google's description: Publish date set by webmaster. See detailed description here:
type String.t · confidence low

`QualityDniDocPreviewRestrictions.firstseenDateSecs`
Rule: The date indexing first saw the page, used when no byline date exists.
Google's description: Firstseen date populated by indexing. It works as fallback to byline_date if it doesn't exist. Note that this will currently only be set for canonical documents where firstseen date could be populated.
type integer() · confidence low

`QualityTimebasedLastSignificantUpdate.source`
Rule: Which component produced the last-significant-update time.
Google's description: The source the signal comes from.
type String.t · confidence low · stages: 7. Freshness & Temporal Signals

`QualityTimebasedLastSignificantUpdateAdjustments.adjustmentSource`
Rule: Which component supplied an adjustment to the last-update time.
Google's description: If the selected LSU has been adjusted, i.e. the maximum passage timestamp, firstseen or contentage were assigned to LSU, the adjustment source is stored here.
type String.t · confidence low · stages: 7. Freshness & Temporal Signals

`QualityTimebasedLastSignificantUpdateAdjustments.unboundedTimestampInSeconds`
Rule: A candidate update time that was thrown out for being too late.
Google's description: The timestamp that was picked up by the component indicated in the LastSignificantUpdateSource but was dropped due to exceeding the upper bound. Set only if it is not equal to final LSU.
type String.t · confidence low · stages: 7. Freshness & Temporal Signals

`QualityTimebasedLastSignificantUpdateAdjustments.unboundedTimestampSource`
Rule: Which component produced the discarded candidate update time.
Google's description: The source that produced the unbounded timestamp.
type String.t · confidence low · stages: 7. Freshness & Temporal Signals

`QualityTimebasedSyntacticDate.fromExplicitTimeZone`
Rule: Whether the date on the page stated its time zone; internal use only.
Google's description: If set to true, the source of the date has explicit time zone specification. Note: This is only used internally and should not be populated in docjoins.
type boolean() · confidence low · stages: 7. Freshness & Temporal Signals

`WWWDocInfo.lastModTime`
Rule: A last-modified time that the serving system does not use.
Google's description: Unused by gws
type String.t · confidence low

Clicks & engagement (17)

`QualityNavboostCrapsCrapsData.country`
Rule: Which country's slice of click data this record covers.
Google's description: The two-letter uppercase country slice of the CrapsData. Examples: "US", "FR", "BR"
type String.t · confidence medium · stages: 5. User Engagement & NavBoost (Click Signals)

`QualityNavboostCrapsCrapsData.device`
Rule: Which device and operating system slice of click data this record covers.
Google's description: The device interface and os slice of the CrapsData.
type GoogleApi.ContentWarehouse.V1.Model.QualityNavboostCrapsCrapsDevice.t · confidence medium · stages: 5. User Engagement & NavBoost (Click Signals)

`QualityNavboostCrapsCrapsData.language`
Rule: Which language slice of click data this record covers.
Google's description: The language slice of the CrapsData. Examples: "en", "fr", "pt-BR",
type String.t · confidence medium · stages: 5. User Engagement & NavBoost (Click Signals)

`QualityNavboostCrapsCrapsData.sliceTag`
Rule: A free-form tag naming how this click data was sliced.
Google's description: This field can be used by the craps pipeline to slice up signals by various attributes such as device type, country, locale etc. The slice_tag can be an arbitrary string, and the CrapsData values for each slice_tag are aggregated separately, together with the default empty slice_tag.
type String.t · confidence medium · stages: 5. User Engagement & NavBoost (Click Signals)

`QualityNavboostCrapsCrapsDevice.uxInterface`
Rule: Whether the request came from a desktop, mobile, or tablet device.
Google's description: An enum taken from GWSLogEntryProto that indicates what type of device a request came from. This includes an entry for DESKTOP(1), MOBILE(2), and TABLET(3) devices. Reference: - gwslog(608): GWSLogEntryProto.ux_interface
type integer() · confidence medium

`QualityNavboostCrapsCrapsDevice.uxTier`
Rule: How modern the requesting browser is, on a three-step scale.
Google's description: Indicates the device browser tier for the given request. 1 means modern browsers, 3 means very old browsers, and 2 is everything in the middle. See Reference: - gwslog(609): GWSLogEntryProto.ux_tier
type integer() · confidence medium

`QualityNavboostCrapsFeatureCrapsData.country`
Rule: Which country this click aggregation covers, if any.
Google's description: Country, like "us". If not present, it's an aggregation for all countries. This is the same format as one used in Glue.
type String.t · confidence medium · stages: 5. User Engagement & NavBoost (Click Signals)

`QualityNavboostCrapsFeatureCrapsData.device`
Rule: Which device type this click aggregation covers, if any.
Google's description: Device, like "m". If not present, it's an aggregation for all devices. "m" - mobile devices. "d" - destop devices.
type String.t · confidence medium · stages: 5. User Engagement & NavBoost (Click Signals)

`QualityNavboostCrapsFeatureCrapsData.language`
Rule: Which language this click aggregation covers, if any.
Google's description: Language, like "en". If not present, it's an aggregation for all languages. This is the same format as one used in Glue.
type String.t · confidence medium · stages: 5. User Engagement & NavBoost (Click Signals)

`QualityNavboostCrapsFeatureCrapsData.locationId`
Rule: Which metro area or city this click aggregation covers.
Google's description: Location id for metro and city. If not present, it's an aggregation for all locations within current country.
type integer() · confidence medium · stages: 5. User Engagement & NavBoost (Click Signals)

`UniversalsearchNewPackerKnowledgeResultSupport.estimatedNaviness`
Rule: An estimate of how likely a user is to click this result, used to break ties.
Google's description: The naviness for this result. Each matcher can determine the naviness to use for the result. This can be copied directly from the result, or estimated based on clicks or some other heuristic. The value should be between 0 and 1 and should indicate the probability that the user will click on that result. The estimated_naviness is used to break ties when between results with the same rank.
type number() · confidence medium

`VideoContentSearchAnchorCommonFeatureSet.retentionScore`
Rule: How likely viewers are to watch this part of the video all the way through.
Google's description: A score that is correlated with retention probability of the interval associated with this anchor (start time to end time). Retention probability of an interval is 1 - (probability the user does not watch the interval all the way through, given they started watching it). This score may be predicted by a model, or calculated from actual retention data.
type number() · confidence medium

`QualityNavboostCrapsCrapsClickSignals.unsquashedClicks`
Rule: A raw click count field that is not filled in under the current format.
Google's description: This is not being populated for the current format - instead two instances of CrapsClickSignals (squashed/unsquashed) are used. We are migrating to the new format where this field will be populated.
type float() · confidence low · stages: 5. User Engagement & NavBoost (Click Signals)

`QualityNavboostCrapsCrapsClickSignals.unsquashedImpressions`
Rule: A raw impression count field that is not filled in under the current format.
Google's description: This is not being populated for the current format - instead two instances of CrapsClickSignals (squashed/unsquashed) are used. We are migrating to the new format where this field will be populated.
type float() · confidence low · stages: 5. User Engagement & NavBoost (Click Signals)

`QualityNavboostCrapsCrapsData.patternLevel`
Rule: How general the URL pattern is that this click record covers.
Google's description: Level of pattern. More general patterns get higher values. For URL patterns this field = 0. For example, if we have "
type integer() · confidence low · stages: 5. User Engagement & NavBoost (Click Signals)

`QualityNavboostCrapsCrapsData.squashed`
Rule: A placeholder for compressed click counts that is not used yet.
Google's description: Not used yet - we will probably move the impressions / clicks / good_clicks bad clicks / last longest clicks into here from top level, and rename those fields to legacy.
type GoogleApi.ContentWarehouse.V1.Model.QualityNavboostCrapsCrapsClickSignals.t · confidence low · stages: 5. User Engagement & NavBoost (Click Signals)

`QualityNavboostCrapsFeatureCrapsData.voterTokenBitmap`
Rule: A compact set of user tokens behind these stats, used for privacy filtering.
Google's description: The set of voter tokens of the sessions that contributed to this feature's stats. Voter tokens are not unique per user, so it is a lower bound on the number of distinct users. Used for privacy-related filtering.
type GoogleApi.ContentWarehouse.V1.Model.QualityNavboostGlueVoterTokenBitmapMessage.t · confidence low · stages: 5. User Engagement & NavBoost (Click Signals)

Links & authority (16)

`ScienceCitation.NumBackwardLinks`
Rule: How many works this article cites, kept for showing to users.
Google's description: for display in gws
type integer() · confidence medium

`ScienceCitation.NumForwardLinks`
Rule: How many works cite this article, kept for showing to users.
Google's description: for display in gws
type integer() · confidence medium

`Anchors.redundantAnchorsDropped`
Rule: Count of duplicate links thrown out in link extraction; not filled by the current pipeline.
Google's description: The *_anchors_dropped fields below are not populated by Alexandria, which uses cdoc.anchor_stats instead. The total # of redundant anchors dropped in linkextractor.
type String.t · confidence low · stages: 4. Link Analysis & Authority Computation

`Anchors.supplementalAnchorsDropped`
Rule: Count of discarded supplemental-tier links; marked deprecated.
Google's description: The total # of supplemental anchors dropped in AnchorAccumulator. ## DEPRECATED.
type String.t · confidence low · stages: 4. Link Analysis & Authority Computation

`AnchorsAnchor.type`
Rule: Old link type field, replaced by link tags.
Google's description: DEPRECATED: Now in link_tags
type integer() · confidence low · stages: 4. Link Analysis & Authority Computation

`AnchorsAnchorSource.doclength`
Rule: Length of the linking page, with its usefulness questioned in the note.
Google's description: necessary for anything?
type integer() · confidence low · stages: 4. Link Analysis & Authority Computation

`AnchorsAnchorSource.pageTags`
Rule: Old property tags for the linking page; deprecated in favour of link tags.
Google's description: Page tags are described by enum PageTag in PerDocData. Page tags are used in anchors to identify properties of the linking page. These are DEPRECATED: in the future, use link_tags instead. DEPRECATED
type list(integer()) · confidence low · stages: 4. Link Analysis & Authority Computation

`GDocumentBase.Pagerank`
Rule: An old PageRank value that is deprecated and no longer maintained.
Google's description: This field is long-deprecated in favour of Pagerank_NS, it is no longer maintained and can break at any moment.
type integer() · confidence low · stages: 2. Document Processing & Indexing

`IndexingDocjoinerAnchorStatistics.scannedAnchorCount`
Rule: How many anchors were read out of storage.
Google's description: The total number of anchors being scanned from storage.
type integer() · confidence low · stages: 4. Link Analysis & Authority Computation

`IndexingDocjoinerAnchorStatisticsPerDupStats.scannedAnchorCount`
Rule: How many anchors were read from storage for this duplicate.
Google's description: The number of anchors being scanned from storage per dupforwarding.
type integer() · confidence low

`KaltixPerDocData.LocalKaltixRank`
Rule: A local variant of the per-document rank value, currently left empty.
Google's description: empty for now
type integer() · confidence low · stages: 4. Link Analysis & Authority Computation

`KaltixPerDocData.SiteKaltixRank`
Rule: A site-level variant of the per-document rank value, currently left empty.
Google's description: empty for now
type integer() · confidence low · stages: 4. Link Analysis & Authority Computation

`PerDocData.ScaledExptIndyRank`
Rule: An old experimental independence-rank value marked deprecated and not for new code.
Google's description: DEPRECATED ---------- Please do not use these fields in any new code. experimental
type integer() · confidence low · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.ScaledExptIndyRank2`
Rule: A second experimental independence-rank value.
Google's description: experimental
type integer() · confidence low · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.ScaledExptIndyRank3`
Rule: A third experimental independence-rank value.
Google's description: experimental
type integer() · confidence low · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.pagerank`
Rule: Deprecated experimental pagerank values; only the pagerank in basic info is used.
Google's description: Experimental pageranks (DEPRECATED; only pagerank in MustangBasicInfo is used).
type number() · confidence low · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

Content quality (15)

`CompressedQualitySignals.nsrConfidence`
Rule: A deprecated confidence number for the site quality score.
Google's description: NSR confidence score: converted from quality_nsr.NsrData. This field is deprecated - use nsr_variance inside nsr_data_proto instead.
type integer() · confidence low · stages: 3. Content Quality & Site-Level Scoring; 8. Spam, Penalty & Trust Systems; 11. Re-Ranking & Twiddler System

`CompressedQualitySignals.nsrOverrideBid`
Rule: A deprecated manual override value for the site quality score.
Google's description: NSR override bid, used in Q* for emergency overrides. This field is deprecated - used the equivalent field inside nsr_data_proto instead.
type number() · confidence low · stages: 3. Content Quality & Site-Level Scoring; 8. Spam, Penalty & Trust Systems; 11. Re-Ranking & Twiddler System

`CompressedQualitySignals.vlqNsr`
Rule: A deprecated site quality score for low quality video pages.
Google's description: NSR for low-quality videos, converted from quality_nsr.NsrData.vlq_nsr. This field is deprecated - used the equivalent field inside nsr_data_proto instead.
type integer() · confidence low · stages: 3. Content Quality & Site-Level Scoring; 8. Spam, Penalty & Trust Systems; 11. Re-Ranking & Twiddler System

`IndexingMobileVoltVoltPerDocData.desktopCwv`
Rule: Holds the desktop page-speed metrics for a document; noted as not filled in one pipeline.
Google's description: Desktop Core Wev Vital metrics. NOTE(yunchengz): This field will not be populated in Muppet.
type GoogleApi.ContentWarehouse.V1.Model.IndexingMobileVoltCoreWebVitals.t · confidence low

`IndexingMobileVoltVoltPerDocData.mobileCwv`
Rule: Holds the mobile page-speed metrics for a document; noted as not filled in one pipeline.
Google's description: Mobile Core Web Vital metrics. NOTE(yunchengz): This field will not be populated in Muppet.
type GoogleApi.ContentWarehouse.V1.Model.IndexingMobileVoltCoreWebVitals.t · confidence low

`MobilePerDocData.transcodedPageScore`
Rule: A retired 0-127 quality rating for a transcoded mobile page.
Google's description: DEPRECATED: The transcoded page quality repesented in 7-bits range from 0 to 127.
type integer() · confidence low

`OceanDocTag.goodTextDetail`
Rule: Rating of how good the scanned text of a book volume is.
Google's description: Text quality as defined in CA_VolumeScoreResult::OACapabilities::TextQualityAssessment Note - This is only populated if good_text() and is_ge_quality() are true
type integer() · confidence low

`OceanDocTag.isGeQuality`
Rule: Flag that the book volume meets the Google Edition quality bar.
Google's description: Set to true if volume has ge quality
type boolean() · confidence low

`PerDocData.hostNsr`
Rule: A deprecated packed site rank value; the same data now lives in the NSR data proto.
Google's description: Site rank computed for host-level sitechunks. This value encodes nsr, site_pr and new_nsr. See quality_nsr::util::ConvertNsrDataToHostNsr and go/nsr. This field is deprecated - used the equivalent field inside nsr_data_proto instead.
type integer() · confidence low · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.nsrIsCovidLocalAuthority`
Rule: Deprecated flag marking the site as a local COVID authority.
Google's description: This field is propagated to shards. In addition, it is populated at serving time by go/web-signal-joins. This field is deprecated - used the equivalent field inside nsr_data_proto instead.
type boolean() · confidence low · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`PerDocData.nsrIsElectionAuthority`
Rule: Deprecated flag marking the site as an election authority.
Google's description: This field is propagated to shards. It will also be populated at serving time by go/web-signal-joins (see b/168114815). This field is deprecated - used the equivalent field inside nsr_data_proto instead.
type boolean() · confidence low · stages: 3. Content Quality & Site-Level Scoring; 7. Freshness & Temporal Signals; 8. Spam, Penalty & Trust Systems; 9. Query Understanding & Intent Classification; 11. Re-Ranking & Twiddler System; 12. Personalization, Local & Context

`QualitySalientTermsSalientTerm.weight`
Rule: An older zero-to-100 importance value for a term, replaced by salience.
Google's description: weight is the importance of the term as a descriptor in [0, 100] (the higher the more important). NOTE: Please do not access this field directly. Use quality_salient_terms::utils::GetSalience() from salient_terms_utils.h instead. DEPRECATED: prefer salience field above.
type integer() · confidence low

`QualityShoppingShoppingAttachment.multiProductScore`
Rule: An older multi-product page score, replaced by newer model scores.
Google's description: From MagicPageTypeAnnotation.multiplicity.confidence_score * 100 Deprecated as of July 2020 when dataset_model_multi_product_score and dataset_model_single_product_score were added.
type integer() · confidence low

`WWWSnippetResponse.isLoginPage`
Rule: Whether the document looks like a login page.
Google's description: True if the document represents a login page.
type boolean() · confidence low · stages: 10. Snippet & Answer Extraction

`WWWSnippetResponse.keyword`
Rule: Keywords listed for the document.
Google's description: Document keywords
type list(String.t) · confidence low · stages: 10. Snippet & Answer Extraction

Site structure (15)

`CompositeDoc.sitemap`
Rule: The set of related links that may be shown alongside this page.
Google's description: Sitelinks: a collection of interesting links a user might be interested in, given they are interested in this document. WARNING: this is different from the crawler Sitemaps (see SitemapsSignals in the attachments).
type GoogleApi.ContentWarehouse.V1.Model.Sitemap.t · confidence medium · stages: 2. Document Processing & Indexing

`CompositeDocLocalizedVariations.webmasterAlternateNames`
Rule: The webmaster-declared alternate language or region versions of a page, used to swap in the right variant.
Google's description: All localized alternate names provided by the webmaster (canonical and dups, indexed and not-indexed). Used on the ranking side for swapping out results based on the webmaster preference.
type list(GoogleApi.ContentWarehouse.V1.Model.IndexingConverterLocalizedAlternateName.t) · confidence medium

`IndexingDocjoinerServingTimeClusterId.clusterId`
Rule: Identifier grouping a page with related pages such as paginated sets.
Google's description: The unique id to distinguish members in cluster. It could be generated in different ways according to reason, e.g. LINK_REL_NEXT_PREVIOUS: it is the fingerprint of the URL of index 0; PAGE_PARAMETER_INFO: it is the fingerprint of pagination pattern (pagination_pattern_fp field in PageParamInfo message).
type String.t · confidence medium

`IndexingDocjoinerServingTimeClusterId.reason`
Rule: Why the page was placed in this cluster.
Google's description: The reason why this document is clustered into this cluster.
type String.t · confidence medium

`IndexingDocjoinerServingTimeClusterIds.clusterId`
Rule: The list of clusters this page belongs to at serving time.
Google's description: The exact cluster metadata for each individual cluster if any.
type list(GoogleApi.ContentWarehouse.V1.Model.IndexingDocjoinerServingTimeClusterId.t) · confidence medium

`BlogPerDocData.blogurlFp`
Rule: A fingerprint of the blog address used to limit how many results one blog gets.
Google's description: used for blogurl crowding.
type String.t · confidence low

`ImageData.onPageAlternateUrl`
Rule: The page address that best matches an image not directly shown on this page.
Google's description: For an image not explicitly visible on this page, the following url is the one which most closely matches it.
type String.t · confidence low

`IndexingConverterLocalizedAlternateName.deviceMatchInfo`
Rule: Guess at which device an alternate URL targets, based on the URL pattern.
Google's description: Device match info calculated only by URL pattern.
type String.t · confidence low

`IndexingDupsComputedLocalizedAlternateNamesLocaleEntry.deviceMatchInfo`
Rule: Whether the URL pattern suggests a desktop or mobile version.
Google's description: Device match info calculated only by URL pattern.
type String.t · confidence low

`IndexingDupsLocalizedLocalizedCluster.sitedupRuleId`
Rule: Identifiers of the site-duplication rules that matched this URL.
Google's description: The list of Sitedup rule IDs for this specific URL. The value is only populated if the URL has at least one localized cluster fulfilling the following conditions: - spans more than one host - does not have filtering enabled due to other input (e.g. due to being a hreflang cluster).
type list(String.t) · confidence low

`IndexingDupsLocalizedLocalizedClusterTargetLinkLink.crossDomain`
Rule: Whether the link crosses from one host to another.
Google's description: For a link A->B where B is represented by this proto, cross_domain := Host(A) != Host(B).
type boolean() · confidence low

`IndexingDupsLocalizedLocalizedClusterTargetLinkSets.directTargetLink`
Rule: Links where the page points straight at the other URL.
Google's description: Direct links are the simplest scenarios where A simply links to B.
type list(GoogleApi.ContentWarehouse.V1.Model.IndexingDupsLocalizedLocalizedClusterTargetLink.t) · confidence low

`IndexingDupsLocalizedLocalizedClusterTargetLinkSets.indirectTargetLink`
Rule: Links reached through another page rather than directly.
Google's description: Repeated field for URLs that are not directly linking to the document TargetLink refers to. We can encounter the following scenario: A -> Links to B -> links to C (i.e. without (A) linking to (C)). In the context of B, indirect_inclusion would include the link to 'C' but not the link back to 'A'.
type list(GoogleApi.ContentWarehouse.V1.Model.IndexingDupsLocalizedLocalizedClusterTargetLink.t) · confidence low

`IndexingDupsLocalizedLocalizedClusterTargetLinkTargetDocData.outgoingLinkData`
Rule: Data about the hreflang links found on the linked page.
Google's description: Repeated field for data about the outgoing hreflang links that appear in the document that the currently processed URL refers to.
type list(GoogleApi.ContentWarehouse.V1.Model.IndexingDupsLocalizedLocalizedClusterTargetLinkLink.t) · confidence low

`Sitemap.sourceOrgfp`
Rule: A fingerprint of the source site that stops sitelinks pointing at other domains.
Google's description: prevents cross-domain forwarding
type String.t · confidence low

infrastructure_non_signal (12)

`AnchorsAnchor.experimental`
Rule: Marks a link as test-only and not to be used in serving.
Google's description: If true, the anchor is for experimental purposes and should not be used in serving.
type boolean() · confidence medium · stages: 4. Link Analysis & Authority Computation

`IndexingSignalAggregatorAggregatedScore.patternLayer`
Rule: Which experimental layer the pattern belongs to.
Google's description: Experimental layer of the corresponding pattern.
type String.t · confidence medium · stages: 2. Document Processing & Indexing; 11. Re-Ranking & Twiddler System

`IndexingSignalAggregatorUrlScore.eligibleExperimentalLayer`
Rule: Marks that the URL may be aggregated into an experimental layer.
Google's description: If this field is set, it indicates the url is eligible to be aggregated to one of the experimental layers.
type String.t · confidence medium

`KnowledgeAnswersIntentQueryArgumentSignals.parsedDueToExperiment`
Rule: The experiment IDs that caused this parse to happen.
Google's description: Experiment ID for experiments that were used to parse this FunctionCall. Empty indicates no experiments used.
type list(String.t) · confidence medium

`AnchorsAnchor.timestamp`
Rule: Old crawl time field for the link, no longer filled.
Google's description: This field is DEPRECATED and no longer filled. For source page crawl timestamp, use Source.crawl_timestamp. Next tag id should be 62.
type String.t · confidence low · stages: 4. Link Analysis & Authority Computation

`AnchorsAnchorSource.ipaddr`
Rule: Old IP address field for the linking page; deprecated.
Google's description: DEPRECATED, use packed_ipaddress
type integer() · confidence low · stages: 4. Link Analysis & Authority Computation

`AnchorsAnchorSource.seglanguage`
Rule: Old segmentation language field; deprecated.
Google's description: DEPRECATED
type integer() · confidence low · stages: 4. Link Analysis & Authority Computation

`BlogPerDocData.microblogQualityExptData`
Rule: Experimental blog quality data that is never filled in for live traffic.
Google's description: Experimental data for quality experiments. This will NOT be populated in prod, but we will use this for experiments.
type GoogleApi.ContentWarehouse.V1.Model.Proto2BridgeMessageSet.t · confidence low

`CompositeDocLiveExperimentInfo.perLiveExperimentInfo`
Rule: The details attached to this document for each live experiment.
Google's description: List of necessary information for each live experiments.
type list(GoogleApi.ContentWarehouse.V1.Model.CompositeDocLiveExperimentInfoPerLiveExperimentInfo.t) · confidence low

`CompositeDocLiveExperimentInfoPerLiveExperimentInfo.experimentId`
Rule: The id of the live experiment this document data belongs to.
Google's description: ID of a live experiment.
type String.t · confidence low

`ScienceCitation.NumRelated2`
Rule: An alternative related-article count used for experiments.
Google's description: for experiments
type integer() · confidence low

`ScienceCitation.NumRelated3`
Rule: Another alternative related-article count used for experiments.
Google's description: for experiments
type integer() · confidence low

Spam & demotions (7)

`AnchorsAnchor.demotionreason`
Rule: Old field for why a link was demoted; deprecated.
Google's description: DEPRECATED
type integer() · confidence low · stages: 4. Link Analysis & Authority Computation

`AnchorsAnchorSource.spamscore1`
Rule: Old spam score for the linking page; deprecated.
Google's description: deprecated, to be removed after October 20. 0-127 scale
type integer() · confidence low · stages: 4. Link Analysis & Authority Computation

`ImageData.adaboostImageFeaturePorn`
Rule: An old adult-content classifier score, replaced by newer safe search scores.
Google's description: Warning: adaboost_image_feature_porn* and imageFeaturePorn fields are DEPRECATED in favor of brain_porn_scores. Please do not use them. Contact safesearch@ for transition advice.
type number() · confidence low

`IndexingDocjoinerAnchorSpamInfo.sampled`
Rule: Whether only a sample of anchors was examined.
Google's description: True if anchors were sampled during observation phrase.
type boolean() · confidence low · stages: 4. Link Analysis & Authority Computation; 8. Spam, Penalty & Trust Systems

`IndexingDocjoinerAnchorSpamInfo.trustedExamples`
Rule: Sample names of trusted sources linking to the page.
Google's description: Examples of trusted sources.
type String.t · confidence low · stages: 4. Link Analysis & Authority Computation; 8. Spam, Penalty & Trust Systems

`OceanDocTag.objectionableContentBitmap`
Rule: Flags marking whether the book content may be objectionable.
Google's description: A bitmap indicating whether content may be objectionable NOTE: Only populated for volume level docs
type integer() · confidence low

`WWWDocInfo.failsSafeSearch`
Rule: A safe-search failure flag that is never filled in production.
Google's description: fails_safe_search is never filled in production.
type String.t · confidence low

Part IV. Key patents (27)

The curated patents that explain the mechanisms most SEO decisions rest on.

Links & authority

US6285999B1 Method for node ranking in a linked database
What it shows: PageRank. Larry Page's original patent, assigned to Stanford and licensed to Google, now expired. Every link-counting mental model in SEO starts here.
Rule: Spend effort earning references from pages that are themselves referenced. Being cited is the game, not link count.
filed 1998 · granted 2001 · assigned to Stanford

US7716225B1 Ranking documents based on user behavior and/or feature data
What it shows: The reasonable surfer. Links do not pass equal weight; the more likely a real user is to click a link, the more it counts.
Rule: Put the links you want to count in the main content where people actually click, not in the footer or sidebar.
filed 2004 · granted 2010

US9165040B1 Producing a ranking for pages using distances in a web-link graph
What it shows: Ranking by link distance from a trusted seed set. Proximity to genuinely trusted sites beats raw link volume.
Rule: Chase links from sites the trusted hubs of your niche already link. One link two hops from a seed beats fifty from nowhere.
filed 2006 · granted 2015

US7308643B1 Anchor tag indexing in a web crawler system
What it shows: Anchor text gets indexed with the page it points to, not the page it sits on.
Rule: Write anchors that describe the target page. Those words become part of how the target is indexed.
filed 2003 · granted 2007

US7565358B2 Agent rank
What it shows: Reputation scores for named authors whose content is digitally signed. The patent SEOs cite when they argue author identity carries weight.
Rule: Publish under a real, consistent author identity everywhere. Same name, same profiles, linked with schema.
filed 2005 · granted 2009

Clicks & behaviour

US8661029B1 Modifying search result ranking based on implicit user feedback
What it shows: Clicks and dwell time reweight rankings. This family is what the leak's NavBoost tables look like in patent form.
Rule: Win the click and the stay: titles that match the page, pages that answer in the first screen.
filed 2006 · granted 2014

US8615514B1 Evaluating website properties by partitioning user feedback
What it shows: User feedback split apart to judge different properties of a website separately, instead of one blended score.
Rule: Audit engagement per template and section, not site-wide averages. Fix or cut the sections that lose people.
filed 2010 · granted 2013

US7962462B1 Deriving and using document and site quality signals from search query streams
What it shows: Quality signals mined from what people type into the search box. People searching your brand by name is a vote.
Rule: Build reasons for people to search your name: a tool, a study, a stance. Brand demand is measurable quality.
filed 2005 · granted 2011

Content & site quality

US8682892B1 Ranking search results
What it shows: A site-wide quality multiplier applied on top of page scores. The inventor is Navneet Panda, the engineer the Panda update is named after.
Rule: Prune or fix your weak pages. Site-wide quality is a multiplier, so bad pages tax your good ones.
filed 2012 · granted 2014

US9031929B1 Site quality score
What it shows: Computing one quality score for a whole site from query and click patterns. Site-level scores are not folklore; they are patented.
Rule: Plan quality site-wide, not page-by-page. Ask of every section: would users pick this by name?
filed 2012 · granted 2015

US9760641B1 Site quality score
What it shows: The continuation of the patent above. Google kept investing in site-level quality scoring for years.
Rule: Nothing new here, and that is the point: treat site-level quality as a standing budget line, not a one-off fix.
filed 2015 · granted 2017

US8195654B1 Prediction of human ratings or rankings of information retrieval quality
What it shows: Models trained to predict what human quality raters would say. Raters do not rank your site; they teach the machine that does.
Rule: Read the Quality Rater Guidelines like a spec. Models are trained to predict those judgements at scale, so build what raters would rate well.
filed 2005 · granted 2012

US11354342B2 Contextual estimation of link information gain
What it shows: Information gain. A document scores on what it adds beyond documents the user already saw. The patent against me-too content.
Rule: Before publishing, name what your page adds that the current top results lack. If the answer is nothing, do not publish.
filed 2018 · granted 2022

Crawling & indexing

US7627613B1 Duplicate document detection in a web crawler system
What it shows: Duplicate clustering at crawl time, with one version chosen to represent the cluster. Canonicalisation existed before you ever set a canonical tag.
Rule: Consolidate near-duplicates yourself, one page per intent, or Google picks the canonical for you.
filed 2003 · granted 2009

US7536408B2 Phrase-based indexing in an information retrieval system
What it shows: Anna Patterson's phrase indexing. Pages understood through meaningful phrases and which related phrases co-occur. Topic completeness has a paper trail.
Rule: Cover the related phrases a real expert would naturally use. Co-occurrence is measurable, thin coverage shows.
filed 2004 · granted 2009

US8037055B2 Sitemap generating client for web crawler
What it shows: Generating sitemaps for the crawler from the site itself. Part of the engineering around the sitemaps protocol Google introduced.
Rule: Keep sitemaps complete, current, and free of junk URLs. They are crawler input, not decoration.
filed 2010 · granted 2011

Query understanding

US7487145B1 Method and system for autocompletion using ranked results
What it shows: Autocomplete, ranked by popularity. The suggestion box is a ranking system of its own.
Rule: Mine autocomplete around your brand and topics. Those suggestions are ranked demand you can build pages for.
filed 2004 · granted 2009

US9037464B1 Computing numeric representations of words in a high-dimensional space
What it shows: word2vec. Words become vectors and similar meanings sit near each other. The ancestor of every neural matching system in search.
Rule: Stop counting exact-match keywords. Write in the natural vocabulary of the topic; the vectors already connect it.
filed 2013 · granted 2015

US10452978B2 Attention-based sequence transduction neural networks
What it shows: The Transformer. Not a ranking patent by class, but it is the architecture behind BERT and the models that rewrite queries and read passages today.
Rule: Write for meaning, not keyword slots. The models reading your page understand context, hedging, and fluff.
filed 2018 · granted 2019

SERP features

US9940367B1 Scoring candidate answer passages
What it shows: Scoring passages, not pages, to answer a query directly. The mechanism behind featured snippets and passage-level ranking.
Rule: Structure direct answers: question as heading, a tight 40-60 word answer, then the depth. Passages are scored on their own.
filed 2015 · granted 2018

US9268820B2 Providing knowledge panels with search results
What it shows: The knowledge panel patent. Entities get a curated box, and ten blue links stopped being the whole page.
Rule: Feed your entity: consistent name and details everywhere, sameAs schema, and the citations that earn a panel.
filed 2012 · granted 2016

Freshness & change

US7346839B2 Information retrieval based on historical data
What it shows: The historical data patent. Document age, link velocity and registration data as signals. Ten years of sandbox arguments trace back to this one.
Rule: Grow steadily. Link spikes and sudden swings read differently from consistent history, and history is a signal.
filed 2003 · granted 2008

US8549014B2 Document scoring based on document content update
What it shows: How much a page changed, how often, and how that feeds scoring. Updating content is measurable, and measured.
Rule: Update meaningfully, not cosmetically, and date honestly. How much changed and how often is measured.
filed 2011 · granted 2013

US8924379B1 Temporal-based score adjustments
What it shows: Scores tuned when interest in a query spikes. Freshness gets a lever when demand does.
Rule: When your topic spikes, ship fresh coverage fast. Freshness weighs more exactly when demand rises.
filed 2010 · granted 2014

US9002867B1 Modifying ranking data based on document changes
What it shows: Rankings revisited when the underlying documents change. What you edit after you rank still matters.
Rule: Keep improving pages that already rank. Edits after ranking are inputs, not cosmetics.
filed 2010 · granted 2015

Trust & spam

US7603350B1 Search result ranking based on trust
What it shows: Ranking adjusted by trust attached to entities and the labels they apply. Trust as a computed input, not a vibe.
Rule: Earn endorsements from sources your niche already trusts. Trust flows through labelled references, not claims about yourself.
filed 2006 · granted 2009

US8554769B1 Identifying gibberish content in resources
What it shows: Language-model scoring to catch spun and stitched-together text. Google patented spun-content detection years before AI slop was a phrase.
Rule: If a page reads assembled rather than written, fix it or kill it. Language-model detection is patented, deployed tech.
filed 2009 · granted 2013

Part V. Rules from all 5,456 patents

One rule per patent, grouped by cluster and ordered by filing date. Clusters are listed largest first.

Search plumbing (715)

US6839702B1 Systems and methods for highlighting search results
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Write for natural spoken language and conversational queries.
filed 2000-12-13, granted 2005-01-04 · leak modules: NlpSaftDocument, PerDocData, SmartphonePerDocData

US7305380B1 Systems and methods for performing in-context searching
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2000-12-13, granted 2007-12-04 · leak modules: CompressedQualitySignals, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData

US7136854B2 Methods and apparatus for providing search results in response to an ambiguous search query
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Write for natural spoken language and conversational queries.
filed 2000-12-26, granted 2006-11-14 · leak modules: NlpSaftDocument, PerDocData, SmartphonePerDocData

US8001118B2 Methods and apparatus for employing usage statistics in document retrieval
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active.
filed 2001-03-02, granted 2011-08-16 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate

US9508101B1 Systems and methods for providing stock ticker information
Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately. Write for intent and meaning. Neural matching connects related concepts.
filed 2001-09-21, granted 2016-11-29 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, SmartphonePerDocData

US7363291B1 Methods and apparatus for increasing efficiency of electronic document delivery to users
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2002-03-29, granted 2008-04-22 · leak modules: IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, ShingleInfoPerDocData, SmartphonePerDocData

US7296016B1 Systems and methods for performing point-of-view searching
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active.
filed 2003-03-12, granted 2007-11-13 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US7319994B1 Document compression scheme that supports searching and partial decompression
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth. Ensure content comprehensively covers the topic with natural vocabulary.
filed 2003-05-23, granted 2008-01-15 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US7836391B2 Document search engine including highlighting of confident results
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2003-06-10, granted 2010-11-16 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US8136025B1 Assigning document identification tags
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2003-07-03, granted 2012-03-13 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US7383258B2 Method and apparatus for characterizing documents based on clusters of related words
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2003-09-30, granted 2008-06-03 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US7437351B2 Method for searching media
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2003-09-30, granted 2008-10-14 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US7505984B1 Systems and methods for information extraction
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2003-09-30, granted 2009-03-17 · leak modules: IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US7730054B1 Systems and methods for providing searchable prior history
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2003-09-30, granted 2010-06-01 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US7925610B2 Determining a meaning of a knowledge item using document-based information
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2003-10-21, granted 2011-04-12 · leak modules: CompressedQualitySignals, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US8914361B2 Methods and systems for determining a meaning of a document to match the document to content
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2003-10-21, granted 2014-12-16 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8521725B1 Systems and methods for improved searching
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2003-12-03, granted 2013-08-27 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US7836038B2 Methods and systems for information extraction
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2003-12-10, granted 2010-11-16 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US7886032B1 Content retrieval from sites that use session identifiers
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2003-12-23, granted 2011-02-08 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate

US20050149499A1 Systems and methods for improving search quality
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Build content for diverse user contexts and search histories. Write for intent and meaning. Neural matching connects related concepts.
filed 2003-12-30 · leak modules: CompressedQualitySignals, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, SmartphonePerDocData

US7346839B2 Information retrieval based on historical data

Rule: Grow steadily. Link spikes and sudden swings read differently from consistent history, and history is a signal. The leak's domainAge, hostAge, lastSignificantUpdate, semanticDate, and timeSensitivity fields are the direct implementation.

filed 2003-12-31, granted 2008-03-18 · leak modules: PerDocData, QualityTimebasedSyntacticDate, QualityNsrNsrData

US7437353B2 Systems and methods for unification of search results

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2003-12-31, granted 2008-10-14 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US8612411B1 Clustering documents using citation patterns

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2003-12-31, granted 2013-12-17 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US7647300B2 Methods and systems for output of search results

Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2004-01-26, granted 2010-01-12 · leak modules: ImageData, ImagePerDocData, PerDocData, SmartphonePerDocData

US7412708B1 Methods and systems for capturing information

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2004-03-31, granted 2008-08-12 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US7707142B1 Methods and systems for performing an offline search

Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth. Avoid AI-stitched content. GibberishScore uses language model scoring.

filed 2004-03-31, granted 2010-04-27 · leak modules: IndexingDocjoinerAnchorSpamInfo, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8131702B1 Systems and methods for browsing historical content

Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2004-03-31, granted 2012-03-06 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US8346777B1 Systems and methods for selectively storing event data

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2004-03-31, granted 2013-01-01 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US7673253B1 Systems and methods for inferring concepts for association with content

Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Write for natural spoken language and conversational queries.

filed 2004-06-30, granted 2010-03-02 · leak modules: NlpSaftDocument, PerDocData, SmartphonePerDocData

US7761439B1 Systems and methods for performing a directory search

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2004-06-30, granted 2010-07-20 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US7783639B1 Determining quality of linked documents

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2004-06-30, granted 2010-08-24 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US7426507B1 Automatic taxonomy generation in search results using phrases
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2004-07-26, granted 2008-09-16 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDocjoinerAnchorStatistics, IndexingDupsLocalizedLocalizedCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US7584175B2 Phrase-based generation of document descriptions
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2004-07-26, granted 2009-09-01 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDocjoinerAnchorStatistics, IndexingDupsLocalizedLocalizedCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US7599914B2 Phrase-based searching in an information retrieval system
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2004-07-26, granted 2009-10-06 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDocjoinerAnchorStatistics, IndexingDupsLocalizedLocalizedCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US7913163B1 Determining semantically distinct regions of a document
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2004-09-22, granted 2011-03-22 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US8099660B1 Tool for managing online content
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2004-09-30, granted 2012-01-17 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US7603355B2 Variably controlling access to content
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2004-10-01, granted 2009-10-13 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo

US7559016B1 System and method for indicating web page modifications
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Structure answers: question heading, tight answer, then depth.
filed 2004-10-07, granted 2009-07-07 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US7853586B1 Highlighting occurrences of terms in documents or search results
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Write for natural spoken language and conversational queries.
filed 2004-10-18, granted 2010-12-14 · leak modules: NlpSaftDocument, PerDocData, SmartphonePerDocData

US8560559B1 Highlighting occurrences of search terms
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Write for natural spoken language and conversational queries.
filed 2004-12-10, granted 2013-10-15 · leak modules: NlpSaftDocument, PerDocData, SmartphonePerDocData

US9195766B2 Providing useful information associated with an item in a document
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Structure answers: question heading, tight answer, then depth. Write for natural spoken language and conversational queries.
filed 2004-12-14, granted 2015-11-24 · leak modules: NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryWebrefEntityJoin, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8090736B1 Enhancing search results using conceptual document relationships
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2004-12-30, granted 2012-01-03 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US8924256B2 System and method for obtaining content based on data from an electronic device
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2005-03-31, granted 2014-12-30 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US9256685B2 Systems and methods for modifying search results based on a user's history
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2005-03-31, granted 2016-02-09 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US8019648B2 Search engines and systems with handheld document data capture devices
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2005-04-01, granted 2011-09-13 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8214387B2 Document enhancement system and method
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2005-04-01, granted 2012-07-03 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8229957B2 Categorizing objects, such as documents and/or clusters, with respect to a taxonomy and data structures derived from such categorization
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2005-04-22, granted 2012-07-24 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US8468048B2 Suggesting targeting information for ads, such as websites and/or categories of websites for example
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2005-04-22, granted 2013-06-18 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US7536641B2 Web page authoring tool for structured documents
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2005-04-29, granted 2009-05-19 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefEntityJoin, SmartphonePerDocData

US8423541B1 Using saved search results for quality feedback
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2005-05-19, granted 2013-04-16 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US7788245B1 Method and system for dynamically generating search links embedded in content
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2005-06-16, granted 2010-08-31 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US7996391B2 Systems and methods for providing search results
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2005-06-20, granted 2011-08-09 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US8631343B1 Automatically designating document items, such as references to other documents, for improved navigation and/or retrieval
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2005-06-30, granted 2014-01-14 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US7716199B2 Aggregating context data for programmable search engines
Rule: For sites with specialized domain expertise, structured data and context annotations can influence how Google processes queries related to your subject matter. Ensure your site's content is well-organized into clear topical collections. Implement structured data that helps Google understand your domain expertise, as context-based query reformulation and re-ranking can affect how your content matches user queries.
filed 2005-08-10, granted 2010-05-11 · leak modules: PerDocData, QualityNsrNsrData, IndexingDupsLocalizedLocalizedCluster, QualitySitemapScoringSignals

US8005720B2 Applying scanned information to identify content
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2005-08-18, granted 2011-08-23 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8417697B2 Permitting users to remove documents
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2005-08-22, granted 2013-04-09 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster

US8271865B1 Detection and utilization of document reading speed
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Structure answers: question heading, tight answer, then depth. Write for natural spoken language and conversational queries.
filed 2005-09-19, granted 2012-09-18 · leak modules: NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US7725530B2 Proxy server collection of data for module incorporation into a container document
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data.
filed 2005-12-12, granted 2010-05-25 · leak modules: CrawlerChangerateUrlChangerate, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US7971137B2 Detecting and rejecting annoying documents
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2005-12-14, granted 2011-06-28 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo

US7769751B1 Method and apparatus for classifying documents based on user inputs
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2006-01-17, granted 2010-08-03 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US7634517B1 System and method for dynamically updating a document repository without interrupting concurrent querying
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2006-02-10, granted 2009-12-15 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US7788293B2 Generating structured information
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2006-03-01, granted 2010-08-31 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US7451120B1 Detecting novel document content
Rule: Write for natural spoken language and conversational queries.
filed 2006-03-20, granted 2008-11-11 · leak modules: NlpSaftDocument, PerDocData

US7933890B2 Propagating useful information among related web pages, such as web pages of a website
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2006-03-31, granted 2011-04-26 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData

US7747749B1 Systems and methods of efficiently preloading documents to client devices
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2006-05-05, granted 2010-06-29 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8429167B2 User-context-based search engine
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2006-08-08, granted 2013-04-23 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US8775684B1 Content request optimization
Rule: Structure answers: question heading, tight answer, then depth.
filed 2006-10-30, granted 2014-07-08 · leak modules: QualityPreviewSnippetRadishFeatures, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8849982B2 Communicating information relating to a network resource
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2007-01-19, granted 2014-09-30 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US7668823B2 Identifying inadequate search content
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2007-04-03, granted 2010-02-23 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US7836012B1 Systems and methods for information extraction
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2007-04-06, granted 2010-11-16 · leak modules: IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US8484192B1 Media search broadening
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2007-04-30, granted 2013-07-09 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US8533761B1 Aggregating media information
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2007-04-30, granted 2013-09-10 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US8676797B2 Managing and accessing data in web notebooks
Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2007-05-10, granted 2014-03-18 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US7890487B1 Facilitating client-side data-management for web-based applications
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2007-05-29, granted 2011-02-15 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US20090024595A1 Automatic expanded language search
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2007-07-20 · leak modules: CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US7752222B1 Finding text on a web page
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2007-07-20, granted 2010-07-06 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8032527B2 Search filtering
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2007-08-29, granted 2011-10-04 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US7962469B1 In-context searching
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2007-10-09, granted 2011-06-14 · leak modules: CompressedQualitySignals, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData

US8296643B1 Running multiple web page experiments on a test page
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2007-10-18, granted 2012-10-23 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US8589366B1 Data extraction using templates
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2007-11-01, granted 2013-11-19 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US7818324B1 Searching indexed and non-indexed resources for content
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2007-11-14, granted 2010-10-19 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US8019748B1 Web search refinement
Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately. Ensure content comprehensively covers the topic with natural vocabulary.
filed 2007-11-14, granted 2011-09-13 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData

US8190618B2 Systems, methods, and media for aggregating electronic document usage information
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2007-12-04, granted 2012-05-29 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US8086600B2 Interleaving search results
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2007-12-06, granted 2011-12-27 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US7801876B1 Systems and methods for customizing behavior of multiple search engines
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Optimize mobile experience. Mobile-specific signals are tracked separately. Earn promotion through authoritative content and entity association.
filed 2008-01-11, granted 2010-09-21 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, SmartphonePerDocData

US8250080B1 Filtering in search engines

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.

filed 2008-01-11, granted 2012-08-21 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US8538989B1 Assigning weights to parts of a document

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active.

filed 2008-02-08, granted 2013-09-17 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US8645409B1 Contextual search term evaluation

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.

filed 2008-04-02, granted 2014-02-04 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US8065309B1 Counting unique search results

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2008-04-21, granted 2011-11-22 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8510325B1 Supplementing search results with information of interest

Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Write for intent and meaning. Neural matching connects related concepts.

filed 2008-04-24, granted 2013-08-13 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US8799307B2 Cross-language information retrieval

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2008-04-29, granted 2014-08-05 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US8688720B1 Method and apparatus for characterizing documents based on clusters of related words

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2008-06-02, granted 2014-04-01 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8190990B2 Annotating webpage content

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2008-06-27, granted 2012-05-29 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, RichsnippetsPageMap

US8005811B2 Systems and media for utilizing electronic document usage information with search engines

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2008-07-01, granted 2011-08-23 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US9684676B1 Method for searching media

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2008-09-08, granted 2017-06-20 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8214361B1 Organizing search results in a topic hierarchy
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2008-09-30, granted 2012-07-03 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US8140449B1 Detecting novel document content
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Write for natural spoken language and conversational queries.
filed 2008-10-03, granted 2012-03-20 · leak modules: NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US20100114874A1 Providing search results
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2008-10-20 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData

US8549008B1 Determining section information of a digital volume
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2008-11-12, granted 2013-10-01 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8788504B1 Web mining to build a landmark database and applications thereof
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2008-11-12, granted 2014-07-22 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US20140108376A1 Enhanced detection of like resources
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2008-11-26 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US8131736B1 System and method for navigating documents
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Optimize video for watch time and retention.
filed 2008-11-26, granted 2012-03-06 · leak modules: PerDocData, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, VideoPerDocData

US8799273B1 Highlighting notebooked web content
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2008-12-12, granted 2014-08-05 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US8086953B1 Identifying transient portions of web pages
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2008-12-19, granted 2011-12-27 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9031216B1 In-conversation search
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2009-03-05, granted 2015-05-12 · leak modules: IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US8346791B1 Search augmentation
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2009-03-16, granted 2013-01-01 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster

US9514235B2 Web page authoring tool for structured documents
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2009-05-19, granted 2016-12-06 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefEntityJoin, SmartphonePerDocData

US9934516B1 Communicating information relating to a network resource
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2009-05-19, granted 2018-04-03 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8423350B1 Segmenting text for searching
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2009-05-21, granted 2013-04-16 · leak modules: IndexingDupsLocalizedLocalizedCluster, PerDocData, ShingleInfoPerDocData, SmartphonePerDocData

US8745168B1 Buffering user interaction data
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2009-07-10, granted 2014-06-03 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US8543599B2 Variably controlling access to content
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2009-07-13, granted 2013-09-24 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo

US8447760B1 Generating a related set of documents for an initial set of documents
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2009-07-20, granted 2013-05-21 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US8209315B2 Search engine with fill-the-blanks capability
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2009-07-21, granted 2012-06-26 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US8498974B1 Refining search results
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2009-08-31, granted 2013-07-30 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US8438148B1 Method and system for generating search shortcuts and inline auto-complete entries
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2009-09-01, granted 2013-05-07 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9183316B2 Providing action links to share web content
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Write for natural spoken language and conversational queries.
filed 2009-09-30, granted 2015-11-10 · leak modules: NlpSaftDocument, PerDocData, SmartphonePerDocData

US9043148B2 Navigation system and methods for generating enhanced search results
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2009-10-15, granted 2015-05-26 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US8713424B1 Asynchronous loading of scripts in web pages
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2009-11-04, granted 2014-04-29 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8229959B1 Sharable search result labels
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2009-11-11, granted 2012-07-24 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefEntityJoin, SmartphonePerDocData

US8739066B1 Systems and methods for inferring concepts for association with content
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Write for natural spoken language and conversational queries.
filed 2010-01-08, granted 2014-05-27 · leak modules: NlpSaftDocument, PerDocData, SmartphonePerDocData

US8316037B1 Providing remedial search operation based on analysis of user interaction with search results
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2010-01-29, granted 2012-11-20 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US8037063B2 Identifying inadequate search content
Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately. Create original, effortful content. Quality is scored at page and site level.
filed 2010-02-03, granted 2011-10-11 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData

US8051063B2 Aggregating context data for programmable search engines
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2010-03-29, granted 2011-11-01 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, IndexingDocjoinerAnchorStatistics, LocalWWWInfo

US8918401B1 Systems and methods for providing searchable prior history
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2010-04-08, granted 2014-12-23 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9305105B2 System and method for aggregating analytics data
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2010-05-25, granted 2016-04-05 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryWebrefEntityJoin, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8438080B1 Learning characteristics for extraction of information from web pages
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2010-05-28, granted 2013-05-07 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8977653B1 Modifying web pages to reduce retrieval latency
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2010-06-17, granted 2015-03-10 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9623119B1 Accentuating search results
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2010-06-29, granted 2017-04-18 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US8364675B1 Recursive algorithm for in-place search for an n:th element in an unsorted array
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Optimize video for watch time and retention.
filed 2010-06-30, granted 2013-01-29 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData, VideoPerDocData

US8176056B1 Determining quality of linked documents

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2010-07-20, granted 2012-05-08 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US8392449B2 Resource search operations

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.

filed 2010-07-20, granted 2013-03-05 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US20120016856A1 Content extractor

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2010-08-18 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US7908284B1 Content reference page

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2010-08-20, granted 2011-03-15 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8521774B1 Dynamically generating pre-aggregated datasets

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2010-08-20, granted 2013-08-27 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US8301615B1 Systems and methods for customizing behavior of multiple search engines

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Optimize mobile experience. Mobile-specific signals are tracked separately. Earn promotion through authoritative content and entity association.

filed 2010-09-20, granted 2012-10-30 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, SmartphonePerDocData

US9251267B2 Generating behavior information for a link

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2010-09-30, granted 2016-02-02 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US8332469B1 Web resource caching

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2010-10-06, granted 2012-12-11 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US8619147B2 Handheld device for capturing text from both a document printed on paper and a document displayed on a dynamic display device

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.

filed 2010-10-06, granted 2013-12-31 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US9189548B2 Document search engine including highlighting of confident results

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2010-10-08, granted 2015-11-17 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US9529517B2 Movable information panels

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.

filed 2010-10-13, granted 2016-12-27 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8489604B1 Automated resource selection process evaluation
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2010-10-26, granted 2013-07-16 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9875306B2 Navigation through a collection of electronic documents
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2010-10-28, granted 2018-01-23 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8762280B1 Method and system for using a network analysis system to verify content on a website
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2010-11-01, granted 2014-06-24 · leak modules: CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US8307076B1 Content retrieval from sites that use session identifiers
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2010-11-03, granted 2012-11-06 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate

US8458584B1 Extraction and analysis of user-generated content
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2010-11-18, granted 2013-06-04 · leak modules: CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9177058B2 Multi-step search result retrieval
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2010-11-18, granted 2015-11-03 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US8856115B1 Framework for suggesting search terms
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2010-12-06, granted 2014-10-07 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9037656B2 Method and system for facilitating interaction with multiple content provider websites
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2010-12-20, granted 2015-05-19 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US8607140B1 Classifying changes to resources
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active. Write in natural vocabulary. ML models understand context and fluff.
filed 2010-12-21, granted 2013-12-10 · leak modules: AnchorsAnchor, AnchorsAnchorSource, IndexingDocjoinerAnchorStatistics, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9245022B2 Context-based person search
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2010-12-30, granted 2016-01-26 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation

US20120192080A1 Tailoring content based on available bandwidth
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-01-21 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US8949242B1 Semantic document analysis
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-01-21, granted 2015-02-03 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US20120191773A1 Caching resources
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-01-26 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US8909619B1 Providing search results tools
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.
filed 2011-02-03, granted 2014-12-09 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData

US8156100B2 Methods and apparatus for employing usage statistics in document retrieval
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active.
filed 2011-02-24, granted 2012-04-10 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData

US8707162B1 Methods and systems for reducing web page transmission bandwidth
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-03-22, granted 2014-04-22 · leak modules: PerDocData, SmartphonePerDocData

US9443014B2 Custom web page themes
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-03-30, granted 2016-09-13 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US8321403B1 Web search refinement
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-04-11, granted 2012-11-27 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8433671B2 Determining a meaning of a knowledge item using document based information
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-04-11, granted 2013-04-30 · leak modules: PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData

US20150199357A1 Selecting primary resources
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active.
filed 2011-04-14 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals

US8521717B2 Propagating information among web pages
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-04-21, granted 2013-08-27 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData

US8364682B1 Identifier mapping from joined data

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.

filed 2011-05-05, granted 2013-01-29 · leak modules: NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData

US8583643B2 Caching electronic document resources in a client device having an electronic resource database

Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.

filed 2011-05-06, granted 2013-11-12 · leak modules: IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, ShingleInfoPerDocData

US8538944B1 Resource catchment areas

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2011-05-10, granted 2013-09-17 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US8898172B2 Parallel generation of topics from documents

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.

filed 2011-05-11, granted 2014-11-25 · leak modules: CrawlerChangerateUrlChangerate, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9208230B2 Enriching search results

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.

filed 2011-05-27, granted 2015-12-08 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US20110302153A1 Service for Aggregating Event Information

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2011-06-03 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8583579B1 Targeting content based on user mode

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2011-06-03, granted 2013-11-12 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US8713034B1 Systems and methods for identifying similar documents

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.

filed 2011-06-03, granted 2014-04-29 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo

US9058377B2 Fixed width encoding document posting lists

Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Write for natural spoken language and conversational queries.

filed 2011-06-03, granted 2015-06-16 · leak modules: NlpSaftDocument, PerDocData, SmartphonePerDocData

US8868549B1 In-context searching

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2011-06-06, granted 2014-10-21 · leak modules: CompressedQualitySignals, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData

US9311403B1 Hashing techniques for data set similarity determination

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2011-06-16, granted 2016-04-12 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US8589784B1 Identifying multiple versions of documents
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Consolidate near-duplicates. Use canonical tags and consistent URLs. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-06-30, granted 2013-11-19 · leak modules: IndexingDupsLocalizedLocalizedCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefEntityJoin, RichsnippetsPageMap

US8839303B2 System and method for enhancing user search results by determining a television program currently being displayed in proximity to an electronic device
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-06-30, granted 2014-09-16 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US8745212B2 Access to network content
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Optimize video for watch time and retention.
filed 2011-07-01, granted 2014-06-03 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData, VideoPerDocData

US8788711B2 Redacting content and inserting hypertext transfer protocol (HTTP) error codes in place thereof
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships.
filed 2011-07-01, granted 2014-07-22 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US9769285B2 Access to network content
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-07-01, granted 2017-09-19 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9203684B1 Reduction of web page load time using HTTP header compression
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-07-14, granted 2015-12-01 · leak modules: PerDocData, SmartphonePerDocData

US20120016759A1 Point of sale data collection
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2011-07-15 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US10635723B2 Search engines and systems with handheld document data capture devices
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2011-07-20, granted 2020-04-28 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8606907B1 Multi-tiered system for receiving and reporting web site traffic data
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-07-20, granted 2013-12-10 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US8775556B1 Automated segmentation and processing of web site traffic data over a rolling window of time
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-07-20, granted 2014-07-08 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9015152B1 Managing search results
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-07-20, granted 2015-04-21 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US8583635B1 Keywords associated with document categories
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-07-26, granted 2013-11-12 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate

US8386480B2 Systems and methods for providing search results
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-08-08, granted 2013-02-26 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US9015143B1 Refining search results
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-08-10, granted 2015-04-21 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US8510285B1 Using pre-search triggers
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-08-18, granted 2013-08-13 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US9432482B2 Asynchronous and synchronous resource links
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-08-31, granted 2016-08-30 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US8655819B1 Predicting user navigation events based on chronological history data
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-09-15, granted 2014-02-18 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8429161B2 Search filtering
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-09-16, granted 2013-04-23 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9020933B2 Identifying inadequate search content
Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately. Create original, effortful content. Quality is scored at page and site level.
filed 2011-09-23, granted 2015-04-28 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData

US8775923B1 Web page restoration
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-09-26, granted 2014-07-08 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8874731B1 Use of timing information for document understanding
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-09-27, granted 2014-10-28 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8370367B1 System and method for navigating documents
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Optimize video for watch time and retention.
filed 2011-09-29, granted 2013-02-05 · leak modules: PerDocData, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, VideoPerDocData

US8589421B1 System and method for navigating documents
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Optimize mobile experience. Mobile-specific signals are tracked separately. Optimize video for watch time and retention.
filed 2011-09-29, granted 2013-11-19 · leak modules: IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityWebanswersVideoTranscriptAnnotations, ShingleInfoPerDocData, SmartphonePerDocData, VideoPerDocData

US20140372873A1 Detecting Main Page Content
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-09-30 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US8239490B2 Exposing resource capabilities to web applications
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-09-30, granted 2012-08-07 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8261192B2 Providing useful information associated with an item in a document
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Structure answers: question heading, tight answer, then depth. Write for natural spoken language and conversational queries.
filed 2011-09-30, granted 2012-09-04 · leak modules: NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryWebrefEntityJoin, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8583663B1 System and method for navigating documents
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Optimize video for watch time and retention.
filed 2011-09-30, granted 2013-11-12 · leak modules: PerDocData, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, VideoPerDocData

US8650196B1 Clustering documents based on common document selections
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-09-30, granted 2014-02-11 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US8756210B1 Aggregating context data for programmable search engines
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-10-05, granted 2014-06-17 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, IndexingDocjoinerAnchorStatistics, LocalWWWInfo

US9104664B1 Access to search results
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-10-07, granted 2015-08-11 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US8478777B2 Gesture-based search
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-10-25, granted 2013-07-02 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate

US8224836B1 Searching in multiple languages
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-11-02, granted 2012-07-17 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData

US8788477B1 Identifying addresses and titles of authoritative web pages by analyzing search queries in query logs
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-11-04, granted 2014-07-22 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate

US8935798B1 Automatically enabling private browsing of a web page, and applications thereof
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2011-11-08, granted 2015-01-13 · leak modules: CompressedQualitySignals, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8572096B1 Selecting keywords using co-visitation information
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-11-16, granted 2013-10-29 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US8321428B1 System and method for navigating documents
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Optimize video for watch time and retention.
filed 2011-11-18, granted 2012-11-27 · leak modules: PerDocData, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, VideoPerDocData

US8639770B1 Separation of mutable and immutable data in a memory cache for improvement of data updates
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships.
filed 2011-11-18, granted 2014-01-28 · leak modules: PerDocData, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US8498984B1 Categorization of search results
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-11-21, granted 2013-07-30 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US8510286B1 Method, architecture and platform for content takedown on the internet
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-11-21, granted 2013-08-13 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations

US8738596B1 Refining search results
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-12-05, granted 2014-05-27 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US9679061B2 Method and apparatus that collect and uploads implicit analytic data
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-12-08, granted 2017-06-13 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US8738597B2 Interleaving search results
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-12-15, granted 2014-05-27 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US8745477B1 Tool for managing online content
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-01-06, granted 2014-06-03 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US8825645B1 Determining quality of linked documents
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-01-06, granted 2014-09-02 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US8880992B2 Method and apparatus for animating transitions between search results
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2012-01-10, granted 2014-11-04 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US9684690B2 Flights search
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-01-12, granted 2017-06-20 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US8805828B1 Providing information regarding prior searches
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.
filed 2012-01-13, granted 2014-08-12 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8918720B1 List of most selected web based application documents
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-01-13, granted 2014-12-23 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US9146992B2 Enriching web resources
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-01-13, granted 2015-09-29 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8793235B2 System and method for improving access to search results
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-01-19, granted 2014-07-29 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US20150193394A1 Scheduling processing of resource requests to ensure fulfillment of foreground requests
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-01-25 · leak modules: CrawlerChangerateUrlChangerate, PerDocData, SmartphonePerDocData, TrawlerFetchReplyData

US8621091B1 System and method for synchronizing settings and state information for a browser component
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-01-27, granted 2013-12-31 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8458194B1 System and method for content-based document organization and filing
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2012-01-31, granted 2013-06-04 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US8886648B1 System and method for computation of document similarity
Rule: Use consistent entity names, sameAs schema, and structured data. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-01-31, granted 2014-11-11 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US8903752B1 Classifying documents based on automatically detected rules
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2012-02-09, granted 2014-12-02 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US20130212089A1 Search Result Categorization
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Build content for diverse user contexts and search histories.
filed 2012-02-10 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US9390077B2 Document division method and system
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-02-10, granted 2016-07-12 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US8548975B1 Systems and methods for browsing historical content
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-03-02, granted 2013-10-01 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US8527489B1 Suggesting a search engine to search for resources
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-03-07, granted 2013-09-03 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefDocumentMetadata

US8352452B2 Methods and apparatus for employing usage statistics in document retrieval
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active.
filed 2012-03-09, granted 2013-01-08 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate

US8775924B1 Processing web pages based on content quality
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Create original, effortful content. Quality is scored at page and site level. Optimize video for watch time and retention.
filed 2012-03-09, granted 2014-07-08 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, VideoPerDocData

US9411889B2 Assigning document identification tags
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2012-03-13, granted 2016-08-09 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US8954524B1 Access to network content
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-03-14, granted 2015-02-10 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US20150293697A1 Assigning keyboard shortcut keys to web page resource locators
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-03-15 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo

US8521665B1 Detecting novel document content
Rule: Use consistent entity names, sameAs schema, and structured data. Write for natural spoken language and conversational queries.
filed 2012-03-19, granted 2013-08-27 · leak modules: NlpSaftDocument, PerDocData, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US9135445B2 Providing information about a web application or extension offered by website based on information about the application or extension gathered from a trusted site
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2012-03-19, granted 2015-09-15 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefEntityJoin, RichsnippetsPageMap, SmartphonePerDocData

US8849818B1 Searching via user-specified ratings
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-03-21, granted 2014-09-30 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9754031B2 Providing content
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-03-22, granted 2017-09-05 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US9064013B1 Application of resource limits to request processing
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-03-27, granted 2015-06-23 · leak modules: CompressedQualitySignals, PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US9148329B1 Resource constraints for request processing
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-03-27, granted 2015-09-29 · leak modules: CompressedQualitySignals, PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US20130262704A1 System and method for improving delivery of content over a network
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-04-03 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US20150160817A1 Search results including supplemental content triggered by user actions when browsing the search results
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-04-10 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate

US8762368B1 Context-based filtering of search results
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-04-30, granted 2014-06-24 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, SmartphonePerDocData, SnippetExtraInfo

US8521772B2 Document enhancement system and method
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2012-05-10, granted 2013-08-27 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US9754040B2 Configuring a content document for users and user groups
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Structure answers: question heading, tight answer, then depth.
filed 2012-05-10, granted 2017-09-05 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US9047012B1 Using information from a user device and a server to suggest an input
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-05-14, granted 2015-06-02 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8719293B2 Generating improved document classification data using historical search results
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active.
filed 2012-05-15, granted 2014-05-06 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US9378191B1 Promoting content
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-05-15, granted 2016-06-28 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9946792B2 Access to network content
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-05-15, granted 2018-04-17 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US8572066B1 Non-cached extraction of data store content
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-05-18, granted 2013-10-29 · leak modules: CompressedQualitySignals, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US8291041B1 Systems and methods for disseminating content to remote devices
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-05-31, granted 2012-10-16 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US8688697B1 Identifying contrarian terms based on website content
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-05-31, granted 2014-04-01 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US20130346197A1 Content category management systems and methods
Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2012-06-20 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData

US8918395B2 Categorizing objects, such as documents and/or clusters, with respect to a taxonomy and data structures derived from such categorization
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-06-20, granted 2014-12-23 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US8972404B1 Methods and systems for organizing content
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-06-22, granted 2015-03-03 · leak modules: PerDocData, SmartphonePerDocData

US8977620B1 Method and system for document classification
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-06-22, granted 2015-03-10 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9002848B1 Automatic incremental labeling of document clusters
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-06-22, granted 2015-04-07 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US9436758B1 Methods and systems for partitioning documents having customer feedback and support content
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-06-22, granted 2016-09-06 · leak modules: CompressedQualitySignals, PerDocData, SmartphonePerDocData

US8578036B1 Providing standardized transparency for cookies and other website data using a server side description file

Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships.

filed 2012-06-25, granted 2013-11-05 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US9767202B2 Linking content files

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.

filed 2012-06-27, granted 2017-09-19 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20150161266A1 Systems and methods for more efficient source code searching

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2012-06-28 · leak modules: CompressedQualitySignals, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8788475B2 System and method of accessing a document efficiently through multi-tier web caching

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2012-06-28, granted 2014-07-22 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate

US9348846B2 User-navigable resource representations

Rule: Use descriptive alt text and surrounding context for images. Structure answers: question heading, tight answer, then depth. Optimize video for watch time and retention.

filed 2012-07-02, granted 2016-05-24 · leak modules: ImageData, ImagePerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo, VideoPerDocData

US8966356B1 Providing views of three-dimensional (3D) object data models

Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2012-07-26, granted 2015-02-24 · leak modules: ImageData, ImagePerDocData, PerDocData, SmartphonePerDocData

US9275018B2 Techniques for analyzing web pages to determine font subsets

Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2012-07-26, granted 2016-03-01 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US8516155B1 Dynamic content conversion

Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately. Optimize video for watch time and retention.

filed 2012-07-30, granted 2013-08-20 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, VideoPerDocData

US9361395B2 System and method for providing offline access in a hosted document service

Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.

filed 2012-07-30, granted 2016-06-07 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8818983B1 Synchronization of search engines

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2012-07-31, granted 2014-08-26 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8880635B1 Selective requesting of cached resources

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2012-07-31, granted 2014-11-04 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9020927B1 Determining resource quality based on resource competition

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2012-07-31, granted 2015-04-28 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US9330093B1 Methods and systems for identifying user input data for matching content to user interests

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2012-08-02, granted 2016-05-03 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US9247020B2 Media content receiving device and distribution of media content utilizing social networks and social circles

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2012-08-07, granted 2016-01-26 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US20150269231A1 Clustered search results

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2012-08-08 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US8887239B1 Access to network content

Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2012-08-08, granted 2014-11-11 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations

US8892596B1 Identifying related documents based on links in documents

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2012-08-08, granted 2014-11-18 · leak modules: PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US9418156B2 Providing local data with search results

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2012-08-10, granted 2016-08-16 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US9569432B1 Evaluating content in a computer networked environment

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2012-08-10, granted 2017-02-14 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9747387B2 Methods and systems for content enhancement

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.

filed 2012-08-14, granted 2017-08-29 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefEntityJoin

US8533206B1 Filtering in search engines

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.

filed 2012-08-17, granted 2013-09-10 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US9223762B2 Encoding information into text for visual representation
Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2012-08-27, granted 2015-12-29 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8694605B1 Systems and methods for disseminating content to remote devices
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-08-29, granted 2014-04-08 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US9870131B2 Exploring information by topic
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-09-07, granted 2018-01-16 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US8909708B2 System and method for confirming authorship of documents
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-09-10, granted 2014-12-09 · leak modules: PerDocData, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US9092488B2 Determination of a desired repository for retrieving search results
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-09-10, granted 2015-07-28 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US9177074B2 System and method for confirming authorship of documents
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data.
filed 2012-09-10, granted 2015-11-03 · leak modules: PerDocData, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US10007731B2 Deduplication in search results
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2012-09-12, granted 2018-06-26 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US8639721B2 Variably controlling access to content
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-09-13, granted 2014-01-28 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo

US8862598B1 Providing an online forum as a search result
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-09-13, granted 2014-10-14 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US8930340B1 Blending content in an output
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-09-13, granted 2015-01-06 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US9268852B2 Search engines and systems with handheld document data capture devices
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2012-09-13, granted 2016-02-23 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US9672232B1 Systems and methods for selectively storing event data
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-09-13, granted 2017-06-06 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20130073558A1 System and program product for providing high performance data lookup
Rule: Monitor this patent for ranking implications.
filed 2012-09-14

US8768958B1 Predictive information retrieval
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-09-14, granted 2014-07-01 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8805872B1 Supplementing search results with information of interest
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Write for intent and meaning. Neural matching connects related concepts.
filed 2012-09-14, granted 2014-08-12 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US8892589B1 Providing remedial search operation based on analysis of user interaction with search results
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-09-14, granted 2014-11-18 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US8977609B1 Providing a search result based on an input including a time specification
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-09-14, granted 2015-03-10 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality

US8977612B1 Generating a related set of documents for an initial set of documents
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-09-14, granted 2015-03-10 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9015159B1 Method for searching media
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-09-14, granted 2015-04-21 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9116992B2 Providing time series information with search results
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-09-14, granted 2015-08-25 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US9418104B1 Refining search results
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-09-14, granted 2016-08-16 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US9449105B1 User-context-based search engine
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-09-14, granted 2016-09-20 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US9141722B2 Access to network content
Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships.
filed 2012-10-02, granted 2015-09-22 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US8890827B1 Selected content refinement mechanisms
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.
filed 2012-10-05, granted 2014-11-18 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData

US8832118B1 Systems and methods of evaluating content in a computer network environment
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-10-10, granted 2014-09-09 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9094461B2 Filtering a stream of content
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-10-19, granted 2015-07-28 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9043464B1 Automatically grouping resources accessed by a user
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-10-23, granted 2015-05-26 · leak modules: PerDocData, SmartphonePerDocData

US10037346B1 Time reservations for ensuring consistent reads in a distributed database without logging
Rule: Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-10-26, granted 2018-07-31 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, RepositoryAnnotationsGeoTopicality

US8943473B1 Consistently delivering a web page having source code with a dynamic instruction
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-10-26, granted 2015-01-27 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US9529916B1 Managing documents based on access context
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-10-30, granted 2016-12-27 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US20140122697A1 Providing content to linked devices associated with a user
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-11-01 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US8886644B1 User control of search filter bubble
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-11-01, granted 2014-11-11 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate

US9298828B1 Supplementing search results with historically selected search results of related queries
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Ensure NAP consistency and local relevance. Distance affects ranking. Build content for diverse user contexts and search histories.
filed 2012-11-01, granted 2016-03-29 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, RepositoryAnnotationsGeoTopicality

US20140129545A1 Sorting search results
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Build content for diverse user contexts and search histories.
filed 2012-11-02 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US8909628B1 Detecting content scraping
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Earn promotion through authoritative content and entity association. Create original, effortful content. Quality is scored at page and site level.
filed 2012-11-02, granted 2014-12-09 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, SmartphonePerDocData

US9135364B1 Delivery of web pages with markup for speed
Rule: Optimize video for watch time and retention.
filed 2012-11-09, granted 2015-09-15 · leak modules: QualityWebanswersVideoTranscriptAnnotations, VideoPerDocData

US8996456B2 Data processing service
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-11-14, granted 2015-03-31 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US10019139B2 System and method for content size adjustment
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-11-15, granted 2018-07-10 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9436766B1 Clustering of documents for providing content
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-11-16, granted 2016-09-06 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9123053B2 Analyzing reading metrics to generate action information
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-12-10, granted 2015-09-01 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20150205761A1 Unloaded content placeholders
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth. Use JSON-LD structured data for entity types and relationships.
filed 2012-12-12 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8825698B1 Showing prominent users for information retrieval requests
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-12-21, granted 2014-09-02 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US9165030B1 Showing prominent users for information retrieval requests
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-12-21, granted 2015-10-20 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9183251B1 Showing prominent users for information retrieval requests
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-12-21, granted 2015-11-10 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US20130212100A1 Estimating rate of change of documents
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-12-26 · leak modules: CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9824151B2 Providing a portion of requested data based upon historical user interaction with the data

Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth. Optimize video for watch time and retention.

filed 2012-12-27, granted 2017-11-21 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo, VideoPerDocData

US20140188861A1 Using scientific papers in web search

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2012-12-28 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US9020956B1 Sentiment and topic based content determination methods and systems

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2012-12-31, granted 2015-04-28 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9244977B2 Using content identification as context for search

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2012-12-31, granted 2016-01-26 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US9558233B1 Determining a quality measure for a resource

Rule: Google scores link quality in three tiers: seeds (high-quality targets), sources (pages linking to seeds), and referenced resources. Links from 'qualified' sources (those that link to high-quality seeds) pass more weight. Links from 'unqualified' sources are discounted. Anchor text n-grams carry the source's quality score, so your anchor text keywords are scored by the quality of the linking page. Focus on earning links from pages that themselves link to authoritative sites. The leak's AnchorsAnchor with pagerankWeight and IndexingDocjoinerAnchorStatistics confirm this tiered link quality system.

filed 2012-12-31, granted 2017-01-31 · leak modules: AnchorsAnchor, Anchors, AnchorsAnchorSource, IndexingDocjoinerAnchorStatistics, AnchorsRedundantAnchorInfo, QualityNsrNsrData, QualityNsrPQData, CompressedQualitySignals

US9280580B1 Customizing search

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.

filed 2013-01-07, granted 2016-03-08 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US8965880B2 Transcoding and serving resources

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2013-01-29, granted 2015-02-24 · leak modules: ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US9275113B1 Language-specific search results

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2013-02-07, granted 2016-03-01 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8719280B1 Person-based information aggregation

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2013-02-08, granted 2014-05-06 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US9916396B2 Methods and systems for content-based search

Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2013-02-19, granted 2018-03-13 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US9177030B2 Systems and methods for providing search results
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-02-25, granted 2015-11-03 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US9251223B2 Alternative web pages suggestion based on language
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-02-26, granted 2016-02-02 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefEntityJoin

US9501460B1 Detecting and processing invisible elements within a web page
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2013-02-28, granted 2016-11-22 · leak modules: IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, ShingleInfoPerDocData, SmartphonePerDocData

US10324952B1 Hosted database
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-03-04, granted 2019-06-18 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9092489B1 Popular media items data set with exponential decay
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2013-03-08, granted 2015-07-28 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9298679B2 System and method providing a binary representation of a web page
Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships.
filed 2013-03-08, granted 2016-03-29 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US20140280079A1 Creating Lists of Digital Content
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-03-13 · leak modules: CrawlerChangerateUrlChangerate, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US9147154B2 Classifying resources using a deep network
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-03-13, granted 2015-09-29 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US9280522B2 Highlighting of document elements
Rule: Use descriptive alt text and surrounding context for images. Structure answers: question heading, tight answer, then depth. Use JSON-LD structured data for entity types and relationships.
filed 2013-03-13, granted 2016-03-08 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SnippetExtraInfo

US9311069B2 Search in application launcher
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Build content for diverse user contexts and search histories.
filed 2013-03-13, granted 2016-04-12 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9015313B2 Providing actionable content to a computing device based on user actions
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-03-14, granted 2015-04-21 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US9313169B2 Providing content to devices in a cluster
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-03-14, granted 2016-04-12 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US10650066B2 Enhancing sitelinks with creative content
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-03-15, granted 2020-05-12 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9063984B1 Methods, systems, and media for providing a media search engine
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-03-15, granted 2015-06-23 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9110981B1 Performance reporting for media content by subject
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Optimize mobile experience. Mobile-specific signals are tracked separately. Optimize video for watch time and retention.
filed 2013-03-15, granted 2015-08-18 · leak modules: CompressedQualitySignals, PerDocData, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, VideoPerDocData

US9189528B1 Searching and tagging media storage with a knowledge database
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2013-03-15, granted 2015-11-17 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US9268844B1 Adding document filters to an existing cluster hierarchy
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Write in natural vocabulary. ML models understand context and fluff.
filed 2013-03-15, granted 2016-02-23 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, ShingleInfoPerDocData, SmartphonePerDocData

US9305053B2 Dynamic sessionization of analytics data
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-03-15, granted 2016-04-05 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US9311361B1 Algorithmically determining the visual appeal of online content
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-03-15, granted 2016-04-12 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityWebanswersVideoTranscriptAnnotations

US9645982B2 System and method for loading a web page
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-03-15, granted 2017-05-09 · leak modules: PerDocData, SmartphonePerDocData

US9659064B1 Obtaining authoritative search results
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-03-15, granted 2017-05-23 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9946746B2 Persist and process analytics data dimensions for server-side sessionization
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-03-15, granted 2018-04-17 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US8782032B2 Minimizing visibility of stale content in web searching including revising web crawl intervals of documents
Rule: Pages that change frequently and get clicked often will be crawled more frequently. Update critical content regularly to trigger 'critical change' detection. Low-traffic, static pages will have their crawl intervals lengthened, so prioritize updates on your most-viewed pages. Monitor CTR in Search Console as it directly influences crawl frequency.
filed 2013-03-22, granted 2014-07-15 · leak modules: crawler, freshness, navboost, indexing, scheduling, tiered_index

US9811776B2 Determining a meaning of a knowledge item using document-based information
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-04-05, granted 2017-11-07 · leak modules: PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData

US8996517B2 Permitting users to remove documents
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-04-08, granted 2015-03-31 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster

US9443250B1 Learning characteristics for extraction of information from web pages
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2013-04-09, granted 2016-09-13 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9600531B1 Method and system for generating search shortcuts and inline auto-complete entries
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2013-04-10, granted 2017-03-21 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9116940B1 Searching for join candidates
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-04-15, granted 2015-08-25 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US9756108B2 Preloading resources of a web page
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-04-23, granted 2017-09-05 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9753901B1 Identifying important document content using geometries of page elements
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2013-05-09, granted 2017-09-05 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo

US9524073B1 Triggering action on a web page
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-05-13, granted 2016-12-20 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8972394B1 Generating a related set of documents for an initial set of documents
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-05-20, granted 2015-03-03 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US10289737B1 Media search broadening
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-06-10, granted 2019-05-14 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US9805406B2 Embeddable media content search widget
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-06-12, granted 2017-10-31 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9792371B1 Automatic synthesis and evaluation of content
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data.
filed 2013-06-19, granted 2017-10-17 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US9135303B2 Using pre-search triggers
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-06-27, granted 2015-09-15 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US9442905B1 Detecting neighborhoods from geocoded web documents
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-06-28, granted 2016-09-13 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate

US20150012524A1 Using models for triggering personal search
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2013-07-02 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US10089394B2 Personal search result identifying a physical location previously interacted with by a user
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-07-11, granted 2018-10-02 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation

US9298779B1 Combining content with a search result
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-07-16, granted 2016-03-29 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US9342839B2 Combining content with a search result
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-07-16, granted 2016-05-17 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US9430519B1 Dynamically generating pre-aggregated datasets
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-07-26, granted 2016-08-30 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US9043425B2 Resource locator remarketing
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-07-29, granted 2015-05-26 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US9697259B1 Refining search results
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-07-29, granted 2017-07-04 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US9576042B2 Categorizing search terms

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2013-08-01, granted 2017-02-21 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US9286397B1 Generating customized content

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data.

filed 2013-08-02, granted 2016-03-15 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US20150039582A1 Providing information in association with a search field

Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2013-08-05 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US10019484B2 Third party search applications for a search system

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.

filed 2013-08-06, granted 2018-07-10 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData

US8990210B2 Propagating information among web pages

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2013-08-15, granted 2015-03-24 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData

US8914358B1 Systems and methods for improved searching

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2013-08-22, granted 2014-12-16 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US9460141B1 Automatic expiring of cached data

Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2013-08-26, granted 2016-10-04 · leak modules: PerDocData, SmartphonePerDocData

US9141703B1 Systems and methods for browsing historical content

Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Create original, effortful content. Quality is scored at page and site level.

filed 2013-08-29, granted 2015-09-22 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, RepositoryAnnotationsGeoTopicality

US9348945B2 Modifying search results based on dismissal action associated with one or more of the search results

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2013-08-29, granted 2016-05-24 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US9355158B2 Modifying search results based on selection of one of the search results

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.

filed 2013-08-29, granted 2016-05-31 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US9086978B2 Caching electronic document resources in a client device having an electronic resource database

Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.

filed 2013-09-24, granted 2015-07-21 · leak modules: IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, ShingleInfoPerDocData

US9349019B2 System and method for associating tags with online content

Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Create original, effortful content. Quality is scored at page and site level. Earn endorsements from trusted sources. Trust is computed and transitive.

filed 2013-10-01, granted 2016-05-24 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, SmartphonePerDocData

US9201935B2 Transition from first search results environment to second search results environment

Rule: Optimize images for smooth transitions from universal search to image search—use high-quality thumbnails that expand cleanly. Ensure your images are properly sized and formatted so they display well in both thumbnail and expanded views. Image results that maintain visual continuity during transitions improve user experience and engagement metrics.

filed 2013-10-10, granted 2015-12-01 · leak modules: ImageSearchImageSelectionInfo, ImageMustangImageLinkSelectionInfo, WWWSnippetResponse, WWWResultInfoSubImageDocInfo, WWWDocInfoRelatedImages, SnapshotImageNode

US9323731B1 Data extraction using templates

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2013-10-15, granted 2016-04-26 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9588675B2 Document scale and position optimization

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2013-10-18, granted 2017-03-07 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US9767076B2 Document scale and position optimization

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2013-10-18, granted 2017-09-19 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US20140122235A1 Determining ad targeting information and/or ad creative information using past search queries

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.

filed 2013-10-24 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US9378299B1 Browsing pages in an electronic document

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2013-11-12, granted 2016-06-28 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9195761B2 System and method for navigating documents

Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Optimize video for watch time and retention.

filed 2013-11-15, granted 2015-11-24 · leak modules: PerDocData, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, VideoPerDocData

US8924850B1 Speeding up document loading

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2013-11-21, granted 2014-12-30 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9367583B1 Systems and methods of generating content performance metrics

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2013-11-21, granted 2016-06-14 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US9715548B2 Surfacing user-specific data records in search

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2013-12-04, granted 2017-07-25 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US9378295B1 Clustering content based on anticipated content trend topics
Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately. Ensure content comprehensively covers the topic with natural vocabulary.
filed 2013-12-23, granted 2016-06-28 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US8838645B2 Variably controlling access to content
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-12-27, granted 2014-09-16 · leak modules: ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20170116284A1 Surfacing Inferred Actions in Search
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-12-30 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US8965919B1 Locating meaningful stopwords or stop-phrases in keyword-based retrieval systems
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2013-12-30, granted 2015-02-24 · leak modules: CompressedQualitySignals, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US8930503B1 Resource locator remarketing
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-12-31, granted 2015-01-06 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US9454621B2 Surfacing navigational search results
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-12-31, granted 2016-09-27 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

USRE45952E1 Method for searching media
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-01-09, granted 2016-03-29 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9773064B1 Contextual search term evaluation
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2014-02-03, granted 2017-09-26 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US9652508B1 Device specific adjustment based on resource utilities
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2014-03-05, granted 2017-05-16 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US10223450B1 Data delivery
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships.
filed 2014-03-14, granted 2019-03-05 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US9930095B2 System for managing extension modifications to web pages
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2014-03-26, granted 2018-03-27 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefEntityJoin, SmartphonePerDocData

US9104746B1 Identifying contrarian terms based on website content
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-03-28, granted 2015-08-11 · leak modules: PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US9805006B1 Asynchronous loading of scripts in web pages
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-03-28, granted 2017-10-31 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9104768B2 Person-based information aggregation
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-04-02, granted 2015-08-11 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US20150293889A1 Perception of page download time by optimized resource scheduling
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2014-04-09 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9530099B1 Access to network content
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Optimize video for watch time and retention.
filed 2014-04-11, granted 2016-12-27 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData, VideoPerDocData

US10282479B1 Resource view data collection
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-05-08, granted 2019-05-07 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9002817B2 Interleaving search results
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2014-05-16, granted 2015-04-07 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US9292609B2 Predictive information retrieval
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-05-20, granted 2016-03-22 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9165038B1 Interpreting adjacent search terms based on a hierarchical relationship
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-05-29, granted 2015-10-20 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US9146116B1 Automatic continued search
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2014-06-04, granted 2015-09-29 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate

US10515133B1 Systems and methods for automatically suggesting metadata for media content
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-06-06, granted 2019-12-24 · leak modules: IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9672285B2 System and method for improving access to search results
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-06-20, granted 2017-06-06 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US10257208B1 Method and system for using a network analysis system to verify content on a website
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-06-23, granted 2019-04-09 · leak modules: CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US9323792B2 Web mining to build a landmark database and applications thereof
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-06-25, granted 2016-04-26 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData

US10970352B1 Selecting content for co-located devices
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-07-02, granted 2021-04-06 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US9195715B1 Providing information regarding prior searches
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.
filed 2014-07-09, granted 2015-11-24 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US10042936B1 Frequency-based content analysis
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-07-11, granted 2018-08-07 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US9843649B1 Providing content based on event related information
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-08-02, granted 2017-12-12 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US10114895B2 System and method for enhancing user search results by determining a streaming media program currently being displayed in proximity to an electronic device
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-08-08, granted 2018-10-30 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US9292509B2 Supplementing search results with information of interest
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Write for intent and meaning. Neural matching connects related concepts.
filed 2014-08-08, granted 2016-03-22 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9703763B1 Automatic document citations by utilizing copied content for candidate sources
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2014-08-14, granted 2017-07-11 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US9946452B1 Movable information panels

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.

filed 2014-08-25, granted 2018-04-17 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US9652530B1 Generating and applying event data extraction templates

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2014-08-27, granted 2017-05-16 · leak modules: IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US9195698B2 Selectively retrieving search results in accordance with different logical relationships

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.

filed 2014-09-09, granted 2015-11-24 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9195764B1 Providing an online forum as a search result

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2014-09-15, granted 2015-11-24 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US9652547B2 Framework for suggesting search terms

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2014-09-15, granted 2017-05-16 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11169988B2 Radar recognition-aided search

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2014-10-01, granted 2021-11-09 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9111000B1 In-context searching

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2014-10-20, granted 2015-08-18 · leak modules: CompressedQualitySignals, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData

US20160125320A1 Grouping flight search results

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2014-10-31 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US9922005B2 Method and apparatus for animating transitions between search results

Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.

filed 2014-11-03, granted 2018-03-20 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US9922122B2 Providing remedial search operation based on analysis of user interaction with search results

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2014-11-11, granted 2018-03-20 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US9916293B2 Module specification for a module to be incorporated into a container document
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data.
filed 2014-11-18, granted 2018-03-13 · leak modules: CrawlerChangerateUrlChangerate, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9218397B1 Systems and methods for improved searching
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-11-26, granted 2015-12-22 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US9971813B2 Categorizing objects, such as documents and/or clusters, with respect to a taxonomy and data structures derived from such categorization
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-12-04, granted 2018-05-15 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US10296654B2 Speeding up document loading
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-12-11, granted 2019-05-21 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US10769191B2 Classifying data objects
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-12-19, granted 2020-09-08 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US9286357B1 Blending content in an output
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-01-05, granted 2016-03-15 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US9147005B1 Consistently delivering a web page having source code with a dynamic instruction
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2015-01-09, granted 2015-09-29 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US9870420B2 Classification and storage of documents
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-01-19, granted 2018-01-16 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9767199B2 Transcoding and serving resources
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2015-02-06, granted 2017-09-19 · leak modules: ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US9875322B2 Saving and retrieving locations of objects
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-02-12, granted 2018-01-23 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20200012727A1 Organizing objects in a feed to facilitate interaction between fans and content owners

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2015-02-13 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9817920B1 Locating meaningful stopwords or stop-phrases in keyword-based retrieval systems

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.

filed 2015-02-23, granted 2017-11-14 · leak modules: CompressedQualitySignals, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US10133788B1 Determining resource quality based on resource competition

Rule: Your ranking is directly influenced by how often users choose your result over competitors in the same SERP. High CTR relative to your position boosts your domain; low CTR demotes it. Optimize titles, meta descriptions, and rich snippets to maximize click-through. Monitor CTR by query in Search Console - underperforming pages relative to competitors will be demoted.

filed 2015-03-17, granted 2018-11-20 · leak modules: demotion, navboost, ranking, quality, click_model

US10108694B1 Content clustering

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2015-04-08, granted 2018-10-23 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9378288B1 Refining search results

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2015-04-09, granted 2016-06-28 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US20160299978A1 Device dependent search experience

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2015-04-13 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US9514227B1 In-conversation search

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2015-04-13, granted 2016-12-06 · leak modules: IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US9582549B2 Computer application data in search results

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2015-04-14, granted 2017-02-28 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US9378249B2 Providing actionable content to a computing device based on user actions

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2015-04-20, granted 2016-06-28 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US9524350B2 Resource locator remarketing

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2015-04-22, granted 2016-12-20 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US10353960B1 Systems and methods for caching structural elements of electronic documents
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2015-05-04, granted 2019-07-16 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9767188B1 Using information from a user device and a server to suggest an input
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-05-14, granted 2017-09-19 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20160344831A1 Proxy service for content requests
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-05-21 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US10527442B2 Navigation system and methods for generating enhanced search results
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2015-05-22, granted 2020-01-07 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US9449054B1 Methods, systems, and media for providing a media search engine
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-05-22, granted 2016-09-20 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9996624B2 Surfacing in-depth articles in search results
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2015-06-26, granted 2018-06-12 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation

US9886475B1 Known item search method
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2015-07-02, granted 2018-02-06 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, RepositoryWebrefEntityJoin

US11194885B1 Incremental document object model updating
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2015-07-07, granted 2021-12-07 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US9836528B1 Data constrained resource access
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2015-07-20, granted 2017-12-05 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9854285B1 Popular media items data set with exponential decay
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2015-07-23, granted 2017-12-26 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9639579B2 Determination of a desired repository for retrieving search results
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-07-27, granted 2017-05-02 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US9542453B1 Systems and methods for promoting search results based on personal information
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-08-07, granted 2017-01-10 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US9756502B2 Reducing data use in limited data environment
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2015-08-13, granted 2017-09-05 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9665650B1 In-context searching
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2015-08-14, granted 2017-05-30 · leak modules: CompressedQualitySignals, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData

US20170061515A1 Systems and methods for setting allocations and prices for content in an online marketplace
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-08-24 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US9449271B2 Classifying resources using a deep network
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-08-24, granted 2016-09-20 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US9760636B1 Systems and methods for browsing historical content
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Create original, effortful content. Quality is scored at page and site level.
filed 2015-08-28, granted 2017-09-12 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, RepositoryAnnotationsGeoTopicality

US10083237B2 Protecting users from inappropriate sensitive or offensive search results
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-08-31, granted 2018-09-25 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US9390150B1 Automatic continued search
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2015-09-04, granted 2016-07-12 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate

US9514241B1 Consistently delivering a web page having source code with a dynamic instruction
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2015-09-25, granted 2016-12-06 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US9569548B2 Systems and methods for highlighting search results
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Write for natural spoken language and conversational queries.
filed 2015-09-25, granted 2017-02-14 · leak modules: NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US10331770B1 System and method for confirming authorship of documents
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data.
filed 2015-10-02, granted 2019-06-25 · leak modules: PerDocData, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US10255323B1 Quantization-based fast inner product search
Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2015-10-08, granted 2019-04-09 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData

US10120903B2 Identifying teachable moments for contextual search
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-10-15, granted 2018-11-06 · leak modules: CompressedQualitySignals, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US9460167B2 Transition from first search results environment to second search results environment
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-10-29, granted 2016-10-04 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US9373004B2 Methods and systems for deleting requested information
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Structure answers: question heading, tight answer, then depth.
filed 2015-10-30, granted 2016-06-21 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US9690842B2 Analyzing frequently occurring data items
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2015-11-09, granted 2017-06-27 · leak modules: PerDocData, SmartphonePerDocData

US10963623B2 Providing useful information associated with an item in a document
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Structure answers: question heading, tight answer, then depth. Write for natural spoken language and conversational queries.
filed 2015-11-23, granted 2021-03-30 · leak modules: NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryWebrefEntityJoin, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US9536006B2 Enriching search results
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-11-23, granted 2017-01-03 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US10277658B2 Reduction of web page load time using http header compression
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2015-11-30, granted 2019-04-30 · leak modules: PerDocData, SmartphonePerDocData

US10102304B1 Multi-stage digital content evaluation
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2015-12-14, granted 2018-10-16 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9875307B2 Using content identification as context for search
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2016-01-22, granted 2018-01-23 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US10049151B2 External action suggestions in search results
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Optimize video for watch time and retention.
filed 2016-01-26, granted 2018-08-14 · leak modules: PerDocData, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, VideoPerDocData

US10216467B2 Systems and methods for automatic content verification
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2016-02-03, granted 2019-02-26 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US9967230B2 Systems and methods for allocating communication resources via information technology infrastructure
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2016-02-04, granted 2018-05-08 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US10394908B1 Systems and methods for modifying search results based on a user's history
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2016-02-08, granted 2019-08-27 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US9830367B2 Predictive information retrieval
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2016-02-16, granted 2017-11-28 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10360291B2 System and method providing a binary representation of a web page
Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships.
filed 2016-02-18, granted 2019-07-23 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US10152554B2 Detecting digital content visibility
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2016-02-23, granted 2018-12-11 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US10572548B2 System and method for improving access to search results
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2016-03-01, granted 2020-02-25 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US9734518B2 Generating customized content
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data.
filed 2016-03-02, granted 2017-08-15 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US9882867B2 Providing content to devices in a cluster
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2016-03-07, granted 2018-01-30 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US9779139B1 Context-based filtering of search results
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2016-03-15, granted 2017-10-03 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, SmartphonePerDocData, SnippetExtraInfo

US10404835B2 Hybrid client-server data provision
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2016-03-17, granted 2019-09-03 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US10140297B2 Supplementing search results with information of interest
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2016-03-21, granted 2018-11-27 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality

US10929883B1 Combining content with a search result
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2016-03-21, granted 2021-02-23 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US10318142B2 Navigating event information

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2016-03-26, granted 2019-06-11 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US10205702B1 Systems and methods for allocating communication resources via information technology infrastructure

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2016-03-30, granted 2019-02-12 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US10025577B2 Search in application launcher

Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Build content for diverse user contexts and search histories.

filed 2016-04-05, granted 2018-07-17 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US10977573B1 Distantly supervised wrapper induction for semi-structured documents

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2016-04-15, granted 2021-04-13 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US10110968B2 Methods, systems and media for interacting with content using a second screen device

Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately. Optimize video for watch time and retention.

filed 2016-04-19, granted 2018-10-23 · leak modules: NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, VideoPerDocData

US10140017B2 Graphical keyboard application with integrated search

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2016-04-20, granted 2018-11-27 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US9965530B2 Graphical keyboard with integrated search features

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2016-04-20, granted 2018-05-08 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US10795926B1 Suppressing personally objectionable content in search results

Rule: Ensure image content is clearly distinguishable as professionally produced—use high-resolution images, proper EXIF metadata, professional watermarks, and consistent branding. Amateur-looking images paired with objectionable content triggers suppression. Avoid low-quality user-submitted images on pages that could be misclassified. Monitor query patterns around personal names to detect reputation issues.

filed 2016-04-22, granted 2020-10-06 · leak modules: ImageSafesearchContentBrainPornAnnotation, ImageQualitySensitiveMediaOrPeopleEntities, SafesearchImageOffensiveAnnotation, AbuseiamContentRestriction, AbuseiamVerdictRestrictionContext, ClassifierPornQueryClassifierOutput, ImageRepositoryUnwantedContent, SearchPolicyRankableSensitivity

US10277650B1 Parallel execution of request tracking and resource delivery

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2016-05-12, granted 2019-04-30 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US9947026B2 Combining content with a search result

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-05-12, granted 2018-04-17 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US10242112B2 Search result filters from resource content

Rule: Use relevant keywords in prominent positions (title, headings, early paragraphs) of your pages — these are more likely to become search result filters. Include clear, distinctive terms that differentiate your content from competitors. Structure content with clear categorical terms that would produce diverse filter results.

filed 2016-06-15, granted 2019-03-26 · leak modules: PerDocData, IndexingDocjoinerAnchorStatistics

US20160299983A1 Programmable search engines
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2016-06-20 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, IndexingDocjoinerAnchorStatistics, LocalWWWInfo

US9699241B2 Providing actionable content to a computing device based on user actions
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2016-06-27, granted 2017-07-04 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US10831777B2 Distributed database configuration
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2016-07-01, granted 2020-11-10 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9910885B1 Automatic continued search
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2016-07-11, granted 2018-03-06 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate

US9919209B2 Document enhancement system and method
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2016-07-18, granted 2018-03-20 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US10216805B1 Dynamically generating pre-aggregated datasets
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2016-08-11, granted 2019-02-26 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US9712618B2 Asynchronous and synchronous resource links
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2016-08-15, granted 2017-07-18 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US10042829B2 Measurement of visibility of overlay content
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2016-08-16, granted 2018-08-07 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20160357788A1 Resource name generation and derivation
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships. Write for natural spoken language and conversational queries.
filed 2016-08-18 · leak modules: NlpSaftDocument, PerDocData, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US20180068324A1 Outputting content based on interests of other users
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2016-09-06 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9690667B2 Automatic expiring of cached data
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2016-09-07, granted 2017-06-27 · leak modules: PerDocData, SmartphonePerDocData

US9785681B2 Methods, systems, and media for providing a media search engine

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2016-09-19, granted 2017-10-10 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US20180096039A1 Systems and methods for context-sensitive data annotation and annotation visualization

Rule: Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2016-09-30 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US10719509B2 Hierarchical quantization for fast inner product search

Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately. Write for intent and meaning. Neural matching connects related concepts.

filed 2016-10-11, granted 2020-07-21 · leak modules: NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData

US10108716B1 Systems and methods for providing stock ticker information

Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately. Write for intent and meaning. Neural matching connects related concepts.

filed 2016-10-18, granted 2018-10-23 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, SmartphonePerDocData

US10643183B2 Search engine

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2016-10-18, granted 2020-05-05 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9946773B2 Graphical keyboard with integrated search features

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2016-10-24, granted 2018-04-17 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US20180113946A1 Method and device for domain web page registration

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2016-10-25 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US10824630B2 Search and retrieval of structured information cards

Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2016-10-26, granted 2020-11-03 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US9854011B2 Analyzing reading metrics to generate action information

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-10-28, granted 2017-12-26 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9866520B2 Method and apparatus for managing internet content

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2016-10-31, granted 2018-01-09 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US10621243B1 In-conversation search

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-11-01, granted 2020-04-14 · leak modules: IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US10445394B2 Resource locator remarketing
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2016-11-16, granted 2019-10-15 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US10713430B2 Systems and methods for applying layout to documents
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2016-11-30, granted 2020-07-14 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US11526554B2 Preventing the distribution of forbidden network content using automatic variant detection
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2016-12-09, granted 2022-12-13 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US10650071B2 Search and retrieval of keyed data maintained using a keyed database
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2016-12-14, granted 2020-05-12 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US10380229B2 Method and system providing contextual functionality in static web pages
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Write for intent and meaning. Neural matching connects related concepts.
filed 2016-12-20, granted 2019-08-13 · leak modules: NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US10051108B2 Contextual information for a notification
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2016-12-22, granted 2018-08-14 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US10891342B2 Content data determination, transmission and storage for local devices
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2017-01-12, granted 2021-01-12 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US10459952B2 Categorizing search terms
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-01-18, granted 2019-10-29 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US11645368B2 Hash-based dynamic restriction of content on information resources
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2017-01-24, granted 2023-05-09 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US20180232295A1 Analyzing large-scale data processing jobs
Rule: Use JSON-LD structured data for entity types and relationships.
filed 2017-02-14 · leak modules: RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US10747825B2 Content search engine
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2017-02-27, granted 2020-08-18 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US10733376B2 Delivering auto-play media content element from cross origin resources
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data.
filed 2017-03-01, granted 2020-08-04 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US10691326B2 Document scale and position optimization
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2017-03-06, granted 2020-06-23 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US10678858B2 Method and system for generating search shortcuts and inline auto-complete entries
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2017-03-17, granted 2020-06-09 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10664509B1 Processing non-uniform datasets
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2017-03-21, granted 2020-05-26 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US20180288168A1 Managing user sessions based on contextual information
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-03-31 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US10387917B2 Suggesting targeting information for ads; such as websites and/or categories of websites for example
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2017-04-07, granted 2019-08-20 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US10360537B1 Generating and applying event data extraction templates
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-04-11, granted 2019-07-23 · leak modules: IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US11036804B1 Device specific adjustment based on resource utilities
Rule: Ensure your pages are categorized for the correct device types. If you have Android-specific or iOS-specific content, properly mark it. Responsive design that works across all device types ensures positive utility classification. Navigational/branded queries bypass device adjustments, so brand strength protects cross-device rankings.
filed 2017-04-20, granted 2021-06-15 · leak modules: PerDocData, CompressedQualitySignals

US10749988B2 Methods, systems, and media for retrieving content associated with links
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2017-04-24, granted 2020-08-18 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US10326820B2 Providing actionable content to a computing device based on user actions
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-05-31, granted 2019-06-18 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US11620347B2 Method and apparatus that collect and uploads implicit analytic data
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2017-06-08, granted 2023-04-04 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US10679068B2 Media contextual information from buffered media data
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2017-06-13, granted 2020-06-09 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US10789283B2 Systems and methods for notifying an author of contextual suggested content
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2017-06-14, granted 2020-09-29 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US10162903B2 Surfacing user-specific data records in search
Rule: Ensure your entity has structured personal data records (events, bookings, reservations) that Google can surface inline. Use schema.org Event, Reservation, and Action markup so Google can trigger display of your content alongside knowledge panels and personal results. Time-sensitive and location-relevant content gets prominence boosts.
filed 2017-06-20, granted 2018-12-25 · leak modules: PerDocData, KnowledgeAnswersIntentQueryPersonalEntity, KnowledgeAnswersIntentQueryArgumentProvenanceAttentionalEntity, RepositoryWebrefPersonalizationContextOutput, RepositoryWebrefPersonalizationContextOutputs, QualityPreviewChosenSnippetInfo, UniversalsearchNewPackerKnowledgeResultSupport

US10783133B1 Time reservations for ensuring consistent reads in a distributed database without logging
Rule: Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-06-23, granted 2020-09-22 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, RepositoryAnnotationsGeoTopicality

US10204162B2 Providing content
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2017-07-28, granted 2019-02-12 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US10019443B2 Generating customized content
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data.
filed 2017-08-03, granted 2018-07-10 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData

US10013502B1 Preloading resources of a web page
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2017-08-04, granted 2018-07-03 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US10149148B2 Reducing data use in limited data environment
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2017-08-04, granted 2018-12-04 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US10887406B2 Dynamic content loading selection
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2017-08-16, granted 2021-01-05 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, RepositoryWebrefEntityJoin, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US11019179B2 Access to network content
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2017-08-18, granted 2021-05-25 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US10599727B2 Transcoding and serving resources
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2017-09-08, granted 2020-03-24 · leak modules: ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US10621187B2 Methods, systems, and media for providing a media search engine
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2017-10-09, granted 2020-04-14 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US10685016B1 Automatic continued search
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2017-10-18, granted 2020-06-16 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate

US10275503B2 Predictive information retrieval
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-10-24, granted 2019-04-30 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11275807B2 React to location changes on web pages
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-11-07, granted 2022-03-15 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10230818B1 Providing content based on event related information
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2017-12-06, granted 2019-03-12 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US10735555B2 Providing content based on event related information
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2017-12-08, granted 2020-08-04 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US11386169B1 Navigation through a collection of electronic documents
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2017-12-11, granted 2022-07-12 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10165020B2 Analyzing reading metrics to generate action information
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-12-13, granted 2018-12-25 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11200375B2 Delivering auto-play media content element from cross origin resources
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data.
filed 2017-12-14, granted 2021-12-14 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US11481455B2 Using content identification as context for search
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2017-12-14, granted 2022-10-25 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US10783189B2 Saving and retrieving locations of objects
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-12-20, granted 2020-09-22 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10164936B2 Providing content to devices in a cluster
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2018-01-05, granted 2018-12-25 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US10754900B2 Access to network content
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2018-01-18, granted 2020-08-25 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US10783194B1 Providing remedial search operation based on analysis of user interaction with search results
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2018-02-01, granted 2020-09-22 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US10909153B2 Methods and systems for identification and correction of controlled system data
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Earn endorsements from trusted sources. Trust is computed and transitive.
filed 2018-02-22, granted 2021-02-02 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US10678429B1 Native search application providing search results of multiple search types
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2018-02-23, granted 2020-06-09 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US10402859B2 Combining content with a search result
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2018-04-12, granted 2019-09-03 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US10250556B2 Systems and methods for allocating communication resources via information technology infrastructure
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2018-04-27, granted 2019-04-02 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US11132406B2 Action indicators for search operation output elements
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2018-05-18, granted 2021-09-28 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US10878037B2 Digital supplement association and retrieval for visual search
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2018-06-21, granted 2020-12-29 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, IndexingDocjoinerAnchorStatistics, LocalWWWInfo

US10872109B2 Generating customized content

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data.

filed 2018-06-26, granted 2020-12-22 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData

US20210103623A1 Media source measurement for incorporation into a censored media corpus

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.

filed 2018-06-29 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US10289618B2 Third party search applications for a search system

Rule: If you operate a data-rich vertical (flights, sports, finance, weather), consider providing structured data feeds to Google. Creating regex-triggered answer patterns with structured data can get your content surfaced as zero-click answers. Ensure your data is available in machine-readable formats and kept updated.

filed 2018-07-09, granted 2019-05-14 · leak modules: RepositoryWebref, PerDocData

US11461336B2 Selecting between global and location-specific search results

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2018-07-31, granted 2022-10-04 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US10664541B2 Personal search result identifying a physical location previously interacted with by a user

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2018-08-14, granted 2020-05-26 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation

US20210165829A1 Intelligent serendipitous document discovery notifications

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.

filed 2018-08-21 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData

US20200074007A1 Timeline of Events in Search Based System

Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately. Optimize video for watch time and retention.

filed 2018-08-30 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, VideoPerDocData

US10528564B2 Identifying teachable moments for contextual search

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2018-09-25, granted 2020-01-07 · leak modules: CompressedQualitySignals, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US10448118B2 Methods, systems and media for interacting with content using a second screen device

Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately. Optimize video for watch time and retention.

filed 2018-10-22, granted 2019-10-15 · leak modules: NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, VideoPerDocData

US10740422B2 Surfacing user-specific data records in search

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2018-12-21, granted 2020-08-11 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US11372941B2 Search result filters from resource content
Rule: Place important descriptive terms in prominent positions within your content (titles, headings, early paragraphs, meta descriptions) since term prominence directly determines whether your content's keywords become search filters. Use diverse, specific descriptive terms throughout content to increase the chance your pages appear in filtered result sets. Avoid overusing common query stop terms that get filtered out.
filed 2019-02-01, granted 2022-06-28 · leak modules: QualitySalientTermsSalientTermSet, QualitySalientTermsSignalData, QualitySalientTermsDocData, PerDocData, QualityPreviewSnippetRadishFeatures

US10666762B2 Providing content based on event related information
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2019-02-28, granted 2020-05-26 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US11468127B1 Data delivery
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships.
filed 2019-03-04, granted 2022-10-11 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US11971897B2 Predictive information retrieval
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2019-03-18, granted 2024-04-30 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10909207B2 Speeding up document loading
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2019-04-05, granted 2021-02-02 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US11120094B1 Resource view data collection
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2019-04-08, granted 2021-09-14 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US11354358B1 Organization and retrieval of conditioned data
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2019-04-10, granted 2022-06-07 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US10693927B2 Parallel execution of request tracking and resource delivery
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2019-04-29, granted 2020-06-23 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US11106707B2 Triggering application information
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2019-04-29, granted 2021-08-31 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US20230122031A1 Systems and methods for providing content candidates
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2019-06-26 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US11037198B2 Suggesting targeting information for ads, such as websites and/or categories of websites for example

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.

filed 2019-06-28, granted 2021-06-15 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US11170057B1 Systems and methods for caching structural elements of electronic documents

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.

filed 2019-07-15, granted 2021-11-09 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11194557B2 System and method providing a binary representation of a web page

Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships.

filed 2019-07-22, granted 2021-12-07 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US11361344B2 Combining content with a search result

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2019-07-22, granted 2022-06-14 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US11190621B2 Hybrid client-server data provision

Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2019-08-29, granted 2021-11-30 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US11310046B2 Systems and methods for secure identification retrieval

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.

filed 2019-09-03, granted 2022-04-19 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10891349B2 Resource locator remarketing

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2019-09-09, granted 2021-01-12 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US10917371B2 Methods and apparatus for determining non-textual reply content for inclusion in a reply to an electronic communication

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2019-09-26, granted 2021-02-09 · leak modules: ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US11734375B2 Automatic navigation of interactive web documents

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.

filed 2019-09-27, granted 2023-08-22 · leak modules: NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11228816B2 Methods, systems and media for interacting with content using a second screen device

Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately. Optimize video for watch time and retention.

filed 2019-10-14, granted 2022-01-18 · leak modules: NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, VideoPerDocData

US11210331B2 Cross-platform content muting
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2019-10-23, granted 2021-12-28 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US11366698B2 Efficient use of computing resources in responding to content requests
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2019-11-04, granted 2022-06-21 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US11232162B1 In-conversation search
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2020-03-03, granted 2022-01-25 · leak modules: IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US11580175B2 Transcoding and serving resources
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2020-03-19, granted 2023-02-14 · leak modules: ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US11496603B2 Providing content based on event related information
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2020-04-08, granted 2022-11-08 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US11372866B2 Methods, systems, and media for providing a media search engine
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2020-04-13, granted 2022-06-28 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US11494451B2 Personal search result identifying a physical location previously interacted with by a user
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2020-04-16, granted 2022-11-08 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation

US11463541B2 Providing content based on event related information
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2020-05-06, granted 2022-10-04 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US10891287B1 Automatic continued search
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2020-05-14, granted 2021-01-12 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate

US11514126B2 Systems and methods for saving and surfacing content
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2020-05-19, granted 2022-11-29 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, SmartphonePerDocData, SnippetExtraInfo

US11836209B2 Tamper-proof interaction data
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2020-05-22, granted 2023-12-05 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11190563B2 Parallel execution of request tracking and resource delivery
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2020-06-18, granted 2021-11-30 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US12488178B2 Verifying content and interactions within webviews
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2020-06-24, granted 2025-12-02 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US11809503B2 Surfacing user-specific data records in search
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2020-07-01, granted 2023-11-07 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US11321523B2 Systems and methods for applying layout to documents
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2020-07-13, granted 2022-05-03 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US11562138B2 Delivering auto-play media content element from cross origin resources
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data.
filed 2020-08-03, granted 2023-01-24 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US11520753B1 Time reservations for ensuring consistent reads in a distributed database without logging
Rule: Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2020-08-13, granted 2022-12-06 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, RepositoryAnnotationsGeoTopicality

US11233877B2 Methods, systems, and media for retrieving content associated with links
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2020-08-17, granted 2022-01-25 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US11960519B2 Classifying data objects
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2020-08-20, granted 2024-04-16 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US12130799B2 Data integrity optimization
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2020-08-21, granted 2024-10-29 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US11741150B1 Suppressing personally objectionable content in search results
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2020-10-05, granted 2023-08-29 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US11556561B2 Distributed database configuration
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2020-10-19, granted 2023-01-17 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11238058B2 Search and retrieval of structured information cards
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2020-11-02, granted 2022-02-01 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US11640431B2 Digital supplement association and retrieval for visual search
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2020-11-19, granted 2023-05-02 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, IndexingDocjoinerAnchorStatistics, LocalWWWInfo

US20210081084A1 User-configurable delivery of informational events
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2020-11-30 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US11386180B2 Resource locator remarketing
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2020-12-07, granted 2022-07-12 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US11720920B1 Combining content with a search result
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2021-01-20, granted 2023-08-08 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US11809511B2 Speeding up document loading
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2021-02-01, granted 2023-11-07 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US11475088B2 Providing notifications based on geofencing search results
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2021-02-05, granted 2022-10-18 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US11366868B1 Notification of change of value in stale content
Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2021-03-11, granted 2022-06-21 · leak modules: ImageData, ImagePerDocData, PerDocData, SmartphonePerDocData

US11544342B1 Selecting content for co-located devices
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2021-03-11, granted 2023-01-03 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US12164527B2 Search result annotations
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2021-05-24, granted 2024-12-10 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US11797653B2 Hash-based dynamic restriction of content on information resources
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2021-07-19, granted 2023-10-24 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US11768904B1 Resource view data collection
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2021-08-11, granted 2023-09-26 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US20230281258A1 Systems and Methods for Generating Contextual Dynamic Content
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2021-09-23 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo

US12047429B2 Parallel execution of request tracking and resource delivery
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2021-10-25, granted 2024-07-23 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US12019690B1 Systems and methods for caching structural elements of electronic documents
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2021-11-08, granted 2024-06-25 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11586663B2 Cross-platform content muting
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2021-11-23, granted 2023-02-21 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US20220382783A1 High Bandwidth Content Addressable Memory (CAM) Based Hardware Architecture For Datacenter Networking
Rule: Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2021-12-10 · leak modules: PerDocData, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US11842150B2 Delivering auto-play media content element from cross origin resources
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data.
filed 2021-12-13, granted 2023-12-12 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US11755666B2 In-conversation search
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2021-12-22, granted 2023-09-12 · leak modules: IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US11936960B2 Methods, systems and media for interacting with content using a second screen device
Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately. Optimize video for watch time and retention.
filed 2022-01-14, granted 2024-03-19 · leak modules: NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, VideoPerDocData

US11991262B2 Methods, systems, and media for retrieving content associated with links
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2022-01-24, granted 2024-05-21 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US11784817B2 Systems and methods for secure identification retrieval
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2022-03-21, granted 2023-10-10 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11720613B2 Contextual estimation of link information gain
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2022-04-22, granted 2023-08-08 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US11727206B2 Systems and methods for applying layout to documents

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2022-05-02, granted 2023-08-15 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US20240296480A1 Methods and apparatus for generating custom content responsive to a received search query

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2022-05-26 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US11809510B2 Notification of change of value in stale content

Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2022-06-15, granted 2023-11-07 · leak modules: ImageData, ImagePerDocData, PerDocData, SmartphonePerDocData

US11687602B2 Efficient use of computing resources in responding to content requests

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.

filed 2022-06-16, granted 2023-06-27 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US12130824B2 Precision of content matching systems at a platform

Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.

filed 2022-06-16, granted 2024-10-29 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US11734289B2 Methods, systems, and media for providing a media search engine

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2022-06-24, granted 2023-08-22 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US11797626B2 Search result filters from resource content

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.

filed 2022-06-27, granted 2023-10-24 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US20240045845A1 Unstructured Data Analytics in Traditional Data Warehouses

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2022-08-06 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US12007956B1 Time reservations for ensuring consistent reads in a distributed database without logging

Rule: Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2022-10-13, granted 2024-06-11 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, RepositoryAnnotationsGeoTopicality

US12488061B2 Mitigating latency and/or resource usage in triggering actionable suggestions related to rendered content

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2022-10-20, granted 2025-12-02 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US12019695B2 Using content identification as context for search

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2022-10-24, granted 2024-06-25 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US12450293B2 Assisted searching of non-document items
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2022-11-21, granted 2025-10-21 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US20230185856A1 Database Container Orchestrator
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2022-12-13 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US11874866B2 Multiscale quantization for fast similarity search
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2022-12-14, granted 2024-01-16 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US11907258B2 Distributed database configuration
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2022-12-28, granted 2024-02-20 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US12026454B2 Storage of content associated with a resource locator
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2022-12-30, granted 2024-07-02 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US12524748B2 Authorizing usage of media content of a content owner by other content creators
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2023-02-03, granted 2026-01-13 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US12032633B2 Digital supplement association and retrieval for visual search
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2023-03-24, granted 2024-07-09 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, IndexingDocjoinerAnchorStatistics, LocalWWWInfo

US12271416B2 Suppressing personally objectionable content in search results
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2023-05-11, granted 2025-04-08 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US12158907B1 Thematic search
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2023-05-16, granted 2024-12-03 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, IndexingDocjoinerAnchorStatistics, LocalWWWInfo

US12222995B2 Web page transformer for structure information extraction
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2023-05-23, granted 2025-02-11 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US12254490B1 Combining content with a search result
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2023-06-14, granted 2025-03-18 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US12244716B2 Systems and methods for secure identification retrieval
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2023-06-26, granted 2025-03-04 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US12332948B2 In-conversation search
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2023-08-11, granted 2025-06-17 · leak modules: IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US12153642B2 Automatic navigation of interactive web documents
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2023-08-16, granted 2024-11-26 · leak modules: NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US12468769B2 Search result filters from resource content
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2023-09-08, granted 2025-11-11 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US20240119052A1 Tuning Approximate Nearest Neighbor Search Engines for Speed-Recall Tradeoffs Via Lagrange Multiplier Methods
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2023-09-26 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US20250094022A1 Determining eligibility of content locators from previous computing device
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2023-10-17 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US12153571B2 Radar recognition-aided search
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2023-10-26, granted 2024-11-26 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US12223008B2 Tamper-proof interaction data
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2023-10-26, granted 2025-02-11 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20250156488A1 Determination of user post items related to search content using creator context data
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2023-11-15 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US20240184831A1 Interactive data connectors for an electronic document at a platform
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2023-11-30 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US12386856B2 Distributed database configuration
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2024-01-20, granted 2025-08-12 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20240378256A1 Artificial Intelligence Generated Badges for Search
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2024-04-16 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US12468667B1 Time reservations for ensuring consistent reads in a distributed database without logging
Rule: Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2024-04-23, granted 2025-11-11 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, RepositoryAnnotationsGeoTopicality

US12511325B2 Reranking documents based on graph representations of the documents
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2024-04-26, granted 2025-12-30 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US12579194B2 Digital supplement association and retrieval for visual search
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2024-06-06, granted 2026-03-17 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, IndexingDocjoinerAnchorStatistics, LocalWWWInfo

US20260037584A1 Methods and Systems for Restricting Cloud Resource Names
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2024-07-30 · leak modules: CrawlerChangerateUrlChangerate, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, TrawlerFetchReplyData

US20250061137A1 File search
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2024-08-16 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US20260056696A1 Excluding content during screensharing session
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2024-08-20 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US20250217373A1 Contextualized token retriever
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2024-12-30 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US20250307872A1 Digital Content Creation With Dynamic Targeting
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2025-03-13 · leak modules: NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20250390532A1 Adaptive sample selection for data item processing
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2025-06-24 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US20260127228A1 Progressing Search Instances in Weak Search Signal Instances
Rule: For image-based search, ensure your images are high-quality, unambiguous, and clearly depict identifiable objects. Poor image quality or ambiguous content triggers weak-signal detection, causing Google to prompt users for clarification rather than showing your results. Use clear, well-lit product images with single focal subjects. Provide comprehensive alt text and structured data so image classification models can confidently identify your content.
filed 2025-10-06 · leak modules: ImageUnderstandingIndexingAnnotation, ImageUnderstandingIndexingLabel, ImageUnderstandingIndexingLabelGroup, ImageUnderstandingIndexingImageRegion, KnowledgeAnswersIntentQueryParsingSignals, UniversalsearchNewPackerKnowledgeResultSupport, ImageRegionsImageRegion, QualityRankembedMustangMustangRankEmbedInfo

Everything else (580)

US7158961B1 Methods and apparatus for estimating similarity

Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.

filed 2001-12-31, granted 2007-01-02 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US6941293B1 Methods and apparatus for determining equivalent descriptions for an information need

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2002-02-01, granted 2005-09-06 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US8255417B2 System and method for providing definitions

Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2003-06-27, granted 2012-08-28 · leak modules: IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, ShingleInfoPerDocData

US8280946B1 Reduction of perceived DNS lookup latency

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.

filed 2003-09-10, granted 2012-10-02 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US7346615B2 Using match confidence to adjust a performance threshold

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.

filed 2003-11-14, granted 2008-03-18 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US8433703B1 Recording user actions

Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2003-12-22, granted 2013-04-30 · leak modules: ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures

US8051104B2 Editing a network of interconnected concepts

Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Ensure content comprehensively covers the topic with natural vocabulary.

filed 2003-12-30, granted 2011-11-01 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US7516118B1 Methods and systems for assisted network browsing

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2003-12-31, granted 2009-04-07 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo

US7827254B1 Automatic generation of rewrite rules for URLs

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2003-12-31, granted 2010-11-02 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, ShingleInfoPerDocData

US8150824B2 Systems and methods for direct navigation to specific portion of target document

Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.

filed 2003-12-31, granted 2012-04-03 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US8099407B2 Methods and systems for processing media files

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2004-03-31, granted 2012-01-17 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures

US7689924B1 Link annotation for keyboard navigation
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Write for natural spoken language and conversational queries.
filed 2004-04-12, granted 2010-03-30 · leak modules: NlpSaftDocument, PerDocData, SmartphonePerDocData

US7702653B1 Methods and systems for triggering actions
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2004-06-30, granted 2010-04-20 · leak modules: IndexingDupsLocalizedLocalizedCluster, PerDocData, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, ShingleInfoPerDocData, SmartphonePerDocData

US8065296B1 Systems and methods for determining a quality of provided items
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2004-09-29, granted 2011-11-22 · leak modules: CompressedQualitySignals, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo

US8495061B1 Automatic metadata identification
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use JSON-LD structured data for entity types and relationships.
filed 2004-09-29, granted 2013-07-23 · leak modules: NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefEntityJoin, RichsnippetsPageMap

US8051096B1 Methods and systems for augmenting a token lexicon
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2004-09-30, granted 2011-11-01 · leak modules: CrawlerChangerateUrlChangerate, PerDocData, QualityPreviewSnippetRadishFeatures, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo, TrawlerFetchReplyData

US8412812B1 Client-side measurement of load times
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2004-12-30, granted 2013-04-02 · leak modules: IndexingDupsLocalizedLocalizedCluster, PerDocData, ShingleInfoPerDocData, SmartphonePerDocData

US20060224583A1 Systems and methods for analyzing a user's web history
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2005-03-31 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US20060224608A1 Systems and methods for combining sets of favorites
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2005-03-31 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US7783631B2 Systems and methods for managing multiple user accounts
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2005-03-31, granted 2010-08-24 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US8438142B2 Suggesting and refining user input based on original user input
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2005-05-04, granted 2013-05-07 · leak modules: NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US7584177B2 Determination of a desired repository
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2005-06-29, granted 2009-09-01 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US8972444B2 Nonstandard locality-based text entry

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2005-06-30, granted 2015-03-03 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US7392244B1 Methods and apparatus for determining equivalent descriptions for an information need

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2005-07-13, granted 2008-06-24 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9118774B2 Dispatch system to remote devices

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2005-07-21, granted 2015-08-25 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores

US7739314B2 Scalable user clustering based on set similarity

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2005-08-15, granted 2010-06-15 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8589391B1 Method and system for generating web site ratings for a user

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2005-09-15, granted 2013-11-19 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8670319B2 Traffic prediction for web sites

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.

filed 2005-09-19, granted 2014-03-11 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8290962B1 Determining the relationship between source code bases

Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2005-09-28, granted 2012-10-16 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, ShingleInfoPerDocData

US7730074B1 Accelerated large scale optimization

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.

filed 2005-11-04, granted 2010-06-01 · leak modules: NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US7499996B1 Systems and methods for detecting a memory condition and providing an alert

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2005-12-01, granted 2009-03-03 · leak modules: CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, TrawlerFetchReplyData

US8375325B2 Customizable, multi-function button

Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2005-12-30, granted 2013-02-12 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality

US7844894B2 Starting landing page experiments
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2006-05-22, granted 2010-11-30 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US7979425B2 Server-side match
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2006-10-25, granted 2011-07-12 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US8533226B1 System and method for verifying and revoking ownership rights with respect to a website in a website indexing system
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2006-12-27, granted 2013-09-10 · leak modules: CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US8966407B2 Expandable homepage modules
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2007-01-17, granted 2015-02-24 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US8015067B2 Deleted account handling for hosted services
Rule: Structure answers: question heading, tight answer, then depth.
filed 2007-02-13, granted 2011-09-06 · leak modules: QualityPreviewSnippetRadishFeatures, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8219678B2 Application verification for hosted services
Rule: Structure answers: question heading, tight answer, then depth.
filed 2007-02-13, granted 2012-07-10 · leak modules: QualityPreviewSnippetRadishFeatures, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8601374B2 Account administration for hosted services
Rule: Structure answers: question heading, tight answer, then depth.
filed 2007-02-13, granted 2013-12-03 · leak modules: QualityPreviewSnippetRadishFeatures, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8065275B2 Systems and methods for cache optimization
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2007-02-15, granted 2011-11-22 · leak modules: IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, ShingleInfoPerDocData

US8301754B1 Statistics overlay
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2007-04-10, granted 2012-10-30 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8799952B2 Virtual channels
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2007-04-24, granted 2014-08-05 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US9497028B1 System and method for remote storage auditing
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2007-05-03, granted 2016-11-15 · leak modules: CompressedQualitySignals, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8255819B2 Web notebook tools
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Build quality links with relevant anchor text. Penguin is still active.
filed 2007-05-10, granted 2012-08-28 · leak modules: AnchorsAnchor, AnchorsAnchorSource, ImageData, ImagePerDocData, IndexingDocjoinerAnchorStatistics, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US8739123B2 Incorporating gadget functionality on webpages
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2007-05-28, granted 2014-05-27 · leak modules: IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8914366B1 Evaluating clustering based on metrics
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2007-06-29, granted 2014-12-16 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US8046355B2 Word decompounder
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2007-09-04, granted 2011-10-25 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8010344B2 Dictionary word and phrase determination
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2007-10-10, granted 2011-08-30 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US9224149B2 External referencing by portable program modules
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2007-10-15, granted 2015-12-29 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US8798988B1 Identifying related terms in different languages
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2007-10-23, granted 2014-08-05 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US8190628B1 Phrase generation
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2007-11-30, granted 2012-05-29 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US8849838B2 Bloom filter for storing file access history
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2008-01-15, granted 2014-09-30 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations

US7836108B1 Clustering by previous representative
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2008-03-31, granted 2010-11-16 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDocjoinerAnchorStatistics, IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, PerDocData

US8661069B1 Predictive-based clustering with representative redirect targets
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2008-03-31, granted 2014-02-25 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate

US8359326B1 Contextual n-gram analysis
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2008-04-02, granted 2013-01-22 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US20120060147A1 Client input method
Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2008-04-08 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData

US7990854B1 Isolation of traffic composition change in performance metrics
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2008-05-30, granted 2011-08-02 · leak modules: CompressedQualitySignals, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US9349134B1 Detecting illegitimate network traffic
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2008-06-02, granted 2016-05-24 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US9256694B2 Named URL entry
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2008-06-27, granted 2016-02-09 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US8010510B1 Method and system for tokenized stream compression
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2008-07-02, granted 2011-08-30 · leak modules: PerDocData, SmartphonePerDocData

US8312032B2 Dictionary suggestions for partial user entries
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2008-07-10, granted 2012-11-13 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US8447748B2 Processing digitally hosted volumes
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2008-07-11, granted 2013-05-21 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US8677018B2 Parallel, side-effect based DNS pre-caching
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2008-08-25, granted 2014-03-18 · leak modules: CrawlerChangerateUrlChangerate, PerDocData, SmartphonePerDocData, TrawlerFetchReplyData

US20120239834A1 Automatic correction of user input using transliteration
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Ensure content comprehensively covers the topic with natural vocabulary.
filed 2008-08-29 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8229732B2 Automatic correction of user input based on dictionary
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Ensure content comprehensively covers the topic with natural vocabulary.
filed 2008-08-29, granted 2012-07-24 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8667417B1 Methods and systems for pop-up suppression
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2008-09-01, granted 2014-03-04 · leak modules: CompressedQualitySignals, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8826145B1 Unified web and application framework
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2008-11-11, granted 2014-09-02 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US20090144272A1 Rating raters
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2008-12-02 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefEntityJoin

US8131751B1 Algorithms for selecting subsequences
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2008-12-03, granted 2012-03-06 · leak modules: PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, SmartphonePerDocData

US8121991B1 Identifying transient paths within websites
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2008-12-19, granted 2012-02-21 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8037073B1 Detection of bounce pad sites
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2008-12-29, granted 2011-10-11 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDocjoinerAnchorSpamInfo, IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US8874565B1 Detection of proxy pad sites
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2008-12-29, granted 2014-10-28 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDocjoinerAnchorSpamInfo, IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9225794B2 Adaptive DNS pre-resolution
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2009-03-31, granted 2015-12-29 · leak modules: CrawlerChangerateUrlChangerate, PerDocData, SmartphonePerDocData, TrawlerFetchReplyData

US8725793B2 Architectural pattern for persistent web application design
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2009-04-03, granted 2014-05-13 · leak modules: IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20100268757A1 Pseudo Pipelining of Client Requests
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2009-04-17 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8255523B1 Server side disambiguation of ambiguous statistics
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2009-04-24, granted 2012-08-28 · leak modules: PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, SmartphonePerDocData

US8954955B2 Standard commands for native commands
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Optimize video for watch time and retention.
filed 2009-06-16, granted 2015-02-10 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData, VideoPerDocData

US8756212B2 Techniques for web site integration
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2009-07-06, granted 2014-06-17 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US9086775B1 Minimizing software based keyboard
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2009-07-10, granted 2015-07-21 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US8266133B2 Determination of a desired repository
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2009-07-28, granted 2012-09-11 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US8825646B1 Scalable system for determining short paths within web link network
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active.
filed 2009-08-07, granted 2014-09-02 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster, PerDocData

US8832548B2 System and method for selective control of zoom adjustments in an electronic publication
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Optimize mobile experience. Mobile-specific signals are tracked separately. Write for natural spoken language and conversational queries.
filed 2009-09-21, granted 2014-09-09 · leak modules: NlpSaftDocument, PerDocData, RepositoryWebrefEntityJoin, SmartphonePerDocData

US20140372119A1 Compounded Text Segmentation
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2009-09-28 · leak modules: NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US20110078602A1 Client Utility Interaction Analysis
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2009-09-30 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8543907B1 Context-sensitive optimization level selection
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2009-10-16, granted 2013-09-24 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US20110106819A1 Identifying a group of related instances
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2009-10-29 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US8374983B1 Distributed object classification
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2009-11-23, granted 2013-02-12 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8656020B1 Delta compression of files in web applications
Rule: Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2010-01-21, granted 2014-02-18 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8180778B1 Generating action trails from web history
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2010-02-05, granted 2012-05-15 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US8473857B1 Link annotation for keyboard navigation
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Write for natural spoken language and conversational queries.
filed 2010-02-12, granted 2013-06-25 · leak modules: NlpSaftDocument, PerDocData, SmartphonePerDocData

US8064700B2 Method and system for character recognition
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2010-03-10, granted 2011-11-22 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8930351B1 Grouping of users
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2010-03-31, granted 2015-01-06 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US9602444B2 Participant suggestion system
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data.
filed 2010-04-29, granted 2017-03-21 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US7962529B1 Scalable user clustering based on set similarity
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2010-05-06, granted 2011-06-14 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20110288931A1 Microsite models
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2010-05-20 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8307005B1 Determining reachability
Rule: Your page's ranking is influenced by the quality and engagement of pages you link to. Link to high-quality, high-engagement content—pages with high CTR and long median access time boost your reachability score. This works multi-hop: the quality of pages your linked pages link to also matters. Avoid linking to low-engagement or dead-end pages. The leak's ToolBarPerDocData (which tracks user interactions) and NavboostCraps click signals confirm reachability scoring is live. Internal linking to your own high-performing pages can boost your primary content.
filed 2010-06-30, granted 2012-11-06 · leak modules: QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, CompressedQualitySignals, PerDocData, ToolBarPerDocData, ImagePerDocData, VideoPerDocData, SdrPageAnchorsDocInfo

US8433704B2 Local item extraction
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2010-09-23, granted 2013-04-30 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8583808B1 Automatic generation of rewrite rules for URLs
Rule: Google automatically generates URL rewrite rules to strip irrelevant parameters. Proactively use canonical tags to speed up this process. Audit your URL parameter space - any parameter that doesn't change content (tracking, session, display options) should be canonicalized away. Use consistent URL structures across your site so Google's rewrite rules can apply site-wide.
filed 2010-09-23, granted 2013-11-12 · leak modules: canonicalization, crawler, indexing, dedup

US8359313B2 Extensible custom variables for tracking user traffic
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2010-10-19, granted 2013-01-22 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US9870539B2 Establishing communication in a rich media notice board
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2010-12-03, granted 2018-01-16 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations

US20110153599A1 Meta-Web
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2010-12-15 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US8473841B2 Independent visual element configuration
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-03-10, granted 2013-06-25 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8548851B2 Conversion path performance measures and reports
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2011-03-23, granted 2013-10-01 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US9245279B2 Conversion path performance measures and reports
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2011-03-23, granted 2016-01-26 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US8645364B2 Providing definitions that are sensitive to the context of a text
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-03-31, granted 2014-02-04 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8185561B1 Scalable user clustering based on set similarity
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-04-01, granted 2012-05-22 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData

US20120260185A1 Path length selector
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-04-11 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures

US9201955B1 Unambiguous noun identification
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-04-15, granted 2015-12-01 · leak modules: CompressedQualitySignals, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US8732569B2 Predicting user navigation events
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Build content for diverse user contexts and search histories.
filed 2011-05-04, granted 2014-05-20 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9141346B2 Layout management in a rapid application development tool
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-05-12, granted 2015-09-22 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US8577914B2 APIS discovery service
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships.
filed 2011-05-18, granted 2013-11-05 · leak modules: CrawlerChangerateUrlChangerate, PerDocData, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData, TrawlerFetchReplyData

US8600803B1 Incentivizing behavior to address pricing, tax, and currency issues in an online marketplace for digital goods
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-05-18, granted 2013-12-03 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8601468B1 Web-based applications using web extents
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-05-18, granted 2013-12-03 · leak modules: PerDocData, SmartphonePerDocData

US8725585B1 Automatic vetting of web applications to be listed in a marketplace for web applications
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-05-18, granted 2014-05-13 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US8849957B1 Installable web applications
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-05-18, granted 2014-09-30 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US8881055B1 HTML pop-up control
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-05-18, granted 2014-11-04 · leak modules: PerDocData, SmartphonePerDocData

US9477534B2 Inter-extension messaging
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Earn endorsements from trusted sources. Trust is computed and transitive.
filed 2011-05-18, granted 2016-10-25 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, SmartphonePerDocData

US9015243B1 Automated metadata updates
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships. Optimize video for watch time and retention.
filed 2011-06-08, granted 2015-04-21 · leak modules: PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData, VideoPerDocData

US8626791B1 Predictive model caching
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2011-06-14, granted 2014-01-07 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US20120331389A1 Download area management
Rule: Monitor this patent for ranking implications.
filed 2011-06-22

US8533260B1 Reduction of perceived DNS lookup latency

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.

filed 2011-06-30, granted 2013-09-10 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9277275B1 System and method for associating individual household members with television programs viewed

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2011-06-30, granted 2016-03-01 · leak modules: PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData

US9569788B1 Systems and methods for associating individual household members with web sites visited

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2011-06-30, granted 2017-02-14 · leak modules: PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData

US8601119B1 Providing navigation statistics

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2011-07-01, granted 2013-12-03 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefEntityJoin

US8650139B2 Predicting user navigation events

Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Build content for diverse user contexts and search histories.

filed 2011-07-01, granted 2014-02-11 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US8612459B1 Lead generation system and methods

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2011-07-13, granted 2013-12-17 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US8566696B1 Predicting user navigation events

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.

filed 2011-07-14, granted 2013-10-22 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US8655907B2 Multi-channel conversion path position reporting

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2011-07-18, granted 2014-02-18 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US8412517B2 Dictionary word and phrase determination

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2011-07-26, granted 2013-04-02 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US8788502B1 Annotating articles

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2011-07-26, granted 2014-07-22 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8959427B1 System and method for JavaScript based HTML website layouts
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships.
filed 2011-08-05, granted 2015-02-17 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US20130047100A1 Link Disambiguation For Touch Screens
Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2011-08-17 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8190609B1 Identifying alternative products
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-08-18, granted 2012-05-29 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9176944B1 Selectively processing user input
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2011-08-23, granted 2015-11-03 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality

US10133737B2 Conversion of input text strings
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-08-26, granted 2018-11-20 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US20130054632A1 Caching Input Suggestions
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-08-30 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9129033B1 Caching efficiency using a metadata cache
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Earn promotion through authoritative content and entity association. Use JSON-LD structured data for entity types and relationships.
filed 2011-08-31, granted 2015-09-08 · leak modules: CompressedQualitySignals, PerDocData, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US8521746B1 Detection of bounce pad sites
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2011-09-07, granted 2013-08-27 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDocjoinerAnchorSpamInfo, IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US8380693B1 System and method for automatically identifying classified websites
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-09-08, granted 2013-02-19 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US8285781B1 Reduction of perceived DNS lookup latency
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2011-09-12, granted 2012-10-09 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8812737B2 Context-specific unicode characters in shortened URLs
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-09-19, granted 2014-08-19 · leak modules: PerDocData, SmartphonePerDocData

US8195663B1 Identifying alternative products
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-09-26, granted 2012-06-05 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8307278B1 Tiling mechanism to combine web services
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-09-26, granted 2012-11-06 · leak modules: PerDocData, SmartphonePerDocData

US8335722B2 Local shopping and inventory
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-09-26, granted 2012-12-18 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8504613B1 Reduction of perceived DNS lookup latency
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2011-09-26, granted 2013-08-06 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8255280B1 Automatic vetting of web applications to be listed in a marketplace for web applications
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-09-27, granted 2012-08-28 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US8255494B1 Installable web applications
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-09-27, granted 2012-08-28 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US8266544B1 HTML pop-up control
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-09-27, granted 2012-09-11 · leak modules: PerDocData, SmartphonePerDocData

US8234348B1 Determining if an application is cached
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2011-09-30, granted 2012-07-31 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData, SnippetExtraInfo

US8321510B1 Automated metadata updates
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships. Optimize video for watch time and retention.
filed 2011-09-30, granted 2012-11-27 · leak modules: PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData, VideoPerDocData

US8370143B1 Selectively processing user input
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2011-09-30, granted 2013-02-05 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality

US8521517B2 Providing definitions that are sensitive to the context of a text
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-09-30, granted 2013-08-27 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8583648B1 Merging semantically similar clusters based on cluster labels
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-09-30, granted 2013-11-12 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, ShingleInfoPerDocData, SmartphonePerDocData

US9507801B2 Enforcing category diversity
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-10-04, granted 2016-11-29 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate

US9021348B1 Composition of templates using transclusion that guarantee a valid HTML document
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth. Use JSON-LD structured data for entity types and relationships.
filed 2011-10-05, granted 2015-04-28 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US9058329B1 Deriving associations between assets
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-10-06, granted 2015-06-16 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US8566095B2 Segmenting words using scaled probabilities
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2011-10-11, granted 2013-10-22 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8413046B1 System and method for optimizing rich web applications
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-10-12, granted 2013-04-02 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US8471871B1 Authoritative text size measuring
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2011-10-17, granted 2013-06-25 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8583636B1 Systems and methods for determining a quality of provided items
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-10-18, granted 2013-11-12 · leak modules: CompressedQualitySignals, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo

US9652529B1 Methods and systems for augmenting a token lexicon
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2011-10-19, granted 2017-05-16 · leak modules: CrawlerChangerateUrlChangerate, PerDocData, QualityPreviewSnippetRadishFeatures, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo, TrawlerFetchReplyData

US8661060B2 Editing a network of interconnected concepts
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Write for intent and meaning. Neural matching connects related concepts.
filed 2011-10-20, granted 2014-02-25 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, RepositoryAnnotationsGeoTopicality

US8452814B1 Gathering context in action to support in-context localization
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-10-24, granted 2013-05-28 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US8862602B1 Systems and methods for improved readability of URLs
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-10-25, granted 2014-10-14 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US9129009B2 Related links
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-11-02, granted 2015-09-08 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo

US9083761B1 Reduced latency for subresource transfer
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships. Optimize video for watch time and retention.
filed 2011-11-09, granted 2015-07-14 · leak modules: PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData, VideoPerDocData

US8788507B1 Watermarking of structured results and watermark generation
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-11-15, granted 2014-07-22 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US8812517B1 Watermarking of structured results and watermark detection
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-11-15, granted 2014-08-19 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US8402367B1 Smart reload pages
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships.
filed 2011-11-16, granted 2013-03-19 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US9015505B2 Delayed shut down of computer
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-11-18, granted 2015-04-21 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US10007736B2 Methods performed by electronic devices that facilitate navigating a webpage
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-12-01, granted 2018-06-26 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US20150161255A1 Systems and Methods for Deriving and Using an Interaction Profile
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-12-01 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US9223868B2 Deriving and using interaction profiles
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-12-01, granted 2015-12-29 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US8612851B2 Edition grid layout
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-12-06, granted 2013-12-17 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData

US8521128B1 Method, system, and computer program product for obtaining crowd-sourced location information
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-12-09, granted 2013-08-27 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US8626739B2 Methods and systems for processing media files
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-12-12, granted 2014-01-07 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures

US8600973B1 Removing substitution rules
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-01-03, granted 2013-12-03 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US9529487B1 Method of providing fast switching to web apps
Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-01-10, granted 2016-12-27 · leak modules: ImageData, ImagePerDocData, PerDocData, SmartphonePerDocData

US20130179770A1 Creating optimal interactive elements
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-01-11 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9055120B1 Device capability filtering
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-01-30, granted 2015-06-09 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8458192B1 System and method for determining topic interest
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-01-31, granted 2013-06-04 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US8458193B1 System and method for determining active topics
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-01-31, granted 2013-06-04 · leak modules: CompressedQualitySignals, PerDocData, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US8458195B1 System and method for determining similar users
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-01-31, granted 2013-06-04 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata

US8458196B1 System and method for determining topic authority
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-01-31, granted 2013-06-04 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US9026900B1 Invisible overlay for dynamic annotation
Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2012-02-10, granted 2015-05-05 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData, SnippetExtraInfo

US9065859B1 Server side disambiguation of ambiguous statistics
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2012-02-13, granted 2015-06-23 · leak modules: PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US9117195B2 Synchronized consumption modes for e-books
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-02-13, granted 2015-08-25 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US20180285148A1 Automate control of media playback based on visibility
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2012-02-17 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8838613B1 Identifying trends from micro-posts
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2012-02-17, granted 2014-09-16 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US9075517B2 Web input through drag and drop
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-02-21, granted 2015-07-07 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US8527516B1 Identifying similar digital text volumes
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2012-02-24, granted 2013-09-03 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US8745027B1 Jslayout cascading style sheets optimization
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth. Use JSON-LD structured data for entity types and relationships.
filed 2012-02-28, granted 2014-06-03 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US20130238970A1 Uniquely Identifying Script Files
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships.
filed 2012-03-07 · leak modules: PerDocData, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefEntityJoin, RichsnippetsPageMap, SmartphonePerDocData

US9053199B2 Uniquely identifying script files by appending a unique identifier to a URL
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships.
filed 2012-03-07, granted 2015-06-09 · leak modules: PerDocData, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefEntityJoin, RichsnippetsPageMap, SmartphonePerDocData

US9274913B2 Event pages for web applications and extensions
Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-03-08, granted 2016-03-01 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData

US9058392B1 Client state result de-duping
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-03-22, granted 2015-06-16 · leak modules: IndexingDupsLocalizedLocalizedCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US8769400B1 Accelerating view transitions
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2012-03-26, granted 2014-07-01 · leak modules: PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8682907B1 Evaluation of substitute terms

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.

filed 2012-03-30, granted 2014-03-25 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9031951B1 Associating interest and disinterest keywords with similar and dissimilar users

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.

filed 2012-04-02, granted 2015-05-12 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8504562B1 Evaluation of substitute terms

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.

filed 2012-04-03, granted 2013-08-06 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8898148B1 Targeting to physical environment

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2012-04-10, granted 2014-11-25 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US8965875B1 Removing substitution rules based on user interactions

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2012-04-10, granted 2015-02-24 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US20160210667A1 Greedy Contextualized Rewrites

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.

filed 2012-04-20 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US9053185B1 Generating a representative model for a plurality of models identified by similar feature data

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2012-04-30, granted 2015-06-09 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8554729B2 System and method for synchronization of actions in the background of an application

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Avoid AI-stitched content. GibberishScore uses language model scoring.

filed 2012-05-03, granted 2013-10-08 · leak modules: IndexingDocjoinerAnchorSpamInfo, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SpamBrainData, SpamBrainScore

US9418335B1 Method and apparatus for selecting links to include in a probabilistic generative model for text

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2012-05-04, granted 2016-08-16 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, TrawlerFetchReplyData

US8713172B1 Identification of websites that break out of frames

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2012-05-10, granted 2014-04-29 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US8560549B1 Generating action trails from web history
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-05-14, granted 2013-10-15 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US8700625B1 Identifying alternative products
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-05-16, granted 2014-04-15 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8788506B1 Methods and systems for estimating a count of unique items
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-05-21, granted 2014-07-22 · leak modules: PerDocData, SmartphonePerDocData

US20150207800A1 Systems and methods for enabling access to a web application
Rule: Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-05-25 · leak modules: PerDocData, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US8463796B1 System and method for providing noted items
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-05-25, granted 2013-06-11 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US8538979B1 Generating phrase candidates from text string entries
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-05-25, granted 2013-09-17 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US9201876B1 Contextual weighting of words in a word grouping
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-05-29, granted 2015-12-01 · leak modules: NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US20150205866A1 Phrase restricted substitute terms
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2012-05-31 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20150234798A1 System and method for changing a web ui application appearance based on state through css selector cascading
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-06-01 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US8626763B1 Server-side suggestion of preload operations
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-06-04, granted 2014-01-07 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9971829B1 Inferring membership in a group
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2012-06-07, granted 2018-05-15 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US8856131B1 Computer networked selecting of consumer interests
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-06-14, granted 2014-10-07 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9600458B2 Overriding style rules for styling web elements
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2012-06-14, granted 2017-03-21 · leak modules: CompressedQualitySignals, PerDocData

US9674247B1 Generating manifest files for digital media items
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2012-06-21, granted 2017-06-06 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US9110984B1 Methods and systems for constructing a taxonomy based on hierarchical clustering
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Optimize video for watch time and retention.
filed 2012-06-22, granted 2015-08-18 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, VideoPerDocData

US20150254475A1 Private browsing mode access for developers
Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2012-06-25 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData, SnippetExtraInfo

US9418243B2 Invoking a private browsing mode by selection of a visual control element within a browser tab
Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2012-06-25, granted 2016-08-16 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData, SnippetExtraInfo

US8538972B1 Context-dependent similarity measurements
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-06-26, granted 2013-09-17 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9342615B2 Reducing redirects
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-06-27, granted 2016-05-17 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US20140019424A1 Identifier validation and debugging
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-07-11 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US8874589B1 Adjust similar users identification based on performance feedback
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2012-07-16, granted 2014-10-28 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US9245014B2 Clustering of feedback reports
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2012-07-19, granted 2016-01-26 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo

US8386237B2 Automatic correction of user input based on dictionary
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Ensure content comprehensively covers the topic with natural vocabulary.
filed 2012-07-23, granted 2013-02-26 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8713047B2 System and method for providing definitions
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-07-24, granted 2014-04-29 · leak modules: IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, ShingleInfoPerDocData, SmartphonePerDocData, SnippetExtraInfo

US8826298B2 Method and apparatus that enables a web-based client-server application to be used offline
Rule: Use JSON-LD structured data for entity types and relationships.
filed 2012-07-24, granted 2014-09-02 · leak modules: RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US9164968B2 Techniques for server-side determination of font subsets
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-07-26, granted 2015-10-20 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US20150193395A1 Predictive link pre-loading
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-07-30 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US8744995B1 Alias disambiguation
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-07-30, granted 2014-06-03 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8990172B1 Key management in smart pixel based tracking
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-08-02, granted 2015-03-24 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US20140047360A1 Background application page architecture for web applications
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-08-09 · leak modules: PerDocData, SmartphonePerDocData

US8849804B1 Distributing interest categories within a hierarchical classification
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-08-10, granted 2014-09-30 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US8832288B1 Transitions between remotely cached and live versions of a webpage
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-08-16, granted 2014-09-09 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8903946B1 Reduction in redirect navigation latency via speculative preconnection
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-08-29, granted 2014-12-02 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8843514B1 Identifier matching exchange
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-08-31, granted 2014-09-23 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo

US9047356B2 Synchronizing multiple reading positions in electronic books
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth. Use JSON-LD structured data for entity types and relationships.
filed 2012-09-05, granted 2015-06-02 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8768917B1 Method and apparatus for automatically identifying compounds
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-09-13, granted 2014-07-01 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, TrawlerFetchReplyData

US8838649B1 Determining reachability
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-09-13, granted 2014-09-16 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US9020924B2 Suggesting and refining user input based on original user input
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-09-13, granted 2015-04-28 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate

US20140081973A1 Spike classification
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-09-14 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8606912B1 Statistics overlay
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-09-14, granted 2013-12-10 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8661051B1 Contextual n-gram analysis
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2012-09-14, granted 2014-02-25 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US8825637B1 Recording user actions
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-09-14, granted 2014-09-02 · leak modules: ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures

US8965874B1 Dynamic aggregation of users
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Ensure NAP consistency and local relevance. Distance affects ranking. Avoid AI-stitched content. GibberishScore uses language model scoring.
filed 2012-09-14, granted 2015-02-24 · leak modules: IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, RepositoryAnnotationsGeoTopicality, SpamBrainData

US8959151B1 Establishing per-page multi-party communication sessions
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-10-04, granted 2015-02-17 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US20150212695A1 Suggested action feedback
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.
filed 2012-10-05 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US9069549B2 Machine processor
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-10-10, granted 2015-06-30 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US9721032B2 Contextual URL suggestions
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-10-18, granted 2017-08-01 · leak modules: PerDocData, SmartphonePerDocData

US8856141B1 Providing posts from an extended network
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-10-23, granted 2014-10-07 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US9460218B2 Indicating location status
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2012-10-24, granted 2016-10-04 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US8990389B2 Using a router based web meter in a mixed mode configuration
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2012-10-25, granted 2015-03-24 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefEntityJoin

US9170834B2 Metadata-based virtual machine configuration
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships.
filed 2012-10-31, granted 2015-10-27 · leak modules: PerDocData, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US20130326040A1 System and method for disambiguating cookies
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-12-06 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9203908B2 System an method for disambiguating cookies
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-12-06, granted 2015-12-01 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9762698B2 Computer application promotion
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-12-14, granted 2017-09-12 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US20170242898A1 Profiles in suggest
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-12-27 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9665547B1 Facilitating webpage navigation
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-12-27, granted 2017-05-30 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality

US9208232B1 Generating synthetic descriptive text
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-12-31, granted 2015-12-08 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorStatistics, NlpSaftDocument, PerDocData

US9152247B2 Computing device with force-triggered non-visual responses
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-01-02, granted 2015-10-06 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US8930399B1 Determining word boundary likelihoods in potentially incomplete text
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2013-01-11, granted 2015-01-06 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US8949228B2 Identification of new sources for topics
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-01-15, granted 2015-02-03 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US20150193061A1 User's computing experience based on the user's computing activity
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Optimize mobile experience. Mobile-specific signals are tracked separately. Optimize video for watch time and retention.
filed 2013-01-29 · leak modules: PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, VideoPerDocData

US8655970B1 Automatic entertainment caching for impending travel
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-01-29, granted 2014-02-18 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US8682882B2 System and method for automatically identifying classified websites
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-01-29, granted 2014-03-25 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US8966179B1 Volatile memory storage for private web browsing
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-02-04, granted 2015-02-24 · leak modules: PerDocData, SmartphonePerDocData

US8656265B1 Low-latency transition into embedded web view
Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-02-05, granted 2014-02-18 · leak modules: ImageData, ImagePerDocData, PerDocData, SmartphonePerDocData

US20160239490A1 Using Alternate Words As an Indication of Word Sense
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-02-08 · leak modules: NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9495455B2 Programming a dynamic digital media queue
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-02-11, granted 2016-11-15 · leak modules: CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US8943099B1 Determining codomain relationship measures
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-02-22, granted 2015-01-27 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US9323721B1 Quotation identification
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-02-27, granted 2016-04-26 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo

US9838444B2 Systems and methods for dynamically providing fonts based on language settings
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2013-03-04, granted 2017-12-05 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9477644B1 Identifying referral pages based on recorded URL requests
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-03-05, granted 2016-10-25 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US20130297776A1 Techniques for delay processing to support offline hits
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-03-15 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US20130297777A1 Server-side configuration driven sessionization to provide consistent business logic
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-03-15 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US20130297778A1 System for centralized analytics tracking via server-side sessionization
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-03-15 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US9268802B2 System and method for end-to-end exposure of exported representations of native data types to third-party applications
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships. Write for natural spoken language and conversational queries.
filed 2013-03-15, granted 2016-02-23 · leak modules: NlpSaftDocument, PerDocData, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US9280592B1 Zombie detector and handler mechanism for accounts, apps, and hardware devices
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-03-15, granted 2016-03-08 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US9372927B1 Original authorship identification of electronic publications
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships.
filed 2013-03-15, granted 2016-06-21 · leak modules: PerDocData, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefEntityJoin, RichsnippetsPageMap, SmartphonePerDocData

US9613101B2 Promoting an original version of a copyrighted media item over an authorized copied version of the copyrighted media item in a search query
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-03-15, granted 2017-04-04 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US9628414B1 User state based engagement
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-03-15, granted 2017-04-18 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9826018B1 Facilitating secure web browsing on untrusted networks
Rule: Earn endorsements from trusted sources. Trust is computed and transitive.
filed 2013-03-15, granted 2017-11-21 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData

US9208170B1 Classifying natural mapping features
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use descriptive alt text and surrounding context for images.
filed 2013-03-29, granted 2015-12-08 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US9454544B1 Selecting destinations or points of interest
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use descriptive alt text and surrounding context for images.
filed 2013-03-29, granted 2016-09-27 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US9268839B2 Methods and systems for editing a network of interconnected concepts
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-04-12, granted 2016-02-23 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US8996537B1 System and method for providing noted items
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-04-15, granted 2015-03-31 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9807099B2 Utility portals for managing demand-response events
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-04-19, granted 2017-10-31 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US9239823B1 Identifying common co-occurring elements in lists
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2013-05-14, granted 2016-01-19 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9830401B2 Automatically selecting optimization filters for optimizing webpages
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-05-16, granted 2017-11-28 · leak modules: CompressedQualitySignals, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9516092B2 Precompiling locally-stored instructions for a web application
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-05-31, granted 2016-12-06 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9317415B2 Application analytics reporting
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-06-03, granted 2016-04-19 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US9330195B1 Inducing command inputs from property sequences
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data.
filed 2013-06-25, granted 2016-05-03 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US20150006499A1 Systems, Methods, and Computer-Readable Media for Locating Real-World Objects Using Computer-Implemented Searching

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2013-06-28 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9134978B1 Context-sensitive optimization level selection

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2013-07-02, granted 2015-09-15 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US9846841B1 Predicting object identity using an ensemble of predictors

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.

filed 2013-07-08, granted 2017-12-19 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8812996B1 Methods and apparatus for processing application windows

Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Avoid AI-stitched content. GibberishScore uses language model scoring.

filed 2013-07-19, granted 2014-08-19 · leak modules: IndexingDocjoinerAnchorSpamInfo, PerDocData, SmartphonePerDocData, SpamBrainData, SpamBrainScore

US10789259B2 Broad match control

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2013-07-22, granted 2020-09-29 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations

US9558234B1 Automatic metadata identification

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2013-07-22, granted 2017-01-31 · leak modules: NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefEntityJoin, RichsnippetsPageMap, SmartphonePerDocData

US9275022B2 Mechanism for setting ascent and baseline for HTML elements

Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2013-07-31, granted 2016-03-01 · leak modules: ImageData, ImagePerDocData, PerDocData, SmartphonePerDocData

US8903899B1 Reduction of perceived DNS lookup latency

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.

filed 2013-08-02, granted 2014-12-02 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9348897B2 Method and system for providing scheduling suggestions

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2013-08-07, granted 2016-05-24 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US9384303B2 Evaluation of substitution contexts

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2013-08-07, granted 2016-07-05 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo

US9483581B2 Evaluation of substitution contexts
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-08-07, granted 2016-11-01 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo

US9171086B1 Website duration performance based on category durations
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2013-08-12, granted 2015-10-27 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US9015117B2 System and method for synchronization of actions in the background of an application
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Avoid AI-stitched content. GibberishScore uses language model scoring.
filed 2013-09-06, granted 2015-04-21 · leak modules: IndexingDocjoinerAnchorSpamInfo, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SpamBrainData, SpamBrainScore

US9621636B1 Distributed processing system throttling
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-09-10, granted 2017-04-11 · leak modules: IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US9092338B1 Multi-level caching event lookup
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-09-26, granted 2015-07-28 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US9619117B2 Multi-channel conversion path position reporting
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-09-26, granted 2017-04-11 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US20190171753A1 Diversification of items for representation to user
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-09-27 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10007726B2 Resolving a host expression to an internet protocol address
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-09-30, granted 2018-06-26 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9466029B1 Demographic inference calibration
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-10-15, granted 2016-10-11 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US9292341B2 RPC acceleration based on previously memorized flows
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships.
filed 2013-10-16, granted 2016-03-22 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US9330074B2 Style sheet speculative preloading
Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships.
filed 2013-10-31, granted 2016-05-03 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US9053078B1 Statistics overlay
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-11-04, granted 2015-06-09 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9336301B2 Merging semantically similar clusters based on cluster labels
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-11-08, granted 2016-05-10 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, ShingleInfoPerDocData, SmartphonePerDocData

US20180052807A1 Endorsement smearing among related webpages
Rule: Social endorsements (likes, shares, +1s) are consolidated across canonical URL variants - ensure your canonical tags are correct so endorsements aren't fragmented across duplicate URLs. Build two-way relationships between your social profiles and canonical webpages. Social signals from related pages contribute to your page's total endorsement count.
filed 2013-11-14 · leak modules: link_analysis, canonicalization, social_signals, ranking, entity

US9910830B2 Edition grid layout
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-11-15, granted 2018-03-06 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData

US9384258B1 Identifying top fans
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2013-11-19, granted 2016-07-05 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefDocumentMetadata

US9037976B2 Account administration for hosted services
Rule: Structure answers: question heading, tight answer, then depth.
filed 2013-12-02, granted 2015-05-19 · leak modules: QualityPreviewSnippetRadishFeatures, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US20150161093A1 Event payload type checking
Rule: Write in natural vocabulary. ML models understand context and fluff. Use JSON-LD structured data for entity types and relationships.
filed 2013-12-09 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US9378275B1 Lead generation system and methods
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-12-11, granted 2016-06-28 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US9311408B2 Methods and systems for processing media files
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-12-12, granted 2016-04-12 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures

US9298831B1 Approximating a user location
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-12-13, granted 2016-03-29 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20150178289A1 Identifying Semantically-Meaningful Text Selections
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-12-20 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures

US9607009B2 Automatically branding topics using color
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-12-20, granted 2017-03-28 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US9846842B2 Predicting user navigation events
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Build content for diverse user contexts and search histories.
filed 2013-12-20, granted 2017-12-19 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9696874B2 Providing media to a user based on a triggering event
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.
filed 2013-12-31, granted 2017-07-04 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US9378276B1 Systems and methods for generating navigation filters
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2014-01-03, granted 2016-06-28 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals

US9443197B1 Predicting user navigation events
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-01-10, granted 2016-09-13 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US10169448B2 Serving font glyphs
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2014-01-14, granted 2019-01-01 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US9449052B1 Trend based distribution parameter suggestion
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-01-21, granted 2016-09-20 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9058562B2 Traffic prediction for web sites
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2014-01-30, granted 2015-06-16 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9064026B2 Parallel, side-effect based DNS pre-caching
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-02-07, granted 2015-06-23 · leak modules: CrawlerChangerateUrlChangerate, PerDocData, SmartphonePerDocData, TrawlerFetchReplyData

US9489447B2 Scalable current leaderboard using delta-encoded updates
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-02-21, granted 2016-11-08 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US10068270B1 Automatic vetting of web applications to be listed in a marketplace for web applications

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2014-04-03, granted 2018-09-04 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US9613009B2 Predicting user navigation events

Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Build content for diverse user contexts and search histories.

filed 2014-04-04, granted 2017-04-04 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9531839B1 Systems and methods for request isolation protection

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2014-04-10, granted 2016-12-27 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US9692701B1 Throttling client initiated traffic

Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2014-04-10, granted 2017-06-27 · leak modules: PerDocData, SmartphonePerDocData

US9665617B1 Methods and systems for generating a stable identifier for nodes likely including primary content within an information resource

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.

filed 2014-04-16, granted 2017-05-30 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US10817884B2 Building topic-oriented audiences

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2014-05-08, granted 2020-10-27 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US20170140022A1 Identifying an assumption about a user, and determining a veracity of the assumption

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2014-05-28 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US9928223B1 Methods for prerendering and methods for managing and configuring prerendering operations

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2014-06-16, granted 2018-03-27 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US9819744B1 Multi-modal communication

Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Earn endorsements from trusted sources. Trust is computed and transitive. Write for natural spoken language and conversational queries.

filed 2014-06-23, granted 2017-11-14 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityNsrNsrData, SmartphonePerDocData

US9015618B2 Methods and apparatus for processing application windows

Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Avoid AI-stitched content. GibberishScore uses language model scoring.

filed 2014-07-15, granted 2015-04-21 · leak modules: IndexingDocjoinerAnchorSpamInfo, PerDocData, SmartphonePerDocData, SpamBrainData, SpamBrainScore

US9727647B1 Annotating articles

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2014-07-18, granted 2017-08-08 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20160028798A1 Peer Network Membership Management
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Earn endorsements from trusted sources. Trust is computed and transitive.
filed 2014-07-23 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US10162806B1 Unified web and application framework
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-07-28, granted 2018-12-25 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US9307067B2 Contextually updating wireless device firmware
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-07-30, granted 2016-04-05 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20160292149A1 Word sense disambiguation using hypernyms
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-08-02 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality

US9880978B2 System and method for selective control of zoom adjustments in an electronic publication
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2014-08-04, granted 2018-01-30 · leak modules: NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryWebrefEntityJoin, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US9536011B1 Applying string matching algorithms to finding ad unit availability interval
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-08-18, granted 2017-01-03 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10567543B1 Installable web applications
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-08-25, granted 2020-02-18 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9641590B2 Resuming session states
Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-08-27, granted 2017-05-02 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData

US9400849B1 Scalable system for determining short paths within web link network
Rule: Confirms that Google's link graph analysis operates at massive scale with parallel computation. Your site's position in the link graph relative to trusted seeds is computed across distributed systems. Focus on being linked from sites that are themselves well-connected to the trusted seed set - the distance metric is computed globally.
filed 2014-08-29, granted 2016-07-26 · leak modules: link_analysis, pagerank, ranking, link_graph, infrastructure

US10007717B2 Clustering communications based on classification
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-09-18, granted 2018-06-26 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9424340B1 Detection of proxy pad sites
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2014-10-22, granted 2016-08-23 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDocjoinerAnchorSpamInfo, IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9729654B1 Reduction in redirect navigation latency via speculative preconnection
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-10-31, granted 2017-08-08 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9239888B1 Determining word boundary likelihoods in potentially incomplete text
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2014-12-04, granted 2016-01-19 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US10216837B1 Selecting pattern matching segments for electronic communication clustering
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-12-29, granted 2019-02-26 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US9363310B2 Standard commands for native commands
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Optimize video for watch time and retention.
filed 2015-01-07, granted 2016-06-07 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData, VideoPerDocData

US20160205173A1 Digital magazine distribution using feeds
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Structure answers: question heading, tight answer, then depth.
filed 2015-01-14 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SnippetExtraInfo

US20160132783A1 User Modelling by Domain Adaptation
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2015-02-05 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US10445325B2 Proximity detection
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-02-13, granted 2019-10-15 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US20160239155A1 Adaptive media
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2015-02-18 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US9454207B2 Delayed shut down of computer
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-03-23, granted 2016-09-27 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9411906B2 Suggesting and refining user input based on original user input
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-04-14, granted 2016-08-09 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate

US10102216B2 System for associating related digital assets
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Earn endorsements from trusted sources. Trust is computed and transitive.
filed 2015-04-21, granted 2018-10-16 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, SmartphonePerDocData

US9787557B2 Determining semantic place names from location reports
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-04-28, granted 2017-10-10 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US10042684B2 Software development kit platform
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-04-29, granted 2018-08-07 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US9807155B1 Device capability filtering
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-05-08, granted 2017-10-31 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US20190018893A1 Determining tone differential of a segment
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-05-11 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US9294588B2 Account administration for hosted services
Rule: Structure answers: question heading, tight answer, then depth.
filed 2015-05-14, granted 2016-03-22 · leak modules: QualityPreviewSnippetRadishFeatures, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US9621670B2 Parallel, side-effect based DNS pre-caching
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2015-05-21, granted 2017-04-11 · leak modules: CrawlerChangerateUrlChangerate, PerDocData, SmartphonePerDocData, TrawlerFetchReplyData

US9652458B1 Deriving associations between assets
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2015-05-27, granted 2017-05-16 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US9811324B2 Code caching system
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2015-05-29, granted 2017-11-07 · leak modules: PerDocData, SmartphonePerDocData

US10089579B1 Predicting user navigation events
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2015-06-05, granted 2018-10-02 · leak modules: PerDocData, SmartphonePerDocData

US9507873B1 Reduced latency for subresource transfer
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships. Optimize video for watch time and retention.
filed 2015-06-11, granted 2016-11-29 · leak modules: PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData, VideoPerDocData

US9933938B1 Minimizing software based keyboard
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2015-06-16, granted 2018-04-03 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US9998475B2 Streamlined utility portals for managing demand-response events
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-06-17, granted 2018-06-12 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US9460230B1 Method and system for visual styling of visual elements for applications
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships.
filed 2015-06-20, granted 2016-10-04 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US10642843B2 World knowledge triggers
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-06-26, granted 2020-05-05 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US9916294B2 Synchronized consumption modes for e-books
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-07-15, granted 2018-03-13 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US20160350136A1 Assist layer with automated extraction
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2015-07-21 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US10380614B1 User reset voting to identify unwanted settings values in client software
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-08-07, granted 2019-08-13 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9792628B2 Dispatch system to remote devices
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-08-14, granted 2017-10-17 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores

US9952839B2 Layout management in a rapid application development tool
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2015-08-14, granted 2018-04-24 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US10216694B2 Generic scheduling
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-08-24, granted 2019-02-26 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo

US9864735B1 In-domain webpage editing
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2015-08-27, granted 2018-01-09 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData

US9514194B1 Website duration performance based on category durations
Rule: Focus on increasing dwell time relative to your topical categories. Build direct traffic (brand searches, bookmarks, typed URLs) — these visits are boosted in duration scoring. Ensure users don't bounce back to search quickly. Being strong in multiple relevant categories compounds your duration performance score.
filed 2015-09-24, granted 2016-12-06 · leak modules: QualityNavboostCrapsCrapsClickSignals, QualityNsrNsrData, CompressedQualitySignals

US9519345B2 Computing device with force-triggered non-visual responses
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-09-24, granted 2016-12-13 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US9692843B1 System and method for associating individual household members with television programs viewed

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2015-10-01, granted 2017-06-27 · leak modules: PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData

US10503733B2 Assistive browsing using context

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2015-10-13, granted 2019-12-10 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9798566B2 Metadata-based virtual machine configuration

Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships.

filed 2015-10-23, granted 2017-10-24 · leak modules: PerDocData, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US9773166B1 Identifying longform articles

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2015-11-03, granted 2017-09-26 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US10666750B2 Enhanced push messaging

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2015-11-17, granted 2020-05-26 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9674179B2 Pass through service login to application login

Rule: Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.

filed 2015-11-17, granted 2017-06-06 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US10244073B2 Adaptive DNS pre-resolution

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2015-11-23, granted 2019-03-26 · leak modules: CrawlerChangerateUrlChangerate, PerDocData, SmartphonePerDocData, TrawlerFetchReplyData

US10387512B2 Deriving and using interaction profiles

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2015-12-11, granted 2019-08-20 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US10565627B2 Systems and methods for automatically generating remarketing lists

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2015-12-30, granted 2020-02-18 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures

US10360586B2 Conversion path performance measures and reports

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.

filed 2016-01-21, granted 2019-07-23 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US20170214954A1 Media Program Moments Guide

Rule: Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.

filed 2016-01-25 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefDocumentMetadata

US9712869B1 System and method for associating individual household members with television programs viewed

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2016-02-16, granted 2017-07-18 · leak modules: PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData

US9661093B2 Device control profile for a fabric network

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.

filed 2016-02-22, granted 2017-05-23 · leak modules: IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata

US9747304B2 Approximating a user location

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2016-02-26, granted 2017-08-29 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9405849B1 Inducing command inputs from property sequences

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data.

filed 2016-03-29, granted 2016-08-02 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US9858171B2 Application analytics reporting

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2016-03-30, granted 2018-01-02 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US10394909B2 Reducing redirects

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2016-04-15, granted 2019-08-27 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US10078673B2 Determining graphical elements associated with text

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-04-20, granted 2018-09-18 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US20170308290A1 Iconographic suggestions within a keyboard

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-04-20 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9798446B2 Standard commands for native commands

Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Optimize video for watch time and retention.

filed 2016-05-11, granted 2017-10-24 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData, VideoPerDocData

US10354292B1 Systems and methods for generating navigation filters

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.

filed 2016-05-25, granted 2019-07-16 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals

US11068584B2 Systems and methods for deploying countermeasures against unauthorized scripts interfering with the rendering of content elements on information resources

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2016-05-27, granted 2021-07-20 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US10785327B2 Client device application interaction monitoring

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-06-02, granted 2020-09-22 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US10282751B1 Lead generation system and methods

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-06-20, granted 2019-05-07 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US10846353B2 Dynamic radius threshold selection

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2016-06-22, granted 2020-11-24 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US20170372348A1 Device configuration-based function delivery

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2016-06-22 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US9830668B1 Identifying top fans

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.

filed 2016-06-29, granted 2017-11-28 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefDocumentMetadata

US10346425B2 Distributed storage system with replica location selection

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2016-07-01, granted 2019-07-09 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9984062B1 Generating author vectors

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-07-11, granted 2018-05-29 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US10198561B2 Systems, methods, and media for media session concurrency management with recurring license renewals

Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships. Optimize video for watch time and retention.

filed 2016-07-19, granted 2019-02-05 · leak modules: NlpSaftDocument, PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData, VideoPerDocData

US10592576B1 Crowdsourcing descriptor selection

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-07-25, granted 2020-03-17 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures

US10501093B2 Application execution while operating vehicle
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2016-08-03, granted 2019-12-10 · leak modules: NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US10083157B2 Text classification and transformation based on author
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2016-08-05, granted 2018-09-25 · leak modules: NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefEntityJoin, SmartphonePerDocData

US9582538B1 Trend based distribution parameter suggestion
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2016-08-11, granted 2017-02-28 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20170053008A1 Time series explorer
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2016-08-18 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9720933B2 Indicating location status
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2016-09-01, granted 2017-08-01 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US10218599B2 Identifying referral pages based on recorded URL requests
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2016-09-13, granted 2019-02-26 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US20170075547A1 Systems and methods for determining application zoom levels
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2016-09-14 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US10498676B2 Contextual automatic grouping
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2016-10-12, granted 2019-12-03 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10531227B2 Time-delimited action suggestion system
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2016-10-19, granted 2020-01-07 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US9712327B1 System and method for remote storage auditing
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2016-10-21, granted 2017-07-18 · leak modules: CompressedQualitySignals, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10515144B2 On-demand auto-fill
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2016-10-27, granted 2019-12-24 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US10338884B2 Computing device with force-triggered non-visual responses
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2016-11-01, granted 2019-07-02 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US10846612B2 Actionable suggestions for activities
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2016-11-01, granted 2020-11-24 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9798532B1 Precompiling locally-stored instructions for a web application
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2016-11-01, granted 2017-10-24 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US10469560B1 Reduced latency for subresource transfer
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships. Optimize video for watch time and retention.
filed 2016-11-02, granted 2019-11-05 · leak modules: PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData, VideoPerDocData

US10397163B2 Third party application configuration for issuing notifications
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2016-11-07, granted 2019-08-27 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10318571B2 Programming a dynamic digital media queue
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2016-11-14, granted 2019-06-11 · leak modules: CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US9674277B1 Systems and methods for request isolation protection
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2016-11-16, granted 2017-06-06 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US10075449B2 Reducing latency
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2016-12-14, granted 2018-09-11 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9838399B2 Reducing latency
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2016-12-19, granted 2017-12-05 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20180181676A1 Nodes in directed acyclic graph
Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2016-12-22 · leak modules: ImageData, ImagePerDocData, PerDocData, SmartphonePerDocData

US10459957B2 User-guided term suggestions
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2016-12-29, granted 2019-10-29 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US20180189399A1 Systems and methods for identifying and characterizing signals contained in a data stream

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.

filed 2016-12-29 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9846722B1 Trend based distribution parameter suggestion

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2017-01-12, granted 2017-12-19 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10735533B2 Client device application interaction monitoring

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2017-02-01, granted 2020-08-04 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US10757204B2 Client device application interaction monitoring

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2017-02-01, granted 2020-08-25 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US10778789B2 Client device application interaction monitoring

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2017-02-01, granted 2020-09-15 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US10140286B2 Optimized graph traversal

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Write in natural vocabulary. ML models understand context and fluff.

filed 2017-02-22, granted 2018-11-27 · leak modules: NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US10333852B1 Distributed processing system throttling

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2017-02-27, granted 2019-06-25 · leak modules: IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US10165078B1 Parallel, side-effect based DNS pre-caching

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2017-03-13, granted 2018-12-25 · leak modules: CrawlerChangerateUrlChangerate, PerDocData, SmartphonePerDocData, TrawlerFetchReplyData

US9996530B1 Determining NCCs and/or using the NCCs to adapt performance of computer-based action(s)

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.

filed 2017-03-30, granted 2018-06-12 · leak modules: NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefDocumentMetadata

US10896285B2 Predicting user navigation events

Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Build content for diverse user contexts and search histories.

filed 2017-04-04, granted 2021-01-19 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US10489395B2 Methods and systems for providing functional extensions with a landing page of a creative

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2017-05-02, granted 2019-11-26 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures

US10296403B2 Software development kit platform

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2017-05-11, granted 2019-05-21 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US10303781B1 Deriving associations between assets

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2017-05-15, granted 2019-05-28 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11055332B1 Adaptive sorting of results

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2017-05-31, granted 2021-07-06 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefEntityJoin

US11275483B2 Providing media to a user based on a triggering event

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.

filed 2017-05-31, granted 2022-03-15 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US9893896B1 System and method for remote storage auditing

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2017-06-13, granted 2018-02-13 · leak modules: CompressedQualitySignals, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US10268770B2 Indicating location status

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.

filed 2017-06-23, granted 2019-04-23 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US10154310B2 System and method for associating individual household members with television programs viewed

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2017-06-27, granted 2018-12-11 · leak modules: PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData

US10540158B2 Post-install application interaction

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.

filed 2017-07-06, granted 2020-01-21 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US10545970B1 Providing posts from an extended network

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2017-07-21, granted 2020-01-28 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US10332009B2 Predicting user navigation events

Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Build content for diverse user contexts and search histories.

filed 2017-07-24, granted 2019-06-25 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US11368556B2 Computer application promotion
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-07-24, granted 2022-06-21 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US10498849B1 Reduction in redirect navigation latency via speculative preconnection
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2017-08-07, granted 2019-12-03 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US10534802B2 Nonstandard locality-based text entry
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2017-08-08, granted 2020-01-14 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US10592573B1 Interactively suggesting network location
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2017-08-21, granted 2020-03-17 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10956498B2 Scanning for bindings in webpage markup
Rule: Ensure your structured data and markup bindings are recognized by Google's processing pipeline. Complex or non-standard markup patterns may be skipped if they're not in Google's predetermined binding list. Keep markup standard and parseable. Excessive binding complexity can trigger time-threshold pauses that may affect how completely Google processes your page's structured data.
filed 2017-09-29, granted 2021-03-23 · leak modules: structured_data, rendering, indexing, web_vitals

US10289275B2 Standard commands for native commands
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Optimize video for watch time and retention.
filed 2017-10-20, granted 2019-05-14 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData, VideoPerDocData

US10324909B2 Omega names: name generation and derivation utilizing nested three or more attributes
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships. Write for natural spoken language and conversational queries.
filed 2017-10-31, granted 2019-06-18 · leak modules: NlpSaftDocument, PerDocData, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US10412150B2 Facilitating secure web browsing on untrusted networks
Rule: Earn endorsements from trusted sources. Trust is computed and transitive.
filed 2017-10-31, granted 2019-09-10 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData

US10581862B2 Utility portals for managing demand-response events
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2017-10-31, granted 2020-03-03 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US20180129685A1 Omega names: name generation and derivation
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships. Write for natural spoken language and conversational queries.
filed 2017-10-31 · leak modules: NlpSaftDocument, PerDocData, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US11782996B2 On-demand location visit conversion metrics
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2017-11-09, granted 2023-10-10 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20180137436A1 Rich media notice board
Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2018-01-12 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, SnippetExtraInfo

US11032388B2 Methods for prerendering and methods for managing and configuring prerendering operations

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2018-02-13, granted 2021-06-08 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US10740072B2 Layout management in a rapid application development tool

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.

filed 2018-04-23, granted 2020-08-11 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US10657165B1 Inferring membership in a group

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs.

filed 2018-04-27, granted 2020-05-19 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US10503521B2 Delayed shut down of a computer

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2018-04-30, granted 2019-12-10 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US10599770B1 Generating author vectors

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2018-05-29, granted 2020-03-24 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US10367819B2 Streamlined utility portals for managing demand-response events

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2018-06-12, granted 2019-07-30 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US20210004682A1 Adapting a sequence model for use in predicting future device interactions with a computing system

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2018-06-27 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US11343254B2 Reducing latency

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2018-08-02, granted 2022-05-24 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20200050707A1 Website representation vector

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2018-08-10 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US11347812B2 Percentile linkage clustering

Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2018-08-30, granted 2022-05-31 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US10789428B2 Optimized graph traversal
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Write in natural vocabulary. ML models understand context and fluff.
filed 2018-10-17, granted 2020-09-29 · leak modules: NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US10747944B1 Unified web and application framework
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2018-12-11, granted 2020-08-18 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US10783456B2 Training encoder model and/or using trained encoder model to determine responsive action(s) for natural language input
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2018-12-14, granted 2020-09-22 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US10887418B1 Parallel, side-effect based DNS pre-caching
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2018-12-21, granted 2021-01-05 · leak modules: CrawlerChangerateUrlChangerate, PerDocData, SmartphonePerDocData, TrawlerFetchReplyData

US20200349210A1 Context Aware Skim-Read Friendly Text View
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2019-01-10 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US10552587B2 Systems, methods, and media for media session concurrency management with recurring license renewals
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships. Optimize video for watch time and retention.
filed 2019-02-04, granted 2020-02-04 · leak modules: NlpSaftDocument, PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData, VideoPerDocData

US11151311B2 Systems and methods for autofill field classification
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Write for natural spoken language and conversational queries.
filed 2019-03-04, granted 2021-10-19 · leak modules: NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US11487830B2 Indicating location status
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2019-03-06, granted 2022-11-01 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US11171872B1 Distributed processing system throttling using a timestamp
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2019-06-07, granted 2021-11-09 · leak modules: IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US10832266B2 Streamlined utility portals for managing demand-response events
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2019-07-24, granted 2020-11-10 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US20210064644A1 Yaml configuration modeling
Rule: Use descriptive alt text and surrounding context for images. Structure answers: question heading, tight answer, then depth. Use JSON-LD structured data for entity types and relationships.
filed 2019-08-30 · leak modules: ImageData, ImagePerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US10938935B1 Reduction in redirect navigation latency via speculative preconnection
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2019-10-18, granted 2021-03-02 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US20210118063A1 Multi-tier scalable media analysis
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2019-10-18 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US11487757B2 Assistive browsing using context
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2019-10-29, granted 2022-11-01 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US11567987B2 Methods, systems, and media for identifying abusive user accounts based on playlists
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2019-11-06, granted 2023-01-31 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US11202167B2 Time-delimited action suggestion system
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2019-11-12, granted 2021-12-14 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US11003432B2 Post-install application interaction
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2019-12-09, granted 2021-05-11 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US11971269B2 Constrained navigation and route planning
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2019-12-19, granted 2024-04-30 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US11153397B2 Enhanced push messaging
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2020-01-23, granted 2021-10-19 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11604856B2 Systems, methods, and media for media session concurrency management with recurring license renewals
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships. Optimize video for watch time and retention.
filed 2020-02-03, granted 2023-03-14 · leak modules: NlpSaftDocument, PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData, VideoPerDocData

US11216852B2 Systems and methods for automatically generating remarketing lists
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2020-02-11, granted 2022-01-04 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures

US11252252B2 Installable web applications
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2020-02-18, granted 2022-02-15 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US11308508B2 Utility portals for managing demand-response events
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2020-03-02, granted 2022-04-19 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US11275895B1 Generating author vectors
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2020-03-19, granted 2022-03-15 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US11256700B2 World knowledge triggers
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2020-03-31, granted 2022-02-22 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US11256853B2 Parameterizing network communication paths
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2020-04-14, granted 2022-02-22 · leak modules: AnchorsAnchor, AnchorsAnchorSource, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US11675975B2 Word classification based on phonetic features
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2020-06-29, granted 2023-06-13 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US11461419B2 Discovering alternate online service providers
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data.
filed 2020-07-09, granted 2022-10-04 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US12008307B1 Unified web and application framework
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2020-07-15, granted 2024-06-11 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US11842253B2 Training encoder model and/or using trained encoder model to determine responsive action(s) for natural language input
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2020-08-17, granted 2023-12-12 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US12169494B2 Systems and methods for identifying events that share a characteristic
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2020-08-18, granted 2024-12-17 · leak modules: IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US11551003B2 Optimized graph traversal
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Write in natural vocabulary. ML models understand context and fluff.
filed 2020-08-25, granted 2023-01-10 · leak modules: NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US11775589B2 Systems and methods for weighted quantization
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2020-08-25, granted 2023-10-03 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsGeoTopicality

US12248854B2 Robust model performance across disparate sub-groups within a same group
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2020-09-30, granted 2025-03-11 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US11755674B2 Dynamic radius threshold selection
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2020-11-23, granted 2023-09-12 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US11887016B2 Actionable suggestions for activities
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2020-11-23, granted 2024-01-30 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11544739B1 Conversion path performance measures and reports
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2020-11-30, granted 2023-01-03 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US11249741B2 Post-install application interaction
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2020-12-08, granted 2022-02-15 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US11620035B2 Streamlined hosted applications
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2021-06-10, granted 2023-04-04 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, ShingleInfoPerDocData, SmartphonePerDocData

US11907360B2 Systems and methods for deploying countermeasures against unauthorized scripts interfering with the rendering of content elements on information resources
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2021-07-16, granted 2024-02-20 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US20230054017A1 Distributed Generic Cacheability Analysis
Rule: Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2021-08-20 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, RepositoryAnnotationsGeoTopicality

US20240354504A1 Structural encoding and attention paradigms for sequence modeling
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.
filed 2021-08-25 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US11604921B2 Systems and methods for autofill field classification
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Write for natural spoken language and conversational queries.
filed 2021-09-13, granted 2023-03-14 · leak modules: NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US12112144B2 API specification generation
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2021-10-15, granted 2024-10-08 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData, SnippetExtraInfo

US11520471B1 Systems and methods for identifying a set of characters in a media file
Rule: Structure answers: question heading, tight answer, then depth. Use JSON-LD structured data for entity types and relationships. Optimize video for watch time and retention.
filed 2021-10-31, granted 2022-12-06 · leak modules: NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US11689466B2 Distributed processing system throttling using a timestamp
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2021-11-04, granted 2023-06-27 · leak modules: IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US12052633B2 Time-delimited action suggestion system
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2021-12-13, granted 2024-07-30 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US12260433B2 Systems and methods for automatically generating remarketing lists
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2022-01-04, granted 2025-03-25 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures

US11663217B2 World knowledge triggers
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2022-01-28, granted 2023-05-30 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US11799984B2 Installable web applications
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2022-02-14, granted 2023-10-24 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US11868724B2 Generating author vectors
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2022-03-14, granted 2024-01-09 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US12067215B2 Providing media to a user based on a triggering event
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.
filed 2022-03-14, granted 2024-08-20 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US11941660B1 Conversion path performance measures and reports
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2022-03-17, granted 2024-03-26 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US11861634B2 Utility portals for managing demand-response events

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2022-04-18, granted 2024-01-02 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US11750683B2 Computer application promotion

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2022-05-11, granted 2023-09-05 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US12099544B2 Systems and methods for generating stories for live events using a scalable pipeline

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2022-07-21, granted 2024-09-24 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US20240048607A1 Links for web-based applications

Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2022-08-08 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US11714867B2 Discovering alternate online service providers

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data.

filed 2022-09-30, granted 2023-08-01 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US12386901B2 Selecting a device to respond to device-agnostic user requests

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2022-09-30, granted 2025-08-12 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US11921803B2 Indicating location status

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.

filed 2022-10-28, granted 2024-03-05 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US12072958B2 Systems, methods, and media for media session concurrency management with recurring license renewals

Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships. Optimize video for watch time and retention.

filed 2022-11-01, granted 2024-08-27 · leak modules: NlpSaftDocument, PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData, VideoPerDocData

US11720200B2 Systems and methods for identifying a set of characters in a media file

Rule: Structure answers: question heading, tight answer, then depth. Use JSON-LD structured data for entity types and relationships. Optimize video for watch time and retention.

filed 2022-11-23, granted 2023-08-08 · leak modules: NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US12236322B2 Training and/or utilizing a model for predicting measures reflecting both quality and popularity of content

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2022-12-05, granted 2025-02-25 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation

US11829579B2 Immersive, multi-state Uni-Card

Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2022-12-06, granted 2023-11-28 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US11714857B2 Learning to select vocabularies for categorical features

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2022-12-07, granted 2023-08-01 · leak modules: NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US12001799B1 Optimized graph traversal

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Write in natural vocabulary. ML models understand context and fluff.

filed 2023-01-09, granted 2024-06-04 · leak modules: NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US12038965B2 Methods, systems, and media for identifying abusive user accounts based on playlists

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2023-01-30, granted 2024-07-16 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US12045302B2 Determining whether and/or how to implement request to prevent provision of search result(s)

Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.

filed 2023-02-07, granted 2024-07-23 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsGeoTopicality

US12050607B2 World knowledge triggers

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2023-04-13, granted 2024-07-30 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US12099674B2 Systems and methods for identifying a set of characters in a media file

Rule: Structure answers: question heading, tight answer, then depth. Use JSON-LD structured data for entity types and relationships. Optimize video for watch time and retention.

filed 2023-06-14, granted 2024-09-24 · leak modules: NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US12292939B2 Discovering alternate online service providers

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data.

filed 2023-07-28, granted 2025-05-06 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US12380175B2 Dynamic radius threshold selection

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2023-07-28, granted 2025-08-05 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US20250054405A1 System for Providing Step-by-Step Explanations of Pedagogical Exercises Using Machine-Learned Models

Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2023-08-08 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US12339910B2 Systems and methods for weighted quantization

Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2023-08-28, granted 2025-06-24 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsGeoTopicality

US20250077776A1 Golden Prompt Generation based on Authoritative Publications
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2023-09-05 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20240394545A1 Universal Self-Adaptive Prompting
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2023-10-06 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US12086389B2 Immersive, multi-state uni-card
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2023-10-17, granted 2024-09-10 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US12499144B2 LLM latency reduction via bridging multiple LLMS of differing sizes
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2023-12-07, granted 2025-12-16 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US12124524B1 Generating prompts for user link notes
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2023-12-21, granted 2024-10-22 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US12468965B2 Actionable suggestions for activities
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2024-01-29, granted 2025-11-11 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US12211065B1 Conversion path performance measures and reports
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2024-02-20, granted 2025-01-28 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US12332957B2 Indicating location status
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2024-03-04, granted 2025-06-17 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US12342244B2 Time-delimited action suggestion system
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2024-06-13, granted 2025-06-24 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US12380112B2 World knowledge triggers
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2024-06-14, granted 2025-08-05 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US12608434B2 Determining whether and/or how to implement request to prevent provision of search result(s)
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2024-07-08, granted 2026-04-21 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsGeoTopicality

US20240427581A1 Debugging applications for delivery via an application delivery server
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2024-09-06 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData

US20250348526A1 Exposing App Functionality using System-level LLM Agent Services
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.
filed 2025-05-12 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US20260050952A1 Finding and Filtering Elements of a Visual Scene
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2025-07-16 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US20260023928A1 Inference Methods For Word Or Wordpiece Tokenization
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Write for natural spoken language and conversational queries.
filed 2025-10-01 · leak modules: NlpSaftDocument, PerDocData, SmartphonePerDocData

Query understanding (543)

US7249121B1 Identification of semantic units from within a search query
Rule: Build quality links with relevant anchor text. Penguin is still active. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2000-12-05, granted 2007-07-24 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US8856163B2 System and method for providing a user interface with search query broadening
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Build content for diverse user contexts and search histories.
filed 2003-07-28, granted 2014-10-07 · leak modules: NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US7505964B2 Methods and systems for improving a search ranking using related queries
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2003-09-12, granted 2009-03-17 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US8086619B2 System and method for providing search query refinements
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2003-09-22, granted 2011-12-27 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo

US8046350B1 Systems and methods for detecting commercial queries
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Write for intent and meaning. Neural matching connects related concepts.
filed 2003-09-24, granted 2011-10-25 · leak modules: NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US7467131B1 Method and system for query data caching and optimization in a search engine system
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2003-09-30, granted 2008-12-16 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US7281008B1 Systems and methods for constructing a query result set
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2003-12-31, granted 2007-10-09 · leak modules: IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, ShingleInfoPerDocData, SmartphonePerDocData

US7664734B2 Systems and methods for generating multiple implicit search queries
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2004-03-31, granted 2010-02-16 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation

US7840547B1 Methods and systems for efficient query rewriting
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2004-03-31, granted 2010-11-23 · leak modules: NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8631001B2 Systems and methods for weighting a search query result
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2004-03-31, granted 2014-01-14 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation

US7565630B1 Customization of search results for search queries received from third party sites
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2004-06-15, granted 2009-07-21 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US8407239B2 Multi-stage query processing system and method for use with tokenspace repository
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2004-08-13, granted 2013-03-26 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US8065316B1 Systems and methods for providing search query refinements
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2004-09-30, granted 2011-11-22 · leak modules: CompressedQualitySignals, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8838562B1 Methods and apparatus for providing query parameters to a search engine
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2004-10-22, granted 2014-09-16 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures

US7440968B1 Query boosting based on classification
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2004-11-30, granted 2008-10-21 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US8346757B1 Determining query terms of little significance
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2005-03-28, granted 2013-01-01 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US7565345B2 Integration of multiple query revision models
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2005-03-29, granted 2009-07-21 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US7617205B2 Estimating confidence for query revision models
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2005-03-30, granted 2009-11-10 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US7636714B1 Determining query term synonyms within query context
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2005-03-31, granted 2009-12-22 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo

US7725485B1 Generating query suggestions using contextual information
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2005-08-01, granted 2010-05-25 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8117531B1 Interpreted language translation system and method
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2005-09-23, granted 2012-02-14 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US7870147B2 Query revision using known highly-ranked queries
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2005-11-22, granted 2011-01-11 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US8489614B2 Ranking academic event related search results using event member metrics
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2005-12-14, granted 2013-07-16 · leak modules: NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefEntityJoin, SmartphonePerDocData

US8078607B2 Generating website profiles based on queries from webistes and user activities on the search results
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2006-03-30, granted 2011-12-13 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US7475063B2 Augmenting queries with synonyms selected using language statistics
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2006-04-19, granted 2009-01-06 · leak modules: IndexingDupsLocalizedLocalizedCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8762358B2 Query language determination using query terms and interface language
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2006-04-19, granted 2014-06-24 · leak modules: IndexingDupsLocalizedLocalizedCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9110975B1 Search result inputs using variant generalized queries
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2006-11-02, granted 2015-08-18 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8051071B2 Document scoring based on query analysis
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active.
filed 2006-11-22, granted 2011-11-01 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, IndexingDocjoinerAnchorStatistics, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8661012B1 Ensuring that a synonym for a query phrase does not drop information present in the query phrase
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2006-12-29, granted 2014-02-25 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US7593939B2 Generating specialized search results in response to patterned queries
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2007-03-30, granted 2009-09-22 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US8782030B1 Using a query log from a two-box interface to identify standalone locations
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2007-04-20, granted 2014-07-15 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US8126874B2 Systems and methods for generating statistics from search engine query logs
Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2007-05-08, granted 2012-02-28 · leak modules: ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData

US8019742B1 Identifying related queries
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2007-05-31, granted 2011-09-13 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8321410B1 Identification of semantic units from within a search query
Rule: Build quality links with relevant anchor text. Penguin is still active. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2007-06-18, granted 2012-11-27 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US7890493B2 Translating a search query into multiple languages
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2007-07-20, granted 2011-02-15 · leak modules: CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US8655862B1 System and method for query re-issue in search engines
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2007-10-17, granted 2014-02-18 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US7953746B1 Contextual query revision
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2007-12-07, granted 2011-05-31 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US8676824B2 Automatic search query correction
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2007-12-14, granted 2014-03-18 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US8024316B2 Providing content using stored query information
Rule: Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2008-01-30, granted 2011-09-20 · leak modules: PerDocData, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US7890521B1 Document-based synonym generation
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active.
filed 2008-02-07, granted 2011-02-15 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CrawlerChangerateUrlChangerate, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US8892552B1 Dynamic specification of custom search engines at query-time, and applications thereof
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Optimize mobile experience. Mobile-specific signals are tracked separately. Earn promotion through authoritative content and entity association.
filed 2008-03-11, granted 2014-11-18 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, VideoPerDocData

US8321406B2 Media object query submission and response
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2008-03-31, granted 2012-11-27 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US7991780B1 Performing multiple related searches
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2008-05-07, granted 2011-08-02 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8099412B2 Query statistics provider
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2008-05-21, granted 2012-01-17 · leak modules: IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US8380563B2 Using previous user search query to target advertisements
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2008-06-25, granted 2013-02-19 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US9183323B1 Suggesting alternative query phrases in query results
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2008-06-27, granted 2015-11-10 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8001136B1 Longest-common-subsequence detection for common synonyms
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2008-07-02, granted 2011-08-16 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US8463774B1 Universal scores for location search queries
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2008-07-15, granted 2013-06-11 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation

US8122022B1 Abbreviation detection for common synonym generation
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2008-08-06, granted 2012-02-21 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US8645390B1 Reordering search query results in accordance with search context specific predicted performance functions
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2008-08-07, granted 2014-02-04 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US8849865B1 Querying a data store of impressions
Rule: Structure answers: question heading, tight answer, then depth. Optimize video for watch time and retention. Write for natural spoken language and conversational queries.
filed 2008-08-15, granted 2014-09-30 · leak modules: NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo, VideoPerDocData

US8140524B1 Estimating confidence for query revision models
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2008-08-19, granted 2012-03-20 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US8996509B1 Query results enhancement through agent interaction
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2008-09-03, granted 2015-03-31 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefEntityJoin, SmartphonePerDocData

US8595252B2 Suggesting alternative queries in query results
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2008-09-12, granted 2013-11-26 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US9069841B1 Estimating confidence for query revision models
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2008-10-02, granted 2015-06-30 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US8725756B1 Session-based query suggestions
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2008-11-11, granted 2014-05-13 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US8392429B1 Informational book query
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2008-11-26, granted 2013-03-05 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US8484190B1 Prompt for query clarification
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2008-12-18, granted 2013-07-09 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US8024326B2 Methods and systems for improving a search ranking using related queries
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2009-01-09, granted 2011-09-20 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US8326826B1 Navigational resources for queries
Rule: This original patent (predecessor to US9213774B2) establishes the core navigational resource mechanism. Your site becomes a navigational resource when it dominates traffic proportion and CTR for specific queries. Users who find your site through revised queries train Google to associate your site with their original (poorer) queries. Build strong navigational intent: ensure users who search for your brand or related terms click through to your site at high rates. The leak's OfficialPagesOfficialKey module confirms Google maintains a registry of official/navigational pages.
filed 2009-01-15, granted 2012-12-04 · leak modules: QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, OfficialPagesOfficialKey, SdrPageAnchorsSitelink, SdrPageAnchorsDocInfo, CompressedQualitySignals

US8972411B1 Selection of sponsored content using multiple sets of query terms
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2009-01-27, granted 2015-03-03 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9128945B1 Query augmentation
Rule: Ensure your business/entity listings in structured data sources use consistent anchor text across many documents (exceeding frequency thresholds) to trigger synthetic query generation. Use entity names as anchor text in internal and external links. Register your entity's home page in structured data. If your domain name matches your entity name or key terms, this directly triggers your site as a result for synthetic queries. Build query performance signals so your pages perform well when selected for specific entity queries.
filed 2009-03-16, granted 2015-09-08 · leak modules: AnchorsAnchor, AnchorsAnchorSource, IndexingDocjoinerAnchorStatistics, PerDocData, QualityNavboostCrapsCrapsClickSignals

US8683329B2 Web translation with display replacement
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2009-03-18, granted 2014-03-25 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US8145623B1 Query ranking based on query clustering and categorization
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2009-05-01, granted 2012-03-27 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9192684B1 Customization of search results for search queries received from third party sites
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2009-06-05, granted 2015-11-24 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US8577909B1 Query translation using bilingual search refinements
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2009-06-09, granted 2013-11-05 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US8171021B2 Query identification and association
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2009-06-17, granted 2012-05-01 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US20100324993A1 Promotional content presentation based on search query
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2009-06-19 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US8521761B2 Transliteration for query expansion
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2009-07-15, granted 2013-08-27 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation

US8458171B2 Identifying query aspects
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2009-07-30, granted 2013-06-04 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US8402021B2 Providing posts to discussion threads in response to a search query
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2009-07-31, granted 2013-03-19 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo

US9098568B2 Query suggestions from documents
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2009-08-04, granted 2015-08-04 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9183297B1 Method and apparatus for generating lexical synonyms for query terms
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2009-08-24, granted 2015-11-10 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US8204897B1 Interactive search querying
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2009-09-09, granted 2012-06-19 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US8244749B1 Generating sibling query refinements
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2009-09-10, granted 2012-08-14 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument

US8156102B2 Inferring search category synonyms
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Build content for diverse user contexts and search histories.
filed 2009-10-19, granted 2012-04-10 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, RepositoryAnnotationsGeoTopicality

US9275164B2 Grouping and presenting search query results
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2009-12-10, granted 2016-03-01 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData

US20120278308A1 Custom search query suggestion tools
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2009-12-30 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, RepositoryWebrefEntityJoin

US8849785B1 Search query reformulation using result term occurrence count
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2010-01-15, granted 2014-09-30 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US8392443B1 Refining search queries
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2010-03-17, granted 2013-03-05 · leak modules: NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefDocumentMetadata

US8930350B1 Autocompletion using previously submitted query data
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2010-03-23, granted 2015-01-06 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US8015199B1 Generating query suggestions using contextual information
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2010-04-15, granted 2011-09-06 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8838587B1 Propagating query classifications
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2010-04-19, granted 2014-09-16 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8478699B1 Multiple correlation measures for measuring query similarity
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2010-04-30, granted 2013-07-02 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US8453049B1 Delayed code parsing for reduced startup latency
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2010-05-19, granted 2013-05-28 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US8655901B1 Translation-based query pattern mining
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2010-06-23, granted 2014-02-18 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US8560562B2 Predictive query suggestion caching
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2010-07-22, granted 2013-10-15 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US8868548B2 Determining user intent from query patterns
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2010-07-22, granted 2014-10-21 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US10061850B1 Using recent queries for inserting relevant search results for navigational queries
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2010-07-27, granted 2018-08-28 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US9703871B1 Generating query refinements using query components
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2010-07-30, granted 2017-07-11 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US9135277B2 Architecture for responding to a visual query
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2010-08-04, granted 2015-09-15 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation

US10496714B2 State-dependent query response
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2010-08-06, granted 2019-12-03 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality

US8977639B2 Actionable search results for visual queries
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2010-08-11, granted 2015-03-10 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation

US9852156B2 Hybrid use of location sensor data and visual query to return local listings for visual query
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2010-08-12, granted 2017-12-26 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation

US8478519B2 Providing results to parameterless search queries
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2010-08-30, granted 2013-07-02 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US8583675B1 Providing result-based query suggestions
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2010-08-30, granted 2013-11-12 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US8375049B2 Query revision using known highly-ranked queries
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2010-09-07, granted 2013-02-12 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US8433705B1 Facet suggestion for search query augmentation
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2010-09-29, granted 2013-04-30 · leak modules: NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8346792B1 Query generation using structural similarity between documents
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2010-11-09, granted 2013-01-01 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8631026B1 Methods and systems for efficient query rewriting
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2010-11-22, granted 2014-01-14 · leak modules: NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8631030B1 Query suggestions with high diversity
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2010-12-28, granted 2014-01-14 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData

US8478773B1 Interpreting search queries
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-01-14, granted 2013-07-02 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US8639725B1 Methods for efficiently querying multiple points in an indexed quadtree
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Earn promotion through authoritative content and entity association.
filed 2011-01-31, granted 2014-01-28 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US8650210B1 Identifying non-search actions based on a search query
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-02-08, granted 2014-02-11 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8688667B1 Providing intent sensitive search results
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-02-08, granted 2014-04-01 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US8161041B1 Document-based synonym generation
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active.
filed 2011-02-10, granted 2012-04-17 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CrawlerChangerateUrlChangerate, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US9164987B2 Translating a search query into multiple languages
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-02-10, granted 2015-10-20 · leak modules: CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US8321201B1 Identifying a synonym with N-gram agreement for a query phrase
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2011-03-17, granted 2012-11-27 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US8060517B2 Methods and systems for adjusting a scoring measure based on query breadth
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-04-08, granted 2011-11-15 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8583672B1 Displaying multiple spelling suggestions
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-04-14, granted 2013-11-12 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US8688727B1 Generating query refinements
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-04-26, granted 2014-04-01 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation

US8639679B1 Generating query suggestions
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-05-05, granted 2014-01-28 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US8626785B2 Contextual query revision
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-05-17, granted 2014-01-07 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US8577913B1 Generating midstring query refinements
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-05-27, granted 2013-11-05 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US9384266B1 Predictive generation of search suggestions
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2011-06-13, granted 2016-07-05 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US8601027B2 Query-based user groups in social networks
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-06-20, granted 2013-12-03 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityWebanswersVideoTranscriptAnnotations

US8209347B1 Generating query suggestions using contextual information
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2011-07-13, granted 2012-06-26 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8738612B1 Resolving ambiguous queries
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-07-27, granted 2014-05-27 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US8341138B2 Providing content using stored query information

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2011-08-12, granted 2012-12-25 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US8756218B1 Query classification based on search engine results

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2011-08-16, granted 2014-06-17 · leak modules: NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8380705B2 Methods and systems for improving a search ranking using related queries

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2011-08-18, granted 2013-02-19 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US20120047025A1 Query stem advertising

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.

filed 2011-08-19 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US8694499B1 Systems and methods for determining query similarity by query distribution comparison

Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Write for natural spoken language and conversational queries.

filed 2011-08-19, granted 2014-04-08 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US8700653B2 Predictive query completion and predictive search results

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.

filed 2011-08-19, granted 2014-04-15 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US8706750B2 Predictive query completion and predictive search results

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.

filed 2011-08-19, granted 2014-04-22 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US8515935B1 Identifying related queries

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2011-08-26, granted 2013-08-20 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US8538742B2 Feed translation for a social network

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Structure answers: question heading, tight answer, then depth.

filed 2011-09-13, granted 2013-09-17 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SnippetExtraInfo

US8458211B2 Search query hash

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2011-09-16, granted 2013-06-04 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8185522B2 Document scoring based on query analysis
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active.
filed 2011-09-26, granted 2012-05-22 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, IndexingDocjoinerAnchorStatistics, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8239378B2 Document scoring based on query analysis
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active.
filed 2011-09-26, granted 2012-08-07 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, IndexingDocjoinerAnchorStatistics, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8244723B2 Document scoring based on query analysis
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active.
filed 2011-09-26, granted 2012-08-14 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, IndexingDocjoinerAnchorStatistics, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8266143B2 Document scoring based on query analysis
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active.
filed 2011-09-26, granted 2012-09-11 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, IndexingDocjoinerAnchorStatistics, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8412512B1 Feed translation for a social network
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Structure answers: question heading, tight answer, then depth.
filed 2011-09-26, granted 2013-04-02 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SnippetExtraInfo

US20120330919A1 Determining cross-language query suggestion based on query translations
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2011-09-29 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8504286B2 Providing results to parameterless search queries
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-09-29, granted 2013-08-06 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US8458585B1 Delayed code parsing for reduced startup latency
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-09-30, granted 2013-06-04 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US8577901B2 Document scoring based on query analysis
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active.
filed 2011-09-30, granted 2013-11-05 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, IndexingDocjoinerAnchorStatistics, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8504584B1 Systems and methods for providing search query refinements
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-10-11, granted 2013-08-06 · leak modules: CompressedQualitySignals, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8510289B1 Systems and methods for detecting commercial queries
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Write for intent and meaning. Neural matching connects related concepts.
filed 2011-10-20, granted 2013-08-13 · leak modules: NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US20150120773A1 Infrequent query variants for use as query suggestions
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Optimize mobile experience. Mobile-specific signals are tracked separately. Write for intent and meaning. Neural matching connects related concepts.
filed 2011-10-26 · leak modules: IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, ShingleInfoPerDocData, SmartphonePerDocData

US8577897B2 Rich results relevant to user search queries for books
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-10-26, granted 2013-11-05 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US8645407B2 System and method for providing search query refinements
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2011-11-04, granted 2014-02-04 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo

US8612414B2 Grouped search query refinements
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-11-21, granted 2013-12-17 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US20130067307A1 User interface for translation webpage
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-11-29 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US8521739B1 Creation of inferred queries for use as query suggestions
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-11-29, granted 2013-08-27 · leak modules: IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, ShingleInfoPerDocData

US8775409B1 Query ranking based on query clustering and categorization
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2011-12-08, granted 2014-07-08 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US20130311506A1 Method and apparatus for user query disambiguation
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-01-09 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US8606826B2 Augmenting queries with synonyms from synonyms map
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2012-01-12, granted 2013-12-10 · leak modules: IndexingDupsLocalizedLocalizedCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9104759B1 Identifying stem variants of search query terms
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-01-13, granted 2015-08-11 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US8392413B1 Document-based synonym generation
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active.
filed 2012-01-17, granted 2013-03-05 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CrawlerChangerateUrlChangerate, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US9213735B1 Flow control in very large query result sets using a release message to confirm that a client computer is ready to receive the data associated with a data collection operation
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-01-25, granted 2015-12-15 · leak modules: PerDocData, SmartphonePerDocData

US9311303B2 Interpreted language translation system and method
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-01-30, granted 2016-04-12 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US8812518B1 Synonym identification based on search quality
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-02-02, granted 2014-08-19 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9262767B2 Systems and methods for generating statistics from search engine query logs
Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2012-02-14, granted 2016-02-16 · leak modules: ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData

US8719282B1 Restricted-locality synonyms
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-02-23, granted 2014-05-06 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US8972388B1 Demotion of already observed search query completions
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-02-29, granted 2015-03-03 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US20150178278A1 Identifying recently submitted query variants for use as query suggestions
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-03-13 · leak modules: IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, ShingleInfoPerDocData

US8725862B1 Server selection based upon time and query dependent hashing
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-03-22, granted 2014-05-13 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8452758B2 Methods and systems for improving a search ranking using related queries
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-04-03, granted 2013-05-28 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US8601019B1 Presenting autocomplete suggestions
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2012-04-03, granted 2013-12-03 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US8768910B1 Identifying media queries
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-04-13, granted 2014-07-01 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US20150066967A1 Determining search query hints using content keywords
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-04-18 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData

US8639690B2 Document scoring based on query analysis
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active.
filed 2012-04-24, granted 2014-01-28 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, IndexingDocjoinerAnchorStatistics, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US10019455B2 Selecting search queries for display with data associated with search terms
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-04-27, granted 2018-07-10 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US8631003B2 Query identification and association
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-04-30, granted 2014-01-14 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US8819000B1 Query modification
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-05-01, granted 2014-08-26 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US9152701B2 Query classification
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-05-02, granted 2015-10-06 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate

US8433719B1 Accelerating find in page queries within a web browser
Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2012-05-08, granted 2013-04-30 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8732158B1 Method and system for matching queries to documents
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-05-09, granted 2014-05-20 · leak modules: CrawlerChangerateUrlChangerate, PerDocData, SmartphonePerDocData, TrawlerFetchReplyData

US8849799B1 Content selection using boolean query expressions
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-05-09, granted 2014-09-30 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US8838603B1 Interactive search querying
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-05-18, granted 2014-09-16 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US8868592B1 Providing customized autocomplete data
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-05-18, granted 2014-10-21 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9009146B1 Ranking search results based on similar queries
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-05-21, granted 2015-04-14 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US8868587B1 Determining correction of queries with potentially inaccurate terms
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active.
filed 2012-05-29, granted 2014-10-21 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData

US8620951B1 Search query results based upon topic
Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2012-06-01, granted 2013-12-31 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData

US9411890B2 Graph-based search queries using web content metadata
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2012-06-05, granted 2016-08-09 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US9009188B1 Drawing-based search queries
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-06-12, granted 2015-04-14 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9275147B2 Providing query suggestions
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-06-18, granted 2016-03-01 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US8838567B1 Customization of search results for search queries received from third party sites
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-07-16, granted 2014-09-16 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US8538946B1 Creating model or list to identify queries
Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2012-07-18, granted 2013-09-17 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8589383B2 Media object query submission and response
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-07-18, granted 2013-11-19 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US20140358904A1 Synonym identification based on selected search result
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2012-08-02 · leak modules: NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8392445B1 Generating sibling query refinements
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2012-08-09, granted 2013-03-05 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument

US8856138B1 Faster substring searching using hybrid range query data structures
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-08-09, granted 2014-10-07 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9305092B1 Search query auto-completions based on social graph
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2012-08-10, granted 2016-04-05 · leak modules: PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9147001B1 Automatic user-based query generation and execution
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-08-15, granted 2015-09-29 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US20160048873A1 Obtaining an ad related to a search query
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Build content for diverse user contexts and search histories.
filed 2012-09-04 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9110993B1 Search query reformulation using result term occurrence count
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-09-13, granted 2015-08-18 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US20170220680A1 Search query reformulation using a user-selected term
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-09-14 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8719262B1 Identification of semantic units from within a search query
Rule: Build quality links with relevant anchor text. Penguin is still active. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-09-14, granted 2014-05-06 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US8732153B1 Identifying related queries
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2012-09-14, granted 2014-05-20 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8799270B1 Determining query terms of little significance
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-09-14, granted 2014-08-05 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US9092479B1 Query generation using structural similarity between documents
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2012-09-14, granted 2015-07-28 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9213774B2 Navigational resources for queries
Rule: Google learns which sites are navigational destinations for specific queries by analyzing how users revise their searches. If users consistently revise from a generic query to one that leads to your site, Google associates your site with the generic query and promotes it. Build strong brand/navigation signals: high traffic proportion from search, high CTR, and strong cross-linkage. The leak's OfficialPagesOfficialKey and SdrPageAnchorsSitelink modules confirm Google tracks official/navigational page associations. Ensure your site is the clear navigational destination for your brand-related queries.
filed 2012-09-14, granted 2015-12-15 · leak modules: QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, OfficialPagesOfficialKey, SdrPageAnchorsSitelink, SdrPageAnchorsDocInfo, CompressedQualitySignals

US9395876B2 Receiving a search query that does not include one or more words that name any geographical location
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-09-14, granted 2016-07-19 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US9569527B2 Machine translation for query expansion
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2012-09-14, granted 2017-02-14 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US9400848B2 Techniques for context-based grouping of messages for translation
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-09-26, granted 2016-07-26 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8930393B1 Referent based search suggestions
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.
filed 2012-10-05, granted 2015-01-06 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, RepositoryWebrefDocumentMetadata

US9652556B2 Search suggestions based on viewport content
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.
filed 2012-10-05, granted 2017-05-16 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, RepositoryWebrefDocumentMetadata

US8713042B1 Processing autocomplete suggestions
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-10-11, granted 2014-04-29 · leak modules: IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate

US9514220B1 Generating content placement criteria based on a search query
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-10-19, granted 2016-12-06 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals

US9229974B1 Classifying queries
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-10-22, granted 2016-01-05 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US9430571B1 Generating travel queries in response to free text queries
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-10-24, granted 2016-08-30 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9342600B1 Populating query suggestion database using chains of related search queries
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Ensure NAP consistency and local relevance. Distance affects ranking. Build content for diverse user contexts and search histories.
filed 2012-11-01, granted 2016-05-17 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, RepositoryAnnotationsGeoTopicality

US9405834B1 System and method for identifying search results satisfying a search query
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Ensure NAP consistency and local relevance. Distance affects ranking. Build content for diverse user contexts and search histories.
filed 2012-11-01, granted 2016-08-02 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, RepositoryAnnotationsGeoTopicality

US9384267B2 Providing suggestion and translation thereof in accordance with a partial user entry
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-11-05, granted 2016-07-05 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US9116918B1 Methods, systems, and media for interpreting queries
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-11-14, granted 2015-08-25 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityOrbitAsteroidBeltDocumentIntentScores

US20140171133A1 Query response
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Structure answers: question heading, tight answer, then depth.
filed 2012-12-18 · leak modules: IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20130179467A1 Calculating Property Caching Exclusions In A Graph Evaluation Query Language
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-12-21 · leak modules: CompressedQualitySignals, PerDocData, SmartphonePerDocData

US9111011B2 Local query suggestions
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-12-27, granted 2015-08-18 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US9135307B1 Selectively generating alternative queries
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-12-27, granted 2015-09-15 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US9292622B2 Systems and methods for providing search suggestions
Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately. Build content for diverse user contexts and search histories.
filed 2012-12-27, granted 2016-03-22 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, VideoPerDocData

US9244972B1 Identifying navigational resources for informational queries
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2012-12-31, granted 2016-01-26 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US9449095B1 Revising search queries
Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately. Ensure content comprehensively covers the topic with natural vocabulary.
filed 2012-12-31, granted 2016-09-20 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData

US9355191B1 Identification of query completions which change users' original search intent
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-01-24, granted 2016-05-31 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores

US9037598B1 Variable query generation
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-01-25, granted 2015-05-19 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9020926B1 Refining search queries
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-01-31, granted 2015-04-28 · leak modules: NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefDocumentMetadata

US9317606B1 Spell correcting long queries
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2013-02-01, granted 2016-04-19 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US9235654B1 Query rewrites for generating auto-complete suggestions
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-02-05, granted 2016-01-12 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9594851B1 Determining query suggestions
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-02-07, granted 2017-03-14 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo

US8762370B1 Document-based synonym generation
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active.
filed 2013-02-08, granted 2014-06-24 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CrawlerChangerateUrlChangerate, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US9135250B1 Query completions in the context of a user's own document
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-02-22, granted 2015-09-15 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9342601B1 Query formulation and search in the context of a displayed document
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2013-02-22, granted 2016-05-17 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9218420B1 Detecting new businesses with unrecognized query terms
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-02-26, granted 2015-12-22 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US20160140228A1 Reformatting queries for search engines and data repositories
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.
filed 2013-02-27 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US9158860B2 Interactive query completion templates
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-02-27, granted 2015-10-13 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores

US9830353B1 Determining match type for query tokens
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-02-27, granted 2017-11-28 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US20160103833A1 Ranking query results using biometric parameters
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2013-02-28 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9824147B1 Query language filter for cross-language information retrieval
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-02-28, granted 2017-11-21 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US8880536B1 Providing book information in response to queries
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-03-01, granted 2014-11-04 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US9122727B1 Identification of related search queries that represent different information requests
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-03-01, granted 2015-09-01 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores

US8667003B1 Generating sibling query refinements
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2013-03-04, granted 2014-03-04 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument

US20140280040A1 Search results for descriptive search queries
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-03-12 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US9063983B1 Detecting name-triggering queries
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-03-12, granted 2015-06-23 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US20140280294A1 Connecting users in search services based on received queries
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2013-03-13 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20150149482A1 Using Live Information Sources To Rank Query Suggestions
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-03-14 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US20160217181A1 Annotating Query Suggestions With Descriptions
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-03-14 · leak modules: CompressedQualitySignals, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate

US9141674B2 Prominent display of selective results of book search queries
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-03-14, granted 2015-09-22 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9268880B2 Using recent media consumption to select query suggestions
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2013-03-14, granted 2016-02-23 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData

US10055462B2 Providing search results using augmented search queries
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-03-15, granted 2018-08-21 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US20150213041A1 Search suggestion rankings
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-03-15 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8983956B1 Category generalization for search queries
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Write for intent and meaning. Neural matching connects related concepts.
filed 2013-03-15, granted 2015-03-17 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9081832B2 Providing leaf page sublinks in response to a search query
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2013-03-15, granted 2015-07-14 · leak modules: CompressedQualitySignals, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US9317585B2 Search query suggestions based on personal information
Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2013-03-15, granted 2016-04-19 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData

US9146967B2 Multi-stage query processing system and method for use with tokenspace repository
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-03-26, granted 2015-09-29 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US9323806B2 Clustering query refinements by inferred user intent
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2013-04-01, granted 2016-04-26 · leak modules: ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US9251206B2 Generalized edit distance for queries
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-04-03, granted 2016-02-02 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9652494B2 Accelerating find in page queries within a web browser
Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2013-04-05, granted 2017-05-16 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US9230023B2 Search suggestion and display environment
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-04-16, granted 2016-01-05 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US9122376B1 System for improving autocompletion of text input
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2013-04-18, granted 2015-09-01 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US8918382B1 Learning common spelling errors through content matching
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-05-08, granted 2014-12-23 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations

US9703761B2 Delayed code parsing for reduced startup latency
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-05-09, granted 2017-07-11 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US9116952B1 Query refinements using search data
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-05-31, granted 2015-08-25 · leak modules: CompressedQualitySignals, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US9336277B2 Query suggestions based on search data
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-05-31, granted 2016-05-10 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US9152676B2 Identifying query aspects
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-06-03, granted 2015-10-06 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US9529856B2 Query suggestion templates
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-06-03, granted 2016-12-27 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8825571B1 Multiple correlation measures for measuring query similarity
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-06-04, granted 2014-09-02 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US9448992B2 Natural language search results for intent queries
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data.
filed 2013-06-04, granted 2016-09-20 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US20170308519A1 Learning semantic parsing
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-06-20 · leak modules: NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US9092505B1 Parsing rule generalization by n-gram span clustering
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2013-06-25, granted 2015-07-28 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9117452B1 Exceptions to action invocation from parsing rules
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-06-25, granted 2015-08-25 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US9251202B1 Corpus specific queries for corpora from search query
Rule: Google generates corpus-specific query variants and only shows results from corpora that score well for a given query. Ensure your content is properly classified by corpus type (images, videos, products) and optimized for corpus-specific query reformulations. If your video content doesn't score well for a hypothesis, it won't appear in results at all. The leak's ImagePerDocData, VideoPerDocData, and IndexingMlVertical modules confirm per-corpus scoring. Use appropriate structured data and corpus-specific metadata to ensure your content qualifies for its corpus hypothesis.
filed 2013-06-25, granted 2016-02-02 · leak modules: ImagePerDocData, VideoPerDocData, IndexingMlVerticalVerticalItem, QualityShoppingShoppingAttachment, QualityOrbitAsteroidBeltDocumentIntentScores, QualityNsrNsrData, PerDocData

US20180357238A1 Providing information based on intent
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2013-06-27 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US9298852B2 Reranking query completions
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-06-27, granted 2016-03-29 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US20150012532A1 User models for implicit intents in search
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2013-07-02 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US20150012558A1 Using models to annotate search queries
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2013-07-02 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US8898129B1 Prompt for query clarification
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2013-07-03, granted 2014-11-25 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US20130346396A1 Automatically updating a query
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-07-12 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US10073882B1 Semantically equivalent query templates
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2013-07-15, granted 2018-09-11 · leak modules: IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9165028B1 Ranking modifications of a previous query
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-07-24, granted 2015-10-20 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9286395B1 Modifying query in discourse context
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-07-25, granted 2016-03-15 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US9495443B1 Systems and methods for providing search query refinements
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-07-31, granted 2016-11-15 · leak modules: CompressedQualitySignals, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9135271B1 Finding similar cities using geo-related queries
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-08-05, granted 2015-09-15 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US9212915B2 Providing results to parameterless search queries
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-08-05, granted 2015-12-15 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US9031949B1 Creation of inferred queries for use as query suggestions
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-08-26, granted 2015-05-12 · leak modules: IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, ShingleInfoPerDocData

US9519638B2 Feed translation for a social network
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-08-29, granted 2016-12-13 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US20170308517A1 Automatic generation of templates for parsing electronic documents
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-09-11 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9323808B1 Predictive query suggestion caching
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2013-09-13, granted 2016-04-26 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US20150088648A1 Determining commercial intent
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2013-09-24 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9443035B2 Method and system for autocompletion for languages having ideographs and phonetic characters
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-09-24, granted 2016-09-13 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US20160048567A1 Search query refinement interface
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-09-27 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, RepositoryAnnotationsGeoTopicality

US9600543B1 Query-based stream
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-09-27, granted 2017-03-21 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations

US10152535B1 Query phrasification
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-10-18, granted 2018-12-11 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData

US10002117B1 Translating annotation tags into suggested markup
Rule: Use Google's Structured Data Markup Helper (which implements this patent) to generate schema.org markup for your pages. Ensure markup correctly wraps the intended content elements. Validate with the Rich Results Test. Properly nested schema.org markup that maps to visible page content is preferred over markup in separate blocks.
filed 2013-10-24, granted 2018-06-19 · leak modules: structured_data, schema, knowledge_graph, snippet

US9092459B2 Media object query submission and response
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-10-24, granted 2015-07-28 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US8843528B1 Query-based user groups in social networks
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-10-25, granted 2014-09-23 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityWebanswersVideoTranscriptAnnotations

US9483565B2 Associating a task with a user based on user selection of a query suggestion
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-10-28, granted 2016-11-01 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US8886585B1 Search suggestions based on viewport content
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-11-01, granted 2014-11-11 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US9092490B2 Rich results relevant to user search queries for books
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-11-01, granted 2015-07-28 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US10482139B2 Structured user graph to support querying and predictions
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-11-05, granted 2019-11-19 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US9092528B1 Providing result-based query suggestions
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-11-08, granted 2015-07-28 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US9031928B2 Grouped search query refinements
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-11-15, granted 2015-05-12 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US9053156B1 Search query results based upon topic
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-11-26, granted 2015-06-09 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8996554B2 Contextual query revision
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-12-05, granted 2015-03-31 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US9396235B1 Search ranking based on natural language query patterns
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-12-13, granted 2016-07-19 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US20150186377A1 Dynamically Sharing Intents
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-12-27 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US9116993B2 System and method for query re-issue in search engines

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2013-12-27, granted 2015-08-25 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9361406B1 Query completions

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2013-12-27, granted 2016-06-07 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9536522B1 Training a natural language processing model with information retrieval model annotations

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2013-12-30, granted 2017-01-03 · leak modules: CompressedQualitySignals, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9460209B1 Identifying non-search actions based on a search query

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2014-01-08, granted 2016-10-04 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US20140129538A1 User interface for query engine

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2014-01-09 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US9165033B1 Efficient query rewriting

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2014-01-13, granted 2015-10-20 · leak modules: NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9208260B1 Query suggestions with high diversity

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2014-01-13, granted 2015-12-08 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData

US9552388B2 System and method for providing search query refinements

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.

filed 2014-01-31, granted 2017-01-24 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo

US10534808B2 Architecture for responding to visual query

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.

filed 2014-02-18, granted 2020-01-14 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation

US20150261880A1 Techniques for translating user interfaces of web-based applications

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2014-03-15 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, RichsnippetsPageMap

US9183277B1 Providing intent sensitive search results
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-03-27, granted 2015-11-10 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US9524347B1 Automatically implementing an application in response to a search query
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2014-04-01, granted 2016-12-20 · leak modules: NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9104764B1 Session-based query suggestions
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2014-04-02, granted 2015-08-11 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US9547424B1 Querying publisher goal and customizing suggestions to achieve goal
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2014-04-02, granted 2017-01-17 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations

US9471709B1 Processing autocomplete suggestions
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-04-28, granted 2016-10-18 · leak modules: IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate

US9146964B2 Providing results to parameterless search queries
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-05-06, granted 2015-09-29 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US9721014B2 System and method of processing database queries
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-05-08, granted 2017-08-01 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US9734246B1 Using “sign in with” status as an oracle for selecting third party intents
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-05-09, granted 2017-08-15 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US9830391B1 Query modification based on non-textual resource context
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2014-06-24, granted 2017-11-28 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US9251262B1 Identifying media queries
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-06-27, granted 2016-02-02 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9852224B2 Promoting preferred content in a search query
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2014-07-03, granted 2017-12-26 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US10169458B1 Using a query log from a two-box interface to identify standalone locations
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-07-14, granted 2019-01-01 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US10963909B2 Handling search queries
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-07-29, granted 2021-03-30 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US9483530B1 Determining query terms of little significance
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-08-04, granted 2016-11-01 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US9690847B2 Selecting content using query-independent scores of query segments
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-08-07, granted 2017-06-27 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10318586B1 Systems and methods for editing and replaying natural language queries
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-08-19, granted 2019-06-11 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate

US9619582B1 Interactive search querying
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2014-08-20, granted 2017-04-11 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US9659097B1 Propagating query classifications
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2014-08-21, granted 2017-05-23 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US9129016B1 Methods and apparatus for providing query parameters to a search engine
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2014-09-15, granted 2015-09-08 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures

US9547690B2 Query rewriting using session information
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-09-15, granted 2017-01-17 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9378272B1 Determining correction of queries with potentially inaccurate terms
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active.
filed 2014-09-19, granted 2016-06-28 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9152721B1 Prompt for query clarification
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2014-10-21, granted 2015-10-06 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US9436781B2 Method and system for autocompletion for languages having ideographs and phonetic characters
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-10-27, granted 2016-09-06 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US9740780B1 Autocompletion using previously submitted query data
Rule: Optimize for branded and navigational queries that users have previously engaged with positively — these will appear in autocompletion. Create content that generates long clicks (users stay on your page) for queries related to your brand, as this positive dwell-time data feeds back into autocompletion rankings.
filed 2014-12-01, granted 2017-08-22 · leak modules: QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9501583B2 Referent based search suggestions
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.
filed 2015-01-05, granted 2016-11-22 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, RepositoryWebrefDocumentMetadata

US9779179B2 Referent based search suggestions
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.
filed 2015-01-05, granted 2017-10-03 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, RepositoryWebrefDocumentMetadata

US9372920B2 Identifying textual terms in response to a visual query
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2015-01-13, granted 2016-06-21 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation

US10296645B2 Systems and methods for content selection based on search query correlation with broadcast media
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2015-02-13, granted 2019-05-21 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US9477767B1 Demotion of already observed search query completions
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2015-02-18, granted 2016-10-25 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US10169488B2 Methods, systems, and media for providing search suggestions based on content ratings of search results
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2015-02-20, granted 2019-01-01 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US20160246805A1 Methods, systems, and media for providing search suggestions
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2015-02-20 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US9390183B1 Identifying navigational resources for informational queries
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2015-02-25, granted 2016-07-12 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US20150169589A1 Adjusting Result Rankings For Broad Queries
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-02-26 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US9305113B2 Contextual query revision
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-03-09, granted 2016-04-05 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US10353964B2 Evaluating semantic interpretations of a search query
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-03-11, granted 2019-07-16 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US9501571B1 Category generalization for search queries
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Write for intent and meaning. Neural matching connects related concepts.
filed 2015-03-16, granted 2016-11-22 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9507805B1 Drawing based search queries
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-04-13, granted 2016-11-29 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US10204104B2 Methods, systems, and media for processing queries relating to presented media content
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-04-14, granted 2019-02-12 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations

US9542476B1 Refining search queries
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2015-04-22, granted 2017-01-10 · leak modules: NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefDocumentMetadata

US9563692B1 Providing result-based query suggestions
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-04-24, granted 2017-02-07 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US10635676B2 Algorithmic radio for arbitrary textual queries
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-05-07, granted 2020-04-28 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US10474671B2 Interpreting user queries based on nearby locations

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2015-05-08, granted 2019-11-12 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US20160335275A1 Privacy-sensitive query for localization area description file

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2015-05-11 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData

US9342565B1 Variable query generation

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2015-05-15, granted 2016-05-17 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9594806B1 Detecting name-triggering queries

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2015-05-21, granted 2017-03-14 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US10152543B1 Generating content labels according to user search queries

Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Write for intent and meaning. Neural matching connects related concepts.

filed 2015-06-02, granted 2018-12-11 · leak modules: NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, SmartphonePerDocData

US9697249B1 Estimating confidence for query revision models

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2015-06-12, granted 2017-07-04 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US9959354B2 Utilizing user co-search behavior to identify search queries seeking inappropriate content

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2015-06-23, granted 2018-05-01 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US9436747B1 Query generation using structural similarity between documents

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.

filed 2015-06-25, granted 2016-09-06 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9489378B1 Parsing rule generalization by N-gram span clustering

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.

filed 2015-07-06, granted 2016-11-08 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9858358B1 Session-based query suggestions

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.

filed 2015-07-10, granted 2018-01-02 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US9430585B2 Media object query submission and response
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2015-07-13, granted 2016-08-30 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US9626454B1 Local query suggestions
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-07-13, granted 2017-04-18 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US9275034B1 Exceptions to action invocation from parsing rules
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-07-22, granted 2016-03-01 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US9916366B1 Query augmentation
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active.
filed 2015-07-28, granted 2018-03-13 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US9727603B1 Query refinements using search data
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-07-30, granted 2017-08-08 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9589071B2 Query suggestions from documents
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2015-08-03, granted 2017-03-07 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9646050B1 Methods, systems, and media for interpreting queries
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-08-03, granted 2017-05-09 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityOrbitAsteroidBeltDocumentIntentScores

US10719667B1 Providing a natural language based application program interface
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-08-05, granted 2020-07-21 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US10019485B2 Search query based form populator
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-08-14, granted 2018-07-10 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityWebanswersVideoTranscriptAnnotations

US10185746B2 Interpreting user queries based on device orientation
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2015-08-20, granted 2019-01-22 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US9430466B1 Techniques for crowd sourcing human translations to provide translated versions of web pages with additional content

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2015-08-26, granted 2016-08-30 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US20170068683A1 Context based instant search suggestions

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2015-09-04 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US20170068720A1 Systems and methods for classifying data queries based on responsive data sets

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2015-09-04 · leak modules: NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US10366422B2 Quality visit measure for controlling computer response to query associated with physical location

Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2015-09-09, granted 2019-07-30 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, RepositoryAnnotationsGeoTopicality

US9323866B1 Query completions in the context of a presented document

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2015-09-09, granted 2016-04-26 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10169421B1 Automatic user-based query generation and execution

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2015-09-15, granted 2019-01-01 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9665652B2 Providing results to parameterless search queries

Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2015-09-25, granted 2017-05-30 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US10146829B2 Query composition system

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2015-09-28, granted 2018-12-04 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US9940398B1 Customization of search results for search queries received from third party sites

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2015-10-06, granted 2018-04-10 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US11169989B1 Ranking modifications of a previous query

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2015-10-13, granted 2021-11-09 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US10685017B1 Methods and systems for efficient query rewriting
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2015-10-16, granted 2020-06-16 · leak modules: NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9514223B1 Synonym identification based on categorical contexts
Rule: Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2015-10-26, granted 2016-12-06 · leak modules: PerDocData, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US10204138B1 Navigational resources for queries
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-11-10, granted 2019-02-12 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US9953185B2 Identifying query patterns and associated aggregate statistics among search queries
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-11-24, granted 2018-04-24 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9842167B2 Search suggestion and display environment
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2015-12-01, granted 2017-12-12 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US20170177738A1 Dynamic intent registry
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-12-21 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9552398B1 Presenting search query results
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2016-01-13, granted 2017-01-24 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData

US11151129B1 Modifying query in discourse context
Rule: Optimize for multi-query sessions by ensuring your content covers contextual attributes (time, location, weather) that Google can inject into follow-up queries. Structure content to answer discourse-contextual queries—e.g., if a user searches 'weather in Atlanta' then 'restaurants,' ensure your restaurant page includes Atlanta location signals so it ranks for the discourse-modified query.
filed 2016-02-04, granted 2021-10-19 · leak modules: KnowledgeAnswersIntentQueryArgumentProvenancePreviousQuery, KnowledgeAnswersIntentQueryArgumentProvenancePreviousResponseMeaning, KnowledgeAnswersIntentQueryArgumentProvenancePreviousTaskState, KnowledgeAnswersIntentQueryCoreference, KnowledgeAnswersIntentQuerySemanticAnnotationSignals, QualityGenieComplexQueriesComplexQueriesOutputRewrite, NlpSemanticParsingQRefAnnotationEntityRelationship

US10496718B2 State-dependent query response
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2016-02-05, granted 2019-12-03 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores

US10599729B2 State-dependent query response
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2016-02-05, granted 2020-03-24 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality

US20170242914A1 Customized Query-Action Mappings for an Offline Grammar Model
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2016-02-24 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US9524336B1 Search query auto-completions based on social graph
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2016-02-26, granted 2016-12-20 · leak modules: PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9684693B1 On-device query rewriting
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2016-04-05, granted 2017-06-20 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo

US10222957B2 Keyboard with a suggested search query region
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2016-04-20, granted 2019-03-05 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo

US10305828B2 Search query predictions by a keyboard
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2016-04-20, granted 2019-05-28 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US9639581B1 Query completions
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2016-06-06, granted 2017-05-02 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US10474719B2 Auto-completion of widely shared search queries
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2016-06-08, granted 2019-11-12 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US9953076B2 Predictive query completion and predictive search results
Rule: Since Google proactively fetches results for predicted queries before users finish typing, optimize for partial query matches and common query stems. Ensure high click-through rates on your results when they appear for query suggestions, as valid prediction rate (selection/impression ratio) directly affects whether your content gets pre-fetched. Build authority signals so your pages rank well in the authority graph, as this is an explicit quality factor for predictive results.
filed 2016-07-14, granted 2018-04-24 · leak modules: QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, PerDocData

US10467256B2 Automatic query pattern generation
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2016-08-03, granted 2019-11-05 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US10423684B2 Generating travel queries in response to free-text search queries
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2016-08-12, granted 2019-09-24 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9727558B2 Techniques for crowd sourcing human translations to provide translated versions of web pages with additional content

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2016-08-17, granted 2017-08-08 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9917904B1 Identifying non-search actions based on a search-query

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-09-02, granted 2018-03-13 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US9875297B2 Associating a task with a user based on user selection of a query suggestion

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-10-04, granted 2018-01-23 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US9720955B1 Search query predictions by a keyboard

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2016-10-10, granted 2017-08-01 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US10635717B2 Query suggestion templates

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-10-18, granted 2020-04-28 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10223439B1 Systems and methods for providing search query refinements

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2016-11-10, granted 2019-03-05 · leak modules: CompressedQualitySignals, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US10437887B1 Determining intent of text entry

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-12-08, granted 2019-10-08 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US10409459B2 Querying publisher goal and customizing suggestions to achieve goal

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.

filed 2016-12-20, granted 2019-09-10 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations

US10410270B2 Granular selection and scheduling of queries

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-12-22, granted 2019-09-10 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US10915580B2 Providing query suggestions and/or search results based on determined user activity

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-12-29, granted 2021-02-09 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate

US20180196854A1 Application extension for generating automatic search queries
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-01-11 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US10387437B2 Query rewriting using session information
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2017-01-13, granted 2019-08-20 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US10031941B1 Structured search query generation and use in a computer network environment
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2017-01-18, granted 2018-07-24 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10459989B1 Providing result-based query suggestions
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2017-02-02, granted 2019-10-29 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US10585927B1 Determining a set of steps responsive to a how-to query
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2017-03-02, granted 2020-03-10 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData

US10545954B2 Determining search queries for obtaining information during a user experience of an event
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-03-15, granted 2020-01-28 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US10621253B2 State-dependent query response
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2017-03-16, granted 2020-04-14 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality

US20180032741A1 Privacy aware intent resolution with external sources
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-03-24 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US10698901B1 Query completions
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2017-03-27, granted 2020-06-30 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US10678871B2 Accelerating find in page queries within a web browser
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2017-04-18, granted 2020-06-09 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality

US10394824B2 Providing results to parameterless search queries
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2017-04-27, granted 2019-08-27 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US11210289B2 Methods, systems, and media for interpreting queries

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2017-05-05, granted 2021-12-28 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityOrbitAsteroidBeltDocumentIntentScores

US10346493B1 On-device query rewriting

Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.

filed 2017-05-15, granted 2019-07-09 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo

US10909124B2 Predicting intent of a search for a particular context

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2017-05-18, granted 2021-02-02 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US9875738B2 Query rewrite corrections

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2017-06-05, granted 2018-01-23 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US10691680B1 Query refinements using search data

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2017-06-08, granted 2020-06-23 · leak modules: CompressedQualitySignals, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US10346463B2 Hybrid use of location sensor data and visual query to return local listings for visual query

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.

filed 2017-06-15, granted 2019-07-09 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation

US10489399B2 Query language identification

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2017-07-11, granted 2019-11-26 · leak modules: NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US10740369B2 Predictive generation of search suggestions

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.

filed 2017-07-12, granted 2020-08-11 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US10572561B1 Performing multiple related searches

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2017-07-21, granted 2020-02-25 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10049109B2 Techniques for crowd sourcing human translations to provide translated versions of web pages with additional content

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2017-08-04, granted 2018-08-14 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, VideoPerDocData

US10592571B1 Query modification based on non-textual resource context
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2017-10-23, granted 2020-03-17 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US20180113919A1 Graphical user interface rendering predicted query results to unstructured queries
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-10-24 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US10423678B1 Query language filter for cross-language information retrieval
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-10-30, granted 2019-09-24 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US10846346B2 Search suggestion and display environment
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2017-11-10, granted 2020-11-24 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US10776407B2 Associating a task with a user based on user selection of a query suggestion
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-11-28, granted 2020-09-15 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US11487806B2 Media item matching using search query analysis
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-12-12, granted 2022-11-01 · leak modules: IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10270862B1 Identifying non-search actions based on a search query
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2018-01-31, granted 2019-04-23 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US10929487B1 Customization of search results for search queries received from third party sites
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2018-02-22, granted 2021-02-23 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US11620318B2 Predictive query completion and predictive search results
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2018-03-08, granted 2023-04-04 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US10296510B2 Search query based form populator
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2018-03-30, granted 2019-05-21 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityWebanswersVideoTranscriptAnnotations

US10509830B2 Rich results relevant to user search queries
Rule: Rich results require your page to be the dominant result for its query (score must significantly exceed competitors). Ensure structured data is accurate and consistent across your site and external sources (Google Books, Wikipedia, etc.). Conflicting data between sources may cause Google to select the 'most popular variant' - maintain consistency everywhere.
filed 2018-05-08, granted 2019-12-17 · leak modules: structured_data, rich_results, ranking, knowledge_graph, entity

US10754850B2 Query composition system

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2018-10-17, granted 2020-08-25 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US10380160B2 Dynamic language model

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2018-11-26, granted 2019-08-13 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US11593432B2 Methods, systems, and media for providing search suggestions based on content ratings of search results

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.

filed 2018-12-31, granted 2023-02-28 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US10922321B2 Interpreting user queries based on device orientation

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2019-01-07, granted 2021-02-16 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US10977320B2 Systems and methods for cancelling a query

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2019-01-29, granted 2021-04-13 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData

US11429665B2 Consolidation of responses from queries to disparate data sources

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.

filed 2019-02-06, granted 2022-08-30 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10984038B2 Methods, systems, and media for processing queries relating to presented media content

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2019-02-11, granted 2021-04-20 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations

US10521479B2 Evaluating semantic interpretations of a search query

Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2019-05-20, granted 2019-12-31 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US11366812B2 Using live data streams and/or search queries to determine information about developing events

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2019-06-25, granted 2022-06-21 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US11120090B2 On-device query rewriting

Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.

filed 2019-07-08, granted 2021-09-14 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo

US10803067B2 Providing results to parameterless search queries

Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2019-07-26, granted 2020-10-13 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US11334916B2 Providing advertisements from related search queries
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2019-07-31, granted 2022-05-17 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US11361041B2 Generating travel queries in response to free-text search queries
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2019-07-31, granted 2022-06-14 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US11397784B2 Structured user graph to support querying and predictions
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2019-08-14, granted 2022-07-26 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US20210334314A1 Sibling search queries
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2019-08-21 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US11868417B2 Identification and issuance of repeatable queries
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2019-11-06, granted 2024-01-09 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US11514896B2 Interfacing with applications via dynamically updating natural language processing
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2019-11-27, granted 2022-11-29 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US20200211537A1 Scalable dynamic class language modeling
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2019-12-31 · leak modules: NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US11580181B1 Query modification based on non-textual resource context
Rule: Optimize images with descriptive alt text and labels that clearly identify entities depicted, as these labels and entity descriptions can be used by Google to disambiguate spoken queries in visual contexts. Ensure video content includes structured data and readable text overlays, since OCR-extracted text from video frames feeds query modification. Structured data markup for entities in visual content increases the chance your content is surfaced when queries are modified based on visual context.
filed 2020-02-07, granted 2023-02-14 · leak modules: PerDocData, QualityNsrNsrData, RepositoryWebrefEntityJoin, QualityOrbitAsteroidBeltDocumentIntentScores

US11216522B2 State-dependent query response
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2020-03-05, granted 2022-01-04 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US11514035B1 Query refinements using search data
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2020-06-01, granted 2022-11-29 · leak modules: CompressedQualitySignals, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US20210390153A1 Formulated Query On Portable Device
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.
filed 2020-06-16 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US11693863B1 Query completions
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2020-06-29, granted 2023-07-04 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US11625392B2 Query composition system
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2020-07-17, granted 2023-04-11 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US11768864B2 Associating a task with a user based on user selection of a query suggestion
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2020-07-30, granted 2023-09-26 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US20200364219A1 Search query refinement
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Write in natural vocabulary. ML models understand context and fluff.
filed 2020-08-03 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, RepositoryWebrefDocumentMetadata

US11301501B2 Predictive generation of search suggestions
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2020-08-10, granted 2022-04-12 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US12045278B2 Intelligent systems and methods for visual search queries
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2020-09-18, granted 2024-07-23 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation

US11494427B2 Generating a personal database entry for a user based on natural language user interface input of the user and generating output based on the entry in response to further natural language user interface input of the user
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2020-09-21, granted 2022-11-08 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US12216983B2 Document mark-up and navigation using natural language processing
Rule: Use consistent entity names, sameAs schema, and structured data. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2020-09-29, granted 2025-02-04 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US11675794B2 Providing results to parameterless search queries
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2020-10-09, granted 2023-06-13 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US20210026846A1 Querying a data graph using natural language queries
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2020-10-13 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11790005B2 Methods and systems for presenting privacy friendly query activity based on environmental signal(s)

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2020-11-30, granted 2023-10-17 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US11461342B2 Predicting intent of a search for a particular context

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2020-12-10, granted 2022-10-04 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US11526570B2 Page-based prediction of user intent

Rule: Use consistent entity names, sameAs schema, and structured data. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2020-12-30, granted 2022-12-13 · leak modules: NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US11605107B2 Handling search queries

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2021-02-26, granted 2023-03-14 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US12204569B2 Natural language understanding clarifications

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2021-03-05, granted 2025-01-21 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US11942094B2 Hybrid multilingual text-dependent and text-independent speaker verification

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2021-03-24, granted 2024-03-26 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US11556599B2 Systems and methods for cancelling a query

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2021-04-02, granted 2023-01-17 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData

US12216700B2 Methods, systems, and media for processing queries relating to presented media content

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2021-04-19, granted 2025-02-04 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations

US11983175B1 Ranking modifications of a previous query

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2021-10-08, granted 2024-05-14 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US11836171B2 Demand based dynamic creation of data analytics query accelerators

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2021-10-15, granted 2023-12-05 · leak modules: CompressedQualitySignals, PerDocData, SmartphonePerDocData

US12147482B1 Modifying query in discourse context
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2021-10-18, granted 2024-11-19 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US12001429B2 Methods, systems, and media for interpreting queries
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2021-12-23, granted 2024-06-04 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityOrbitAsteroidBeltDocumentIntentScores

US12158917B2 State-dependent query response
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2021-12-29, granted 2024-12-03 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality

US11960547B2 Single view presentation of multiple queries in a data visualization application
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2022-01-11, granted 2024-04-16 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US20220171942A1 Natural Language Processing with an N-Gram Machine
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2022-02-15 · leak modules: NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata

US11893061B2 Systems and methods for editing and replaying natural language queries
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2022-02-18, granted 2024-02-06 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate

US11914627B1 Parsing natural language queries without retraining
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2022-04-11, granted 2024-02-27 · leak modules: IndexingDocjoinerAnchorSpamInfo, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefDocumentMetadata

US12554750B2 Predictive generation of search suggestions
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2022-04-12, granted 2026-02-17 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US11854044B2 Providing advertisements from related search queries
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2022-04-19, granted 2023-12-26 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US11574131B2 Machine-learned language models which generate intermediate textual analysis in service of contextual text generation
Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2022-05-20, granted 2023-02-07 · leak modules: NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations

US11748402B2 Consolidation of responses from queries to disparate data sources

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.

filed 2022-06-14, granted 2023-09-05 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US12124457B2 Triggering local extensions based on inferred intent

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2022-06-22, granted 2024-10-22 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US20240028661A1 Augmented ui generated through natural language query and semantic data connections

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2022-07-21 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US11688191B2 Contextually disambiguating queries

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2022-09-09, granted 2023-06-27 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US12147470B2 Handling contradictory queries on a shared device

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2022-10-06, granted 2024-11-19 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo

US11907276B2 Generating a personal database entry for a user based on natural language user interface input of the user and generating output based on the entry in response to further natural language user interface input of the user

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2022-10-12, granted 2024-02-20 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US12141132B1 Query refinements using search data

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2022-11-28, granted 2024-11-12 · leak modules: CompressedQualitySignals, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US12013911B2 Page-based prediction of user intent

Rule: Use consistent entity names, sameAs schema, and structured data. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2022-12-09, granted 2024-06-18 · leak modules: NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US20230214094A1 Methods and apparatus for related search within a browser

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2022-12-16 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US12056196B2 Systems and methods for cancelling a query

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2023-01-04, granted 2024-08-06 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData

US12067040B2 Determining responsive content for a compound query based on a set of generated sub-queries

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2023-01-30, granted 2024-08-20 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US12579136B2 Semantic parsing using embedding space representations of example natural language queries

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2023-01-31, granted 2026-03-17 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US12026194B1 Query modification based on non-textual resource context

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.

filed 2023-02-13, granted 2024-07-02 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US12093317B2 Methods, systems, and media for providing search suggestions based on content ratings of search results

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.

filed 2023-02-27, granted 2024-09-17 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US12554786B2 Matching search queries to application content

Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Write for intent and meaning. Neural matching connects related concepts.

filed 2023-03-21, granted 2026-02-17 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US12013846B2 Query composition system

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2023-03-29, granted 2024-06-18 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US12566784B2 Predictive query completion and predictive search results

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.

filed 2023-03-31, granted 2026-03-03 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US20230259537A1 Subquery generation from a query

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2023-04-28 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US20230281193A1 Generating query variants using a trained generative model

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2023-05-12 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US12591605B2 Computer session management using a language model

Rule: Ensure your page content is easily summarizable by language models—use clear headings, structured content, and descriptive titles. Since Google's language model summarizes tab content based on engagement signals, pages with high engagement get summarized and cached in context. Optimize page titles and meta descriptions for natural language queries about tab content. Structured, scannable content is more likely to be accurately summarized.

filed 2023-06-09, granted 2026-03-31 · leak modules: KnowledgeAnswersIntentQueryArgumentProvenanceInjectedContextualSchema, KnowledgeAnswersIntentQueryArgumentProvenanceCurrentQuery, KnowledgeAnswersIntentQueryArgumentProvenancePreviousTaskState, HtmlrenderWebkitHeadlessProtoDOMTreeNodeAttribute, HtmlrenderWebkitHeadlessProtoImage, PerDocData

US12050613B1 Query completions
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2023-07-03, granted 2024-07-30 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US12373487B2 Consolidation of responses from queries to disparate data sources
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2023-07-24, granted 2025-07-29 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US12380169B2 Methods and systems for presenting privacy friendly query activity based on environmental signal(s)
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2023-08-29, granted 2025-08-05 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US11907674B1 Generating multi-modal response(s) through utilization of large language model(s)
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2023-09-20, granted 2024-02-20 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US20250103826A1 Processing documents in cloud storage using query embeddings
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2023-09-21 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US20260064743A1 Systems and Methods for Prompt-Based Query Generation for Diverse Retrieval
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2023-09-21 · leak modules: CompressedQualitySignals, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US12164546B2 Associating a task with a user based on user selection of a query suggestion
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2023-09-25, granted 2024-12-10 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US20250103648A1 Method and system for understanding and fulfilling verb-based succinct queries
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2023-09-26 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo

US12554942B2 Large language model-based responses to targeted UI elements
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2023-10-04, granted 2026-02-17 · leak modules: NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US20240135187A1 Method for Training Large Language Models to Perform Query Intent Classification
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2023-10-23 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US20250139379A1 Generating multi-modal response(s) through utilization of large language model(s) and other generative model(s)

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2023-10-30 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US12292941B2 Identification and issuance of repeatable queries

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2023-11-22, granted 2025-05-06 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US20250173519A1 Query Intent Understanding and Search Result Generation

Rule: Optimize for intent clusters rather than just keywords—Google maps your content to embedding clusters based on query intent. Create comprehensive topic clusters that cover related intents, so your content maps to the right embedding cluster. Since the model uses multi-turn context, ensure your content answers progressive questions within a topic. Use structured data to help Google understand entity relationships and intent attributes. Multimodal content (text + images) is processed together for intent matching.

filed 2023-11-29 · leak modules: QualityRankembedMustangMustangRankEmbedInfo, QualityRankembedMustangMustangRankEmbedInfoCompressedEmbedding, KnowledgeAnswersIntentQueryParsingSignals, KnowledgeAnswersIntentQueryConceptSignals, KnowledgeAnswersIntentQueryRelatednessSignals, KnowledgeAnswersIntentQueryGroundingSignals, RepositoryWebrefEntityAnnotations, NlpSaftEntity

US20240362257A1 Enterprise Application Query Enrichment For Language Models

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2023-12-06 · leak modules: NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US12437016B2 Fine-tuning large language model(s) using reinforcement learning with search engine feedback

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2023-12-07, granted 2025-10-07 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US20250200294A1 System And Methods For Multi-User Large Language Model Execution

Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Structure answers: question heading, tight answer, then depth.

filed 2023-12-15 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US12142277B2 Generating a composite response to natural language input using a trained generative model and based on responsive content from disparate content agents

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2024-01-08, granted 2024-11-12 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US20250217423A1 Systems and Methods for Generating Benchmark Queries

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.

filed 2024-03-11 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US12361035B1 Ranking modifications of a previous query

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2024-04-15, granted 2025-07-15 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US20240377932A1 Language model assistance provided by an operating system

Rule: Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2024-05-09 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US20250036670A1 Large Language Models in Cloud Database Platforms
Rule: Use consistent entity names, sameAs schema, and structured data. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2024-06-07 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US12321401B1 Multimodal query prediction
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2024-06-10, granted 2025-06-03 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US20250165543A1 Automatic generation of facets using a large language model
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2024-06-12 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US12339896B2 Intelligent systems and methods for visual search queries
Rule: Ensure product images are clearly recognizable by object detection models — use clean backgrounds, proper lighting, and clear product framing. Structure product data with rich entity associations (reviews, related items) to appear in visual search interest graphs. Optimize images for OCR if they contain text.
filed 2024-06-14, granted 2025-06-24 · leak modules: RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, QualitySalientTermsSalientTermSet, PerDocData

US12468692B2 Query composition system
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2024-06-14, granted 2025-11-11 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US12499152B1 Query modification based on non-textual resource context
Rule: Optimize images with descriptive alt text, captions, and structured data labels — these may be extracted to modify voice queries. Ensure entity recognition signals are strong in your visual content (schema markup, descriptive filenames, surrounding text). For visual-first platforms, ensure your images clearly convey entity information that could disambiguate voice queries.
filed 2024-06-14, granted 2025-12-16 · leak modules: PerDocData, RepositoryWebref

US12530420B2 Page-based prediction of user intent
Rule: Use consistent entity names, sameAs schema, and structured data. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2024-06-14, granted 2026-01-20 · leak modules: NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US20260023951A1 Methods and systems for customized query responses using artificial intelligence
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2024-07-22 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US12405961B1 Query completions
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2024-07-29, granted 2025-09-02 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US20250094456A1 Large language model output entailment
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2024-09-17 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US12608369B2 Combining parameters of multiple search queries that share a line of inquiry
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2025-01-31, granted 2026-04-21 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US20260023940A1 Multilingual generative model(s)
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2025-07-21 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsGeoTopicality

SERP & browsing UI (438)

US8600963B2 System and method for presenting multiple sets of search results for a single query
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2003-08-14, granted 2013-12-03 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8140965B2 Identifying related information given content and/or presenting related information in association with content-related advertisements
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2003-12-29, granted 2012-03-20 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US7447678B2 Interface for a universal search engine
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2003-12-31, granted 2008-11-04 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US7424478B2 System and method for selecting content for displaying over the internet based upon some user input
Rule: Monitor this patent for ranking implications.
filed 2004-03-10, granted 2008-09-09

US7272601B1 Systems and methods for associating a keyword with a user interface area
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2004-03-31, granted 2007-09-18 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US7558822B2 Accelerating user interfaces by predicting user actions
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2004-06-30, granted 2009-07-07 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US7853606B1 Alternate methods of displaying search results
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2004-09-14, granted 2010-12-14 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US9031898B2 Presentation of search results based on document structure
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2004-09-27, granted 2015-05-12 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US8364668B2 User interfaces for a document search engine
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2004-09-29, granted 2013-01-29 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, RepositoryWebrefEntityJoin

US7664751B2 Variable user interface based on document access privileges
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2004-09-30, granted 2010-02-16 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures

US7694212B2 Systems and methods for providing a graphical display of search activity
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2005-03-31, granted 2010-04-06 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US7805495B2 Method and system for transferring web browser data between web browsers
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2005-03-31, granted 2010-09-28 · leak modules: PerDocData, SmartphonePerDocData

US8515816B2 Aggregate analysis of text captures performed by multiple users from rendered documents
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2005-04-01, granted 2013-08-20 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8713418B2 Adding value to a rendered document
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2005-04-12, granted 2014-04-29 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation

US8583627B1 Display-content alteration for user interface devices
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2005-07-01, granted 2013-11-12 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US7548928B1 Data compression of large scale data stored in sparse tables
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2005-08-05, granted 2009-06-16 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US8166028B1 Method, system, and graphical user interface for improved searching via user-specified annotations
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2005-09-15, granted 2012-04-24 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8990193B1 Method, system, and graphical user interface for improved search result displays via user-specified annotations
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2005-09-15, granted 2015-03-24 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US8949154B2 Content feed user interface with gallery display of same-type items
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use descriptive alt text and surrounding context for images.
filed 2005-10-07, granted 2015-02-03 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityWebanswersVideoTranscriptAnnotations

US7925649B2 Method, system, and graphical user interface for alerting a computer user to new results for a prior search
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2005-12-30, granted 2011-04-12 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US8010523B2 Dynamic search box for web browser
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2005-12-30, granted 2011-08-30 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US8407665B1 Rendering contextual related content with a document, such as on a web page for example

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2005-12-30, granted 2013-03-26 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData

US7739594B2 Encoding and displaying default landing page content

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2006-08-17, granted 2010-06-15 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9569556B2 Software architecture for displaying information content from plug-in modules in a user interface

Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships.

filed 2006-08-21, granted 2017-02-14 · leak modules: ImageData, ImagePerDocData, PerDocData, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US7890499B1 Presentation of search results with common subject matters

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use descriptive alt text and surrounding context for images.

filed 2006-12-01, granted 2011-02-15 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US9703889B2 Providing auto-focus for a search field in a user interface

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2007-02-14, granted 2017-07-11 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8631440B2 Program guide user interface

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2007-04-30, granted 2014-01-14 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US9256676B2 Presenting search result information

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2007-05-10, granted 2016-02-09 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorStatistics, NlpSemanticParsingQRefAnnotation, PerDocData

US8429247B1 Method and apparatus for fulfilling browser requests from a local cache

Rule: Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2007-05-30, granted 2013-04-23 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, RepositoryAnnotationsGeoTopicality

US8245130B1 Performing an estimation on content to be presented

Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2007-05-31, granted 2012-08-14 · leak modules: ImageData, ImagePerDocData, PerDocData, SmartphonePerDocData

US7873632B2 Systems and methods for associating a keyword with a user interface area

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2007-08-06, granted 2011-01-18 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US9084025B1 System and method for displaying both multimedia events search results and internet search results

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2007-08-06, granted 2015-07-14 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US8060634B1 Determining and displaying a count of unread items in content feeds

Rule: Structure answers: question heading, tight answer, then depth.

filed 2007-09-26, granted 2011-11-15 · leak modules: QualityPreviewSnippetRadishFeatures, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US7836041B1 System and method for displaying both time information search results and internet search results

Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2008-03-10, granted 2010-11-16 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality

US8694526B2 Apparatus and method for displaying search results using tabs

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2008-07-24, granted 2014-04-08 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US8301656B2 Interface for a universal search

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2008-08-28, granted 2012-10-30 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US8146156B2 Archive of text captures from rendered documents

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2008-09-02, granted 2012-03-27 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US20100115613A1 Cacheable Mesh Browsers

Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately. Optimize video for watch time and retention.
filed 2008-10-31 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, VideoPerDocData

US8352967B2 Safe browser plugins using native code modules

Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Earn endorsements from trusted sources. Trust is computed and transitive.
filed 2008-11-10, granted 2013-01-08 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US20100185651A1 Retrieving and displaying information from an unstructured electronic document collection

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Consolidate near-duplicates. Use canonical tags and consistent URLs. Write in natural vocabulary. ML models understand context and fluff.
filed 2009-01-16 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US8412749B2 Populating a structured presentation with new values

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2009-01-16, granted 2013-04-02 · leak modules: NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8977645B2 Accessing a search interface in a structured presentation

Rule: Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2009-01-16, granted 2015-03-10 · leak modules: ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US8745018B1 Search application and web browser interaction

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2009-07-10, granted 2014-06-03 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US8442331B2 Capturing text from rendered documents using supplemental information

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2009-08-18, granted 2013-05-14 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8447144B2 Data capture from rendered documents using handheld device

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2009-08-18, granted 2013-05-21 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8417892B1 Differential storage and eviction for information resources from a browser cache
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships.
filed 2009-08-31, granted 2013-04-09 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US8762879B1 Tab management in a browser
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2009-09-01, granted 2014-06-24 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US20150193554A1 System and method for selecting information for display
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2009-10-05 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9323426B2 System and method for selecting information for display based on past user interactions
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2009-10-05, granted 2016-04-26 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8332748B1 Multi-directional auto-complete menu
Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately. Write for natural spoken language and conversational queries.
filed 2009-10-22, granted 2012-12-11 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, SmartphonePerDocData

US8706802B1 Latency-guided web content retrieval, serving, and rendering
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2009-11-24, granted 2014-04-22 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8364712B2 Variable user interface based on document access privileges
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2009-11-30, granted 2013-01-29 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality

US8392841B1 Web page composition tool with browser page size guide
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2009-12-08, granted 2013-03-05 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US9875671B2 Cloud-based user interface augmentation
Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately. Build content for diverse user contexts and search histories.
filed 2009-12-17, granted 2018-01-23 · leak modules: NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData

US8407590B1 On-screen user-interface graphic
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2010-02-12, granted 2013-03-26 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData, SnippetExtraInfo

US20110145730A1 Utilization of Browser Space
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2010-03-19 · leak modules: PerDocData, SmartphonePerDocData

US8874504B2 Processing techniques for visual capture data from a rendered document
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2010-03-22, granted 2014-10-28 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8914813B1 Interfaces to enable cross-browser extensions and applications thereof
Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2010-04-09, granted 2014-12-16 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8812500B2 System and method of displaying related sites
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2010-04-23, granted 2014-08-19 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US8346755B1 Iterative off-line rendering process
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2010-05-04, granted 2013-01-01 · leak modules: CrawlerChangerateUrlChangerate, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, TrawlerFetchReplyData

US8781864B2 Anticipatory presentation of travel information
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2010-05-04, granted 2014-07-15 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US8793162B2 Adding information or functionality to a rendered document via association with an electronic counterpart
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2010-05-05, granted 2014-07-29 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US20150074561A1 Customizable themes for browsers and web content
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Write for natural spoken language and conversational queries.
filed 2010-05-07 · leak modules: NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8549019B2 Dynamically generating aggregate tables
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2010-05-25, granted 2013-10-01 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryWebrefEntityJoin, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8635205B1 Displaying local site name information with search results
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2010-06-18, granted 2014-01-21 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US8539372B1 Pre-scrolling a search results page
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2010-07-21, granted 2013-09-17 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores

US9170705B1 System and method for determining an interest in promotional content displayed at a mobile communication device
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2010-09-16, granted 2015-10-27 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US9203932B2 Optimization guided by connection classification in a web browser extension
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2010-09-28, granted 2015-12-01 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US8799303B2 Establishing an interactive environment for rendered documents
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2010-10-13, granted 2014-08-05 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8806187B1 Protecting browser-viewed content from piracy
Rule: Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately. Build content for diverse user contexts and search histories.
filed 2010-12-01, granted 2014-08-12 · leak modules: PerDocData, QualityNsrNsrData, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US8010525B1 Unified search interface
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use descriptive alt text and surrounding context for images.
filed 2011-01-12, granted 2011-08-30 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8543715B1 Redirect control in web browsers
Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2011-01-12, granted 2013-09-24 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8694491B2 Method, system, and graphical user interface for alerting a computer user to new results for a prior search
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-03-08, granted 2014-04-08 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US9069855B2 Modifying a hierarchical data structure according to a pseudo-rendering of a structured document by annotating and merging nodes
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-03-21, granted 2015-06-30 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US8560635B1 User experience of content rendering with time budgets
Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2011-03-30, granted 2013-10-15 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US20110252335A1 Zooming in a Web Browser
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-04-12 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8413045B2 Rich browser-based word processor
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-04-12, granted 2013-04-02 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US8438474B1 Speculative rendering during cache revalidation
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-04-27, granted 2013-05-07 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8682904B1 System of intuitive sorting of a table based on a column containing fractions
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2011-04-27, granted 2014-03-25 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8825747B2 Managing multiple logins from a single browser
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-05-06, granted 2014-09-02 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9691086B1 Adaptive content rendering
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-05-13, granted 2017-06-27 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US10580051B2 Background element associated with an application installed in a browser application
Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-05-18, granted 2020-03-03 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData

US20150205489A1 Browser interface for installed applications
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-05-18 · leak modules: PerDocData, SmartphonePerDocData

US8650481B1 Stable and secure use of content scripts in browser extensions
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-05-18, granted 2014-02-11 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8667487B1 Web browser extensions
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2011-05-18, granted 2014-03-04 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8732474B1 Safe installation of browser extensions
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-05-18, granted 2014-05-20 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9021468B1 Bundling extension installation with web browser installation
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2011-05-18, granted 2015-04-28 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8954427B2 Search result previews
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-05-31, granted 2015-02-10 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US8286076B1 Rendering approximate webpage screenshot client-side
Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-06-27, granted 2012-10-09 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData

US8464148B1 Intelligent editing of web pages in a web browser
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2011-06-29, granted 2013-06-11 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8744988B1 Predicting user navigation events in an internet browser
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-07-15, granted 2014-06-03 · leak modules: PerDocData, SmartphonePerDocData

US8892994B2 System, method, and architecture for displaying a document
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2011-08-10, granted 2014-11-18 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9436928B2 User graphical interface for displaying a belonging-related stream
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-08-30, granted 2016-09-06 · leak modules: ImageData, ImagePerDocData, PerDocData, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US8384726B1 Selective rendering of off-screen content
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth. Optimize video for watch time and retention.
filed 2011-08-31, granted 2013-02-26 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo, VideoPerDocData

US8775458B1 System and method for retrieving and displaying search results without navigating from a currently displayed window
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-08-31, granted 2014-07-08 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US20150019985A1 Systems and methods for customizing a web browser
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Build content for diverse user contexts and search histories.
filed 2011-09-02 · leak modules: PerDocData, QualityNsrNsrData, SmartphonePerDocData

US8725849B1 Browser cache pre-population
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-09-14, granted 2014-05-13 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData, TrawlerFetchReplyData

US8600921B2 Predicting user navigation events in a browser using directed graphs
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2011-09-15, granted 2013-12-03 · leak modules: PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8225197B1 Rendering approximate webpage screenshot client-side
Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-09-26, granted 2012-07-17 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData

US8314809B1 Selective rendering of off-screen content
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth. Optimize video for watch time and retention.
filed 2011-09-26, granted 2012-11-20 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo, VideoPerDocData

US8341245B1 Content-facilitated speculative preparation and rendering
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-09-26, granted 2012-12-25 · leak modules: PerDocData, SmartphonePerDocData

US8341519B1 Tab assassin
Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-09-26, granted 2012-12-25 · leak modules: ImageData, ImagePerDocData, PerDocData, SmartphonePerDocData

US8341543B2 Method and apparatus of scrolling a document displayed in a browser window
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-09-26, granted 2012-12-25 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData

US8412728B1 User interface (UI) for presentation of match quality in auto-complete suggestions
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-09-26, granted 2013-04-02 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9881096B1 Web page tab management based on location
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-09-26, granted 2018-01-30 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US8176321B1 Safe installation of browser extensions
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-09-27, granted 2012-05-08 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8200962B1 Web browser extensions
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2011-09-27, granted 2012-06-12 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8407584B1 Stable and secure use of content scripts in browser extensions
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-09-27, granted 2013-03-26 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9110568B2 Browser tab management
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Ensure content comprehensively covers the topic with natural vocabulary.
filed 2011-09-27, granted 2015-08-18 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US8856644B1 System, method, and architecture for displaying a document
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2011-09-28, granted 2014-10-07 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8522128B1 Systems and methods for modifying the order of links presented in a document
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-09-30, granted 2013-08-27 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US8484187B1 Enhanced presentation mode for search results
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-10-05, granted 2013-07-09 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US8191007B1 Organizing a browser environment
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Create original, effortful content. Quality is scored at page and site level.
filed 2011-10-07, granted 2012-05-29 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8650492B1 System and method for page navigation visualization and analysis
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-10-11, granted 2014-02-11 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US8935422B1 Embedded streams user interface
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-10-11, granted 2015-01-13 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefEntityJoin, SmartphonePerDocData

US8606955B1 Pause-button content rendering
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-10-21, granted 2013-12-10 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8745161B2 Determining and displaying a count of unread items in content feeds
Rule: Structure answers: question heading, tight answer, then depth.
filed 2011-10-27, granted 2014-06-03 · leak modules: QualityPreviewSnippetRadishFeatures, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8938441B2 Presenting search results for gallery web pages
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2011-10-28, granted 2015-01-20 · leak modules: ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8365061B1 Cross-origin communication in browser frames
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships.
filed 2011-11-02, granted 2013-01-29 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US8645360B2 Previewing search results
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-11-03, granted 2014-02-04 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8225191B1 Synchronizing web browsers
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2011-11-07, granted 2012-07-17 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US8984414B2 Function extension for browsers or documents
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Write for intent and meaning. Neural matching connects related concepts. Use JSON-LD structured data for entity types and relationships.
filed 2011-11-07, granted 2015-03-17 · leak modules: NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US8245125B1 Hybrid rendering for webpages
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-11-30, granted 2012-08-14 · leak modules: PerDocData, SmartphonePerDocData

US9395868B2 Graphical user interface window spacing mechanisms
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2011-12-06, granted 2016-07-19 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8726189B2 Multiple tab stack user interface
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-12-07, granted 2014-05-13 · leak modules: PerDocData, SmartphonePerDocData

US9201952B1 User interface for activity status and history
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-12-21, granted 2015-12-01 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US20150205454A1 Systems and methods for displaying preview data
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-01-10 · leak modules: PerDocData, SmartphonePerDocData

US9661073B2 Web browser synchronization with multiple simultaneous profiles
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Build content for diverse user contexts and search histories.
filed 2012-01-13, granted 2017-05-23 · leak modules: PerDocData, QualityNsrNsrData, SmartphonePerDocData

US8935755B1 Managing permissions and capabilities of web applications and browser extensions based on install location
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-02-06, granted 2015-01-13 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefEntityJoin, SmartphonePerDocData

US9672525B2 Identifying related information given content and/or presenting related information in association with content-related advertisements
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-02-13, granted 2017-06-06 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US8505090B2 Archive of text captures from rendered documents
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2012-02-20, granted 2013-08-06 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8291312B1 Displaying web pages without downloading static files
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2012-03-06, granted 2012-10-16 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US20150205766A1 Serving combined resources to a browser client
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-03-08 · leak modules: CrawlerChangerateUrlChangerate, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData, TrawlerFetchReplyData

US9891727B2 Simulating multi-touch events on a browser system
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-03-08, granted 2018-02-13 · leak modules: PerDocData, SmartphonePerDocData

US8549042B1 Systems and methods for sorting and displaying search results in multiple dimensions
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-03-16, granted 2013-10-01 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US9317605B1 Presenting forked auto-completions
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-03-21, granted 2016-04-19 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US8832275B1 Offline web browser
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-03-30, granted 2014-09-09 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9454531B1 Media content presentation by categorizing and formatting media types
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-04-03, granted 2016-09-27 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US9106762B2 Associating content with a graphical interface window using a fling gesture
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-04-04, granted 2015-08-11 · leak modules: PerDocData, SmartphonePerDocData

US20130268837A1 Method and system to manage interactive content display panels
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2012-04-10 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8898560B1 Fixing problems with a user interface
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-04-25, granted 2014-11-25 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8971642B2 Testing the rendering of visual content
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-04-26, granted 2015-03-03 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo

US9100697B1 Intelligent full window web browser transparency
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-04-30, granted 2015-08-04 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US20150212694A1 Internet browser zooming
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-05-02 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9678618B1 Using an expanded view to display links related to a topic
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-05-31, granted 2017-06-13 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, SnippetExtraInfo

US20150100878A1 Method and system for rendering a web page
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-06-05 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9367524B1 Systems and methods for selecting web page layouts including content slots for displaying content items based on predicted click likelihood
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-06-06, granted 2016-06-14 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US9043441B1 Methods and systems for providing network content for devices with displays having limited viewing area
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-06-13, granted 2015-05-26 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US9146906B1 Modifying a webpage display based on a referrer web address
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-06-19, granted 2015-09-29 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9417759B1 Synchronizing data across multiple browser tabs or windows
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-06-19, granted 2016-08-16 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9003310B2 Modifying running of plug-in on hidden tab of browser
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Optimize video for watch time and retention.
filed 2012-06-20, granted 2015-04-07 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, VideoPerDocData

US9146656B1 Notifications user interface
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-06-20, granted 2015-09-29 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20120272176A1 Persistent User Interface for Providing Navigational Functionality
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-06-21 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US20150161203A1 Searching for user interface objects
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-07-05 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality

US9342619B2 Renderer-assisted webpage navigating tool
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-07-24, granted 2016-05-17 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US8990710B1 Building native computer applications using a browser platform
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-08-30, granted 2015-03-24 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9135313B2 Providing a search display environment on an online resource
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-08-31, granted 2015-09-15 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US9953093B1 Mapping system for a browser history stack and a browser page state
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-09-04, granted 2018-04-24 · leak modules: PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo

US8732613B2 Dynamic user interface for navigating among GUI elements
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-09-06, granted 2014-05-20 · leak modules: PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo

US8892543B1 Iterative off-line rendering process
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-09-13, granted 2014-11-18 · leak modules: CrawlerChangerateUrlChangerate, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, TrawlerFetchReplyData

US9607078B1 Presenting related communications
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-09-13, granted 2017-03-28 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo

US20130198177A1 Displaying compact and expanded data items
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-09-14 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityOrbitAsteroidBeltDocumentIntentScores

US9058389B2 Interface for a universal search
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2012-09-14, granted 2015-06-16 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9268872B2 Displaying web pages without downloading static files
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2012-09-14, granted 2016-02-23 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US9098174B1 Expanding the functionality of the browser URL box
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2012-09-19, granted 2015-08-04 · leak modules: NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8935245B1 Presenting search results
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-09-20, granted 2015-01-13 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US8892591B1 Presenting search results
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-10-01, granted 2014-11-18 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores

US9317391B2 Logging graphical user interface events
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-10-09, granted 2016-04-19 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US20150278367A1 Determination and Presentation of Content from Connections
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-10-23 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate

US9201925B2 Search result previews
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-11-08, granted 2015-12-01 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8762490B1 Content-facilitated speculative preparation and rendering
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-11-19, granted 2014-06-24 · leak modules: PerDocData, SmartphonePerDocData

US9009739B2 Safe browser plugins using native code modules
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Earn endorsements from trusted sources. Trust is computed and transitive.
filed 2012-11-29, granted 2015-04-14 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9087096B1 Systems, methods, and media for controlling the presentation of search results with advertisement indications
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-12-05, granted 2015-07-21 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, VideoPerDocData

US9792383B2 Unload and display content placeholders with versions
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth. Use JSON-LD structured data for entity types and relationships.
filed 2012-12-12, granted 2017-10-17 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US9483534B2 User interfaces for a document search engine
Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2012-12-26, granted 2016-11-01 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, SmartphonePerDocData

US20150310100A1 Presenting user-generated content in search results
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data.
filed 2013-01-09 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData

US9589303B2 Selective presentation of content types and sources in search
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-01-09, granted 2017-03-07 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9665553B1 Techniques for determining a location of user interaction within a table of a web page
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth. Use JSON-LD structured data for entity types and relationships.
filed 2013-01-10, granted 2017-05-30 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8832150B2 Variable user interface based on document access privileges
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-01-11, granted 2014-09-09 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality

US9870567B2 User interface tailored by subscription state for a channel of a content sharing platform
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-01-30, granted 2018-01-16 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US20150199076A1 System and method for providing web content for display based on physical dimension requirements
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2013-02-15 · leak modules: IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityPreviewSnippetRadishFeatures, ShingleInfoPerDocData, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US9092531B2 Customized content consumption interface
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-02-25, granted 2015-07-28 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US9448978B2 Rich browser-based word processor
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-02-26, granted 2016-09-20 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US9535887B2 Creation of a content display area on a web page
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-02-26, granted 2017-01-03 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9336285B2 Presenting content in accordance with a placement designation
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-02-28, granted 2016-05-10 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US8831365B2 Capturing text from rendered documents using supplement information
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2013-03-11, granted 2014-09-09 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US20150199387A1 Systems and methods for determining capabilities of a web browser
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-03-14 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9195720B2 Requesting search results by user interface gesture combining display objects
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-03-14, granted 2015-11-24 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US9881058B1 Methods, systems, and media for displaying information related to displayed content upon detection of user attention
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-03-14, granted 2018-01-30 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9070211B1 Webview tag for a sandboxed multiprocess browser
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2013-03-15, granted 2015-06-30 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US9224228B1 Data presentation
Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2013-03-15, granted 2015-12-29 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8924409B1 Presentation of match quality in auto-complete suggestions
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-03-22, granted 2014-12-30 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US8943275B1 Differential storage and eviction for information resources from a browser cache
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships.
filed 2013-03-25, granted 2015-01-27 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US8924436B1 Populating a structured presentation with new values
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-04-01, granted 2014-12-30 · leak modules: NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9858273B2 Speculative rendering during cache revalidation
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-05-06, granted 2018-01-02 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9298778B2 Presenting related content in a stream of content
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-05-14, granted 2016-03-29 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US9158794B2 System and method for presentation of media related to a context
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-05-24, granted 2015-10-13 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US20130283375A1 Browser System and Method for Warning Users of Potentially Fraudulent Websites
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Structure answers: question heading, tight answer, then depth.
filed 2013-06-11 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8914355B1 Display-content alteration for user interface devices
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2013-06-17, granted 2014-12-16 · leak modules: NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, SnippetExtraInfo

US9471649B1 Enhanced presentation mode for search results
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-06-18, granted 2016-10-18 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20150169704A1 Reversible user interface
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-06-27 · leak modules: NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData

US8775965B1 Immersive mode for a web browser
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-06-27, granted 2014-07-08 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo

US9727212B2 Full screen content viewing interface entry
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2013-07-09, granted 2017-08-08 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US9811728B2 Adding value to a rendered document
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2013-08-13, granted 2017-11-07 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation

US9514102B2 User interface for presentation of a document
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-09-27, granted 2016-12-06 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US9575941B1 Dynamic adjustment to content for a display device
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-10-01, granted 2017-02-21 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US9310988B2 Scroll end effects for websites and content
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2013-10-04, granted 2016-04-12 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8862529B1 Predicting user navigation events in a browser using directed graphs
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2013-10-09, granted 2014-10-14 · leak modules: PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9459863B2 System for assessing an application for tablet compatibility and quality
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-10-11, granted 2016-10-04 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9589033B1 Presenting results from multiple search engines
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-10-14, granted 2017-03-07 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9658988B2 Systems and methods to segment text for layout and rendering
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Structure answers: question heading, tight answer, then depth.
filed 2013-10-16, granted 2017-05-23 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US20150113399A1 Landing Page for Presenting Media Items
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-10-22 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US20180365295A1 Tuning Parameters for Presenting Content
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-11-04 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData

US9767504B1 Method, system, and computer readable medium for rendering a graphical user interface visually indicating search results including related suggested products
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2013-11-04, granted 2017-09-19 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsGeoTopicality

US10185770B2 System and method for presenting multiple sets of search results for a single query
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2013-11-12, granted 2019-01-22 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9098530B2 Scalable rendering of large spatial databases
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-11-14, granted 2015-08-04 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9485543B2 Methods, systems, and media for presenting suggestions of media content
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2013-12-03, granted 2016-11-01 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US20140172853A1 Generating and displaying tasks
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-12-05 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US9510058B2 Program guide user interface
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-12-05, granted 2016-11-29 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US20160078005A1 Adaptive table rendering
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-12-06 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9275106B2 Dynamic search box for web browser
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-12-18, granted 2016-03-01 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US9720896B1 Synthesizing union tables from the web
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-12-30, granted 2017-08-01 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US9727587B2 Previewing search results
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-02-03, granted 2017-08-08 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9323846B2 Method, system, and graphical user interface for alerting a computer user to new results for a prior search
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-02-13, granted 2016-04-26 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US20150169154A1 User interface for an application displaying pages
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2014-02-19 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9710566B2 User interface for webpage permission requests
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2014-02-19, granted 2017-07-18 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9456237B2 Methods, systems, and media for presenting supplemental information corresponding to on-demand media content
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-02-26, granted 2016-09-27 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US9530229B2 Data object visualization using graphs
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-03-24, granted 2016-12-27 · leak modules: IndexingDocjoinerAnchorSpamInfo, PerDocData, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData, SpamBrainData, SpamBrainScore

US9247028B1 Latency-guided web content retrieval, serving, and rendering
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-03-26, granted 2016-01-26 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData

US9767189B2 Custom electronic message presentation based on electronic message category
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-03-28, granted 2017-09-19 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate

US9075778B1 Predicting user navigation events within a browser
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-04-11, granted 2015-07-07 · leak modules: PerDocData, SmartphonePerDocData

US9298786B1 Deferred content presentation
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-04-29, granted 2016-03-29 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US9602620B1 Content-facilitated speculative preparation and rendering
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-05-15, granted 2017-03-21 · leak modules: PerDocData, SmartphonePerDocData

US9720571B2 Immersive mode for a web browser
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-05-27, granted 2017-08-01 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo

US10152464B2 Determining and displaying a count of unread items in content feeds
Rule: Structure answers: question heading, tight answer, then depth.
filed 2014-06-02, granted 2018-12-11 · leak modules: QualityPreviewSnippetRadishFeatures, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US9613145B2 Generating contextual search presentations
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-06-18, granted 2017-04-04 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US20150370446A1 Application Specific User Interfaces
Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships.
filed 2014-06-20 · leak modules: NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US9124560B1 Protecting browser-viewed content from piracy
Rule: Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately. Build content for diverse user contexts and search histories.
filed 2014-07-02, granted 2015-09-01 · leak modules: PerDocData, QualityNsrNsrData, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US9154493B2 Managing multiple logins from a single browser
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-08-01, granted 2015-10-06 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9224004B2 Variable user interface based on document access privileges
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-08-07, granted 2015-12-29 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality

US9640100B2 Displaying always on display-related content
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2014-08-15, granted 2017-05-02 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8984612B1 Method of identifying an electronic device by browser versions and cookie scheduling
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2014-09-04, granted 2015-03-17 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryWebrefEntityJoin, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US10013497B1 Background reloading of currently displayed content
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-09-17, granted 2018-07-03 · leak modules: PerDocData, SmartphonePerDocData

US9785720B2 Script optimized browser rendering process
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2014-10-22, granted 2017-10-10 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9736212B2 Optimized browser rendering process
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2014-11-03, granted 2017-08-15 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9355074B1 Display-content alteration for user interface devices
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2014-11-14, granted 2016-05-31 · leak modules: NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US10157130B1 Differential storage and eviction for information resources from a browser cache
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-12-05, granted 2018-12-18 · leak modules: CrawlerChangerateUrlChangerate, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US9323847B1 Presenting search results
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2014-12-11, granted 2016-04-26 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US10417299B1 Embedded streams user interface
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-12-12, granted 2019-09-17 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefEntityJoin, SmartphonePerDocData

US9635393B2 Methods, systems, and media for remote rendering of web content on a television device
Rule: Use descriptive alt text and surrounding context for images. Optimize video for watch time and retention. Write for natural spoken language and conversational queries.
filed 2014-12-30, granted 2017-04-25 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityWebanswersVideoTranscriptAnnotations, VideoPerDocData

US20150220219A1 Content feed user interface with gallery display of same type items
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use descriptive alt text and surrounding context for images.
filed 2015-02-02 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityWebanswersVideoTranscriptAnnotations

US9703846B1 Methods, systems, and media for presenting content based on a generic rating
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-02-03, granted 2017-07-11 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores

US20190050490A1 Presenting contextual user suggestions
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2015-02-05 · leak modules: PerDocData, SmartphonePerDocData

US9280588B2 Search result previews
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-02-09, granted 2016-03-08 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US11829373B2 Methods, systems, and media for presenting search results

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2015-02-20, granted 2023-11-28 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US10248912B1 Systems and methods for displaying travel itineraries

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2015-03-06, granted 2019-04-02 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures

US9529861B2 Method, system, and graphical user interface for improved search result displays via user-specified annotations

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2015-03-20, granted 2016-12-27 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20190065502A1 Providing information related to a table of a document in response to a search query

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2015-04-21 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US10275536B2 Systems, methods, and computer-readable media for displaying content

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2015-05-01, granted 2019-04-30 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US9342563B2 Interface for a universal search

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.

filed 2015-05-29, granted 2016-05-17 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9710472B2 Customized content consumption interface

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2015-06-18, granted 2017-07-18 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US9466072B1 Systems, methods, and media for controlling the presentation of search results with advertisement indications

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2015-07-13, granted 2016-10-11 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, VideoPerDocData

US9858348B1 System and method for presentation of media related to a context

Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.

filed 2015-08-25, granted 2018-01-02 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations

US10218670B2 Presenting tasks in email application and calendar application

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2015-09-23, granted 2019-02-26 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US10613713B2 Integration of content in non-browser applications

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2015-10-07, granted 2020-04-07 · leak modules: PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, SmartphonePerDocData

US10326827B1 System and method for determining an interest in promotional content displayed at a mobile communication device

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.

filed 2015-10-21, granted 2019-06-18 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US10110524B1 User interface for activity status and history

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2015-10-29, granted 2018-10-23 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US20170364598A1 Methods, systems, and media for presenting links to media content

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2015-11-19 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate

US9507648B1 Separate plug-in processes in browsers and applications thereof

Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2015-11-20, granted 2016-11-29 · leak modules: PerDocData, SmartphonePerDocData

US20160078374A1 Graphical user interface for hotel search systems

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2015-11-25 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9659067B2 Providing a search results document that includes a user interface for performing an action in connection with a web page identified in the search results document

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2015-12-28, granted 2017-05-23 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US20160148326A1 Graphical user interface comprising multiple, interrelated components generated in response to a data request

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-01-29 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9852191B2 Presenting search result information

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2016-02-05, granted 2017-12-26 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorStatistics, PerDocData, QualityNsrNsrData

US11138207B2 Integrated dynamic interface for expression-based retrieval of expressive media content

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2016-02-14, granted 2021-10-05 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US10127287B1 Presenting related content in a stream of content

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2016-02-16, granted 2018-11-13 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US10088999B2 Scroll end effects for websites and content

Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.

filed 2016-03-03, granted 2018-10-02 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US10311135B1 System and method for selecting information for display based on past user interactions

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-03-16, granted 2019-06-04 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20170269893A1 Remote rendering of locally displayed content

Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2016-03-16 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US10169110B2 Navigation application programming interface

Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2016-03-25, granted 2019-01-01 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US10210242B1 Presenting forked auto-completions

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2016-04-01, granted 2019-02-19 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US20170286537A1 Content browsing on a vehicular display

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Optimize mobile experience. Mobile-specific signals are tracked separately. Build content for diverse user contexts and search histories.

filed 2016-04-01 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, SmartphonePerDocData

US10289712B2 Method, system, and graphical user interface for alerting a computer user to new results for a prior search

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-04-07, granted 2019-05-14 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US9542475B2 Presenting content in accordance with a placement designation

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2016-04-11, granted 2017-01-10 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US10409865B2 Interface for a universal search

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-04-15, granted 2019-09-10 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US10216388B2 Graphical user interface window spacing mechanisms

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.

filed 2016-04-21, granted 2019-02-26 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US10212061B2 Dynamically rendering interaction statistics data for content elements of an information resource using visual styles

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2016-04-25, granted 2019-02-19 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US9799131B1 Rendering interaction statistics data for content elements of an information resource by identifying client device segments

Rule: Use consistent entity names, sameAs schema, and structured data. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2016-04-25, granted 2017-10-24 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US9864732B2 User interfaces that facilitate management of formatting of digital content
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2016-05-02, granted 2018-01-09 · leak modules: PerDocData, SmartphonePerDocData

US10776571B2 Dispatch of user input to multiple input fields in a user interface
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2016-05-04, granted 2020-09-15 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9870623B2 Segmenting content displayed on a computing device into regions based on pixels of a screenshot image that captures the content
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2016-05-14, granted 2018-01-16 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US10025871B2 Setting and displaying a read status for items in content feeds
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Structure answers: question heading, tight answer, then depth.
filed 2016-05-18, granted 2018-07-17 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US11036743B2 Methods, systems, and media for presenting content organized by category
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2016-05-23, granted 2021-06-15 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11003667B1 Contextual information for a displayed resource
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2016-05-27, granted 2021-05-11 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US9848215B1 Methods, systems, and media for identifying and presenting users with multi-lingual media content items
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2016-06-21, granted 2017-12-19 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US20180032491A1 Web page display systems and methods
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2016-07-26 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US10452750B2 Systems and methods for interactively presenting a visible portion of a rendering surface on a user device
Rule: Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2016-08-04, granted 2019-10-22 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData, SnippetExtraInfo

US20180046470A1 Methods, systems, and media for presenting a user interface customized for a predicted user activity
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2016-08-11 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US10191995B2 Providing auto-focus for a search field in a user interface
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2016-08-12, granted 2019-01-29 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US10481861B2 Using user input to adapt search results provided for presentation to the user
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2016-08-30, granted 2019-11-19 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US9712878B2 Methods, systems, and media for presenting supplemental information corresponding to on-demand media content
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2016-09-21, granted 2017-07-18 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US11023550B2 User interfaces for a document search engine
Rule: Use descriptive alt text and surrounding context for images. Structure answers: question heading, tight answer, then depth. Ensure content comprehensively covers the topic with natural vocabulary.
filed 2016-10-26, granted 2021-06-01 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, SnippetExtraInfo

US9794636B2 Methods, systems, and media for presenting suggestions of media content
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2016-10-28, granted 2017-10-17 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US10528650B2 User interface for presentation of a document
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use descriptive alt text and surrounding context for images. Ensure content comprehensively covers the topic with natural vocabulary.
filed 2016-11-01, granted 2020-01-07 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US10430901B2 Graphical user interface for displaying current and future data
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2016-11-04, granted 2019-10-01 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US11553033B2 Adaptive user interface with reduced payload
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2016-11-04, granted 2023-01-10 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10140256B2 Location specific content visualization formatting
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2016-11-14, granted 2018-11-27 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US20170255594A1 Location specific content visualizations
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2016-11-14 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US10262057B2 Presenting content in accordance with a placement designation
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2016-12-01, granted 2019-04-16 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US10356200B2 Optimizing user interface data caching for future actions
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2016-12-28, granted 2019-07-16 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US11170014B2 Method and system for preview of search engine processing
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2016-12-29, granted 2021-11-09 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US10331749B2 Selective presentation of content types and sources in search
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-01-24, granted 2019-06-25 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US20180191798A1 Methods and systems for server-side rendering of native content for presentation
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2017-01-25 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US10509817B2 Displaying search results on a one or two dimensional graph
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2017-02-06, granted 2019-12-17 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US10394841B2 Generating contextual search presentations
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-03-09, granted 2019-08-27 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US10902495B2 Presenting search results in a dynamically formatted graphical user interface
Rule: Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2017-03-30, granted 2021-01-26 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData

US10127849B2 Displaying always on display-related content
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2017-04-12, granted 2018-11-13 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US10075761B2 Methods, systems, and media for remote rendering of web content on a television device
Rule: Use descriptive alt text and surrounding context for images. Optimize video for watch time and retention. Write for natural spoken language and conversational queries.
filed 2017-04-24, granted 2018-09-11 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityWebanswersVideoTranscriptAnnotations, VideoPerDocData

US10332160B2 Identifying related information given content and/or presenting related information in association with content-related advertisements
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2017-05-12, granted 2019-06-25 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US10725611B1 Optimizing presentation of interactive graphical elements based on contextual relevance
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2017-05-22, granted 2020-07-28 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US10606858B2 Methods, systems, and media for presenting content based on a generic rating
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-07-10, granted 2020-03-31 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores

US9998795B2 Methods, systems, and media for presenting supplemental information corresponding to on-demand media content
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2017-07-13, granted 2018-06-12 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US10698578B1 User interface for an application displaying page permissions
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2017-07-17, granted 2020-06-30 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10284623B2 Optimized browser rendering service
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2017-08-09, granted 2019-05-07 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US10402889B1 Method, system, and computer readable medium for rendering a graphical user interface visually indicating search results including related suggested products
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2017-09-01, granted 2019-09-03 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsGeoTopicality

US10242471B2 Rendering interaction statistics data for content elements of an information resource by identifying client device segments
Rule: Use consistent entity names, sameAs schema, and structured data. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2017-09-26, granted 2019-03-26 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US10713330B2 Optimized browser render process
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2017-10-03, granted 2020-07-14 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData, TrawlerFetchReplyData

US10341741B2 Methods, systems, and media for presenting suggestions of media content
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2017-10-16, granted 2019-07-02 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US10521438B2 Presenting search result information
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2017-11-17, granted 2019-12-31 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorStatistics, PerDocData, QualityNsrNsrData

US10147197B2 Segment content displayed on a computing device into regions based on pixels of a screenshot image that captures the content
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2017-12-12, granted 2018-12-04 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US11170067B2 Methods, systems, and media for updating a webpage rendered with cached content
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2017-12-13, granted 2021-11-09 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US10313713B2 Methods, systems, and media for identifying and presenting users with multi-lingual media content items

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2017-12-14, granted 2019-06-04 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores

US11069110B2 Systems and methods for authoring cross-browser HTML 5 motion path animation

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.

filed 2017-12-14, granted 2021-07-20 · leak modules: ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US10210219B2 Methods, systems, and media for displaying information related to displayed content upon detection of user attention

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2017-12-15, granted 2019-02-19 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20180357301A1 Rendering Graphical User Interfaces To Present Grouped Search Results

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2018-02-13 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US10896213B2 Interface for a distributed network system

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2018-03-07, granted 2021-01-19 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US11227688B2 Interface for patient-provider conversation and auto-generation of note or summary

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2018-05-24, granted 2022-01-18 · leak modules: NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US10448110B2 Methods, systems, and media for presenting supplemental information corresponding to on-demand media content

Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2018-06-11, granted 2019-10-15 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US10963696B2 Visual menu

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2018-07-09, granted 2021-03-30 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10706121B2 Setting and displaying a read status for items in content feeds

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Structure answers: question heading, tight answer, then depth.

filed 2018-07-17, granted 2020-07-07 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US10572911B2 Identifying related information given content and/or presenting related information in association with content-related advertisements

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2018-08-03, granted 2020-02-25 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US10997261B2 System and method that facilitate display responsive message rendering on client device

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2018-09-06, granted 2021-05-04 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20200082022A1 Display responsive communication system and method

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2018-09-06 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10631048B2 Methods, systems, and media for remote rendering of web content on a television device

Rule: Use descriptive alt text and surrounding context for images. Optimize video for watch time and retention. Write for natural spoken language and conversational queries.

filed 2018-09-10, granted 2020-04-21 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityWebanswersVideoTranscriptAnnotations, VideoPerDocData

US11100147B2 Electronic list user interface

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2018-09-12, granted 2021-08-24 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US11170033B2 Electronic list user interface

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Write for natural spoken language and conversational queries.

filed 2018-09-12, granted 2021-11-09 · leak modules: NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US20190187864A1 Providing optimized displays on user interfaces based on user generated lists of items

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2018-09-14 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US20190171689A1 Optimizing item display on graphical user interfaces

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2018-12-05 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US11210302B2 Methods, systems, and media for displaying information related to displayed content upon detection of user attention

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2019-02-15, granted 2021-12-28 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11263248B2 Presenting content in accordance with a placement designation

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2019-02-27, granted 2022-03-01 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US11102275B2 System and method for determining an interest in promotional content displayed at a mobile communication device

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.

filed 2019-03-05, granted 2021-08-24 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US11238242B2 Generating output for presentation in response to user interface input, where the input and/or the output include chatspeak

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2019-03-21, granted 2022-02-01 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US11288440B1 System and method for selecting information for display based on past user interactions

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2019-04-22, granted 2022-03-29 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20190340199A1 Methods and Systems for Identifying, Selecting, and Presenting Media-Content Items Related to a Common Story

Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.

filed 2019-05-06 · leak modules: NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, SnippetExtraInfo

US11288331B2 Selective presentation of content types and sources in search

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2019-05-15, granted 2022-03-29 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US10560546B2 Optimizing user interface data caching for future actions

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2019-05-30, granted 2020-02-11 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US10880613B2 Methods, systems, and media for presenting suggestions of media content

Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.

filed 2019-07-01, granted 2020-12-29 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US11314822B2 Interface for a universal search

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.

filed 2019-07-22, granted 2022-04-26 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US11138286B2 Browser-based navigation suggestions for task completion

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.

filed 2019-07-24, granted 2021-10-05 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US11049201B2 Graphical user interface for displaying current and future data

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2019-09-24, granted 2021-06-29 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US11074038B2 Using user input to adapt search results provided for presentation to the user

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2019-10-02, granted 2021-07-27 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US10992993B2 Methods, systems, and media for presenting supplemental information corresponding to on-demand media content

Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2019-10-14, granted 2021-04-27 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US11341180B2 Displaying search results on a one or two dimensional graph

Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2019-11-06, granted 2022-05-24 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US11775535B2 Presenting search result information
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2019-12-11, granted 2023-10-03 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorStatistics, PerDocData, QualityNsrNsrData

US11310331B2 Optimizing user interface data caching for future actions
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2019-12-18, granted 2022-04-19 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US11367112B2 Identifying related information given content and/or presenting related information in association with content-related advertisements
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2020-01-15, granted 2022-06-21 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US11809435B2 Efficient storage, retrieval, and/or rendering of personal entries
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2020-02-17, granted 2023-11-07 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US11151303B2 Integration of content in non-browser applications
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2020-03-23, granted 2021-10-19 · leak modules: PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, SmartphonePerDocData

US11308111B2 Methods, systems, and media for presenting content based on a generic rating
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2020-03-30, granted 2022-04-19 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores

US11470387B2 Methods, systems, and media for remote rendering of Web content on a television device
Rule: Use descriptive alt text and surrounding context for images. Optimize video for watch time and retention. Write for natural spoken language and conversational queries.
filed 2020-04-20, granted 2022-10-11 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityWebanswersVideoTranscriptAnnotations, VideoPerDocData

US11714851B2 Media contextual information for a displayed resource
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2020-06-03, granted 2023-08-01 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US20210311611A1 Dynamic multi-axis graphical user interface
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2020-12-15 · leak modules: CompressedQualitySignals, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo

US11381880B2 Methods, systems, and media for presenting suggestions of media content
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2020-12-23, granted 2022-07-05 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US11886531B2 Presenting indicators associated with network-associated content
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2020-12-30, granted 2024-01-30 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData

US11625160B2 Content navigation method and user interface
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2020-12-31, granted 2023-04-11 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11575733B2 System and method for determining an interest in promotional content displayed at a mobile communication device
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2021-01-13, granted 2023-02-07 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, SmartphonePerDocData

US11599518B2 Efficient embedding table storage and lookup
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2021-01-13, granted 2023-03-07 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US11630878B2 System and method for creating responsive display content
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2021-01-20, granted 2023-04-18 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US11651408B2 Presenting search results in a dynamically formatted graphical user interface
Rule: Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2021-01-25, granted 2023-05-16 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData

US20210209665A1 Dynamically Formatted Graphical User Interface
Rule: Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2021-01-25 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData

US11709881B2 Visual menu
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2021-03-12, granted 2023-07-25 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20220269889A1 Visual tag classification for augmented reality display
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2021-04-13 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US12063419B2 Methods, systems, and media for presenting supplemental information corresponding to on-demand media content
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2021-04-26, granted 2024-08-13 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US12504862B2 Browser tab management
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2021-05-12, granted 2025-12-23 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US11741434B2 Systems and methods for visualizing categorized sets of calendar events
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2021-05-21, granted 2023-08-29 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11803604B2 User interfaces for a document search engine
Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2021-06-01, granted 2023-10-31 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, SmartphonePerDocData

US11875086B2 Using user input to adapt search results provided for presentation to the user
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2021-06-30, granted 2024-01-16 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US11854135B2 Systems and methods for authoring cross-browser HTML 5 motion path animation
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2021-07-07, granted 2023-12-26 · leak modules: ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US20210349926A1 Method of presenting excluded keyword categories in keyword suggestions
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2021-07-21 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData

US20250005823A1 Automated Generation of Immersive Interfaces
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2021-07-22 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US12231489B2 System and method for determining an interest in promotional content displayed at a mobile communication device
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2021-08-02, granted 2025-02-18 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US11861679B2 Electronic list user interface
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Write for natural spoken language and conversational queries.
filed 2021-08-04, granted 2024-01-02 · leak modules: NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US11734323B2 Electronic list user interface
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2021-08-05, granted 2023-08-22 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US11727076B2 Browser-based navigation suggestions for task completion
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2021-09-09, granted 2023-08-15 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US11995393B2 Integration of content in non-browser applications
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2021-09-17, granted 2024-05-28 · leak modules: PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, SmartphonePerDocData

US12093273B2 Method and system for preview of search engine processing
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2021-10-22, granted 2024-09-17 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US11995089B2 Methods, systems, and media for displaying information related to displayed content upon detection of user attention
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2021-12-23, granted 2024-05-28 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11726641B1 Encoding/decoding user interface interactions
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2022-02-14, granted 2023-08-15 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo

US11860962B1 System and method for selecting information for display based on past user interactions
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2022-02-18, granted 2024-01-02 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11934474B2 Selective presentation of content types and sources in search
Rule: Create timely, trending-topic content on social platforms that can appear in Google search results. For trending queries, Google blends social graph content with web results. Ensure your social content is topical, recent, and connected to your audience's social graph for inclusion in blended results.
filed 2022-03-08, granted 2024-03-19 · leak modules: QualityOrbitAsteroidBeltDocumentIntentScores, QualityNavboostCrapsCrapsClickSignals, PerDocData, QualityTimebasedPageType

US11755607B2 Methods, systems, and media for presenting content based on a generic rating
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2022-04-18, granted 2023-09-12 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores

US20220286743A1 Systems and Methods for Detecting Improper Implementation of Presentation of Content Items by Applications Executing on Client Devices
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2022-05-23 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11783003B2 User interfaces for surfacing web browser history data
Rule: Create content that supports multi-session search journeys — users who return to your content across multiple visits build engagement scores that increase your prominence in browser history surfaces. Ensure your pages have clear topic associations through structured data. Support journey continuation with clear, resumable content structures.
filed 2022-08-05, granted 2023-10-10 · leak modules: QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, PerDocData

US12547968B2 Independently presenting status of order
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2022-08-19, granted 2026-02-10 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US11941244B1 Presenting suggestions from search corpora
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2022-10-03, granted 2024-03-26 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US12267329B2 Proactive rendering of credentials according to determined location characteristics
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2022-10-10, granted 2025-04-01 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US20230152946A1 Methods and apparatus for search of an area rendered within a browser
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2022-11-16 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US12003573B2 Adaptive user interface with reduced payload
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2022-12-06, granted 2024-06-04 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20230216858A1 Dynamic permissions interface
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2022-12-13 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US12189700B2 Presenting related content while browsing and searching content
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2022-12-15, granted 2025-01-07 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US12230030B2 Object filtering and information display in an augmented-reality experience
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2022-12-20, granted 2025-02-18 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US11954167B1 Techniques for presenting graphical content in a search result
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2022-12-21, granted 2024-04-09 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US20230214435A1 Methods and apparatus for persistent search within a browser
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2022-12-30 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US11892998B2 Efficient embedding table storage and lookup
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2023-01-30, granted 2024-02-06 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US11978103B2 Presenting search results in a dynamically formatted graphical user interface
Rule: Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2023-04-10, granted 2024-05-07 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData

US12461985B2 System and method for creating responsive display content
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2023-04-17, granted 2025-11-04 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US12469185B2 Application customization using an interface-guided diffusion model
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2023-06-14, granted 2025-11-11 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US12236064B2 Encoding/decoding user interface interactions
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2023-07-05, granted 2025-02-25 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo

US12475166B2 Media contextual information for a displayed resource
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2023-07-31, granted 2025-11-18 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US12346395B2 Browser-based navigation suggestions for task completion
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2023-08-14, granted 2025-07-01 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US12197525B2 Techniques for presenting graphical content in a search result
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2023-09-06, granted 2025-01-14 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US12596722B2 Efficient storage, retrieval, and/or rendering of personal entries
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2023-10-02, granted 2026-04-07 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US12547668B2 User interfaces for surfacing web browser history data
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2023-10-06, granted 2026-02-10 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US20240104126A1 Methods, systems, and media for modifying the presentation of contextually relevant documents in browser windows of a browsing application
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2023-12-11 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US20240111805A1 Predictively Presenting Search Capabilities
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2023-12-14 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate

US12229108B2 Efficient embedding table storage and lookup
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2023-12-20, granted 2025-02-18 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US20250209308A1 Risk Analysis and Visualization for Sequence Processing Models
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2023-12-22 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US12260154B2 Using user input to adapt search results provided for presentation to the user
Rule: Monitor and manage sentiment around your brand/entity name in voice assistant interactions. Negative user sentiment expressed about your brand attribute in follow-up queries causes Google to suppress additional results from your source. Build positive brand sentiment to maintain visibility in conversational search.
filed 2024-01-03, granted 2025-03-25 · leak modules: QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsFeatureCrapsData, RepositoryWebrefEntityJoin, PerDocData

US12468771B2 Selective presentation of content types and sources in search
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2024-02-07, granted 2025-11-11 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US12443667B2 Contextual search tool in a browser interface
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2024-03-07, granted 2025-10-14 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US20240311436A1 Displaying browser history in a user interface
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2024-03-14 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US12354145B2 Presenting search results in a dynamically formatted graphical user interface
Rule: Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2024-04-04, granted 2025-07-08 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData

US12293145B2 Integration of content in non-browser applications
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2024-05-21, granted 2025-05-06 · leak modules: PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, SmartphonePerDocData

US20250028774A1 Dynamically generated comparison user interface
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2024-07-19 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US20250110978A1 Automated Content Presentation Based on a Determined Keyword
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2024-07-26 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US20250068305A1 Tab grouping for an application
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2024-08-23 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation

US20260119584A1 Identifying Domains for Priority Rendering at User Interface Output Devices
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use descriptive alt text and surrounding context for images.
filed 2024-10-31 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US20250217428A1 Web Browser with Integrated Vector Database
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2024-12-16 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US20250225717A1 User-Context Aware Rendering Dataset Selection
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2025-03-25 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

Ads & content delivery (375)

US8527337B1 Internet based system and apparatus for paying users to view content and receiving micropayments
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2000-07-18, granted 2013-09-03 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US7136875B2 Serving advertisements based on content
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2003-02-26, granted 2006-11-14 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US8352499B2 Serving advertisements using user request information and user information
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2003-06-02, granted 2013-01-08 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US7647299B2 Serving advertisements using a search of advertiser web information
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2003-06-30, granted 2010-01-12 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US8438154B2 Generating information for online advertisements from internet data and traditional media data
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2003-09-29, granted 2013-05-07 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US7533090B2 System and method for rating electronic documents
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2004-03-30, granted 2009-05-12 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US7840648B1 Web-page authoring tool for automatic enrollment in advertising program
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2005-06-30, granted 2010-11-23 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefEntityJoin, RichsnippetsPageMap

US8849692B1 Online advertising program enrollment program
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2005-06-30, granted 2014-09-30 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US7590691B2 Indirect subscriptions to top N lists of content feeds
Rule: Monitor this patent for ranking implications.
filed 2005-10-07, granted 2009-09-15

US20070198500A1 User distributed search results
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2006-05-30 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US7734624B2 Serving advertisements based on content
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2006-07-14, granted 2010-06-08 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US7657611B2 Content request optimization
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2006-10-30, granted 2010-02-02 · leak modules: CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, ShingleInfoPerDocData, TrawlerFetchReplyData

US8239491B1 Content request optimization
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2006-10-30, granted 2012-08-07 · leak modules: CompressedQualitySignals, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20080256109A1 Dynamic Podcast Content Delivery
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2007-04-13 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8326806B1 Content item parameter filter
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2007-05-11, granted 2012-12-04 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US20080040221A1 Interest Targeting
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Write in natural vocabulary. ML models understand context and fluff.
filed 2007-08-08 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, RepositoryWebrefDocumentMetadata

US9064024B2 Bundle generation
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Structure answers: question heading, tight answer, then depth.
filed 2007-08-21, granted 2015-06-23 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SnippetExtraInfo

US8504711B1 Integrating web services with a content item
Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2007-12-07, granted 2013-08-06 · leak modules: ImageData, ImagePerDocData, PerDocData, SmartphonePerDocData

US8055655B1 User interaction based related digital content items
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2008-02-15, granted 2011-11-08 · leak modules: IndexingDocjoinerAnchorSpamInfo, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US8984147B1 Velocity based content delivery
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2008-04-18, granted 2015-03-17 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US20080270364A1 Expansion rule evaluation
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2008-04-22 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US20190259057A1 Graduated feature transitions
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2008-04-28 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US7987194B1 Targeting advertisements based on cached contents
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2008-04-28, granted 2011-07-26 · leak modules: CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8150842B2 Reputation of an author of online content
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2008-12-11, granted 2012-04-03 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US8214350B1 Pre-computed impression lists
Rule: Structure answers: question heading, tight answer, then depth. Write for natural spoken language and conversational queries.
filed 2009-01-02, granted 2012-07-03 · leak modules: NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8095546B1 Book content item search
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2009-01-09, granted 2012-01-10 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US8316032B1 Book content item search
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2009-01-09, granted 2012-11-20 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US20100211960A1 Characterizing User Information
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2009-02-17 · leak modules: IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US8219638B2 Editing information configured for use in selecting content regarding at least one content provider
Rule: Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2009-03-10, granted 2012-07-10 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US9170995B1 Identifying context of content items
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2009-03-19, granted 2015-10-27 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8463783B1 Advertisement selection data clustering
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2009-07-06, granted 2013-06-11 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8370202B2 Audience segment estimation
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2009-08-28, granted 2013-02-05 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9754285B1 Identifying alternate content distribution locations
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2009-09-30, granted 2017-09-05 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US20110029319A1 Impression forecasting and reservation analysis
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2009-10-21 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US8219640B2 Content request optimization
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2010-01-11, granted 2012-07-10 · leak modules: CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, ShingleInfoPerDocData, TrawlerFetchReplyData

US8843465B2 Distributing content
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2010-01-29, granted 2014-09-23 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8527493B1 Distributing content
Rule: Structure answers: question heading, tight answer, then depth. Optimize video for watch time and retention.
filed 2010-03-16, granted 2013-09-03 · leak modules: QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo, VideoPerDocData

US7937405B2 Serving advertisements based on content
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2010-03-27, granted 2011-05-03 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US8275771B1 Non-text content item search
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2010-06-07, granted 2012-09-25 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US8423551B1 Clustering internet resources
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Ensure content comprehensively covers the topic with natural vocabulary.
filed 2010-11-05, granted 2013-04-16 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8176120B2 Web-page authoring tool for automatic enrollment in advertising program
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2010-11-23, granted 2012-05-08 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefEntityJoin, RichsnippetsPageMap

US9390139B1 Presentation of content items in view of commerciality
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2010-12-29, granted 2016-07-12 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData

US8375025B1 Language-specific search results
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2010-12-30, granted 2013-02-12 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20150248698A1 Distributing content items
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-01-14 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData

US20120259854A1 Conversion Path Based Segmentation
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-04-11 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData

US8504551B2 Serving advertisements based on content
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-04-11, granted 2013-08-06 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US20150317398A1 Presenting non-suggested content items to a user of a social network account
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-04-19 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US8868570B1 Selection and display of online content items
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-06-14, granted 2014-10-21 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US9026545B1 Targeting advertisements based on cached contents
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-06-16, granted 2015-05-05 · leak modules: CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9147199B2 Advertisements in view
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-06-17, granted 2015-09-29 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US8515968B1 Tie breaking rules for content item matching
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-08-12, granted 2013-08-20 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US8250070B1 User interaction based related digital content items
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-08-24, granted 2012-08-21 · leak modules: IndexingDocjoinerAnchorSpamInfo, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US8515811B2 Online advertising valuation apparatus and method
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-08-29, granted 2013-08-20 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US8650188B1 Retargeting in a search environment
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-08-31, granted 2014-02-11 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata

US20120022954A1 Cloud based media player and offline media access
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Earn endorsements from trusted sources. Trust is computed and transitive.
filed 2011-09-27 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, VideoPerDocData

US20140324583A1 Suggestion box for input keywords
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-09-27 · leak modules: PerDocData, SmartphonePerDocData

US8510314B1 Book content item search
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-10-06, granted 2013-08-13 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US8438155B1 Impressions-weighted coverage monitoring for search results
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-11-10, granted 2013-05-07 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8375036B1 Book content item search
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-11-17, granted 2013-02-12 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US20190119145A1 Methods, apparatuses, and computer-readable media for automatic ad campaign creation
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Write for natural spoken language and conversational queries.
filed 2011-12-01 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US8984098B1 Organizing a stream of content
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2011-12-17, granted 2015-03-17 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo

US8990352B1 Stream of content for a channel
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2011-12-17, granted 2015-03-24 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo

US9146993B1 Content keyword identification
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Optimize video for watch time and retention.
filed 2012-03-16, granted 2015-09-29 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, VideoPerDocData

US9098598B1 Non-default location support for expandable content item publisher side files
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-05-04, granted 2015-08-04 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US8521718B1 Pre-computed impression lists
Rule: Structure answers: question heading, tight answer, then depth. Write for natural spoken language and conversational queries.
filed 2012-05-17, granted 2013-08-27 · leak modules: NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8688669B1 Content item searching and retrieval
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-05-22, granted 2014-04-01 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US20120239511A1 Automated Graphical Advertisement Size Compatibility and Link Insertion
Rule: Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2012-05-29 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US20130332267A1 Selecting content based on data analysis
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Structure answers: question heading, tight answer, then depth.
filed 2012-06-06 · leak modules: PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US20150186937A1 Obtaining alternative keywords
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2012-06-12 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9213769B2 Providing a modified content item to a user
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-06-13, granted 2015-12-15 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US9396458B2 Controlling content distribution
Rule: Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2012-06-14, granted 2016-07-19 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US9317487B1 Expansion of high performing placement criteria
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-06-25, granted 2016-04-19 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US9990431B2 Rich web page generation
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2012-06-27, granted 2018-06-05 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate

US20150169722A1 Generatring n-gram clusters associated with events
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth. Ensure content comprehensively covers the topic with natural vocabulary.
filed 2012-06-28 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US9501572B2 Content placement criteria expansion
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-06-29, granted 2016-11-22 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US9043699B1 Determining expansion directions for expandable content item environments
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-07-05, granted 2015-05-26 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9047254B1 Detection and validation of expansion types of expandable content items
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-07-05, granted 2015-06-02 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9276757B1 Generating viral metrics
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-07-13, granted 2016-03-01 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality

US9449094B2 Navigating among content items in a set
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-07-13, granted 2016-09-20 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US8560533B1 User interaction based related digital content items
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-07-19, granted 2013-10-15 · leak modules: IndexingDocjoinerAnchorSpamInfo, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US9552436B2 Serving expandable content items
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2012-07-20, granted 2017-01-24 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US10007645B2 Modifying the presentation of a content item
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-07-25, granted 2018-06-26 · leak modules: PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US20170323230A1 Evaluating keyword performance
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-08-21 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9400871B1 Selecting content for devices specific to a particular user
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-08-27, granted 2016-07-26 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US10055755B2 Targeting content based on receipt of partial terms
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-09-04, granted 2018-08-21 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US8856125B1 Non-text content item search
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-09-07, granted 2014-10-07 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US20140074588A1 Determining content item engagement
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-09-10 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US20130110627A1 Generating and Presenting Advertisements Based on Context Data for Programmable Search Engines
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-09-13 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, IndexingDocjoinerAnchorStatistics, LocalWWWInfo

US10198744B2 User-targeted advertising
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use JSON-LD structured data for entity types and relationships.
filed 2012-09-14, granted 2019-02-05 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US9043338B1 Book content item search
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-09-14, granted 2015-05-26 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US9069864B2 Prioritizing a content item for a user
Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-09-21, granted 2015-06-30 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData

US20140089124A1 Dynamic Product Content Generation
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-09-26 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9330072B1 Serving content for a space based on a probabilistic distribution
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-09-26, granted 2016-05-03 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US9442912B1 Estimating the number of content items to fill an online space before distributing the content items to the online space
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-09-26, granted 2016-09-13 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11544750B1 Overlaying content items with third-party reviews
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-10-02, granted 2023-01-03 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US9870344B2 Reassigning ordinal positions of content item slots according to viewport information during resource navigation

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2012-10-02, granted 2018-01-16 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US10013152B2 Content selection disambiguation

Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.

filed 2012-10-05, granted 2018-07-03 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US9355186B2 Launching applications from webpages

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2012-10-31, granted 2016-05-31 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US9477376B1 Prioritizing content based on user frequency

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2012-12-19, granted 2016-10-25 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, VideoPerDocData

US8639575B2 Audience segment estimation

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.

filed 2012-12-31, granted 2014-01-28 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9009170B1 Book content item search

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2013-02-11, granted 2015-04-14 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US9262405B1 Systems and methods of serving a content item to a user in a specific language

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2013-02-28, granted 2016-02-16 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US9436767B2 Serving content items based on device rotational orientation

Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2013-03-07, granted 2016-09-06 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality

US9881091B2 Content item audience selection

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2013-03-08, granted 2018-01-30 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US9235804B1 System and method for selecting and serving content items based on sensor data from mobile devices

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2013-03-12, granted 2016-01-12 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9547697B2 Aggregating interactions for content items

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.

filed 2013-03-14, granted 2017-01-17 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9549017B2 Predicting content performance with interest data

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2013-03-14, granted 2017-01-17 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US9043302B1 Campaign and competitive analysis and data visualization based on search interest data

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.

filed 2013-03-15, granted 2015-05-26 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US9213749B1 Content item selection based on presentation context

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2013-03-15, granted 2015-12-15 · leak modules: PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9325653B1 Notifying content owners of related posts

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.

filed 2013-03-15, granted 2016-04-26 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9760547B1 Monetization of online content

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.

filed 2013-03-15, granted 2017-09-12 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US20140281901A1 Estimating visibility of content items

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2013-03-28 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US9135292B1 Selecting a template for a content item

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2013-04-08, granted 2015-09-15 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US9143555B1 Smart asset management for a content item

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.

filed 2013-05-24, granted 2015-09-22 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9418114B1 Augmenting a content item using search results content

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2013-06-19, granted 2016-08-16 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US9160800B1 Integrating web services with a content item

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2013-07-15, granted 2015-10-13 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9319451B2 Systems and methods for selecting an accounting technique for interactions with electronic content

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2013-07-15, granted 2016-04-19 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US9336210B2 Determining a likelihood and degree of derivation among media content items

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2013-07-15, granted 2016-05-10 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US8819029B1 Tie breaking rules for content item matching

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2013-07-24, granted 2014-08-26 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US8832070B1 Pre-computed impression lists

Rule: Structure answers: question heading, tight answer, then depth. Write for natural spoken language and conversational queries.

filed 2013-07-29, granted 2014-09-09 · leak modules: NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US9152718B2 Serving advertisements based on content

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2013-08-06, granted 2015-10-06 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US20140372230A1 Systems and methods for selecting and serving content items based on view probability

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2013-08-21 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US9521189B2 Providing contextual data for selected link units

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.

filed 2013-08-21, granted 2016-12-13 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US20170316314A1 Automatic optimization of content items

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2013-08-23 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US10354294B2 Methods and systems for providing third-party content on a web page

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2013-08-28, granted 2019-07-16 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US9779065B1 Displaying graphical content items based on textual content items

Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.

filed 2013-08-29, granted 2017-10-03 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations

US9372939B2 Internet based system and apparatus for paying users to view content and receiving micropayments

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2013-08-30, granted 2016-06-21 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US8972392B2 User interaction based related digital content items

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2013-09-12, granted 2015-03-03 · leak modules: IndexingDocjoinerAnchorSpamInfo, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US20150081420A1 Methods and systems for identifying relationships between online content items

Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.

filed 2013-09-18 · leak modules: IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, ShingleInfoPerDocData, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US9767489B1 Content item impression effect decay

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2013-09-19, granted 2017-09-19 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US9424345B1 Contextual content distribution

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2013-09-25, granted 2016-08-23 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9514199B1 Sorting and pagination of content item streams

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.

filed 2013-09-27, granted 2016-12-06 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefEntityJoin, SmartphonePerDocData

US9703837B1 Predicting interest of a user of a social networking service in a content item

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2013-09-27, granted 2017-07-11 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9703757B2 Automatically determining a size for a content item for a web page

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2013-10-08, granted 2017-07-11 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US10185780B1 Resharing content items

Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2013-10-15, granted 2019-01-22 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate

US9569502B1 Search lift remarketing

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2013-11-08, granted 2017-02-14 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9275133B1 Content request identification via a computer network
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-11-13, granted 2016-03-01 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US9767196B1 Content selection
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-11-20, granted 2017-09-19 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US9460159B1 Detecting visibility of a content item using tasks triggered by a timer
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-12-10, granted 2016-10-04 · leak modules: IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US20150178281A1 Determining whether a user has seen a content item
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-12-20 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US9672530B2 Supporting voting-based campaigns in search
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-12-30, granted 2017-06-06 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US20150348097A1 Autocreated campaigns for hashtag keywords
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-01-24 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality

US20160328752A1 Native creative generation using hashtagged user generated content
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-01-24 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US9679314B1 Content selection using distribution parameter data
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-01-31, granted 2017-06-13 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US9177329B2 Retargeting in a search environment
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-02-10, granted 2015-11-03 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata

US9721012B1 Providing social presence information for content items
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-02-21, granted 2017-08-01 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefDocumentMetadata

US9727818B1 Impression effect modeling for content items
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-02-23, granted 2017-08-08 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US20150278860A1 Dynamically determining a search radius to select online content
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-03-25 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9317873B2 Automatic verification of advertiser identifier in advertisements
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2014-03-28, granted 2016-04-19 · leak modules: IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9875287B1 Providing a notification in place of content item
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-05-21, granted 2018-01-23 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US9465887B2 Systems and methods for optimizing content layout using behavior metrics
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-06-19, granted 2016-10-11 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US10402852B2 Content item slot location suggestions
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-07-24, granted 2019-09-03 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US20180285940A1 Web advertisement placement
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-07-24 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US9325842B1 Systems and methods for associating a string with a content item
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-07-28, granted 2016-04-26 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9984159B1 Providing information about content distribution
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-08-12, granted 2018-05-29 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US20160050129A1 Performance assessment based on analysis of resources
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2014-08-15 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9892430B1 System and method for providing content items with format elements
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2014-08-18, granted 2018-02-13 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US9129306B1 Tie breaking rules for content item matching
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-08-20, granted 2015-09-08 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US9794211B2 Event-based comment grouping for content items
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-08-20, granted 2017-10-17 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US9235852B2 Distributing content
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-08-21, granted 2016-01-12 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10049170B1 Methods and systems for selectively preventing third-party content from being displayed using undefined publisher identifier
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2014-08-26, granted 2018-08-14 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9767169B1 Enhancing search results for improved readability
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2014-09-26, granted 2017-09-19 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US9940646B2 Dynamic insertion of content items into resources
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-10-09, granted 2018-04-10 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US9317868B2 Advertisements in view
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-11-03, granted 2016-04-19 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US20150081430A1 System and method of providing targeted advertisements from subscribers of directory services
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Build content for diverse user contexts and search histories.
filed 2014-11-20 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US20160210689A1 Content item configuration optimization
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-01-16 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US9900358B1 Organizing a stream of content

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.

filed 2015-01-26, granted 2018-02-20 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo

US9280212B1 Velocity based content delivery

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2015-02-11, granted 2016-03-08 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US9723044B1 Stream of content for a channel

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.

filed 2015-02-13, granted 2017-08-01 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo

US10318574B1 Generating moments

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2015-03-16, granted 2019-06-11 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US9342583B2 Book content item search

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2015-03-16, granted 2016-05-17 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US9595045B2 Advertisements in view

Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2015-04-15, granted 2017-03-14 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US10108979B2 Advertisement effectiveness measurements

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.

filed 2015-04-30, granted 2018-10-23 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US20150235261A1 Advertisement effectiveness measurement

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.

filed 2015-04-30 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US9569523B2 Bundle generation

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Structure answers: question heading, tight answer, then depth.

filed 2015-06-12, granted 2017-02-14 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SnippetExtraInfo

US9984049B2 Systems and methods for providing call context to content providers

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2015-07-14, granted 2018-05-29 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US9529852B1 Selecting a template for a content item

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2015-08-11, granted 2016-12-27 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US20190087852A1 Re-messaging with alternative content items in an online remarketing campaign
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-08-24 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US10223742B2 Systems and methods for selecting third party content based on feedback
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-08-26, granted 2019-03-05 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US9456058B1 Smart asset management for a content item
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2015-09-02, granted 2016-09-27 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US11709889B1 Content keyword identification
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Optimize video for watch time and retention.
filed 2015-09-14, granted 2023-07-25 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, VideoPerDocData

US10324960B1 Determining a number of unique viewers of a content item
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-09-18, granted 2019-06-18 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US9530153B2 Retargeting in a search environment
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2015-09-30, granted 2016-12-27 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata

US20160034946A1 System and method for providing and managing third party content with call functionality
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-10-09 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9805102B1 Content item selection based on presentation context
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2015-11-10, granted 2017-10-31 · leak modules: PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9898758B2 Providing a modified content item to a user
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2015-11-11, granted 2018-02-20 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US10572491B2 Methods, systems, and media for presenting related media content items
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2015-11-19, granted 2020-02-25 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US10394777B2 Fast orthogonal projection

Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.

filed 2015-11-25, granted 2019-08-27 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US10892968B2 Systems and methods for latency reduction in content item interactions using client-generated click identifiers

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2015-12-18, granted 2021-01-12 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate

US10075815B1 System and method for selecting and serving content items based on sensor data from mobile devices

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2015-12-21, granted 2018-09-11 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20170199888A1 Detecting visibility of a content item in a content item slot on a resource

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-01-13 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US10423674B1 Serving multiple content items responsive to a single request

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2016-02-03, granted 2019-09-24 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US10289296B1 Systems and methods for dynamically appending supplemental content to an information resource responsive to scroll activity

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2016-02-08, granted 2019-05-14 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US20170236148A1 Efficient Content Distribution

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-02-16 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US10402869B2 System and methods for automatic verification of advertiser identifier in advertisements

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.

filed 2016-02-26, granted 2019-09-03 · leak modules: IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US10572750B2 Publisher tool for controlling sponsored content quality across mediation platforms

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.

filed 2016-03-03, granted 2020-02-25 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9607314B1 Expansion of high performing placement criteria

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2016-03-10, granted 2017-03-28 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US11087359B1 Immersive content items
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2016-03-22, granted 2021-08-10 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US10084854B2 Response latency reduction in fixed allocation content selection infrastructure
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2016-04-22, granted 2018-09-25 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US11314831B2 Allocating communication resources via information technology infrastructure
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2016-04-25, granted 2022-04-26 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US10452724B2 Attribution model for content item conversions
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2016-05-18, granted 2019-10-22 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals

US11694237B2 Systems and methods for dynamically inserting content item slots in an information resource
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2016-05-27, granted 2023-07-04 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US10606907B1 Interactive browsable content items
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2016-06-01, granted 2020-03-31 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US10366414B1 Presentation of content items in view of commerciality
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2016-06-10, granted 2019-07-30 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData

US11580186B2 Reducing latency of digital content delivery over a network
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2016-06-14, granted 2023-02-14 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US10394911B1 Selecting content for devices specific to a particular user
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2016-06-17, granted 2019-08-27 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9799050B2 Controlling content distribution
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2016-06-20, granted 2017-10-24 · leak modules: PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US10303925B2 Optimization processes for compressing media content
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2016-06-24, granted 2019-05-28 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US10254935B2 Systems and methods of providing content selection
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2016-06-29, granted 2019-04-09 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US10839310B2 Selecting content items using reinforcement learning
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2016-07-15, granted 2020-11-17 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US11080762B1 Content item auction bidding
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2016-07-15, granted 2021-08-03 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US9852189B1 Augmenting a content item using search results content
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2016-07-18, granted 2017-12-26 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US9811587B1 Contextual content distribution
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2016-08-01, granted 2017-11-07 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US10210278B2 Optimized digital components
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2016-08-29, granted 2019-02-19 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10146743B2 Systems and methods for optimizing content layout using behavior metrics
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2016-09-09, granted 2018-12-04 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US10129309B2 Mixing content into a content stream
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2016-10-07, granted 2018-11-13 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US10482105B1 External verification of content popularity
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2016-10-10, granted 2019-11-19 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US10057345B2 Optimization of a multi-channel system using a feedback loop
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2016-10-11, granted 2018-08-21 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10289748B2 Optimizing transmission of digital components

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2016-10-14, granted 2019-05-14 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US10346492B2 Content placement criteria expansion

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2016-10-20, granted 2019-07-09 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US10102545B2 Retargeting in a search environment

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2016-11-28, granted 2018-10-16 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata

US10404814B1 Systems and methods for arranging and storing content selection parameters based on selection policies

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.

filed 2016-12-01, granted 2019-09-03 · leak modules: IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US9767079B1 Serving expandable content items

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.

filed 2016-12-12, granted 2017-09-19 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US10789276B2 Network based content transmission based on client device parameters

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2016-12-14, granted 2020-09-29 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US10397056B2 Optimized digital component analysis system

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2016-12-16, granted 2019-08-27 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US20190317970A1 Content search engine

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2016-12-16 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9928308B1 Selecting a template for a content item

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-12-20, granted 2018-03-27 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US10671690B2 Search and retrieval of keyed data maintained using a keyed database

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2016-12-29, granted 2020-06-02 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US10409463B2 Forking digital content items between digital topical environments
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2016-12-30, granted 2019-09-10 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US9830382B1 Search lift remarketing
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2017-01-09, granted 2017-11-28 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US10496713B2 Gain adjustment component for computer network routing infrastructure
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2017-02-01, granted 2019-12-03 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US10540055B2 Generating interactive content items based on content displayed on a computing device
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2017-02-08, granted 2020-01-21 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US10311472B1 Expansion of high performing placement criteria
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2017-02-14, granted 2019-06-04 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US11334910B2 Advertisements in view
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2017-02-17, granted 2022-05-17 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US10229439B1 Content selection using distribution parameter data
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2017-05-08, granted 2019-03-12 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US10580033B2 Suggesting and/or providing targeting criteria for advertisements
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2017-05-16, granted 2020-03-03 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US10445406B1 Automatically determining a size for a content item for a web page
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2017-06-13, granted 2019-10-15 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US10943144B2 Web-based data extraction and linkage
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2017-06-29, granted 2021-03-09 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US10762146B2 Content selection and presentation of electronic content
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-07-26, granted 2020-09-01 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US10671927B1 Impression effect modeling for content items
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2017-07-27, granted 2020-06-02 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US10430453B1 Identifying alternate content distribution locations
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-08-04, granted 2019-10-01 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US11086892B1 Search result content item enhancement
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2017-08-23, granted 2021-08-10 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US10645146B2 Transmitting high latency digital components in a low latency environment
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2017-09-06, granted 2020-05-05 · leak modules: NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US10417286B1 Content Selection
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2017-09-14, granted 2019-09-17 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US10726453B1 Content item impression effect decay
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2017-09-14, granted 2020-07-28 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US10559006B2 Controlling content distribution
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2017-09-18, granted 2020-02-11 · leak modules: PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US10445388B2 Selection bias correction for paid search in media mix modeling
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2017-09-19, granted 2019-10-15 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US10747821B2 Optimized digital component analysis system
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-10-11, granted 2020-08-18 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US10715478B2 Event-based comment grouping for content items
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-10-13, granted 2020-07-14 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US11615128B1 Contextual content distribution
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2017-10-25, granted 2023-03-28 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US10095777B1 Search lift remarketing
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2017-10-26, granted 2018-10-09 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US10659544B2 Opt-out compliance
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2017-11-14, granted 2020-05-19 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10528571B2 Augmenting a content item using search results content
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2017-11-20, granted 2020-01-07 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US10657310B2 Ordinal positioning of content items based on viewport
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-12-08, granted 2020-05-19 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US10979284B2 Systems and methods for providing cross-network event attribution
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-12-08, granted 2021-04-13 · leak modules: IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US10565118B2 Systems and methods for prefetching content items
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data.
filed 2017-12-12, granted 2020-02-18 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US11347763B2 Providing a notification in place of content item
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-12-19, granted 2022-05-31 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US10747835B2 Content item audience selection
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2018-01-03, granted 2020-08-18 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US10671798B2 Digital component backdrop rendering
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2018-02-01, granted 2020-06-02 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10535084B1 System and method for providing content items with format elements
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2018-02-12, granted 2020-01-14 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US10242120B1 Selecting a template for a content item
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2018-02-14, granted 2019-03-26 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US10748186B2 Providing a modified content item to a user
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2018-02-17, granted 2020-08-18 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US10671836B2 Optimization processes for compressing media content
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2018-03-01, granted 2020-06-02 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US10706443B2 Dynamic insertion of content items into resources
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2018-04-09, granted 2020-07-07 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US11288336B2 Systems and methods for providing content items in situations involving suboptimal network conditions
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2018-04-18, granted 2022-03-29 · leak modules: PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US10824792B2 Systems and methods for providing call context to content providers
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2018-05-21, granted 2020-11-03 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US11095723B2 Maintaining session identifiers across multiple webpages for content selection
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2018-06-15, granted 2021-08-17 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US10250681B2 Optimization of a multi-channel system using a feedback loop
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2018-06-22, granted 2019-04-02 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11164208B2 Presenting options for content delivery
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2018-07-11, granted 2021-11-02 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US11089437B2 System and method for selecting and serving content items based on sensor data from mobile devices
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2018-08-21, granted 2021-08-10 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20190050913A1 Methods, apparatuses, and computer-readable media for content creation and distribution

Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Write for natural spoken language and conversational queries.

filed 2018-08-31 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US20200090220A1 Creating content items

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2018-09-17 · leak modules: PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US11216449B2 Content item profiles

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2018-09-27, granted 2022-01-04 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US11036813B2 Content placement criteria expansion

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2018-09-28, granted 2021-06-15 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US20190320222A1 Content item display states on user devices

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2018-10-04 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US10666589B2 Identifying transitions within media content items

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2018-10-30, granted 2020-05-26 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US10997270B2 Optimized digital components

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2019-01-04, granted 2021-05-04 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11132718B1 Content selection using distribution parameter data

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2019-01-24, granted 2021-09-28 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US10997265B1 Selecting a template for a content item

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2019-02-14, granted 2021-05-04 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US10817931B2 Systems and methods for selecting third party content based on feedback

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2019-02-20, granted 2020-10-27 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US10609142B2 Optimization of a multi-channel system using a feedback loop

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2019-02-21, granted 2020-03-31 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11971921B2 Comprehensibility-based identification of educational content of multiple content types

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2019-03-01, granted 2024-04-30 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10747940B2 Displaying graphical content items

Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.

filed 2019-03-07, granted 2020-08-18 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations

US20210406333A1 Causal selection preferences

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data.

filed 2019-03-18 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US10515132B2 Optimizing transmission of digital components

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2019-03-29, granted 2019-12-24 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US10838609B2 Systems and methods for dynamically appending supplemental content to an information resource responsive to scroll activity

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2019-04-09, granted 2020-11-17 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US10943259B1 Expansion of high performing placement criteria

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2019-04-19, granted 2021-03-09 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US10671267B2 Systems and methods for presentation of content items relating to a topic

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.

filed 2019-05-06, granted 2020-06-02 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefDocumentMetadata

US12026790B2 Methods, systems, and media for identifying abusive content items

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2019-05-13, granted 2024-07-02 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US11809502B1 Selecting content for devices specific to a particular user

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2019-07-24, granted 2023-11-07 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US11144588B2 Determining and utilizing actual purpose of user-assembled collection of electronic content items

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2019-07-30, granted 2021-10-12 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US11461803B2 Content item slot location suggestions
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2019-08-27, granted 2022-10-04 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US11176575B2 Dynamic content aggregation
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2019-09-09, granted 2021-11-16 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData

US11586684B2 Serving multiple content items responsive to a single request
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2019-09-10, granted 2023-02-21 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US11120194B2 Automatically determining a size for a content item for a web page
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2019-09-27, granted 2021-09-14 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US10817904B2 Controlling content distribution
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2019-10-17, granted 2020-10-27 · leak modules: PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US11361048B2 Conditional interpretation of a single style definition identifier on a resource
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2019-10-28, granted 2022-06-14 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US10917310B2 Modeling distribution of digital components over a network
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2019-10-29, granted 2021-02-09 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData

US11379540B2 Gain adjustment component for computer network routing infrastructure
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2019-10-30, granted 2022-07-05 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US11138210B2 Augmenting a content item using search results content
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2019-11-26, granted 2021-10-05 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US11625472B2 Converting static content items into interactive content items
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2019-12-03, granted 2023-04-11 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US11550598B2 Systems and methods for adding digital content during an application opening operation

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2019-12-13, granted 2023-01-10 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US12216726B2 Provision of different content pages based on varying user interactions with a single content item

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data.

filed 2020-02-06, granted 2025-02-04 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US11727050B2 Publisher tool for controlling sponsored content quality across mediation platforms

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.

filed 2020-02-07, granted 2023-08-15 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11354245B2 Systems and methods for prefetching content items

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data.

filed 2020-02-17, granted 2022-06-07 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US10965747B2 Optimization of a multi-channel system using a feedback loop

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2020-02-21, granted 2021-03-30 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11816111B2 Methods, systems, and media for presenting related media content items

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2020-02-24, granted 2023-11-14 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US12445468B2 Systems and methods for generating and selecting local content to improve security and utilization of network resources

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2020-03-13, granted 2025-10-14 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US11258843B2 Transmitting high latency digital components in a low latency environment

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.

filed 2020-03-24, granted 2022-02-22 · leak modules: NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US11372946B2 Search and retrieval of keyed data maintained using a keyed database

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2020-04-10, granted 2022-06-28 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US11409944B2 Ordinal positioning of content items based on viewport

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2020-04-13, granted 2022-08-09 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US11336737B2 Opt-out compliance

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2020-04-16, granted 2022-05-17 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11743530B2 Systems and methods for improved searching and categorizing of media content items based on a destination for the media content machine learning

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2020-04-30, granted 2023-08-29 · leak modules: ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US11055474B2 Digital component backdrop rendering

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2020-05-29, granted 2021-07-06 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11048402B2 Systems and methods for presentation of content items relating to a topic

Rule: Use consistent entity names, sameAs schema, and structured data. Write in natural vocabulary. ML models understand context and fluff. Structure answers: question heading, tight answer, then depth.

filed 2020-06-01, granted 2021-06-29 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US11443349B2 Dynamic insertion of content items into resources

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2020-07-06, granted 2022-09-13 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US11250479B2 Content item impression effect decay

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2020-07-14, granted 2022-02-15 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US11250087B2 Content item audience selection

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2020-07-16, granted 2022-02-15 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US11663277B2 Content selection and presentation of electronic content

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2020-08-28, granted 2023-05-30 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US12182213B2 Automated user language detection for content selection

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2020-09-14, granted 2024-12-31 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US11201952B2 Controlling content distribution

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2020-09-23, granted 2021-12-14 · leak modules: PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US12062085B2 Adjust ads auction using predicted user response to an in-ad survey
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2020-10-19, granted 2024-08-13 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US11886524B2 Limiting provision and display of redundant digital components on a client device
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2020-11-23, granted 2024-01-30 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US11228663B2 Controlling content distribution
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2020-11-30, granted 2022-01-18 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US11494457B1 Selecting a template for a content item
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2020-11-30, granted 2022-11-08 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US12386925B2 Integrating secure watermarks into content
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2020-11-30, granted 2025-08-12 · leak modules: ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US11481386B2 Content item profiles
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2020-12-02, granted 2022-10-25 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US11609943B2 Contextual content distribution
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2020-12-07, granted 2023-03-21 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US12020286B1 Immersive content items
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2020-12-07, granted 2024-06-25 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US11405454B2 Optimization of a multi-channel system using a feedback loop
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2020-12-08, granted 2022-08-02 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11477103B2 Systems and methods for latency reduction in content item interactions using client-generated click identifiers
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2020-12-10, granted 2022-10-18 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate

US12147480B2 Content keyword identification
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Optimize video for watch time and retention.
filed 2021-01-04, granted 2024-11-19 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, VideoPerDocData

US11917486B2 System and method for selecting and serving content items based on sensor data from mobile devices

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2021-01-11, granted 2024-02-27 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11430003B1 Expansion of high performing placement criteria

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2021-02-03, granted 2022-08-30 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US12477175B2 Systems and methods for adjusting an aspect ratio of a content slot

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2021-04-07, granted 2025-11-18 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11435890B2 Systems and methods for presentation of content items relating to a topic

Rule: Use consistent entity names, sameAs schema, and structured data. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2021-06-28, granted 2022-09-06 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US12580991B2 Maintaining session identifiers across multiple webpages for content selection

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2021-07-22, granted 2026-03-17 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US11899706B2 Content-specific keyword notification system

Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Build content for diverse user contexts and search histories. Avoid AI-stitched content. GibberishScore uses language model scoring.

filed 2021-07-30, granted 2024-02-13 · leak modules: IndexingDocjoinerAnchorSpamInfo, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, SpamBrainData, SpamBrainScore

US11704701B1 Content selection using distribution parameter data

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2021-09-03, granted 2023-07-18 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US11586801B2 Automatically determining a size for a content item for a web page

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2021-09-14, granted 2023-02-21 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US12437318B2 Presenting options for content delivery

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2021-10-01, granted 2025-10-07 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US12614111B2 Machine learning techniques for user group based content distribution

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2021-10-28, granted 2026-04-28 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US12141833B2 Controlling content distribution
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2021-11-12, granted 2024-11-12 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US11789939B2 Content item profiles
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2022-01-03, granted 2023-10-17 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US12038987B2 Content item audience selection
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2022-01-10, granted 2024-07-16 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US12039565B2 Advertisements in view
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2022-05-02, granted 2024-07-16 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US12437021B2 Search and retrieval of keyed data maintained using a keyed database
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2022-05-13, granted 2025-10-07 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US11711424B2 Optimization of a multi-channel system using a feedback loop
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2022-05-26, granted 2023-07-25 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20220329430A1 Systems and methods for authenticated control of content delivery
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2022-06-21 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US11989755B1 Expansion of high performing placement criteria
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2022-08-01, granted 2024-05-21 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US12292936B2 Intelligent asset suggestions based on both previous phrase and whole asset performance
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2022-08-30, granted 2025-05-06 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US12002075B2 Dynamic insertion of content items into resources
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2022-09-13, granted 2024-06-04 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US12363368B2 Event-based content distribution
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2022-10-26, granted 2025-07-15 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US12388910B2 Providing content based on event related information
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2022-11-08, granted 2025-08-12 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US12321410B2 Reducing latency of digital content delivery over a network
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2022-11-30, granted 2025-06-03 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US20250265359A1 Configuration based dataset generation for content serving systems
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2023-01-11 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US20250094508A1 Digital component provision based on contextual feature driven audience interest profiles
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2023-01-18 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US12067342B2 Automatically determining a size for a content item for a web page
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2023-02-21, granted 2024-08-20 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US20230221837A1 Coalescing Notifications Associated with Interactive Digital Content
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2023-03-16 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US12118073B2 Converting static content items into interactive content items
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2023-04-10, granted 2024-10-15 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US12561382B2 Automatically restructuring search campaigns
Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2023-04-17, granted 2026-02-24 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData

US12154144B2 Systems and methods for dynamically inserting content item slots in an information resource
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2023-05-17, granted 2024-11-26 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US20240420203A1 Accessing Information Associated With Digital Components
Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2023-06-14 · leak modules: ImageData, ImagePerDocData, PerDocData, SmartphonePerDocData

US12242531B2 Publisher tool for controlling sponsored content quality across mediation platforms
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2023-06-27, granted 2025-03-04 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US12265524B2 Content item profiles
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2023-08-03, granted 2025-04-01 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US20250225213A1 System and Method for Policy Enforcement
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2023-09-14 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US20250342219A1 Fuzzy cache architecture for serving content items
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2024-05-06 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20240378251A1 Guided content generation using pre-existing media assets
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data.
filed 2024-05-09 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US12321983B2 Adjust ads auction using predicted user response to an in-ad survey
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2024-07-02, granted 2025-06-03 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US20260064787A1 Audience-Based Content Modification
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2024-09-05 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US12554779B1 Generating content items based on source document metadata using a generative neural network
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2024-10-08, granted 2026-02-17 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US20250284741A1 Efficient and scalable data processing and modeling
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2024-10-15 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US20250356395A1 Content Group Generation for Content Delivery Campaigns
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2024-10-21 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefEntityJoin

US20260119512A1 Using Adjusted Retrieval-Augmented Generation for Content Selection for Incorporation in Large Language Model Output
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2024-12-19 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US20250245284A1 Predicting non-observable parameters for digital components
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data.
filed 2025-01-27 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

Images, video & media (331)

US7801738B2 System and method for rating documents comprising an image
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2004-05-10, granted 2010-09-21 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals

US7788258B1 Automatic determination of whether a document includes an image gallery
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2004-06-21, granted 2010-08-31 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8065611B1 Method and system for mining image searches to associate images with concepts
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2004-06-30, granted 2011-11-22 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US7877382B1 System and methods for detecting images distracting to a user
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2004-12-31, granted 2011-01-25 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US7796837B2 Processing an image map for display on computing device
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2005-09-22, granted 2010-09-14 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US7890858B1 Transferring, processing and displaying multiple images using single transfer request
Rule: Use descriptive alt text and surrounding context for images. Use JSON-LD structured data for entity types and relationships.
filed 2005-10-11, granted 2011-02-15 · leak modules: ImageData, ImagePerDocData, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US8566712B1 Image management
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2006-01-04, granted 2013-10-22 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US8577166B1 Optimizing web site images using a focal point
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2006-03-31, granted 2013-11-05 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US7979785B1 Recognizing table of contents in an image sequence
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2006-10-04, granted 2011-07-12 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8090222B1 Selection of an image or images most representative of a set of images
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2006-11-15, granted 2012-01-03 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US8645860B2 Automatic display of resized images
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2007-03-15, granted 2014-02-04 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9710553B2 Graphical user interface for management of remotely stored videos, and captions or subtitles thereof

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships.

filed 2007-05-25, granted 2017-07-18 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData, VideoPerDocData

US8160923B2 Video advertisements

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2008-01-18, granted 2012-04-17 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US8181197B2 System and method for voting on popular video intervals

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships.

filed 2008-02-06, granted 2012-05-15 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US8112702B2 Annotating video intervals

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2008-02-19, granted 2012-02-07 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8504547B1 Customizing image search for user attributes

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2008-04-23, granted 2013-08-06 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate

US8538943B1 Providing images of named resources in response to a search query

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.

filed 2008-07-24, granted 2013-09-17 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US8775436B1 Image selection for news search

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2008-08-20, granted 2014-07-08 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9396258B2 Recommending video programs

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2009-01-22, granted 2016-07-19 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US8132200B1 Intra-video ratings

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2009-03-30, granted 2012-03-06 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8209330B1 Ordering image search results

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2009-05-29, granted 2012-06-26 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US8856124B2 Co-selected image classification

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2009-06-03, granted 2014-10-07 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US8478052B1 Image classification
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2009-07-17, granted 2013-07-02 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US8918330B1 Display of videos based on referrers
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2010-01-20, granted 2014-12-23 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US8326836B1 Providing time series information with search results
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2010-07-13, granted 2012-12-04 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US8312012B1 Automatic determination of whether a document includes an image gallery
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2010-07-23, granted 2012-11-13 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8064727B2 Adaptive image maps
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2010-08-20, granted 2011-11-22 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8352465B1 Grouping of image search results
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2010-09-03, granted 2013-01-08 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US20120096344A1 Rendering or resizing of text and images for display on mobile / small screen devices
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2010-10-19 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefEntityJoin, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US20150169571A1 Social Image Search
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2010-10-21 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US8788514B1 Triggering music answer boxes relevant to user search queries
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2010-10-28, granted 2014-07-22 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US8533134B1 Graph-based fusion for video classification
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2010-11-02, granted 2013-09-10 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US8761512B1 Query by image
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2010-12-03, granted 2014-06-24 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US8732571B2 Methods and systems for generating and displaying a preview image of a content area
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2011-03-31, granted 2014-05-20 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US20140300811A1 Methods and Systems for Providing and Playing Videos Having Multiple Tracks of Timed Text Over A Network
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships. Optimize video for watch time and retention.
filed 2011-04-18 · leak modules: PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData, VideoPerDocData

US10108620B2 Associating still images and videos
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-04-29, granted 2018-10-23 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation

US9652462B2 Identifying responsive resources across still images and videos
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-04-29, granted 2017-05-16 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation

US8994748B2 Anchors for displaying image sprites, sub-regions and 3D images
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-05-10, granted 2015-03-31 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US8346815B2 Dynamic image display area and image display within web search results
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-05-12, granted 2013-01-01 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US8407207B1 Measuring video content of web domains
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-05-12, granted 2013-03-26 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US9465814B2 Annotating search results with images
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-05-12, granted 2016-10-11 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US9098529B1 Determining content for delivery based on imagery
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-05-25, granted 2015-08-04 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US8688514B1 Ad selection using image data
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-06-24, granted 2014-04-01 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US8458174B1 Semantic image label synthesis
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-09-02, granted 2013-06-04 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US8250228B1 Pausing or terminating video portion while continuing to run audio portion of plug-in on browser

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Optimize video for watch time and retention.

filed 2011-09-27, granted 2012-08-21 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, VideoPerDocData

US8605119B1 Scaling image modules

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2011-09-30, granted 2013-12-10 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US8510644B2 Optimization of web page content including video

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2011-10-20, granted 2013-08-13 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US20150309970A1 State information in a stateless environment

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.

filed 2011-11-18 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US8498490B1 Selection of an image or images most representative of a set of images

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2011-11-21, granted 2013-07-30 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US9563646B1 Method and system for mining image searches to associate images with concepts

Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2011-11-21, granted 2017-02-07 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US8880535B1 System and method for selecting user generated content related to a point of interest

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2011-11-29, granted 2014-11-04 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US20130151937A1 Selective image loading in mobile browsers

Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2011-12-08 · leak modules: ImageData, ImagePerDocData, PerDocData, SmartphonePerDocData

US9684644B2 Annotating video intervals

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2011-12-30, granted 2017-06-20 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20160062845A1 Populating Image Metadata By Cross-Referencing Other Images

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2012-01-09 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US10171522B1 Video commentary

Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth. Optimize video for watch time and retention.

filed 2012-01-13, granted 2019-01-01 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo, VideoPerDocData

US9044183B1 Intra-video ratings

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2012-01-31, granted 2015-06-02 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8913068B1 Displaying video on a browser
Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships.
filed 2012-02-06, granted 2014-12-16 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData, VideoPerDocData

US9037676B1 Systems and methods for scheduled online publishing of internet media
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-03-02, granted 2015-05-19 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData

US9188444B2 3D object positioning in street view
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-03-07, granted 2015-11-17 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US20150153933A1 Navigating Discrete Photos and Panoramas
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-03-16 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US8799257B1 Searching based on audio and/or visual features of documents
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2012-03-19, granted 2014-08-05 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData

US20140365900A1 Replication of mobile rendering using computer-readable images
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-03-30 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US8826320B1 System and method for voting on popular video intervals
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships.
filed 2012-04-03, granted 2014-09-02 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US8782077B1 Query image search
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-04-04, granted 2014-07-15 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals

US8983939B1 Query image search
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-04-04, granted 2015-03-17 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals

US9002831B1 Query image search
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-04-04, granted 2015-04-07 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals

US9031960B1 Query image search
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-04-04, granted 2015-05-12 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals

US8280197B1 Managed background loading of image resources in resource-constrained devices
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Write for natural spoken language and conversational queries.
filed 2012-05-01, granted 2012-10-02 · leak modules: NlpSaftDocument, PerDocData, SmartphonePerDocData

US8949253B1 Low-overhead image search result generation
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-05-24, granted 2015-02-03 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9483556B1 Aggregating photos captured at an event
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-05-25, granted 2016-11-01 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US20130321461A1 Method and System for Navigation to Interior View Imagery from Street Level Imagery
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-05-29 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US8745059B1 Clustering queries for image search
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-05-29, granted 2014-06-03 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US9218546B2 Choosing image labels
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-06-01, granted 2015-12-22 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US9158857B2 Identifying landing pages for images
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-06-05, granted 2015-10-13 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate

US20150088859A1 Click magnet images
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-06-21 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US8898139B1 Systems and methods for dynamic visual search engine
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-06-22, granted 2014-11-25 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US8566331B1 Ordering image search results
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-06-25, granted 2013-10-22 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9098584B1 Image search privacy protection techniques
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-07-19, granted 2015-08-04 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8819004B1 Ranking image search results using hover data
Rule: For image SEO, Google tracks not just clicks but hover behavior—images that get hovered over but not clicked can be a negative signal (low click-to-hover ratio demotes). Ensure your images are genuinely clickworthy: use compelling thumbnails that accurately represent the target page. For rare queries (low total clicks), hover behavior matters more. The leak's ImagePerDocData and NavboostCraps click signals confirm image-level user interaction tracking. Optimize image file names, alt text, and surrounding context to ensure the right users click through.
filed 2012-08-15, granted 2014-08-26 · leak modules: QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, ImagePerDocData, ImageData, ImageRepositoryCrawlStatusInfo, CompressedQualitySignals, PerDocData

US9128993B2 Presenting secondary music search result links
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2012-08-15, granted 2015-09-08 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US8898150B1 Collecting image search event information
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2012-09-10, granted 2014-11-25 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9274678B2 Identifying a thumbnail image to represent a video
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-09-13, granted 2016-03-01 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US8843478B1 Grouping of image search results
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-09-14, granted 2014-09-23 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US8914349B2 Dynamic image display area and image display within web search results
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-09-14, granted 2014-12-16 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US8924372B2 Dynamic image display area and image display within web search results
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-09-14, granted 2014-12-30 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US9244917B1 Generating a layout
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-09-28, granted 2016-01-26 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9372871B2 Combining unstructured image and 3D search results for interactive search and exploration
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2012-10-09, granted 2016-06-21 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US8838609B1 IDF weighting of LSH bands for live reference ingestion
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-10-10, granted 2014-09-16 · leak modules: NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US9235325B2 Persona chooser
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-10-17, granted 2016-01-12 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9497276B2 Trackable sharing of on-line video content
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-10-17, granted 2016-11-15 · leak modules: CrawlerChangerateUrlChangerate, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, SnippetExtraInfo

US8732165B1 Automatic determination of whether a document includes an image gallery
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-11-12, granted 2014-05-20 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9251171B2 Propagating image signals to images
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2012-11-30, granted 2016-02-02 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US8891907B2 System and method of identifying visual objects
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-12-04, granted 2014-11-18 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US9116926B2 Sharing photos
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.
filed 2012-12-20, granted 2015-08-25 · leak modules: ImageData, ImagePerDocData, PerDocData, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US8996516B2 Adjacent search results exploration
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2013-01-02, granted 2015-03-31 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US9690762B1 Manipulating image content items through determination and application of multiple transparency values to visually merge with other content as part of a web page
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-01-14, granted 2017-06-27 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9171018B2 System and method for associating images with semantic entities
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-01-16, granted 2015-10-27 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US9645981B1 Extraction of business-relevant image content from the web
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-01-18, granted 2017-05-09 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20140226907A1 Automated detection of distracting images
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2013-02-14 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9256637B2 Suggesting media content based on an image capture
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-02-22, granted 2016-02-09 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US9591050B1 Image recommendations for thumbnails for online media items based on user activity
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-02-28, granted 2017-03-07 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9424279B2 Presenting image search results
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-03-08, granted 2016-08-23 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US20150169177A1 Classifying particular images as primary images
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2013-03-12 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9183312B2 Image display within web search results
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-03-12, granted 2015-11-10 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US9015163B1 Using cross-matching between users and matching against reference data to facilitate content identification
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-03-13, granted 2015-04-21 · leak modules: IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, RichsnippetsPageMap

US9195717B2 Image result provisioning based on document classification
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-03-13, granted 2015-11-24 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9110988B1 Methods, systems, and media for aggregating and presenting multiple videos of an event
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-03-14, granted 2015-08-18 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate

US9305025B2 Content selection based on image content
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-03-14, granted 2016-04-05 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US9507860B1 Systems, methods, and media for identifying content
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2013-03-14, granted 2016-11-29 · leak modules: NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, SnippetExtraInfo

US9135317B2 Image-centric documents
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-03-15, granted 2015-09-15 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US8949229B1 Measuring video content of web domains
Rule: Build a substantial video library on your domain (quantity score) and ensure videos generate long-click engagement (quality score). Use video-indicative URL patterns. Domains with higher video content volume and engagement get boosted in video search operations.
filed 2013-03-25, granted 2015-02-03 · leak modules: QualityNsrNsrData, PerDocData, QualityNavboostCrapsCrapsClickSignals, IndexingCrawlerIdServingDocumentIdentifier

US9076079B1 Selecting photographs for a destination
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use descriptive alt text and surrounding context for images.
filed 2013-03-29, granted 2015-07-07 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US9483495B1 Selecting photographs for a destination or point of interest
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use descriptive alt text and surrounding context for images.
filed 2013-03-29, granted 2016-11-01 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US9123330B1 Large-scale speaker identification
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2013-05-01, granted 2015-09-01 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsGeoTopicality

US8938449B1 Semantic image label synthesis
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-05-03, granted 2015-01-20 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US8787683B1 Image classification
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-06-05, granted 2014-07-22 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US9304648B2 Video segments for a video related to a task
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-06-26, granted 2016-04-05 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData

US8744191B1 Selection of an image or images most representative of a set of images
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-07-29, granted 2014-06-03 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US8782029B1 Customizing image search for user attributes
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-08-05, granted 2014-07-15 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate

US9465871B1 External verification of content popularity
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-08-07, granted 2016-10-11 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US9026526B1 Providing images of named resources in response to a search query
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2013-09-06, granted 2015-05-05 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US9501499B2 Methods and systems for creating image-based content based on text-based content
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-10-21, granted 2016-11-22 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate

US9135224B1 Display of videos based on referrers
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-12-28, granted 2015-09-15 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US9690768B2 Annotating video intervals
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-12-31, granted 2017-06-27 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US9208243B2 Systems and methods for processing machine readable codes by a locked device
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2014-01-07, granted 2015-12-08 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US9639634B1 Identifying related videos based on relatedness of elements tagged in the videos
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2014-01-28, granted 2017-05-02 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US9619470B2 Adaptive music and video recommendations
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-02-04, granted 2017-04-11 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9727582B2 Providing photo heat maps
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-02-18, granted 2017-08-08 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9971790B2 Generating descriptive text for images in documents using seed descriptors
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data.
filed 2014-03-14, granted 2018-05-15 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US20150169708A1 Providing recently selected images
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-04-10 · leak modules: ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US9424338B2 Clustering queries for image search
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-04-29, granted 2016-08-23 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US10216737B1 Subscribe to people in videos
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-05-21, granted 2019-02-26 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US9268795B2 Selection of an image or images most representative of a set of images
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-05-28, granted 2016-02-23 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US9418482B1 Discovering visited travel destinations from a set of digital images
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2014-05-28, granted 2016-08-16 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9613061B1 Image selection for news search
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-05-28, granted 2017-04-04 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9773057B2 Content item usage based song recommendation
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Optimize video for watch time and retention.
filed 2014-06-06, granted 2017-09-26 · leak modules: PerDocData, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, VideoPerDocData

US9201903B2 Query by image
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2014-06-10, granted 2015-12-01 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9195741B2 Triggering music answer boxes relevant to user search queries
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-06-24, granted 2015-11-24 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US9208420B2 Techniques for generating customized two-dimensional barcodes
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-06-25, granted 2015-12-08 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo

US9569498B2 Using image features to extract viewports from images
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2014-06-27, granted 2017-02-14 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9594826B2 Co-selected image classification
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-06-30, granted 2017-03-14 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US9146997B2 Customizing image search for user attributes
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-07-14, granted 2015-09-29 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate

US9183226B2 Image classification
Rule: Ensure text near images on your pages accurately describes the image content using exact-match keywords, since the image classification system assigns labels based on associated text and image features. Use descriptive file names, alt text, captions, and surrounding text that match target query terms. Exact label-to-query matches receive the highest relevance boost. Images with high selection rates for specific n-grams are used as positive training data, so building click engagement for image results reinforces classification.
filed 2014-07-21, granted 2015-11-10 · leak modules: PerDocData, CompressedQualitySignals, QualityNavboostCrapsCrapsClickSignals

US9218369B2 Ranking image search results using hover data
Rule: Avoid 'click magnet' images that attract hovers but don't convert to clicks — Google explicitly demotes images with high hover-to-click ratios. Ensure your images deliver on user expectations so hover-to-click conversion is strong. For less popular queries, hover engagement matters more (higher weighting), so optimize image thumbnails for visual appeal in less competitive niches. Build genuine click engagement rather than just visual attention, as the click-to-hover ratio directly affects ranking.
filed 2014-07-29, granted 2015-12-22 · leak modules: QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNavboostCrapsFeatureCrapsData, QualityNavboostCrapsStatsWithWeightsProto

US9798708B1 Annotating relevant content in a screen capture image
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2014-08-04, granted 2017-10-24 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData

US9811352B1 Replaying user input actions using screen capture images
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2014-08-04, granted 2017-11-07 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData

US9916328B1 Providing user assistance from interaction understanding
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2014-08-04, granted 2018-03-13 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData

US9208154B1 IDF weighting of LSH bands for live reference ingestion
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-08-13, granted 2015-12-08 · leak modules: NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US9317468B2 Personal content streams based on user-topic profiles
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-08-27, granted 2016-04-19 · leak modules: PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US9116921B2 Grouping of image search results
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-09-22, granted 2015-08-25 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US20160019620A1 Dynamic image sitelinks
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2014-09-29 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9460160B1 System and method for selecting user generated content related to a point of interest
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-10-08, granted 2016-10-04 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US9442950B2 Systems and methods for dynamic visual search engine
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-10-29, granted 2016-09-13 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US9442957B2 System and method of identifying visual objects
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2014-11-14, granted 2016-09-13 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US9177046B2 Refining image relevance models
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2014-11-17, granted 2015-11-03 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9471695B1 Semantic image navigation experiences
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2014-12-02, granted 2016-10-18 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9503456B1 Video chat abuse detection based on external context
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-01-22, granted 2016-11-22 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US9189498B1 Low-overhead image search result generation
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-01-30, granted 2015-11-17 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US10496650B1 Fine-grained video classification
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2015-02-25, granted 2019-12-03 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US20160267569A1 Providing Search Results Comprising Purchase Links For Products Associated With The Search Results
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2015-03-10 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US9361318B2 Adjacent search results exploration
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2015-03-27, granted 2016-06-07 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US9589027B2 Using cross-matching between users and matching against reference data to facilitate content identification
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2015-04-09, granted 2017-03-07 · leak modules: IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, RichsnippetsPageMap

US9189554B1 Providing images of named resources in response to a search query
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2015-05-04, granted 2015-11-17 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US10013436B1 Image annotation based on label consensus
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2015-06-17, granted 2018-07-03 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US10162889B2 Methods, systems, and media for searching for video content
Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately. Write for intent and meaning. Neural matching connects related concepts.
filed 2015-06-18, granted 2018-12-25 · leak modules: ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, VideoPerDocData

US10282430B1 Determining content for delivery based on imagery
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-07-21, granted 2019-05-07 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US9881085B2 Methods, systems, and media for aggregating and presenting multiple videos of an event
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-07-24, granted 2018-01-30 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate

US10467284B2 Establishment anchoring with geolocated imagery
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-08-03, granted 2019-11-05 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US9965494B2 Sharing photos
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.
filed 2015-08-13, granted 2018-05-08 · leak modules: ImageData, ImagePerDocData, PerDocData, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US10165332B2 Contextual video browsing
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2015-08-14, granted 2018-12-25 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US10387431B2 Video recommendation based on video titles
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-08-24, granted 2019-08-20 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US9990425B1 Presenting secondary music search result links
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2015-08-24, granted 2018-06-05 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US20170068870A1 Using image similarity to deduplicate video suggestions based on thumbnails
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-09-03 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate

US9600496B1 System and method for associating images with semantic entities
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-09-10, granted 2017-03-21 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US10089402B1 Display of videos based on referrers
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2015-09-11, granted 2018-10-02 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US9411827B1 Providing images of named resources in response to a search query
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2015-10-15, granted 2016-08-09 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US9396413B2 Choosing image labels
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-11-13, granted 2016-07-19 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US9792304B1 Query by image
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2015-11-19, granted 2017-10-17 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9552427B2 Suggesting media content based on an image capture
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-12-09, granted 2017-01-24 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US10116603B1 Methods, systems, and media for identifying and presenting video objects linked to a source video
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-12-10, granted 2018-10-30 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US20170177577A1 Biasing scrubber for digital content
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-12-18 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20170185670A1 Generating labels for images associated with a user
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-12-28 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US9881236B2 Organizing images associated with a user
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-12-28, granted 2018-01-30 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US11308148B2 Identifying a thumbnail image to represent a video
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2016-02-12, granted 2022-04-19 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9436737B2 Content selection based on image content
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2016-02-25, granted 2016-09-06 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US10296640B1 Video segments for a video related to a task
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2016-02-26, granted 2019-05-21 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData

US20160283483A1 Providing selected images from a set of images
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2016-03-26 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData

US10007867B2 Systems and methods for identifying entities directly from imagery
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2016-04-04, granted 2018-06-26 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US20160335664A1 Image pixel status determination
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2016-04-05 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US10318581B2 Video metadata association recommendation
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2016-04-13, granted 2019-06-11 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US10489410B2 Mapping images to search queries
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2016-04-18, granted 2019-11-26 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US10003847B2 Watch-time clustering for improving video searches, selection and provision
Rule: Create video content that drives sustained watch time across multiple audience segments. Videos that perform well across diverse viewer categories (age, geography, behavior) get broader distribution. Understand which audience segments watch longest and optimize content for those clusters.
filed 2016-04-22, granted 2018-06-19 · leak modules: QualityNavboostCrapsCrapsData, QualityNavboostCrapsFeatureCrapsData, QualityNavboostCrapsStatsWithWeightsProto, PerDocData

US10430476B2 Annotation of videos using aggregated user session data
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2016-06-02, granted 2019-10-01 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US10210181B2 Searching and annotating within images
Rule: Ensure objects within your images are clearly depicted and visually distinct so they can be matched to query-responsive images. Use high-quality, well-lit product images with clean backgrounds. Image annotations create clickable links to your site—optimize the landing pages for these annotations. Submit images to Google's image index with proper alt text and surrounding context. Objects in your images that match top-ranked image search results get annotated and linked.
filed 2016-06-28, granted 2019-02-19 · leak modules: ImageUnderstandingIndexingAnnotation, ImageUnderstandingIndexingImageRegion, ImageRegionsImageRegion, ImageRegionsImageRegions, ImageContentQueryBoost, ImageContentQueryBoostQueryBoost, ImageSearchImageSelectionInfo, ImageMustangImageLinkSelectionInfo

US10353950B2 Visual recognition using user tap locations
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2016-06-28, granted 2019-07-16 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US10802671B2 Contextual information for a displayed resource that includes an image
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2016-07-11, granted 2020-10-13 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US10691746B2 Images for query answers
Rule: For pages that may serve as answer sources, include high-quality, prominently-positioned images that are relevant to the answer content. Avoid low-information images (clipart, icons, buttons) near answer text. Use descriptive image labels and alt text matching the answer topic. Ensure your page ranks in the top-N results for question queries, since image selection is constrained to top-ranked pages. Place key images in prominent rendered positions on the page.
filed 2016-07-12, granted 2020-06-23 · leak modules: SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo, QualityPreviewSnippetRadishFeatures, PerDocData

US11055335B2 Contextual based image search results
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2016-07-15, granted 2021-07-06 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US9753951B1 Presenting image search results
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2016-07-27, granted 2017-09-05 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US20180039854A1 Personalized image collections
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2016-08-02 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US10346417B2 Optimizing digital video distribution
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data.
filed 2016-08-18, granted 2019-07-09 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US10664157B2 Image search query predictions by a keyboard
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2016-08-24, granted 2020-05-26 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US10198457B2 System and method of identifying visual objects
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2016-08-25, granted 2019-02-05 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US10860650B1 Determining which videos are newsworthy events
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2016-09-01, granted 2020-12-08 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US10212113B2 Uniform resource identifier and image sharing for contextual information display
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2016-09-19, granted 2019-02-19 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US9805428B2 Selecting photographs for a destination or point of interest
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use descriptive alt text and surrounding context for images.
filed 2016-09-29, granted 2017-10-31 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US10339175B2 Aggregating photos captured at an event
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2016-10-31, granted 2019-07-02 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US10176254B2 Systems, methods, and media for identifying content
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2016-11-28, granted 2019-01-08 · leak modules: NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, SnippetExtraInfo

US10482126B2 Determination of similarity between videos using shot duration correlation
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2016-11-30, granted 2019-11-19 · leak modules: IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10268703B1 System and method for associating images with semantic entities
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2016-12-08, granted 2019-04-23 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US10347247B2 Modulation of packetized audio signals
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2016-12-30, granted 2019-07-09 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US10146776B1 Method and system for mining image searches to associate images with concepts
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2017-01-05, granted 2018-12-04 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US10162824B2 Using cross-matching between users and matching against reference data to facilitate content identification
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2017-03-06, granted 2018-12-25 · leak modules: IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, RichsnippetsPageMap

US10747802B1 Image recommendations for thumbnails for online media items based on user activity
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2017-03-06, granted 2020-08-18 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US10430452B2 Adaptive music and video recommendations
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2017-03-08, granted 2019-10-01 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11086961B2 Visual leaf page identification and processing
Rule: Design product/image pages as true 'visual leaf pages'—images should dominate the page visually (central position, large size proportion, grid structures for multiple images). Ensure hub pages (category pages, galleries) link to these leaf pages through image-based links. Maintain consistent URL depth across pages linked from the same hub. Pages classified as visual leaf pages get search score boosts for image-search queries related to specific activities (e.g., shopping, recipes, travel).
filed 2017-04-05, granted 2021-08-10 · leak modules: ImageSearchImageIndexingInfo, ImageSearchImageSelectionInfo, ImagePerDocData, ImageData, HtmlrenderWebkitHeadlessProtoImage, IndexingEmbeddedContentEmbeddedContentInfo, PerDocData, CompressedQualitySignals

US10331733B2 System and method for presenting condition-specific geographic imagery
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2017-04-20, granted 2019-06-25 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US11030239B2 Audio based entity-action pair based selection
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2017-05-02, granted 2021-06-08 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData

US10853659B2 Methods, systems, and media for adaptive presentation of a video content item based on an area of interest

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2017-05-05, granted 2020-12-01 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores

US20180336045A1 Determining agents for performing actions based at least in part on image data

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2017-05-23 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation

US10600409B2 Balance modifications of audio-based computer program output including a chatbot selected based on semantic processing of audio

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.

filed 2017-06-09, granted 2020-03-24 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US10614122B2 Balance modifications of audio-based computer program output using a placeholder field based on content

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.

filed 2017-06-09, granted 2020-04-07 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US10652170B2 Modification of audio-based computer program output

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.

filed 2017-06-09, granted 2020-05-12 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US10657173B2 Validate modification of audio-based computer program output

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.

filed 2017-06-09, granted 2020-05-19 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US20180357309A1 Balance modifications of audio-based computer program output

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.

filed 2017-06-09 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US10573322B2 Establishment of audio-based network sessions with non-registered resources

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2017-06-13, granted 2020-02-25 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US11100538B1 Image recognition based content item selection

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2017-06-29, granted 2021-08-24 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10296592B2 Spherical video in a web browser

Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.

filed 2017-08-07, granted 2019-05-21 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US10321193B2 Sharing a user-selected video in a group communication

Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2017-09-01, granted 2019-06-11 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US11231832B2 Efficiently augmenting images with related content
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-09-13, granted 2022-01-25 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US10268663B1 Displaying graphical content items based on an audio input
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.
filed 2017-09-26, granted 2019-04-23 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations

US10846331B2 Music recommendations from trending queries
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2017-09-28, granted 2020-11-24 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate

US10671660B2 Contextual ranking of photos and text in search
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2017-09-29, granted 2020-06-02 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10510129B2 Selecting photographs for a destination or point of interest
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use descriptive alt text and surrounding context for images.
filed 2017-10-12, granted 2019-12-17 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US11403337B2 Identifying videos with inappropriate content by processing search logs
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2017-12-05, granted 2022-08-02 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US10248662B2 Generating descriptive text for images in documents using seed descriptions
Rule: Maintain consistent document structure across your website — use the same HTML layout patterns for images and their accompanying descriptive text, since Google generates templates from structural patterns. Use descriptive text near images that includes target query terms as seed descriptors. Ensure image features (aspect ratio, file names, display size) are consistent across similar product/content pages. The system requires template patterns to appear in multiple documents on your site, so standardize your image-text layout.
filed 2018-03-20, granted 2019-04-02 · leak modules: PerDocData, CompressedQualitySignals, IndexingEmbeddedContentPageSizeInfo, QualityPreviewSnippetRadishFeatures

US20180210895A1 Generating descriptive text for images
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data.
filed 2018-03-20 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US11379516B2 Similar medical image search
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2018-03-29, granted 2022-07-05 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US10185725B1 Image annotation based on label consensus
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2018-05-31, granted 2019-01-22 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US11109097B2 Watch-time accumulation for video searches
Rule: Build a consistent body of video work under a single creator identity. Google computes a creator-level authority score from accumulated watch time across all videos, weighted by viewer category quality. Creators with higher accumulated engagement get their videos ranked higher in search results.
filed 2018-06-15, granted 2021-08-31 · leak modules: QualityNavboostCrapsCrapsData, QualityNavboostCrapsFeatureCrapsData, QualityNsrNsrData, RepositoryWebrefEntityJoin, PerDocData

US10244369B1 Screen capture image repository for a user
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2018-09-14, granted 2019-03-26 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData

US10394878B2 Associating still images and videos
Rule: Ensure your images and videos on the same topic share visual features (similar compositions, color palettes, objects) so Google can associate them. Use consistent text descriptions and categories across image alt text and video titles/descriptions—category matching drives feature extraction. Cross-linking images and videos with matching visual content strengthens entity associations.
filed 2018-09-21, granted 2019-08-27 · leak modules: ImageRepositoryFrameLevelStarburstEmbeddings, ImageRepositorySUPFrameLevelEmbedding, ImageRepositoryFramePerdoc, ImageRepositoryFramePerdocs, ImageContentStarburstVersionGroup, ImageDataMultibangEntities, ImageMustangImageLinkSelectionInfo, VideoContentSearchFrameStarburstData

US11409812B1 Method and system for mining image searches to associate images with concepts
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2018-11-29, granted 2022-08-09 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US10990621B2 Using cross-matching between users and matching against reference data to facilitate content identification
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2018-12-21, granted 2021-04-27 · leak modules: IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, RichsnippetsPageMap

US10880247B2 Uniform resource identifier and image sharing for contextaul information display
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2019-01-07, granted 2020-12-29 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US11803589B2 Systems, methods, and media for identifying content
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2019-01-07, granted 2023-10-31 · leak modules: IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, ShingleInfoPerDocData

US10409855B2 System and method of identifying visual objects
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2019-01-09, granted 2019-09-10 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US20200159765A1 Performing image search using content labels
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2019-01-31 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US20200201915A1 Ranking image search results using machine learning models
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2019-01-31 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US11586927B2 Training image and text embedding models
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2019-02-01, granted 2023-02-21 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US10592261B1 Automating user input from onscreen content
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2019-04-01, granted 2020-03-17 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData

US11727681B2 Media annotation with product source linking
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2019-04-08, granted 2023-08-15 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11354356B1 Video segments for a video related to a task
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2019-05-20, granted 2022-06-07 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData

US10657144B2 Optimizing digital video distribution
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data.
filed 2019-05-24, granted 2020-05-19 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US10877982B1 Detection of popular content with narrow appeal
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth. Earn promotion through authoritative content and entity association.
filed 2019-06-03, granted 2020-12-29 · leak modules: CompressedQualitySignals, PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData, SnippetExtraInfo

US10664519B2 Visual recognition using user tap locations
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2019-06-07, granted 2020-05-26 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US10922350B2 Associating still images and videos
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2019-07-15, granted 2021-02-16 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation

US11232149B2 Establishment anchoring with geolocated imagery
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2019-07-24, granted 2022-01-25 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US10552476B2 System and method of identifying visual objects
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2019-09-06, granted 2020-02-04 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US11188603B2 Annotation of videos using aggregated user session data
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2019-09-27, granted 2021-11-30 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US11269897B2 Mapping images to search queries
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2019-10-18, granted 2022-03-08 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US11263712B2 Selecting photographs for a destination or point of interest
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2019-11-14, granted 2022-03-01 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US10855627B2 Modification of audio-based computer program output
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2019-11-25, granted 2020-12-01 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US10902055B2 System and method of identifying visual objects
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2020-01-16, granted 2021-01-26 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US11475900B2 Establishment of audio-based network sessions with non-registered resources
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2020-02-03, granted 2022-10-18 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US11782998B2 Embedding based retrieval for image search
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2020-02-28, granted 2023-10-10 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US11068489B2 Optimizing digital video distribution
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data.
filed 2020-04-21, granted 2021-07-20 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US12236514B2 Efficiently rendering video having dynamic components
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2020-05-14, granted 2025-02-25 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US11190471B2 Methods, systems, and media for identifying and presenting video objects linked to a source video
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2020-05-22, granted 2021-11-30 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US11461386B2 Visual recognition using user tap locations
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2020-05-22, granted 2022-10-04 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US20200320119A1 Association of data items and objects
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2020-06-22 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US11973819B1 Image recommendations for thumbnails for online media items based on user activity
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2020-08-17, granted 2024-04-30 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US20230178114A1 Generating multiple versions of an audio digital component
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Build content for diverse user contexts and search histories.
filed 2020-09-10 · leak modules: NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, VideoPerDocData

US11507253B2 Contextual information for a displayed resource that includes an image
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2020-10-05, granted 2022-11-22 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US11582169B2 Modification of audio-based computer program output
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2020-11-10, granted 2023-02-14 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US11704359B2 Music recommendations from trending queries
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2020-11-23, granted 2023-07-18 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate

US11580740B2 Methods, systems, and media for adaptive presentation of a video content item based on an area of interest
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2020-11-30, granted 2023-02-14 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores

US11425071B2 Uniform resource identifier and image sharing for contextual information display
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2020-12-14, granted 2022-08-23 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US11615136B2 System and method of identifying visual objects
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2021-01-25, granted 2023-03-28 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US11727045B2 Using cross-matching between users and matching against reference data to facilitate content identification
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2021-04-26, granted 2023-08-15 · leak modules: IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, RichsnippetsPageMap

US20220374585A1 User interfaces and tools for facilitating interactions with video content
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2021-05-19 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US11442948B2 Optimizing digital video distribution
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data.
filed 2021-06-16, granted 2022-09-13 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US11935199B2 Augmented reality depth detection through object recognition
Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2021-07-26, granted 2024-03-19 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US11593906B2 Image recognition based content item selection
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2021-08-04, granted 2023-02-28 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US11570512B2 Watch-time clustering for video searches
Rule: Design video content to achieve minimum watch time thresholds that qualify it for broader viewer category distribution. Videos that fail to meet watch time thresholds for key demographics get restricted distribution. Optimize the opening segments to retain viewers past critical drop-off points.
filed 2021-08-06, granted 2023-01-31 · leak modules: QualityNavboostCrapsCrapsData, QualityNavboostCrapsFeatureCrapsData, QualityNavboostCrapsStatsWithWeightsProto, PerDocData

US11977599B2 Matching video content to podcast episodes
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2021-10-04, granted 2024-05-07 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US11841893B2 Coordination of parallel processing of audio queries across multiple devices
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2021-10-11, granted 2023-12-12 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US12008057B2 Determining a visual theme in a collection of media items
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2021-10-25, granted 2024-06-11 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US11997062B2 Methods, systems, and media for identifying and presenting video objects linked to a source video
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2021-11-30, granted 2024-05-28 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US20230113883A1 Digital Signal Processor-Based Continued Conversation
Rule: Use consistent entity names, sameAs schema, and structured data. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2021-12-15 · leak modules: NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefDocumentMetadata

US11747960B2 Efficiently augmenting images with related content
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2021-12-28, granted 2023-09-05 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US12147486B2 System and method for representing busyness using content specific to a place
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2021-12-30, granted 2024-11-19 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US11670061B2 Generating augmented reality prerenderings using template images
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2022-02-14, granted 2023-06-06 · leak modules: ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US11734287B2 Mapping images to search queries
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2022-02-21, granted 2023-08-22 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US11681761B1 Method and system for mining image searches to associate images with concepts
Rule: Ensure images on your pages are conceptually tagged and associated with clear topics. Google mines user image search interactions to build concept-image associations — images that generate more user interactions get stronger concept associations and broader distribution. Use descriptive filenames and alt text aligned with search behavior.
filed 2022-03-21, granted 2023-06-20 · leak modules: QualitySalientTermsSalientTermSet, QualitySalientTermsSignalData, QualityNavboostCrapsCrapsClickSignals, PerDocData

US11775595B1 Method and system for mining image searches to associate images with concepts
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2022-03-21, granted 2023-10-03 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US12271420B1 Video segments for a video related to a task
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2022-06-06, granted 2025-04-08 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData

US11714820B2 Optimizing digital video distribution
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data.
filed 2022-08-01, granted 2023-08-01 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US11893350B2 Detecting continuing conversations with computing devices
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2022-09-02, granted 2024-02-06 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US11966433B1 Subscribe to people in videos
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2022-09-26, granted 2024-04-23 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US11948572B2 Modulation of packetized audio signals
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2022-10-24, granted 2024-04-02 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US12229127B2 Assistive browsing using context
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2022-10-31, granted 2025-02-18 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US20260023800A1 Systems and Methods for Place Search in Augmented Reality
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2022-12-30 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores

US11861908B2 Methods, systems, and media for adaptive presentation of a video content item based on an area of interest
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2023-02-13, granted 2024-01-02 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores

US12038970B2 Training image and text embedding models
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2023-02-20, granted 2024-07-16 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US20250191267A1 Processing video and text inputs using co-tokenization
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2023-03-07 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US20230229690A1 Systems and methods of image searching
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2023-03-20 · leak modules: ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11886489B2 System and method of identifying visual objects
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2023-03-24, granted 2024-01-30 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US12346386B2 Visual and audio multimodal searching system
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2023-04-25, granted 2025-07-01 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US12164562B1 Background audio identification for query disambiguation
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2023-05-01, granted 2024-12-10 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US12223950B2 Detecting conversations with computing devices
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2023-05-08, granted 2025-02-11 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US12033351B2 Use of image sensors to query real world for geo-reference information
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2023-05-15, granted 2024-07-09 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US20240411791A1 Semantic Annotation and Framework in a Storage API
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2023-06-12 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US12298985B2 Mapping images to search queries
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2023-06-29, granted 2025-05-13 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US12118184B2 Efficiently augmenting images with related content
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2023-07-18, granted 2024-10-15 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US12086198B2 Embedding based retrieval for image search
Rule: Ensure images and their surrounding landing page context are semantically aligned — Google embeds both together as a pair. Optimize image alt text, surrounding text, and page topic to create strong image-page semantic coherence. The embedding model rewards pages where image content matches page content.
filed 2023-09-05, granted 2024-09-10 · leak modules: QualitySalientTermsSalientTermSet, QualitySalientTermsSignalData, QualitySalientTermsDocData, PerDocData, IndexingEmbeddedContentPageSizeInfo

US20240004927A1 Providing a video preview of search results
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2023-09-18 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US20240152546A1 Processing Diagrams as Search Input
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2023-11-06 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US12346371B2 System and method of identifying visual objects
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2023-12-06, granted 2025-07-01 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US20250190503A1 Video Query Contextualization
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2023-12-11 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US12367679B2 Methods, systems, and media for adaptive presentation of a video content item based on an area of interest
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2023-12-29, granted 2025-07-22 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores

US20250272350A1 Multimodal Web Navigation
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2024-02-22 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20240330381A1 User-Specific Content Generation Using Text-To-Image Machine-Learned Models
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2024-03-29 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US12321413B2 Matching video content to podcast episodes
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2024-05-06, granted 2025-06-03 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US12608418B2 Training image and text embedding models
Rule: Ensure your images are associated with related images users frequently click together in search. Google's image embedding model learns from co-click behavior — images that are contextually related to other high-performing images get better embedding alignment. Use structured data to link related image content.
filed 2024-06-12, granted 2026-04-21 · leak modules: QualitySalientTermsSalientTermSet, QualitySalientTermsSignalData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, PerDocData

US12314313B2 Image query analysis
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2024-06-18, granted 2025-05-27 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US12566797B1 Retrieval, annotation, and selection of video segments
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2024-10-10, granted 2026-03-03 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US20250166241A1 Generating images by expanding prompts using generative neural networks
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2024-11-18 · leak modules: ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US12596744B2 Multimodal search on wearable smart devices
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2024-11-27, granted 2026-04-07 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US20250124075A1 Visual Search Determination for Text-To-Image Replacement
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2024-12-23 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US20250348498A1 Providing context for an image
Rule: Ensure images on your pages are surrounded by rich, relevant text content—salient terms from the source document generate the context query that determines which documents are retrieved as image context. Use descriptive alt text, captions, and surrounding paragraphs. For original images, ensure your page is the first appearance (origin source) to get attribution. Include schema.org/ImageObject markup with license info. Visually similar images with shared semantic terms strengthen context query matching.
filed 2025-05-07 · leak modules: ImageSearchImageIndexingInfo, ImageSearchImageLicenseInfo, ImageUnderstandingIndexingAnnotation, ImageUnderstandingIndexingLabel, ImageContentQueryBoost, ImageExactBoost, ImageExactBoostNavQuery, RepositoryWebrefImageMention

Entities & structured data (321)

US7536382B2 Query rewriting with entity detection
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2004-03-31, granted 2009-05-19 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality

US7996419B2 Query rewriting with entity detection
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2004-03-31, granted 2011-08-09 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US9009153B2 Systems and methods for identifying a named entity
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2004-03-31, granted 2015-04-14 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation

US7788274B1 Systems and methods for category-based search
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2004-06-30, granted 2010-08-31 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US8131754B1 Systems and methods for determining an article association measure
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2004-06-30, granted 2012-03-06 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8386453B2 Providing search information relating to a document
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2004-09-30, granted 2013-02-26 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8682913B1 Corroborating facts extracted from multiple sources
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active.
filed 2005-03-31, granted 2014-03-25 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US7487447B1 Web page zoom feature
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2005-05-25, granted 2009-02-03 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US7769579B2 Learning facts from semi-structured text
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2005-05-31, granted 2010-08-03 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefEntityJoin, SmartphonePerDocData, SnippetExtraInfo

US7693830B2 Programmable search engine

Rule: This patent reveals that Google's search architecture supports programmable query modification, collection selection, and result re-ranking through context files. For SEO, this means your site's topical authority and structured data can influence which document collections your content appears in, how queries are reformulated to match your content, and how results are clustered and ranked. Ensure clear topical organization and structured data markup.

filed 2005-08-10, granted 2010-04-06 · leak modules: PerDocData, QualityNsrNsrData, IndexingDupsLocalizedLocalizedCluster, QualitySitemapScoringSignals

US7933900B2 Search over structured data

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2005-10-23, granted 2011-04-26 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals

US7774328B2 Browseable fact repository

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.

filed 2006-02-17, granted 2010-08-10 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US8055674B2 Annotation framework

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2006-02-17, granted 2011-11-08 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, RichsnippetsPageMap

US8825471B2 Unsupervised extraction of facts

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2006-03-31, granted 2014-09-02 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US7778952B2 Displaying facts on a linear graph

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.

filed 2006-09-27, granted 2010-08-17 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityNsrNsrData, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, ShingleInfoPerDocData

US8954412B1 Corroborating facts in electronic documents

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.

filed 2006-09-28, granted 2015-02-10 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US8122026B1 Finding and disambiguating references to entities on web pages

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2006-10-20, granted 2012-02-21 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US7783644B1 Query-independent entity importance in books

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2006-12-13, granted 2010-08-24 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures

US8463830B2 Keyword-based content suggestions

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Write in natural vocabulary. ML models understand context and fluff.

filed 2007-01-05, granted 2013-06-11 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData

US7657514B2 Content identification expansion

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2007-01-30, granted 2010-02-02 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8005842B1 Inferring attributes from search queries

Rule: Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.

filed 2007-05-18, granted 2011-08-23 · leak modules: NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US8185830B2 Configuring a content document for users and user groups

Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Structure answers: question heading, tight answer, then depth.

filed 2007-06-01, granted 2012-05-22 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US7966291B1 Fact-based object merging

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.

filed 2007-06-26, granted 2011-06-21 · leak modules: AnchorsAnchor, AnchorsAnchorSource, IndexingDocjoinerAnchorStatistics, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata

US9058608B2 Placement attribute targeting

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2007-09-12, granted 2015-06-16 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US7792837B1 Entity name recognition

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2007-11-14, granted 2010-09-07 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US8812435B1 Learning objects and facts from documents

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2007-11-16, granted 2014-08-19 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefDocumentMetadata

US8417713B1 Sentiment detection as a ranking signal for reviewable entities

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data.

filed 2007-12-05, granted 2013-04-09 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US8844800B2 Ratings using machine-readable representations

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2008-03-31, granted 2014-09-30 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US20140129549A1 Attribute-Based Rating of Authors and Content

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2008-11-13 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefEntityJoin, SmartphonePerDocData

US8756240B2 Attribute based rating of authors and content

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2008-11-13, granted 2014-06-17 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefEntityJoin, SmartphonePerDocData

US8126882B2 Credibility of an author of online content

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.

filed 2008-12-11, granted 2012-02-28 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US8615707B2 Adding new attributes to a structured presentation

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2009-01-16, granted 2013-12-24 · leak modules: NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefEntityJoin

US8078617B1 Model based ad targeting

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2009-01-20, granted 2011-12-13 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US8112432B2 Query rewriting with entity detection

Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2009-04-08, granted 2012-02-07 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality

US8805861B2 Methods and systems to train models to extract and integrate information from data sources

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2009-05-15, granted 2014-08-12 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US8600993B1 Determining resource attributes from site address attributes

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2009-08-26, granted 2013-12-03 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefEntityJoin

US8515975B1 Search entity transition matrix and applications of the transition matrix

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2009-12-07, granted 2013-08-20 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument

US20150112981A1 Entity Review Extraction

Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2009-12-14 · leak modules: IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8171011B2 Content identification expansion

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2010-01-13, granted 2012-05-01 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8316040B2 Programmable search engine
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2010-02-16, granted 2012-11-20 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, IndexingDocjoinerAnchorStatistics, LocalWWWInfo

US8645367B1 Predicting data for document attributes based on aggregated user behavior data
Rule: Maintain consistent URL patterns across your site—Google aggregates click behavior data across URLs sharing the same pattern to predict quality for uncrawled pages. Use clean, predictable URL structures (e.g., /category/product-id) so behavioral signals transfer to new pages. High engagement (dwell time, click-to-impression ratio) on existing pages in a pattern class directly boosts predicted quality of new pages in the same pattern. Avoid chaotic URL structures that prevent pattern matching.
filed 2010-03-08, granted 2014-02-04 · leak modules: PerDocData, QualityNsrNsrData, QualityNsrPQData, CompressedQualitySignals, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsStatsWithWeightsProto, QualityTimebasedLastSignificantUpdate, IndexingSignalAggregatorAggregatedScore

US8645384B1 Updating taxonomy based on webpage
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2010-05-05, granted 2014-02-04 · leak modules: CrawlerChangerateUrlChangerate, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8688681B1 Identifying internet protocol addresses for internet hosting entities
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2010-06-17, granted 2014-04-01 · leak modules: CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US7958128B2 Query-independent entity importance in books
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2010-07-15, granted 2011-06-07 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures

US8356029B2 Method and system for reconstruction of object model data in a relational database
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2010-08-06, granted 2013-01-15 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8886747B1 Verifying domain ownership
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2010-09-15, granted 2014-11-11 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US8554699B2 Method and system for detecting anomalies in time series data
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2010-10-19, granted 2013-10-08 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo

US8583584B2 Method and system for using web analytics data for detecting anomalies
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2010-10-19, granted 2013-11-12 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo

US8078573B2 Identifying the unifying subject of a set of facts
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2010-11-04, granted 2011-12-13 · leak modules: AnchorsAnchor, AnchorsAnchorSource, IndexingDocjoinerAnchorStatistics, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20130282699A1 Using Authority Website to Measure Accuracy of Business Information
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-01-14 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefDocumentMetadata

US8781990B1 Crowdsensus: deriving consensus information from statements made by a crowd of users
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-02-25, granted 2014-07-15 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8402032B1 Generating context-based spell corrections of entity names
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-03-24, granted 2013-03-19 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9002866B1 Generating context-based spell corrections of entity names
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-03-24, granted 2015-04-07 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8538916B1 Extracting instance attributes from text
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-04-11, granted 2013-09-17 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8762372B2 Search over structured data
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-04-22, granted 2014-06-24 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals

US8756617B1 Schema validation for secure development of browser extensions
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Build content for diverse user contexts and search histories.
filed 2011-05-18, granted 2014-06-17 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US10068022B2 Identifying topical entities
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-06-03, granted 2018-09-04 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US20160026727A1 Generating additional content
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-06-03 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US8676804B1 Managing information about entities using observations generated from user modified values
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-07-12, granted 2014-03-18 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8521764B2 Query rewriting with entity detection
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-07-14, granted 2013-08-27 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US8688688B1 Automatic derivation of synonym entity names
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-08-05, granted 2014-04-01 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US8682881B1 System and method for extracting structured data from classified websites
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-09-07, granted 2014-03-25 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9037563B1 Search result claiming
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-09-09, granted 2015-05-19 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US9501465B2 Use of templates as message pruning schemas
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth. Use JSON-LD structured data for entity types and relationships.
filed 2011-09-26, granted 2016-11-22 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8473489B1 Identifying entities using search results
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-09-27, granted 2013-06-25 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US8775439B1 Identifying entities using search results
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-09-27, granted 2014-07-08 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US8843466B1 Identifying entities using search results
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-09-27, granted 2014-09-23 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US8856099B1 Identifying entities using search results
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-09-27, granted 2014-10-07 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US9146939B1 Generating and using result suggest boost factors
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-09-30, granted 2015-09-29 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US8898169B2 Automated product attribute selection
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-11-09, granted 2014-11-25 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US8983142B1 Programmatic silhouette attribute determination
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-11-10, granted 2015-03-17 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US9110998B2 Hierarchical behavioral profile
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Structure answers: question heading, tight answer, then depth.
filed 2011-12-22, granted 2015-08-18 · leak modules: CompressedQualitySignals, PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SnippetExtraInfo

US8484203B1 Cross media type recommendations for media items based on identified entities
Rule: Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships.
filed 2012-01-04, granted 2013-07-09 · leak modules: PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, RichsnippetsPageMap, SmartphonePerDocData

US8667013B1 Systems and methods for determining an article association measure
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-01-27, granted 2014-03-04 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8751498B2 Finding and disambiguating references to entities on web pages
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-02-01, granted 2014-06-10 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US8452799B2 Query rewriting with entity detection
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-02-06, granted 2013-05-28 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality

US8688713B1 Resource identification from organic and structured content
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Write in natural vocabulary. ML models understand context and fluff.
filed 2012-03-22, granted 2014-04-01 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, RepositoryWebrefEntityJoin

US8589399B1 Assigning terms of interest to an entity
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2012-03-26, granted 2013-11-19 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9965529B2 Maintaining search context
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.
filed 2012-05-02, granted 2018-05-08 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations

US8671106B1 Indicators for entities corresponding to search suggestions
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-05-30, granted 2014-03-11 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9444909B2 Application verification for hosted services
Rule: Structure answers: question heading, tight answer, then depth.
filed 2012-06-05, granted 2016-09-13 · leak modules: QualityPreviewSnippetRadishFeatures, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8688714B1 Resource identification from organic and structured content
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Write in natural vocabulary. ML models understand context and fluff.
filed 2012-06-07, granted 2014-04-01 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, RepositoryWebrefEntityJoin

US8688715B1 Resource identification from organic and structured content
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Write in natural vocabulary. ML models understand context and fluff.
filed 2012-06-07, granted 2014-04-01 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, RepositoryWebrefEntityJoin

US9477758B1 Automatic identification of related entities
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-07-19, granted 2016-10-25 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9268820B2 Providing knowledge panels with search results
Rule: Feed your entity: consistent name and details everywhere, sameAs schema, and the citations that earn a panel. The leak's webrefEntities field in PerDocData confirms Knowledge Graph entity extraction and association.
filed 2012-08-03, granted 2016-02-23 · leak modules: PerDocData, RepositoryWebref, QualityNsrNsrData

US20160063106A1 Related Entity Search
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-08-08 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US9031937B2 Programmable search engine
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-09-14, granted 2015-05-12 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, IndexingDocjoinerAnchorStatistics, LocalWWWInfo

US20150112961A1 User Submission of Search Related Structured Data
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2012-09-18 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefEntityJoin

US20150113019A1 Obtaining Access-Restricted Search Related Structured Data
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-09-18 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US8825671B1 Referent determination from selected content
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.
filed 2012-10-05, granted 2014-09-02 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US9305108B2 Semantic selection and purpose facilitation
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.
filed 2012-10-05, granted 2016-04-05 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US9390175B2 Systems and methods for determining the operating hours of an entity
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-10-09, granted 2016-07-12 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US9336311B1 Determining the relevancy of entities
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-10-15, granted 2016-05-10 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata

US8886625B1 Methods and computer-readable media for providing recommended entities based on a user's social graph

Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2012-10-31, granted 2014-11-11 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US8812509B1 Inferring attributes from search queries

Rule: Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.

filed 2012-11-02, granted 2014-08-19 · leak modules: NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US9047278B1 Identifying and ranking attributes of entities

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2012-11-09, granted 2015-06-02 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata

US9256682B1 Providing search results based on sorted properties

Rule: Use consistent entity names, sameAs schema, and structured data. Write in natural vocabulary. ML models understand context and fluff. Write for natural spoken language and conversational queries.

filed 2012-12-05, granted 2016-02-09 · leak modules: NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US10235423B2 Ranking search results based on entity metrics

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2012-12-12, granted 2019-03-19 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US9098502B1 Identifying documents for dissemination by an entity

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2012-12-27, granted 2015-08-04 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9009099B1 Method and system for reconstruction of object model data in a relational database

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2013-01-15, granted 2015-04-14 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20150169701A1 Providing customized content in knowledge panels

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2013-01-25 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US9092463B2 Keyword generation

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2013-02-06, granted 2015-07-28 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US20150161127A1 Ranking entity realizations for information retrieval

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.

filed 2013-02-13 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9275152B2 Related entities

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2013-02-22, granted 2016-03-01 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US9424353B2 Related entities
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-02-22, granted 2016-08-23 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US20140115010A1 Propagating information through networks
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-02-27 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata

US9536259B2 Entity-based searching with content selection
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data.
filed 2013-03-05, granted 2017-01-03 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument

US10713261B2 Generating insightful connections between graph entities
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-03-13, granted 2020-07-14 · leak modules: CrawlerChangerateUrlChangerate, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US20160224621A1 Associating A Search Query With An Entity
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-03-13 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US9336211B1 Associating an entity with a search query
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-03-13, granted 2016-05-10 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US9355140B1 Associating an entity with a search query
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-03-13, granted 2016-05-31 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US10922326B2 Triggering knowledge panels
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-03-14, granted 2021-02-16 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster

US9460457B1 Automatically annotating content items with an entity
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-03-14, granted 2016-10-04 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9037568B1 Factual query pattern learning
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-03-15, granted 2015-05-19 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US9146984B1 Enhancing queries for data tables with nested fields
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth. Write for intent and meaning. Neural matching connects related concepts.
filed 2013-03-15, granted 2015-09-29 · leak modules: NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US9311362B1 Personal knowledge panel interface

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2013-03-15, granted 2016-04-12 · leak modules: CompressedQualitySignals, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US9471606B1 Obtaining information to provide to users

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2013-03-15, granted 2016-10-18 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US9477711B2 Knowledge panel

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2013-03-15, granted 2016-10-25 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US9811780B1 Identifying subjective attributes by analysis of curation signals

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2013-03-15, granted 2017-11-07 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo

US10372715B1 Multiple entity merge for ranking objects in a social network

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs.

filed 2013-04-03, granted 2019-08-06 · leak modules: IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9317559B1 Sentiment detection as a ranking signal for reviewable entities

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data.

filed 2013-04-08, granted 2016-04-19 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US9183499B1 Evaluating quality based on neighbor features

Rule: Your site's quality score is influenced by the quality of sites that link to you—links from high-quality neighbors boost you more. But the mechanism is bidirectional and iterative: quality 'smears' across the graph over time, so link neighborhood matters holistically. Also, layout quality and user engagement metrics (CTR, dwell time) are direct inputs to the quality model. Improve your page layout and user engagement signals alongside link building.

filed 2013-04-19, granted 2015-11-10 · leak modules: QualityNsrNsrData, QualityNsrPQData, AnchorsAnchor, Anchors, IndexingDocjoinerAnchorStatistics, CompressedQualitySignals, PerDocData, QualityNsrNsrDataClusterUplift

US9311363B1 Personalized entity rankings

Rule: Google personalizes entity rankings based on review text analysis—characteristics mentioned in reviews influence quality scores, and these are re-weighted per user based on their personal preferences. Ensure your product/service reviews mention specific positive characteristics (e.g., 'durable,' 'fast,' 'reliable'), as these characteristics directly influence the quality score. The leak's ShoppingAnnotationProductRating and GoogleLabelData modules confirm review text analysis feeds into ranking. Structured review markup helps Google extract these characteristics.

filed 2013-05-15, granted 2016-04-12 · leak modules: QualityShoppingShoppingAttachmentProduct, ShoppingWebentityShoppingAnnotationProductRating, QualityLabelsGoogleLabelData, QualitySherlockKnexAnnotation, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, QualityNsrPQData, PerDocData

US9607317B2 Keyword-based content suggestions

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Write in natural vocabulary. ML models understand context and fluff.

filed 2013-05-20, granted 2017-03-28 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData

US8805867B2 Query rewriting with entity detection
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-05-24, granted 2014-08-12 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality

US8856111B1 Cross media type recommendations for media items based on identified entities
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-06-10, granted 2014-10-07 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US9311414B2 Systems and methods of selecting content based on aggregate entity co-occurrence
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2013-06-14, granted 2016-04-12 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9411857B1 Grouping related entities
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-06-28, granted 2016-08-09 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US9953085B1 Feed upload for search entity based content selection
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-07-01, granted 2018-04-24 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData

US9305307B2 Selecting content associated with a collection of entities
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-07-15, granted 2016-04-05 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US9336330B2 Associating entities based on resource associations
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2013-07-17, granted 2016-05-10 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata

US9400789B2 Associating resources with entities
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2013-07-17, granted 2016-07-26 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate

US9367529B1 Selecting content based on entities
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-07-31, granted 2016-06-14 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US9436946B2 Selecting content based on entities present in search results
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.
filed 2013-07-31, granted 2016-09-06 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US9268824B1 Search entity transition matrix and applications of the transition matrix
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-08-13, granted 2016-02-23 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument

US9792333B2 Adding attributes and labels to structured data
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-08-23, granted 2017-10-17 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals

US9047339B2 Query rewriting with entity detection
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-08-26, granted 2015-06-02 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US10303684B1 Resource scoring adjustment based on entity selections
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2013-08-27, granted 2019-05-28 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US8682816B2 Method and system for detecting anomalies in time series data
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-09-10, granted 2014-03-25 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo

US8983898B1 Extracting instance attributes from text
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-09-16, granted 2015-03-17 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9342626B1 Query suggestions based on entity collections of one or more past queries
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-09-26, granted 2016-05-17 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US8972332B2 Method and system for detecting anomalies in web analytics data
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-10-10, granted 2015-03-03 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo

US20140280133A1 Structured Data to Aggregate Analytics
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-10-24 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US20170091263A1 Event-based entity and object creation
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships.
filed 2013-10-25 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US9384224B2 Information retrieval system for archiving multiple document versions
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active.
filed 2013-11-18, granted 2016-07-05 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US9727545B1 Selecting textual representations for entity attribute values
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-12-04, granted 2017-08-08 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US9830361B1 Facilitating content entity annotation while satisfying joint performance conditions
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2013-12-04, granted 2017-11-28 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9135361B1 Updating taxonomy based on webpage
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2013-12-27, granted 2015-09-15 · leak modules: CrawlerChangerateUrlChangerate, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9342597B1 Associating an event attribute with a user based on a group of electronic messages associated with the user
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-12-31, granted 2016-05-17 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US9396236B1 Ranking users based on contextual factors
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-12-31, granted 2016-07-19 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US8990211B1 Managing information about entities using observations
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-01-30, granted 2015-03-24 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9317588B1 Resource identification from organic and structured content
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Write in natural vocabulary. ML models understand context and fluff.
filed 2014-02-06, granted 2016-04-19 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, RepositoryWebrefEntityJoin

US9875286B1 Scoring messages based on sender attributes
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2014-02-06, granted 2018-01-23 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9158861B1 Indicators for entities corresponding to search suggestions
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-02-18, granted 2015-10-13 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9679018B1 Document ranking based on entity frequency
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-02-19, granted 2017-06-13 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US9727926B2 Entity page recommendation based on post content
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2014-03-03, granted 2017-08-08 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9727617B1 Systems and methods for searching quotes of entities using a database
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-03-10, granted 2017-08-08 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US9507830B2 Tailoring user experience for unrecognized and new users
Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2014-03-12, granted 2016-11-29 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US9436742B1 Ranking search result documents based on user attributes
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2014-03-14, granted 2016-09-06 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US9754036B1 Adapting third party applications
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-03-26, granted 2017-09-05 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US9710549B2 Entity normalization via name normalization
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2014-03-28, granted 2017-07-18 · leak modules: IndexingDupsLocalizedLocalizedCluster, PerDocData, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, ShingleInfoPerDocData, SmartphonePerDocData

US9501530B1 Systems and methods for selecting content
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-04-01, granted 2016-11-22 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9542450B1 Selecting content using entity properties
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-04-01, granted 2017-01-10 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9805142B2 Ranking suggestions based on user attributes
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2014-04-11, granted 2017-10-31 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9348663B1 Schema validation for secure development of browser extensions
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Build content for diverse user contexts and search histories.
filed 2014-05-01, granted 2016-05-24 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US9075862B1 Ranking user content by content attributes
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-05-02, granted 2015-07-07 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefEntityJoin, SmartphonePerDocData

US9262406B1 Semantic frame identification with distributed word representations
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-05-07, granted 2016-02-16 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo

US9760570B2 Finding and disambiguating references to entities on web pages
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-06-09, granted 2017-09-12 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US9852188B2 Contextual search on multimedia content
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-06-23, granted 2017-12-26 · leak modules: CompressedQualitySignals, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US9582482B1 Providing an annotation linking related entities in onscreen content
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2014-08-04, granted 2017-02-28 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData

US9558186B2 Unsupervised extraction of facts
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-08-14, granted 2017-01-31 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US9747347B1 Providing posts from an extended network
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-09-03, granted 2017-08-29 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US9672251B1 Extracting facts from documents
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-09-29, granted 2017-06-06 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US10019487B1 Method and computer-readable media for providing recommended entities based on a user's social graph
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-10-13, granted 2018-07-10 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US9864795B1 Identifying entity attributes
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2014-10-28, granted 2018-01-09 · leak modules: NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9858291B1 Detection of related local entities

Rule: Build consistent NAP (Name, Address, Phone) data across all local listings so your entity is properly resolved and compared with nearby competitors. Optimize for query terms that also drive clicks to related local entities, as shared query-selection patterns increase similarity rankings. Earn clicks for queries containing terms that your nearby competitors also rank for, as selection concentration ratios determine which entities are considered 'related.' Monitor which query terms have location bias vs. navigational bias to understand how your entity is classified.

filed 2014-10-30, granted 2018-01-02 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, PerDocData

US20160125034A1 Annotate Apps with Entities by Fusing Heterogeneous Signals

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2015-02-05 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsGeoTopicality

US9785686B2 Corroborating facts in electronic documents

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.

filed 2015-02-06, granted 2017-10-10 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US9940381B1 Managing information about entities using observations

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.

filed 2015-02-06, granted 2018-04-10 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US10162895B1 Generating context-based spell corrections of entity names

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2015-02-09, granted 2018-12-25 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11049029B2 Identifying content appropriate for children algorithmically without human intervention

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2015-02-22, granted 2021-06-29 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20180329873A1 Automated data extraction system based on historical or related data

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2015-04-08 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CrawlerChangerateUrlChangerate, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate

US9852225B2 Associating features with entities, such as categories of web page documents, and/or weighting such features

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.

filed 2015-04-23, granted 2017-12-26 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US10185751B1 Identifying and ranking attributes of entities

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2015-05-06, granted 2019-01-22 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata

US10146852B1 Search result claiming

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2015-05-07, granted 2018-12-04 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US9678945B2 Automated reading comprehension
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-05-11, granted 2017-06-13 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9898512B1 Factual query pattern learning
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2015-05-13, granted 2018-02-20 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US9773055B2 Query rewriting with entity detection
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-05-29, granted 2017-09-26 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US10621191B2 Surfacing live events in search results
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2015-06-08, granted 2020-04-14 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US9454776B2 Placement attribute targeting
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2015-06-12, granted 2016-09-27 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US10102291B1 Computerized systems and methods for building knowledge bases using context clouds
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-07-06, granted 2018-10-16 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US10198491B1 Computerized systems and methods for extracting and storing information regarding entities
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-07-06, granted 2019-02-05 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, IndexingDocjoinerAnchorStatistics, LocalWWWInfo

US10068027B2 Systems and methods for selecting content based on linked devices
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-07-22, granted 2018-09-04 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20170032257A1 Modeling personal entities
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-07-29 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US10102482B2 Factorized models
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-08-07, granted 2018-10-16 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20170061005A1 Automatic Background Information Retrieval and Profile Updating
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-08-25 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10496691B1 Clustering search results
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-09-08, granted 2019-12-03 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData

US9569560B2 Indicators for entities corresponding to search suggestions
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-09-28, granted 2017-02-14 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10110701B2 Structured entity information page
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2015-11-16, granted 2018-10-23 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US10178527B2 Personalized entity repository
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2015-12-08, granted 2019-01-08 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US10402410B2 Contextualizing knowledge panels
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2015-12-16, granted 2019-09-03 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US10257660B2 Systems and methods of sourcing hours of operation for a location entity
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2015-12-22, granted 2019-04-09 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US9454611B2 Providing knowledge panels with search results
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2016-01-20, granted 2016-09-27 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US10289952B2 Semantic frame identification with distributed word representations
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2016-01-28, granted 2019-05-14 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo

US9875320B1 Providing search results based on sorted properties
Rule: Use consistent entity names, sameAs schema, and structured data. Write in natural vocabulary. ML models understand context and fluff. Write for natural spoken language and conversational queries.
filed 2016-02-08, granted 2018-01-23 · leak modules: NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US10270791B1 Search entity transition matrix and applications of the transition matrix
Rule: Anchor text is evaluated for quality based on its relationship to queries that naturally lead to the target document. Generic or irrelevant anchor text that doesn't match the document's natural query transition pattern may be devalued. Build natural anchor text that matches what users actually search for when seeking your content. Avoid manipulative anchor text patterns.
filed 2016-02-19, granted 2019-04-23 · leak modules: link_analysis, anchor_text, spam_detection, ranking, navboost, quality

US9830390B2 Related entities

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2016-02-26, granted 2017-11-28 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US10534810B1 Computerized systems and methods for enriching a knowledge base for search queries

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2016-02-29, granted 2020-01-14 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9594474B2 Semantic selection and purpose facilitation

Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.

filed 2016-03-01, granted 2017-03-14 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US10475074B2 Selecting content associated with a collection of entities

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-03-07, granted 2019-11-12 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US10394830B1 Sentiment detection as a ranking signal for reviewable entities

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data.

filed 2016-03-11, granted 2019-08-27 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US9589028B1 Resource identification from organic and structured content

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Write in natural vocabulary. ML models understand context and fluff.

filed 2016-03-16, granted 2017-03-07 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, RepositoryWebrefEntityJoin

US10331681B1 Crowdsourced evaluation and refinement of search clusters

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-04-11, granted 2019-06-25 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US20170293696A1 Related entity discovery

Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2016-04-11 · leak modules: IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo

US9870423B1 Associating an entity with a search query

Rule: Ensure your entity has a clear, unambiguous canonical name and consistent properties across the web. The entity search query is determined by which query retrieves the most entity-specific, high-quality results—so optimize your primary keyword to be unambiguous (low ratio of non-entity documents). Build entity authority by ensuring top-ranked results for your entity's name are high-quality and about your entity. Use schema.org Entity, Organization, and Person markup with trusted names. Entity ambiguity directly affects which query Google assigns to your entity.

filed 2016-04-13, granted 2018-01-16 · leak modules: RepositoryWebrefEntityAnnotations, RepositoryWebrefEntityLinkMetadata, RepositoryWebrefDetailedEntityScores, RepositoryWebrefEntityNameRatings, RepositoryWebrefAggregatedEntityNameScores, NlpSaftEntity, NlpSaftEntityProfile, UniversalsearchNewPackerKnowledgeResultSupport

US9990235B2 Determining tasks to be performed by a modular entity

Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.

filed 2016-04-15, granted 2018-06-05 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US9507836B1 Associating an event attribute with a user based on a group of one or more electronic messages associated with the user

Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2016-04-26, granted 2016-11-29 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US10346519B1 Selecting content based on entities

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2016-05-16, granted 2019-07-09 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US10360225B1 Query suggestions based on entity collections of one or more past queries

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2016-05-16, granted 2019-07-23 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US20170337284A1 Determining and using attributes of message exchange thread participants

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2016-05-17 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US10133790B1 Ranking users based on contextual factors

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-06-16, granted 2018-11-20 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US10180965B2 User attribute resolution of unresolved terms of action queries

Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-07-07, granted 2019-01-15 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality

US10318567B2 Providing knowledge panels with search results

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2016-08-10, granted 2019-06-11 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US9830341B2 Resource name generation and derivation utilizing attribute space monikers and their associated context

Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships. Write for natural spoken language and conversational queries.

filed 2016-08-18, granted 2017-11-28 · leak modules: NlpSaftDocument, PerDocData, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US9916384B2 Related entities

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2016-08-18, granted 2018-03-13 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US9679309B2 Placement attribute targeting

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2016-08-23, granted 2017-06-13 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US10289625B2 Providing context facts

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2016-09-15, granted 2019-05-14 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US10019495B2 Knowledge panel

Rule: Optimize your entity's presence in Google's Knowledge Graph by ensuring consistent structured data (Organization, LocalBusiness schema), Wikipedia/Wikidata entries, and authoritative references. The knowledge panel content is ranked - higher-quality, more-referenced content about your entity will appear. Monitor and claim your knowledge panel.

filed 2016-09-21, granted 2018-07-10 · leak modules: knowledge_graph, structured_data, entity, ranking, snippet

US11055312B1 Selecting content using entity properties

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2016-10-19, granted 2021-07-06 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9749274B1 Associating an event attribute with a user based on a group of one or more electronic messages associated with the user

Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2016-10-25, granted 2017-08-29 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US10580039B1 Entity-based searching with content selection

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data.

filed 2016-11-16, granted 2020-03-03 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument

US9922125B1 Selecting content using entity properties

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2016-12-02, granted 2018-03-20 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US10331727B2 Graphical user interface to display commonly categorized entities

Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2016-12-07, granted 2019-06-25 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US10136191B1 Methods, systems, and media for recommending media content based on attribute grouped viewing sessions

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships.

filed 2016-12-15, granted 2018-11-20 · leak modules: NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US10997227B2 Systems and methods for processing a natural language query in data tables

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2017-01-18, granted 2021-05-04 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US10949468B2 Indicators for entities corresponding to search suggestions

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2017-01-26, granted 2021-03-16 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10585962B2 Systems and methods for selecting content based on linked devices

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2017-02-01, granted 2020-03-10 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10657192B2 Systems and methods for selecting content based on linked devices
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2017-02-01, granted 2020-05-19 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10657193B2 Systems and methods for selecting content based on linked devices
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2017-02-01, granted 2020-05-19 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10007703B1 Feed upload for search entity based content selection
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2017-05-02, granted 2018-06-26 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData

US11568274B2 Surfacing unique facts for entities
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2017-07-12, granted 2023-01-31 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US10825110B2 Entity page recommendation based on post content
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2017-08-03, granted 2020-11-03 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10685073B1 Selecting textual representations for entity attribute values
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-08-04, granted 2020-06-16 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US10691702B1 Generating ranked lists of entities
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-08-31, granted 2020-06-23 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US20180089208A1 Configuring a content document for users and use groups
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Structure answers: question heading, tight answer, then depth.
filed 2017-09-01 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US11328218B1 Identifying subjective attributes by analysis of curation signals
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-11-06, granted 2022-05-10 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo

US11204927B2 Contextual search on multimedia content
Rule: For video content, ensure entity-rich metadata and on-screen text at key timestamps. Google extracts entities from what's displayed at the moment of a voice query, so include visible text overlays, captions, and structured metadata naming people, places, and objects. Video transcripts and caption annotations directly feed query rewriting.
filed 2017-11-16, granted 2021-12-21 · leak modules: VideoContentSearchEntityAnnotations, VideoContentSearchNamedEntity, VideoContentSearchVideoAnchor, ImageUnderstandingIndexingAnnotation, ImageUnderstandingIndexingLabel, KnowledgeAnswersIntentQueryMediaEntitySignals, NlpSemanticParsingRelatedEntity, RepositoryWebrefEntityAnnotations

US11042553B2 Facilitating content entity annotation while satisfying joint performance conditions

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.

filed 2017-11-21, granted 2021-06-22 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10789309B1 Associating an entity with a search query

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2017-12-11, granted 2020-09-29 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US20190179878A1 Generating organization-specific tags for communications from various sources of an organization using an expanded knowledge graph for organization-specific content

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2017-12-12 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US10467233B2 Scoring messages based on sender attributes

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.

filed 2017-12-19, granted 2019-11-05 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10846293B1 Factual query pattern learning

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2018-01-16, granted 2020-11-24 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US10216851B1 Selecting content using entity properties

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2018-02-02, granted 2019-02-26 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US11714815B2 Method and computer-readable media for providing recommended entities based on a user's social graph

Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2018-03-28, granted 2023-08-01 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US11288351B2 Delayed two-factor authentication in a networked environment

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2018-04-25, granted 2022-03-29 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US10831845B2 Generation of enhanced search results

Rule: Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.

filed 2018-04-30, granted 2020-11-10 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US20180246978A1 Providing actions for onscreen entities

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2018-05-01 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US10587726B2 Structured entity information page
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2018-10-12, granted 2020-03-10 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US11252479B2 Methods, systems, and media for recommending media content based on attribute grouped viewing sessions
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships.
filed 2018-11-19, granted 2022-02-15 · leak modules: NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US10956425B2 User attribute resolution of unresolved terms of action queries
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2018-12-11, granted 2021-03-23 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality

US11847176B1 Generating context-based spell corrections of entity names
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2018-12-20, granted 2023-12-19 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11089457B2 Personalized entity repository
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2019-01-07, granted 2021-08-10 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US11553063B2 System and method for selecting and providing available actions from one or more computer applications to a user
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2019-01-10, granted 2023-01-10 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US11341143B2 Providing context facts
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2019-04-03, granted 2022-05-24 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US20200320168A1 Discovering Customer Support Chat Sessions
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2019-04-03 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefDocumentMetadata

US11005654B2 Outsourcing exponentiation in a private group
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Earn endorsements from trusted sources. Trust is computed and transitive.
filed 2019-05-14, granted 2021-05-11 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, SmartphonePerDocData

US11093539B2 Providing knowledge panels with search results
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2019-05-15, granted 2021-08-17 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US11853307B1 Query suggestions based on entity collections of one or more past queries
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2019-06-28, granted 2023-12-26 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US20210004438A1 Identifying entity attribute relations
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2019-07-05 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata

US11244352B2 Selecting content associated with a collection of entities
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2019-10-07, granted 2022-02-08 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US11216503B1 Clustering search results
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2019-11-26, granted 2022-01-04 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData

US20220414538A1 Determining dynamic interaction condition(s) for triggering provision of related content interface notification
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2019-12-13 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US11886533B2 Transferable neural architecture for structured data extraction from web documents
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2020-01-29, granted 2024-01-30 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US20230014465A1 A Transferable Neural Architecture for Structured Data Extraction From Web Documents
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2020-01-29 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US10929416B2 Surfacing live events in search results
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2020-03-02, granted 2021-02-23 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US11301536B2 Systems and methods for selecting content based on linked devices
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2020-03-06, granted 2022-04-12 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10917499B2 Personalized entity information page
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2020-03-09, granted 2021-02-09 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US11568004B2 Comparative search within user-generated content
Rule: Encourage and structure user reviews with keyword-rich content. Since reviews are indexed by an inverted index and ranked by term occurrence plus quality/length/recency, longer detailed reviews with relevant keywords rank higher. Use review schema markup so Google can extract snippets. Keep reviews fresh—newly posted reviews are crawled and indexed continuously.
filed 2020-07-14, granted 2023-01-31 · leak modules: RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsDataObject, RichsnippetsPageMap, QualityPreviewSnippetDocumentFeatures, QualityPreviewSnippetQueryTermCoverageFeatures, SnippetExtraInfoSnippetCandidateInfo, WWWSnippetResponse

US11531655B2 Automatically improving data quality
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2020-07-30, granted 2022-12-20 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData

US11294970B1 Associating an entity with a search query
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2020-09-16, granted 2022-04-05 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US11487719B2 Single table multi-schema data store in a key value store
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2020-12-18, granted 2022-11-01 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US11218568B2 Personalized entity information page
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2020-12-30, granted 2022-01-04 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US12045248B2 Surfacing live events in search results
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2021-01-08, granted 2024-07-23 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US12061594B2 Verified entity attributes
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.
filed 2021-02-04, granted 2024-08-13 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefDocumentMetadata

US12130827B1 Triggering knowledge panels
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2021-02-10, granted 2024-10-29 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster

US11681712B2 User attribute resolution of unresolved terms of action queries
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2021-03-05, granted 2023-06-20 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality

US11716600B2 Personalized entity repository
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2021-08-09, granted 2023-08-01 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US11836177B2 Providing knowledge panels with search results
Rule: Ensure your entity has rich, multi-source data in Google's Knowledge Graph. Maintain active social media profiles for entities. Provide high-quality images through structured data and Wikimedia. Multiple disparate sources feeding the knowledge panel means broader web presence improves panel richness.
filed 2021-08-12, granted 2023-12-05 · leak modules: RepositoryWebref, PerDocData

US11847124B2 Contextual search on multimedia content
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2021-11-19, granted 2023-12-19 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US11706318B2 Structured entity information page
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2021-11-30, granted 2023-07-18 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US20220138591A1 Developing event-specific provisional knowledge graphs
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2022-01-11 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation

US11720577B2 Contextualizing knowledge panels
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2022-01-26, granted 2023-08-08 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US11921833B2 Delayed two-factor authentication in a networked environment
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2022-03-04, granted 2024-03-05 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US20240054392A1 Transfer machine learning for attribute prediction
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2022-04-01 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US11874891B2 Systems and methods for selecting content based on linked devices
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2022-04-12, granted 2024-01-16 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US12130875B2 Distributing digital components based on predicted attributes
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2022-06-02, granted 2024-10-29 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US20230004598A1 Storing semi-structured data
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2022-09-07 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality

US20240126978A1 Determining attributes for elements of displayable content and adding them to an accessibility tree
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2022-10-14 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US12001443B2 Authoritative factual service for blockchain smart contracts
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2022-12-08, granted 2024-06-04 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US11831738B2 System and method for selecting and providing available actions from one or more computer applications to a user
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2022-12-15, granted 2023-11-28 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US12111872B2 Comparative search within user-generated content
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2023-01-27, granted 2024-10-08 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData

US20230289626A1 Knowledge Graph Completion and Multi-Hop Reasoning in Knowledge Graphs at Scale
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2023-03-14 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US12086147B2 User attribute resolution of unresolved terms of action queries
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2023-05-12, granted 2024-09-10 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality

US12149599B2 Structured entity information page
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2023-06-06, granted 2024-11-19 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US12182142B2 Contextualizing knowledge panels
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2023-06-22, granted 2024-12-31 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US20250037144A1 Recommending Configurable Controls to an Entity
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2023-07-25 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US12108314B2 Personalized entity repository
Rule: Build strong topical authority in specific niches. Google creates personalized entity repositories on user devices based on their interaction patterns — if users in your niche have your entities downloaded, you get on-device recognition priority. Create content that establishes clear topical entity associations.
filed 2023-07-28, granted 2024-10-01 · leak modules: RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, RepositoryWebrefGlobalNameInfo, PerDocData, RepositoryAnnotationsGeoTopicality

US12126700B2 System and method for selecting and providing available actions from one or more computer applications to a user
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2023-10-24, granted 2024-10-22 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US20240153297A1 Zero-shot form entity query framework
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2023-11-03 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US12216669B1 Query suggestions based on entity collections of one or more past queries
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2023-11-27, granted 2025-02-04 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US20240289395A1 Factuality of generated responses
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2023-12-04 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US12579152B2 Contextual search on multimedia content
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2023-12-06, granted 2026-03-17 · leak modules: CompressedQualitySignals, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US12373509B2 Systems and methods for selecting content based on linked devices
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2023-12-07, granted 2025-07-29 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US12566805B1 Generating context-based spell corrections of entity names
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2023-12-15, granted 2026-03-03 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US12353427B2 Rendering suggestion for searching entity within application in response to determining the entity is associated with the application
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2023-12-20, granted 2025-07-08 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US20240119086A1 Media comsumption history
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.
filed 2023-12-20 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US12361014B2 Authoritative factual service for blockchain smart contracts
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2024-04-25, granted 2025-07-15 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US12493624B2 Surfacing live events in search results
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2024-06-14, granted 2025-12-09 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US12547607B2 Verified entity attributes
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.
filed 2024-08-09, granted 2026-02-10 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefDocumentMetadata

US12461924B2 User attribute resolution of unresolved terms of action queries
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2024-09-05, granted 2025-11-04 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality

US12579158B1 Dynamic organization of search results
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2024-12-19, granted 2026-03-17 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

Ranking machinery (308)

US7096214B1 System and method for supporting editorial opinion in the ranking of search results
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2000-12-13, granted 2006-08-22 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US7089237B2 Interface and system for providing persistent contextual relevance for commerce activities in a networked environment

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.

filed 2001-01-26, granted 2006-08-08 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US6526440B1 Ranking search results by reranking the results based on local inter-connectivity

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2001-01-30, granted 2003-02-25 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate

US7716161B2 Methods and apparatus for serving relevant advertisements

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2002-12-06, granted 2010-05-11 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US6725259B1 Ranking search results by reranking the results based on local inter-connectivity

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2003-01-27, granted 2004-04-20 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate

US7451129B2 System and method for providing preferred language ordering of search results

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2003-04-03, granted 2008-11-11 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8868543B1 Finding web pages relevant to multimedia streams

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2003-04-08, granted 2014-10-21 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US8090717B1 Methods and apparatus for ranking documents

Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.

filed 2003-06-30, granted 2012-01-03 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8548995B1 Ranking of documents based on analysis of related documents

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2003-09-10, granted 2013-10-01 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US7454417B2 Methods and systems for improving a search ranking using population information

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2003-09-12, granted 2008-11-18 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US7231399B1 Ranking documents based on large data sets

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.

filed 2003-11-14, granted 2007-06-12 · leak modules: NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8676790B1 Methods and systems for improving search rankings using advertising data
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2003-12-05, granted 2014-03-18 · leak modules: NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US7814085B1 System and method for determining a composite score for categorized search results
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2004-02-26, granted 2010-10-12 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8577893B1 Ranking based on reference contexts
Rule: Use consistent entity names, sameAs schema, and structured data. Build quality links with relevant anchor text. Penguin is still active. Structure answers: question heading, tight answer, then depth.
filed 2004-03-15, granted 2013-11-05 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData

US7698266B1 Meaning-based advertising and document relevance determination
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2004-03-24, granted 2010-04-13 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo

US7693825B2 Systems and methods for ranking implicit search results
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2004-03-31, granted 2010-04-06 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityOrbitAsteroidBeltDocumentIntentScores

US7716216B1 Document ranking based on semantic distance between terms in a document
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2004-03-31, granted 2010-05-11 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate

US8954420B1 Methods and systems for improving a search ranking using article information
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2004-03-31, granted 2015-02-10 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US7698626B2 Enhanced document browsing with automatically generated links to relevant information
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2004-06-30, granted 2010-04-13 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US7028029B2 Adaptive computation of ranking
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2004-08-23, granted 2006-04-11 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8874570B1 Search boost vector based on co-visitation information
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2004-11-30, granted 2014-10-28 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US8843536B1 Methods and systems for providing relevant advertisements or other content for inactive uniform resource locators using search queries

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2004-12-31, granted 2014-09-23 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo

US7756887B1 System and method for modulating search relevancy using pointer activity monitoring

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2005-02-16, granted 2010-07-13 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US8195654B1 Prediction of human ratings or rankings of information retrieval quality

Rule: Read the Quality Rater Guidelines like a spec. Models are trained to predict those judgements at scale, so build what raters would rate well. The leak's chard, keto, rhubarb, and tofu quality predictors are the ML implementations.

filed 2005-07-13, granted 2012-06-05 · leak modules: QualityNsrNsrData, QualityNsrPQData, CompressedQualitySignals, PerDocData

US7565358B2 Agent rank

Rule: Publish under a real, consistent author identity everywhere. Same name, same profiles, linked with schema. The leak's authorObfuscatedGaiaStr field in PerDocData confirms Google associates content with Google Account IDs for E-E-A-T author identity matching.

filed 2005-08-08, granted 2009-07-21 · leak modules: PerDocData, QualityNsrNsrData, RepositoryWebref

US8005825B1 Identifying relevant portions of a document

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.

filed 2005-09-27, granted 2011-08-23 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US8370342B1 Display of relevant results

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.

filed 2005-09-27, granted 2013-02-05 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US9262446B1 Dynamically ranking entries in a personal data book

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2005-12-29, granted 2016-02-16 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData

US7587391B1 Method and apparatus for generating a preference ranking

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2006-06-13, granted 2009-09-08 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US7386543B1 System and method for supporting editorial opinion in the ranking of search results

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2006-06-30, granted 2008-06-10 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US7840572B2 Document scoring based on document inception date

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active.

filed 2006-11-20, granted 2010-11-23 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, IndexingDocjoinerAnchorStatistics, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US7613693B1 Preferential ranking of code search results

Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Write for natural spoken language and conversational queries.

filed 2006-11-30, granted 2009-11-03 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US8316029B2 Document scoring based on traffic associated with a document
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active.
filed 2006-11-30, granted 2012-11-20 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, IndexingDocjoinerAnchorStatistics, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US7987185B1 Ranking custom search results
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2006-12-29, granted 2011-07-26 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US7966309B2 Providing relevance-ordered categories of information
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2007-01-17, granted 2011-06-21 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US7610282B1 Rank-adjusted content items
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2007-03-30, granted 2009-10-27 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US8086594B1 Bifurcated document relevance scoring
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2007-03-30, granted 2011-12-27 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData

US8166045B1 Phrase extraction using subphrase scoring
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2007-03-30, granted 2012-04-24 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData

US7743050B1 Model generation for ranking documents based on large data sets
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2007-04-18, granted 2010-06-22 · leak modules: NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8768932B1 Method and apparatus for ranking search results
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2007-05-14, granted 2014-07-01 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate

US8892549B1 Ranking expertise
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2007-06-29, granted 2014-11-18 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US8572094B2 Ranking social network objects
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2007-08-17, granted 2013-10-29 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8694511B1 Modifying search result ranking based on populations
Rule: Google weights country+language-specific click data more than language-wide or global click data. If your content targets a specific country, ensure your click performance is strong in that specific locale—global CTR matters less than locale-specific CTR. Dwell time weights each click, so longer visits in your target country amplify your ranking signal. The leak's NavboostCraps modules with per-country/per-language aggregation and the LocalizedCluster module confirm this population-based click weighting is live. The prior model means even new pages can get estimated quality scores.
filed 2007-08-20, granted 2014-04-08 · leak modules: QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNavboostCrapsStatsWithWeightsProto, QualityNavboostCrapsFeatureCrapsData, CompressedQualitySignals, PerDocData, IndexingDupsLocalizedLocalizedCluster

US7953741B2 Online ranking metric
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2007-10-18, granted 2011-05-31 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate

US7984049B2 Generic online ranking system and method suitable for syndication
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2007-10-18, granted 2011-07-19 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate

US8312004B2 Online ranking protocol
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2007-10-18, granted 2012-11-13 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate

US8051076B1 Demotion of repetitive search results
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2007-12-13, granted 2011-11-01 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US7840563B2 Collective ranking of digital content
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2008-02-01, granted 2010-11-23 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US7996379B1 Document ranking using word relationships
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2008-02-01, granted 2011-08-09 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8306972B2 Ordering of search results based on language and/or country of the search results
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2008-10-21, granted 2012-11-06 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8090713B2 Methods and systems for improving a search ranking using population information
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2008-11-18, granted 2012-01-03 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US8396865B1 Sharing search engine relevance data between corpora
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2008-12-10, granted 2013-03-12 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo

US8209331B1 Context sensitive ranking
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2009-03-09, granted 2012-06-26 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9098582B1 Identifying relevant document languages through link context
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active. Write in natural vocabulary. ML models understand context and fluff.
filed 2009-04-10, granted 2015-08-04 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US20100306223A1 Rankings in Search Results with User Corrections
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2009-06-01 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo

US8224826B2 Agent rank
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2009-07-21, granted 2012-07-17 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate

US8060497B1 Framework for evaluating web search scoring functions
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2009-07-23, granted 2011-11-15 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster

US8195637B1 Rank-adjusted content items
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2009-07-31, granted 2012-06-05 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US8150860B1 Ranking authors and their content in the same framework
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2009-08-12, granted 2012-04-03 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US8321463B2 Objective and subjective ranking of comments
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2009-08-12, granted 2012-11-27 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8620929B2 Context based resource relevance
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2009-08-14, granted 2013-12-31 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US8972391B1 Recent interest based relevance scoring
Rule: Google infers your users' short-term interests from their recent search sessions and re-ranks results based on category-specific click patterns. If your content gets clicked disproportionately by users with a specific interest category, it gets boosted for similar users. Optimize for category-relevant engagement: ensure your content satisfies users who arrive with a specific topical intent. The leak's NavboostCraps click signals and OrbitAsteroidBelt intent scores confirm this category-based re-ranking is live. Dwell time weighting means longer engagement amplifies the boost.
filed 2009-10-02, granted 2015-03-03 · leak modules: QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNavboostCrapsStatsWithWeightsProto, QualityFringeFringeQueryPriorPerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, CompressedQualitySignals, PerDocData

US8965883B2 Ranking user generated web content
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2009-10-29, granted 2015-02-24 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US8874555B1 Modifying scoring data based on historical changes
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2009-11-20, granted 2014-10-28 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8131786B1 Training scoring models optimized for highly-ranked results
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2009-11-23, granted 2012-03-06 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US8135619B2 Increasing a number of relevant advertisements using a relaxed match
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2009-11-30, granted 2012-03-13 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US8244722B1 Ranking documents
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2010-01-05, granted 2012-08-14 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDocjoinerAnchorSpamInfo, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate

US8380710B1 Ordering of ranked documents
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2010-01-15, granted 2013-02-19 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9135239B1 Meaning-based advertising and document relevance determination
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2010-01-22, granted 2015-09-15 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo

US8386914B2 Enhanced document browsing with automatically generated links to relevant information
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2010-02-23, granted 2013-02-26 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US8452763B1 Extracting and scoring class-instance pairs
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2010-03-19, granted 2013-05-28 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8171034B2 Methods and apparatus for serving relevant advertisements
Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately. Earn promotion through authoritative content and entity association.
filed 2010-03-22, granted 2012-05-01 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData

US8060501B1 Document ranking based on semantic distance between terms in a document
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2010-03-23, granted 2011-11-15 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate

US8620943B2 System and method for modulating search relevancy using pointer activity monitoring
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2010-03-31, granted 2013-12-31 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US8606792B1 Scoring authors of posts
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2010-04-07, granted 2013-12-10 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals

US8386495B1 Augmented resource graph for scoring resources
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2010-04-23, granted 2013-02-26 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9116976B1 Ranking documents based on large data sets
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2010-05-11, granted 2015-08-25 · leak modules: NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8494992B1 Ranking and vote scheduling using statistical confidence intervals
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2010-08-26, granted 2013-07-23 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US8595240B1 Labeling objects by propagating scores in a graph
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2010-09-24, granted 2013-11-26 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8145618B1 System and method for determining a composite score for categorized search results
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2010-10-11, granted 2012-03-27 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8521749B2 Document scoring based on document inception date
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active.
filed 2010-10-12, granted 2013-08-27 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, IndexingDocjoinerAnchorStatistics, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8504437B1 Dynamically selecting and presenting content relevant to user input
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2010-11-04, granted 2013-08-06 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US8676828B1 Selecting and presenting content relevant to user input
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2010-11-04, granted 2014-03-18 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US8645362B1 Using resource load times in ranking search results
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2010-11-12, granted 2014-02-04 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US20150142767A1 Scoring authors of social network content
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2010-12-07 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US8407211B1 Determining relevance scores for locations
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2010-12-16, granted 2013-03-26 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US8548981B1 Providing relevance- and diversity-influenced advertisements including filtering
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2010-12-28, granted 2013-10-01 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9002867B1 Modifying ranking data based on document changes
Rule: Keep improving pages that already rank. Edits after ranking are inputs, not cosmetics. The leak's lastSignificantUpdate and goodClicks/badClicks fields confirm post-change quality is re-evaluated.
filed 2010-12-30, granted 2015-04-07 · leak modules: QualityNavboostCrapsCrapsClickSignals, PerDocData, QualityTimebasedSyntacticDate

US8676812B1 Dynamic weighting of indicator values for item scoring
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-01-28, granted 2014-03-18 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US8359309B1 Modifying search result ranking based on corpus search statistics
Rule: Diversify your content into multiple corpus types (video, images) because Google tracks per-corpus click-through rates and boosts results from corpora that users select disproportionately for a given query. Language and country alignment of your content matters: the system weights language-specific CTR data more than country-specific data, so ensure your content language matches the target audience's query language.
filed 2011-02-07, granted 2013-01-22 · leak modules: QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNavboostCrapsStatsWithWeightsProto, CompressedQualitySignals, PerDocData, VideoPerDocData, ImagePerDocData

US8645393B1 Ranking clusters and resources in a cluster
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2011-04-15, granted 2014-02-04 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US8645394B1 Ranking clusters and resources in a cluster
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2011-04-15, granted 2014-02-04 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US8180782B2 Online ranking metric
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-05-09, granted 2012-05-15 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate

US20160117324A1 Ranking labeled instances extracted from text
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-05-11 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8296293B2 Agent rank
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-05-11, granted 2012-10-23 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US8620914B1 Ranking of digital goods in a marketplace
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-05-18, granted 2013-12-31 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US9223897B1 Adjusting ranking of search results based on utility
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data.
filed 2011-05-26, granted 2015-12-29 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US8468197B2 Generic online ranking system and method suitable for syndication
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-06-16, granted 2013-06-18 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate

US8280891B1 System and method for the calibration of a scoring function
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-06-17, granted 2012-10-02 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US10783177B2 Providing relevance-ordered categories of information
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-06-20, granted 2020-09-22 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US8606776B2 Affinity based ranked for search and display
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-06-27, granted 2013-12-10 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US8321409B1 Document ranking using word relationships
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-06-30, granted 2012-11-27 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8849817B1 Ranking custom search results
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-06-30, granted 2014-09-30 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US8930359B1 Ranking custom search results
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-06-30, granted 2015-01-06 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9002856B2 Agent rank
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-08-05, granted 2015-04-07 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate

US8478751B1 Demotion of repetitive search results
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-09-23, granted 2013-07-02 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US8224827B2 Document ranking based on document classification
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active.
filed 2011-09-26, granted 2012-07-17 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, IndexingDocjoinerAnchorStatistics, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US9020965B1 Social networking relationship ranking and decay
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-09-29, granted 2015-04-28 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate

US8402033B1 Phrase extraction using subphrase scoring
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-10-14, granted 2013-03-19 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData

US9406077B1 System and method for ad keyword scoring
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Create original, effortful content. Quality is scored at page and site level.
filed 2011-10-19, granted 2016-08-02 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData

US8606778B1 Document ranking based on semantic distance between terms in a document
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-10-20, granted 2013-12-10 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate

US8843477B1 Onsite and offsite search ranking results
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-10-31, granted 2014-09-23 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US8572075B1 Framework for evaluating web search scoring functions
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-11-14, granted 2013-10-29 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster

US8843479B1 Methods and apparatus for ranking documents
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-11-18, granted 2014-09-23 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US9208231B1 Identifying languages relevant to resources
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-11-28, granted 2015-12-08 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US8510294B2 Methods and systems for improving a search ranking using population information
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-12-15, granted 2013-08-13 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US8515951B2 Methods and systems for improving a search ranking using population information
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-12-15, granted 2013-08-20 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US8600981B1 Using activity status to adjust activity rankings
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-12-21, granted 2013-12-03 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US8429212B1 Training scoring models optimized for highly-ranked results
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-01-03, granted 2013-04-23 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US8843491B1 Ranking and ordering items in stream
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-01-24, granted 2014-09-23 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8396879B1 Ranking authors and their content in the same framework
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-02-28, granted 2013-03-12 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US8850301B1 Linking to relevant content from an ereader
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-03-05, granted 2014-09-30 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US8983970B1 Ranking content using content and content authors
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2012-04-16, granted 2015-03-17 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8635212B1 Floating ranking of product results
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-04-24, granted 2014-01-21 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US9323850B1 Potential social recipient ranking for maximal viral content distribution
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-05-30, granted 2016-04-26 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, VideoPerDocData

US8751470B1 Context sensitive ranking
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-05-31, granted 2014-06-10 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US8972402B1 Ranking users and posts in social networking services
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2012-05-31, granted 2015-03-03 · leak modules: CrawlerChangerateUrlChangerate, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, TrawlerFetchReplyData

US8700639B1 Rank-adjusted content items
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-06-01, granted 2014-04-15 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US9116945B1 Prediction of human ratings or rankings of information retrieval quality
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-06-04, granted 2015-08-25 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US8484343B2 Online ranking metric
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-06-12, granted 2013-07-09 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate

US9087107B2 Rank-specific search results
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-06-14, granted 2015-07-21 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US8645396B2 Reputation scoring of an author
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2012-06-21, granted 2014-02-04 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US8818977B1 Context sensitive ranking
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-07-12, granted 2014-08-26 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US8756241B1 Determining rewrite similarity scores
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-08-06, granted 2014-06-17 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9390174B2 Search result ranking and presentation
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-08-08, granted 2016-07-12 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8924380B1 Changing a rank of a document by applying a rank transition function
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-08-13, granted 2014-12-30 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDocjoinerAnchorSpamInfo, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate

US9503411B1 Ranking posts based on a prioritized list of recipients
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-08-30, granted 2016-11-22 · leak modules: ImageData, ImagePerDocData, PerDocData, RepositoryWebrefEntityJoin, SmartphonePerDocData

US20150066940A1 Providing relevant online content
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-09-10 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US9767484B2 Defining relevant content area based on category density
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2012-09-11, granted 2017-09-19 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US8825644B1 Adjusting a ranking of search results
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-09-13, granted 2014-09-02 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9355169B1 Phrase extraction using subphrase scoring
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-09-13, granted 2016-05-31 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData

US8589457B1 Training scoring models optimized for highly-ranked results
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-09-14, granted 2013-11-19 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US8738654B2 Objective and subjective ranking of comments
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-09-14, granted 2014-05-27 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8838619B1 Ranking authors and their content in the same framework
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-09-14, granted 2014-09-16 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US8898152B1 Sharing search engine relevance data
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-09-14, granted 2014-11-25 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo

US8898153B1 Modifying scoring data based on historical changes
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-09-14, granted 2014-11-25 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US9152713B1 Finding web pages relevant to multimedia streams
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-09-14, granted 2015-10-06 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US9183311B2 Ordering of search results based on language and/or country of the search results
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-09-14, granted 2015-11-10 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9767478B2 Document scoring based on traffic associated with a document
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active.
filed 2012-09-14, granted 2017-09-19 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, IndexingDocjoinerAnchorStatistics, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US9053154B1 System and method for generating a compatibility score for web content
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2012-09-20, granted 2015-06-09 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US8682892B1 Ranking search results
Rule: Prune or fix your weak pages. Site-wide quality is a multiplier, so bad pages tax your good ones. The leak's nsr, pandaDemotion, babyPandaV2Demotion, and siteQualityStddev fields are the direct implementation.
filed 2012-09-28, granted 2014-03-25 · leak modules: QualityNsrNsrData, CompressedQualitySignals, QualityNsrPQData

US8874559B1 Ranking and ordering items in user-streams
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-10-01, granted 2014-10-28 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefEntityJoin

US9152714B1 Selecting score improvements
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-10-01, granted 2015-10-06 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate

US8793255B1 Generating a reputation score based on user interactions
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-10-15, granted 2014-07-29 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData

US8977617B1 Computing social influence scores for users
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-10-31, granted 2015-03-10 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9779140B2 Ranking signals for sparse corpora
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-11-16, granted 2017-10-03 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate

US20150242512A1 Systems and Methods for Ranking Search Results Based on User Identification of Items of Interest
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-12-11 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9058332B1 Blended ranking of dissimilar populations using an N-furcated normalization technique
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-12-28, granted 2015-06-16 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US9208233B1 Using synthetic descriptive text to rank search results
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-12-31, granted 2015-12-08 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorStatistics, NlpSaftDocument, PerDocData

US9684697B1 Ranking search results
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-12-31, granted 2017-06-20 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US8756220B1 Modifying search result ranking based on corpus search statistics
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-01-14, granted 2014-06-17 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US9697205B2 Enhanced document browsing with automatically generated links to relevant information
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-02-21, granted 2017-07-04 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US8812520B1 Augmented resource graph for scoring resources
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-02-25, granted 2014-08-19 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9235625B2 Ranking media content sources
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-02-25, granted 2016-01-12 · leak modules: PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefDocumentMetadata

US9116957B1 Substitute term scoring
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2013-03-05, granted 2015-08-25 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo

US9495456B2 Selecting, ranking, and/or presenting microsite content
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-03-11, granted 2016-11-15 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US9424360B2 Ranking events
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-03-12, granted 2016-08-23 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US9098176B1 Scoring content based on social interaction
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-03-13, granted 2015-08-04 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9141906B2 Scoring concept terms using a deep network
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-03-13, granted 2015-09-22 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefDocumentMetadata

US9053129B1 Content item relevance based on presentation data
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-03-14, granted 2015-06-09 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US9183257B1 Using web ranking to resolve anaphora
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-03-14, granted 2015-11-10 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US9146972B2 Ranking of presentation modes for particular content
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-03-15, granted 2015-09-29 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US9213745B1 Methods, systems, and media for ranking content items using topics
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2013-03-15, granted 2015-12-15 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9483528B1 Social ranking metric
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-03-20, granted 2016-11-01 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US8738602B1 Determining relevance scores for locations
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-03-25, granted 2014-05-27 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US9405803B2 Ranking signals in mixed corpora environments
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-04-23, granted 2016-08-02 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate

US9348922B2 Ranking channels in search
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-05-17, granted 2016-05-24 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityOrbitAsteroidBeltDocumentIntentScores

US8744989B1 Ranking and vote scheduling using statistical confidence intervals
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-07-01, granted 2014-06-03 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US8949013B2 Ranking nearby destinations based on visit likelihoods and predicting future visits to places from location history

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.

filed 2013-07-31, granted 2015-02-03 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsGeoTopicality, RepositoryWebrefEntityJoin

US9201929B1 Ranking a search result document based on data usage to load the search result document

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2013-08-09, granted 2015-12-01 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9256688B2 Ranking content items using predicted performance

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.

filed 2013-08-09, granted 2016-02-09 · leak modules: PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US9483524B1 Ranking communities based on content

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2013-09-04, granted 2016-11-01 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9858317B1 Ranking communities based on members

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2013-09-04, granted 2018-01-02 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US9734209B1 Chronological sorting of items ranked according to a score

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2013-09-27, granted 2017-08-15 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9244891B2 Adjusting search result rankings based on multiple user highlighting of documents

Rule: Use descriptive alt text and surrounding context for images. Structure answers: question heading, tight answer, then depth. Use JSON-LD structured data for entity types and relationships.

filed 2013-10-03, granted 2016-01-26 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SnippetExtraInfo

US9081823B2 Ranking social network objects

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2013-10-09, granted 2015-07-14 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9262541B2 Distance based search ranking demotion

Rule: Ensure your Google Business Profile location data is accurate and consistent with your physical location. If you serve multiple locations, create location-specific pages. High location-independent relevance (strong content, links, authority) can protect you from distance-based demotion even for distant users — build strong non-local authority signals to trigger the preservation threshold. Local intent queries without explicit location phrases still trigger demotion, so local SEO matters even for generic queries.

filed 2013-10-18, granted 2016-02-16 · leak modules: PerDocData, QualityNsrNsrData, CompressedQualitySignals, LocalWWWInfo

US9141644B2 Affinity based ranked for search and display

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2013-11-11, granted 2015-09-22 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US8983974B1 Scoring authors of posts

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2013-11-13, granted 2015-03-17 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals

US9767158B1 Ranking algorithms using unique bucketing identifiers

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2013-11-14, granted 2017-09-19 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8965891B1 Training scoring models optimized for highly-ranked results

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2013-11-18, granted 2015-02-24 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US9532176B1 Smoothed activity signals for suggestion ranking

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2013-11-26, granted 2016-12-27 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US10831348B1 Ranking and selecting task components based on frequency of completions

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2013-12-13, granted 2020-11-10 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US9229989B1 Using resource load times in ranking search results

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2013-12-30, granted 2016-01-05 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9436755B1 Determining and scoring task indications

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.

filed 2014-01-26, granted 2016-09-06 · leak modules: NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata

US9361381B2 Selecting and presenting content relevant to user input

Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.

filed 2014-03-17, granted 2016-06-07 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US20150287099A1 Method to compute the prominence score to phone numbers on web pages and automatically annotate/attach it to ads

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2014-04-07 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US9501549B1 Scoring criteria for a content item

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.

filed 2014-04-28, granted 2016-11-22 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9002894B2 Objective and subjective ranking of comments
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-05-16, granted 2015-04-07 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US20190102393A1 Ranking contact entries
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2014-06-04 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9594808B2 Determining relevance of points of interest to a user
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2014-06-04, granted 2017-03-14 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US9767159B2 Ranking search results
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-06-13, granted 2017-09-19 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US9817907B1 Using place of accommodation as a signal for ranking reviews and point of interest search results
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-06-18, granted 2017-11-14 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US9747263B1 Dynamic page classifier for ranking content
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-06-27, granted 2017-08-29 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US9223849B1 Generating a reputation score based on user interactions
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2014-07-11, granted 2015-12-29 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData

US9223835B1 Ranking and ordering items in stream
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-07-28, granted 2015-12-29 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10229164B1 Adjusting a relevancy score of a keyword cluster—time period—event category combination based on event related information
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-08-02, granted 2019-03-12 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9779144B1 Identifying a level of relevancy of a keyword cluster related to an event category for a given time period relative to the event
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-08-02, granted 2017-10-03 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9875313B1 Ranking authors and their content in the same framework
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-08-08, granted 2018-01-23 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US9454582B1 Ranking search results
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2014-08-18, granted 2016-09-27 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US9286356B1 Adjusting a ranking of search results
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-08-29, granted 2016-03-15 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9342609B1 Ranking custom search results
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-09-10, granted 2016-05-17 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US9477714B1 Methods and apparatus for ranking documents
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2014-09-17, granted 2016-10-25 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9449053B2 Ranking nearby destinations based on visit likelihoods and predicting future visits to places from location history
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2014-11-18, granted 2016-09-20 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsGeoTopicality, RepositoryWebrefEntityJoin

US10372768B1 Ranking content using sharing attribution
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2014-12-31, granted 2019-08-06 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals

US9390143B2 Recent interest based relevance scoring
Rule: This continuation confirms Google's ongoing use of short-term interest categories for re-ranking. Your content gets boosted when users with a specific recent interest category click it disproportionately. The 'similar users' mechanism means if users with similar profiles to the searcher have selected your result, it gets boosted. Optimize for category-relevant engagement and ensure your content satisfies specific intent categories. The leak's FringeQueryPrior and OrbitAsteroidBelt modules track per-document intent scores that feed this system.
filed 2015-01-22, granted 2016-07-12 · leak modules: QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNavboostCrapsStatsWithWeightsProto, QualityFringeFringeQueryPriorPerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, CompressedQualitySignals, PerDocData

US10423679B2 Methods and systems for improving a search ranking using article information
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-02-10, granted 2019-09-24 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US10671616B1 Selectively modifying scores of youth-oriented content search results
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2015-02-22, granted 2020-06-02 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US9569438B1 Ranking content using content and content authors

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.

filed 2015-02-27, granted 2017-02-14 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9442989B1 Scoring authors of posts

Rule: Author authority is computed from engagement quality, not just quantity. Encourage substantive, high-quality comments on your content—comment length and substance directly feed the author's reputation score. Building a subscriber base of high-reputation authors amplifies your score. The social graph distance means content from closely-connected authors ranks higher for users in that network. The leak's BlogPerDocData and GroupsPerDocData modules confirm per-author signals are tracked at the document level.

filed 2015-03-13, granted 2016-09-13 · leak modules: QualityNsrNsrData, QualityNsrPQData, PerDocData, BlogPerDocData, GroupsPerDocData, QualityLabelsGoogleLabelData, StorageGraphBfgAuthorityFeedbackMetadata, CompressedQualitySignals

US20190155948A1 Re-ranking resources based on categorical quality

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data.

filed 2015-03-31 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US9390144B2 Objective and subjective ranking of comments

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2015-04-06, granted 2016-07-12 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9558222B1 Blended ranking of dissimilar populations using an N-furcated normalization technique

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2015-05-11, granted 2017-01-31 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US10339144B1 Search operation adjustment and re-scoring

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data.

filed 2015-05-21, granted 2019-07-02 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US10169390B2 Ranking social network objects

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2015-05-28, granted 2019-01-01 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10296642B1 Ranking content for user engagement

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2015-06-05, granted 2019-05-21 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US20150363809A1 Endorsements Used in Ranking Ads

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2015-06-22 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefEntityJoin

US9948699B1 Scoring content based on social interaction

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2015-06-26, granted 2018-04-17 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10354308B2 Distinguishing accessories from products for ranking search results

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.

filed 2015-07-02, granted 2019-07-16 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo

US10061817B1 Social ranking for apps

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2015-07-29, granted 2018-08-28 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10055461B1 Ranking documents based on large data sets

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.

filed 2015-07-31, granted 2018-08-21 · leak modules: NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9799052B2 Methods and apparatus for serving relevant advertisements

Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately. Earn promotion through authoritative content and entity association.

filed 2015-08-03, granted 2017-10-24 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData

US9710825B1 Meaning-based advertising and document relevance determination

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2015-09-14, granted 2017-07-18 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo

US10621250B2 Notifying users of relevant content

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2015-09-15, granted 2020-04-14 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9740750B1 Selecting score improvements

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2015-09-17, granted 2017-08-22 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate

US9514405B2 Scoring concept terms using a deep network

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2015-09-21, granted 2016-12-06 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefDocumentMetadata

US9672253B1 Ranking a search result document based on data usage to load the search result document

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2015-11-02, granted 2017-06-06 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9542441B1 Using web ranking to resolve anaphora

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2015-11-06, granted 2017-01-10 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US10055463B1 Feature based ranking adjustment

Rule: Google actively diversifies search results by demoting pages that share features (same author, same topic, same content type) with higher-ranked results. To avoid demotion, differentiate your content from competitors ranking above you. Publish diverse content types and ensure your pages have unique feature combinations. Diversify authors and content formats.

filed 2015-11-19, granted 2018-08-21 · leak modules: demotion, ranking, diversification, twiddler

US10685029B2 Information ranking based on properties of a computing device

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2015-11-23, granted 2020-06-16 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US10223461B1 Identifying languages relevant to resources

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2015-12-04, granted 2019-03-05 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US10031951B1 Distance based search ranking demotion

Rule: For local SEO, ensure your business location is accurate and consistent across all citations. If your business is far from the searcher, your local boost may be demoted in favor of more relevant non-local results. Optimize for relevance signals beyond just proximity - content quality and query relevance can overcome distance-based demotion.

filed 2016-01-07, granted 2018-07-24 · leak modules: demotion, local_search, ranking, navboost

US9959322B1 Ranking channels in search

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-05-13, granted 2018-05-01 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityOrbitAsteroidBeltDocumentIntentScores

US10089393B2 Selecting and presenting content relevant to user input

Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.

filed 2016-06-06, granted 2018-10-02 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US10445328B2 Search result ranking and presentation

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2016-07-08, granted 2019-10-15 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US10332019B2 Ranking nearby destinations based on visit likelihoods and predicting future visits to places from location history

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.

filed 2016-07-11, granted 2019-06-25 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsGeoTopicality, RepositoryWebrefEntityJoin

US9779411B1 System and method for ad keyword scoring

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Create original, effortful content. Quality is scored at page and site level.

filed 2016-07-25, granted 2017-10-03 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData

US10496652B1 Methods and apparatus for ranking documents

Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.

filed 2016-09-12, granted 2019-12-03 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9846728B1 Scoring authors of posts

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2016-09-12, granted 2017-12-19 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals

US10282357B1 Dynamic resampling for ranking viewer experiences

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2016-09-15, granted 2019-05-07 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US10296512B1 Action-based content scoring

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2016-09-23, granted 2019-05-21 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US10394832B2 Ranking search results documents

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.

filed 2016-10-24, granted 2019-08-27 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US9984167B1 Ranking communities based on content

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2016-10-31, granted 2018-05-29 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9792376B1 Selecting, ranking, and/or presenting microsite content

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2016-11-14, granted 2017-10-17 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US10324993B2 Predicting a search engine ranking signal value

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Build quality links with relevant anchor text. Penguin is still active.

filed 2016-12-05, granted 2019-06-18 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorStatistics, NlpSemanticParsingQRefAnnotation, PerDocData

US9977791B2 Smoothed activity signals for suggestion ranking

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-12-23, granted 2018-05-22 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US10185778B1 Ranking content using content and content authors

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.

filed 2016-12-30, granted 2019-01-22 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11475290B2 Structured machine learning for improved whole-structure relevance scoring

Rule: Structure your content as items with attributes (e.g., products with specifications, conditions with symptoms) using rich schema markup. Google's model jointly selects which items and which attributes to display—if your attributes are well-structured and distinctive, they're more likely to be selected. Use schema.org properties to define item-attribute relationships. Ensure attribute values are unique and differentiating, as similarity features between attributes affect selection. The model considers cross-item similarity, so differentiate your offerings from competitors.

filed 2016-12-30, granted 2022-10-18 · leak modules: QualityRankembedMustangMustangRankEmbedInfo, QualityRankembedMustangMustangRankEmbedInfoCompressedEmbedding, QualityPreviewSnippetDocumentFeatures, QualityPreviewChosenSnippetInfo, RichsnippetsDataObject, RichsnippetsDataObjectAttribute, IndexingSignalAggregatorAggregatedScore, PerDocData

US10083226B1 Using web ranking to resolve anaphora
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-01-09, granted 2018-09-25 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US10430484B2 Determining relevance of points of interest to a user
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2017-01-31, granted 2019-10-01 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US20180357303A1 Determining feature scores for message features
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2017-02-02 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US10055467B1 Ranking search results
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use descriptive alt text and surrounding context for images.
filed 2017-05-30, granted 2018-08-21 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US10579714B1 Dynamic page classifier for ranking content
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-08-08, granted 2020-03-03 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US10552504B1 Selecting, ranking, and/or presenting microsite content
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2017-09-05, granted 2020-02-04 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US10198746B2 Methods and apparatus for serving relevant advertisements
Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately. Earn promotion through authoritative content and entity association.
filed 2017-09-14, granted 2019-02-05 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData

US10713263B2 Ranking algorithms using unique bucketing identifiers
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2017-09-18, granted 2020-07-14 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US10614079B1 Identifying a level of relevancy of a keyword cluster related to an event category for a given time period relative to the event
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2017-09-26, granted 2020-04-07 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US10614483B1 System and method for ad keyword scoring
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Create original, effortful content. Quality is scored at page and site level.
filed 2017-09-29, granted 2020-04-07 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData

US11514353B2 Generating, using a machine learning model, request agnostic interaction scores for electronic communications, and utilization of same

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2017-10-26, granted 2022-11-29 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo

US10877978B1 Ranking communities based on members

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2017-12-11, granted 2020-12-29 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US10949429B1 Scoring authors of posts

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2017-12-18, granted 2021-03-16 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals

US10630763B1 Scoring content based on social interaction

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2018-03-14, granted 2020-04-21 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11194874B1 Ranking communities based on content

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2018-05-25, granted 2021-12-07 · leak modules: ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US11442946B2 Distance based search ranking demotion

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.

filed 2018-07-03, granted 2022-09-13 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US10204145B2 Systems and methods for re-ranking ranked search results

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs.

filed 2018-08-01, granted 2019-02-12 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US10503740B2 Systems and methods for re-ranking ranked search results

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs.

filed 2018-12-14, granted 2019-12-10 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US10970353B1 Ranking content using content and content authors

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.

filed 2019-01-17, granted 2021-04-06 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10991005B2 Methods and apparatus for serving relevant advertisements

Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately. Earn promotion through authoritative content and entity association.

filed 2019-01-28, granted 2021-04-27 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, VideoPerDocData

US11093567B1 Ranking content using sharing attribution
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2019-08-05, granted 2021-08-17 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals

US10970293B2 Ranking search result documents
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.
filed 2019-08-26, granted 2021-04-06 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US11086953B2 Determining relevance of points of interest to a user
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2019-09-03, granted 2021-08-10 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US11403301B2 Search result ranking and presentation
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2019-09-26, granted 2022-08-02 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US10936689B1 Selecting, ranking, and/or presenting microsite content
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2019-12-04, granted 2021-03-02 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US11763339B2 System and method for ad keyword scoring
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2020-03-19, granted 2023-09-19 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11347818B2 Notifying users of relevant content
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2020-04-13, granted 2022-05-31 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20200242180A1 Notifying users of a relevant content
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2020-04-13 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11550798B2 Ranking algorithms using unique bucketing identifiers
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2020-07-13, granted 2023-01-10 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US11334610B2 Providing relevance-ordered categories of information
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2020-08-28, granted 2022-05-17 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US11238120B1 Selecting, ranking, and/or presenting microsite content
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2021-02-19, granted 2022-02-01 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US11775605B2 Determining relevance of points of interest to a user

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2021-08-09, granted 2023-10-03 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US11727071B1 Selecting, ranking, and/or presenting microsite content

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2022-01-31, granted 2023-08-15 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US20230281205A1 Contextually Relevant Suggestions

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2022-03-01 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11709876B2 Providing relevance-ordered categories of information

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2022-04-21, granted 2023-07-25 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US11868357B2 Search result ranking and presentation

Rule: Optimize content for superlative and comparative queries ('best', 'top', 'most') by clearly establishing quality signals and popularity metrics in your content. Implement knowledge graph entity markup so your content is retrievable from Google's knowledge graph for entity-type queries. Structure content with clear property-value pairs that can be used for range filtering. Since presentation format depends on entity type, use schema.org typing to help Google select the right display format for your content.

filed 2022-08-01, granted 2024-01-09 · leak modules: RepositoryWebrefEntityJoin, RepositoryWebrefDocumentMetadata, RepositoryWebrefWebrefEntities, PerDocData, QualityNsrNsrData

US11893034B2 Distance based search ranking demotion

Rule: Ensure your Google Business Profile location data is accurate and consistent with your physical location. If you serve multiple locations, create location-specific pages. High location-independent relevance (strong content, links, authority) can protect you from distance-based demotion even for distant users.

filed 2022-09-12, granted 2024-02-06 · leak modules: PerDocData, QualityNsrNsrData, CompressedQualitySignals

US12443883B2 Generating, using a machine learning model, request agnostic interaction scores for electronic communications, and utilization of same

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2022-11-28, granted 2025-10-14 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo

US20230153346A1 Predicting topics of potential relevance based on retrieved/created digital media files

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2023-01-06 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US12019664B2 Providing relevance-ordered categories of information

Rule: Structure content to clearly fit specific categories that Google can identify. Google orders result categories based on historical query-category correlations — content in categories that historically correlate with your query type gets priority placement. Use clear category signals in structured data.

filed 2023-06-05, granted 2024-06-25 · leak modules: QualityOrbitAsteroidBeltDocumentIntentScores, LocalWWWInfo, RepositoryAnnotationsGeoTopicality, PerDocData

US12001496B1 Selecting, ranking, and/or presenting microsite content

Rule: Consider microsite strategies for location-based and time-based content delivery. Google can trigger microsites based on geographic location, time of day, and even ambient sound. For local businesses, create microsites with geographic and temporal triggers to capture context-aware local search traffic.

filed 2023-07-05, granted 2024-06-04 · leak modules: LocalWWWInfo, RepositoryAnnotationsGeoTopicality, PerDocData, QualityTimebasedPageType, SmartphonePerDocData

US20250045305A1 Adaptive ranking of navigation suggestions
Rule: Optimize page titles and domain names for autocomplete match—title/domain match degree is a direct ranking signal for navigation suggestions. Build return-visit behavior to improve relative visit ratio. Since user cohorts (power vs. casual users) affect ranking, ensure your site appeals to both types. The model runs on-device, so fast-loading, bookmark-worthy pages that users return to frequently will rank higher in navigation autocomplete.
filed 2023-07-31 · leak modules: PerDocData, SdrPageAnchorsDocInfo, SdrPageAnchorsSitelink, QualityNsrNsrData, CompressedQualitySignals, AssistantPrefulfillmentRankerPrefulfillmentSignals, KnowledgeAnswersIntentQueryParsingSignals

US12154129B2 System and method for ad keyword scoring
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2023-08-11, granted 2024-11-26 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US12314275B2 Cross-list learning to rank
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2023-08-14, granted 2025-05-27 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefDocumentMetadata

US12346393B2 Determining relevance of points of interest to a user
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2023-09-29, granted 2025-07-01 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US12298986B2 Search result ranking and presentation
Rule: Optimize entity data in the Knowledge Graph with complete property attributes. For entity-type queries, ensure your content addresses implicit modifiers (best, top, most popular) even if users don't explicitly search for them. Use high-quality thumbnails and structured image data. Popularity signals feed implicit modifier ranking.
filed 2023-11-28, granted 2025-05-13 · leak modules: RepositoryWebref, PerDocData, QualityNsrNsrData

US20250110962A1 Automated Keyword Generation Based on Similarity Score
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2024-07-16 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US20260023790A1 Detecting signal exploitation from consistent ranking patterns
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2024-07-22 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US20250124067A1 Method for Text Ranking with Pairwise Ranking Prompting
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2024-10-11 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US20260105104A1 Learning to Rank with Asymmetric Matching Losses
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2024-10-11 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata

Clicks & personalisation (291)

US7523096B2 Methods and systems for personalized network searching
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2003-12-03, granted 2009-04-21 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US7716225B1 Ranking documents based on user behavior and/or feature data
Rule: Put the links you want to count in the main content where people actually click, not in the footer or sidebar. The leak's pagerankWeight and sourceType fields in AnchorsAnchor confirm link weight varies by source quality and click probability.
filed 2004-06-17, granted 2010-05-11 · leak modules: AnchorsAnchor, QualityNavboostCrapsCrapsClickSignals, IndexingDocjoinerAnchorStatistics, CompressedQualitySignals

US8832132B1 Personalizing search queries based on user membership in social network communities
Rule: Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately. Build content for diverse user contexts and search histories.
filed 2004-06-22, granted 2014-09-09 · leak modules: NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US8321278B2 Targeted advertisements based on user profiles and page profile
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2004-06-24, granted 2012-11-27 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US8825639B2 Endorsing search results
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2004-06-30, granted 2014-09-02 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US7693827B2 Personalization of placed content ordering in search results
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2004-07-13, granted 2010-04-06 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US7580929B2 Phrase-based personalization of searches in an information retrieval system
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2004-07-26, granted 2009-08-25 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDocjoinerAnchorStatistics, IndexingDupsLocalizedLocalizedCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US7716223B2 Variable personalization of search results in a search engine
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2004-12-01, granted 2010-05-11 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US8538970B1 Personalizing search results
Rule: Build authority and trust signals so your pages qualify for the high-quality document set (the intersection of trusted + high PageRank). Earn backlinks from pages that users have in their preference/bias sets, as those links transmit preference weight to your results. Build community following among user groups with shared interests, as collective preferences amplify re-ranking effects. Avoid negative sentiment signals that could place your site in distrusted document sets.
filed 2004-12-30, granted 2013-09-17 · leak modules: PerDocData, QualityNsrNsrData, KaltixPerDocData, QualityNavboostCrapsCrapsClickSignals

US7747632B2 Systems and methods for providing subscription-based personalization
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2005-03-31, granted 2010-06-29 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US8190997B2 Personalized content feed suggestions page
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2005-10-07, granted 2012-05-29 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, PerDocData, ShingleInfoPerDocData, SmartphonePerDocData

US7603619B2 Formatting a user network site based on user preferences and format performance data
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2005-11-29, granted 2009-10-13 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, RepositoryAnnotationsGeoTopicality

US8185819B2 Module specification for a module to be incorporated into a container document
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data.
filed 2005-12-12, granted 2012-05-22 · leak modules: CrawlerChangerateUrlChangerate, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US8799302B2 Recommended alerts
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2005-12-29, granted 2014-08-05 · leak modules: PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US20080010252A1 Bookmarks and ranking
Rule: Build content for diverse user contexts and search histories. Earn promotion through authoritative content and entity association. Write for natural spoken language and conversational queries.
filed 2006-09-29 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityNsrNsrData

US8548991B1 Personalized browsing activity displays
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Build content for diverse user contexts and search histories. Earn promotion through authoritative content and entity association.
filed 2006-09-29, granted 2013-10-01 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityNsrNsrData, SmartphonePerDocData

US9037581B1 Personalized search result ranking
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Build content for diverse user contexts and search histories.
filed 2006-09-29, granted 2015-05-19 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8661029B1 Modifying search result ranking based on implicit user feedback
Rule: Win the click and the stay: titles that match the page, pages that answer in the first screen. The leak's goodClicks, badClicks, lastLongestClicks, and unsquashedClicks fields are the direct implementation of this patent.
filed 2006-11-02, granted 2014-02-25 · leak modules: QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, CompressedQualitySignals

US9305088B1 Personalized search results
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2006-11-30, granted 2016-04-05 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US8312009B1 Obtaining user preferences for query results
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2007-02-14, granted 2012-11-13 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US8938463B1 Modifying search result ranking based on implicit user feedback and a model of presentation bias
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2007-03-12, granted 2015-01-20 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US7730017B2 Open profile content identification
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write for natural spoken language and conversational queries.
filed 2007-03-30, granted 2010-06-01 · leak modules: NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US8321462B2 Custodian based content identification
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2007-03-30, granted 2012-11-27 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo

US8782698B2 Customizable media channels

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2007-04-30, granted 2014-07-15 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US7827170B1 Systems and methods for demoting personalized search results based on personal information

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2007-08-28, granted 2010-11-02 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US8620915B1 Systems and methods for promoting personalized search results based on personal information

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2007-08-28, granted 2013-12-31 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US8943425B2 Method and apparatus for context-aware delivery of informational content on ambient displays

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.

filed 2007-10-30, granted 2015-01-27 · leak modules: CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations

US8065254B1 Presenting a diversity of recommendations

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2008-02-19, granted 2011-11-22 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US8015170B2 Personalized network searching

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2008-04-08, granted 2011-09-06 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8271413B2 Providing digital content based on expected user behavior

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2008-11-25, granted 2012-09-18 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US8200661B1 Dynamic recommendations based on user actions

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2008-12-18, granted 2012-06-12 · leak modules: NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData

US8595228B1 Preferred sites

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2009-01-09, granted 2013-11-26 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US8612306B1 Method, system, and storage device for recommending products utilizing category attributes

Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2009-07-29, granted 2013-12-17 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US8910070B2 Tab pages and bookmark toolbars in a browser
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2009-09-01, granted 2014-12-09 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData

US9703886B2 Formatting a user network site based on user preferences and format performance data
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2009-10-09, granted 2017-07-11 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, RepositoryAnnotationsGeoTopicality

US8543668B1 Click tracking using link styles
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2009-11-24, granted 2013-09-24 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US8615514B1 Evaluating website properties by partitioning user feedback
Rule: Audit engagement per template and section, not site-wide averages. Fix or cut the sections that lose people. The leak's siteQualityStddev field confirms Google measures quality consistency across partitions.
filed 2010-02-03, granted 2013-12-24 · leak modules: QualityNavboostCrapsCrapsClickSignals, QualityNsrNsrData, QualityNsrPQData

US8180776B2 Variable personalization of search results in a search engine
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2010-03-09, granted 2012-05-15 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US8341169B2 Open profile content identification
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write for natural spoken language and conversational queries.
filed 2010-03-18, granted 2012-12-25 · leak modules: NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US8117209B1 Ranking documents based on user behavior and/or feature data
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2010-03-19, granted 2012-02-14 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo

US8412727B1 Generating query refinements from user preference data
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2010-06-04, granted 2013-04-02 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US8316019B1 Personalized query suggestions from profile trees
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2010-06-23, granted 2012-11-20 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US8326861B1 Personalized term importance evaluation in queries
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2010-06-23, granted 2012-12-04 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8392435B1 Query suggestions for a document based on user history
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2010-07-06, granted 2013-03-05 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsGeoTopicality

US8832083B1 Combining user feedback
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Create original, effortful content. Quality is scored at page and site level.
filed 2010-07-23, granted 2014-09-09 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData

US8762326B1 Personalized hot topics
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2010-09-23, granted 2014-06-24 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8688704B1 User feedback in people search clustering
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2010-11-24, granted 2014-04-01 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US8881021B1 Automated toolbar personalization based on user metadata
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2010-12-15, granted 2014-11-04 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US9026479B1 Predicting user interests
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-02-02, granted 2015-05-05 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8732240B1 Scoring stream items with models based on user interests
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-04-29, granted 2014-05-20 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9165305B1 Generating models based on user behavior
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-05-06, granted 2015-10-20 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8589407B2 Automated generation of suggestions for personalized reactions in a social network
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-06-17, granted 2013-11-19 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US8700985B2 Collecting user feedback about web pages
Rule: Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-06-20, granted 2014-04-15 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US8150830B2 Personalized network searching
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-06-30, granted 2012-04-03 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8166017B2 Personalized network searching
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-06-30, granted 2012-04-24 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9075875B1 System and method for recommending television programs based on user search queries
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-06-30, granted 2015-07-07 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US8782034B1 Utilizing information about user-visited places to recommend novel spaces to explore
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-08-17, granted 2014-07-15 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US8826153B2 Speculative actions based on user dwell time over selectable content
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-10-04, granted 2014-09-02 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9268818B1 Determining intent of a recommendation on a URL of a web page or advertisement
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-10-11, granted 2016-02-23 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US9141589B2 Method and an apparatus to provide a personalized page
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-10-25, granted 2015-09-22 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8639693B1 Personalized place recommendations using collections of place listings
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-11-11, granted 2014-01-28 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US9519722B1 Method and system for providing dynamic personalized recommendations for a destination
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2011-11-14, granted 2016-12-13 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US8374985B1 Presenting a diversity of recommendations
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-11-21, granted 2013-02-12 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US8965882B1 Click or skip evaluation of synonym rules
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-11-22, granted 2015-02-24 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8370348B1 Magazine edition recommendations
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-12-06, granted 2013-02-05 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US9055311B1 Providing a program listing
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-01-03, granted 2015-06-09 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US9305099B1 Ranking documents based on user behavior and/or feature data
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-01-10, granted 2016-04-05 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo

US9443022B2 Method, system, and graphical user interface for providing personalized recommendations of popular search queries
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-01-19, granted 2016-09-13 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US8700628B1 Personalized aggregation of annotations
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-02-03, granted 2014-04-15 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US8521735B1 Anonymous personalized recommendation method
Rule: Use consistent entity names, sameAs schema, and structured data. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-02-27, granted 2013-08-27 · leak modules: NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US9727653B2 System and method for identifying and ranking user preferences
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-03-08, granted 2017-08-08 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US9288122B1 Providing recommendations to users in a social network
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-03-15, granted 2016-03-15 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8612415B2 Personalized network searching
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-04-09, granted 2013-12-17 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20150127748A1 Recommendations for enhanced content in social posts
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2012-04-13 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US8874567B2 Variable personalization of search results in a search engine
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-05-04, granted 2014-10-28 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US8918713B2 Module specification for a module to be incorporated into a container document
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data.
filed 2012-05-10, granted 2014-12-23 · leak modules: CrawlerChangerateUrlChangerate, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US8959103B1 Click or skip evaluation of reordering rules
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-05-25, granted 2015-02-17 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8930392B1 Simulated annealing in recommendation systems
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Build content for diverse user contexts and search histories.
filed 2012-06-05, granted 2015-01-06 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9477757B1 Latent user models for personalized ranking
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-06-14, granted 2016-10-25 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, VideoPerDocData

US8484636B2 Generating application recommendations based on user installed applications
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2012-06-22, granted 2013-07-09 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US20140006388A1 Event searching and suggestion
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-06-27 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US8893014B1 Deferred loading of a web page
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-06-28, granted 2014-11-18 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8898296B2 Detection of boilerplate content
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active.
filed 2012-08-01, granted 2014-11-25 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9680959B2 Recommending content based on intersecting user interest profiles
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2012-08-30, granted 2017-06-13 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US9047383B1 Analyzing user profiles
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-09-06, granted 2015-06-02 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate

US8719347B1 Scoring stream items with models based on user interests
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-09-14, granted 2014-05-06 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8745074B1 Method and system for evaluating content via a computer network
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-09-14, granted 2014-06-03 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US8762373B1 Personalized search result ranking
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Build content for diverse user contexts and search histories.
filed 2012-09-14, granted 2014-06-24 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9058364B2 Variable personalization of search results in a search engine
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-09-14, granted 2015-06-16 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US9053424B1 Learning mechanism for recommended reordering of elements based on demographic information
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-10-26, granted 2015-06-09 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US9098551B1 Method and system for ranking content by click count and other web popularity signals
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2012-10-26, granted 2015-08-04 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US8868565B1 Calibrating click duration according to context
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2012-10-30, granted 2014-10-21 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8819013B2 Customer support solution recommendation system
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-11-01, granted 2014-08-26 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20150066915A1 System and Method for Group Recommendation of Objects Using User Comparisons of Object Characteristics
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-12-12 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate

US9213754B1 Personalizing content items
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-12-17, granted 2015-12-15 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US9141672B1 Click or skip evaluation of query term optionalization rule
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-12-27, granted 2015-09-22 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20140188927A1 Presenting recommended content in search pages

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2012-12-28 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US9031954B1 Methods, system, and media for recommending media content

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2012-12-31, granted 2015-05-12 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US9129227B1 Methods, systems, and media for recommending content items based on topics

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2012-12-31, granted 2015-09-08 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US9146966B1 Click or skip evaluation of proximity rules

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2013-01-07, granted 2015-09-29 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US9239865B1 Systems, methods, and computer-readable media for providing recommended entities based on a query-specific subset of a user's social graph

Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2013-01-18, granted 2016-01-19 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US9098572B1 Magazine edition recommendations

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2013-01-30, granted 2015-08-04 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US9208237B2 Personalized content suggestions for subscribers of a channel of a content sharing platform

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2013-01-30, granted 2015-12-08 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US8762392B1 Query suggestions for a document based on user history

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2013-02-22, granted 2014-06-24 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsGeoTopicality

US9253541B2 Method for one-click subscribing to multiple channels of information on a single topic

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.

filed 2013-02-26, granted 2016-02-02 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US9305102B2 Systems and methods for providing personalized search results based on prior user interactions

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2013-02-27, granted 2016-04-05 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate

US20140250177A1 Recommending content based on proxy-based preference indications
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-03-01 · leak modules: PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US20140250105A1 Reliable content recommendations
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-03-04 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US9311384B1 Providing recommendations
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-03-07, granted 2016-04-12 · leak modules: PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US20130346403A1 Signal based recommender
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-03-11 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US9015236B1 Personalized services using a client-side user model
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2013-03-13, granted 2015-04-21 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8918417B1 Generating query refinements from user preference data
Rule: Create content that serves as a quality result for both broad and refined queries—this increases your page's likelihood of appearing for query refinements. Monitor which query refinements Google suggests for your target keywords, as these are derived from user preference data (clicks, dwell time). Optimize for both the head query and its refinements to capture users at different search stages. Geographic specificity in content helps with location-based refinement associations.
filed 2013-03-26, granted 2014-12-23 · leak modules: QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNavboostCrapsStatsWithWeightsProto, PerDocData, CompressedQualitySignals, QualityNsrPQData, IndexingSignalAggregatorAggregatedScore

US20140108445A1 System and Method for Personalizing Query Suggestions Based on User Interest Profile
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-04-10 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US20140317099A1 Personalized digital content search
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-04-23 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US20140317105A1 Live recommendation generation
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize video for watch time and retention.
filed 2013-04-23 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, VideoPerDocData

US9826057B2 Providing digital content based on expected user behavior
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-05-30, granted 2017-11-21 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US8924955B2 Generating application recommendations based on user installed applications
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2013-06-05, granted 2014-12-30 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US9037573B2 Phase-based personalization of searches in an information retrieval system
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2013-06-17, granted 2015-05-19 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDocjoinerAnchorStatistics, IndexingDupsLocalizedLocalizedCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US9608892B2 Client-side click tracking
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-07-12, granted 2017-03-28 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US8745049B2 Anonymous personalized recommendation method
Rule: Use consistent entity names, sameAs schema, and structured data. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-07-22, granted 2014-06-03 · leak modules: NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US8977630B1 Personalizing search results
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-09-16, granted 2015-03-10 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US10031895B2 Click tracking using link styles
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-09-19, granted 2018-07-24 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US10496649B1 Personalized suggestions based on past queries
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-09-26, granted 2019-12-03 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US8914362B1 Personalized browsing activity displays
Rule: Build content for diverse user contexts and search histories. Earn promotion through authoritative content and entity association. Write for intent and meaning. Neural matching connects related concepts.
filed 2013-09-30, granted 2014-12-16 · leak modules: CompressedQualitySignals, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US9336278B2 User experience and user flows for third-party application recommendation in cloud storage systems
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-09-30, granted 2016-05-10 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9390141B2 Systems and methods for determining application installation likelihood based on probabilistic combination of subordinate methods
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2013-09-30, granted 2016-07-12 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US9633081B1 Systems and methods for determining application installation likelihood based on user network characteristics

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2013-09-30, granted 2017-04-25 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9002835B2 Query response using media consumption history

Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2013-10-07, granted 2015-04-07 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US9385972B2 Automated generation of suggestions for personalized reactions in a social network

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2013-10-15, granted 2016-07-05 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US9317563B1 Preferred sites

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2013-10-23, granted 2016-04-19 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US9355415B2 Providing content recommendation to users on a site

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2013-11-04, granted 2016-05-31 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US8886626B2 Personalized network searching

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2013-11-08, granted 2014-11-11 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9767187B2 Content recommendations based on organic keyword analysis

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2013-11-20, granted 2017-09-19 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9396238B2 Systems and methods for determining user preferences

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2013-12-02, granted 2016-07-19 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US9552395B2 Methods, systems, and media for presenting recommended media content items

Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.

filed 2013-12-05, granted 2017-01-24 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo

US20150006290A1 Providing information to a user based on determined user activity

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2013-12-06 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate

US9116963B2 Systems and methods for promoting personalized search results based on personal information

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2013-12-06, granted 2015-08-25 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US9471671B1 Identifying and/or recommending relevant media content
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2013-12-18, granted 2016-10-18 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US10275464B2 Media consumption history
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-12-19, granted 2019-04-30 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US9219736B1 Application programming interface for rendering personalized related content to third party applications
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-12-20, granted 2015-12-22 · leak modules: PerDocData, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US9460221B2 History of reading positions in eBooks
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2013-12-20, granted 2016-10-04 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality

US9535897B2 Content recommendation system using a neural network language model
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-12-20, granted 2017-01-03 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations

US9563695B2 Personalized electronic magazine
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-12-26, granted 2017-02-07 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate

US10198442B2 Media consumption history
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.
filed 2013-12-27, granted 2019-02-05 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US9235627B1 Modifying search result ranking based on implicit user feedback
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-12-30, granted 2016-01-12 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US9043259B2 Systems and methods for recommending media content items
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-03-04, granted 2015-05-26 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9298763B1 Methods for providing a profile completion recommendation module
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-03-05, granted 2016-03-29 · leak modules: ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9542697B1 Customized landing pages
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2014-03-17, granted 2017-01-10 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US9477709B2 Query response using media consumption history
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-03-18, granted 2016-10-25 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US9015192B1 Systems and methods for improved processing of personalized message queries
Rule: Structure answers: question heading, tight answer, then depth.
filed 2014-03-19, granted 2015-04-21 · leak modules: QualityPreviewSnippetRadishFeatures, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US9436757B1 Generating a graph for a user profile
Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately. Create original, effortful content. Quality is scored at page and site level.
filed 2014-03-28, granted 2016-09-06 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityNsrPQData, SmartphonePerDocData

US9979777B1 Scoring stream items with models based on user interests
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-04-03, granted 2018-05-22 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US20150293916A1 Dynamic and Personalized Filtering of Media Content
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.
filed 2014-04-15 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9355080B2 Propagating user feedback on shared posts
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Structure answers: question heading, tight answer, then depth.
filed 2014-04-24, granted 2016-05-31 · leak modules: PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SnippetExtraInfo

US9585229B2 Anticipatory lighting from device screens based on user profile
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Optimize video for watch time and retention.
filed 2014-05-13, granted 2017-02-28 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData, VideoPerDocData

US9703838B1 Multi sender and source recommendation aggregation and prompting system
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2014-05-13, granted 2017-07-11 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US9906621B2 Providing language recommendations
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Build content for diverse user contexts and search histories.
filed 2014-06-03, granted 2018-02-27 · leak modules: NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9177063B2 Endorsing search results
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Optimize mobile experience. Mobile-specific signals are tracked separately. Build content for diverse user contexts and search histories.
filed 2014-06-13, granted 2015-11-03 · leak modules: PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, SmartphonePerDocData

US20150363502A1 Optimizing personalized recommendations with longitudinal data and a future objective

Rule: Use consistent entity names, sameAs schema, and structured data. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2014-06-16 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US20190286745A1 Community-based recommendations

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2014-06-16 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US9098599B1 Query suggestions for a document based on user history

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2014-06-18, granted 2015-08-04 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsGeoTopicality

US9984174B2 Speculative actions based on user dwell time over selectable content

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2014-07-31, granted 2018-05-29 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9779159B2 Customer support solution recommendation system

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2014-08-11, granted 2017-10-03 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11157572B1 Sharing user activity data with other users

Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2014-08-12, granted 2021-10-26 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations

US9489462B1 Personalizing search queries based on user membership in social network communities

Rule: Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately. Build content for diverse user contexts and search histories.

filed 2014-08-25, granted 2016-11-08 · leak modules: NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US9679067B2 Personalized network searching

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2014-10-16, granted 2017-06-13 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9378247B1 Generating query refinements from user preference data

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2014-12-17, granted 2016-06-28 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US10223459B2 Methods, systems, and media for personalizing computerized services based on mood and/or behavior information from multiple data sources

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2015-02-11, granted 2019-03-05 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9734211B1 Personalizing search results
Rule: This continuation patent confirms that Google personalizes rankings via preference-transmitting links. Earn backlinks from pages that are both authoritative (in the high-quality set) AND frequently visited/preferred by your target audience. Avoid backlinks from distrusted sources, as negative preference weights can demote your rankings. Build content that users repeatedly visit and engage with, as browsing history creates positive preference signals that propagate through the link graph to your search results.
filed 2015-02-27, granted 2017-08-15 · leak modules: PerDocData, QualityNsrNsrData, KaltixPerDocData, QualityNavboostCrapsCrapsClickSignals

US10127311B2 Personalized content sharing
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2015-03-05, granted 2018-11-13 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US20190012302A1 Annotations of textual segments based on user feedback
Rule: Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2015-03-06 · leak modules: NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, RichsnippetsPageMap

US9953047B2 Refining location estimates and reverse geocoding based on a user profile
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2015-03-16, granted 2018-04-24 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9952748B1 Contextual recommendations based on interaction within collections of content
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-03-26, granted 2018-04-24 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US9424320B2 Methods, systems, and media for recommending media content
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-04-23, granted 2016-08-23 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US9633312B1 Predicting user interests
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2015-04-28, granted 2017-04-25 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9454583B1 Analyzing user profiles
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-05-15, granted 2016-09-27 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate

US9438961B2 Providing a program listing
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-05-26, granted 2016-09-06 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US20160371737A1 Personalized and contextual notifications of content releases
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-06-18 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9552555B1 Methods, systems, and media for recommending content items based on topics
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2015-08-03, granted 2017-01-24 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US9984075B2 Media consumption context for personalized instant query suggest
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-10-06, granted 2018-05-29 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US20160026651A1 Bookmark-Based Access to Content Feeds
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use descriptive alt text and surrounding context for images.
filed 2015-10-07 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityWebanswersVideoTranscriptAnnotations

US9705999B1 Application programming interface for rendering personalized related content to third party applications
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2015-12-11, granted 2017-07-11 · leak modules: PerDocData, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US10185486B2 Method for one-click subscribing to multiple channels of information on a single topic
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2016-01-05, granted 2019-01-22 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US9811566B1 Modifying search result ranking based on implicit user feedback
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2016-01-11, granted 2017-11-07 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US10681155B1 Presenting user activity timeline in a colloquial style
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2016-01-12, granted 2020-06-09 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US9952752B1 Determining intent of a recommendation on a URL of a web page or advertisement
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2016-02-19, granted 2018-04-24 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US10025868B1 Preferred sites
Rule: Users can explicitly mark your site as a 'preferred site,' which gives you a ranking boost and visual distinction in their search results. Encourage users to bookmark/return to your site—Google uses visit history to recommend your domain as a preferred site. The locally relevant host requirement means your subdomain/URL structure should match the user's locale and language. The leak's BiasingPerDocData and BiasingPerDocData2 modules confirm per-document biasing based on user preferences is live. Build brand loyalty to increase the chance users accept your site as preferred.
filed 2016-03-15, granted 2018-07-17 · leak modules: BiasingPerDocData, BiasingPerDocData2, PerDocData, QualityNsrNsrData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, SdrPageAnchorsSitelink

US10152520B1 Ranking documents based on user behavior and/or feature data
Rule: Not all links are weighted equally - Google models the click probability of links based on their features (anchor text, surrounding context, source page quality). Links in prominent positions with descriptive anchor text and relevant surrounding context get more weight. Build links from pages where users are likely to click them. Avoid links in footers, sidebars, or low-visibility positions.
filed 2016-03-30, granted 2018-12-11 · leak modules: demotion, link_analysis, navboost, ranking, anchor_text, click_model

US10073915B1 Personalized search results
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2016-04-01, granted 2018-09-11 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US10346416B2 User experience and user flows for third-party application recommendation in cloud storage systems

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2016-04-14, granted 2019-07-09 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9565477B2 Bookmarking prospective media content on computer network

Rule: Structure answers: question heading, tight answer, then depth. Optimize video for watch time and retention.

filed 2016-04-14, granted 2017-02-07 · leak modules: QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo, VideoPerDocData

US10303779B2 Media consumption history

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2016-06-06, granted 2019-05-28 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US10289732B2 Server-based conversion of autoplay content to click-to-play content

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2016-06-13, granted 2019-05-14 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US10152521B2 Resource recommendations for a displayed resource

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-06-22, granted 2018-12-11 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US10241772B1 Recommending substitute applications

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2016-07-07, granted 2019-03-26 · leak modules: PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US10423661B2 Methods, systems, and media for providing a media consumption history annotated with social context

Rule: Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately. Optimize video for watch time and retention.

filed 2016-08-08, granted 2019-09-24 · leak modules: NlpSaftDocument, PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData, VideoPerDocData

US9852192B2 Methods, systems, and media for recommending media content

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-08-22, granted 2017-12-26 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US9973826B2 Providing a program listing

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2016-09-02, granted 2018-05-15 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US10860639B2 Query response using media consumption history

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2016-09-16, granted 2020-12-08 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US10169711B1 Generalized engine for predicting actions

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.

filed 2016-09-27, granted 2019-01-01 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals

US9971839B1 Personalizing search queries based on user membership in social network communities
Rule: Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately. Build content for diverse user contexts and search histories.
filed 2016-10-06, granted 2018-05-15 · leak modules: NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US10242006B2 Identifying and/or recommending relevant media content
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2016-10-14, granted 2019-03-26 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US20170031907A1 Indentifying and/or recommending relevant media content
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2016-10-14 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US10509827B1 Allowing users to categorize and visualize content recommendations
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2016-11-09, granted 2019-12-17 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10387559B1 Template-based identification of user interest
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2016-11-22, granted 2019-08-20 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US10430479B1 Method and system for providing dynamic personalized recommendations for a destination
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2016-11-22, granted 2019-10-01 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo

US9934515B1 Content recommendation system using a neural network language model
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2016-12-06, granted 2018-04-03 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations

US9785979B2 Customized landing pages
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2016-12-15, granted 2017-10-10 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US11023542B2 Methods, systems, and media for presenting recommended media content items
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2016-12-19, granted 2021-06-01 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo

US10467300B1 Topical resource recommendations for a displayed resource
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2016-12-22, granted 2019-11-05 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US10489459B1 Query recommendations for a displayed resource

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2016-12-22, granted 2019-11-26 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US10607273B2 System for determining and displaying relevant explanations for recommended content

Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.

filed 2016-12-28, granted 2020-03-31 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US20170161818A1 Explanations for personalized recommendations

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build content for diverse user contexts and search histories.

filed 2017-02-22 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, VideoPerDocData

US20180285448A1 Producing personalized selection of applications for presentation on web-based interface

Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.

filed 2017-04-04 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo

US10832130B2 Recommending a document for a user to access

Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2017-04-05, granted 2020-11-10 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData

US10398898B2 Personalized network searching

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2017-04-20, granted 2019-09-03 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20180336645A1 Using machine learning to recommend live-stream content

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2017-05-22 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US10372778B2 Speculative actions based on user dwell time over selectable content

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2017-06-26, granted 2019-08-06 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US10216808B1 Multi sender and source recommendation aggregation and prompting system

Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.

filed 2017-07-10, granted 2019-02-26 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US10673964B2 Application programming interface for rendering personalized related content to third party applications

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2017-07-10, granted 2020-06-02 · leak modules: PerDocData, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US10089387B1 Content recommendations based on organic keyword analysis

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2017-08-15, granted 2018-10-02 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US10462259B2 Providing digital content based on expected user behavior
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2017-08-15, granted 2019-10-29 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US9836766B1 Customized landing pages
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2017-08-21, granted 2017-12-05 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US10445351B2 Customer support solution recommendation system
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2017-08-31, granted 2019-10-15 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10474688B2 System and method to recommend a bundle of items based on item/user tagging and co-install graph
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-10-06, granted 2019-11-12 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US10229166B1 Modifying search result ranking based on implicit user feedback
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-10-25, granted 2019-03-12 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US10140375B2 Personalized network searching
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2017-12-26, granted 2018-11-27 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10853432B2 Methods and systems for generating search results and recommendations based on multi-sourced two-way correspondence and relative entity prominence
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2018-02-10, granted 2020-12-01 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US10878048B2 Methods and systems for generating search results and recommendations based on multi-sourced two-way correspondence and relative entity prominence
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2018-02-10, granted 2020-12-29 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US11314715B2 Refining location estimates and reverse geocoding based on a user profile
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Build content for diverse user contexts and search histories.
filed 2018-03-01, granted 2022-04-26 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US10444957B1 Determining intent of a recommendation on a URL of a web page or advertisement
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2018-03-21, granted 2019-10-15 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US10706115B1 Personalizing search queries based on user membership in social network communities
Rule: Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately. Build content for diverse user contexts and search histories.
filed 2018-04-09, granted 2020-07-07 · leak modules: NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US10656804B1 Contextual recommendations based on interaction within collections of content
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2018-04-13, granted 2020-05-19 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US10542322B2 Providing a program listing
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2018-05-02, granted 2020-01-21 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US10503814B1 Click tracking using link styles
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2018-06-25, granted 2019-12-10 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US11526569B2 Generating directions and routes with personalized landmarks
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2018-08-24, granted 2022-12-13 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US20200250251A1 Personalized Landmarks
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2018-08-24 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US10691765B1 Personalized search results
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2018-09-10, granted 2020-06-23 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US11135426B2 Personalized network searching
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2018-11-09, granted 2021-10-05 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11147970B2 Personalized network searching
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2018-11-09, granted 2021-10-19 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11369792B2 Personalized network searching
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2018-11-09, granted 2022-06-28 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11061963B2 Personalized content sharing
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2018-11-12, granted 2021-07-13 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US11669431B2 Analytics personalization framework
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2019-01-11, granted 2023-06-06 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US11853346B2 Media consumption history
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.
filed 2019-01-31, granted 2023-12-26 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US20200272953A1 Using a status centric view for alternative recommendations

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2019-02-27 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US12050655B2 Methods, systems, and media for personalizing computerized services based on mood and/or behavior information from multiple data sources

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2019-03-04, granted 2024-07-30 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US11188544B1 Modifying search result ranking based on implicit user feedback

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2019-03-11, granted 2021-11-30 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US11269953B2 Server-based conversion of autoplay content to click-to-play content

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2019-04-04, granted 2022-03-08 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US11574028B2 Annotation and retrieval of personal bookmarks

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2019-04-16, granted 2023-02-07 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo

US11816141B2 Media consumption history

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2019-04-24, granted 2023-11-14 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US11194866B2 Low entropy browsing history for content quasi-personalization

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs.

filed 2019-08-08, granted 2021-12-07 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US12056186B2 Methods, systems, and media for providing a media consumption history annotated with social context

Rule: Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately. Optimize video for watch time and retention.

filed 2019-09-23, granted 2024-08-06 · leak modules: NlpSaftDocument, PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData, VideoPerDocData

US11354372B2 Method and system for providing dynamic personalized recommendations for a destination

Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.

filed 2019-09-24, granted 2022-06-07 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo

US10997358B1 Click tracking using link styles

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2019-11-06, granted 2021-05-04 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US11423441B2 Low entropy browsing history for ads quasi-personalization
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2019-11-27, granted 2022-08-23 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US11102552B2 Providing a program listing
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2020-01-17, granted 2021-08-24 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US11539805B2 Application programming interface for rendering personalized related content to third party applications
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2020-06-01, granted 2022-12-27 · leak modules: PerDocData, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US11379490B2 Dynamic injection of related content in search results
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data.
filed 2020-06-08, granted 2022-07-05 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US20230214425A1 Node Embedding via Hash-Based Projection of Transformed Personalized PageRank
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2020-09-24 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US11960526B2 Query response using media consumption history
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2020-11-09, granted 2024-04-16 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US20220415319A1 On-device generation and personalization of zero-prefix suggestion(s) and use thereof
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2021-06-28 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11947594B2 Personalized content sharing
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2021-07-12, granted 2024-04-02 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US11687597B2 Low entropy browsing history for content quasi-personalization
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2021-07-21, granted 2023-06-27 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US11966432B2 Media consumption context for personalized instant query suggest
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2021-08-23, granted 2024-04-23 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US11979640B2 Providing a program listing
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2021-08-23, granted 2024-05-07 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US11574013B1 Query recommendations for a displayed resource
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2021-09-10, granted 2023-02-07 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US12001782B2 Personalized autonomous spreadsheets
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2021-09-24, granted 2024-06-04 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate

US11816114B1 Modifying search result ranking based on implicit user feedback
Rule: Optimize for voice search and audio consumption — content that performs well when read aloud (clear structure, concise answers, natural language) will receive better implicit feedback signals. Ensure content is accessible and engaging when consumed via screen readers or voice assistants, as listening duration now feeds into ranking.
filed 2021-11-23, granted 2023-11-14 · leak modules: QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US11995128B2 Low entropy browsing history for content quasi-personalization
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2021-11-29, granted 2024-05-28 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US20240200960A1 Using historical user routes to recommend navigational routes
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2022-02-22 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US12086127B2 Refining location estimates and reverse geocoding based on a user profile
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Build content for diverse user contexts and search histories.
filed 2022-04-14, granted 2024-09-10 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US11420059B1 Personalized network searching
Rule: Encourage users to favorite/bookmark your content across Google services. Personalized search objects (favorites) are searched alongside the general index — favorited content gets preferential placement in personalized results. Make content easily savable and re-engaging to drive favorite additions.
filed 2022-05-09, granted 2022-08-23 · leak modules: QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsFeatureCrapsData, PerDocData, KaltixPerDocData

US11547853B2 Personalized network searching
Rule: Build cross-device user journeys that encourage favoriting/saving content. Google synchronizes favorites across devices and uses them in personalized search results from multiple global indices. Content saved on one device appears in personalized results on other devices the user owns.
filed 2022-05-09, granted 2023-01-10 · leak modules: QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsFeatureCrapsData, PerDocData, KaltixPerDocData, IndexingDupsLocalizedLocalizedCluster

US11954705B2 Low entropy browsing history for ads quasi-personalization
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2022-08-22, granted 2024-04-09 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US20240303692A1 System And Method For Personalized Banner Placement
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2022-09-20 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US12468740B2 Category recommendation with implicit item feedback
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2022-12-07, granted 2025-11-11 · leak modules: CompressedQualitySignals, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US12531847B2 User data and personalization models implemented on a blockchain
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2022-12-07, granted 2026-01-20 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US12321404B2 Privacy preserving recommendation system
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2022-12-08, granted 2025-06-03 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US12229695B2 Content provider recommendations to improve targetting and other settings
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2023-04-28, granted 2025-02-18 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US12158916B2 Low entropy browsing history for content quasi-personalization
Rule: Create content that appeals to coherent audience clusters. Google groups users by dimensionally-reduced browsing history — content that resonates with specific cluster profiles gets quasi-personalized distribution. Understand your audience's broader browsing patterns to align with their cluster's content preferences.
filed 2023-08-25, granted 2024-12-03 · leak modules: QualityNavboostCrapsCrapsData, QualityNavboostCrapsFeatureCrapsData, PerDocData, KaltixPerDocData, CompressedQualitySignals

US12210562B2 Media consumption history
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2023-10-04, granted 2025-01-28 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US20250190502A1 Proactive feature outage detection
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2023-12-11 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US12405993B2 Media consumption history
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.
filed 2023-12-20, granted 2025-09-02 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US20250240486A1 Television based consumer or travel recommendations in a media platform
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2024-01-24 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US12511326B2 Keyword filtering for digital content recommendation
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2024-02-22, granted 2025-12-30 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US12481699B2 Personalized content sharing
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2024-04-01, granted 2025-11-25 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US12536221B2 Query response using media consumption history
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2024-04-04, granted 2026-01-27 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US20240378636A1 Asset Audience Gap Recommendation and Insight
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2024-05-09 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US12493738B2 Personalized autonomous spreadsheets
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2024-06-03, granted 2025-12-09 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate

US20250328568A1 Content-Based Feedback Recommendation Systems and Methods
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2025-06-27 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US20260105087A1 Recommendation System Using a Language Model Neural Network
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2025-10-10 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

Social & collaboration (253)

US7359894B1 Methods and systems for requesting and providing information in a social network
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2004-06-30, granted 2008-04-15 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality

US8015019B1 Methods and systems for providing a document
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2004-08-03, granted 2011-09-06 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US7962461B2 Method and system for finding and aggregating reviews for a product
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2004-12-14, granted 2011-06-14 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US7610287B1 System and method for impromptu shared communication spaces
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2005-06-28, granted 2009-10-27 · leak modules: PerDocData, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US8438469B1 Embedded review and rating information
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data.
filed 2005-09-30, granted 2013-05-07 · leak modules: CrawlerChangerateUrlChangerate, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata

US7584268B2 Collaborative web page authoring
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2006-01-31, granted 2009-09-01 · leak modules: CompressedQualitySignals, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8862572B2 Sharing user distributed search results
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2006-03-03, granted 2014-10-14 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8307119B2 Collaborative online spreadsheet application
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth. Write for natural spoken language and conversational queries.
filed 2006-03-31, granted 2012-11-06 · leak modules: NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US7844603B2 Sharing user distributed search results
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2006-05-30, granted 2010-11-30 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8122019B2 Sharing user distributed search results
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2006-05-30, granted 2012-02-21 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US9031964B2 Shared geo-located objects
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2007-04-25, granted 2015-05-12 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8595295B2 Method and system for determining and sharing a user's web presence
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2007-07-02, granted 2013-11-26 · leak modules: PerDocData, SmartphonePerDocData

US7945861B1 Initiating communications with web page visitors and known contacts
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2007-09-04, granted 2011-05-17 · leak modules: NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US8924465B1 Content sharing based on social graphing
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2007-11-06, granted 2014-12-30 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9305087B2 Method and apparatus for acquiring content-based capital via a sharing technology
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2007-12-20, granted 2016-04-05 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US8713429B2 Generating wiki pages from content and transformation objects
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2008-06-30, granted 2014-04-29 · leak modules: CompressedQualitySignals, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US9129008B1 Sentiment-based classification of media content
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2008-11-10, granted 2015-09-08 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US8656266B2 Identifying comments to show in connection with a document
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2008-12-18, granted 2014-02-18 · leak modules: NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefEntityJoin, SmartphonePerDocData

US9100438B2 Shared identity profile management
Rule: Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately. Build content for diverse user contexts and search histories.
filed 2009-03-24, granted 2015-08-04 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US8423610B2 User interface for web comments
Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2009-08-12, granted 2013-04-16 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US8645363B2 Spreading comments to other documents
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2009-08-12, granted 2014-02-04 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US8745067B2 Presenting comments from various sources
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2009-08-12, granted 2014-06-03 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US8756519B2 Techniques for sharing content on a web page
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2009-09-11, granted 2014-06-17 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData

US20110078232A1 Dynamic action links for web content sharing
Rule: Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships.
filed 2009-09-30 · leak modules: NlpSaftDocument, PerDocData, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, RichsnippetsPageMap, SmartphonePerDocData

US9490991B2 System and method for impromptu shared communication spaces
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2009-10-27, granted 2016-11-08 · leak modules: PerDocData, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US8689117B1 Webpages with conditional content
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2009-10-30, granted 2014-04-01 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData, SnippetExtraInfo

US8738634B1 Generating contact suggestions
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2010-04-16, granted 2014-05-27 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8694313B2 Disambiguation of contact information using historical data
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2010-05-19, granted 2014-04-08 · leak modules: NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US8626865B1 Shared worker application
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2010-06-07, granted 2014-01-07 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9032289B1 Providing suggestions to users to write comments
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2010-06-16, granted 2015-05-12 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US9336310B2 Monitoring of negative feedback systems
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Optimize mobile experience. Mobile-specific signals are tracked separately. Avoid AI-stitched content. GibberishScore uses language model scoring.
filed 2010-07-06, granted 2016-05-10 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, SmartphonePerDocData, SpamBrainData

US8670597B2 Facial recognition with social network aiding
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2010-08-05, granted 2014-03-11 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation

US8560605B1 Social affinity on the web
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2010-10-21, granted 2013-10-15 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8849810B2 Sharing user distributed search results
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2010-10-21, granted 2014-09-30 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US9432242B1 Display of social content
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2010-11-05, granted 2016-08-30 · leak modules: PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US8886643B2 Presenting social search results
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2010-12-01, granted 2014-11-11 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US8214373B1 Systems and methods for assignment of human reviewers using probabilistic prioritization
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-02-18, granted 2012-07-03 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8407226B1 Collaborative filtering
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-03-02, granted 2013-03-26 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8839120B2 Initiating communications with web page visitors and known contacts
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-03-11, granted 2014-09-16 · leak modules: NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US8468201B2 Organizing social activity information on a site
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-03-23, granted 2013-06-18 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US9166806B2 Shared communication space invitations
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-03-24, granted 2015-10-20 · leak modules: PerDocData, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US9002935B1 Copying document content through a hosted system
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2011-05-04, granted 2015-04-07 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsGeoTopicality, RepositoryWebrefEntityJoin, SmartphonePerDocData

US8161030B2 Method and system for aggregating reviews and searching within reviews for a product
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-05-05, granted 2012-04-17 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US20150142571A1 System and method for increasing the likelihood of users reviewing advertisements
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth. Ensure content comprehensively covers the topic with natural vocabulary.
filed 2011-05-23 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8639706B1 Shared metadata for media files
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2011-07-01, granted 2014-01-28 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US8719177B2 Methods and systems for providing a document
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-07-28, granted 2014-05-06 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8495058B2 Filtering social search results
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-08-05, granted 2013-07-23 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US8782761B1 Generating authentication challenges based on preferences of a user's contacts
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-08-08, granted 2014-07-15 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20130055056A1 System and Method for Searching Belongings Using Social Graph Information
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.
filed 2011-08-30 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo

US8661124B2 Organizing social activity information on a site
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-09-26, granted 2014-02-25 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US8250632B1 Generating authentication challenges based on preferences of a user's contacts
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-09-27, granted 2012-08-21 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8280821B1 Methods and systems for providing a document
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-09-27, granted 2012-10-02 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8291014B2 User interface for web comments
Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2011-09-30, granted 2012-10-16 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US8386250B2 Disambiguation of contact information using historical data
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-09-30, granted 2013-02-26 · leak modules: NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US9092529B1 Social search endorsements
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use descriptive alt text and surrounding context for images.
filed 2011-11-11, granted 2015-07-28 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US8832233B1 Experience sharing for conveying communication status
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Optimize video for watch time and retention.
filed 2011-12-08, granted 2014-09-09 · leak modules: NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US8934015B1 Experience sharing
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-12-08, granted 2015-01-13 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US9015245B1 Experience sharing with commenting
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2011-12-08, granted 2015-04-21 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US8832570B1 Representing a group identity in a social network
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-12-22, granted 2014-09-09 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8732241B2 Following content providers in a social context
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-12-29, granted 2014-05-20 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9436762B1 Sharing a plug-in instance in a web client
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-01-03, granted 2016-09-06 · leak modules: PerDocData, SmartphonePerDocData

US9294579B2 Centralized registration for distributed social content services
Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-01-04, granted 2016-03-22 · leak modules: ImageData, ImagePerDocData, PerDocData, SmartphonePerDocData

US9406090B1 Content sharing system
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-01-09, granted 2016-08-02 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US9015149B2 Sharing user distributed search results
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2012-01-13, granted 2015-04-21 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8903852B1 Experience sharing system and method
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-01-31, granted 2014-12-02 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US8972398B1 Integrating online search results and social networks
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-02-27, granted 2015-03-03 · leak modules: NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US9106602B1 Differentiated display of messages on social networks
Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2012-02-27, granted 2015-08-11 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8380694B2 Method and system for aggregating reviews and searching within reviews for a product
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-02-28, granted 2013-02-19 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US8825658B1 Organizing indications of approval for collections
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-03-27, granted 2014-09-02 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US8914472B1 Experience sharing for training
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-04-04, granted 2014-12-16 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations

US9275419B1 Method for building, expanding or complementing a social graph based on contact information
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-04-30, granted 2016-03-01 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US8913797B1 Techniques for determining a social distance between two people
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-05-09, granted 2014-12-16 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US8938500B1 Retrieving social network content
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency.
filed 2012-05-09, granted 2015-01-20 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals

US8321364B1 Method and system for including robots into social networks
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-05-14, granted 2012-11-27 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9256343B1 Dynamically modifying an electronic article based on commentary
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-05-14, granted 2016-02-09 · leak modules: IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations

US8799276B1 Displaying social content in search results
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2012-05-30, granted 2014-08-05 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData

US20130325846A1 Latent collaborative retrieval
Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately. Build content for diverse user contexts and search histories.
filed 2012-06-01 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData

US8959083B1 Searching using social context
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-06-13, granted 2015-02-17 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8688450B2 Disambiguation of contact information using historical and context data
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-07-10, granted 2014-04-01 · leak modules: NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US9286263B1 System and method for providing information regarding posts on social networking sites
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-07-12, granted 2016-03-15 · leak modules: PerDocData, SmartphonePerDocData

US9245288B1 Experience sharing for a registry event
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.
filed 2012-07-19, granted 2016-01-26 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefDocumentMetadata

US9436686B1 Claim evaluation system
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-08-07, granted 2016-09-06 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US20150199307A1 Pluggable Architecture For Optimizing Versioned Rendering of Collaborative Documents
Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships.
filed 2012-08-08 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US8924493B1 Prioritized notifications in a social network
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2012-08-08, granted 2014-12-30 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US9083670B1 Notifications about users in a social network
Rule: Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately. Optimize video for watch time and retention.
filed 2012-08-15, granted 2015-07-14 · leak modules: PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData, VideoPerDocData

US11200287B2 Global address list
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-08-23, granted 2021-12-14 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US8997072B1 Compressing dependency graphs in a social network
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-08-23, granted 2015-03-31 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9471682B1 Providing information associated with a profile owner in a social network system
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-08-28, granted 2016-10-18 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, ShingleInfoPerDocData

US9229938B1 System and method for suggesting media content contributions for a collaborative playlist
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-08-31, granted 2016-01-05 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, RepositoryWebrefEntityJoin, SmartphonePerDocData, VideoPerDocData

US8886737B1 Identifying particular parties
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-09-06, granted 2014-11-11 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US9244985B1 Generating search results for people
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency.
filed 2012-09-06, granted 2016-01-26 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals

US9519683B1 Inferring social affinity based on interactions with search results
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2012-09-06, granted 2016-12-13 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData

US9904703B1 Determining content of interest based on social network interactions and information
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-09-06, granted 2018-02-27 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, VideoPerDocData

US9959348B2 Applying social annotations to search results
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Optimize mobile experience. Mobile-specific signals are tracked separately. Create original, effortful content. Quality is scored at page and site level.
filed 2012-09-06, granted 2018-05-01 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US8447886B2 Collaborative online spreadsheet application
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Write for natural spoken language and conversational queries.
filed 2012-09-14, granted 2013-05-21 · leak modules: NlpSaftDocument, PerDocData, SmartphonePerDocData

US8756164B1 Methods and systems for providing a document
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-09-14, granted 2014-06-17 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8762286B1 Methods and systems for providing a document
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-09-14, granted 2014-06-24 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9195969B2 Presenting quick list of contacts to communication application user
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-09-14, granted 2015-11-24 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9425971B1 System and method for impromptu shared communication spaces
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-09-14, granted 2016-08-23 · leak modules: PerDocData, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US9477720B1 Social search endorsements
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use descriptive alt text and surrounding context for images.
filed 2012-09-15, granted 2016-10-25 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US8977654B1 Assigning classes to users of an online community
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Structure answers: question heading, tight answer, then depth.
filed 2012-09-21, granted 2015-03-10 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US9489458B1 Suggesting interaction among members of a social network
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-09-28, granted 2016-11-08 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US9411856B1 Overlay generation for sharing a website
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-10-01, granted 2016-08-09 · leak modules: CompressedQualitySignals, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US8856173B2 User engagement in a social network using indications of acknowledgement
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-10-04, granted 2014-10-07 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US8959166B1 System and method for identifying websites
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-10-04, granted 2015-02-17 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US9348921B2 Determining access to comments
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-10-04, granted 2016-05-24 · leak modules: PerDocData, SmartphonePerDocData

US8918404B1 Displaying content in a user interface in a social network system
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Structure answers: question heading, tight answer, then depth.
filed 2012-10-09, granted 2014-12-23 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SnippetExtraInfo

US8990701B2 Gathering and organizing content distributed via social media
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-10-11, granted 2015-03-24 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData

US9171095B1 Track sharing
Rule: Optimize video for watch time and retention.
filed 2012-10-12, granted 2015-10-27 · leak modules: QualityWebanswersVideoTranscriptAnnotations, VideoPerDocData

US9098819B1 Identifying social network accounts belonging to the same user
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Build content for diverse user contexts and search histories.
filed 2012-10-18, granted 2015-08-04 · leak modules: PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, VideoPerDocData

US8977649B1 Generating suggestions for user groups in social networks
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-10-19, granted 2015-03-10 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US9418370B2 Obtaining event reviews
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-10-23, granted 2016-08-16 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US9141617B1 Social lens for search
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-10-26, granted 2015-09-22 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US20160070770A1 Suggesting social groups from user social graphs
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-10-29 · leak modules: PerDocData, SmartphonePerDocData

US9177062B2 Sorting social profile search results based on computing personal similarity scores
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-10-31, granted 2015-11-03 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US9275421B2 Triggering social pages
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-11-01, granted 2016-03-01 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US9069856B1 Evaluating access control list data for a social network
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2012-11-02, granted 2015-06-30 · leak modules: CompressedQualitySignals, PerDocData, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US20130124504A1 Sharing Digital Content to Discovered Content Streams in Social Networking Services
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2012-11-14 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData

US8655871B2 Sharing of profile information with content providers
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-11-16, granted 2014-02-18 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US8983948B1 Providing electronic content based on a composition of a social network
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data.
filed 2012-12-26, granted 2015-03-17 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US20140189534A1 Customizing shared web resource snapshots
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-12-31 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US9509732B2 System and method for hosting and sharing a live event
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use descriptive alt text and surrounding context for images.
filed 2012-12-31, granted 2016-11-29 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData

US9183259B1 Selecting content based on social significance
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2013-01-11, granted 2015-11-10 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9401947B1 Methods, systems, and media for presenting comments based on correlation with content
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-02-08, granted 2016-07-26 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9280601B1 Modifying search results
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Optimize mobile experience. Mobile-specific signals are tracked separately. Write for natural spoken language and conversational queries.
filed 2013-02-14, granted 2016-03-08 · leak modules: NlpSaftDocument, PerDocData, RepositoryWebrefEntityJoin, SmartphonePerDocData

US9047345B1 Processing annotations for a search result
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-02-19, granted 2015-06-02 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US9146943B1 Determining user content classifications within an online community
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2013-02-26, granted 2015-09-29 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US20130325795A1 Systems and Methods for Dynamically Loading Web Fonts and Dynamic Availability for Fonts Based on Language Settings
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2013-03-04 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20160246789A1 Searching content of prominent users in social networks
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-03-07 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US9934283B2 Social annotations for enhanced search results
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-03-08, granted 2018-04-03 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US10061851B1 Encouraging inline person-to-person interaction
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2013-03-12, granted 2018-08-28 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData

US9830360B1 Determining content classifications using feature frequency
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Avoid AI-stitched content. GibberishScore uses language model scoring.
filed 2013-03-12, granted 2017-11-28 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10346411B1 Automatic target audience suggestions when sharing in a social network
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2013-03-14, granted 2019-07-09 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20140122522A1 Method for providing users with help from their contacts while searching the web
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-03-15 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality

US9294583B1 Updating event posts
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.
filed 2013-03-15, granted 2016-03-22 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9177048B1 Collaborative filtering
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2013-03-26, granted 2015-11-03 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US9081833B1 Providing a tooltip based on search results
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-04-04, granted 2015-07-14 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US8886662B1 Generating word completions based on shared suffix analysis
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2013-04-08, granted 2014-11-11 · leak modules: NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9031945B1 Sharing and using search results
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-04-15, granted 2015-05-12 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US9063920B2 Collaborative online spreadsheet application
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Write for natural spoken language and conversational queries.
filed 2013-05-20, granted 2015-06-23 · leak modules: NlpSaftDocument, PerDocData, SmartphonePerDocData

US9280533B2 Collaborative online spreadsheet application
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Write for natural spoken language and conversational queries.
filed 2013-05-20, granted 2016-03-08 · leak modules: NlpSaftDocument, PerDocData, SmartphonePerDocData

US9201928B2 Assessing quality of reviews based on online reviewer generated content
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-05-31, granted 2015-12-01 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9122756B2 On-line social search
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2013-06-06, granted 2015-09-01 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US9607024B2 Sharing information with other users
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-06-13, granted 2017-03-28 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US8630663B1 Labeling visited locations based on contact information
Rule: Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-07-19, granted 2014-01-14 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, RepositoryAnnotationsGeoTopicality

US9547711B1 Shard data based on associated social relationship
Rule: Monitor this patent for ranking implications.
filed 2013-07-22, granted 2017-01-17

US9495660B2 Inferring social groups through patterns of communication
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Avoid AI-stitched content. GibberishScore uses language model scoring.
filed 2013-07-31, granted 2016-11-15 · leak modules: ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, SpamBrainData

US9426190B1 Crowdsourcing descriptor selection
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-08-01, granted 2016-08-23 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures

US9612996B1 Providing suggestions to a user based on social activity patterns
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2013-09-11, granted 2017-04-04 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata

US8880608B1 Social affinity on the web
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-09-13, granted 2014-11-04 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9665657B2 Dynamically picking content from social shares to display in a user interface
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2013-09-24, granted 2017-05-30 · leak modules: IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US9558242B2 Social where next suggestion
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-10-15, granted 2017-01-31 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9824145B1 User experience in social networks by weighting user interaction patterns
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-10-18, granted 2017-11-21 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US9811535B1 Creating social network groups
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-10-29, granted 2017-11-07 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US8930460B2 Method and system for determining and sharing a user's web presence
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-11-08, granted 2015-01-06 · leak modules: PerDocData, SmartphonePerDocData

US8761812B2 Labeling visited locations based on contact information
Rule: Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-12-17, granted 2014-06-24 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, RepositoryAnnotationsGeoTopicality

US9152677B2 Shared metadata for media files
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2013-12-23, granted 2015-10-06 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US9521172B1 Method and system for sharing online content
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-12-31, granted 2016-12-13 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures

US8977644B2 Collaborative search results
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2014-01-16, granted 2015-03-10 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US9646063B1 Sharing of profile information with content providers
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2014-02-04, granted 2017-05-09 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US9208177B2 Facial recognition with social network aiding
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2014-02-20, granted 2015-12-08 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation

US9870403B1 Estimating social content interactions
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2014-02-26, granted 2018-01-16 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US10223470B1 Methods and systems for providing a document
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-04-08, granted 2019-03-05 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9311415B2 Generating contact suggestions
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-04-18, granted 2016-04-12 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US10049138B1 Reputation and engagement system for online community management
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-04-21, granted 2018-08-14 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate

US9256891B2 Following content providers in a social context
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2014-04-22, granted 2016-02-09 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US10255281B2 Methods and systems for providing a document
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-05-07, granted 2019-04-09 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9344852B2 Labeling visited locations based on contact information
Rule: Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-05-19, granted 2016-05-17 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, RepositoryAnnotationsGeoTopicality

US9418142B2 Overlapping community detection in weighted graphs
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-05-23, granted 2016-08-16 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata

US9253302B2 Populating user contact entries
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-06-04, granted 2016-02-02 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9129026B1 Displaying social content in search results
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2014-06-23, granted 2015-09-08 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData

US9276923B1 Generating authentication challenges based on preferences of a user's contacts
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-07-14, granted 2016-03-01 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9524352B1 Sharing data across partner websites
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-07-15, granted 2016-12-20 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US9342578B1 Organizing indications of approval for collections
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-07-28, granted 2016-05-17 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US9762651B1 Redaction suggestion for sharing screen content
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2014-08-21, granted 2017-09-12 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData

US9432376B2 Method and system for determining and sharing a user's web presence
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-12-12, granted 2016-08-30 · leak modules: PerDocData, SmartphonePerDocData

US9787485B1 Prioritized notifications in a social network for a user-selected view
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2014-12-24, granted 2017-10-10 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata

US9660951B1 Content sharing based on social graphing
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-12-29, granted 2017-05-23 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US9208228B1 Searching using social context
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-01-06, granted 2015-12-08 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US10592513B1 Multiple social streams
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-01-16, granted 2020-03-17 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9798815B1 Assigning classes to users of an online community
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Structure answers: question heading, tight answer, then depth.
filed 2015-02-04, granted 2017-10-24 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US10481762B2 Gathering and organizing content distributed via social media
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2015-02-05, granted 2019-11-19 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData

US9367864B2 Experience sharing with commenting
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2015-03-19, granted 2016-06-14 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US11429689B1 Generating high visibility social annotations
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2015-04-17, granted 2022-08-30 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US9418163B2 Shared geo-located objects
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-05-11, granted 2016-08-16 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9781072B2 Unifying web and phone presence
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2015-07-06, granted 2017-10-03 · leak modules: NlpSaftDocument, PerDocData, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US10284558B2 Systems and methods for managing privacy settings of shared content
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-08-12, granted 2019-05-07 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US9495425B1 Sentiment-based classification of media content
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2015-08-14, granted 2016-11-15 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US10339189B2 Identification of people in common
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-08-24, granted 2019-07-02 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US9870360B1 Shared metadata for media files
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2015-09-04, granted 2018-01-16 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US9418164B2 Shared geo-located objects
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-09-11, granted 2016-08-16 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US10031927B2 Facial recognition with social network aiding
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-11-02, granted 2018-07-24 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US10083468B2 Experience sharing for a registry event
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.
filed 2015-12-21, granted 2018-09-25 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations

US9865006B2 Following content providers in a social context
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2015-12-23, granted 2018-01-09 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9864748B1 System and method for suggesting media content contributions for a collaborative playlist
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2016-01-04, granted 2018-01-09 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, RepositoryWebrefEntityJoin, SmartphonePerDocData, VideoPerDocData

US9678958B2 Populating user contact entries
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2016-01-12, granted 2017-06-13 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US10019589B1 Generating search results for people
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency.
filed 2016-01-13, granted 2018-07-10 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals

US9934286B2 Generating contact suggestions
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2016-03-02, granted 2018-04-03 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US10412026B2 Updating event posts
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.
filed 2016-03-18, granted 2019-09-10 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9571984B2 Labeling visited locations based on contact information
Rule: Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2016-05-04, granted 2017-02-14 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, RepositoryAnnotationsGeoTopicality

US10115118B2 Obtaining event reviews
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2016-05-19, granted 2018-10-30 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US10911390B2 Methods, systems, and media for presenting comments based on correlation with content
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2016-06-22, granted 2021-02-02 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10127288B1 Display of social content
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2016-07-25, granted 2018-11-13 · leak modules: PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US10409818B1 Populating streams of content
Rule: Use consistent entity names, sameAs schema, and structured data. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2016-08-04, granted 2019-09-10 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US10534817B2 Sharing a process in a web client
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2016-09-02, granted 2020-01-14 · leak modules: PerDocData, SmartphonePerDocData

US9875244B1 Sentiment-based classification of media content
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2016-10-06, granted 2018-01-23 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US10095773B1 Processing a social endorsement for an item
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships.
filed 2016-10-13, granted 2018-10-09 · leak modules: PerDocData, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US10306010B2 Inferring social groups through patterns of communication
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Avoid AI-stitched content. GibberishScore uses language model scoring.
filed 2016-10-19, granted 2019-05-28 · leak modules: ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, SpamBrainData

US20170046312A1 Using content structure to socially connect users
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Structure answers: question heading, tight answer, then depth.
filed 2016-10-31 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefDocumentMetadata

US10129323B1 Sharing data across partner websites
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2016-11-30, granted 2018-11-13 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US9986386B2 Labeling visited locations based on contact information
Rule: Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-01-06, granted 2018-05-29 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, RepositoryAnnotationsGeoTopicality

US10445373B2 Social where next suggestion
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-01-09, granted 2019-10-15 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20180165368A1 Demographic Based Collaborative Filtering for New Users
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2017-01-13 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US10152549B1 Providing suggestions to a user based on social activity patterns
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2017-03-07, granted 2018-12-11 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata

US9785625B1 Increasing comment visibility
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2017-03-29, granted 2017-10-10 · leak modules: PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, VideoPerDocData

US10871881B2 Dynamically picking content from social shares to display in a user interface
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2017-04-20, granted 2020-12-22 · leak modules: IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US10009310B1 Content sharing based on social graphing
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2017-05-16, granted 2018-06-26 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US10162823B2 Populating user contact entries
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-05-17, granted 2018-12-25 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US20180082218A1 Tracking Feeds in a Social Network
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-11-28 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10748187B2 Following content providers in a social context
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2017-12-13, granted 2020-08-18 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US10198443B2 System and method for suggesting media content contributions for a collaborative playlist
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2017-12-19, granted 2019-02-05 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, RepositoryWebrefEntityJoin, SmartphonePerDocData, VideoPerDocData

US10191905B2 Sentiment-based classification of media content
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2017-12-21, granted 2019-01-29 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US11269963B1 Estimating social content interactions
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2018-01-12, granted 2022-03-08 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US10579244B2 Simplified sharing of content among computing devices
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2018-02-12, granted 2020-03-03 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10430597B1 Generating search results for people
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency.
filed 2018-05-01, granted 2019-10-01 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals

US10915587B2 Data processing system for generating entries in data structures from network requests
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2018-05-18, granted 2021-02-09 · leak modules: PerDocData, SmartphonePerDocData

US10469994B2 Labeling locations based on contact information
Rule: Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2018-05-27, granted 2019-11-05 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, RepositoryAnnotationsGeoTopicality

US10515114B2 Facial recognition with social network aiding
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2018-07-09, granted 2019-12-24 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US11423034B1 Display of social content
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2018-07-13, granted 2022-08-23 · leak modules: PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US11194818B1 Promoting social media content in search
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2018-08-17, granted 2021-12-07 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US11531808B2 Intelligent document notifications based on user comments
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2018-08-21, granted 2022-12-20 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData

US10698942B2 Sentiment-based classification of media content
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2019-01-28, granted 2020-06-30 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US10462144B2 Systems and methods for managing privacy settings of shared content
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2019-02-22, granted 2019-10-29 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US11301537B1 Methods and systems for providing a document

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2019-03-01, granted 2022-04-12 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10855641B1 Updating event posts

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.

filed 2019-07-24, granted 2020-12-01 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US11669553B2 Context-dependent shared dictionaries

Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.

filed 2019-12-06, granted 2023-06-06 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US11500530B2 Simplified sharing of content among computing devices

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2020-03-02, granted 2022-11-15 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10956482B2 Sentiment-based classification of media content

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2020-06-29, granted 2021-03-23 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US11941352B2 Collaborative online spreadsheet application

Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Write for natural spoken language and conversational queries.

filed 2020-08-10, granted 2024-03-26 · leak modules: NlpSaftDocument, PerDocData, SmartphonePerDocData

US11675859B2 Data processing system for generating entries in data structures from network requests

Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2021-01-28, granted 2023-06-13 · leak modules: PerDocData, SmartphonePerDocData

US11689491B2 Methods, systems, and media for presenting comments based on correlation with content

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2021-02-01, granted 2023-06-27 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11632250B2 Signatures of updates exchanged in a binary data synchronization protocol

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2021-02-25, granted 2023-04-18 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US11379512B2 Sentiment-based classification of media content

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2021-03-22, granted 2022-07-05 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US12093339B1 Methods and systems for providing a document

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2022-03-16, granted 2024-09-17 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11921809B2 Generating high visibility social annotations
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2022-08-29, granted 2024-03-05 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US11861153B2 Simplified sharing of content among computing devices
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2022-11-03, granted 2024-01-02 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11914947B2 Intelligent document notifications based on user comments
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2022-12-19, granted 2024-02-27 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData

US20240311558A1 Comment section analysis of a content sharing platform
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2023-03-13 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo

US12067064B2 Data processing system for generating entries in data structures from network requests
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2023-05-04, granted 2024-08-20 · leak modules: PerDocData, SmartphonePerDocData

US12277309B2 Simplified sharing of content among computing devices
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2023-07-17, granted 2025-04-15 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20250217028A1 Interactive whiteboard using artificial intelligence
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2023-12-28 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US12530418B2 Data processing system for generating entries in data structures from network requests
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2024-07-16, granted 2026-01-20 · leak modules: PerDocData, SmartphonePerDocData

US20260080285A1 Using bayesian inference to predict review decisions in a match graph
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2025-11-21 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

Voice & assistants (222)

US7421432B1 Hypertext browser assistant
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2000-12-13, granted 2008-09-02 · leak modules: NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8595214B1 Systems and methods for article location and retrieval
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2004-03-31, granted 2013-11-26 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US7437364B1 System and method of accessing a document efficiently through multi-tier web caching
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2004-06-30, granted 2008-10-14 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate

US7587398B1 System and method of accessing a document efficiently through multi-tier web caching
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2004-06-30, granted 2009-09-08 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata

US8676922B1 Automatic proxy setting modification
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2004-06-30, granted 2014-03-18 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9026511B1 Call connection via document browsing
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2005-06-29, granted 2015-05-05 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US20080071544A1 Integrating Voice-Enabled Local Search and Contact Lists
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2007-09-14 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate

US8560564B1 Hypertext browser assistant
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2008-07-24, granted 2013-10-15 · leak modules: NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8275790B2 System and method of accessing a document efficiently through multi-tier web caching
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2008-10-14, granted 2012-09-25 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate

US9081868B2 Voice web search
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Write for intent and meaning. Neural matching connects related concepts.
filed 2009-12-16, granted 2015-07-14 · leak modules: NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US8626511B2 Multi-dimensional disambiguation of voice commands
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2010-01-22, granted 2014-01-07 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US8521526B1 Disambiguation of a spoken query term
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2010-07-28, granted 2013-08-27 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8392411B2 Automatic routing of search results
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2010-08-06, granted 2013-03-05 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US9015043B2 Choosing recognized text from a background environment
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2010-10-01, granted 2015-04-21 · leak modules: CompressedQualitySignals, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US8533347B1 Non-modal dialog for managing blocking of multiple web browser cookies
Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-05-26, granted 2013-09-10 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData

US8312042B2 Automatic routing of search results
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-09-26, granted 2012-11-13 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US8316016B1 Hypertext browser assistant
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-09-26, granted 2012-11-20 · leak modules: NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8874730B1 Non-modal dialog for managing blocking of multiple web browser cookies
Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-09-26, granted 2014-10-28 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData

US9098606B1 Activity assistant
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-12-21, granted 2015-08-04 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US9177029B1 Determining activity importance to a user
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-12-21, granted 2015-11-03 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US9110852B1 Methods and systems for extracting information from text
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-07-20, granted 2015-08-18 · leak modules: IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, ShingleInfoPerDocData

US8515746B1 Selecting speech data for speech recognition vocabulary
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2012-08-24, granted 2013-08-20 · leak modules: NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefDocumentMetadata

US20130166526A1 Conversational question and answer
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-09-14 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8688687B1 Hypertext browser assistant
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-09-14, granted 2014-04-01 · leak modules: NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8856007B1 Use text to speech techniques to improve understanding when announcing search results

Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth. Write for intent and meaning. Neural matching connects related concepts.

filed 2012-10-15, granted 2014-10-07 · leak modules: NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US9679568B1 Training a dialog system using user feedback

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.

filed 2012-11-05, granted 2017-06-13 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US9529793B1 Resolving pronoun ambiguity in voice queries

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2013-02-22, granted 2016-12-27 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US8719281B2 Automatic dialing

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2013-03-04, granted 2014-05-06 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US9190054B1 Natural language refinement of voice and text entry

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.

filed 2013-03-13, granted 2015-11-17 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo

US9069825B1 Search dialogue user interface

Rule: Structure content to support conversational/multi-turn search flows. Ensure your pages answer progressive query refinements within a topic cluster. Since Google combines rapid successive queries, make sure related sub-topics on your site are interlinked so they can be surfaced across a search thread session.

filed 2013-03-15, granted 2015-06-30 · leak modules: QualityDialogManagerLocalIntentOptions, KnowledgeAnswersIntentQueryArgumentProvenancePreviousTaskState, KnowledgeAnswersIntentQueryFunctionCall, BlogsearchConversationNode, BlogsearchConversationTree, PerDocData

US9984684B1 Inducing command inputs from high precision and high recall data

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.

filed 2013-06-25, granted 2018-05-29 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US9418177B1 Disambiguation of a spoken query term

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2013-08-05, granted 2016-08-16 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US9959300B1 Systems and methods for article location and retrieval

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2013-10-24, granted 2018-05-01 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US9485140B2 Automatic proxy setting modification

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2014-02-04, granted 2016-11-01 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US20140229469A1 Automatic dialing of search results
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-04-23 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US10083697B2 Local persisting of data for selectively offline capable voice action in a voice-enabled electronic device
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-05-27, granted 2018-09-25 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US9870196B2 Selective aborting of online processing of voice inputs in a voice-enabled electronic device
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-05-27, granted 2018-01-16 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US9922138B2 Dynamically updatable offline grammar model for resource-constrained offline device
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-05-27, granted 2018-03-20 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US9483536B1 Search dialogue user interface
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-06-03, granted 2016-11-01 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US10504509B2 Providing suggested voice-based action queries
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-07-24, granted 2019-12-10 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US9798817B1 Activity assistant
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2015-07-29, granted 2017-10-24 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US9576578B1 Contextual improvement of voice query recognition
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-08-12, granted 2017-02-21 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US10885129B2 Using frames for action dialogs
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2015-10-13, granted 2021-01-05 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo

US10891106B2 Automatic batch voice commands
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Ensure content comprehensively covers the topic with natural vocabulary. Write for natural spoken language and conversational queries.
filed 2015-10-13, granted 2021-01-12 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityNsrNsrData, SmartphonePerDocData

US10133821B2 Search result prefetching of voice queries

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2016-01-06, granted 2018-11-20 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US9740751B1 Application keywords

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Write for intent and meaning. Neural matching connects related concepts.

filed 2016-02-18, granted 2017-08-22 · leak modules: NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US10354647B2 Correcting voice recognition using selective re-speak

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2016-04-28, granted 2019-07-16 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US11227017B2 Providing suggestions for interaction with an automated assistant in a multi-user message exchange thread

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2016-05-17, granted 2022-01-18 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US10311492B2 Enhancing functionalities of virtual assistants and dialog systems via plugin marketplace

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2016-05-27, granted 2019-06-04 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US10282218B2 Nondeterministic task initiation by a personal assistant module

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.

filed 2016-06-07, granted 2019-05-07 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US10346129B1 Gamifying voice search experience for children

Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-06-08, granted 2019-07-09 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityWebanswersVideoTranscriptAnnotations

US11232136B2 Contextual voice search suggestions

Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-06-27, granted 2022-01-25 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US10026398B2 Follow-up voice query prediction

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-07-08, granted 2018-07-17 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US10210267B1 Disambiguation of a spoken query term

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2016-08-10, granted 2019-02-19 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US10185840B2 Conditional disclosure of individual-controlled content in group contexts

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-08-30, granted 2019-01-22 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US10192551B2 Using textual input and user state information to generate reply content to present in response to the textual input

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2016-08-30, granted 2019-01-29 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US9767200B1 Search dialogue user interface

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2016-09-23, granted 2017-09-19 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US10026402B2 Business or personal listing search

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-10-03, granted 2018-07-17 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US10297254B2 Task initiation using long-tail voice commands by weighting strength of association of the tasks and their respective commands based on user feedback

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-10-03, granted 2019-05-21 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US10170106B2 Parameter collection and automatic dialog generation in dialog systems

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-10-21, granted 2019-01-01 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9911418B2 Systems and methods for speech command processing

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2016-11-08, granted 2018-03-06 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US10446144B2 Providing prompt in an automated dialog session based on selected content of prior automated dialog session

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2016-11-21, granted 2019-10-15 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US20180096072A1 Personalization of a virtual assistant

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-11-30 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate

US10552742B2 Proactive virtual assistant

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2016-12-16, granted 2020-02-04 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US10019434B1 Resolving pronoun ambiguity in voice queries

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-12-21, granted 2018-07-10 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US10224031B2 Generating and transmitting invocation request to appropriate third-party agent
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2016-12-30, granted 2019-03-05 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US10431209B2 Feedback controller for data transmissions
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2016-12-30, granted 2019-10-01 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US10630751B2 Sequence dependent data message consolidation in a voice activated computer network environment
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2016-12-30, granted 2020-04-21 · leak modules: IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US10699181B2 Virtual assistant generation of group recommendations
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2016-12-30, granted 2020-06-30 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US10924376B2 Selective sensor polling
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2016-12-30, granted 2021-02-16 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US10909199B2 Automatic routing using search results
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-02-08, granted 2021-02-02 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US10636418B2 Proactive incorporation of unsolicited content into human-to-computer dialogs
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.
filed 2017-03-22, granted 2020-04-28 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US11231943B2 Smart setup of assistant services
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2017-03-24, granted 2022-01-25 · leak modules: CompressedQualitySignals, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10666581B2 Instantiation of dialog process at a particular child node state
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-04-26, granted 2020-05-26 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US9865260B1 Proactive incorporation of unsolicited content into human-to-computer dialogs
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.
filed 2017-05-03, granted 2018-01-09 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US11170285B2 Virtual assistant configured to recommended actions in furtherance of an existing conversation

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.

filed 2017-05-05, granted 2021-11-09 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US10127227B1 Providing access to user-controlled resources by automated assistants

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2017-05-15, granted 2018-11-13 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10269351B2 Systems, methods, and apparatuses for resuming dialog sessions via automated assistant

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2017-05-16, granted 2019-04-23 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures

US10735552B2 Secondary transmissions of packetized data

Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2017-05-24, granted 2020-08-04 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US11017428B2 System and method of data transmission rate adjustment

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2017-05-24, granted 2021-05-25 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US10504521B1 Training a dialog system using user feedback for answers to questions

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2017-05-26, granted 2019-12-10 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US10275651B2 Resolving automated assistant requests that are based on image(s) and/or other sensor data

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2017-06-23, granted 2019-04-30 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US20190018827A1 Electronic content insertion systems and methods

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2017-07-12 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US11005993B2 Computational assistant extension device

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2017-07-14, granted 2021-05-11 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US10366291B2 Systems, methods, and apparatus for providing image shortcuts for an assistant application

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2017-09-09, granted 2019-07-30 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11017037B2 Obtaining responsive information from multiple corpora
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-09-13, granted 2021-05-25 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US11238116B2 Autonomously providing search results post-facto, including in assistant context
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.
filed 2017-09-29, granted 2022-02-01 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US10431219B2 User-programmable automated assistant
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-10-03, granted 2019-10-01 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US10613748B2 Stylus assist
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-10-03, granted 2020-04-07 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US10424302B2 Turn-based reinforcement learning for dialog management
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2017-10-12, granted 2019-09-24 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10929486B1 Activity assistant
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2017-10-23, granted 2021-02-23 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US10311856B2 Synthesized voice selection for computational agents
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data.
filed 2017-11-16, granted 2019-06-04 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US10558426B2 Graphical user interface rendering management by voice-driven computing infrastructure
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2017-12-08, granted 2020-02-11 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US10580412B2 Digital assistant processing of stacked data structures
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-12-08, granted 2020-03-03 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10552489B2 Dynamically updatable offline grammar model for resource-constrained offline device
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2018-02-04, granted 2020-02-04 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US10573298B2 Automated assistants that accommodate multiple age groups and/or vocabulary levels
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2018-04-16, granted 2020-02-25 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US10984784B2 Facilitating end-to-end communications with automated assistants in multiple languages
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2018-04-16, granted 2021-04-20 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US10803860B2 Dependency graph conversation modeling for use in conducting human-to-computer dialog sessions with a computer-implemented automated assistant
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2018-05-01, granted 2020-10-13 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo

US10963492B2 Generation of domain-specific models in networked system
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2018-06-14, granted 2021-03-30 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10635860B1 Resolving pronoun ambiguity in voice queries
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2018-06-15, granted 2020-04-28 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US10679624B2 Personal directory service
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2018-07-16, granted 2020-06-09 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US11455418B2 Assembling and evaluating automated assistant responses for privacy concerns
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2018-08-07, granted 2022-09-27 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US10334080B2 Local persisting of data for selectively offline capable voice action in a voice-enabled electronic device
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2018-08-28, granted 2019-06-25 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US10699710B2 Detecting and suppressing voice queries
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2018-11-21, granted 2020-06-30 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US10635832B2 Conditional disclosure of individual-controlled content in group contexts
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2018-11-28, granted 2020-04-28 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US11347801B2 Multi-modal interaction between users, automated assistants, and other computing services

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2019-01-04, granted 2022-05-31 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US11455176B2 Selecting content to render on display of assistant device

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2019-01-04, granted 2022-09-27 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US10714086B2 Generating and transmitting invocation request to appropriate third-party agent

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2019-01-10, granted 2020-07-14 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US11010428B2 Systems, methods, and apparatuses for providing assistant deep links to effectuate third-party dialog session transfers

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.

filed 2019-01-15, granted 2021-05-18 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US10650820B2 Using textual input and user state information to generate reply content to present in response to the textual input

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2019-01-25, granted 2020-05-12 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US11238857B2 Supplementing voice inputs to an automated assistant according to selected suggestions

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2019-02-07, granted 2022-02-01 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US11314890B2 Threshold-based assembly of remote automated assistant responses

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2019-02-07, granted 2022-04-26 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US11264033B2 Systems, methods, and apparatuses for resuming dialog sessions via automated assistant

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2019-03-20, granted 2022-03-01 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures

US11086741B2 Immersive web-based simulator for digital assistant-based applications

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2019-04-30, granted 2021-08-10 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US11361066B2 Secure digital assistant integration in web pages

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2019-05-06, granted 2022-06-14 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US11487832B2 Analyzing web pages to facilitate automatic navigation

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2019-05-09, granted 2022-11-01 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US10854188B2 Synthesized voice selection for computational agents

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data.

filed 2019-05-20, granted 2020-12-01 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US11238864B2 Selectively generating expanded responses that guide continuance of a human-to-computer dialog

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2019-05-30, granted 2022-02-01 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US10986214B2 Local persisting of data for selectively offline capable voice action in a voice-enabled electronic device

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2019-06-24, granted 2021-04-20 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US20220113935A1 Mobile-enabled voice search of media items for displaying on alternative playback devices

Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately. Build content for diverse user contexts and search histories.

filed 2019-07-01 · leak modules: NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, VideoPerDocData

US11217243B2 Systems and methods to verify trigger keywords in acoustic-based digital assistant applications

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2019-07-17, granted 2022-01-04 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US11322140B2 Providing prompt in an automated dialog session based on selected content of prior automated dialog session

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2019-08-02, granted 2022-05-03 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US20210216384A1 Automated assistant for generating, in response to a request from a user, application input content using application data from other sources

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2019-08-06 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US10643608B2 Feedback controller for data transmissions

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2019-08-19, granted 2020-05-05 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US10573315B1 Tailoring an interactive dialog application based on creator provided content

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2019-10-16, granted 2020-02-25 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US11289096B2 Providing answers to voice queries using user feedback

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2019-11-15, granted 2022-03-29 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US11232155B2 Providing command bundle suggestions for an automated assistant

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2019-11-22, granted 2022-01-25 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11074039B2 Graphical user interace rendering management by voice-driven computing infrastructure

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2020-02-03, granted 2021-07-27 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US11823068B2 Proactive virtual assistant

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2020-02-03, granted 2023-11-21 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US11436417B2 Providing access to user-controlled resources by automated assistants

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2020-02-12, granted 2022-09-06 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US11417092B2 Systems, methods, and apparatus for image-responsive automated assistants

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2020-03-02, granted 2022-08-16 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US10893088B2 Sequence dependent data message consolidation in a voice activated computer network environment

Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2020-03-13, granted 2021-01-12 · leak modules: IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US11264032B2 Using textual input and user state information to generate reply content to present in response to the textual input

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2020-04-20, granted 2022-03-01 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US11475886B2 Feedback controller for data transmissions

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2020-04-20, granted 2022-10-18 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US10937427B2 Generating and transmitting invocation request to appropriate third-party agent

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2020-05-19, granted 2021-03-02 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US11381530B2 Instantiation of dialog process at a particular child node state
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2020-05-22, granted 2022-07-05 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US11562198B2 Virtual assistant generation of group recommendations
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2020-05-27, granted 2023-01-24 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US11341970B2 Personal directory service
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2020-06-08, granted 2022-05-24 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US11086959B2 Secure digital assistant integration in web pages
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2020-06-11, granted 2021-08-10 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US20230245650A1 Using canonical utterances for text or voice communication
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Write in natural vocabulary. ML models understand context and fluff. Write for natural spoken language and conversational queries.
filed 2020-06-11 · leak modules: IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, ShingleInfoPerDocData

US11210345B2 Using distributed state machines for human-to-computer dialogs with automated assistants to protect private data
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Write for intent and meaning. Neural matching connects related concepts.
filed 2020-06-24, granted 2021-12-28 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US11354368B2 Displaying information related to spoken dialogue in content playing on a device
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2020-07-21, granted 2022-06-07 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US20220028379A1 Automated assistant adaptation of a response to an utterance and/or of processing of the utterance, based on determined interaction measure
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2020-08-04 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US11488601B2 Dependency graph conversation modeling for use in conducting human-to-computer dialog sessions with a computer-implemented automated assistant
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2020-10-05, granted 2022-11-01 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo

US20230186911A1 Delivery of Compatible Supplementary Content via a Digital Assistant
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2020-10-21 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US20210089565A1 Streaming real-time dialog management

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.

filed 2020-12-07 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20210096815A1 Systems and methods for enabling user voice interaction with a host computing device

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2020-12-10 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US11558505B2 Computational assistant extension device

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2020-12-15, granted 2023-01-17 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US11714870B2 Using frames for action dialogs

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.

filed 2020-12-15, granted 2023-08-01 · leak modules: NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US11494453B2 Automatic dialing

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2020-12-29, granted 2022-11-08 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US11562742B2 Generating and transmitting invocation request to appropriate third-party agent

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2021-01-15, granted 2023-01-24 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US11562009B2 Generation of domain-specific models in networked system

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2021-03-18, granted 2023-01-24 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US11550846B2 Systems, methods, and apparatuses for providing assistant deep links to effectuate third-party dialog session transfers

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.

filed 2021-05-17, granted 2023-01-10 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US20230267928A1 Indexing Application Actions for Voice-Based Execution

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2021-05-18 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US20220382819A1 Search Results Based Triggering For Understanding User Intent On Assistant

Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Build content for diverse user contexts and search histories.

filed 2021-05-28 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, RepositoryAnnotationsGeoTopicality

US20220406301A1 Passive disambiguation of assistant commands

Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2021-06-16 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US11429346B2 Graphical user interface rendering management by voice-driven computing infrastructure

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2021-06-30, granted 2022-08-30 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US12380885B2 Bit vector-based content matching for third-party digital assistant actions

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data.

filed 2021-07-19, granted 2025-08-05 · leak modules: CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US11556442B2 Immersive web-based simulator for digital assistant-based applications

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2021-08-09, granted 2023-01-17 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US12148421B2 Using large language model(s) in generating automated assistant response(s

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2021-11-22, granted 2024-11-19 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US20230025709A1 Transferring dialog data from an initially invoked automated assistant to a subsequently invoked automated assistant

Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2021-11-22 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US12100395B2 Dynamic assistant suggestions during assistant browsing

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2021-11-30, granted 2024-09-24 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US11960543B2 Providing suggestions for interaction with an automated assistant in a multi-user message exchange thread

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2021-12-03, granted 2024-04-16 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US12347429B2 Specifying preferred information sources to an assistant

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.

filed 2021-12-10, granted 2025-07-01 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US11914660B2 Collaborative search sessions through an automated assistant

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.

filed 2021-12-17, granted 2024-02-27 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US11869504B2 Systems and methods to verify trigger keywords in acoustic-based digital assistant applications

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2022-01-03, granted 2024-01-09 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US20250028744A1 Domain-specific conversational automated assistant
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Structure answers: question heading, tight answer, then depth.
filed 2022-01-07 · leak modules: NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SnippetExtraInfo

US11887595B2 User-programmable automated assistant
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2022-01-10, granted 2024-01-30 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US11971935B2 Autonomously providing search results post-facto, including in assistant context
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.
filed 2022-01-10, granted 2024-04-30 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US12086168B2 Contextual voice search suggestions
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2022-01-24, granted 2024-09-10 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US11869506B2 Selectively generating expanded responses that guide continuance of a human-to-computer dialog
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2022-01-28, granted 2024-01-09 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US12073832B2 Supplementing voice inputs to an automated assistant according to selected suggestions
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2022-01-31, granted 2024-08-27 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US11929072B2 Using textual input and user state information to generate reply content to present in response to the textual input
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2022-02-03, granted 2024-03-12 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US11830499B2 Providing answers to voice queries using user feedback
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2022-02-28, granted 2023-11-28 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US12094471B2 Providing answers to voice queries using user feedback
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2022-03-09, granted 2024-09-17 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US12119006B2 Biasing interpretations of spoken utterance(s) that are received in a vehicular environment
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2022-03-09, granted 2024-10-15 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US12154564B2 Providing prompt in an automated dialog session based on selected content of prior automated dialog session

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2022-04-29, granted 2024-11-26 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US11942082B2 Facilitating communications with automated assistants in multiple languages

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2022-05-26, granted 2024-03-26 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US11797625B2 Displaying information related to spoken dialogue in content playing on a device

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2022-06-03, granted 2023-10-24 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US11829435B2 Secure digital assistant integration in web pages

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2022-06-10, granted 2023-11-28 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US11749278B2 Recommending automated assistant action for inclusion in automated assistant routine

Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2022-07-25, granted 2023-09-05 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, RepositoryAnnotationsGeoTopicality

US12400641B2 Condition dependent scalable utilities for an automated assistant

Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.

filed 2022-08-09, granted 2025-08-26 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsGeoTopicality

US11822695B2 Assembling and evaluating automated assistant responses for privacy concerns

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2022-08-23, granted 2023-11-21 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US11748430B2 Automatic routing using search results

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2022-10-24, granted 2023-09-05 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US11971936B2 Analyzing web pages to facilitate automatic navigation

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2022-10-26, granted 2024-04-30 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US11790004B2 Systems, methods, and apparatuses for providing assistant deep links to effectuate third-party dialog session transfers

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.

filed 2023-01-09, granted 2023-10-17 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US11947984B2 Systems, methods, and apparatus that provide multi-functional links for interacting with an assistant agent

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2023-01-12, granted 2024-04-02 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US11907214B2 Conditional response fulfillment cache for locally responding to automated assistant inputs

Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Write for intent and meaning. Neural matching connects related concepts.

filed 2023-01-30, granted 2024-02-20 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US12530536B2 Mixture-of-expert approach to reinforcement learning-based dialogue management

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2023-02-23, granted 2026-01-20 · leak modules: NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US20240386886A1 Generating and updating a custom automated assistant based on a domain-specific resource

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2023-05-15 · leak modules: CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US12602424B2 Proactive personalization of multimedia content and dialog content through utilization of large language model(s)

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2023-06-12, granted 2026-04-14 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US12455925B2 Real-time voice processing

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Write for natural spoken language and conversational queries.

filed 2023-07-24, granted 2025-10-28 · leak modules: NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US12124523B2 Automatic routing using search results

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2023-08-09, granted 2024-10-22 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US12614548B2 Pre-emptively initializing an automated assistant routine and/or dismissing a scheduled alarm

Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2023-08-14, granted 2026-04-28 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US12505829B2 Augmenting retrieval systems with user-provided phonetic signals

Rule: Optimize content for phonetic similarity to common spoken query patterns. Google can now match spoken audio directly to document tokens without transcription — content with phonetically clear terms and natural spoken-language patterns may rank better in voice search. Use conversational language that matches how people speak.

filed 2023-08-22, granted 2025-12-23 · leak modules: QualitySalientTermsSalientTermSet, QualitySalientTermsSignalData, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores

US12603099B2 Self-adjusting assistant LLMs enabling robust interaction with business LLMs

Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2023-08-22, granted 2026-04-14 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US20250068258A1 Conversational User Interfaces With Multimodal Inputs and Mind Map Like Interactions

Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2023-08-22 · leak modules: PerDocData, SmartphonePerDocData

US20240004608A1 Automatic adjustment of muted response setting

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2023-09-18 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US12430498B2 Drafting assistant for a browser

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2023-10-04, granted 2025-09-30 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US12387110B2 Proactive virtual assistant

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2023-10-09, granted 2025-08-12 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US12093320B2 Systems, methods, and apparatuses for providing assistant deep links to effectuate third-party dialog session transfers

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.

filed 2023-10-16, granted 2024-09-17 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US12475185B2 Secure digital assistant integration in web pages

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2023-10-26, granted 2025-11-18 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US12154557B2 Adaptive interface in a voice-activated network

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2023-11-16, granted 2024-11-26 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US12254333B2 Systems, methods, and apparatus for providing dynamic auto-responses at a mediating assistant application

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2023-12-12, granted 2025-03-18 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US12602539B2 Proactive assistance via a cascade of LLMS

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2023-12-20, granted 2026-04-14 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20250210033A1 Query replay for personalized responses in an llm powered assistant

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2023-12-20 · leak modules: NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US12230269B2 Systems and methods to verify trigger keywords in acoustic-based digital assistant applications

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2024-01-08, granted 2025-02-18 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US12223960B2 Adapting automated assistant functionality based on generated proficiency measure(s)

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2024-03-18, granted 2025-02-11 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US12572601B2 Providing suggestions for interaction with an automated assistant in a multi-user message exchange thread

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2024-04-15, granted 2026-03-10 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US12437012B2 Autonomously providing search results post-facto, including in assistant context

Rule: Ensure your content is indexed in Google's Knowledge Graph so it can be served as authoritative answers in voice/assistant contexts. If your content isn't in the KG when a voice query is asked, you miss the initial answer AND the post-facto follow-up. Build entity-level structured data and schema markup.

filed 2024-04-29, granted 2025-10-07 · leak modules: RepositoryWebrefEntityJoin, RepositoryWebrefDocumentMetadata, QualityNsrPQData, PerDocData

US20240282305A1 Providing related queries to a secondary automated assistant based on past interactions

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2024-04-29 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US12499112B2 Methods, systems, and media for interpreting voice queries

Rule: Ensure your brand/entity names are clearly established in Google's entity tables. Voice queries are interpreted using entity name feasibility — if your brand name matches a known entity, it gets preferential selection over alternative interpretations. Build entity presence through Knowledge Graph, structured data, and consistent NAP.

filed 2024-05-28, granted 2025-12-16 · leak modules: RepositoryWebrefEntityJoin, RepositoryWebrefGlobalNameInfo, RepositoryWebrefWebrefEntities, QualityOrbitAsteroidBeltDocumentIntentScores

US20260010569A1 Generating response(s) to user input(s) for new conversation(s) by selecting and prepending conversational context(s) from prior conversation(s)

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2024-07-03 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo

US20240371375A1 Semi-delegated calling by an automated assistant on behalf of human participant

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2024-07-15 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo

US20240411754A1 Retrieving context from previous sessions

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2024-08-23 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US20240420698A1 Combining responses from multiple automated assistants

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2024-08-30 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US20260080866A1 Entry Points for LLM-Powered Assistants
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2024-09-19 · leak modules: IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US20250037711A1 Using large language model(s) in generating automated assistant response(s)
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2024-10-10 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US20250053596A1 Inferring semantic label(s) for assistant device(s) based on device-specific signal(s)
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2024-10-25 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US20250279088A1 Unstructured description-based chatbot development techniques
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2025-05-20 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US20250349297A1 Selecting between multiple automated assistants based on invocation properties
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Write for natural spoken language and conversational queries.
filed 2025-07-21 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US20260105090A1 Conversational computing device assistant
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2025-10-10 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

Crawling & indexing (196)

US6658423B1 Detecting duplicate and near-duplicate files
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2001-01-24, granted 2003-12-02 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, ShingleInfoPerDocData

US8495483B1 Using text surrounding hypertext links when indexing and generating page summaries
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2003-03-12, granted 2013-07-23 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US7366718B1 Detecting duplicate and near-duplicate files
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2003-06-27, granted 2008-04-29 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, ShingleInfoPerDocData

US7627613B1 Duplicate document detection in a web crawler system
Rule: Consolidate near-duplicates yourself, one page per intent, or Google picks the canonical for you. The leak's anchorMismatchDemotion field confirms mismatched content/anchors across duplicates are penalized.
filed 2003-07-03, granted 2009-12-01 · leak modules: PerDocData, IndexingDocjoinerAnchorStatistics, QualityNsrNsrData

US7174346B1 System and method for searching an extended database
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2003-09-30, granted 2007-02-06 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US7801898B1 Methods and systems for compressing indices
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2003-12-30, granted 2010-09-21 · leak modules: CrawlerChangerateUrlChangerate, PerDocData, SmartphonePerDocData, TrawlerFetchReplyData

US7581227B1 Systems and methods of synchronizing indexes
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships.
filed 2004-03-31, granted 2009-08-25 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US8161053B1 Methods and systems for eliminating duplicate events
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2004-03-31, granted 2012-04-17 · leak modules: IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, ShingleInfoPerDocData

US7725452B1 Scheduler for search engine crawler
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2004-05-20, granted 2010-05-25 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US8042112B1 Scheduler for search engine crawler
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2004-06-30, granted 2011-10-18 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, ShingleInfoPerDocData

US8707312B1 Document reuse in a search engine crawler
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2004-06-30, granted 2014-04-22 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, ShingleInfoPerDocData

US7536408B2 Phrase-based indexing in an information retrieval system
Rule: Cover the related phrases a real expert would naturally use. Co-occurrence is measurable, thin coverage shows. The leak's titlematchScore and bodyWordsToTokensRatio fields confirm content density and topical alignment are scored.
filed 2004-07-26, granted 2009-05-19 · leak modules: PerDocData, QualityNsrNsrData, QualityNsrPQData

US7711679B2 Phrase-based detection of duplicate documents in an information retrieval system
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2004-07-26, granted 2010-05-04 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDocjoinerAnchorStatistics, IndexingDupsLocalizedLocalizedCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US7546370B1 Search engine with multiple crawlers sharing cookies
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency.
filed 2004-08-18, granted 2009-06-09 · leak modules: CrawlerChangerateUrlChangerate, TrawlerFetchReplyData

US7987172B1 Minimizing visibility of stale content in web searching including revising web crawl intervals of documents
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2004-08-30, granted 2011-07-26 · leak modules: CrawlerChangerateUrlChangerate, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, TrawlerFetchReplyData

US8386728B1 Methods and systems for prioritizing a crawl
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2004-09-14, granted 2013-02-26 · leak modules: CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate

US7567959B2 Multiple index based information retrieval system
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active.
filed 2005-01-25, granted 2009-07-28 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US7680809B2 Profile based capture component
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth. Use JSON-LD structured data for entity types and relationships.
filed 2005-02-04, granted 2010-03-16 · leak modules: NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData, SnippetExtraInfo

US8719244B1 Methods and systems for retrieval of information items and associated sentence fragments
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2005-03-23, granted 2014-05-06 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US7680773B1 System for automatically managing duplicate documents when crawling dynamic documents
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2005-03-31, granted 2010-03-16 · leak modules: CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, ShingleInfoPerDocData

US8140505B1 Near-duplicate document detection for web crawling
Rule: Avoid publishing near-duplicate content across URLs - Google's simhash-based detection will suppress both indexing and link extraction from duplicates. If you must have similar content (e.g., product variants), ensure substantial unique content differences beyond template boilerplate. Use canonical tags proactively for known duplicate scenarios.
filed 2005-03-31, granted 2012-03-20 · leak modules: canonicalization, dedup, crawler, indexing, link_analysis

US8666964B1 Managing items in crawl schedule
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2005-04-25, granted 2014-03-04 · leak modules: CrawlerChangerateUrlChangerate, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, TrawlerFetchReplyData

US7509315B1 Managing URLs
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2005-06-24, granted 2009-03-24 · leak modules: CrawlerChangerateUrlChangerate, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, TrawlerFetchReplyData

US7769742B1 Web crawler scheduler that utilizes sitemaps from websites
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2005-06-30, granted 2010-08-03 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US7801881B1 Sitemap generating client for web crawler
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use JSON-LD structured data for entity types and relationships.
filed 2005-06-30, granted 2010-09-21 · leak modules: CrawlerChangerateUrlChangerate, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, TrawlerFetchReplyData

US7979417B1 Embedded communication of link information
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active.
filed 2005-06-30, granted 2011-07-12 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CrawlerChangerateUrlChangerate, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData

US8055626B1 Detecting mirrors on the web
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2005-08-09, granted 2011-11-08 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate

US7617197B2 Combined title prefix and full-word content searching
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2005-08-19, granted 2009-11-10 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US7765209B1 Indexing and retrieval of blogs
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Ensure content comprehensively covers the topic with natural vocabulary.
filed 2005-09-13, granted 2010-07-27 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US7739258B1 Facilitating searches through content which is accessible through web-based forms
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2006-04-05, granted 2010-06-15 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation

US8015162B2 Detecting duplicate and near-duplicate files
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2006-08-04, granted 2011-09-06 · leak modules: IndexingDocjoinerAnchorSpamInfo, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, ShingleInfoPerDocData, SmartphonePerDocData

US7599920B1 System and method for enabling website owners to manage crawl rate in a website indexing system
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2006-10-12, granted 2009-10-06 · leak modules: CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US7725453B1 Custom search index
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2006-12-29, granted 2010-05-25 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US8082242B1 Custom search
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2006-12-29, granted 2011-12-20 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US20140188919A1 Duplicate document detection
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2007-02-14 · leak modules: CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US7693813B1 Index server architecture using tiered and sharded phrase posting lists
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2007-03-30, granted 2010-04-06 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData

US7702614B1 Index updating using segment swapping
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2007-03-30, granted 2010-04-20 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData

US7925655B1 Query scheduling using hierarchical tiers of index servers
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2007-03-30, granted 2011-04-12 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData

US8484566B2 Analyzing a form page for indexing
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2007-10-15, granted 2013-07-09 · leak modules: IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, ShingleInfoPerDocData, SmartphonePerDocData

US8255386B1 Selection of documents to place in search index
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2008-01-30, granted 2012-08-28 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDocjoinerAnchorSpamInfo, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US7865497B1 Sitemap generation where last modified time is not available to a network crawler
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2008-02-21, granted 2011-01-04 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US9275143B2 Detecting duplicate and near-duplicate files
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2008-03-15, granted 2016-03-01 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, ShingleInfoPerDocData

US8296279B1 Identifying results through substring searching
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2008-06-03, granted 2012-10-23 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9081861B2 Uniform resource locator canonicalization
Rule: Use canonical tags to proactively consolidate URL parameter variations. Google's system can auto-detect irrelevant parameters, but explicit canonicalization is faster and more reliable. Audit URL parameters in Google Search Console's URL Parameters tool. Ensure session IDs, tracking parameters, and sorting parameters are marked as irrelevant or handled with canonical tags.
filed 2008-07-21, granted 2015-07-14 · leak modules: canonicalization, crawler, indexing, dedup

US8442994B1 Custom search index data security
Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2008-09-12, granted 2013-05-14 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US8224870B1 Data structure for incremental search

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2009-01-26, granted 2012-07-17 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8001462B1 Updating search engine document index based on calculated age of changed portions in a document

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2009-01-30, granted 2011-08-16 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US7991762B1 Managing URLs

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2009-03-24, granted 2011-08-02 · leak modules: CrawlerChangerateUrlChangerate, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, TrawlerFetchReplyData

US8560550B2 Multiple index based information retrieval system

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active.

filed 2009-07-20, granted 2013-10-15 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US8032518B2 System and method for enabling website owners to manage crawl rate in a website indexing system

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs.

filed 2009-09-04, granted 2011-10-04 · leak modules: CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US7984054B2 Representative document selection for sets of duplicate documents in a web crawler system

Rule: When duplicate content exists across multiple URLs, Google selects the highest-PageRank URL as the representative and indexes only that one. Ensure your preferred URL has the strongest internal and external link profile. Use canonical tags and 301 redirects to consolidate authority to your preferred version. Syndicated content should link back to the original.

filed 2009-12-01, granted 2011-07-19 · leak modules: canonicalization, dedup, indexing, pagerank, crawler

US8090723B2 Index server architecture using tiered and sharded phrase posting lists

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2010-03-02, granted 2012-01-03 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData

US8108412B2 Phrase-based detection of duplicate documents in an information retrieval system

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.

filed 2010-03-04, granted 2012-01-31 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDocjoinerAnchorStatistics, IndexingDupsLocalizedLocalizedCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US9026566B2 Generating equivalence classes and rules for associating content with document identifiers

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2010-03-16, granted 2015-05-05 · leak modules: CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, ShingleInfoPerDocData

US8161033B2 Scheduler for search engine crawler
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2010-05-25, granted 2012-04-17 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US8037054B2 Web crawler scheduler that utilizes sitemaps from websites
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2010-06-25, granted 2011-10-11 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9183224B2 Identifying matching canonical documents in response to a visual query
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2010-08-06, granted 2015-11-10 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument

US8037055B2 Sitemap generating client for web crawler
Rule: Keep sitemaps complete, current, and free of junk URLs. They are crawler input, not decoration. The leak's hostAge and domainAge fields confirm crawl discovery timing feeds into trust signals.
filed 2010-08-23, granted 2011-10-11 · leak modules: PerDocData, QualityNsrNsrData

US8060516B2 Methods and systems for compressing indices
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2010-09-20, granted 2011-11-15 · leak modules: CrawlerChangerateUrlChangerate, PerDocData, SmartphonePerDocData, TrawlerFetchReplyData

US8090708B1 Searching indexed and non-indexed resources for content
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2010-10-18, granted 2012-01-03 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US8032512B1 Generating sitemap where last modified time is not available to a network crawler
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2010-12-29, granted 2011-10-04 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US8266115B1 Identifying duplicate electronic content based on metadata
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-01-14, granted 2012-09-11 · leak modules: IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US8868541B2 Scheduling resource crawls
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-01-21, granted 2014-10-21 · leak modules: CrawlerChangerateUrlChangerate, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US8707313B1 Scheduler for search engine crawler
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2011-02-18, granted 2014-04-22 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, ShingleInfoPerDocData

US8386459B1 Scheduling a recrawl
Rule: Ensure your most important pages have high PageRank/internal authority so they get crawled more frequently. Use sitemap lastmod dates and changefreq hints to signal crawl priority. Maintain consistent update cadences so the crawler's estimated change period matches your actual content refresh rate.
filed 2011-02-22, granted 2013-02-26 · leak modules: crawler, freshness, pagerank, indexing, scheduling

US8655886B1 Selective indexing of content portions

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2011-03-25, granted 2014-02-18 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20120259882A1 Mining for Product Classification Structures for Intenet-Based Product Searching

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2011-04-06 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US9171088B2 Mining for product classification structures for internet-based product searching

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2011-04-06, granted 2015-10-27 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US20130144836A1 Canonicalization of uniform resource identifiers

Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2011-06-02 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, ShingleInfoPerDocData

US8407204B2 Minimizing visibility of stale content in web searching including revising web crawl intervals of documents

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2011-06-22, granted 2013-03-26 · leak modules: CrawlerChangerateUrlChangerate, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, TrawlerFetchReplyData

US8676783B1 Method and apparatus for managing a backlog of pending URL crawls

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2011-06-28, granted 2014-03-18 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US8260766B2 Embedded communication of link information

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active.

filed 2011-07-12, granted 2012-09-04 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CrawlerChangerateUrlChangerate, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData

US8386460B1 Managing URLs

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2011-07-15, granted 2013-02-26 · leak modules: CrawlerChangerateUrlChangerate, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, TrawlerFetchReplyData

US8260781B2 Representative document selection for sets of duplicate documents in a web crawler system

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.

filed 2011-07-19, granted 2012-09-04 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDocjoinerAnchorStatistics, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityPreviewSnippetRadishFeatures

US8423885B1 Updating search engine document index based on calculated age of changed portions in a document

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2011-08-15, granted 2013-04-16 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US8543572B2 Systems and methods for analyzing boilerplate

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.

filed 2011-09-14, granted 2013-09-24 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation

US8290928B1 Generating sitemap where last modified time is not available to a network crawler

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2011-09-23, granted 2012-10-16 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US8280861B1 Identifying duplicate electronic content based on metadata

Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2011-09-30, granted 2012-10-02 · leak modules: IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US8458163B2 System and method for enabling website owner to manage crawl rate in a website indexing system

Rule: In Google Search Console, check if your crawl rate limit is the actual bottleneck before requesting changes. If Google is crawling below the limit, increasing the limit won't help - focus on improving page speed and server capacity instead. Use the Crawl Stats report to monitor bytes downloaded and access times.

filed 2011-10-03, granted 2013-06-04 · leak modules: crawler, scheduling, search_console

US8417686B2 Web crawler scheduler that utilizes sitemaps from websites

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs.

filed 2011-10-11, granted 2013-04-09 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US8713071B1 Detecting mirrors on the web

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2011-11-04, granted 2014-04-29 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate

US8468145B2 Indexing of URLs with fragments

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2011-11-10, granted 2013-06-18 · leak modules: CrawlerChangerateUrlChangerate, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8549000B2 Methods and systems for compressing indices

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2011-11-14, granted 2013-10-01 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, PerDocData, SmartphonePerDocData, TrawlerFetchReplyData

US8886715B1 Dynamically determining a tile budget when pre-fetching data in a client device

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2011-11-16, granted 2014-11-11 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures

US8489628B2 Phrase-based detection of duplicate documents in an information retrieval system

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.

filed 2011-12-01, granted 2013-07-16 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDocjoinerAnchorStatistics, IndexingDupsLocalizedLocalizedCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US8811742B2 Identifying matching canonical documents consistent with visual query structural information

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.

filed 2011-12-01, granted 2014-08-19 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument

US9176986B2 Generating a combination of a visual query and matching canonical document

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.

filed 2011-12-01, granted 2015-11-03 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument

US20130144847A1 De-Duplication of Featured Content

Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2011-12-05 · leak modules: IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9305107B2 Method and apparatus for pre-fetching place page data for subsequent display on a mobile computing device

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2011-12-08, granted 2016-04-05 · leak modules: CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9197713B2 Method and apparatus for pre-fetching remote resources for subsequent display on a mobile computing device

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2011-12-09, granted 2015-11-24 · leak modules: CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8682901B1 Index server architecture using tiered and sharded phrase posting lists

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2011-12-20, granted 2014-03-25 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData

US8631027B2 Integrated external related phrase information into a phrase-based indexing information retrieval system

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2012-01-10, granted 2014-01-14 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, IndexingDupsLocalizedLocalizedCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US8862569B2 Method and techniques for determining crawling schedule

Rule: For classified ads, product listings, or other expiring content, ensure pages return proper 410 Gone or 404 status when content is deleted so Google can detect deletion patterns. Categories with high turnover need more frequent crawling - maintain accurate sitemap updates for these sections. Remove expired content promptly to avoid Google's 'deleted page penalty' in crawl scheduling.

filed 2012-01-11, granted 2014-10-14 · leak modules: crawler, scheduling, freshness, indexing, quality

US8756236B1 System and method for indexing documents

Rule: Use consistent entity names, sameAs schema, and structured data. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2012-01-31, granted 2014-06-17 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US9152736B2 Efficient indexing and searching of access control listed documents

Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2012-03-07, granted 2015-10-06 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US8548972B1 Near-duplicate document detection for web crawling
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-03-16, granted 2013-10-01 · leak modules: CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, ShingleInfoPerDocData, SmartphonePerDocData

US8965873B2 Methods and systems for eliminating duplicate events
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-03-16, granted 2015-02-24 · leak modules: IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, ShingleInfoPerDocData

US8775403B2 Scheduler for search engine crawler
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-04-17, granted 2014-07-08 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US9262420B1 Third-party indexable text
Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately. Write for intent and meaning. Neural matching connects related concepts.
filed 2012-04-23, granted 2016-02-16 · leak modules: NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US9037591B1 Storing term substitution information in an index
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-04-30, granted 2015-05-19 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8650199B1 Document similarity detection
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-06-25, granted 2014-02-11 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8452766B1 Detecting query-specific duplicate documents
Rule: Even if your pages aren't globally duplicate, they may be flagged as query-specific duplicates if their query-relevant portions (snippets) are too similar. Ensure each page has unique, substantive content in the sections most relevant to your target queries. Vary meta descriptions and on-page content around target keywords to avoid query-specific dedup.
filed 2012-07-02, granted 2013-05-28 · leak modules: canonicalization, dedup, ranking, snippet

US10013493B1 Customized search engines
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2012-07-13, granted 2018-07-03 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData, TrawlerFetchReplyData

US8554759B1 Selection of documents to place in search index
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-08-16, granted 2013-10-08 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDocjoinerAnchorSpamInfo, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US8868559B2 Representative document selection for a set of duplicate documents
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2012-08-30, granted 2014-10-21 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDocjoinerAnchorStatistics, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityPreviewSnippetRadishFeatures

US9165079B1 Access controls in a search index
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2012-09-06, granted 2015-10-20 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US8719265B1 Pre-fetching information in anticipation of a user request
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-09-14, granted 2014-05-06 · leak modules: CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US9569550B1 Custom search index
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-09-14, granted 2017-02-14 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US9047378B1 Systems and methods for accessing a multi-organization collection of hosted contacts
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-09-28, granted 2015-06-02 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US20130117252A1 Large-scale real-time fetch service
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-10-04 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, PerDocData, SmartphonePerDocData, TrawlerFetchReplyData

US8972375B2 Adapting content repositories for crawling and serving
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data.
filed 2012-12-20, granted 2015-03-03 · leak modules: CrawlerChangerateUrlChangerate, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US9582588B2 Methods and systems for providing custom crawl-time metadata
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-12-20, granted 2017-02-28 · leak modules: CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US8958817B1 Weighted-distance spatial indexing
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-01-18, granted 2015-02-17 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate

US9189553B2 Methods and systems for prioritizing a crawl
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2013-01-28, granted 2015-11-17 · leak modules: CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate

US9361331B2 Multiple index based information retrieval system
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active.
filed 2013-03-13, granted 2016-06-07 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US8943067B1 Index server architecture using tiered and sharded phrase posting lists
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-03-15, granted 2015-01-27 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData

US9002819B2 Web crawler scheduler that utilizes sitemaps from websites
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2013-04-08, granted 2015-04-07 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9342849B2 Near-duplicate filtering in search engine result page of an online shopping system
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2013-09-19, granted 2016-05-17 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData

US9396263B1 Identifying canonical content items for answering online questions
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-10-14, granted 2016-07-19 · leak modules: IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9262455B2 Indexing quoted text in messages in conversations to support advanced conversation-based searching
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-10-30, granted 2016-02-16 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US10073874B1 Updating inverted indices
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-11-21, granted 2018-09-11 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US10474650B1 In-place updates for inverted indices
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-11-21, granted 2019-11-12 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US9483568B1 Indexing system
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2013-12-16, granted 2016-11-01 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US10235459B1 Creating entries in at least one of a personal cache and a personal index
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data.
filed 2013-12-31, granted 2019-03-19 · leak modules: CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US20150199402A1 Computerized systems and methods for indexing and serving recurrent calendar events
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-01-14 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20190121518A1 Presentation of Stream Content Related to a Canonical Identifier
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-02-20 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US10235454B1 Generating playlist inclusive canonical network addresses
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships.
filed 2014-04-01, granted 2019-03-19 · leak modules: IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, ShingleInfoPerDocData, SmartphonePerDocData, VideoPerDocData

US9513961B1 Monitoring application loading
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-04-02, granted 2016-12-06 · leak modules: CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US9679056B2 Document reuse in a search engine crawler
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2014-04-04, granted 2017-06-13 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, ShingleInfoPerDocData

US9898507B2 Pre-fetching information in anticipation of a user request
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-05-13, granted 2018-02-20 · leak modules: CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US10621241B2 Scheduler for search engine crawler
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-07-07, granted 2020-04-14 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US20140337368A1 Duplicate Content Search
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2014-07-25 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US9087235B2 Identifying matching canonical documents consistent with visual query structural information
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2014-07-29, granted 2015-07-21 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument

US9984130B2 Batch-optimized render and fetch architecture utilizing a virtual clock
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2014-10-22, granted 2018-05-29 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData

US9307045B2 Dynamically determining a tile budget when pre-fetching data in a client device
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2014-11-10, granted 2016-04-05 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures

US9223877B1 Index server architecture using tiered and sharded phrase posting lists
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-01-07, granted 2015-12-29 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData

US9355177B2 Web crawler scheduler that utilizes sitemaps from websites
Rule: Submit comprehensive XML sitemaps with accurate lastmod dates. The priority and changefreq tags are read but may or may not influence scheduling. Ensure sitemap indexes have accurate lastmod dates so Google knows when to re-download them. Filter out error URLs from sitemaps.
filed 2015-01-27, granted 2016-05-31 · leak modules: crawler, scheduling, indexing, sitemap

US10180980B2 Methods and systems for eliminating duplicate events
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2015-02-23, granted 2019-01-15 · leak modules: IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, ShingleInfoPerDocData

US9864767B1 Storing term substitution information in an index
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2015-05-01, granted 2018-01-09 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9569505B2 Phrase-based searching in an information retrieval system
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2015-05-15, granted 2017-02-14 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDocjoinerAnchorStatistics, IndexingDupsLocalizedLocalizedCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US10558715B2 Automatic crawling of applications
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2015-06-15, granted 2020-02-11 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US10013496B2 Indexing actions for resources
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-06-18, granted 2018-07-03 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US9479508B2 Efficient indexing and searching of access control listed documents
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2015-08-24, granted 2016-10-25 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US9836544B2 Methods and systems for prioritizing a crawl
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2015-11-13, granted 2017-12-05 · leak modules: CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate

US9491255B2 Method and apparatus for pre-fetching remote resources for subsequent display on a mobile computing device
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-11-23, granted 2016-11-08 · leak modules: CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9652483B1 Index server architecture using tiered and sharded phrase posting lists
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-11-23, granted 2017-05-16 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData

US10275606B2 Indexing and searching documents with restricted portions
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2015-12-10, granted 2019-04-30 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US10031920B1 Third-party indexable text
Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately. Write for intent and meaning. Neural matching connects related concepts.
filed 2015-12-29, granted 2018-07-24 · leak modules: NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US9872072B2 Systems and methods for identifying non-canonical sessions
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2016-03-21, granted 2018-01-16 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US9813521B2 Method and apparatus for pre-fetching place page data for subsequent display on a mobile computing device
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2016-04-04, granted 2017-11-07 · leak modules: CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9607331B2 Near-duplicate filtering in search engine result page of an online shopping system
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2016-04-20, granted 2017-03-28 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData

US9817825B2 Multiple index based information retrieval system
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active.
filed 2016-06-03, granted 2017-11-14 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US10229213B1 Protocol based computer network content transmission based on client computing device parameters
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2016-12-14, granted 2019-03-12 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US10430490B1 Methods and systems for providing custom crawl-time metadata
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2016-12-27, granted 2019-10-01 · leak modules: CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US9990421B2 Phrase-based searching in an information retrieval system
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2017-02-02, granted 2018-06-05 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDocjoinerAnchorStatistics, IndexingDupsLocalizedLocalizedCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US10776435B2 Canonicalized online document sitelink generation
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2017-04-19, granted 2020-09-15 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US10216847B2 Document reuse in a search engine crawler
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2017-06-08, granted 2019-02-26 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, ShingleInfoPerDocData

US10909100B2 Object identifier index
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-09-05, granted 2021-02-02 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US10389866B2 Advanced content retrieval
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-09-08, granted 2019-08-20 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10671676B2 Multiple index based information retrieval system
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active.
filed 2017-11-10, granted 2020-06-02 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US10575054B2 Systems and methods for identifying non-canonical sessions
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2017-12-13, granted 2020-02-25 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US11157488B2 Reinforcement learning techniques to improve searching and/or to conserve computational and network resources
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2017-12-13, granted 2021-10-26 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US10984000B2 Pre-fetching information in anticipation of a user request
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2018-02-16, granted 2021-04-20 · leak modules: CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations

US10754908B2 Indexing actions for resources
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2018-04-18, granted 2020-08-25 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US11328114B2 Batch-optimized render and fetch architecture
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2018-05-28, granted 2022-05-10 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData

US10983956B1 Third-party indexable text
Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately. Write for intent and meaning. Neural matching connects related concepts.
filed 2018-07-23, granted 2021-04-20 · leak modules: NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US11755662B1 Creating entries in at least one of a personal cache and a personal index
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data.
filed 2019-03-06, granted 2023-09-12 · leak modules: CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US11188602B2 Protocol based computer network content transmission based on client computing device parameters
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2019-03-11, granted 2021-11-30 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US11307897B2 Resource pre-fetch using age threshold
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2019-08-20, granted 2022-04-19 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US10963472B2 Systems and methods for indexing content
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2019-11-22, granted 2021-03-30 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US11423029B1 Index-side stem-based variant generation
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2019-12-11, granted 2022-08-23 · leak modules: CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US11194776B2 Selecting content using a location feature index
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2020-01-07, granted 2021-12-07 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US11055369B2 Automatic crawling of applications
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2020-01-22, granted 2021-07-06 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US12141214B2 Adversarial bandits policy for crawling highly dynamic content
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data.
filed 2020-03-30, granted 2024-11-12 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CrawlerChangerateUrlChangerate, IndexingDocjoinerAnchorStatistics, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US11630876B2 Indexing actions for resources
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2020-07-29, granted 2023-04-18 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US11853381B2 Hybrid fetching using a on-device cache
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2020-11-13, granted 2023-12-26 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US11995119B2 Privacy controls for sharing embeddings for searching and indexing media content
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2021-01-06, granted 2024-05-28 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US11347760B2 Systems and methods for indexing content
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2021-02-12, granted 2022-05-31 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US11599499B1 Third-party indexable text
Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately. Write for intent and meaning. Neural matching connects related concepts.
filed 2021-04-19, granted 2023-03-07 · leak modules: NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US11874887B2 Protocol based computer network content transmission based on client computing device parameters
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2021-11-30, granted 2024-01-16 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US11720407B2 Resource pre-fetch using age threshold
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2022-04-15, granted 2023-08-08 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US12353483B2 Identifying non-fungible tokens and related content for indexing
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data.
filed 2022-06-30, granted 2025-07-08 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US12373443B2 Retrieval and composition of modular spaces
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2022-06-30, granted 2025-07-29 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US20240303733A1 Generating a Comprehensive Non- Fungible Token Search Index
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2022-06-30 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US20240303734A1 Deduplication of Non-Fungible Tokens in Search Index
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2022-06-30 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US20250231997A1 Private pre-fetching of search results
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2022-08-05 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US12013885B2 Canonicalizing search queries to natural language questions
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2022-08-17, granted 2024-06-18 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US12045241B2 Index-side stem-based variant generation
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2022-08-19, granted 2024-07-23 · leak modules: CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US20250165469A1 Machine Learning Models as a Differentiable Search Index for Directly Predicting Resource Retrieval Results
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2023-02-09 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US12135755B2 Indexing actions for resources
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2023-03-28, granted 2024-11-05 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US12020071B2 Resource pre-fetch using age threshold
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2023-06-22, granted 2024-06-25 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US12254051B2 Crawl algorithm
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2023-09-25, granted 2025-03-18 · leak modules: CrawlerChangerateUrlChangerate, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, TrawlerFetchReplyData

US20240037096A1 Searchable index
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2023-10-06 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US12141216B2 Protocol based computer network content transmission based on client computing device parameters
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2023-12-08, granted 2024-11-12 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US12561387B2 Indexing actions for resources
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2024-10-03, granted 2026-02-24 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US12585712B2 Adversarial bandits policy for crawling highly dynamic content
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data.
filed 2024-11-04, granted 2026-03-24 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CrawlerChangerateUrlChangerate, IndexingDocjoinerAnchorStatistics, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

Apps & mobile (169)

US7643822B2 Method and system for processing queries initiated by users of mobile devices
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2004-09-30, granted 2010-01-05 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality

US8996514B1 Mobile to non-mobile document correlation
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2005-06-15, granted 2015-03-31 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US8849752B2 Overloaded communication session
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2005-07-21, granted 2014-09-30 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores

US8781532B2 Customized data retrieval applications for mobile devices providing interpretation of markup language data
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2005-09-19, granted 2014-07-15 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US7788577B2 Displaying information on a mobile device
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth. Ensure content comprehensively covers the topic with natural vocabulary.
filed 2005-09-23, granted 2010-08-31 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US7653617B2 Mobile sitemaps
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency.
filed 2006-05-01, granted 2010-01-26 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US7779370B2 User interface for mobile devices
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2006-06-30, granted 2010-08-17 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores

US8041506B2 Mapping in mobile devices
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2006-11-07, granted 2011-10-18 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US8965885B2 System and method for browsing web pages on a mobile communication device
Rule: Monitor this patent for ranking implications.
filed 2006-11-14, granted 2015-02-24

US8326858B2 Synchronization of fixed and mobile data
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2007-01-17, granted 2012-12-04 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US7984068B2 Providing profile information to partner content providers
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2007-05-25, granted 2011-07-19 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US7962477B2 Blending mobile search results
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2008-01-24, granted 2011-06-14 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US9081860B2 Integration of device location into search
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2009-08-24, granted 2015-07-14 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US8306511B2 Method and system for processing queries initiated by users of mobile devices
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2010-01-05, granted 2012-11-06 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US8234266B2 Mobile SiteMaps
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency.
filed 2010-01-25, granted 2012-07-31 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US20110196887A1 Light-Weight Network Traffic Cache
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2010-02-08 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8694533B2 Presenting mobile content based on programming context
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2010-05-19, granted 2014-04-08 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US8880467B1 Smart sync—refreshing application state based on user migration patterns
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2010-06-14, granted 2014-11-04 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8271459B2 Displaying information on a mobile device
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2010-07-29, granted 2012-09-18 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US9165081B2 Hovercard pivoting for mobile devices
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-01-04, granted 2015-10-20 · leak modules: PerDocData, SmartphonePerDocData

US8370332B2 Blending mobile search results
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-01-18, granted 2013-02-05 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US8713473B2 Mobile browser context switching
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Avoid AI-stitched content. GibberishScore uses language model scoring.
filed 2011-04-26, granted 2014-04-29 · leak modules: IndexingDocjoinerAnchorSpamInfo, PerDocData, SmartphonePerDocData, SpamBrainData, SpamBrainScore

US8694520B1 Method and apparatus for providing virtual content to a mobile device
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth. Ensure content comprehensively covers the topic with natural vocabulary.
filed 2011-04-28, granted 2014-04-08 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8341147B2 Blending mobile search results
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-05-11, granted 2012-12-25 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US8316043B2 Providing profile information to partner content providers
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-06-16, granted 2012-11-20 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US8819582B2 Mobile browser context switching
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Avoid AI-stitched content. GibberishScore uses language model scoring.
filed 2011-09-28, granted 2014-08-26 · leak modules: IndexingDocjoinerAnchorSpamInfo, PerDocData, SmartphonePerDocData, SpamBrainData, SpamBrainScore

US8359290B1 Smart sync—refreshing application state based on user migration patterns
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-09-30, granted 2013-01-22 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8612418B2 Mobile web browser for pre-loading web pages
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-10-20, granted 2013-12-17 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9251126B1 System and method for using pre-defined character ranges to denote document features
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2011-11-16, granted 2016-02-02 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8732447B2 Contextual history of computing objects
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2011-11-18, granted 2014-05-20 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US9536012B2 Presentation of the media content on mobile devices
Rule: Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-12-05, granted 2017-01-03 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData

US20130145258A1 Incremental Synchronization for Magazines
Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2011-12-06 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8392617B1 Browsing context continuation across multiple devices
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Write for natural spoken language and conversational queries.
filed 2011-12-06, granted 2013-03-05 · leak modules: NlpSaftDocument, PerDocData, SmartphonePerDocData

US9262413B2 Mobile user interface for contextual browsing while playing digital content
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2012-06-06, granted 2016-02-16 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8655864B1 Mobile SiteMaps
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency.
filed 2012-06-11, granted 2014-02-18 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9087058B2 Method and apparatus for enabling a searchable history of real-world user experiences
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-06-11, granted 2015-07-21 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate

US8903897B2 System and method for providing interactive content to non-native application environments

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2012-07-06, granted 2014-12-02 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData

US9237386B2 Aiding discovery of program content by providing deeplinks into most interesting moments via social media

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2012-08-31, granted 2016-01-12 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8805345B2 Method and system for processing queries initiated by users of mobile devices

Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.

filed 2012-09-14, granted 2014-08-12 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US8707201B1 Systems and methods for prioritizing notifications on mobile devices

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.

filed 2012-10-10, granted 2014-04-22 · leak modules: NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8631097B1 Methods and systems for finding a mobile and non-mobile page pair

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2012-10-11, granted 2014-01-14 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8954092B2 Pre-caching data related to a travel destination

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2012-10-22, granted 2015-02-10 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US20150207854A1 External application request

Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2012-10-29 · leak modules: ImageData, ImagePerDocData, PerDocData, SmartphonePerDocData

US9002821B2 Indexing application pages of native applications

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2013-01-16, granted 2015-04-07 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo

US8996520B2 Ranking of native application content

Rule: If you have a native app, ensure it has corresponding web resources so Google can normalize scores and rank your app alongside web results. App scores use link signals (both app-to-app and web-to-app links), download counts, and app page content. Deep linking from your web pages to app content creates web-to-app links that boost your app's score. The leak's QualityCalypsoAppsLink module confirms Google tracks app linking. Ensure your app content is indexed via Firebase App Indexing or equivalent. Cross-attribute quality signals between your web and app properties.

filed 2013-03-15, granted 2015-03-31 · leak modules: QualityCalypsoAppsLink, IndexingEmbeddedContentPageSizeInfo, SmartphonePerDocData, IndexingConverterLocalizedAlternateName, IndexingCrawlerIdServingDocumentIdentifier, PerDocData, CompressedQualitySignals

US20150193396A1 Loading device specific content automatically when switching between devices

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2013-04-16 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefEntityJoin, SmartphonePerDocData

US9135346B2 Index data for native applications
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-06-07, granted 2015-09-15 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData

US9377523B2 Determining wireless access point locations using clustered data points
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-06-07, granted 2016-06-28 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US20140201195A1 Unified searchable storage for resource-constrained and other devices
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2013-08-20 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9311407B2 Native application search results
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-09-05, granted 2016-04-12 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9996509B1 System for creating managing and displaying multi-frame based ads on mobile devices
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-09-30, granted 2018-06-12 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9674640B2 Method and system for vicarious downloading or uploading of information
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-10-28, granted 2017-06-06 · leak modules: PerDocData, SmartphonePerDocData

US10079920B2 Customized data retrieval applications for mobile devices providing interpretation of markup language data
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-01-30, granted 2018-09-18 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9608870B1 Deep link verification for native applications
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2014-02-28, granted 2017-03-28 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9251224B2 Triggering and ranking of native applications
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2014-03-04, granted 2016-02-02 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9740696B2 Presenting mobile content based on programming context
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-03-10, granted 2017-08-22 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US10061796B2 Native application content verification
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2014-03-11, granted 2018-08-28 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8904163B2 Contextual history of computing objects
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2014-04-02, granted 2014-12-02 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US9817869B2 Systems and methods for prioritizing notifications on mobile devices
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2014-04-22, granted 2017-11-14 · leak modules: NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9875242B2 Dynamic current results for second device
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Optimize video for watch time and retention.
filed 2014-06-03, granted 2018-01-23 · leak modules: PerDocData, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, VideoPerDocData

US10582030B2 Customized data retrieval applications for mobile devices providing interpretation of markup language data
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-07-11, granted 2020-03-03 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9372898B2 Enabling event prediction as an on-device service for mobile interaction
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-07-17, granted 2016-06-21 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US9451428B2 Method and system for processing queries initiated by users of mobile devices
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2014-07-23, granted 2016-09-20 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality

US9824079B1 Providing actions for mobile onscreen content
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2014-08-04, granted 2017-11-21 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData

US9886461B1 Indexing mobile onscreen content
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2014-08-04, granted 2018-02-06 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData

US9965559B2 Providing automatic actions for mobile onscreen content
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-08-21, granted 2018-05-08 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9338813B2 Overloaded communication session
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-08-22, granted 2016-05-10 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores

US9542451B2 Mobile application search ranking
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-09-05, granted 2017-01-10 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US9355082B2 Contextual history of computing objects
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2014-10-28, granted 2016-05-31 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US9891803B2 Simplified projection of content from computer or mobile devices into appropriate videoconferences

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2014-11-13, granted 2018-02-13 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9401000B2 Protecting content on a mobile device from mining

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2014-12-15, granted 2016-07-26 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US10216852B1 Indexing access limited native applications

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2015-01-22, granted 2019-02-26 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10474685B1 Mobile to non-mobile document correlation

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2015-02-13, granted 2019-11-12 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US9262459B2 Indexing application pages of native applications

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2015-02-27, granted 2016-02-16 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo

US9645980B1 Verification of native applications for indexing

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2015-03-19, granted 2017-05-09 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9788179B1 Detection and ranking of entities from mobile onscreen content

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.

filed 2015-05-14, granted 2017-10-10 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData

US9703541B2 Entity action suggestion on a mobile device

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2015-05-29, granted 2017-07-11 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US10210263B1 Native application search results

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2015-06-23, granted 2019-02-19 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo

US9881095B2 Search results for native applications

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2015-06-23, granted 2018-01-30 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData

US9892190B1 Search suggestions based on native application history

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.

filed 2015-06-23, granted 2018-02-13 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US10073911B2 Deep links for native applications
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2015-06-25, granted 2018-09-11 · leak modules: CrawlerChangerateUrlChangerate, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US10268732B2 Ranking native applications and native application deep links
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-06-29, granted 2019-04-23 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US9348671B1 Monitoring application loading
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-07-23, granted 2016-05-24 · leak modules: CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US10803391B2 Modeling personal entities on a mobile device using embeddings
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-07-29, granted 2020-10-13 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US9846745B2 Index data for native applications
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2015-08-10, granted 2017-12-19 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData

US10361936B2 Filtering content based on user mobile network and data-plan
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2015-08-19, granted 2019-07-23 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US10223460B2 Application partial deep link to a corresponding resource
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2015-08-25, granted 2019-03-05 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US10579687B2 Providing native application search results with web search results
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2015-09-01, granted 2020-03-03 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData

US9311168B1 Deeplinking to multiple native applications
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth. Use JSON-LD structured data for entity types and relationships.
filed 2015-09-30, granted 2016-04-12 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US10970646B2 Action suggestions for user-selected content
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-10-01, granted 2021-04-06 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US20170103129A1 Scoring content within native applications
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active.
filed 2015-10-12 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData

US10120949B2 Indexing native application data

Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2015-10-29, granted 2018-11-06 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US9655158B2 Overloaded communication session

Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2015-10-29, granted 2017-05-16 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations

US20170140008A1 Suggestion-based differential diagnostics

Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.

filed 2015-11-13 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo

US10055390B2 Simulated hyperlinks on a mobile device based on user intent and a centered selection of text

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data.

filed 2015-11-18, granted 2018-08-21 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US10146842B2 Pre-instantiating native applications in background

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2015-11-19, granted 2018-12-04 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9514195B2 Triggering and ranking of native applications

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2015-12-09, granted 2016-12-06 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9436531B1 Monitoring application loading

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2015-12-22, granted 2016-09-06 · leak modules: CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US9767162B2 Aiding discovery of program content by providing deeplinks into most interesting moments via social media

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-01-04, granted 2017-09-19 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US9652550B2 Indexing application pages of native applications

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2016-01-11, granted 2017-05-16 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo

US10268343B2 Mobile user interface for contextual browsing while playing digital content

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.

filed 2016-02-12, granted 2019-04-23 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US9465676B1 Deeplinking to multiple native applications

Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth. Use JSON-LD structured data for entity types and relationships.

filed 2016-03-04, granted 2016-10-11 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US9547721B2 Native application search results
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2016-03-04, granted 2017-01-17 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9651921B1 Metronome embedded in search results page and unaffected by lock screen transition
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Earn endorsements from trusted sources. Trust is computed and transitive. Write for natural spoken language and conversational queries.
filed 2016-03-04, granted 2017-05-16 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityNsrNsrData, SmartphonePerDocData

US10176192B2 System and method for detecting and integrating with native applications enabled for web-based storage
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2016-03-14, granted 2019-01-08 · leak modules: PerDocData, RepositoryWebrefEntityJoin, SmartphonePerDocData

US10635725B2 Providing app store search results
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2016-04-06, granted 2020-04-28 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate

US10678605B2 Reducing latency in downloading electronic resources using multiple threads
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2016-04-12, granted 2020-06-09 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US10248698B2 Native application search result adjustment based on user specific affinity
Rule: For app deeplink indexing: drive user engagement with your app to increase affinity scores. The more time users spend in your app relative to other apps, the higher your deep links rank. Ensure your app is properly indexed with Firebase App Indexing. Build app content that encourages repeat usage and longer sessions.
filed 2016-04-14, granted 2019-04-02 · leak modules: PerDocData, QualityNavboostCrapsCrapsData

US20160314215A1 Generating mobile-friendliness scores for resources
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2016-04-20 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US9921853B2 Contextual history of computing objects
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2016-04-29, granted 2018-03-20 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US9563671B2 Protecting content on a mobile device from mining
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2016-06-22, granted 2017-02-07 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US9836597B2 Protecting content on a mobile device from mining
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2016-12-14, granted 2017-12-05 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US9965522B2 Mobile application search ranking
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-01-06, granted 2018-05-08 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US10650068B2 Search engine
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-01-09, granted 2020-05-12 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US10795741B2 Generating deeplinks for applications based on multi-level referrer data
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2017-02-15, granted 2020-10-06 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US10212566B2 Method and system for vicarious downloading or uploading of information
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2017-05-31, granted 2019-02-19 · leak modules: PerDocData, SmartphonePerDocData

US10013552B2 Protecting content on a mobile device from mining
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2017-07-03, granted 2018-07-03 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US10509815B2 Presenting mobile content based on programming context
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-07-20, granted 2019-12-17 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US10491660B1 Sharing screen content in a mobile environment
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2017-08-18, granted 2019-11-26 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData

US10080114B1 Detection and ranking of entities from mobile onscreen content
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2017-08-31, granted 2018-09-18 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData

US11144557B2 Aiding discovery of program content by providing deeplinks into most interesting moments via social media
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-09-18, granted 2021-10-12 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10248440B1 Providing a set of user input actions to a mobile device to cause performance of the set of user input actions
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2017-10-13, granted 2019-04-02 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData

US10713324B2 Search results for native applications
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2017-12-21, granted 2020-07-14 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData

US10402432B1 Search suggestions based on native application history
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2018-01-02, granted 2019-09-03 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US10448313B2 Identifying users via mobile hotspots
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2018-04-10, granted 2019-10-15 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US10678993B1 System for creating managing and displaying multi-frame based ads on mobile devices
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2018-06-11, granted 2020-06-09 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US11403271B2 Native application content verification
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2018-08-13, granted 2022-08-02 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US10915592B2 Indexing native application data
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2018-09-26, granted 2021-02-09 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US11500952B1 Indexing access limited native applications
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2019-01-11, granted 2022-11-15 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11003728B1 Native application search results
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2019-02-08, granted 2021-05-11 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo

US10652706B1 Entity disambiguation in a mobile environment
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2019-03-14, granted 2020-05-12 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData

US10928983B2 Mobile user interface for contextual browsing while playing digital content
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2019-04-22, granted 2021-02-23 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US11166155B2 Secure communication in mobile digital pages
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2019-05-06, granted 2021-11-02 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US11218390B2 Filtering content based on user mobile network and data-plan
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2019-06-11, granted 2022-01-04 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US11188578B1 Search suggestions based on native application history
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2019-07-22, granted 2021-11-30 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US10826967B1 Passing of data via deep links for application components
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2019-08-20, granted 2020-11-03 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US11470478B2 Secure communication in mobile digital pages
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2019-08-27, granted 2022-10-11 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US11178596B2 Identifying users via mobile hotspots
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2019-10-10, granted 2021-11-16 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US10917494B2 Dynamic application content analysis
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2019-10-22, granted 2021-02-09 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US10963630B1 Sharing screen content in a mobile environment
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2019-10-29, granted 2021-03-30 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData

US11265403B2 Customized data retrieval applications for mobile devices providing interpretation of markup language data
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2020-02-20, granted 2022-03-01 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11550638B2 Reducing latency in downloading electronic resources using multiple threads
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2020-03-31, granted 2023-01-10 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US11354367B2 Search engine
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2020-04-02, granted 2022-06-07 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US11704136B1 Automatic reminders in a mobile environment
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2020-05-11, granted 2023-07-18 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData

US11087073B1 System for creating managing and displaying multi-frame based ads on mobile devices
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2020-06-08, granted 2021-08-10 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US11425569B2 Secure communication in mobile digital pages
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2020-06-15, granted 2022-08-23 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US11442797B2 Generating deeplinks for applications based on multi-level referrer data
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2020-10-06, granted 2022-09-13 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US12026593B2 Action suggestions for user-selected content
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2020-10-15, granted 2024-07-02 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US11089085B2 Passing of data via deep links for application components
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2020-11-02, granted 2021-08-10 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US11500953B2 Indexing native application data
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2021-01-11, granted 2022-11-15 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US11388254B2 Dynamic application content analysis
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2021-01-13, granted 2022-07-12 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US11347385B1 Sharing screen content in a mobile environment
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2021-03-04, granted 2022-05-31 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData

US11741110B2 Aiding discovery of program content by providing deeplinks into most interesting moments via social media
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2021-10-08, granted 2023-08-29 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11836167B2 Search suggestions based on native application history
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2021-11-29, granted 2023-12-05 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US11573810B1 Sharing screen content in a mobile environment
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2022-05-27, granted 2023-02-07 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData

US11722575B2 Dynamic application content analysis
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2022-06-27, granted 2023-08-08 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US11782776B2 Generating deeplinks for applications based on multi-level referrer data
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2022-09-13, granted 2023-10-10 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US11924644B2 Secure communication in mobile digital pages
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2022-10-10, granted 2024-03-05 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US11989247B2 Indexing access limited native applications
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2022-10-14, granted 2024-05-21 · leak modules: CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US11748429B2 Indexing native application data
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2022-10-19, granted 2023-09-05 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US11907739B1 Annotating screen content in a mobile environment
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2023-02-06, granted 2024-02-20 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData

US20240419677A1 Methods and apparatus for donating in-app search queries, events, and/or actions
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2023-06-15 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityWebanswersVideoTranscriptAnnotations

US12222951B2 Aiding discovery of program content by providing deeplinks into most interesting moments via social media
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2023-08-28, granted 2025-02-11 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US12067433B2 Generating deeplinks for applications based on multi-level referrer data
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2023-09-07, granted 2024-08-20 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US12430368B2 Search suggestions based on native application history
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2023-10-26, granted 2025-09-30 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US12147652B1 Annotating screen content in a mobile environment
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2023-12-18, granted 2024-11-19 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData

US12475303B2 Secure communication in mobile digital pages
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2024-03-04, granted 2025-11-18 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US12505384B2 Action suggestions for user-selected content
Rule: Maintain high search rankings for queries that map to your mobile app — Google uses high-ranking website positions as training data for app action suggestions. Ensure your website and mobile app are properly linked and mapped. High web rankings translate to mobile app action suggestions through this training pipeline.
filed 2024-05-20, granted 2025-12-23 · leak modules: QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsFeatureCrapsData, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores

US20250315463A1 Deep linking using generative artificial intelligence
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2025-06-16 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

Local search (157)

US7827176B2 Methods and systems for endorsing local search results
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2004-06-30, granted 2010-11-02 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US8595225B1 Systems and methods for correlating document topicality and popularity
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Build content for diverse user contexts and search histories.
filed 2004-09-30, granted 2013-11-26 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, RepositoryAnnotationsGeoTopicality

US7925676B2 Data object visualization using maps
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2006-01-27, granted 2011-04-12 · leak modules: IndexingDocjoinerAnchorSpamInfo, PerDocData, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData, SpamBrainData, SpamBrainScore

US8122013B1 Title based local search ranking
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2006-01-27, granted 2012-02-21 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US10198521B2 Processing ambiguous search requests in a geographic information system
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2006-06-27, granted 2019-02-05 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US7746343B1 Streaming and interactive visualization of filled polygon data in a geographic information system
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2006-06-27, granted 2010-06-29 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US7933395B1 Virtual tour of user-defined paths in a geographic information system
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2006-06-27, granted 2011-04-26 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US7933929B1 Network link for providing dynamic data layer in a geographic information system
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2006-06-27, granted 2011-04-26 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US7933897B2 Entity display priority in a distributed geographic information system
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data.
filed 2006-10-11, granted 2011-04-26 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US7840407B2 Business listing search
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2006-10-13, granted 2010-11-23 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US7945546B2 Local search and mapping for mobile devices
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2006-11-07, granted 2011-05-17 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US8347202B1 Determining geographic locations for place names in a fact repository
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2007-03-14, granted 2013-01-01 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US20080301643A1 Map Gadgets
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2007-05-28 · leak modules: IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8484199B1 Ranking of geographic information
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2007-12-11, granted 2013-07-09 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US8745041B1 Ranking of geographic information
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2007-12-11, granted 2014-06-03 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9378571B1 Browsing large geocoded datasets using nested shapes
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2008-05-29, granted 2016-06-28 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US8301639B1 Location based query suggestion
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2010-01-29, granted 2012-10-30 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate

US8812585B2 Trusted maps: updating map locations using trust-based social graphs
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2010-03-29, granted 2014-08-19 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US20100191768A1 Search query categorization for business listings search
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2010-04-08 · leak modules: ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8392394B1 Merging search results
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2010-05-04, granted 2013-03-05 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8332408B1 Date-based web page annotation
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2010-08-23, granted 2012-12-11 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US8489586B2 Methods and systems for endorsing local search results
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2010-10-26, granted 2013-07-16 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US8831930B2 Business listing search
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2010-10-27, granted 2014-09-09 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US8396888B2 Location-based searching using a search area that corresponds to a geographical location of a computing device
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2010-11-24, granted 2013-03-12 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8473491B1 Systems and methods of detecting keyword-stuffed business titles
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2010-12-03, granted 2013-06-25 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsGeoTopicality

US8949277B1 Semantic geotokens
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2010-12-30, granted 2015-02-03 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US8290942B2 Entity display priority in a distributed geographic information system
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data.
filed 2011-02-17, granted 2012-10-16 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US8515973B1 Identifying geographic features from query prefixes
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-03-25, granted 2013-08-20 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9798831B2 Processing data in a MapReduce framework
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-04-01, granted 2017-10-24 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US9129029B1 Ranking content using location-based query log analysis
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-05-19, granted 2015-09-08 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US9870572B2 System and method of providing information based on street address
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Build content for diverse user contexts and search histories.
filed 2011-05-20, granted 2018-01-16 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9043325B1 Collecting useful user feedback about geographical entities
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images.
filed 2011-06-24, granted 2015-05-26 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US20130007575A1 Managing Map Data in a Composite Document
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-06-29 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8280414B1 Map tile data pre-fetching based on mobile device generated event analysis
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-09-26, granted 2012-10-02 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US8626766B1 Systems and methods for ranking and importing business listings
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-09-28, granted 2014-01-07 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US20120023088A1 Location-Based Searching
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-09-30 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8386514B2 Location-based searching using a search area that corresponds to a geographical location of a computing device
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-09-30, granted 2013-02-26 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US20150248192A1 Semi-Automated Generation of Address Components of Map Features
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-10-03 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US20130124563A1 Controlling pre-fetching of map data tiles based on selectable parameters
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-11-16 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8711181B1 Pre-fetching map data using variable map tile radius
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-11-16, granted 2014-04-29 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures

US9063951B1 Pre-fetching map data based on a tile budget
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-11-16, granted 2015-06-23 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures

US8893010B1 Experience sharing in location-based social networking
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-12-08, granted 2014-11-18 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityOrbitAsteroidBeltDocumentIntentScores

US20150169626A1 System and method for identifying a new geographical area name
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-01-09 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8650197B2 Authoritative document identification
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active.
filed 2012-03-09, granted 2014-02-11 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData

US9224118B2 Dynamic display of content consumption by geographic location
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-03-12, granted 2015-12-29 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8468156B2 Determining a geographic location relevant to a web page
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-03-19, granted 2013-06-18 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation

US20150139482A1 System and method for updating geographic data
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-04-12 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US9069793B2 Dynamic highlighting of geographic entities on electronic maps
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-04-25, granted 2015-06-30 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US8350849B1 Dynamic view-based data layer in a geographic information system
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2012-05-14, granted 2013-01-08 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US8621623B1 Method and system for identifying business records
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-07-06, granted 2013-12-31 · leak modules: IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9389088B2 Method of pre-fetching map data for rendering and offline routing
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-07-11, granted 2016-07-12 · leak modules: CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US8954860B1 Method and apparatus for generating and displaying tourist maps
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2012-08-27, granted 2015-02-10 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US8812031B2 Map tile data pre-fetching based on mobile device generated event analysis
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-08-30, granted 2014-08-19 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US9880699B2 Multiple views of a geographic area on a mobile device
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-08-31, granted 2018-01-30 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US8880516B2 Endorsing local search results
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-09-14, granted 2014-11-04 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US9684727B1 Ranking of geographic information
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-09-14, granted 2017-06-20 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US8626754B2 Entity display priority in a distributed geographic information system
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data.
filed 2012-09-15, granted 2014-01-07 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US9164655B2 Multiple views of a geographic area on a mobile device
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-09-15, granted 2015-10-20 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US20140095303A1 Apparatus and Method for Personalizing Maps
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-10-01 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US8676777B1 Determining geographic areas of interest for a query
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-10-03, granted 2014-03-18 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US8554873B1 Custom event and attraction suggestions
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-10-05, granted 2013-10-08 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9547722B2 System and method for providing a temporally or geographically relevant item
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-10-26, granted 2017-01-17 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US20150058357A1 Ratings weighted or filtered by context
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-11-02 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate

US8972368B1 Systems, methods, and computer-readable media for providing search results having contacts from a user's social graph

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2012-12-07, granted 2015-03-03 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate

US9892132B2 Determining geographic locations for place names in a fact repository

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2012-12-31, granted 2018-02-13 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US9471625B2 Dynamic view-based data layer in a geographic information system

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.

filed 2013-01-08, granted 2016-10-18 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US8874594B2 Search with my location history

Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2013-02-06, granted 2014-10-28 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US10467296B2 Location-based searching using a search area that corresponds to a geographical location of a computing device

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2013-02-21, granted 2019-11-05 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US9081797B2 Systems and methods for associating microposts with geographic locations

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2013-03-06, granted 2015-07-14 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US20140279261A1 Destination and point of interest search

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use descriptive alt text and surrounding context for images.

filed 2013-03-29 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US20140310266A1 Systems and Methods for Suggesting Places for Persons to Meet

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2013-04-10 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US8698824B1 Computing systems, devices and methods for rendering maps remote from a host application

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2013-04-12, granted 2014-04-15 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo

US9792330B1 Identifying local experts for local search

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2013-04-30, granted 2017-10-17 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US9135625B1 Systems and methods of detecting keyword-stuffed business titles
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2013-05-09, granted 2015-09-15 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsGeoTopicality

US9201898B2 Efficient fetching of map tile data
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-05-15, granted 2015-12-01 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US9378504B2 Highlighting related points of interest in a geographical region
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2013-05-21, granted 2016-06-28 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US9563641B1 Suggestion refinement
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-06-26, granted 2017-02-07 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US9110982B1 Method, system, and computer program product for obtaining crowd-sourced location information
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-07-17, granted 2015-08-18 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US9317528B1 Identifying geographic features from query prefixes
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-07-29, granted 2016-04-19 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8990344B2 Custom event and attraction suggestions
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-09-05, granted 2015-03-24 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20150095195A1 Content discovery via tags
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-09-27 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US9313113B2 Determining unique vistors to a network location
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-10-10, granted 2016-04-12 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US9451041B2 Caching geographic data according to server-specified policy
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-11-25, granted 2016-09-20 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8965884B2 Entity display priority in a distributed geographic information system
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data.
filed 2013-11-26, granted 2015-02-24 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US8973097B1 Method and system for identifying business records
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-12-19, granted 2015-03-03 · leak modules: IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9043304B1 Determining geographic areas of interest for a query
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-03-14, granted 2015-05-26 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US9569463B1 Pre-fetching map data using variable map tile radius
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2014-03-17, granted 2017-02-14 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures

US9690805B1 Ranking of geographic information
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2014-06-02, granted 2017-06-27 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9633116B2 Endorsing local search results
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2014-06-19, granted 2017-04-25 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US9245046B2 Map tile data pre-fetching based on mobile device generated event analysis
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2014-06-24, granted 2016-01-26 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US10061854B2 Trusted maps: updating map locations using trust-based social graphs
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-07-01, granted 2018-08-28 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US9747346B1 Attention spots in a map interface
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2014-08-06, granted 2017-08-29 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9460712B1 Business listing search
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-08-07, granted 2016-10-04 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US9602589B1 Systems and methods for determining room types for regions of a map
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-08-07, granted 2017-03-21 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9323738B2 Classification of ambiguous geographic references
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-09-26, granted 2016-04-26 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US9864783B1 Systems and methods for identifying outlying point of interest search results
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2014-12-02, granted 2018-01-09 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US9582548B1 Semantic geotokens
Rule: Include specific geographic location references (at minimum city and state) prominently in your page content. Avoid mentioning too many different locations on a single page (which prevents geotoken assignment). Build backlinks with anchor text containing your target location terms, as off-page anchor text references directly boost geographic relevance scoring. Use specific location expressions rather than generic ones, since the system evaluates different expressions and selects the highest likelihood.
filed 2014-12-29, granted 2017-02-28 · leak modules: RepositoryAnnotationsGeoTopicality, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, AnchorsAnchor

US9471691B1 Systems, methods, and computer-readable media for providing search results having contacts from a user's social graph
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2015-01-23, granted 2016-10-18 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate

US20180232398A1 Automatically embedding digital maps in software applications
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2015-02-09 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US9715530B2 Entity display priority in a distributed geographic information system
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data.
filed 2015-02-23, granted 2017-07-25 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US9305063B1 Determining geographic areas of interest for a query
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-04-28, granted 2016-04-05 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US10127324B2 Dynamically integrating offline and online suggestions in a geographic application
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Write for intent and meaning. Neural matching connects related concepts.
filed 2015-06-18, granted 2018-11-13 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US10274324B2 Dynamic highlighting of geographic entities on electronic maps
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2015-06-29, granted 2019-04-30 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US20160379117A1 Location-based delivery of structured content
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-06-29 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US10311119B1 Determining location-based contextual hashtags

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2015-08-21, granted 2019-06-04 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US9547696B2 Ranking content using location-based query log analysis

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2015-09-04, granted 2017-01-17 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US10055414B2 Access points for maps

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2015-11-13, granted 2018-08-21 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US20170169025A1 Estimating Geographic Entity Capacity

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2015-12-14 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US10242029B2 Dynamic display of content consumption by geographic location

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2015-12-28, granted 2019-03-26 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US10437861B1 Inferring geographic locations for entities appearing in search queries

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.

filed 2016-02-29, granted 2019-10-08 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US9442990B1 Determining geographic areas of interest for a query

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-02-29, granted 2016-09-13 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US9843648B2 Caching geographic data according to server-specified policy

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-09-19, granted 2017-12-12 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10254120B2 API for obtaining geographic location data

Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2016-10-11, granted 2019-04-09 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US10496724B2 Intelligent distributed geographic information system

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.

filed 2016-10-17, granted 2019-12-03 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US10289648B2 Enforcing category diversity
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2016-11-14, granted 2019-05-14 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate

US9785648B2 Entity display priority in a distributed geographic information system
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data.
filed 2016-12-22, granted 2017-10-10 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US10102297B2 System and method for providing a temporally or geographically relevant item
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2016-12-28, granted 2018-10-16 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9870409B2 Entity display priority in a distributed geographic information system
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data.
filed 2017-02-01, granted 2018-01-16 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US10671685B1 Suggestion refinement
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2017-02-03, granted 2020-06-02 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US10102222B2 Semantic geotokens
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2017-02-27, granted 2018-10-16 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US10393539B2 API for obtaining geographic location data
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2017-05-05, granted 2019-08-27 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US10360228B1 Ranking of geographic information
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2017-06-01, granted 2019-07-23 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US10565237B2 Logical segmentation data processing system
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2017-06-05, granted 2020-02-18 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US20180143998A1 Electronic map interfaces
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2017-08-21 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US11100114B1 Attention spots in a map interface
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2017-08-28, granted 2021-08-24 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US10929409B2 Identifying local experts for local search
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-10-16, granted 2021-02-23 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US11315296B2 Dynamically varying visual properties of indicators on a digital map
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-11-22, granted 2022-04-26 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10521957B2 Methods and systems for reconstructing GIS scenes
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships.
filed 2017-12-05, granted 2019-12-31 · leak modules: PerDocData, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US10592537B2 Entity display priority in a distributed geographic information system
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data.
filed 2017-12-12, granted 2020-03-17 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US10459955B1 Determining geographic locations for place names
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2018-02-08, granted 2019-10-29 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US11907497B2 Multiple views of a geographic area on a mobile device
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2018-05-08, granted 2024-02-20 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US20200265089A1 Optimizing geographic region selection
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2018-05-30 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US10896217B2 Access points for maps
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2018-07-31, granted 2021-01-19 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US20200097564A1 Selecting points of interest for display on a personalized digital map
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2018-09-25 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, RepositoryAnnotationsGeoTopicality

US12435984B2 Discovery and ranking of locations for use by geographic context applications
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2019-01-07, granted 2025-10-07 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US10990638B2 Processing ambiguous search requests in a geographic information system
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2019-02-04, granted 2021-04-27 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US10977312B2 Apparatus and method for inferring an origin

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2019-02-14, granted 2021-04-13 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, RepositoryAnnotationsGeoTopicality

US10866974B2 Dynamic display of content consumption by geographic location

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2019-03-25, granted 2020-12-15 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US11941009B1 Ranking of geographic information

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2019-07-10, granted 2024-03-26 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US11176181B2 Inferring geographic locations for entities appearing in search queries

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.

filed 2019-10-07, granted 2021-11-16 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US10795958B2 Intelligent distributed geographic information system

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.

filed 2019-10-22, granted 2020-10-06 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US11386167B2 Location-based searching using a search area that corresponds to a geographical location of a computing device

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2019-11-04, granted 2022-07-12 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US12141136B2 Systems and methods for low-latency provision of content

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2019-12-05, granted 2024-11-12 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US11366837B2 Logical segmentation data processing system

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2020-01-07, granted 2022-06-21 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US11288292B2 Entity display priority in a distributed geographic information system

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data.

filed 2020-03-16, granted 2022-03-29 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US11461003B1 User interface for presenting suggestions from a local search corpus

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2020-04-30, granted 2022-10-04 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US11914661B2 Integration of web and media snippets into map applications
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2020-09-02, granted 2024-02-27 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US11860857B2 MUSS—map user submission states
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2020-10-23, granted 2024-01-02 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US20220129440A1 MUSS - Map User Submission States
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2020-10-23 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US12026190B2 Access points for maps
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2020-12-10, granted 2024-07-02 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US12072898B1 Attention spots in a map interface
Rule: Encourage user activity/check-ins at your business location to create map attention spots. Google boosts content items near high-activity locations — businesses in areas with high map engagement get ranking bonuses. Drive foot traffic and map interactions to build attention spot authority for your location.
filed 2021-08-20, granted 2024-08-27 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, QualityNavboostCrapsCrapsClickSignals, RepositoryAnnotationsGeoTopicality, PerDocData

US20240152564A1 Temporal Map for Neighborhood and Regional Business Vibe and Social Semantics
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2022-03-25 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US20240232273A1 Cluster Search on Maps Applications
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Write for natural spoken language and conversational queries.
filed 2022-07-05 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US12001492B2 Location-based searching using a search area that corresponds to a geographical location of a computing device
Rule: Optimize for multiple geographic granularity levels (neighborhood, city, region). Google uses nested search areas and ranks results differently at each level. Ensure your local SEO targets are appropriate for each nesting level. User behavior with area selection feeds back into area scoring.
filed 2022-07-11, granted 2024-06-04 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, RepositoryAnnotationsGeoTopicality, PerDocData, IndexingDupsLocalizedLocalizedCluster

US12306891B2 Integration of web and media snippets into map applications
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2024-01-22, granted 2025-05-20 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US12554766B2 Access points for maps
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2024-05-22, granted 2026-02-17 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US12579207B2 Location-based searching using a search area that corresponds to a geographical location of a computing device

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2024-05-30, granted 2026-03-17 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

Snippets & answers (135)

US7587309B1 System and method for providing text summarization for use in web-based content

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Create original, effortful content. Quality is scored at page and site level.

filed 2003-12-01, granted 2009-09-08 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US7310633B1 Methods and systems for generating textual information

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2004-03-31, granted 2007-12-18 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US7917480B2 Document compression system and method for use with tokenspace repository

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.

filed 2004-08-13, granted 2011-03-29 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US7587387B2 User interface for facts query engine with snippets from information sources that include query terms and answer terms

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2005-03-31, granted 2009-09-08 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US7953720B1 Selecting the best answer to a fact query from among a set of potential answers

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.

filed 2005-03-31, granted 2011-05-31 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US8631006B1 System and method for personalized snippet generation

Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2005-04-14, granted 2014-01-14 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US8145617B1 Generation of document snippets based on queries and search results

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2005-11-18, granted 2012-03-27 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US8073830B2 Expanded text excerpts

Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2006-03-31, granted 2011-12-06 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8595619B1 In response to a search result query providing a snippet of a document including an element previously highlighted by a user

Rule: Use descriptive alt text and surrounding context for images. Structure answers: question heading, tight answer, then depth. Use JSON-LD structured data for entity types and relationships.

filed 2007-01-31, granted 2013-11-26 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SnippetExtraInfo

US9323827B2 Identifying key terms related to similar passages

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2008-01-30, granted 2016-04-26 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US8725716B1 Customized web summaries and alerts based on custom search engines

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2008-05-30, granted 2014-05-13 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate

US20090327234A1 Updating answers with references in forums

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2008-06-27 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US20090327235A1 Presenting references with answers in forums

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2008-06-27 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US8135712B1 Posting questions from search queries

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2008-06-27, granted 2012-03-13 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US8065290B2 User interface for facts query engine with snippets from information sources that include query terms and answer terms

Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2009-08-24, granted 2011-11-22 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US8423392B2 Trusted participants of social network providing answers to questions through on-line conversations

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2010-04-01, granted 2013-04-16 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8468143B1 System and method for directing questions to consultants through profile matching

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.

filed 2010-04-07, granted 2013-06-18 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument

US8538982B2 Framework for selecting and presenting answer boxes relevant to user input as query suggestions

Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.

filed 2010-08-31, granted 2013-09-17 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US20110258192A1 Providing question and answer services

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.

filed 2010-11-29 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo

US20130304730A1 Automated answers to online questions

Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.

filed 2011-01-18 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US8655866B1 Returning factual answers in response to queries
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-02-10, granted 2014-02-18 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8321445B2 Generating content snippets using a tokenspace repository
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2011-03-03, granted 2012-11-27 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US9355175B2 Triggering answer boxes
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-05-27, granted 2016-05-31 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US8380707B1 Session-based dynamic search snippets
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-07-11, granted 2013-02-19 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US8589235B2 Method of answering questions by trusted participants
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2011-07-21, granted 2013-11-19 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8224802B2 User interface for facts query engine with snippets from information sources that include query terms and answer terms
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-08-09, granted 2012-07-17 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US8255381B2 Expanded text excerpts
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-10-27, granted 2012-08-28 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8768920B1 Posting questions from search queries
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-02-06, granted 2014-07-01 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate

US8959111B2 Providing answer box functionality to third party search engines
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2012-02-06, granted 2015-02-17 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US9299059B1 Generating a summary of social media content
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-06-07, granted 2016-03-29 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US8650175B2 User interface for facts query engine with snippets from information sources that include query terms and answer terms
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-07-13, granted 2014-02-11 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US9116996B1 Reverse question answering
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-07-24, granted 2015-08-25 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US8527491B2 Expanded text excerpts
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2012-08-27, granted 2013-09-03 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US9087130B2 Generating snippets for prominent users for information retrieval queries
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-10-04, granted 2015-07-21 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US9098501B2 Generating content snippets using a tokenspace repository
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2012-11-26, granted 2015-08-04 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US9274683B2 Interactive answer boxes for user search queries
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-12-07, granted 2016-03-01 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US9087131B1 Auto-summarization for a multiuser communication session
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-12-18, granted 2015-07-21 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9390456B2 Summary view of a profile
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-12-26, granted 2016-07-12 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US20160110360A1 Providing numerical answers to queries
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-02-05 · leak modules: ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US9158849B2 Synthesis of webpage snippets using sub-pages of the webpage
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-02-19, granted 2015-10-13 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9235626B2 Automatic generation of snippets based on context and user interest
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-03-13, granted 2016-01-12 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US9081831B2 Methods and systems for presenting document-specific snippets
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-03-14, granted 2015-07-14 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US9213748B1 Generating related questions for search queries
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-03-14, granted 2015-12-15 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US9336269B1 Determining question and answer alternatives
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-03-14, granted 2016-05-10 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US10108700B2 Question answering to populate knowledge base
Rule: Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2013-03-15, granted 2018-10-23 · leak modules: NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US9477759B2 Question answering using entity references in unstructured data
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2013-03-15, granted 2016-10-25 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation

US20140344259A1 Answering people-related questions
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-06-11 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US9514113B1 Methods for automatic footnote generation
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-07-29, granted 2016-12-06 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefEntityJoin

US20150039607A1 Providing a summary presentation
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-07-31 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US9881077B1 Relevance determination and summary generation for news objects
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-08-08, granted 2018-01-30 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US9110995B2 Framework for selecting and presenting answer boxes relevant to user input as query suggestions
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2013-09-16, granted 2015-08-18 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9824161B1 Providing third party answers
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-11-08, granted 2017-11-21 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US9898554B2 Implicit question query identification
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data.
filed 2013-11-18, granted 2018-02-20 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US8819006B1 Rich content for query answers
Rule: Use descriptive alt text, captions, and surrounding context for images near answer content on your pages. Images that contextualize both the question and answer topic are more likely to appear in rich answer results. Ensure your images are properly labeled with relevant entity information.
filed 2013-12-31, granted 2014-08-26 · leak modules: PerDocData, RepositoryWebref

US9766998B1 Determining a user habit
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-12-31, granted 2017-09-19 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US9418118B2 System and method for personalized snippet generation
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-01-13, granted 2016-08-16 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US9971746B2 Identifying information using referenced text
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-01-30, granted 2018-05-15 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US9959315B1 Context scoring adjustments for answer passages
Rule: Use proper heading hierarchy (H1 > H2 > H3) to structure your content. Answer passages under deeper, more specific headings get boosted. Ensure your headings contain keywords relevant to the questions users ask—heading text similarity to the query directly adjusts answer scores. Put answers in sections where the passage covers a substantial portion of the text block. Include question-style sentences before answer passages. For how-to queries, format answers as lists. The leak's SnippetRadishFeatures and RanklabSnippet modules confirm heading-based context scoring is active.
filed 2014-01-31, granted 2018-05-01 · leak modules: QualityPreviewSnippetRadishFeatures, QualityPreviewRanklabSnippet, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo, QualityPreviewSnippetBrainFeatures, QualityPreviewSnippetQualityFeatures, QualityPreviewSnippetQueryFeatures, QualitySalientTermsSalientTermSet

US9323853B1 Customized web summaries and alerts based on custom search engines
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2014-04-22, granted 2016-04-26 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9805125B2 Displaying a summary of media content items
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-06-20, granted 2017-10-31 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US9336318B2 Rich content for query answers
Rule: Label images and rich media on answer-source pages with terms from BOTH the question context and the answer content. Images labeled with only question-related or only answer-related terms get demoted. Use entity detection-compatible markup (structured data, schema.org) so Google can identify entities depicted in your images. Ensure image labels encompass the full contextual relationship between the question and its answer.
filed 2014-07-25, granted 2016-05-10 · leak modules: SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo, QualityPreviewSnippetRadishFeatures, PerDocData, RepositoryWebrefEntityJoin

US9607087B1 Providing answer boxes based on query results
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-08-01, granted 2017-03-28 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US9842166B1 Semi structured question answering system
Rule: Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2014-08-08, granted 2017-12-12 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US9965474B2 Dynamic summary generator
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-10-02, granted 2018-05-08 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US10311114B2 Displaying stylized text snippets with search engine results
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2014-11-03, granted 2019-06-04 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US9672291B2 Summarizing social interactions between users
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-02-13, granted 2017-06-06 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US20160132501A1 Determining answers to interrogative queries using web resources
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-05-11 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US10372735B2 Generating activity summaries
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-05-20, granted 2019-08-06 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US20150341454A1 Backend Pipeline for Story Generation
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-05-20 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9569487B1 Using an entity database to answer entity-triggering questions
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2015-06-16, granted 2017-02-14 · leak modules: ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US9652544B2 Generating snippets for prominent users for information retrieval queries
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2015-07-01, granted 2017-05-16 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US9396268B2 Framework for selecting and presenting answer boxes relevant to user input as query suggestions
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2015-07-17, granted 2016-07-19 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9619565B1 Generating content snippets using a tokenspace repository
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2015-08-03, granted 2017-04-11 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US10019513B1 Weighted answer terms for scoring answer passages
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data.
filed 2015-08-12, granted 2018-07-10 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US10180964B1 Candidate answer passages
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-08-12, granted 2019-01-15 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US9916348B1 Answer facts from structured content
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data.
filed 2015-08-12, granted 2018-03-13 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US9940367B1 Scoring candidate answer passages
Rule: Structure direct answers: question as heading, a tight 40-60 word answer, then the depth. Passages are scored on their own. The leak's titlematchScore and contentEffort fields confirm passage-level quality is assessed independently.
filed 2015-08-12, granted 2018-04-10 · leak modules: PerDocData, QualityNsrNsrData, QualityNsrPQData

US9697256B2 Synthesis of webpage snippets using sub-pages of the webpage
Rule: Optimize sub-page titles on your website as they can directly generate snippets for parent pages. Write clear, descriptive, appropriately-lengthed titles for all sub-pages. Implement well-structured meta descriptions on your main pages to ensure the text-based snippet wins when it's more relevant. Maintain clear site hierarchy and internal linking structure, as sub-page selection depends on link structure and website hierarchy. Keep sub-page quality high since quality scores affect snippet generation.
filed 2015-10-09, granted 2017-07-04 · leak modules: SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo, QualityPreviewSnippetRadishFeatures, SdrPageAnchorsDocInfo, QualitySitemapScoringSignals

US9720913B1 Automatic generation of snippets based on context and user interest
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2015-11-24, granted 2017-08-01 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US9679027B1 Generating related questions for search queries
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2015-12-14, granted 2017-06-13 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US10353554B2 Interactive answer boxes for user search queries
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2015-12-16, granted 2019-07-16 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US20170262529A1 Sponsor answers and user-approved, system-suggested links in a social search engine
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2016-03-07 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US10289729B2 Question and answer interface based on contextual information
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2016-03-17, granted 2019-05-14 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US10346415B1 Determining question and answer alternatives
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2016-04-01, granted 2019-07-09 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US9805110B2 Triggering answer boxes
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2016-05-02, granted 2017-10-31 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US10592540B2 Generating elements of answer-seeking queries and elements of answers
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2016-06-28, granted 2020-03-17 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores

US9805116B2 System and method for personalized snippet generation
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2016-07-28, granted 2017-10-31 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US10339190B2 Question answering using entity references in unstructured data
Rule: Implement structured entity markup (schema.org Person, Place, Event, etc.) on your pages so Google can identify entity references in your content. Ensure entities appear frequently and with clear topical relevance to the document, since weighted frequency and topicality scores determine entity ranking. Keep content fresh, as freshness scores factor into entity selection. Since answers replace search results, being the source entity for answers means zero-click visibility dominance.
filed 2016-08-30, granted 2019-07-02 · leak modules: RepositoryWebrefEntityJoin, RepositoryWebrefDocumentMetadata, RepositoryWebrefWebrefEntities, PerDocData, QualityTimebasedLastSignificantUpdate

US10503803B2 Animated snippets for search results
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2016-11-23, granted 2019-12-10 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9786279B2 Answering questions using environmental context
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2017-01-19, granted 2017-10-10 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US10515076B1 Generating query answers from a user's history
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2017-01-31, granted 2019-12-24 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation

US10275420B2 Summarizing social interactions between users
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-05-11, granted 2019-04-30 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US10572489B2 Synthesis of webpage snippets using sub-pages of the webpage
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-06-08, granted 2020-02-25 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US10394684B1 Determining a user habit
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-08-11, granted 2019-08-27 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US10762152B2 Displaying a summary of media content items
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-10-06, granted 2020-09-01 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US11093813B2 Answer to question neural networks
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2017-10-18, granted 2021-08-17 · leak modules: CompressedQualitySignals, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US10146849B2 Triggering answer boxes
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2017-10-25, granted 2018-12-04 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US10346485B1 Semi structured question answering system
Rule: Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2017-11-08, granted 2019-07-09 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US10698888B1 Answer facts from structured content
Rule: Use well-structured HTML tables with clear column headers and row values for factual data. Ensure tables have descriptive column names that match common question query terms. Position tables prominently in top-ranked pages since only top-ranked subsets are processed. Use proper table markup (<th>, <td>) so Google can parse row attributes and column names to extract fact answers for question queries.
filed 2018-01-26, granted 2020-06-30 · leak modules: SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo, PerDocData, QualitySalientTermsSalientTermSet

US10783156B1 Scoring candidate answer passages
Rule: For featured snippets/answer passages: (1) Keep answer text within a single section—passages spanning multiple sections are penalized. (2) Don't start answer passages with discourse boundary terms like 'however' or 'therefore.' (3) Include exact query terms and answer terms in your passages. (4) Your site's overall quality score and reputation directly boost your answer passage scores. (5) Structure content with tables and lists—structured content is eligible for answer passages alongside prose. (6) Ensure entities in your answer match the expected entity type for the question. The leak's SnippetRadishFeatures and RanklabSnippet modules confirm these scoring features are live.
filed 2018-02-22, granted 2020-09-22 · leak modules: QualityPreviewSnippetRadishFeatures, QualityPreviewRanklabSnippet, QualityPreviewRanklabTitle, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo, QualityPreviewSnippetBrainFeatures, QualityPreviewSnippetQualityFeatures, QualityPreviewSnippetQueryFeatures

US11409748B1 Context scoring adjustments for answer passages
Rule: Structure answer content with proper heading hierarchy (H1→H2→H3) — deeper nesting under relevant headings boosts answer selection. Use bold text for key terms near answers. Format step-by-step instructions as HTML lists. Ensure answer passages are appropriately sized relative to their surrounding text block. Include the question before the answer in your content.
filed 2018-03-16, granted 2022-08-09 · leak modules: PerDocData, QualityNsrPQData

US10878176B2 Identifying information using referenced text
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2018-04-11, granted 2020-12-29 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US20190130251A1 Neural question answering system
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2018-10-31 · leak modules: NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US11042577B2 Question and answer interface based on contextual information
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2019-04-30, granted 2021-06-22 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US11762923B1 Displaying stylized text snippets with search engine results
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2019-04-30, granted 2023-09-19 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US11176212B2 Question answering using entity references in unstructured data
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2019-05-22, granted 2021-11-16 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation

US11016638B2 Interactive answer boxes for user search queries
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2019-05-30, granted 2021-05-25 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US11120055B2 Generating activity summaries
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2019-08-05, granted 2021-09-14 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US11016872B1 Determining a user habit
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2019-08-22, granted 2021-05-25 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US11188533B1 Generating query answers from a user's history
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2019-11-12, granted 2021-11-30 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation

US11249993B2 Answer facts from structured content
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data.
filed 2020-05-22, granted 2022-02-15 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US11580177B2 Identifying information using referenced text
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2020-10-07, granted 2023-02-14 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US11681604B1 Determining a user habit
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2021-05-18, granted 2023-06-20 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US11928168B2 Question answering using entity references in unstructured data
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2021-11-03, granted 2024-03-12 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation

US12164515B2 Generating query answers from a user's history
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2021-11-29, granted 2024-12-10 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation

US11789946B2 Answer facts from structured content
Rule: Use HTML tables with clear column headers and structured data to enable answer fact extraction. Ensure your tables have a clear subject attribute (first column or title) with unique values. Match table column headers to common question terms. Use schema.org/Table markup. Tables in top-ranked documents are scanned for fact extraction, so table content quality and relevance to common questions directly enables answer box placement.
filed 2022-02-14, granted 2023-10-17 · leak modules: RichsnippetsDataObject, RichsnippetsPageMap, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryAnnotationsRdfaBreadcrumbs, QualityPreviewChosenSnippetInfo, QualityPreviewSnippetDocumentFeatures, SnippetExtraInfoSnippetCandidateInfo, WWWSnippetResponse

US11709994B1 Contextual answer generation in spreadsheets
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2022-03-04, granted 2023-07-25 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US12248529B2 Multi source extraction and scoring of short query answers
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2022-03-09, granted 2025-03-11 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US12038997B2 Generating a snippet packet based on a selection of a portion of a web page
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2022-12-15, granted 2024-07-16 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US12056197B2 Identifying information using referenced text
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2023-01-05, granted 2024-08-06 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US11769017B1 Generative summaries for search results
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2023-03-20, granted 2023-09-26 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US12182002B1 Determining a user habit
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2023-05-09, granted 2024-12-31 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US12367246B2 Displaying stylized text snippets with search engine results
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2023-06-26, granted 2025-07-22 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US12164862B2 Contextual answer generation in spreadsheets
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2023-07-25, granted 2024-12-10 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US12346366B2 Providing generative answers including citations to source documents
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2023-09-21, granted 2025-07-01 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US12079292B1 Proactive query and content suggestion with generative model generated question and answer

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2023-10-13, granted 2024-09-03 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US20250139105A1 Query-aware extractive hierarchical summarization

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.

filed 2023-10-27 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US20250111674A1 Summarizing Events Over a Time Period

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2024-01-23 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US20250278443A1 Content snippet generation and storage with generative model content grouping

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2024-02-29 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US12591629B2 Question answering using entity references in unstructured data

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.

filed 2024-03-08, granted 2026-03-31 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation

US12488057B2 Proactive query and content suggestion with generative model generated question and answer

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2024-07-26, granted 2025-12-02 · leak modules: CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US20250045316A1 Systems and Methods for Generating Instruction Fine-tuning Dataset for a General Purpose Embedding Model

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2024-07-30 · leak modules: CompressedQualitySignals, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US20260080183A1 Adaption of Large Language Model Answers

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2024-09-17 · leak modules: NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US12579141B2 Generating query answers from a user's history

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2024-10-16, granted 2026-03-17 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation

US20250245270A1 Identifying search results in a history repository

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data.

filed 2025-01-30 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation

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US8006197B1 Method and apparatus for output of search results
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2003-09-29, granted 2011-08-23 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US7941439B1 Methods and systems for information capture
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth. Use JSON-LD structured data for entity types and relationships.
filed 2004-03-31, granted 2011-05-10 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US7581173B1 Methods and apparatuses for formatting web pages
Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2004-09-24, granted 2009-08-25 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityPreviewSnippetRadishFeatures, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US7401072B2 Named URL entry
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2004-09-30, granted 2008-07-15 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US8176414B1 Document division method and system
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2005-09-30, granted 2012-05-08 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US8364467B1 Content-based classification
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2006-03-31, granted 2013-01-29 · leak modules: CompressedQualitySignals, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US8037068B2 Searching through content which is accessible through web-based forms
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2006-04-05, granted 2011-10-11 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation

US7730109B2 Message catalogs for remote modules
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data.
filed 2006-06-06, granted 2010-06-01 · leak modules: CrawlerChangerateUrlChangerate, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8239751B1 Data from web documents in a spreadsheet
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2007-05-16, granted 2012-08-07 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US8429243B1 Web analytics event tracking system
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Optimize video for watch time and retention.
filed 2008-12-12, granted 2013-04-23 · leak modules: PerDocData, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, VideoPerDocData

US8621012B2 Active e-mails
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2010-02-08, granted 2013-12-31 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US8069223B2 Transferring data between applications
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2010-08-26, granted 2011-11-29 · leak modules: PerDocData, SmartphonePerDocData

US8521823B1 System and method for targeting information based on message content in a reply
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2010-09-03, granted 2013-08-27 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US8688857B1 Filtering messages based on pruning profile generated from pruning profile schema
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2010-09-15, granted 2014-04-01 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US8386487B1 Clustering internet messages
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2010-11-05, granted 2013-02-26 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData

US8635215B2 Method and web platform for brokering know-how
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Ensure content comprehensively covers the topic with natural vocabulary. Use JSON-LD structured data for entity types and relationships.
filed 2010-12-20, granted 2014-01-21 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US8739061B1 Method and apparatus for output of search results
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2011-06-17, granted 2014-05-27 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US8700576B2 Method, system, and program for archiving files
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-07-18, granted 2014-04-15 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8700990B1 PDF plugin to view any page
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-09-20, granted 2014-04-15 · leak modules: PerDocData, SmartphonePerDocData

US20120173507A1 Searching through content which is accesible through web-based forms
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-09-27 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation

US8335837B2 Transferring data between applications
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2011-10-28, granted 2012-12-18 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US20120131441A1 Multi-Mode Web Browsing
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2011-11-18 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9633114B1 Inbox for task management
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2011-12-08, granted 2017-04-25 · leak modules: NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo

US20130110748A1 Policy Violation Checker
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-08-30 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9116986B1 Interactive calendar with an integrated journal
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-10-12, granted 2015-08-25 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US9870554B1 Managing documents based on a user's calendar
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-10-23, granted 2018-01-16 · leak modules: PerDocData, SmartphonePerDocData

US20150205767A1 Link appearance formatting based on target content
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2012-11-12 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryWebrefEntityJoin, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US9384285B1 Methods for identifying related documents
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-12-18, granted 2016-07-05 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9317592B1 Content-based classification
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2013-01-29, granted 2016-04-19 · leak modules: CompressedQualitySignals, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US9245025B2 Systems, methods, and computer-readable media for searching for events from a computer-implemented calendar
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Write for intent and meaning. Neural matching connects related concepts.
filed 2013-02-07, granted 2016-01-26 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9934224B2 Document editor with research citation insertion tool
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-02-25, granted 2018-04-03 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US10353938B1 Aggregating activity data for multiple users
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency.
filed 2013-02-27, granted 2019-07-16 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals

US8700722B1 User-aware cloud to device messaging systems and methods
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2013-03-15, granted 2014-04-15 · leak modules: NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US9183238B2 Providing task-based information
Rule: For task-oriented queries (repair, DIY, cooking), structure your content to clearly identify required tools, parts, and supplies. Google indexes 'implements' from interactive sessions and serves them as task-specific search results. Creating structured content about what tools/parts are needed for specific tasks can help your content surface in task-based search results. Product reliability data from repair content feeds Google's failure rate profiles.
filed 2013-03-15, granted 2015-11-10 · leak modules: QualityShoppingShoppingAttachment, QualityShoppingShoppingAttachmentProduct, QualityShoppingShoppingAttachmentOffer, ShoppingWebentityShoppingAnnotationOfferAvailabilityInfo, ShoppingWebentityShoppingAnnotationProductRating, IndexingMlVerticalVerticalItem, PerDocData

US20150193391A1 Clustering of Time and Date Annotations in a Document
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2013-04-17 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US9596195B1 System and method for targeting information based on message content in a reply
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2013-07-30, granted 2017-03-14 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US9378196B1 Associating information with a task based on a category of the task
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-10-28, granted 2016-06-28 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US9553835B2 Active e-mails
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-11-26, granted 2017-01-24 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US20140237342A1 System and method for information gathering utilizing form identifiers
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2013-12-30 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US20150186455A1 Systems and methods for automatic electronic message annotation
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-12-30 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate

US20150106358A1 Systems and methods for conveying search results within an email application
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-12-31 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US20150186494A1 Systems and methods for message categorization management
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-12-31 · leak modules: ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US9548951B2 Providing additional information related to a vague term in a message
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Write for natural spoken language and conversational queries.
filed 2013-12-31, granted 2017-01-17 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US9760600B2 Serving recurrent calendar events
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-01-14, granted 2017-09-12 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20140143657A1 Generation of topical subjects from alert search terms
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Earn promotion through authoritative content and entity association. Create original, effortful content. Quality is scored at page and site level.
filed 2014-01-24 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, SmartphonePerDocData

US9391947B1 Automatic delivery channel determination for notifications
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-02-05, granted 2016-07-12 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US9602574B1 Prerendering time zone aware layouts for rich web apps
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2014-02-26, granted 2017-03-21 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US9734197B2 Determining corresponding terms written in different formats
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active.
filed 2014-03-06, granted 2017-08-15 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CrawlerChangerateUrlChangerate, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation

US9400824B2 Systems and methods for sorting data
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2014-03-19, granted 2016-07-26 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US9369765B2 Virtual channels
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-08-04, granted 2016-06-14 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US9563689B1 Generating and applying data extraction templates
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-08-27, granted 2017-02-07 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US9785705B1 Generating and applying data extraction templates
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-10-16, granted 2017-10-10 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US11494447B2 Distributed file system for virtualized computing clusters
Rule: Monitor this patent for ranking implications.
filed 2015-01-13, granted 2022-11-08

US9959416B1 Systems and methods for joining online meetings
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2015-03-27, granted 2018-05-01 · leak modules: CompressedQualitySignals, PerDocData, SmartphonePerDocData

US20160314184A1 Classifying documents by cluster
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-04-27 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9940362B2 Predicting user needs for a particular context
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-05-26, granted 2018-04-10 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US10002201B2 Named URL entry
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-09-23, granted 2018-06-19 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US9311174B2 Providing task-based information
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data.
filed 2015-10-30, granted 2016-04-12 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation

US20170147710A1 Multi-format search results in dialer application
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-11-20 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US20160140192A1 Systems, methods and computer-readable media for searching for events from a computer-implemented calendar
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Write for intent and meaning. Neural matching connects related concepts.
filed 2016-01-25 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US10417661B2 Dynamic content aggregation
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2016-03-10, granted 2019-09-17 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData

US9436745B2 Providing task-based information
Rule: For product/repair content: clearly list required tools and parts in your content so they can be extracted. Create detailed how-to sessions (video, interactive) that demonstrate specific tasks. Structure product repair content with explicit part names and tool lists for extraction into task-based indexes.
filed 2016-03-15, granted 2016-09-06 · leak modules: RepositoryWebref, PerDocData

US10474745B1 Systems and methods for a knowledge-based form creation platform
Rule: Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2016-04-27, granted 2019-11-12 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US10263933B2 Incorporating selectable application links into message exchange threads
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2016-05-17, granted 2019-04-16 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US10769155B2 Automatically augmenting message exchange threads based on tone of message
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2016-05-17, granted 2020-09-08 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US10795947B2 Unified message search
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2016-05-17, granted 2020-10-06 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US9967228B2 Time variant data profile for a fabric network

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.

filed 2016-06-20, granted 2018-05-08 · leak modules: IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata

US10521450B2 Distributed storage system with replica selection

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2016-07-01, granted 2019-12-31 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10346448B2 System and method for classifying an alphanumeric candidate identified in an email message

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-07-13, granted 2019-07-09 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US9947319B1 Forming chatbot output based on user state

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2016-09-27, granted 2018-04-17 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US10091147B2 Providing additional information related to a vague term in a message

Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Write for natural spoken language and conversational queries.

filed 2016-12-08, granted 2018-10-02 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US9690773B1 Associating one or more terms in a message trail with a task entry

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-12-27, granted 2017-06-27 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US10216838B1 Generating and applying data extraction templates

Rule: Google extracts structured data from templated content (emails, receipts, confirmations) using pattern recognition. Ensure your transactional emails and structured communications follow consistent templates with clear semantic structure. Use schema.org markup in emails (e.g., Order, Reservation schemas) to help Google's extraction templates work more efficiently.

filed 2016-12-29, granted 2019-02-26 · leak modules: structured_data, entity, knowledge_graph, data_extraction

US10013978B1 Sequence dependent operation processing of packet based data message transmissions

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-12-30, granted 2018-07-03 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US9911144B1 System and method for targeting information based on message content in a reply

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.

filed 2017-02-10, granted 2018-03-06 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US10949448B1 Determining additional features for a task entry based on a user habit

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2017-03-31, granted 2021-03-16 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US10853747B2 Selection of computational agent for task performance
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data.
filed 2017-11-16, granted 2020-12-01 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US11663535B2 Multi computational agent performance of tasks
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data.
filed 2017-11-16, granted 2023-05-30 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US10650005B2 Predicting user needs for a particular context
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2018-02-28, granted 2020-05-12 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US10699311B1 System and method for targeting information based on message content in a reply
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2018-03-01, granted 2020-06-30 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US10853403B2 Document editor with research citation insertion tool
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2018-03-05, granted 2020-12-01 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US10157615B2 Forming chatbot output based on user state
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2018-03-08, granted 2018-12-18 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US11909729B2 Auto-form fill based website authentication
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2018-04-26, granted 2024-02-20 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US10515635B2 Forming chatbot output based on user state
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2018-11-06, granted 2019-12-24 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US11057333B2 Incorporating selectable application links into message exchange threads
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2019-03-29, granted 2021-07-06 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US11238080B2 Aggregating activity data for multiple users
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency.
filed 2019-05-30, granted 2022-02-01 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals

US11222030B2 Automatically augmenting message exchange threads based on tone of message
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2019-12-11, granted 2022-01-11 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US11322143B2 Forming chatbot output based on user state
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2019-12-12, granted 2022-05-03 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US11562036B2 Unified message search
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2020-01-10, granted 2023-01-24 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US12229501B2 Populating values in a spreadsheet using semantic cues
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2020-04-13, granted 2025-02-18 · leak modules: CompressedQualitySignals, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US11100543B1 System and method for targeting information based on message content in a reply
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2020-06-03, granted 2021-08-24 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US12164584B2 System and method for identifying places using contextual information
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2020-09-08, granted 2024-12-10 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US11556231B1 Selecting an action member in response to input that indicates an action class
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2020-10-02, granted 2023-01-17 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US11481195B2 Synthesizing programs in a spreadsheet programming language
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2020-12-15, granted 2022-10-25 · leak modules: NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US11734311B1 Determining additional features for a task entry based on a user habit
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2021-03-12, granted 2023-08-22 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US12007868B2 Generating action elements suggesting content for ongoing tasks
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2021-07-06, granted 2024-06-11 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US11682051B1 System and method for targeting information based on message content in a reply
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2021-07-16, granted 2023-06-20 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US12547823B2 Dynamically and selectively updated spreadsheets based on knowledge monitoring and natural language processing
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2021-09-24, granted 2026-02-10 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US11762865B2 Automatically augmenting message exchange threads based on tone of message
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2021-10-08, granted 2023-09-19 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US11699030B1 Systems and methods for a knowledge-based form creation platform
Rule: Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2022-07-08, granted 2023-07-11 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US12073194B2 Synthesizing programs in a spreadsheet programming language
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2022-10-24, granted 2024-08-27 · leak modules: NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US12321398B2 Distributed file system for virtualized computing clusters
Rule: Monitor this patent for ranking implications.
filed 2022-10-24, granted 2025-06-03

US11947603B2 Unified message search
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2023-01-23, granted 2024-04-02 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US12468897B2 Self-improving LLMs through consistency-based self-generated demonstrations
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2023-03-30, granted 2025-11-11 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US12079845B1 System and method for targeting information based on message content in a reply
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2023-05-09, granted 2024-09-03 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US12093270B2 Automatically augmenting message exchange threads based on tone of message
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2023-08-21, granted 2024-09-17 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US12393875B2 Training encoder model and/or using trained encoder model to determine responsive action(s) for natural language input
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2023-11-01, granted 2025-08-19 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US20250181655A1 Interactive Validation Of Freeform Entries In Web Forms
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2023-12-04 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US20250209127A1 Search System Having Task-Based Machined-Learned Models
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2023-12-20 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US12495026B2 Auto-form fill based website authentication
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff.
filed 2024-02-07, granted 2025-12-09 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US12287836B2 Unified message search
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2024-04-01, granted 2025-04-29 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US12386645B2 Automating semantically-related computing tasks across contexts
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Write in natural vocabulary. ML models understand context and fluff.
filed 2024-04-11, granted 2025-08-12 · leak modules: NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US20250355948A1 Instruction-based dual encoder retrieval system
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2024-05-17 · leak modules: NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US20260003902A1 Generative ai search formats
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2025-06-27 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US20260099668A1 Autonomous spreadsheet creation
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2025-12-11 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

Freshness & news (107)

US7568148B1 Methods and apparatus for clustering news content
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2003-06-30, granted 2009-07-28 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo

US7577655B2 Systems and methods for improving the ranking of news articles
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2003-09-16, granted 2009-08-18 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo

US8676837B2 Systems and methods for personalizing aggregated news content
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Build content for diverse user contexts and search histories.
filed 2003-12-31, granted 2014-03-18 · leak modules: NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9251519B1 Systems and methods for monetizing subscription and archival news content
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2003-12-31, granted 2016-02-02 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US7725508B2 Methods and systems for information capture and retrieval
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships.
filed 2004-06-30, granted 2010-05-25 · leak modules: PerDocData, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US7797316B2 Systems and methods for determining document freshness
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2004-06-30, granted 2010-09-14 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20140053053A1 Methods and systems for real-time extraction of user-specified information
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2005-03-31 · leak modules: CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US7529739B2 Temporal ranking scheme for desktop searching
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2005-08-19, granted 2009-05-05 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8244720B2 Ranking blog documents
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2005-09-13, granted 2012-08-14 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData

US7739254B1 Labeling events in historic news
Rule: Write for natural spoken language and conversational queries.
filed 2005-09-30, granted 2010-06-15 · leak modules: NlpSaftDocument, PerDocData

US7734610B2 Decentralised web annotation
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2005-12-12, granted 2010-06-08 · leak modules: IndexingDocjoinerAnchorSpamInfo, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US8880499B1 Personalizing aggregated news content
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2005-12-28, granted 2014-11-04 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8117195B1 Providing blog posts relevant to search results
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Optimize mobile experience. Mobile-specific signals are tracked separately. Earn promotion through authoritative content and entity association.
filed 2006-03-22, granted 2012-02-14 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, VideoPerDocData

US8453052B1 Real-time document sharing and editing
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2006-08-16, granted 2013-05-28 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US8112426B2 Document scoring based on document content update
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active.
filed 2006-11-21, granted 2012-02-07 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, IndexingDocjoinerAnchorStatistics, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8935235B1 Search history, email checking, and blog posting widgets
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth. Build content for diverse user contexts and search histories.
filed 2007-02-21, granted 2015-01-13 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8909655B1 Time based ranking
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2007-10-11, granted 2014-12-09 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US8375073B1 Identification and ranking of news stories of interest
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2007-11-12, granted 2013-02-12 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US8225190B1 Methods and apparatus for clustering news content
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2008-12-24, granted 2012-07-17 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo

US8412699B1 Fresh related search suggestions
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2009-06-12, granted 2013-04-02 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US8126876B2 Systems and methods for improving the ranking of news articles
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2009-07-10, granted 2012-02-28 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo

US8229873B1 News topic-interest-based recommendations twiddling
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2009-09-18, granted 2012-07-24 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8700610B1 Systems and methods for providing news alerts
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Write for natural spoken language and conversational queries.
filed 2009-09-29, granted 2014-04-15 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9129012B2 Information search system with real-time feedback
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Build content for diverse user contexts and search histories.
filed 2010-02-03, granted 2015-09-08 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US8924379B1 Temporal-based score adjustments
Rule: When your topic spikes, ship fresh coverage fast. Freshness weighs more exactly when demand rises. The leak's isHotdoc, freshboxArticleScores, and timeSensitivity fields confirm trending/freshness detection.
filed 2010-03-05, granted 2014-12-30 · leak modules: PerDocData, QualityTimebasedSyntacticDate, QualityNsrNsrData

US8332399B1 Labeling events in historic news
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Write for natural spoken language and conversational queries.
filed 2010-05-04, granted 2012-12-11 · leak modules: NlpSaftDocument, PerDocData, SmartphonePerDocData

US8095528B2 Decentralised web annotation
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2010-06-07, granted 2012-01-10 · leak modules: IndexingDocjoinerAnchorSpamInfo, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US8082244B2 Systems and methods for determining document freshness
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2010-08-11, granted 2011-12-20 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8843856B2 Presenting real-time search results
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2010-12-03, granted 2014-09-23 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9043319B1 Generating real-time search results
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2010-12-03, granted 2015-05-26 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, IndexingDupsLocalizedLocalizedCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US9158775B1 Scoring stream items in real time
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-04-29, granted 2015-10-13 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8234273B2 Document scoring based on document content update
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active.
filed 2011-06-30, granted 2012-07-31 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, IndexingDocjoinerAnchorStatistics, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8527524B2 Document scoring based on document content update
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active.
filed 2011-06-30, granted 2013-09-03 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, IndexingDocjoinerAnchorStatistics, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8775941B1 System for monitoring and reporting deviations of real-time analytics data from expected analytics data
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2011-07-20, granted 2014-07-08 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8782162B1 System for merging and comparing real-time analytics data with conventional analytics data

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2011-07-20, granted 2014-07-15 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8996629B1 Generating a stream of content for a channel

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2011-09-02, granted 2015-03-31 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US8549014B2 Document scoring based on document content update

Rule: Update meaningfully, not cosmetically, and date honestly. How much changed and how often is measured. The leak's lastSignificantUpdate and trustSyntacticDateInRanking fields confirm update magnitude is tracked.

filed 2011-09-14, granted 2013-10-01 · leak modules: PerDocData, QualityTimebasedSyntacticDate, QualityNsrNsrData

US9032309B2 Temporal task-based tab management

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2011-09-26, granted 2015-05-12 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US20130144869A1 Trending

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2011-12-06 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate

US8849829B2 Trending search magazines

Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.

filed 2011-12-06, granted 2014-09-30 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US8515952B2 Systems and methods for determining document freshness

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2011-12-19, granted 2013-08-20 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8412704B1 Providing blog posts relevant to search results

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Optimize mobile experience. Mobile-specific signals are tracked separately. Earn promotion through authoritative content and entity association.

filed 2011-12-29, granted 2013-04-02 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, VideoPerDocData

US8463767B2 Decentralised web annotation

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.

filed 2012-01-04, granted 2013-06-11 · leak modules: IndexingDocjoinerAnchorSpamInfo, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US9569429B2 Translated news

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2012-02-03, granted 2017-02-14 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US8332382B2 Systems and methods for improving the ranking of news articles

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2012-02-24, granted 2012-12-11 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo

US9984155B2 Inline discussions in search results around real-time clusterings
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use descriptive alt text and surrounding context for images.
filed 2012-06-07, granted 2018-05-29 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityNsrPQData, RepositoryWebrefEntityJoin, SmartphonePerDocData

US9111218B1 Method and system for remediating topic drift in near-real-time classification of customer feedback
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-06-22, granted 2015-08-18 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9361369B1 Method and apparatus for clustering news online content based on content freshness and quality of content source
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-07-13, granted 2016-06-07 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo

US8682826B2 News topic-interest-based recommendations twiddling
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-07-23, granted 2014-03-25 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8832088B1 Freshness-based ranking
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-07-30, granted 2014-09-09 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US8874558B1 Promoting fresh content for authoritative channels
Rule: Build topical authority by consistently publishing high-quality content on a specific topic. New content from authoritative channels gets a freshness boost, but only if the channel has sufficient content volume—channels with too few items are penalized. The topic distribution of your content items determines what terms you're authoritative for. The leak's QualityAuthorityTopicEmbeddings and NsrDataClusterUplift modules confirm Google tracks per-channel topical authority and applies cluster-level uplift. Publish regularly on your core topic to maintain and grow authority.
filed 2012-09-11, granted 2014-10-28 · leak modules: QualityNsrNsrData, QualityNsrNsrDataClusterUplift, QualityNsrNsrDataMetadata, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityAuthorityTopicEmbeddingsVersionedItem, QualitySalientTermsSalientTermSet, QualitySalientTermsSignalData

US8645368B2 Systems and methods for improving the ranking of news articles
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2012-09-14, granted 2014-02-04 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo

US8655906B1 Method and system for displaying real time trends
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-09-14, granted 2014-02-18 · leak modules: CompressedQualitySignals, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8838604B1 Labeling events in historic news
Rule: Write for natural spoken language and conversational queries.
filed 2012-09-14, granted 2014-09-16 · leak modules: NlpSaftDocument, PerDocData

US9300757B1 Personalizing aggregated news content
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-09-14, granted 2016-03-29 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9760629B1 Systems and methods for implementing a news round table
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Write for natural spoken language and conversational queries.
filed 2012-09-14, granted 2017-09-12 · leak modules: NlpSaftDocument, PerDocData, SmartphonePerDocData

US8667037B1 Identification and ranking of news stories of interest
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-02-08, granted 2014-03-04 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US9098511B1 Watch time based ranking
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-03-06, granted 2015-08-04 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US9384242B1 Discovery of news-related content
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-03-14, granted 2016-07-05 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo

US8782071B1 Fresh related search suggestions
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-03-15, granted 2014-07-15 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals

US9189526B1 Freshness based ranking
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-03-15, granted 2015-11-17 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US9430454B2 Real-time document sharing and editing
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-05-02, granted 2016-08-30 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9852120B2 Collaborative access spreadsheet with a real-time visual indication identifying last edit user
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth. Write for natural spoken language and conversational queries.
filed 2013-05-20, granted 2017-12-26 · leak modules: NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8918384B2 Decentralized web annotation
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2013-05-31, granted 2014-12-23 · leak modules: IndexingDocjoinerAnchorSpamInfo, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US9146980B1 Temporal content selection
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2013-06-24, granted 2015-09-29 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20170116314A1 Integrating real-time news with historic events
Rule: Use consistent entity names, sameAs schema, and structured data. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-07-08 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityNsrPQData, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefDocumentMetadata

US9875221B1 Real-time document sharing and editing
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-08-15, granted 2018-01-23 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US8898166B1 Temporal content selection
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2013-12-20, granted 2014-11-25 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9037575B2 Systems and methods for improving the ranking of news articles
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-12-24, granted 2015-05-19 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo

US9569547B2 Generating a news timeline
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-12-31, granted 2017-02-14 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9405805B2 Identification and ranking of news stories of interest
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-01-24, granted 2016-08-02 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US10387507B2 Systems and methods for personalizing aggregated news content
Rule: Build content for diverse user contexts and search histories. Write for natural spoken language and conversational queries.
filed 2014-03-17, granted 2019-08-20 · leak modules: NlpSaftDocument, PerDocData, QualityNsrNsrData

US9514232B1 Systems and methods for providing news alerts
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Write for natural spoken language and conversational queries.
filed 2014-03-31, granted 2016-12-06 · leak modules: NlpSaftDocument, PerDocData, SmartphonePerDocData

US9626407B2 Real-time saved-query updates for a large graph
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-06-17, granted 2017-04-18 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9535999B1 Trending search magazines
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-08-25, granted 2017-01-03 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityWebanswersVideoTranscriptAnnotations

US9672277B2 Presenting real-time search results
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2014-09-22, granted 2017-06-06 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9477715B1 Personalizing aggregated news content
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2014-10-30, granted 2016-10-25 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9152678B1 Time based ranking
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2014-12-08, granted 2015-10-06 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US9858308B2 Real-time content recommendation system
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2015-01-16, granted 2018-01-02 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US10162896B1 Event stream architecture for syncing events
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-02-13, granted 2018-12-25 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9712588B1 Generating a stream of content for a channel
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-02-23, granted 2017-07-18 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US10459926B2 Systems and methods for improving the ranking of news articles
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2015-04-27, granted 2019-10-29 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo

US9507826B1 Generating real-time search results
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-05-22, granted 2016-11-29 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, IndexingDupsLocalizedLocalizedCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US10133809B1 Watch time based ranking
Rule: Optimize video content for retention and watch time — longer engagement directly boosts ranking via watch-time multipliers. Use chapter markers and progressive disclosure to keep viewers watching. For web pages, design for dwell time as the same attention-time mechanism applies to page review duration.
filed 2015-06-29, granted 2018-11-20 · leak modules: QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNavboostCrapsFeatureCrapsData, QualityNavboostCrapsStatsWithWeightsProto, PerDocData

US10257140B1 Content sharing to represent user communications in real-time collaboration sessions
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-08-04, granted 2019-04-09 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9858275B1 Scoring stream items in real time
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-09-01, granted 2018-01-02 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9679043B1 Temporal content selection
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2015-09-25, granted 2017-06-13 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10389723B1 Systems and methods for monetizing subscription and archival news content
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Structure answers: question heading, tight answer, then depth.
filed 2015-12-17, granted 2019-08-20 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10095752B1 Methods and apparatus for clustering news online content based on content freshness and quality of content source

Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-05-03, granted 2018-10-09 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo

US10235428B2 Discovery of news-related content

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2016-06-28, granted 2019-03-19 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US10599738B1 Real-time generation of an improved graphical user interface for overlapping electronic content

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2016-06-28, granted 2020-03-24 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11086883B2 Systems and methods for suggesting content to a writer based on contents of a document

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.

filed 2016-08-31, granted 2021-08-10 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US10078702B1 Personalizing aggregated news content

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2016-09-23, granted 2018-09-18 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9792336B1 Generating real-time search results

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2016-11-28, granted 2017-10-17 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, IndexingDupsLocalizedLocalizedCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US10397359B2 Streaming media cache for media streaming service

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2016-12-30, granted 2019-08-27 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9996581B2 Real-time saved-query updates for a large graph

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2017-04-17, granted 2018-06-12 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US10628453B1 Temporal content selection

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.

filed 2017-05-18, granted 2020-04-21 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10742435B2 Proactive provision of new content to group chat participants

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2017-06-29, granted 2020-08-11 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US10678807B1 Generating real-time search results
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2017-09-01, granted 2020-06-09 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, IndexingDupsLocalizedLocalizedCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US10740551B2 Collaborative access spreadsheet with a real-time visual indication identifying last edit user
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Write for natural spoken language and conversational queries.
filed 2017-12-12, granted 2020-08-11 · leak modules: NlpSaftDocument, PerDocData, SmartphonePerDocData

US10417319B1 Real-time document sharing and editing
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2018-01-22, granted 2019-09-17 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US11194868B1 Providing supplemental information in news search
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Write for intent and meaning. Neural matching connects related concepts. Optimize video for watch time and retention.
filed 2018-09-28, granted 2021-12-07 · leak modules: NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, VideoPerDocData

US11443202B2 Real-time on the fly generation of feature-based label embeddings via machine learning
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2019-08-27, granted 2022-09-13 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US11347821B2 Real-time generation of an improved graphical user interface for overlapping electronic content
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2020-02-13, granted 2022-05-31 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11552814B2 Proactive provision of new content to group chat participants
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2020-07-13, granted 2023-01-10 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US20260094124A1 Generative Model for Generating Personalized Newsletters for a Publisher
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2024-09-27 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData

US12561366B2 Real-time micro-profile generation using a dynamic tree structure
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2024-10-17, granted 2026-02-24 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

ML systems (63)

US7886217B1 Identification of web sites that contain session identifiers
Rule: Consolidate near-duplicates. Use canonical tags and consistent URLs. Write in natural vocabulary. ML models understand context and fluff. Structure answers: question heading, tight answer, then depth.
filed 2003-09-29, granted 2011-02-08 · leak modules: CompressedQualitySignals, IndexingDupsLocalizedLocalizedCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US7222127B1 Large scale machine learning systems and methods

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Avoid AI-stitched content. GibberishScore uses language model scoring.

filed 2003-12-15, granted 2007-05-22 · leak modules: IndexingDocjoinerAnchorSpamInfo, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SpamBrainData

US7769763B1 Large scale machine learning systems and methods

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2007-04-17, granted 2010-08-03 · leak modules: IndexingDocjoinerAnchorSpamInfo, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8024372B2 Method and apparatus for learning a probabilistic generative model for text

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2007-04-27, granted 2011-09-20 · leak modules: CompressedQualitySignals, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8255812B1 Embedding user-selected content feed items in a webpage

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Structure answers: question heading, tight answer, then depth.

filed 2008-03-17, granted 2012-08-28 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US8180725B1 Method and apparatus for selecting links to include in a probabilistic generative model for text

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2008-07-21, granted 2012-05-15 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, TrawlerFetchReplyData

US8296257B1 Comparing models

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2009-04-08, granted 2012-10-23 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8868402B2 Construction of text classifiers

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2009-12-30, granted 2014-10-21 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US8195674B1 Large scale machine learning systems and methods

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2010-06-24, granted 2012-06-05 · leak modules: IndexingDocjoinerAnchorSpamInfo, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9110922B2 Joint embedding for item association

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2011-02-01, granted 2015-08-18 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US8356030B2 Domain-specific sentiment classification

Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2011-06-17, granted 2013-01-15 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8364618B1 Large scale machine learning systems and methods

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Avoid AI-stitched content. GibberishScore uses language model scoring.

filed 2012-06-04, granted 2013-01-29 · leak modules: IndexingDocjoinerAnchorSpamInfo, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SpamBrainData

US9213946B1 Comparing models

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.

filed 2012-10-16, granted 2015-12-15 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US9507858B1 Selectively merging clusters of conceptually related words in a generative model for text

Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.

filed 2012-11-27, granted 2016-11-29 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US8688705B1 Large scale machine learning systems and methods

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Avoid AI-stitched content. GibberishScore uses language model scoring.

filed 2013-01-28, granted 2014-04-01 · leak modules: IndexingDocjoinerAnchorSpamInfo, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SpamBrainData

US9104982B2 Prevention of invalid selections based on machine learning of user-specific latency

Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2013-03-15, granted 2015-08-11 · leak modules: ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US9760644B2 Embedding event creation link in a document

Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.

filed 2013-04-17, granted 2017-09-12 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US9317564B1 Construction of text classifiers

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2014-10-20, granted 2016-04-19 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityOrbitAsteroidBeltDocumentIntentScores

US20180314948A1 Generating multiple language training data for seach classifier

Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.

filed 2015-06-25 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US20170228414A1 Generating feature embeddings from a co-occurrence matrix

Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.

filed 2017-02-03 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US11188824B2 Cooperatively training and/or using separate input and subsequent content neural networks for information retrieval

Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.

filed 2017-03-31, granted 2021-11-30 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations

US10579682B1 Machine-learned disambiguation of user action data
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-04-20, granted 2020-03-03 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10642830B2 Context aware chat history assistance using machine-learned models
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2017-06-12, granted 2020-05-05 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US10482394B2 Large-scale in-database machine learning with pure SQL
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2017-06-13, granted 2019-11-19 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US10567845B2 Embeddable media content search widget
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2017-10-30, granted 2020-02-18 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo

US11488054B2 Systems and methods for distributed on-device learning with data-correlated availability
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2017-12-06, granted 2022-11-01 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US20190148023A1 Machine-Learned Epidemiology
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2018-01-12 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11568003B2 Refined search with machine learning
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2018-06-13, granted 2023-01-31 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US12165032B2 Neural networks with area attention
Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately. Write for natural spoken language and conversational queries.
filed 2019-09-27, granted 2024-12-10 · leak modules: NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData

US11899733B2 Method and system for activity prediction, prefetching and preloading of computer assets by a client-device
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2020-01-14, granted 2024-02-13 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US11361046B2 Machine learning classification of an application link as broken or working
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2020-02-21, granted 2022-06-14 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US20230111978A1 Cross-example softmax and/or cross-example negative mining
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2020-03-18 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US20230267277A1 Systems and methods for using document activity logs to train machine-learned models for determining document relevance
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2020-06-15 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US12566997B2 Processing of machine learning modeling data to improve accuracy of categorization
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2021-04-09, granted 2026-03-03 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US12579150B2 Hybrid and hierarchical multi-trial and OneShot neural architecture search on datacenter machine learning accelerators
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2022-04-15, granted 2026-03-17 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US12165024B2 Systems and methods for distributed on-device learning with data-correlated availability
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2022-10-17, granted 2024-12-10 · leak modules: PerDocData, SmartphonePerDocData

US20250272485A1 Deep learning system for navigating feedback
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2022-10-20 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation

US12524476B2 Refined search with machine learning
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2022-12-21, granted 2026-01-13 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US20250252137A1 Zero-Shot Multi-Modal Data Processing Via Structured Inter-Model Communication
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2023-03-31 · leak modules: ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US20230273923A1 Generating and/or utilizing a machine learning model in response to a search request
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2023-04-06 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US20250356166A1 Recurrence in transformer architecture
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2023-05-23 · leak modules: PerDocData, SmartphonePerDocData

US20240289384A1 Local Node Embeddings for Heterogeneous Graphs
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2023-05-25 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20250013915A1 Reinforcement Learning with Information Retrieval Feedback
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2023-07-07 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityPreviewSnippetRadishFeatures

US20250021842A1 Monitoring generative model quality
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2023-07-13 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData

US20250068396A1 Automatic generation of no-code embedded data-driven application views
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2023-08-22 · leak modules: CompressedQualitySignals, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefEntityJoin, RichsnippetsPageMap, SmartphonePerDocData

US20250078484A1 Multimodal embeddings
Rule: Write in natural vocabulary. ML models understand context and fluff. Write for intent and meaning. Neural matching connects related concepts. Write for natural spoken language and conversational queries.
filed 2023-09-05 · leak modules: NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityRankembedMustangMustangRankEmbedInfo

US12111834B1 Ambient multi-device framework for agent companions
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2023-12-20, granted 2024-10-08 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US20240256965A1 Instruction Fine-Tuning Machine-Learned Models Using Intermediate Reasoning Steps
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2024-01-26 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores

US20250265285A1 Computing Tool Retrieval Using Sequence Processing Models
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2024-02-16 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US20240289407A1 Search with stateful chat
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2024-02-27 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US20260080025A1 Resource Locator Prediction for Shortcut Generation
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2024-05-02 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US20250086246A1 Landing Page Optimization Using Machine-Learning Techniques
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2024-07-16 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData

US20260023747A1 Utilizing previous intermediate model output for generating responses
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2024-07-16 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US12566764B2 Ambient multi-device framework for agent companions
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2024-09-04, granted 2026-03-03 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US20260072965A1 Controlled content diversity in retrieval for generative search
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2024-09-06 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US20260080012A1 Preemptive generation of generative model output(s)
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2024-09-17 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US20250094521A1 Generative navigational corpus
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Write in natural vocabulary. ML models understand context and fluff.
filed 2024-09-18 · leak modules: NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US20260127232A1 Merging generative model prompts based on context
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2024-11-01 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US12536233B1 AI-generated content page tailored to a specific user
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2025-01-03, granted 2026-01-27 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US20250355958A1 On-Demand Generative Response Simplification
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2025-05-14 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US20260064780A1 Dynamic attribution and/or modification of responsive content that is generated using a retrieval augmented generation (rag) process
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2025-06-18 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US20260064984A1 Dynamic attribution and/or modification of ai-generated content based on contextual data
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2025-06-18 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures

US20260050742A1 Ambient Sensor Representations for Contextual Inputs in Machine-Learned Sequence Processing Models
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2025-08-15 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

Links & anchors (51)

US9092535B1 E-mail embedded textual hyperlink object
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 1999-09-21, granted 2015-07-28 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US6754873B1 Techniques for finding related hyperlinked documents using link-based analysis
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2000-04-06, granted 2004-06-22 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US7213198B1 Link based clustering of hyperlinked documents
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2000-08-10, granted 2007-05-01 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, RepositoryAnnotationsGeoTopicality

US7305610B1 Distributed crawling of hyperlinked documents
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2000-08-14, granted 2007-12-04 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, ShingleInfoPerDocData, SmartphonePerDocData

US7089490B1 Identifying navigation bars and objectionable navigation bars
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2000-11-30, granted 2006-08-08 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData, SnippetExtraInfo

US8127220B1 Scoring links in a document
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2000-12-13, granted 2012-02-28 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals

US7146358B1 Systems and methods for using anchor text as parallel corpora for cross-language information retrieval
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Write for natural spoken language and conversational queries.
filed 2001-08-28, granted 2006-12-05 · leak modules: NlpSaftDocument, PerDocData, SmartphonePerDocData

US7308643B1 Anchor tag indexing in a web crawler system
Rule: Write anchors that describe the target page. Those words become part of how the target is indexed. The leak's anchorCount, offdomainAnchorCount, and spamLog10Odds in IndexingDocjoinerAnchorStatistics confirm anchor text is a primary indexing input.
filed 2003-07-03, granted 2007-12-11 · leak modules: IndexingDocjoinerAnchorStatistics, AnchorsAnchor, PerDocData

US7634716B1 Techniques for finding related hyperlinked documents using link-based analysis
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2003-12-30, granted 2009-12-15 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US20050149851A1 Generating hyperlinks and anchor text in HTML and non-HTML documents
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2003-12-31 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, RepositoryAnnotationsGeoTopicality, RepositoryWebrefEntityJoin

US7260573B1 Personalizing anchor text scores in a search engine
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active.
filed 2004-05-17, granted 2007-08-21 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDocjoinerAnchorStatistics, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate

US7509344B1 Method for detecting link spam in hyperlinked databases
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2004-08-18, granted 2009-03-24 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDocjoinerAnchorSpamInfo, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, ShingleInfoPerDocData

US9569555B2 Method for adding a user-selectable function to a hyperlink
Rule: Use descriptive alt text and surrounding context for images. Structure answers: question heading, tight answer, then depth.
filed 2005-01-21, granted 2017-02-14 · leak modules: ImageData, ImagePerDocData, QualityPreviewSnippetRadishFeatures, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US9208229B2 Anchor text summarization for corroboration
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2006-03-31, granted 2015-12-08 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata

US7814103B1 Systems and methods for using anchor text as parallel corpora for cross-language information retrieval
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Write for natural spoken language and conversational queries.
filed 2006-08-30, granted 2010-10-12 · leak modules: NlpSaftDocument, PerDocData, SmartphonePerDocData

US9165040B1 Producing a ranking for pages using distances in a web-link graph
Rule: Chase links from sites the trusted hubs of your niche already link to. One link two hops from a seed beats fifty from nowhere. The leak's homepagePagerankNs and siteAuthority confirm proximity-to-trust scoring.
filed 2006-10-12, granted 2015-10-20 · leak modules: PerDocData, AnchorsAnchor, CompressedQualitySignals, QualityNsrNsrData

US8738643B1 Learning synonymous object names from anchor texts
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2007-08-02, granted 2014-05-27 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate

US8266134B1 Distributed crawling of hyperlinked documents
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2007-10-24, granted 2012-09-11 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, ShingleInfoPerDocData, SmartphonePerDocData

US8484548B1 Anchor tag indexing in a web crawler system
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2007-11-07, granted 2013-07-09 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US7809697B1 Compressing hyperlinks in a hyperlink-based document
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2007-11-09, granted 2010-10-05 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US8732187B1 Link-based ranking of objects that do not include explicitly defined links
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2008-04-08, granted 2014-05-20 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorStatistics, NlpSaftDocument, PerDocData

US7953763B2 Method for detecting link spam in hyperlinked databases
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2009-03-24, granted 2011-05-31 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDocjoinerAnchorSpamInfo, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, ShingleInfoPerDocData

US8788490B1 Link based locale identification for domains and domain content
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2009-06-26, granted 2014-07-22 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate

US8615508B2 Artificial anchor for a document
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2009-07-24, granted 2013-12-24 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US8782507B2 Framework for providing visual context to WWW hyperlinks
Rule: Monitor this patent for ranking implications.
filed 2010-03-09, granted 2014-07-15

US8959093B1 Ranking search results based on anchors
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2010-03-15, granted 2015-02-17 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorStatistics, LocalWWWInfo, LocalWWWInfoCluster

US9047259B1 System and method for renaming hyperlinks
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth. Use JSON-LD structured data for entity types and relationships.
filed 2010-07-21, granted 2015-06-02 · leak modules: PerDocData, QualityPreviewSnippetRadishFeatures, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData, SnippetExtraInfo, SnippetExtraInfoSnippetCandidateInfo

US7996402B1 Systems and methods for using anchor text as parallel corpora for cross-language information retrieval
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Write for natural spoken language and conversational queries.
filed 2010-08-31, granted 2011-08-09 · leak modules: NlpSaftDocument, PerDocData, SmartphonePerDocData

US8407231B2 Document scoring based on link-based criteria
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build quality links with relevant anchor text. Penguin is still active.
filed 2010-10-01, granted 2013-03-26 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, IndexingDocjoinerAnchorStatistics, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData

US8516357B1 Link based clustering of hyperlinked documents
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2010-12-02, granted 2013-08-20 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, RepositoryAnnotationsGeoTopicality

US8719276B1 Ranking nodes in a linked database based on node independence
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data.
filed 2011-01-04, granted 2014-05-06 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorStatistics, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US8612846B2 Framework for providing visual context to www hyperlink
Rule: Monitor this patent for ranking implications.
filed 2011-01-19, granted 2013-12-17

US8250114B2 Method for detecting link spam in hyperlinked databases
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2011-05-31, granted 2012-08-21 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDocjoinerAnchorSpamInfo, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, ShingleInfoPerDocData

US8190608B1 Systems and methods for using anchor text as parallel corpora for cross-language information retrieval
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Write for natural spoken language and conversational queries.
filed 2011-06-30, granted 2012-05-29 · leak modules: NlpSaftDocument, PerDocData, SmartphonePerDocData

US8595270B2 Artificial anchor for a document
Rule: Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2011-09-29, granted 2013-11-26 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US9305091B2 Anchor tag indexing in a web crawler system
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2011-11-18, granted 2016-04-05 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US20150193389A1 Presenting updated hyperlink information on a webpage
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-03-06 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US8631010B1 Systems and methods for using anchor text as parallel corpora for cross-language information retrieval
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Write for natural spoken language and conversational queries.
filed 2012-05-18, granted 2014-01-14 · leak modules: NlpSaftDocument, PerDocData, SmartphonePerDocData

US9053177B1 Sitelinks based on visual location
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-06-11, granted 2015-06-09 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US8812478B1 Distributed crawling of hyperlinked documents
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2012-09-10, granted 2014-08-19 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, ShingleInfoPerDocData, SmartphonePerDocData

US20150193390A1 Visited hyperlink appearance based on user activity
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Write in natural vocabulary. ML models understand context and fluff.
filed 2012-12-03 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, SmartphonePerDocData

US9298777B2 Personalization of web search results using term, category, and link-based user profiles
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-01-07, granted 2016-03-29 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData

US9430578B2 System and method for anchoring third party metadata in a document
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-03-15, granted 2016-08-30 · leak modules: ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9448994B1 Grammar extraction using anchor text
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately. Write for natural spoken language and conversational queries.
filed 2013-09-27, granted 2016-09-20 · leak modules: NlpSaftDocument, PerDocData, SmartphonePerDocData

US9953049B1 Producing a ranking for pages using distances in a web-link graph
Rule: This is TrustRank-style distance-based ranking. Pages closer to trusted seed sites rank higher. Build links from sites that are themselves close to seed/trusted sites. Links from pages with fewer outgoing links carry more weight (each link's value is diluted by total outbound links). Focus on getting links from authoritative, well-connected sites rather than many links from obscure sites.
filed 2015-10-19, granted 2018-04-24 · leak modules: link_analysis, pagerank, ranking, trust, link_graph

US9977816B1 Link-based ranking of objects that do not include explicitly defined links
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2015-12-10, granted 2018-05-22 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorStatistics, NlpSaftDocument, PerDocData

US10839029B2 Personalization of web search results using term, category, and link-based user profiles
Rule: Google personalizes rankings using link-based user profiles - pages linked from documents a user has visited are more likely to rank for that user. Build content networks where your pages link to each other contextually. User behavior history (visited pages, clicked links) creates link-based personalization signals. Focus on building topical authority clusters where interlinked content reinforces user interest profiles.
filed 2016-03-03, granted 2020-11-17 · leak modules: link_analysis, personalization, ranking, navboost, user_profile

US10210256B2 Anchor tag indexing in a web crawler system
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2016-04-01, granted 2019-02-19 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData

US10795959B2 Conversion latency reduction in online chat-based communication infrastructure
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2016-06-01, granted 2020-10-06 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US20250103662A1 Unifying transformers with link based ranking
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2023-09-21 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20250148024A1 Independent content hosting system
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2024-11-05 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, NlpSaftDocument, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

Quality & spam (44)

US7302645B1 Methods and systems for identifying manipulated articles
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2003-12-10, granted 2007-11-27 · leak modules: IndexingDocjoinerAnchorSpamInfo, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefEntityJoin, SmartphonePerDocData

US7962462B1 Deriving and using document and site quality signals from search query streams
Rule: Build reasons for people to search your name: a tool, a study, a stance. Brand demand is measurable quality. The leak's chromeInTotal, directFrac, and pnav fields confirm brand/direct traffic as site-level quality signals.
filed 2005-05-31, granted 2011-06-14 · leak modules: QualityNsrNsrData, QualityNavboostCrapsCrapsClickSignals, CompressedQualitySignals

US7743045B2 Detecting spam related and biased contexts for programmable search engines
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2005-08-10, granted 2010-06-22 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, IndexingDocjoinerAnchorStatistics, LocalWWWInfo

US8290964B1 Method and apparatus for obtaining recommendations from trusted sources
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build content for diverse user contexts and search histories.
filed 2006-01-17, granted 2012-10-16 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US7603350B1 Search result ranking based on trust
Rule: Earn endorsements from sources your niche already trusts. Trust flows through labelled references, not claims about yourself. The leak's unauthoritativeScore, scamness, and siteAuthority fields confirm trust is quantified.
filed 2006-05-09, granted 2009-10-13 · leak modules: CompressedQualitySignals, QualityNsrNsrData, PerDocData

US8131712B1 Regional indexes
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2007-10-15, granted 2012-03-06 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US8752184B1 Spam detection for user-generated multimedia items based on keyword stuffing
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2008-01-17, granted 2014-06-10 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication

US8745056B1 Spam detection for user-generated multimedia items based on concept clustering
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2008-03-31, granted 2014-06-03 · leak modules: IndexingDocjoinerAnchorSpamInfo, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US8554769B1 Identifying gibberish content in resources
Rule: If a page reads assembled rather than written, fix it or kill it. Language-model detection is patented, deployed tech. The leak's GibberishScore, KeywordStuffingScore, DocLevelSpamScore, and spambrainTotalDocSpamScore fields are the direct implementation.
filed 2009-06-17, granted 2013-10-08 · leak modules: PerDocData, CompressedQualitySignals, QualityNsrNsrData

US8392823B1 Systems and methods for detecting hidden text and hidden links
Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately. Write for natural spoken language and conversational queries.
filed 2009-08-25, granted 2013-03-05 · leak modules: NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData

US8352467B1 Search result ranking based on trust
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data.
filed 2009-09-02, granted 2013-01-08 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, ImageData, ImagePerDocData, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US8078629B2 Detecting spam documents in a phrase based information retrieval system
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2009-10-13, granted 2011-12-13 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, IndexingDocjoinerAnchorStatistics, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData

US8452746B2 Detecting spam search results for context processed search queries
Rule: Google uses annotation files to flag spam in programmable search contexts—your site can be removed from results if flagged as spam in these annotations. Avoid patterns that trigger spam detection in third-party/vertical search contexts. The leak's SpamBrainScore, SpamBrainData, and AnchorSpamInfo modules confirm Google actively tracks spam signals at the anchor and document level. Ensure your anchor text patterns and linking behavior don't trigger spam classifiers.
filed 2010-04-30, granted 2013-05-28 · leak modules: SpamBrainScore, SpamBrainData, SpamMuppetjoinsMuppetSignals, IndexingDocjoinerAnchorSpamInfo, IndexingDocjoinerAnchorPhraseSpamInfo, SpamCookbookAction, QualityLabelsGoogleLabelData, QualitySherlockKnexAnnotation

US20150154301A1 Method and system for testing spam result detection algorithms
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2011-04-25 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US8738557B1 Detection of spam using contextual analysis of data sources
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-07-26, granted 2014-05-27 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8612436B1 Reverse engineering circumvention of spam detection algorithms
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2011-09-27, granted 2013-12-17 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, RepositoryAnnotationsGeoTopicality

US9177065B1 Quality score for posts in social networking services
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2012-02-09, granted 2015-11-03 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8620950B1 Regional indexes
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2012-02-17, granted 2013-12-31 · leak modules: CompressedQualitySignals, CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData

US8850305B1 Automatic detection and manipulation of calls to action in web pages
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-02-28, granted 2014-09-30 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData

US8818982B1 Deriving and using document and site quality signals from search query streams
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-04-25, granted 2014-08-26 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityTimebasedLastSignificantUpdate

US8639680B1 Hidden text detection for search result scoring
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-05-07, granted 2014-01-28 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US9002832B1 Classifying sites as low quality sites
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2012-06-04, granted 2015-04-07 · leak modules: AnchorsAnchor, AnchorsAnchorSource, CompressedQualitySignals, ImageData, ImagePerDocData, IndexingDocjoinerAnchorSpamInfo, IndexingDocjoinerAnchorStatistics, NlpSemanticParsingQRefAnnotation

US9031929B1 Site quality score
Rule: Plan quality site-wide, not page-by-page. Ask of every section: would users pick this by name? The leak's nsr, impressions, and absoluteImpressions fields confirm site-level query/click aggregation.
filed 2012-06-27, granted 2015-05-12 · leak modules: QualityNsrNsrData, QualityNavboostCrapsCrapsClickSignals, CompressedQualitySignals

US9390193B2 Delay the persistent memory storage of website data to control data that is stored in the browser
Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2012-07-16, granted 2016-07-12 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData

US8762394B1 Method and apparatus for obtaining recommendations from trusted sources
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Build content for diverse user contexts and search histories.
filed 2012-09-13, granted 2014-06-24 · leak modules: CompressedQualitySignals, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US8818995B1 Search result ranking based on trust
Rule: Google tracks trust relationships derived from user browsing behavior and uses them to weight annotations/labels on your content. Encourage repeat visits and user engagement on your site—trust decays for sites users don't revisit. Also, getting annotations/labels from trusted entities (expert reviewers, authoritative sites) for your content can boost your relevance scores. The leak's Weboftrust and GoogleLabelData modules confirm this trust-annotation system is live.
filed 2012-09-13, granted 2014-08-26 · leak modules: QualityLabelsGoogleLabelData, QualitySherlockKnexAnnotation, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, WeboftrustLiveResultDocBoostData, WeboftrustLiveResultsDocAttachments, StorageGraphBfgAuthorityFeedbackMetadata, CompressedQualitySignals

US9483566B2 System and method for determining the legitimacy of a listing
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-01-23, granted 2016-11-01 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US8949174B2 Detection and prevention of unwanted content on cloud-hosted services
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2013-02-22, granted 2015-02-03 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9043398B2 Detection and prevention of unwanted content on cloud-hosted services
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2013-02-22, granted 2015-05-26 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9369483B2 Detection and prevention of unwanted content on cloud-hosted services
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2013-02-22, granted 2016-06-14 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9195944B1 Scoring site quality
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-02-25, granted 2015-11-24 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US9767157B2 Predicting site quality
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-03-15, granted 2017-09-19 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9727644B1 Determining a quality score for a content item
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2013-09-25, granted 2017-08-08 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

US9372896B1 Reverse engineering circumvention of spam detection algorithms
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs. Use consistent entity names, sameAs schema, and structured data.
filed 2013-11-26, granted 2016-06-21 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, RepositoryAnnotationsGeoTopicality

US9336279B2 Hidden text detection for search result scoring
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2014-01-27, granted 2016-05-10 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US8909591B1 Detection of spam using contextual analysis of data sources
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-04-16, granted 2014-12-09 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US9208157B1 Spam detection for user-generated multimedia items based on concept clustering
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2014-05-15, granted 2015-12-08 · leak modules: IndexingDocjoinerAnchorSpamInfo, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap

US10268641B1 Search result ranking based on trust
Rule: Build entity-level trust signals. If your brand/entity provides annotations or structured data that Google uses, your trust rank affects how your content ranks for users who trust you. Encourage users to interact with your content and brand entity. Author authority and entity expertise signals (E-E-A-T) feed into trust-based ranking adjustments.
filed 2014-07-29, granted 2019-04-23 · leak modules: demotion, ranking, trust, personalization, entity, annotation

US9569504B1 Deriving and using document and site quality signals from search query streams
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2014-08-22, granted 2017-02-14 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9225741B1 Detection and prevention of unwanted content on cloud-hosted services
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Consolidate near-duplicates. Use canonical tags and consistent URLs.
filed 2015-01-16, granted 2015-12-29 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, IndexingDupsLocalizedLocalizedCluster, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityNsrPQData

US9760641B1 Site quality score
Rule: Nothing new here, and that is the point: treat site-level quality as a standing budget line, not a one-off fix. The leak confirms NSR scoring is still active with 58+ fields in QualityNsrNsrData.
filed 2015-05-11, granted 2017-09-12 · leak modules: QualityNsrNsrData, QualityNavboostCrapsCrapsClickSignals, CompressedQualitySignals

US10133765B1 Quality score for posts in social networking services
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-09-21, granted 2018-11-20 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US10402457B1 Methods and systems for correlating connections between users and links between articles
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2016-01-26, granted 2019-09-03 · leak modules: IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US10565267B1 Determining a quality score for a content item
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images.
filed 2017-07-26, granted 2020-02-18 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityNsrPQData

Security & privacy (37)

US7031954B1 Document retrieval system with access control
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 1997-09-10, granted 2006-04-18 · leak modules: NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryWebrefEntityJoin, SmartphonePerDocData

US8056128B1 Systems and methods for detecting potential communications fraud
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2004-09-30, granted 2011-11-08 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US8407250B2 Distribution of content document to varying users with security customization and scalability
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Structure answers: question heading, tight answer, then depth.
filed 2007-06-01, granted 2013-03-26 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US8291492B2 Authentication of a contributor of online content
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches.
filed 2008-12-11, granted 2012-10-16 · leak modules: CompressedQualitySignals, IndexingDocjoinerAnchorSpamInfo, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData

US8732855B2 Launching a cached web application based on authentication status
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2010-09-30, granted 2014-05-20 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9116991B2 Searching encrypted electronic books
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-08-16, granted 2015-08-25 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US8528084B1 Systems and methods for detecting potential communications fraud
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-09-23, granted 2013-09-03 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US8615802B1 Systems and methods for detecting potential communications fraud
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2011-09-23, granted 2013-12-24 · leak modules: CompressedQualitySignals, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US8402554B2 Launching a cached web application based on authentication status
Rule: Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2011-09-30, granted 2013-03-19 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality, SmartphonePerDocData

US9195840B2 Application-specific file type generation and use
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately. Use JSON-LD structured data for entity types and relationships.
filed 2012-04-23, granted 2015-11-24 · leak modules: PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RichsnippetsPageMap, SmartphonePerDocData

US20120222128A1 Distribution of content document with security, customization and scalability
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Structure answers: question heading, tight answer, then depth.
filed 2012-05-10 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US9208298B2 Pass through service login to application login
Rule: Use consistent entity names, sameAs schema, and structured data. Optimize mobile experience. Mobile-specific signals are tracked separately. Structure answers: question heading, tight answer, then depth.
filed 2012-06-18, granted 2015-12-08 · leak modules: PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities, SmartphonePerDocData, SnippetExtraInfo

US9141656B1 Searching using access controls
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2012-09-06, granted 2015-09-22 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US8832151B2 Distribution of content document to varying users with security, customization and scalability
Rule: Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking. Structure answers: question heading, tight answer, then depth.
filed 2012-09-14, granted 2014-09-09 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US9069776B1 Serving encrypted and plain data from a low latency non-volatile memory
Rule: Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2013-03-15, granted 2015-06-30 · leak modules: PerDocData, SmartphonePerDocData

US9195722B1 Privacy-protected search
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-04-01, granted 2015-11-24 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US9448980B1 Organization ownership verification in a web application
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2013-10-29, granted 2016-09-20 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin

US10445769B2 Systems and methods for audience measurement
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2013-12-24, granted 2019-10-15 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate

US20160380992A1 Authentication specific data
Rule: Maintain clean sitemaps. Crawl scheduling uses PageRank + change frequency. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-02-11 · leak modules: CrawlerChangerateUrlChangerate, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations

US9894050B1 Server based settings for client software with asymmetric signing
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2015-08-11, granted 2018-02-13 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate

US10268739B1 User-list management
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2017-08-17, granted 2019-04-23 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US10218505B1 Server based settings for client software with asymmetric signing
Rule: Write titles that match content. goodClicks and lastLongestClicks are key NavBoost signals. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2018-02-09, granted 2019-02-26 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNavboostCrapsCrapsClickSignals, QualityNavboostCrapsCrapsData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate

US20200084213A1 Low-latency differential access controls in a time-series prediction system
Rule: Publish under real, consistent author identity. Agent Rank scores authors. Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data.
filed 2018-09-07 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US11163737B2 Storage and structured search of historical security data
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2018-11-21, granted 2021-11-02 · leak modules: PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US20210350014A1 Encrypted Search over Encrypted Data with Reduced Volume Leakage
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2020-05-05 · leak modules: CompressedQualitySignals, PerDocData, SmartphonePerDocData

US20220318644A1 Privacy preserving machine learning predictions
Rule: Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Use descriptive alt text and surrounding context for images. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2020-10-14 · leak modules: ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, NlpSaftDocument, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures

US12111950B2 Privacy preserving data collection and analysis
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2021-02-08, granted 2024-10-08 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityPreviewSnippetRadishFeatures, QualityTimebasedLastSignificantUpdate

US11843672B2 Privacy preserving centroid models using secure multi-party computation
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2021-09-16, granted 2023-12-12 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US12244693B2 Multi-key information retrieval
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes.
filed 2021-10-19, granted 2025-03-04 · leak modules: CompressedQualitySignals, ImageData, ImagePerDocData, PerDocData, QualityNsrNsrData, QualityNsrPQData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US12393563B2 Storage and structured search of historical security data
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2021-10-20, granted 2025-08-19 · leak modules: PerDocData, QualityNsrNsrData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsRdfaRdfaRichSnippetsApplication, RepositoryWebrefDocumentMetadata, RepositoryWebrefEntityJoin, RepositoryWebrefWebrefEntities

US12380170B2 Performance optimizations for row-level security filters
Rule: Avoid triggers for demotion: low quality, spam patterns, anchor mismatches. Ensure NAP consistency and local relevance. Distance affects ranking. Write in natural vocabulary. ML models understand context and fluff.
filed 2021-12-01, granted 2025-08-05 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality

US20230177543A1 Privacy preserving machine learning expansion models
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2021-12-06 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, NlpSemanticParsingQRefAnnotation, PerDocData, QualityNsrNsrData, QualityOrbitAsteroidBeltDocumentIntentScores, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo

US11720708B2 Privacy preserving data collection and analysis
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2022-03-21, granted 2023-08-08 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, QualityWebanswersVideoTranscriptAnnotations, RepositoryAnnotationsGeoTopicality, RepositoryWebrefDocumentMetadata

US20240273401A1 Privacy preserving transfer learning
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2023-02-14 · leak modules: LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate, QualityTimebasedSyntacticDate, RepositoryAnnotationsGeoTopicality

US20230254126A1 Encrypted search with a public key
Rule: Use descriptive alt text and surrounding context for images. Optimize mobile experience. Mobile-specific signals are tracked separately.
filed 2023-04-18 · leak modules: ImageData, ImagePerDocData, PerDocData, SmartphonePerDocData

US12149594B2 Privacy preserving centroid models using secure multi-party computation
Rule: Use consistent entity names, sameAs schema, and structured data. Update content meaningfully. lastSignificantUpdate tracks magnitude of changes. Ensure NAP consistency and local relevance. Distance affects ranking.
filed 2023-10-30, granted 2024-11-19 · leak modules: CompressedQualitySignals, LocalWWWInfo, LocalWWWInfoCluster, PerDocData, QualityNsrNsrData, QualityPreviewSnippetRadishFeatures, QualityRankembedMustangMustangRankEmbedInfo, QualityTimebasedLastSignificantUpdate

US20240346104A1 Automated decisioning system for acceptability of user privacy notices and controls on a user interface

Rule: Write in natural vocabulary. ML models understand context and fluff. Optimize mobile experience. Mobile-specific signals are tracked separately.

filed 2024-04-12 · leak modules: PerDocData, QualityNsrNsrData, QualityRankembedMustangMustangRankEmbedInfo, SmartphonePerDocData

Unclustered (5)

US6285999B1 Method for node ranking in a linked database

Rule: Spend effort earning references from pages that are themselves referenced. Being cited is the game, not link count. The leak confirms pagerank and pagerankNs are still actively computed for every indexed document.

filed 1998, granted 2001 · leak modules: PerDocData, QualityNsrNsrData, CompressedQualitySignals, AnchorsAnchor

US7487145B1 Method and system for autocompletion using ranked results

Rule: Mine autocomplete around your brand and topics. Those suggestions are ranked demand you can build pages for. The leak's impressions and voterTokenCount fields confirm query popularity aggregation.

filed 2004, granted 2009 · leak modules: QualityNsrNsrData, QualityNavboostCrapsCrapsClickSignals

US9037464B1 Computing numeric representations of words in a high-dimensional space

Rule: Stop counting exact-match keywords. Write in the natural vocabulary of the topic; the vectors already connect it. The leak's titlematchScore and page2vecLq fields confirm vector-based quality assessment.

filed 2013, granted 2015 · leak modules: PerDocData, QualityNsrNsrData, QualityNsrPQData

US10452978B2 Attention-based sequence transduction neural networks

Rule: Write for meaning, not keyword slots. The models reading your page understand context, hedging, and fluff. The leak's contentEffort and page2vecLq fields confirm ML-based content quality assessment.

filed 2018, granted 2019 · leak modules: PerDocData, QualityNsrNsrData, QualityNsrPQData

US11354342B2 Contextual estimation of link information gain

Rule: Before publishing, name what your page adds that the current top results lack. If the answer is nothing, do not publish. The leak's OriginalContentScore in PerDocData directly measures content originality.

filed 2018, granted 2022 · leak modules: PerDocData, QualityNsrPQData, CompressedQualitySignals